

C. JAKE BARLOW

PART ONE

A PRETTY GOOD GUIDE
TO THE PRIMARY EXAM

LIFE IN THE FAST LANE

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Contents

Part One	1
Layout	1
Acknowledgements + Technical Stuff	2
About the Author	3
Major Contributors	3
Copyright + Legal	3
How to Pass	5
Plan for Success	5
Be Strategic	6
Suggested Approach	8
The Bottom Line	10
Prizewinners Perspective	10
References	11
How to Fail	12
References	13
The SAQ	14
Answering the Question	15
The Viva	16
Understanding the Viva	16
Curriculum	18
Research Methods and Statistics	19
Evidence-Based Medicine	20
What is Evidence-Based Medicine?	20
Levels of Evidence	21
Grades of Evidence	24

Study Types	25
Randomised Control Trial	25
Strengths	25
Weaknesses	26
Systematic Review	26
Meta - Analysis	26
Stages of a Meta - Analysis and Systematic Review:	27
Heterogeneity	27
Forest Plots	28
Funnel Plots	29
Strengths and Weaknesses of Meta - analyses	29
References	30
Clinical Trial Design	31
References	32
Data Types	33
Qualitative Data	33
Quantitative Data	33
References	35
Bias and Confounding	36
Bias	36
Confounder	37
Controlling For Confounding	38
References	39
Frequency Distributions	40
The Normal Distribution	40
The Standard Normal Distribution	41
Binomial Distribution	42
Measures of Central Tendency	42
Measures of Dispersion	43
Basic Measures of Deviation	43
Variance	43
Standard Error and Confidence Intervals	45
References	46
Sample Size Calculation	47
Samples	47
Sampling Error	47
Sample Size Calculation	48

References	48
Statistical Tests	50
Parametric Tests	50
Z test	51
Student's T Test	51
F Test	52
Analysis of Variance (ANOVA)	52
Non-Parametric Tests	52
Mann-Whitney U Test/Wilcoxon Rank Sum Test	53
Wilcoxon Signed Ranks Test	53
References	54
Statistical Terms	55
Error, Sensitivity, Specificity, and Predictive Values	55
Error	56
Sensitivity	56
Specificity	57
Positive and Negative Predictive Values	57
Receiver Operating Characteristic	58
Remembering the Difference	59
Likelihood Ratios	60
References	60
Risk and Odds	62
Risk	62
Odds	63
Risk versus Odds	63
References	65
Significance Testing	66
p -values	66
Problems with p -values	67
References	68
Drug Approval and Development	70
Drug Approval	70
Research	70
Clinical Use	71
Phases of Drug Development	71
References	72

Pharmacology	73
Pharmacokinetics	74
Additives	75
References	76
Isomerism	77
Types of Isomerism	77
Structural Isomers	78
Stereoisomers	78
Preparations	80
References	80
Pharmacokinetic Terms	81
Volume of Distribution (V_D)	81
Significance of the Volume of Distribution	82
Half-life	83
Significance of Half-Life	84
Time Constant (τ)	85
Clearance and Rate of Elimination	85
Significance of Clearance	86
Total Clearance	86
Clearance Kinetics	87
Michaelis-Menten	88
References	88
Pharmacokinetic Models	90
One-Compartment Models	90
Steady State	91
Multiple Compartment Models	92
Clearance in Multi-Compartment Models	94
The Effect Site	95
Non-Compartmental Models	97
References	97
Absorption	98
Bioavailability	98
Significance of Bioavailability	99
Enteric Administration	99
Gastric pH	99

First-pass Metabolism	100
References	100
Distribution	102
Diffusion Characteristics	102
Protein Binding	104
Regional Blood Flow	106
References	106
Metabolism	107
Hepatic Metabolism	107
Phase I	108
CYP450 System	108
Key CYP450 Enzymes	109
Phase II	111
Extraction Ratio	111
Organ Independent Metabolism	113
References	114
Elimination and Excretion	115
Renal Excretion	115
Hepatic Excretion	116
Enterohepatic Recirculation	117
References	117
Bolus and Infusion Kinetics	118
Continuous Infusions	118
Stopping an Infusion	120
Context-Sensitive Half-Time	122
Context-Sensitive Decrement Time	123
References	123
Drug Monitoring	125
Therapeutic Index	125
Indications	126
Timing of Samples	127
Interpretation	128
References	128
Routes of Administration	130
Intravenous	130
Oral	130

Epidural	131
Subarachnoid/Intrathecal	132
Inhalation	133
Transdermal	133
Subcutaneous	134
Sublingual	134
Rectal	134
Intramuscular	135
References	135
Target - Controlled Infusions	136
Principles of TCI	136
Plasma Targeting C_P	137
Effect - Site Targeting	137
TCI Models for Propofol	138
References	140
Pharmacodynamics	141
Receptor Theory	142
Dissociation Constants	142
Properties of Drugs in Receptor Theory	143
Drug - Receptor Interactions	144
Agonists	144
Partial Agonist	144
Inverse Agonist	145
Antagonist	145
Indirect Antagonist	145
Mixed Agonist - Antagonist	146
Allosteric Modulator	146
References	146
Receptor Types	148
Intracellular Receptors	148
Enzyme-linked Receptors	149
Ion-channel Receptors	149
Ligand-gated Channels	149
Voltage-gated Channels	150
G-protein Coupled Receptors	151
References	152

Dose - Response Curves	153
Log - Dose Response Curves	153
References	154
Mechanisms of Action	156
Enzyme Interaction	156
Enzyme Antagonism	157
False Substrates	157
Physicochemical	157
References	158
Variability in Drug Response	159
Adverse Effects	160
Type A Adverse Reactions	160
Type B Adverse Reactions	161
Type C Adverse Reactions	162
References	162
Safety of Drugs in Pregnancy	163
Categories	163
References	164
Drug Interactions	165
Physicochemical	165
Pharmacokinetic	165
Absorption	166
Distribution	166
Metabolism	166
Elimination	167
Pharmacodynamic	167
References	168
Alterations to Drug Response	169
Key Terms	170
Tolerance	170
Pharmacokinetic Tolerance	171
Pharmacodynamic Tolerance	171
Patient Factors Affecting Drug Response	172
Pregnancy	172
Foetus	173

Neonates	174
Elderly	175
Disease Factors Affecting Drug Response	176
Cardiac Disease	176
Hepatic Disease	177
Renal Disease	177
Obesity	178
Non-Specific Alterations to Drug Response	178
References	179
Pharmacogenetics	180
References	180
Pharmacogenetic Disorders	181
Pseudocholinesterase Deficiency	181
G6PD	182
Malignant Hyperthermia	183
Porphyria	184
References	185
 Cellular Physiology	 186
The Cell Membrane	187
Ion Permeability	187
References	188
Membrane Transport	189
Diffusion	189
Active Transport	190
Exo- and Endocytosis	191
Gibbs-Donnan Effect	192
Gibbs-Donnan Equilibrium	193
References	193
Organelles	194
Mitochondria	194
References	195
Excitable Cells	196
Membrane Potential	196
Nernst Equation	197

Goldman - Hodgkin - Katz Equation	198
Action Potential	198
Generating an Action Potential	199
Key Players in the Action Potential	200
Phases of the Action Potential	200
Propagation of the Action Potential	201
Classification of Nerve Fibres	202
References	202
Fluid Compartments	204
Intracellular Fluid	204
Extracellular Fluid	205
Variations	205
Measuring Volumes of Fluid Compartments	206
References	206
Cell Homeostasis	208
Glycolysis	208
Citric Acid Cycle/Kreb's Cycle	209
Electron Transport Chain	209
References	210
Respiratory Physiology	211
Anatomy	212
Airway and Alveolar Anatomy	213
Upper Airway	213
Nasal Cavity	214
Pharyngeal Structures	214
Lower Airway	215
Conducting Zone	215
Respiratory Zone	217
Alveolus	218
References	219
Chest Wall and Diaphragm	220
Inspiration	221
Expiration	221
Respiratory Mechanics in Spinal Injury	222

References	223
Variations in Upper Airway Anatomy	224
Neonates and Children	224
References	225
 Mechanics	 226
Control of Breathing	227
Inputs	227
Peripheral Chemoreceptors	228
Central Chemoreceptors	228
Mechanism of CO ₂ Retention	229
Mechanoreceptors	229
Other Stimulants	229
Integration and Control	230
References	230
 Intrapleural Pressure	 232
Inspiration	233
Expiration	234
References	234
 Compliance	 235
Compliance of the Respiratory System	235
Measurement of Lung and Chest Wall Compliance	236
Static Compliance	237
Dynamic Compliance	237
Specific Compliance	238
Hysteresis	238
Changes in Compliance	239
Increased Lung Compliance	239
Decreased Lung Compliance	239
Increased Chest Wall Compliance	240
Decreased Chest Wall Compliance	240
References	240
 Time Constants	 241
Clinical Significance	243
References	244

Resistance	245
Types of Flow	245
Reynolds' Number	245
Laminar Flow	246
Turbulent Flow	248
Transitional Flow	249
References	249
Surfactant	250
Surface Tension	250
Surfactant	251
References	252
Volumes and Capacities	253
Volumes	254
Capacities	254
Functional Residual Capacity	255
Measurement of Lung Volumes and Capacities	256
Gas Dilution	257
Body Plethysmography	258
Closing Capacity	258
Measurement of Closing Capacity	259
References	260
Spirometry	261
Basic Spirometry	262
Flow - Volume Loops	263
References	266
Work of Breathing	267
Determinants of Work of Breathing	267
Graphing Work of Breathing	268
Minimising Work of Breathing	269
References	270
Diffusion of Gases	271
Oxygen Cascade	272
Atmospheric Gas	273
Humidified Tracheal Gas	273
Alveolar Gas	274

Arterial Blood	275
Mitochondria	276
Venous Blood	276
References	276
Diffusion and Diffusing Capacity	277
Pressure Gradient	278
Lung Factors	278
Substance Factors	279
Diffusion and Perfusion Limitation	280
Oxygen	280
Carbon Dioxide	281
Other Gases	282
Diffusing Capacity	282
References	283
V/Q Relationships	284
Basics of V/Q Matching	285
Distribution of Ventilation	285
Distribution of Perfusion	286
V/Q Ratios	286
V/Q Mismatch and Etymology	286
References	287
West's Zones	288
References	289
Dead Space	290
Types of Dead Space	290
Calculation of Dead Space	291
Fowler's Method	292
Bohr's Method	293
Physiological Consequences of Increased Dead Space	294
Relationship between Alveolar Ventilation and PaCO_2	295
References	295
Shunt	296
Factors Contributing to Shunt	296
Calculation of Shunt	297

Physiological Consequences of Shunt	298
Effect on Carbon Dioxide	298
Effect on Oxygen	298
References	299
Gas Transport	300
Oxygen Storage	301
Oxygen-Haemoglobin Dissociation Curve	301
Comparison Curve	303
Haemoglobin Species	303
Oxygen Saturation	305
Myoglobin	305
References	306
Carbon Dioxide	307
Carbon Dioxide in Blood	308
Bicarbonate	308
Dissolved Gas	309
Carbamino Compounds	309
CO ₂ Dissociation Curve	310
Comparison Curve	311
Removal of CO ₂	311
Distribution of Carbon Dioxide	312
CO ₂ Cascade	313
References	313
Applied Respiratory Physiology	315
Positive Pressure Ventilation	316
Respiratory Effects	317
Cardiovascular Effects	317
End-Organ Effects	319
References	319
Hypoxia	321
Classifications and Causes of Hypoxia	322
Hypoxaemic Hypoxia	322
Anaemic Hypoxia	324
Ischaemic Hypoxia	325

Histotoxic Hypoxia	325
Effects of Hypoxia	325
Acid-Base Changes	326
CO ₂ retention	326
References	326
Hypo and Hypercapnea	327
Hypercapnea	327
Hypocapnea	328
References	328
Position and Ventilation	329
Factors Affecting Distribution of Blood Flow	329
Factors Affecting Distribution of Ventilation	330
Lateral Decubitus	330
Thoracotomy	331
References	332
Humidification	333
Mechanism	333
References	334
Cough Reflex	335
Sensation	335
Integration	335
Effector	336
References	336
Non-Respiratory Functions	337
Filtration	337
Metabolism	338
Blood Reservoir	338
Defence	338
Drug Delivery	339
Endocrine	339
References	340
Altitude	341
Pressure Effects	341
Respiratory	341
Cardiovascular	342
Haematological	343

References	343
Obesity	344
Effects of Obesity on the Respiratory System	344
References	346
Neonates and Children	347
Transition at Birth	347
Neonates and Children	348
References	349
Cardiovascular Physiology	350
Structure and Function	351
Cardiac Anatomy	352
Echocardiographic Anatomy	352
Coronary Supply	353
References	355
Coronary Circulation	356
Vascular Anatomy	357
Coronary Artery Anatomy	357
Venous Anatomy	359
Coronary Blood Flow	360
Control of Coronary Blood Flow	361
References	362
Cardiac Cycle	363
Phases of the Cardiac Cycle	364
Slow Ventricular Filling (Diastasis)	364
Atrial Contraction	365
Isovolumetric Ventricular Contraction	365
Ejection	366
Isovolumetric Relaxation	366
Rapid Ventricular Filling	367
References	367

Electrical Properties	369
Cardiac Action Potential	370
Pacemaker Potential	370
Phases of the Pacemaker Potential	371
Ventricular Action Potential	372
Phases of the Ventricular Action Potential	373
Propagation of the Cardiac Action Potential	373
Autonomic Control	375
Cardiac Transplant	376
References	376
 Cardiac Output	 377
Determinants of Cardiac Output	378
Preload	379
Determinants of Preload	379
Preload and the Respiratory Cycle	380
Frank-Starling Mechanism	380
Afterload	381
End-Diastolic Volume	382
Myocardial Wall Thickness	382
Ventricular Transmural Systolic Pressure	382
Contractility	384
Determinants of Contractility	384
Measuring Contractility	385
Further Reading	386
References	386
 Venous Return	 387
Factors Affecting Venous Return	389
Interaction between Venous Return and Cardiac Function Curves	390
References	390
 Myocardial Oxygen Supply and Demand	 392
Myocardial Oxygen Supply	392
Coronary Blood Flow	393
Myocardial Oxygen Demand	394
Major Determinants	394
Minor Determinants	395
References	396

Pressure - Volume Relationships	397
Basic Pressure-Volume Loops	400
Increased Preload	400
Increased Afterload	401
Increased Contractility	401
Advanced Pressure Volume Loops	402
Aortic Stenosis	402
Aortic Regurgitation	403
Mitral Stenosis	404
Mitral Regurgitation	404
Right Ventricular Pressure-Volume Loop	405
Further Reading	406
References	406
Cardiac Reflexes	407
References	408
Peripheral Circulation	409
Starling Forces	410
Organ-Specific Values	411
Causes of Oedema	412
References	413
Variations in Blood Pressure	414
Changes by Site of Measurement	415
Respiratory Variation	416
Pulse Pressure Variation	418
Stroke Volume Variation	418
Circulatory Factors	419
Pathological Changes	420
References	420
Pulmonary Circulation	422
Differences between Pulmonary and Systemic Circulations	423
Blood Pressure	423
Pulmonary Vascular Resistance	424
Response to Substances	426
Distribution of Pulmonary Flow	426
West's Zones	427
Hypoxic Pulmonary Vasoconstriction	428

References	428
Cerebral Blood Flow	429
Factors Affecting Perfusion Pressure	430
Factors Affecting Vessel Radius	431
Cerebral Metabolism	431
PaCO ₂	432
PaO ₂	433
Neurohormonal	434
Factors Affecting Blood Viscosity	434
References	434
Hepatic Blood Flow	435
Regulation of Flow	436
Intrinsic Autoregulation	436
Extrinsic Autoregulation	436
External Factors	437
Microvasculature	437
References	438
Circulatory Control	439
Baroreceptors	440
Baroreceptor Control	441
Baroreceptor Activity	441
Hormonal control	442
References	443
Valsalva Manoeuvre	444
Phases	444
Phase I	444
Phase II	445
Phase III	445
Phase IV	445
Abnormal Responses	445
Cardiac Failure	446
Autonomic Neuropathy	446
References	446
Cardiovascular Changes with Obesity	447
Hormonal Changes	447

Cardiovascular Changes	448
References	449
Cardiovascular Effects of Ageing	450
Cardiac Changes	450
Vascular Changes	451
Autonomic Changes	451
References	451
Adrenoreceptors	452
Adrenoreceptor Subtypes	452
References	454
Renal System	455
Renal Physiology	456
Functional Anatomy and Control of Renal Blood Flow	457
Functional Anatomy	457
Control of Renal Blood flow	457
Autoregulation	458
Neuronal Control	459
Hormonal Control	459
References	460
Glomerular Filtration and Tubular Function	461
Glomerulus	461
Glomerular Filtration Rate	461
Tubular Function	463
Proximal Tubule	463
Loop of Henle	464
Distal tubules	465
Collecting Ducts	465
References	466
Handling of Organic Substances	467
Methods of Reabsorption	467
Rate Limitation	468
Glucose	468
Consequences of Glycosuria	469

Nitrogenous Products	470
Urea	471
pH Dependent Drug Reabsorption	472
References	472
Measurement of GFR	473
Clearance and GFR	473
Serum Creatinine	474
Estimating Creatinine Clearance	475
References	476
Endocrine Functions of the Kidney	477
Renin - Angiotensin - Aldosterone System	477
Vitamin D	478
Erythropoietin	479
References	479
Acid - Base Balance	480
Bicarbonate and the Kidney	481
Reabsorption of Bicarbonate	481
Ammonia	482
Bound to Filtered Buffers	483
References	483
Dialysis	484
Indications	484
Physical Mechanisms	484
Implementation	485
Differences	486
Pharmacokinetics of RRT	487
References	488
Fluids and Electrolytes	489
Sodium and Water	490
Function of Sodium	490
Regulation of Sodium and Water	491
Sodium Elimination	491
Sodium Reabsorption	492
Water Homeostasis	493
References	493

Potassium	494
Function and Dysfunction	494
Hyperkalaemia	495
Hypokalaemia	496
Regulation	496
Intake	497
Sequestration	497
Elimination	497
References	499
 Acid - Base	 500
Principles of Acid - Base Physiology	501
pH	501
pKa	501
Systemic Effects of Acid - Base Disorders	503
Change with Temperature	504
References	506
 Compensation	 507
Metabolic Acidosis	507
References	508
 Buffers	 509
Buffer Systems	510
Bicarbonate Buffer System	510
Protein Buffer System	511
Haemoglobin Buffer System	512
Phosphate Buffer System	513
Further Reading	513
References	514
 Nervous System	 515
 Blood - Brain Barrier	 516
Anatomy	516
Movement of Substances	517
References	518

Cerebrospinal Fluid	519
Functions	519
Formation	520
Factors Affecting Formation	520
Contents	520
Circulation	521
Reabsorption	522
References	522
Spinal Cord Anatomy	523
Spinal Cord Anatomy	523
Spinal Cord Blood Supply	523
Spinal Cord Syndromes	524
Complete Transection	525
Central Cord Syndrome	525
Anterior Cord Syndrome	525
Brown-Sequard Syndrome	525
Cauda Equina	526
References	526
Intracranial Pressure	527
Monroe Kellie Doctrine	528
Physiological Responses to an Increase in ICP	528
Physiological Basis of Treatment	529
Brain	529
CSF	529
Blood	530
References	530
Intraocular Pressure	532
Determinants of Intraocular Pressure	532
References	533
Sleep	534
Stages of Sleep	534
Respiratory Effects	535
References	535
Pain	537

Key Definitions	538
Peripheral Nociception	539
Nociception	539
Pain Pathways and Site of Action of Analgesics	540
Neuropathic Pain	542
Mechanisms of Neuropathic Pain	542
Features of Neuropathic Pain	542
Chronic Regional Pain Syndrome	543
Pain in the Elderly	543
References	544
Autonomic Nervous System	545
Central Control	546
Central Anatomy	547
Sympathetic Nervous System	547
Effect	548
Parasympathetic Nervous System	549
Ganglion Blockade	551
Enteric Plexus	552
References	552
Acetylcholine Receptors	553
Synthesis, Release, and Metabolism	553
ACh Receptor Subtypes	554
References	555
Endocrine System	556
Hormones	557
References	558
Insulin, Glucagon, and Somatostatin	559
Insulin	559
Actions of Insulin	559
Glucose Tolerance	561
Glucagon	561
Somatostatin	562
References	562

Control of Blood Glucose	563
Regulation	564
Hormonal: Short Term	564
Hormonal: Long Term	565
Neuronal	565
References	565
Pituitary Hormones	566
Anterior Pituitary Hormones	566
Posterior Pituitary Hormones	567
References	568
Hypothalamic Hormones	569
Hypothalamic - Pituitary Axis	570
References	570
Thyroid	571
Synthesis	571
Secretion and Metabolism	572
Physiological Effects	573
References	574
Adrenal Hormones	575
Steroid Hormones	575
Mineralocorticoids	576
Glucocorticoids	576
Catecholamines	577
References	578
Calcium Homeostasis	579
Functions of Calcium	579
Regulation of Calcium	580
References	581
Histamine	582
Histamine Receptors and Effects	582
References	584
Prostanoids	585
Synthesis	585
References	588

Musculoskeletal	589
Skeletal Muscle	590
Structure and Contents	590
Muscle Connective Tissue	591
Muscle Cells	591
Sarcomere	592
Excitation - Contraction Coupling	593
Muscular Relaxation:	595
References	595
Skeletal Muscle Innervation	596
Motor Units	596
Force of Contraction	596
Proprioception	597
Muscle Spindles	597
Golgi Tendon Organs	598
Reflexes	598
Twitch and Tetany	599
References	599
Nutrition and Metabolism	600
Basal Metabolic Rate	601
Factors Affecting Metabolic Rate	601
Measurement of BMR using Indirect Calorimetry	603
Errors in Indirect Calorimetry	604
References	604
Fat Metabolism	606
Digestion	606
Absorption	606
Storage	607
Metabolism	608
References	608
Carbohydrate Metabolism	609
Storage	609
Glycolysis	609
Gluconeogenesis	610
References	610

Protein Metabolism	611
Metabolism	611
References	612
Fasting	613
Fasting	613
Starvation	614
Refeeding Syndrome	615
References	615
Anaerobic Metabolism	617
Lactate	617
Ketones	618
References	618
Body Temperature	619
Principles	620
Radiative	620
Conduction	621
Convection	621
Evaporation	621
Temperature Sensation and Regulation	622
Central Sensation	622
Peripheral Sensation	622
Regulation	623
Effector Responses	623
Effect of Anaesthesia	625
References	626
Immunology	627
Inflammation	628
Process of Inflammation	628
References	629
Innate Immunity	630
Components	630
Physicochemical Barriers	630
Innate Humoral Immunity	631
Innate Cellular Immunity	632
References	633

Adaptive Immunity	634
Components	635
Adaptive Cellular Immunity	635
Adaptive Humoral Immunity	636
Adaptive Immune Response	636
Primary Immune Response	637
Secondary Immune Response	638
References	638
Hypersensitivity	639
Type I Hypersensitivity	641
Type II Hypersensitivity	642
Type III Hypersensitivity	642
Type IV Hypersensitivity	643
References	643
Microbiology	644
Bacteria	645
Bacterial Subclassification	646
References	647
Viruses	648
References	649
Eukaryotes	650
References	651
Antimicrobial Resistance	652
Mechanisms	652
References	653
Obstetrics	654
Respiratory Changes of Pregnancy	655
Anatomical Changes	655
Volumes and Capacities	656
Ventilation	656
Labour and Postpartum	656
References	657

Cardiovascular Changes of Pregnancy	658
Magnitude of Changes by Trimester	660
Other Changes	662
Changes During Labour	662
References	663
Foetal Circulation	664
In Utero	664
Changes at Birth	665
References	666
The Placenta	668
Nutrient and Waste Exchange Functions	669
Oxygen Transfer	669
Carbon Dioxide Transfer	670
Nutrient Diffusion	671
Immunological Function	671
Endocrine Function	672
References	672
Gastrointestinal System	673
Oesophagus	674
Lower Oesophageal Sphincter	674
References	676
Gastric Secretions	677
Composition of Secretions	678
Salivary Secretions	678
Gastric Secretions	679
Pancreatic Secretions	680
Small Bowel Secretions	681
References	682
Control of Gastric Emptying	683
Determinants of Gastric Emptying	683
Stomach Factors	684
Duodenal Factors	684
Systemic Factors	684
Drug Factors	685
References	685

Swallowing	686
References	687
Vomiting	688
Stimulation	688
Postoperative Nausea and Vomiting	689
Process of Vomiting	690
References	691
Functions of the Liver	692
Storage	692
Synthetic	693
Metabolic	693
Carbohydrate Metabolism	693
Lipid Metabolism	694
Protein Metabolism	694
Endocrine	694
Immune Function	695
Acid-Base Balance	695
References	696
Liver Function Assessment	697
Synthetic Function	697
Metabolic Function	698
Obstructive Tests	698
References	699
Bile	700
References	701
Haematology	702
Erythrocytes	703
Production	704
Function	704
Elimination	705
Haemoglobin Metabolism	705
References	707
Iron Homeostasis	708
Absorption	708

Regulation	709
References	709
Platelets	710
Production	710
Contents	711
Activation	711
References	712
Blood Groups	713
ABO	713
Rhesus	714
Compatibility Testing	715
Indirect Coomb's Test	716
References	716
Blood Product Storage	717
Red Blood Cells	717
Storage Lesion	718
Platelets	718
Fresh Frozen Plasma	719
Cryoprecipitate	720
Whole Blood	720
References	720
Transfusion Reactions	721
Complications of Massive Transfusion	724
References	725
Haemostasis	726
Primary Haemostasis	726
Secondary Haemostasis	727
Extrinsic Pathway	728
Intrinsic Pathway	729
Common Pathway	729
The Cell-Based Model of Coagulation	730
References	730
Haemostatic Regulation	732
Endothelial Regulation	732
Clot Regulation	733
Clot Lysis	734

References	734
Coagulation Testing	735
Prothrombin Time/INR	735
(Activated) Partial Thromboplastin Time	736
Activated Clotting Time	737
Platelet Function Tests	737
References	738
Viscoelastic Testing	739
Thromboelastography	740
Rotational Thromboelastometry	741
Interpretation	742
References	744
Physics and Measurement	745
System of Units	746
Base Units	746
Derived Units	749
References	750
Electrical Safety	751
Electrical Injury	753
Electrical Shock	753
Principles of Electrical Safety	754
Electrical Dangers	754
Methods of Electrical Safety	755
Classification of Electrically Safe Equipment	758
Classification of Electrically Safe Areas	758
References	759
Wheatstone Bridge	760
Mechanism	760
References	761
Neuromuscular Monitoring	762
Nerve Stimulator	763
Stimulation Patterns	764
References	766

Pressure Transduction	767
Static Calibration	768
Dynamic Calibration	768
Resonance	769
Damping	769
Dynamic Response	770
Principles of Waveform Analysis	771
Testing Dynamic Response	771
Optimising Dynamic Response	772
References	772
Non-Invasive Blood Pressure	774
DINAMAP	774
Von Recklinghausen's Oscillotonometer	775
References	776
Cardiac Output Measurement	777
Thermodilution	778
Cardiac Output Measurement by Thermodilution	778
Errors in Thermodilution	779
Fick Principle	780
References	781
Pressure Waveform Analysis	782
Principles	782
Devices	783
Common Devices in Use	784
References	785
Pulse Oximetry	786
Method	787
The Isobestic Point	788
Limitations	788
References	789
Oxygen Analysis	790
Paramagnetic Analysis	790
Pressure Method	791
Temperature Method	791
Pros	792
Cons	792

Fuel Cells	792
Method	792
Pros	793
Cons	793
References	793
End-Tidal Gas Analysis	794
Principles	794
Infrared Spectroscopy	794
Measurement of CO ₂	795
Equipment Errors	796
Normal E _T CO ₂ Waveform	798
Waveform Variations	798
Changes in E _T CO ₂	800
High E _T CO ₂	800
Low E _T CO ₂	801
Discrepancy between E _T CO ₂ , PACO ₂ , and PaCO ₂	801
Colourimetric Methods	802
References	802
Blood Gas Measurement	803
Oxygen Tension	804
Method	805
Calibration, Limitations, and Accuracy	805
pH Measurement	806
Method	807
Calibration, Limitations, and Accuracy	807
Carbon Dioxide Tension	808
Method	809
Calibration, Limitations, and Accuracy	809
References	809
Gas Flow	811
Flowmeters	812
Variable Orifice Flowmeters	812
Fixed Orifice Flowmeters	813
Hot Wire Flowmeter	814
References	815
Principles of Ultrasound	816
Basic Terms	817

Effect of Tissues on Ultrasound	817
Modes	818
Doppler Effect	819
Calculation of Cardiac Output	819
References	820
Temperature	821
Temperature Measurement	821
Liquid Expansion Thermometry	822
Electrical Thermometry	823
References	823
Humidity	825
Measurement of Humidity	825
References	826
Humidifiers	827
Passive Humidifiers	827
Active Humidifiers	828
References	829
Electrocardiography	830
Components	830
Leads	831
Method	832
Sources of Error	833
Risks	833
References	834
Supplemental Oxygen	835
Variable Performance Devices	835
Fixed Performance Devices	836
References	837
Bispectral Index	838
Method	838
Pros	839
Cons	840
References	840

Medical Gas Supply	841
Production	841
Fractional Distillation	841
Oxygen Concentrator	843
Storage	843
Medical Gas Cylinders	844
Cylinder Manifolds	844
Vacuum Insulated Evaporator	845
Safety in Medical Gas Supply	846
References	848
Vapourisers	849
Variable Bypass Vapouriser	849
Comparison of Plenum and Draw-Over Vapourisers	851
Measured Flow Vapourisers	852
General Safety Features of Vapourisers	853
Other Factors Affecting Vapourisers	854
References	855
Breathing Systems	856
Mapleson System	857
Mapleson A	857
Mapleson B and C	858
Mapleson D	858
Mapleson E	859
Mapleson F	859
References	860
Circle System	861
Components	861
Soda Lime	863
Pros	864
Cons	864
References	865
Bag-Valve-Mask	866
Components	866
Method	867
Pros	868
Cons	868

Scavenging	870
Methods of Scavenging	870
References	871
Diathermy	872
Diathermy Types	872
Diathermy Modes	873
Risks	873
References	874
Lasers	875
Principles	875
Pros	876
Cons	876
References	877
Procedural Anatomy	878
Internal Jugular Vein	879
Borders	879
Relationships	880
Surface Anatomy	880
Ultrasound Anatomy	880
Approaches	881
References	881
Subclavian Vein	882
Borders	882
Surface Anatomy	883
Ultrasound Anatomy	883
Approaches	883
References	884
Intercostal Catheter	885
Surface Anatomy	885
Layers of Dissection	886
References	886
Antecubital Fossa	887
Borders	887
Contents	888

References	888
Tracheostomy	889
Trachea	889
Relationships	890
Surface Anatomy	890
Layers of Dissection	890
References	891
Neuraxial Anatomy	892
Borders	893
Surface Anatomy	893
Layers	894
References	894
Pharmacopoeia	895
Toxicology	896
Tricyclic Antidepressant Overdose	897
Toxicity	897
Cardiac toxicity	897
Central toxicity	898
Management	898
Cardiac toxicity	899
Central toxicity	899
References	900
Organophosphate Poisoning	901
Toxicity	901
Management	901
References	902
Toxic Alcohols	903
Ethanol	903
Methanol	904
Ethylene Glycol	904
References	905

Naloxone	906
References	908
Flumazenil	909
References	911
 Respiratory Pharmacology	 912
Oxygen	914
References	916
Helium	917
References	919
Anti - Asthma Drugs	920
References	921
Beta Agonists	922
References	926
Antimuscarinics (Respiratory)	928
References	931
Phosphodiesterase Inhibitors / Methylxanthines	933
References	935
Leukotriene Antagonists	937
References	938
Corticosteroids	940
References	948
Pulmonary Vasodilators	950
References	955
 Cardiovascular Pharmacology	 956
Inotropes	958
Classes of Positive Inotrope	958
References	962

Antiarrhythmics	964
Class I	965
Class Ia	965
Class Ib	966
Class Ic	967
Class II	975
Class III	980
Class IV	985
Alternatives to Vaughan Williams	989
References	994
Adrenergic Vasoactives	996
Structure-Activity Relationships of Sympathomimetics	1007
Dopamine	1008
Noradrenaline	1009
Adrenaline	1009
Metaraminol	1010
Ephedrine	1010
References	1011
Non-Adrenergic Vasoactives	1013
References	1018
Centrally Acting Anti-Hypertensives	1019
References	1022
Calcium Channel Blockers	1024
References	1029
Direct Vasodilators	1030
Nitrate Toxicity	1035
Cyanide Toxicity	1035
Thiocyanate Toxicity	1036
Methaemoglobinaemia	1036
References	1037
Angiotensin Converting Enzyme Inhibitors	1039
Indications	1039
Classification	1040
References	1050

Angiotensin Receptor Blockers	1052
References	1059
Angiotensin Receptor Neprilysin Inhibitors	1060
Indications	1060
References	1061
Potassium Channel Activators	1062
References	1064
Sodium Channel Blockers	1065
References	1070
Beta - Blockers	1071
Indications	1071
References	1080
Amiodarone	1082
References	1085
Sotalol	1086
References	1088
Digoxin	1089
References	1092
Adenosine	1093
References	1094
Magnesium	1096
References	1098
Atropine	1099
References	1101
 Renal Pharmacology	 1102
Diuretics	1104
References	1117
Intravenous Fluids	1118
References	1119

Neuropharmacology	1120
Neurotransmitters	1121
GABA	1121
NMDA	1122
References	1122
Anticonvulsants	1123
Sodium Channel Blockers	1123
GABA Mediators	1123
Glutamate Blockers	1124
Other Agents	1124
References	1125
Local Anaesthetics	1126
Mechanism of Action	1126
Chemical Structure of Local Anaesthetics	1127
Key Characteristics of Local Anaesthetics	1128
Toxicity	1131
Cardiac Toxicity	1133
CNS Toxicity	1134
Treatment	1135
References	1136
Neuraxial Blockade	1137
Respiratory Responses	1137
Cardiovascular Responses	1137
CNS Responses	1138
References	1138
Opioids	1139
Classification of Opioids	1139
Opioid Receptor Classification	1140
References	1141
Inhalational Anaesthetics	1143
Key Principles of Inhalational Agents	1144
Minimum Alveolar Concentration (MAC)	1145
Partition Coefficients	1148
Pharmacokinetics of Inhalational Agents	1149
Wash-out of Inhalational Agents	1153
Alteration to Pharmacokinetics	1154

Alteration to Pharmacokinetics with Special Methods of Administration	1155
Mechanism of Action of Inhaled Anaesthetic Agents	1156
Incomplete Theories of the Mechanism of Action of General Anaesthetic Agents	1157
Structure - Activity Relationships of Inhaled Anaesthetics	1158
The Ideal Inhaled Anaesthetic Agent	1158
References	1160
Propofol	1161
References	1164
Barbiturates	1166
References	1168
Ketamine	1170
References	1173
Dexmedetomidine	1174
References	1176
Local Anaesthetic Agents	1178
Pharmaceutics of Topical Local Anaesthetics	1188
References	1189
Benzodiazepines	1191
References	1197
Antidepressants	1198
Individual Antidepressants	1202
Serotonin Syndrome	1215
References	1215
Antipsychotics	1217
Neuroleptic Malignant Syndrome	1218
Extra - Pyramidal Side Effects	1218
References	1224
Anticonvulsants	1225
References	1232
GABA Analogues	1233
References	1236

Inhalational Anaesthetic Agents	1238
Comparison of Common Inhalational Agents	1241
Comparison of Uncommon Inhalational Agents	1247
References	1251
Nitrous Oxide	1253
Entonox	1255
References	1256
Opioids	1257
References	1273
COX Inhibitors	1275
Mechanism of Action	1275
Adverse Effects	1276
References	1288
Tramadol	1289
References	1291
Paracetamol	1292
Toxicity	1294
Features of Overdose	1295
Treatment of Overdose	1295
References	1296
Anticholinesterases	1297
References	1300
Antimuscarinics (Cardiac)	1302
References	1305
Neuromuscular Blockers	1307
Factors Affecting Neuromuscular Blockade	1307
References	1310
Depolarising NMBs	1311
Adverse Effects	1313
Phase I and Phase II Blockade	1314
Malignant Hyperthermia	1314
Suxamethonium Apnoea	1315
References	1315

Non - Depolarising Neuromuscular Blockers	1317
References	1324
Dantrolene	1326
References	1327
Sugammadex	1329
References	1330
 Haematological Pharmacology	 1331
Anticoagulants	1333
HITs	1337
Protamine	1337
References	1338
Direct Thrombin Inhibitors	1340
References	1342
Antifibrinolytics	1343
References	1344
Antiplatelets	1346
Classification of Antiplatelet Agents	1346
References	1355
 Antimicrobials	 1356
Antiseptics	1358
Key Definitions	1358
References	1360
Penicillins	1361
Mechanisms of Resistance	1362
Individual Penicillins	1363
References	1373
Glycopeptides	1375
References	1377

Aminoglycosides	1378
References	1380
Lincosamides	1382
References	1383
Metronidazole	1385
References	1387
Antifungals	1389
Mechanisms of Antifungal Resistance	1389
References	1397
 Endocrine Pharmacology	 1398
Insulins	1400
Types of Insulin	1400
Fast Acting	1401
Intermediate Acting	1402
Long Acting	1402
References	1408
Oral Hypoglycaemics	1410
References	1416
 Obstetric Pharmacology	 1417
Oxytocics	1419
References	1425
Tocolytics	1427
References	1428
 Gastrointestinal Pharmacology	 1429
Acid Suppression	1430
References	1437
Antiemetics	1439
References	1445

Other Drugs	1446
Intravenous Contrast	1448
Adverse Reactions	1449
General Adverse Reactions	1449
Contrast Nephropathy	1450
References	1450
Appendices	1451
Definitions	1451
A	1451
B	1453
C	1454
D	1456
E	1458
F	1459
G	1459
H	1460
I	1461
J	1462
L	1462
M	1463
N	1463
O	1464
P	1465
R	1466
S	1466
T	1467
V	1468
Key Graphs	1470
Pharmacology	1471
Models	1473
Pharmacokinetics	1475
Pharmacodynamics	1476
Statistics	1478
Cellular Physiology	1479
Action Potential	1479

Respiratory	1480
Oxygen	1480
Carbon Dioxide	1485
Anatomical and Physiological Interactions	1487
Spirometry	1494
Flow - Volume Loops	1495
Anaesthetic Agents	1498
Cardiovascular	1499
Action Potentials	1505
Basic Pressure - Volume Loops	1508
Advanced - Pressure Volume Loops	1510
Antiarrhythmics	1512
CNS	1515
Renal & Acid - Base	1518
Haematology	1521
Other	1523
Equipment & Measurement	1524
Gas Analysis	1526
Capnography	1527
References	1528
Laws and Equations	1529
General Laws	1529
Cell Physiology	1530
Respiratory Laws	1531
Gas Laws	1531
Cardiovascular Equations	1532
Equipment	1533
References	1533
Structures for SAQs	1534
Regulation of Physiological Responses	1534
Change in Level of a Substance	1534
Compare and Contrast (Drugs)	1535
Describe Why Two Drugs are Used in Combination	1535
Describe the Physiology of...	1536
Anatomical Structure	1538
Physics and Measurement	1538
References	1538

Part One

Part One is a free and open-access reference for trainees preparing for the CICM and ANZCA Primary Exams.

- Part One **is**:
 - Designed to cover the **assessed** sections of the CICM and ANZCA curricula **in enough detail to pass**
 - A rough guide for the expected depth of knowledge required on a topic
 - A tool to correct your written answers
 - A source of information you might find difficult to find elsewhere
- Part One **is not**:
 - A textbook
 - The definitive guide to the primary exam
 - A complete reference

There will be both omissions and errors. If you find any, please let me know.

Layout

The book is divided into three sections:

- **Curriculum**

Covers statistics, physiology, equipment and measurement, and anatomy.

 - Pages are laid out using the section title, topic titles, and order from the CICM curriculum

- When a topic is only examinable in the ANZCA curriculum, it has been slotted in somewhere sensible
- Topics covered by the page are listed at the beginning of each page

A purple block indicates a topic is from both the ANZCA and CICM curricula

A blue block indicates a topic is ONLY from the CICM curriculum

A maroon block indicates a topic is ONLY from the ANZCA curriculum

- **Pharmacopoeia**

Covers both the common features of major drug classes and the specifics of individual agents. Note that the *general principles* of pharmacology (e.g. pharmacokinetics) are covered in the curriculum.

- **Appendices**

Includes the key definitions, graphs, and equations you should know, as well as sample structures for SAQs.

- **Exam Answers**

High quality model answers to the ANZCA primary.

Acknowledgements + Technical Stuff

Part One is built and made pretty with Quarto.

- Equations are written in LaTeX
- Graphs are:

- Written in PGF/Tikz using texworks
- Saved as vector graphics with dvisvgm
- Stripped of extraneous parts using svgo
- (Some graphs have been taken from open-source sites such as Wikimedia Commons. These have been credited where used, and gone through the svgo crash diet to keep it lightweight.)
- Chemical structures have been built in MarvinSketch

About the Author

Jake Barlow is an Anaesthetic and Intensive Care Registrar Fellow Consultant from Melbourne, Australia. Interested in cardiac critical care, as well as biotech, physical computing, teaching, data analytics, and outcome prediction. Send all comments, criticism, and complaints about Part One to him [here](#).

Major Contributors

Stuart Watson is an anaesthetic registrar and Renton Prize winner from St. Vincent's Hospital, Melbourne. He has kindly provided his brilliant primary exam question answers to be included in Part One (the originals can be found on his website [Ketamine Nightmares](#)).

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How to Pass

The first part exam is:

- Painful
The knowledge demanded is huge, and cannot be avoided.
- **Eminently achievable**
Remember, it is not impossible - **everybody before you has completed it.**

Plan for Success

This is not an exam you want to have to sit more than once
- try to give yourself the best chance of success the first time round:

- Commit yourself early
Decide when you are going to sit:
 - Pick a date ~9 months in advance
 - 6 months is probably pushing it
 - 9 months is achievable
 - 12 months is almost too long - you will lose motivation and knowledge will fade.
 - Accept that the time between now and the exam is not going to be the best time of your life
 - Consider paying the money as soon as possible
 - lock yourself in
 - Your family and friends will forgive you, eventually
- Don't lose faith

- There will be times that you question why you have to learn this
- Those are very legitimate feelings
- Accept that part of **this exam is an academic hazing you must pass** through on your path to fellowship
- There is no substitute for knowledge

Be Strategic

The curriculum provided is overwhelming, and probably not achievable for most of us. Have a plan about how you will approach it:

- Have a timetable
 - Content to cover each week
 - I found setting a weekly goal would allow me to plan around day-to-day variations (finishing late, good days, bad days, etc)
 - A daily timetable was often mangled by life, creating unnecessary stress
 - Time to start viva practice
Aim to start before the written.
 - Topics that you can't explain, you probably don't understand fully
This may not be apparent until you try and explain it.
- Know the enemy
 - Syllabus
Read through it so you appreciate the breadth of knowledge required.
 - Know the style
This allows you to give answers **efficiently** - the key metric for both the vivas and the SAQs is **marks per unit time**.
 - Style of exam questions
Including the style of answers - see the SAQ.

- Style of vivas
 - Do practice vivas
 - Record yourself, so you know your tics
 - Do a dress rehearsal
Make sure your suit still fits before the day.
 - Graphs
Be able to draw them while talking about them.

- **Do past questions**

I cannot stress this enough. This is **the** key to preparing for this exam.

- Past questions:
 - Teach you appropriate structure
 - Teach you to write to time
 - Ensure you learn the content in the way it will be recalled
 - Ensure you don't waste time learning things that are unlikely to be examined
When I sat the CICM exam, I had done almost all the past questions, which covered ~60% of the curriculum. There was 1 (out of 24) of the SAQs on a topic I had not answered an SAQ on before.
- Do questions to time
Keeping to time is **vital**.
- It is almost impossible to write a perfect answer in 10 minutes
- In many cases you will need to move on to the next question despite still having things to say
- Remember that the marking follows a sigmoid distribution
 - The first 30% of marks for a question are easy to get
 - The last 30% of marks are very difficult to get
 - Therefore, the most efficient use of your time is to aim to get ~60-70% of marks for each question.

- **Remember the pass mark is 50%**
 - You are not expected to know everything
 - Breadth tends to be rewarded over depth
 - It is normal to sit the exam and have a question you have not thought about before

Suggested Approach

There are many equally valid ways to approach these exams. This is how I would do it, if I had to do it again:

1. **Read a general physiology and pharmacology textbook**

This will help you understand the scope of the undertaking. I would recommend spending 2-3 weeks reading:

- Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
In my opinion, this is **the** general physiology text. I believe that if you knew everything in this book, you would pass the physiology component of both exams.
- Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
The first few chapters are a good introduction to pharmaceuticals and pharmacokinetics, which will help you put later information from more complete texts into context.

2. **Start doing practice questions:**

This is the key to the exam. I suggest:

- Start doing one question at a time
In the beginning, you will not know enough to write for 10 minutes.
 - After doing the question, check your answer against available past answers

This forces active learning, and is far more efficient than reading. Look at:

- Structure
How did you structure your answer?
What was the example structure?
- Content
What did you miss? Are the numbers/graphs you used correct?
- Then study the curriculum areas that question covered, and make notes
This would take me ~1-2 hours for a new curriculum area.
- Once you start doing questions which you know something about (having answered one similar previously), move up to three questions in 24 minutes.
This teaches you to keep time, which is vital for success in the SAQ.
 - Still check each answer afterwards, look over that area of the curriculum, and revise and refine your notes
- When you find yourself running out of time before you run out of things to write, give yourself 9 minutes per question
I would suggest not going beyond this - you need to allocate your time strategically on the day, and writing to time is critical.
- As this gets easier, start doing 6 or more questions at a time to train your writing hand
- Do one or two full exams to time before game day

3. **Do a lot of flashcards**

Flashcards are less demanding than doing questions, and a simple form of revision.

- They are the absolute best way of rote learning facts (in my opinion)
- I used anki, but use whatever works for you
- My anki deck is available here

4. **Do practice vivas**

Start before the written. There is a lot of crossover of skills between the viva and the written. Both require a

structured approach, and good content knowledge.

- Remember to take a break after the written exams, it is exhausting

The Bottom Line

- Pick a date, and commit to it
- Work out which times work best for you with respect to study
Different times will be better for different things. I found:
 - Days off (including weekends) were best for learning new content
 - Work days were for revising
 - Post night shift was a write-off
- Maintain a positive attitude
Study groups are good for this - share the suffering!
- Split large topics into manageable chunks
- Don't lose your head
Set aside time for relaxation, and don't feel guilty about it.
- You don't have to know everything
The pass mark is 50%

Prizewinners Perspective

Key thoughts from those who won the prize; for those of you who are:



- Feeling anxious
- Wanting to excel
- Renton-inclined

Again, this is a really long time and is not necessary for those only seeking to pass.

- Spend 12 months learning content, and 6 months revising
 - Manage expectations of friends, family, and partner early
 - Learn from good quality texts and credible online resources
 - Preference and use multiple references for core topics, and minimise time spent on peripheral topics
 - Engage in rote-learning on a regular basis
 - Reinforce knowledge in other ways besides re-reading
 - Heed the examiners reports
 - Focus on the SAQ
- Practice:
- Under exam conditions
 - Recently asked questions
 - Poorly answered questions
 - Repeatedly
-

References

1. A talk I gave at the 2016 VPECC Course, still raw from the CICM primary.
2. Periodic updates from talking to colleagues, and guiding junior trainees through the academic crucible.

How to Fail

Alternatively, for those of you who wish to donate large sums of money to the college coffers in a tax-deductible way, there are a series of tried-and-true techniques for failing the primary.

These factors are common among trainees who have had repeated unsuccessful attempts:

- Leave gaps in the curriculum
 - Don't distinguish between the things you ought to know in great detail and the things where a superficial knowledge is sufficient
 - Don't identify where the "one thousand hours" of study is going to come from
 - Don't use several resources
 - Don't ask questions when you don't know the answers
 - Accept other people's answers for MCQs
 - Don't practise SAQs
 - Don't practise vivas until after you get an invitation
 - Only ask "nice" people for assessments
 - Don't seek out a past or current primary examiner in your region
 - Don't buy any books
 - Buy lots of books but don't read them
 - Mistake familiarity for knowledge
 - Don't have a plan or, if you do, don't stick to it
 - Don't decide on the purpose of a study group before embarking on it
 - Don't think about how any of what you study might influence your clinical practice
 - Don't cultivate any curiosity you might have about your chosen specialty
-

References

1. Reeves, M. A Primer for the Primary FANZCA Examination. 2018.

The SAQ

The bottom line is that a good response will:

- Cover the major points in reasonable detail
- Generally focus on principles rather than specifics

Marks become progressively harder to acquire:

- The first one or two marks on a question should be easy
- Going from an 8/10 to a 10/10 will require time which you likely cannot spare

Constructing a good response to an SAQ requires two things:

- **Structure**

Developing a structured approach to answering SAQs is essential to succeeding in this section of the exam.

A structured approach:

- Is easily digested by the examiner
- Reduces the amount of filler you need to write, meaning you can write more facts
- Typically lends itself to bullet points rather than paragraphs
- Allows you to recall more information than you would otherwise

Especially if you learnt it **in the same format**. This is particularly important for pharmacology.

- **Knowledge**

Obviously.

Additionally, a good response will:

- **Answer the question**

This is stated *repeatedly* in examiner reports. If the question asks for a discussion of the respiratory changes of pregnancy, no marks will be awarded for cardiovascular changes.

- **Be legible**

- **Not be perfect**

This is often - overlooked.

- Examiner reports (and some model answers), assume a perfect response
- This not feasible given the time allowed
- It is also not actually expected - remember that **the pass mark is 50%**

Answering the Question

- You have exactly **10 minutes** per SAQ
 - You should practice to **8-9 minutes** per SAQ
In many cases, the last question (or questions) goes unanswered. This demonstrates poor time management, as easy marks were thrown away by candidates reaching for harder marks on earlier questions.
 - During reading time, you should evaluate each question to:
 - Decide what part of the curriculum it is assessing
 - Work out the context, if any
 - Decide what structure would be most appropriate
-

The Viva

The viva is the part of the exam most candidates seem most stressed about. However:

- If you make it through the written, you will most likely pass
Very few people succeed in the written exams to fail at the viva.
 - The knowledge is there
 - Examiners want you to pass
They will redirect you if you're off track.
 - This makes it easier to make up marks than on the written, where you can easily go off down the rabbit hole, haemorrhaging time and marks

Understanding the Viva

To do well at the viva:

- Understand **the viva is a performance piece**
The viva is a ritualised conversation. Success requires you to know and understand the language and structure used, just like the SAQ.
- **Structure your answer**
As with the SAQ, categorise your answer.
 - Have a good opening statement
Don't answer more than is asked.
 - Start broad
Often the viva will go into depth on only one or two areas of a topic. If you start going into detail on only parts of a topic, it makes it hard for

the examiner to redirect you and scores you no marks.

- Be confident
Enjoy it if you can.
 - Learn to think on your feet
The viva assesses knowledge in a different way to the SAQs.
 - The knowledge will be there, but it may require a different approach to access it
This requires **practice**.
 - This is also important for delivering a sound answer based on incomplete knowledge
 - It's okay to say "I don't know"
But probably not on the first question.
 - If you don't know immediately, can you work it out from first principles?
 - Don't get angry
 - With yourself
 - With the examiner
Don't argue.
 - Don't apologise
Apologies:
 - Make you lose confidence
 - Don't get you marks
Remember, marks per unit-time.
 - Don't talk over the examiner
They are interrupting you because what you are saying is gaining no marks. If you keep talking, you will:
 - Not be getting marks
 - Irritate them
Potentially losing future marks.
-

Curriculum

Research Methods and Statistics

Evidence-Based Medicine

Describe the features of evidence-based medicine, including levels of evidence (e.g. NHMRC), ~~meta-analysis, and systematic review~~

What is Evidence-Based Medicine?

Evidence-based Medicine (EBM):

- Is **"the conscientious, explicit, and judicious use and appraisal of current best evidence in making decisions about the care of individual patients."**
- Aims to provide a framework for acquiring knowledge and making optimal decisions around medical care
- Includes **"integrating individual clinical expertise with the best available external clinical evidence from systematic research."**
- Has **five stages**:
 1. Ask an answerable question
 2. Search
 3. Critically appraise the evidence
 4. **Integrate the evidence with the patients unique circumstances** and values
 5. Evaluate the result

Levels of Evidence

Grade studies on likelihood of bias and internal validity.

The NHMRC defines 6 levels of evidence, graded from I-IV:

- Level I is evidence from a **systematic review** of RCTs
- Level II is evidence from **at least one good RCT**
- Level III-1 is evidence from a **pseudo-RCT**
- Level III-2 is evidence from a **comparative study with concurrent controls**, such as a cohort or case-control study
- Level III-3 is evidence from a **comparative study without concurrent controls**, such as a cohort study with historical controls
- Level IV is evidence from a case-series

Other organisations and institutions have different grading systems, most are broadly similar.

Note that expert opinion is **not** part of NHMRC levels of evidence, though it is included on the Oxford Centre for Evidence Based Medicine system, used by the NHS.

Table 1: NHMRC Levels of Evidence by Research Question

Level	Intervention	Diagnostic Accuracy	Prognostic	Aetiology	Screening
I	A systematic review of level II studies	A systematic review of level II studies	A systematic review of level II studies	A systematic review of level II studies	A systematic review of level II studies
II	A randomised controlled trial	A study of test accuracy with: an independent, blinded comparison with a valid reference standard, among consecutive persons with a defined clinical presentation	A prospective cohort study	A prospective cohort study	A randomised controlled trial
III - 1	A pseudoran-domised controlled trial	A study of test accuracy with: an independent, blinded comparison with a valid reference standard, among non - consecutive persons with a defined clinical presentation	All or none	All or none	A pseudoran-domised controlled trial

Level	Intervention	Diagnostic Accuracy	Prognostic	Aetiology	Screening
III - 2	A comparative study with concurrent controls	A comparison with reference standard that does not meet the criteria required for Level II and III - 1 evidence	Analysis of prognostic factors amongst persons in a single arm of a randomised controlled trial	A retrospective cohort study	A comparative study with concurrent controls
III - 3	A comparative study without concurrent controls	Diagnostic case-control study	A retrospective cohort study	A case-control study	A comparative study without concurrent controls
IV	Case series with either post-test or pre-test/post-test outcomes	Study of diagnostic yield (no reference standard)	Case series, or cohort study of persons at different stages of disease	A cross-sectional study or case series	Case series

Grades of Evidence

Evidence is graded to “indicate the strength of the body of evidence underpinning a recommendation” (e.g. in a clinical guideline). The NHMRC grades recommendations from A to D as follows:

- **A:** Body of evidence can be trusted to guide practice
 - **B:** Body of evidence can be trusted to guide practice, in most situations
 - **C:** Body of evidence provides some support, but care should be taken in its application
 - **D:** Body of evidence is weak and recommendation must be applied with caution
-

Study Types

Describe the features of evidence-based medicine, including ~~levels of evidence~~ (e.g. NHMRC), meta-analysis, and systematic review

Randomised Control Trial

A prospective randomised controlled trial is the **gold standard** of experimental research.

It involves allocating patients randomly to either an intervention or a reference (control) group, and measuring the outcome of interest. Allocation can be performed in three ways:

- Simple
Individuals allocated randomly. This may lead to uneven group sizes.
- Block
Allocation is performed within blocks such that group sizes will remain close in size
- Stratified
Groups are randomised within a category (i.e. men and women are randomised separately).

Strengths

- Only study design which can establish causation

- Eliminates confounding
Randomisation controls for both known and *unknown* confounding factors, as these should be randomly allocated between groups.
- Blinding can be performed in a standardised fashion
- Decreases selection bias

Weaknesses

- Costly
- Time-consuming
- **Not appropriate for all study designs**
 - Ethical concerns
e.g. Adrenaline in ALS
 - Practical concerns
Small patient population or uncommon disease may cause recruitment difficulties

Systematic Review

The process of evaluating all of the (quality) literature to answer a specific clinical question. These can be either:

- Meta-analysis
Statistical analysis of multiple trials to generate a combined estimate of effect.
- Narrative reviews
A review of several studies on a topic, aiming to draw conclusions from the totality of the evidence.

Meta-Analysis

Mathematical technique of combining the results of different trials to derive a single pooled estimate of effect. Can be performed by:

- Pooling the results of each trial
- Combining all of the raw data and conducting a re-analysis
 - Meta-analyses usually use **random-effects models**, which assumes there will be a variety of similar treatment effects
 - Individual trials are **summarised with an odds ratio**, and **weighted**, usually predominantly by sample size

Stages of a Meta-Analysis and Systematic Review:

1. Inclusion and exclusion criteria are predefined
2. Search: including online databases, reference lists, citations, and experts
3. Validation of potentially eligible trials (critique of interval validity, i.e. trial quality)
4. Analysis
For a meta-analysis, this includes heterogeneity analysis.
5. Reliability of result determined
i.e. Consistency across studies, statistical significance, large effect size, biological plausibility.
6. Sensitivity analysis
Repeating the analysis with alternative statistical models, or excluding borderline trials or outliers. If the result is unchanged despite these changes, then the findings are more **robust**.

Heterogeneity

For the pooling of results to be valid, the trials need to be similar. Differences between trials is known as **heterogeneity**. Heterogeneity can be either:

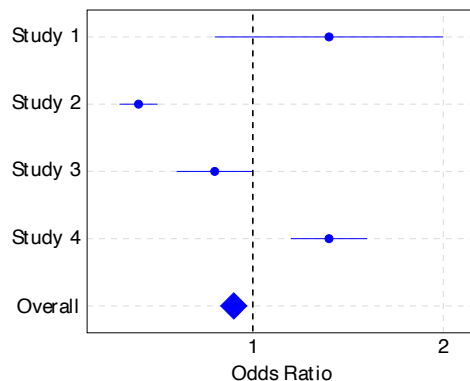
- **Statistical**; where the effects of the intervention are more different than would be expected to occur

through chance alone. Heterogeneity analysis affects the type of model that can be used (**fixed** or **mixed effects**) and highly heterogeneous data is not appropriate for meta-analysis.

- **Clinical**; where, due to trial design, it would be inappropriate to pool the results. For example, conducting a meta-analysis on the effects of the same drug in a paediatric and adult population would be inappropriate, as these are two different populations.
- **Methodological**; Where the methods used in different trials are too different to allow pooling of the data.

Forest Plots

Results of meta-analyses are presented in a blobbogram, or more boringly, a **Forest Plot**.



Where:

- The **x-axis** plots the odds ratio
- The **y-axis** lists the studies included, and the overall summary statistic
- The **circles** indicate the **point estimate** (the x-location) and the **weight** (the size of the dot) given to the study
- The **horizontal lines** indicate the upper and lower bounds of the **confidence interval** for each point estimate

NB: Remember that an OR of 1 indicates **no effect** of the intervention on the outcome.

- The **diamond** indicates the **overall point estimate** and (by its width) the **confidence interval** for the point estimate
- The result of the heterogeneity test should also be displayed
 $P < 0.1$ indicates significant heterogeneity.

Funnel Plots

A graphical tool to detect publication bias. Due to statistical power, larger studies should be a closer representation of the true effect. When evaluating an number of studies, one would expect that large studies cluster around the 'true effect' and smaller studies to have more scatter.

Strengths and Weaknesses of Meta-analyses

- Because of these weaknesses, positive meta-analyses should be considered largely hypothesis-generating, and should be confirmed by (a large) RCT. Negative meta-analyses can probably be accepted.

Table 1: Practical Strengths and Limitations of Meta-Analysis

Strengths	Weaknesses
Enhanced precision of estimates of effect	Publication bias
Useful when large trials have not been done or are not feasible	Duplicate publication
Generate clinically relevant measures (NNT, NNH)	Heterogeneity
	Inclusion of outdated studies

References

1. Myles PS, Gin T. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001.

Clinical Trial Design

Describe the stages in design of a clinical trial

1. Determine research question
 - Literature review
 - Current evidence on topic
 - Unanswered questions
 - Define the question
 - Aim
 - Significance
 - Is the outcome of interest, and is the result implementable?
 - Endpoint
 - Hard endpoints
 - Patient - centered
 - Surrogate
 - Composite
2. Determine target population
 - Inclusion and exclusion criteria, and justifications for each.
3. Study methodology
 - Protocol
 - Should be strict and highly prescriptive to minimise number of protocol violations.
 - Statistical analysis
 - Randomisation
 - Need for control group
 - Treatment allocation and concealment
 - Missing data analysis plan
 - Sensitivity analysis

- Adverse event reporting method
- Sample size estimation
Should be powered adequately for desired outcome.
- Identify sources of:
 - Confounding
 - Bias
- Pre-specify analysis plan
- Data collection and storage
 - Issue with multi-centre, international trials
 - National Privacy Principles compliance
- Funding
- Ethics committee review and approval
- 4. Register study in an online database
- 5. Perform pilot study
- 6. Refine methodology based on pilot study
Determine if doing the definitive study is feasible. Potential issues:
 - Patient recruitment
 - Event rate as predicted
 - Group separation
 - Feasibility of methodology
- 7. Perform definitive study
- 8. Publish
CONSORT methodology or similar.

References

1. PS Myles, T Gin. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001.

Data Types

Describe the different types of data

Data are a series of observations or measurements, and can be either **qualitative** or **quantitative**.

Qualitative Data

Using words as data rather than numbers, evaluating meaning and process. Common in the social sciences.

Quantitative Data

Uses numbers, or can be coded numerically. Divided into multiple types, each with multiple subtypes.

- **Categorical**

Data exist in **discrete categories without intrinsic order**.

- e.g. Medical specialty (intensive care, emergency medicine, orthopaedics, cardiology)
- Descriptive statistics for categorical data can be reported using the absolute number for each category, percentages, or proportions

- **Ordinal**

Data exists in **discrete categories with an intrinsic order**, e.g. age groups (0-5, 6-10, 11-15...)

- Descriptive statistics for ordinal data are the same for categorical data, but they can also be summarised by the median and the range (e.g. median age group, age group range).

- **Numeric**

Data is an actual number. Can be subdivided into discrete or continuous:

- **Discrete**

Can only be recorded as an integer (whole number), e.g. number of hospital admissions.

- **Dichotomous** or **binary** data, which occurs when there are only two categories

- **Continuous**

Where data can assume any value (including fractions), e.g. white cell count.

- Continuous data can be further subdivided into **interval** or **ratio** data:

- **Ratio data**

Are expressed with reference to a **rational zero**, which is where zero means **no measurement**.

- e.g. Temperature in °K is a ratio variable, whilst temperature in °C is not
This is because 0°K means no temperature, whilst 0°C does not; e.g. 50°K is half the temperature of 100°K, but 50°C is not half the temperature of 100°C.
- Ratio variables can (unsurprisingly) be expressed as ratios, whilst interval variables can not

- **Interval data**

Do not have a rational 0 - this is just another point on the line (e.g. temperature in °C).

References

1. Myles PS, Gin T. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001.

Bias and Confounding

Describe bias, ~~types of error~~, confounding factors and sample size calculations, and the factors that influence them

Bias

Bias is a **systematic deviation from truth**, and causes a study to lack **internal validity**.

In a research study, an observed difference between groups may be due to:

- A true difference between groups
- Error

Error can be due to:

- Normal **random variation**, i.e. **chance**
- A **systematic difference**, i.e. **bias**

Unlike error due to chance, the effect of bias cannot be reduced by increasing the sample size.

Table 1: Types of Bias and Methods of Prevention

Type of Bias	Description	Prevention
Selection	Where subject allocation results in treatment groups that are systematically different, apart from in the intervention being studied	Randomisation
Detection	Where measurements are taken differently between treatment groups	Blinding
Observer	Where the data collector is able to be subjective about the outcome	Blinding, Hard outcomes
Publication	When negative studies are less likely to be submitted or published than positive ones	Clinical trial registries
Recall	Altered reporting of symptoms by patients depending on which group they have been allocated to	Blinding
Response	When patients who enroll for a trial differ from the population, limiting generalisability	Random sampling
Hawthorne Effect	When the process of actually doing the study improves the outcome	Control group, masking study intent from patients and observers

Confounder

A confounder is:

- **"A variable that, if removed, results in a change in the outcome variable by a clinically significant amount."**

- A type of bias which will result in a *distortion* of the measured effect.

A confounding factor must be:

- **Associated with the exposure** but **not a consequence of it**
 - A confounding factor cannot be on the causal pathway between exposure and disease
 - It must be present unevenly between groups to cause distortion of the measured effect
- **An independent predictor of outcome**

The confounding factor must also be a risk factor for the disease, but independently from exposure.

Controlling For Confounding

Confounding can be controlled for:

- By **design**:
 - **Randomisation**

All confounders (known and unknown) are distributed evenly between groups.
 - **Restriction**

Restricts participants to remove confounders.
 - Results in reduced generalisability and does not control all factors
 - **Matching**

Pairing of similar subjects between groups.

 - May introduce additional confounding, and matching by multiple characteristics is difficult
- By **analysis**:
 - **Standardisation**

Adjust for differences by transforming data.
 - **Stratification**

Analyse the data in subgroups for each potential confounding factor.

References

1. Sackett, D. L. (1979). Bias in analytic research. *Journal of Chronic Diseases* 32 (1–2): 51–63.
2. PS Myles, T Gin. *Statistical methods for anaesthesia and intensive care*. 1st ed. Oxford: Butterworth-Heinemann, 2001.
3. Stats notes from my MPh (University of Sydney). Probably a Timothy Schlub lecture, circa 2014.

Frequency Distributions

Describe frequency distributions and measures of central tendency and dispersion

Frequency distributions are a method of tabulating or graphically displaying a number of observations.

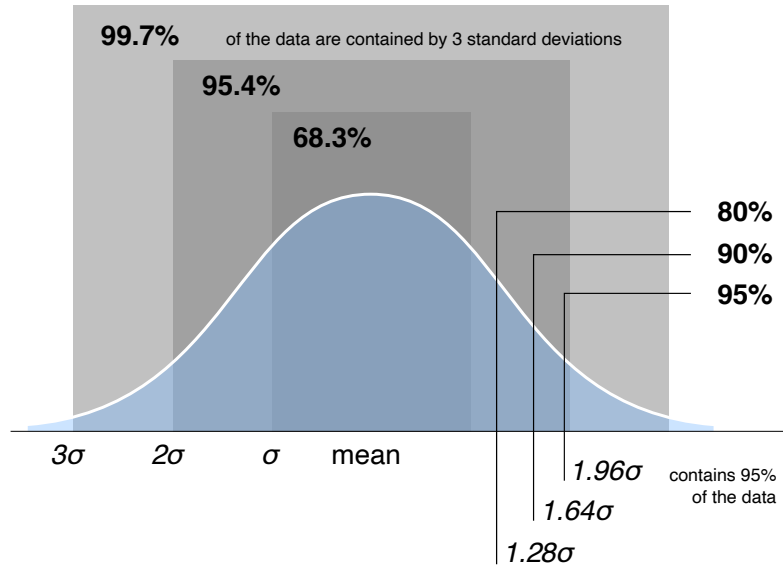
The Normal Distribution

The normal distribution is a **gaussian distribution**, where the majority of values cluster around the mean, whilst more extreme values become progressively less frequent.

The normal distribution is common in medicine for two reasons:

1. Much of the variation in biology follows a normal distribution
2. When multiple random samples are taken from a population, the **mean of these samples follows a normal distribution, even if the characteristic being measured is not normally distributed**

This is known as the **central limit theorem**, and is useful because many statistical tests are only valid when the data follow a normal distribution.



The formula for the normal distribution is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

From this, it can be seen the two variables which will determine the shape of the normal distribution are:

- μ (mu): The mean
- σ (sigma): The standard deviation

The Standard Normal Distribution

The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1. The equation for the standard normal distribution is much simpler, which is why it is used:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Any normal distribution can be transformed to fit a standard normal distribution using a **z transformation**:

$$z = \frac{x - \mu}{SD}$$

The value of z then gives a **standardised score**, i.e. the number of standard deviations from the mean in a standardised curve. This can then be used to determine probability.

Binomial Distribution

Where observations belong to one of two mutually exclusive categories, i.e.:

- If: $P(A) = x$
- Then: $P(B) = 1 - x$

If the number of observations is very large and the probability of an event is small, a **Poisson distribution** can be used to approximate a binomial distribution.

Measures of Central Tendency

As noted above in the **normal distribution**, results tend to cluster around a central value. Quantification of the degree of clustering can be done using **measures of central tendency**, of which there are three:

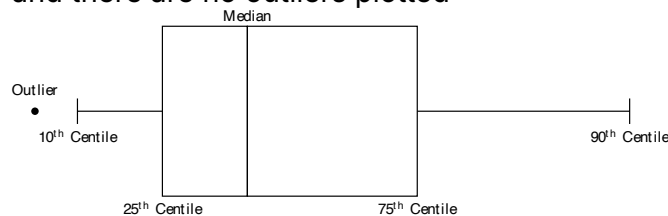
- **Mode**
The most common value in the sample.
- **Median**
The middle value when the sample is ranked from lowest to highest.
 - The median is the best measure of central tendency when the data is **skewed**
- **Arithmetic mean**
The average, i.e: $\bar{x} = \frac{\sum x}{n}$
The mean is common and reliable, though inaccurate if the distribution is **skewed**.

Measures of Dispersion

Measures of variability describe the degree of dispersion around the central value.

Basic Measures of Deviation

- **Range:** The lowest and highest values in the sample
Highly influenced by outliers
- **Percentiles:** Rank observations into 100 equal parts, so that the median becomes the 50% percentile.
Better measure of spread than range.
- **Interquartile range:** The 25th to 75th centile
A **box-and-whisker plot** graphically demonstrates the mean, 25th centile, 75th centile, and (usually), the 10th and 90th centiles.
 - Outliers are represented by dots
 - Occasionally the range is plotted by the whiskers, and there are no outliers plotted



Variance

Variance is a better measure of variability than the above methods. Variance:

- Evaluates how far each observation is from the mean, and penalises observations more the further they lie from the mean
- **Sums the squares** of each difference and divides by the number of observations i.e:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

- $n - 1$ is used (instead of n) because the mean of the sample is known and therefore the last observation calculated must taken on a known quantity
- This is known as a **degrees of freedom**, which is a mathematical restriction used when using one statistical test in order to estimate another
- It is a confusing topic best illustrated with an example:
 - You have been given a sample of two observations (say, ages of two individuals), and you know nothing about them
 - The degrees of freedom is **two**, since those observations can take on any value.
 - Alternatively, imagine you have been given the same sample, but this time I tell you that the mean age of the sample is 20
 - The degrees of freedom is **one**, since if I tell you the value of one of the observations is 30, you know that the other must be 10
Therefore, only one of the observations is free to vary - as soon as its value is known then the value of the other observation is known as well.
- Different statistical tests may result in additional losses in degrees of freedom.

Standard Deviation

The standard deviation is the **positive square root of the variance**.

In a sample of normal distribution:

- 1 σ either side of the mean should include ~68% of results
- 2 σ either side of the mean should include ~95% of results

- 3 σ either side of the mean should include ~99.7% of results

Standard Error and Confidence Intervals

Standard error of the mean is:

- A measure of the precision of the estimate of the mean
- Calculated from the standard deviation and the sample size
As the sample size grows, the SEM decreases (as the estimate becomes more precise).
- Given by the formula:
$$SE = \frac{\sigma}{\sqrt{n}}$$
- Used to calculate the **confidence interval**

Confidence Interval

The **confidence interval**:

- Gives a range **in which the true population parameter is likely to lie**
The width of the interval is related to the **standard error**, and the degree of confidence (typically 95%):
 - 95% *CI of $\bar{x}$* = $\bar{x} \pm (1.96 \times SE)$
 - 99% *CI of $\bar{x}$* = $\bar{x} \pm (2.58 \times SE)$
- Is a function of the sample statistic (in this case the mean), rather than the actual observations
- Has several benefits over the p -value:
 - Indicates magnitude of the difference in a meaningful way
 - Indicates the precision of the estimate
The smaller the confidence interval, the more precise the estimate.
 - Allows statistical significance to be calculated
If the confidence interval crosses 1, then the result is insignificant. Note that the **inverse is not true**; a result is not necessarily significant because the CI does not cross 1.

References

1. "Normal distribution". Licensed under Attribution 3.0 Unported (CC BY 3.0) via SubSurfWiki.
2. Myles PS, Gin T. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001
3. Course notes from "Introduction to Biostats", University of Sydney, School of Public Health, circa 2013.

Sample Size Calculation

Describe bias, types of error, confounding factors and **sample size calculations, and the factors that influence them**

Samples

A sample is a subset of a population that we wish to investigate. We take measurements on our sample with the aim to make inferences on the general population. An optimal sample (in quantitative research) will be **representative**, that is, it has the same characteristics of the population it is drawn from.

Sampling Error

Due to chance, the sample mean for any given variable will not equal the population mean. This is called **sampling error**, and is a form of random error. A larger sample will more closely approximate the population mean, reducing random error leading to more accurate point estimates and narrower confidence intervals.

This is why large sample sizes are desirable in research. However, larger studies are also more costly and time consuming to run. Sample-size calculations are performed to find a happy medium.

Sample Size Calculation

All sample size calculations depend on:

- Acceptable **risk of Type I error (α)**, typically set at 0.05
A **smaller α** (lower false positive risk) requires a larger sample size.
- Acceptable **risk of Type II error (β)**, typically set at 0.20
A **smaller β** (lower false negative risk) requires a larger sample size.
- Expected **effect size**
A **smaller effect size** requires a larger sample size, as the difference between groups will be smaller and harder to detect.
- Population **variance**
A **larger population variance** requires a larger sample size, as there is more 'noise' in the sample.
- **Study design**
Certain trial designs (e.g. multiple arms) require a larger sample size for a given effect size and power.
- Practical considerations
 - **Cost**
Increasing sample size increases the cost of a study.
 - **Participant availability**
Sample size is limited when the number of eligible participants for a study is small (e.g. rare diseases)

Different formulas for sample size calculations exist for different studies, and can be adjusted for particular study designs, such as multiple or unequal groups.

References

1. Myles PS, Gin T. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001.

2. Course notes from "Introduction to Biostats", University of Sydney, School of Public Health, circa 2013.

Statistical Tests

Describe the appropriate selection of non-parametric and parametric tests and tests that examine relationships (e.g. correlation, regression)

Parametric Tests

Parametric tests are used when data is:

- Continuous and numerical
 - Normally distributed
 - Remember that due to the central limit theorem - large data sets ($n > 100$) are typically amenable to parametric analysis, as sample means will follow a normal distribution
 - Non-normal data can be transformed so that they follow a normal distribution
 - Samples are taken randomly
 - Samples have the same variance
 - Observations within the group are independent
- Independent results are those when one value is not expected to influence another value.
- A common example is repeated measures: when serial measures are taken from a patient or a hospital, the results cannot be treated as independent
 - **Paired tests** are used when two dependent samples are compared

- **Unpaired test** are used when two independent samples are compared

Tests may be one-tailed or two-tailed:

- A **two-tailed test** evaluates whether the sample mean is significantly greater or less than the population mean
- A **one-tailed test** only evaluates the relationship in one direction
This doubles the power of the test to detect a difference, but should only be performed if there is a very good reason that the effect could only occur in one direction.

Common parametric tests include:

- Z test
- Student's T test
- F test
- ANOVA

Z test

Used to test whether the mean of a particular sample ($\bar{x}$) differs from the population mean (μ) by random variation.

Assumptions:

- Large sample
 $n > 100$.
- Data is normally distributed
- Population standard deviation is known

Student's T Test

This is a variant of the Z test, used when the population standard deviation is not known.

The results from Student's T test approximate the results of the Z test when $n > 100$

F Test

Compares the ratio of variances ($\frac{Var_1}{Var_2}$) for two samples. If F deviates significantly from 1, then there is a significant difference in group variances.

Analysis of Variance (ANOVA)

ANOVA tests for significant differences between means of multiple groups, in a more efficient manner than multiple comparisons (doing lots of T tests).

There are several types of ANOVA tests used in different situations, which are not discussed here.

Non-Parametric Tests

Non-parametric tests are used when the assumptions for parametric tests are not met. Non-parametric tests:

- Do not assume the data follows any particular distribution
 - This is required when:
 - Non-normality is obvious
e.g. Multiple observations of 0.
 - Possible non-normality
Typically small sample sizes.
 - Data is ordinal
- Are not as powerful as parametric tests (a larger sample size is required to achieve the same error rate)
- Are more broadly applicable than parametric tests as they do not require the same assumptions

Non-parametric tests still require that data:

- Is continuous or ordinal
- Within-group observations are independent
- Samples are taken randomly

In general, non-parametric tests:

- Take each observation and rank them in order
- Calculations are then performed based on rank order (rather than absolute value) to find the test statistic

Common non-parametric tests include:

- Mann-Whitney U Test/Wilcoxon Rank Sum Test
- Wilcoxon Signed Ranks Test

Mann-Whitney U Test/Wilcoxon Rank Sum Test

Alternative to the unpaired T-test for non-parametric data.

Process:

- Observations from both groups are combined, ordered, and given ranks
Tied data are given identical ranks, where that rank is equal to the average rank of the tied observations.
- The data are then separated into their original group
- Ranks in each group are added to give a test statistic for each group
- A statistical test is performed to see if the sum of ranks in one group is different to another

Wilcoxon Signed Ranks Test

Alternative to the paired T-test for non-parametric data.

Process:

- Observations are ranked as above (for the Wilcoxon Rank Sum Test), except absolute difference between paired observations are ranked
The sign (i.e. positive or negative) is preserved.

- The sum of positive ranks is then compared with the sum of negative ranks
 - If there is no difference between groups, we would expect the net value to be 0
-

References

1. Myles PS, Gin T. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001.

Statistical Terms

Understand the terms sensitivity, specificity, positive and negative predictive value and how these are affected by the prevalence of the disease in question

Describe bias, types of error, confounding factors and sample size calculations, and the factors that influence them

Error, Sensitivity, Specificity, and Predictive Values



All of these terms refer to characteristics of **diagnostic tests**. The easiest way to approach this is via a 2×2 table, and has been recommended in previous exams as an approach to questions on this topic.

Table 1: Components of a 2×2 table

Result	Disease Positive	Disease Negative	Total
Test Positive	True Positives	False Positives	All Test Positives
Test Negative	False Negatives	True Negatives	All Test Negatives
Total	All Disease Positives	All Disease Negatives	Total

In this table:

- **Positive** or **negative** refers to the **test result**
- **True** or **false** refers to whether the test was **correct**

Error

From the table, we can see that there can be both false positive and false negative results:

- A **Type I error** is a **false positive**, when we incorrectly reject the null hypothesis
 - The type I error rate can be decreased by decreasing α
- A **Type II error** is a **false negative**, when we incorrectly accept the null hypothesis
 - The type II error rate can be decreased by decreasing β , usually expressed as increasing **power**. Power is the chance of detecting a difference if it exists. Power is equal to $1 - \beta$.

Sensitivity

Sensitivity is the probability those with the disease test positive, i.e. the **true positive rate**, and expressed mathematically as:

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{All Disease Positives}} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Sensitivity:

- Refers to the ability of a test to **detect the condition**
- A highly sensitive test will likely be positive if the condition is present
- Therefore, a **negative** result on a **sensitive test** gives a **high likelihood the disease is not present**
 - The mnemonic for this is **SNOUT** - Sensitive, Negative, rule OUT

Specificity

Specificity is the probability those without the disease test negative, i.e. the **true negative rate**, and expressed mathematically as:

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{All Disease Negatives}} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

Specificity:

- Refers to the ability of a test to **detect absence of the condition**
- A highly specific test will likely be negative if the condition is not present
- Therefore a **positive** result on a **specific test** gives a **high likelihood the disease is present**
 - The mnemonic for this is **SPIN** - Specific, Positive, rule IN

Positive and Negative Predictive Values

- Positive and negative predictive values describe **the proportion of test results which are true**
- A high value indicates accuracy of the test
- Because of how they are derived, **they are dependent on population prevalence of the disease**

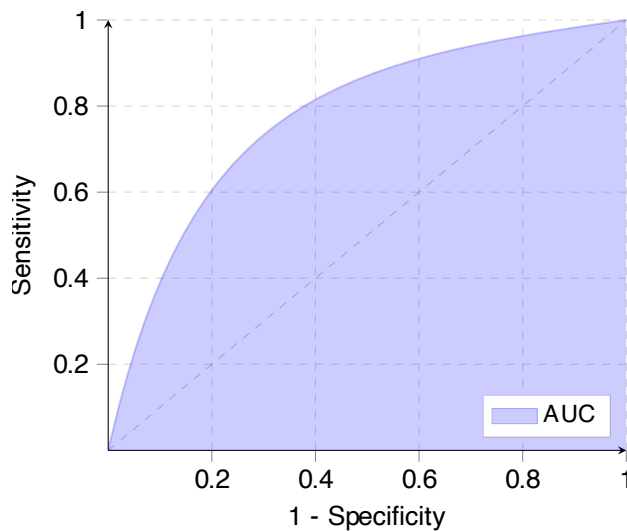
- **Positive Predictive Value (PPV)** is the probability that the **disease is present** when the **test is positive**:

$$\text{Positive Predictive Value} = \frac{\text{Disease Positives}}{\text{All Test Positives}} = \frac{\text{Disease Positives}}{\text{Disease Positives} + \text{False Positives}}$$

- **Negative Predictive Value (NPV)** is the probability that the **disease is absent** when the **test is negative**:

$$\text{Negative Predictive Value} = \frac{\text{Disease Negatives}}{\text{All Test Negatives}} = \frac{\text{Disease Negatives}}{\text{Disease Negatives} + \text{False Negatives}}$$

Receiver Operating Characteristic



The ROC curve:

- Plots the diagnostic ability of any **binary classifier**
- Graphically demonstrates the relationship between true and false positive rates at variable threshold settings

This allows cut-points to be made at different levels, depending on the relative trade-off between sensitivity and false-positive rate.

- Has an area under the curve (AUC), which estimates the test performance

The higher the AUC, the better the test:

- 0.5 is no better than chance

- 0.7-0.8 is acceptable
- 0.8-0.9 is excellent
- 1 is perfect prediction

Remembering the Difference

Table 2: Relationships Between Diagnostic-Test Results and Derived Measures

Test Result	Disease Positive	Disease Negative	Derived Variable
Test Positive	TP	FP	PPV
Test Negative	FN	TN	NPV
Derived Variable	Sensitivity	Specificity	

- Rote learning these formulas is hard
- Just remember:
 - The above table
 - That the **numerator** will **always** be (for the type of stats that is assessed) **true**, i.e. either a true positives or true negatives
- The equations are easy to derive from here
- Other hints:
 - Sensitivity and specificity are **the same** for any given prevalence of disease
Therefore they look at columns (disease positive or disease negative).
 - PPV and NPV **depend on the population**
Therefore they look at rows (test positive or test negative).

Likelihood Ratios

The weakness of PPV and NPV as tools of evaluating the utility of a test in clinical practice is that they do not take into account the population prevalence, i.e. the *prior probability*, of a condition.

A classic example is the urine bHCG, which has a high positive predictive value for pregnancy. Tested on an exclusively male group however, the true positive rate will be 0 (since there are no pregnancies), and so all test positives will be false positives.

Therefore:

- The actual utility of a test in decision making is dependent upon the **prior probability** of the disease being present
- Likelihood Ratios relate the **pre-test odds** to the **post-test odds**

They are useful because (unlike the above values) they do not assume that the patient you are applying them to is identical to the sample from which the statistic was derived.

- The likelihood ratio multiplied by the pre-test odds gives the post-test odds of the disease being present
 - A **positive likelihood ratio** is used when the test is positive:

$$LR(+) = \frac{\text{sensitivity}}{1 - \text{specificity}}$$
 - A **negative likelihood ratio** is used when the test is negative:

$$LR(-) = \frac{1 - \text{sensitivity}}{\text{specificity}}$$

References

1. Myles PS, Gin T. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-

Heinemann, 2001.

2. Course notes from "Introduction to Biostats", University of Sydney, School of Public Health, circa 2013.

Risk and Odds

Understand the concepts of risk and Odds Ratio

Risk

Risk has **two** broad subtypes:

- **Absolute Risk** is the risk of an event occurring in the exposed group: $\frac{\text{Exposed individuals with outcome}}{\text{Number of exposed}}$
- **Relative Risk** (or risk ratio) is the risk of an event occurring in the exposed group relative to the unexposed group: $\text{Relative Risk} = \frac{\text{Risk in Exposed}}{\text{Risk in Unexposed}} = \frac{\frac{\text{Exposed individuals with outcome}}{\text{Number of exposed}}}{\frac{\text{Unexposed individuals with outcome}}{\text{Number of unexposed}}}$
- **Absolute Risk Reduction** is the decrease in risk provided by an exposure: $\text{ARR} = \text{Risk in Exposed} - \text{Risk in Unexposed}$
 - Is a clinically useful measure of the value of an intervention because contextualises the efficacy of the intervention with the prevalence of the outcome

ARR is better (more intuitively) expressed as the **Number Needed to Treat (NNT)**, which is the number of individuals who must receive a treatment to prevent one event: $\text{NNT} = \frac{1}{\text{ARR}}$
- **Relative Risk Reduction** is the decrease in incidence provided by treatment.
It is not as useful a measure of the value of an intervention, but drug companies like it because the numbers are bigger than absolute risk reduction.

Odds

Odds are the probability of an event happening compared to the probability of it not happening, and is usually expressed as a fraction.

The **Odds Ratio**:

- Is the ratio of the odds of the outcome occurring in the exposed compared to the odds of it occurring in the unexposed: $OR = \frac{\text{Odds of the outcome in the exposed}}{\text{Odds of the outcome in the unexposed}}$
 - An $OR < 1$ suggests the risk is lower in the exposed group
 - An $OR > 1$ suggests the risk is higher in the exposed group
 - An $OR = 1$ suggests that the groups are equivalent
- Generally overstates risk compared to the RR
- The OR is approximately equal to the RR when the outcome is rare ($< 10\%$)

This is because **when the event rate is small, the number of non-events in a group is very similar to the overall number of individuals in the group.**
- Is used when:
 - The denominator is uncertain
e.g. Retrospective designs, such as case-control studies when patients with the disease were identified, and then exposures ascertained.
 - When it statistically appropriate
 - Multivariate regression
 - Systematic Reviews

ORs are much easier to use in many statistical tests than RR because the log-odds ratio is normally distributed – a mathematically useful property – and so tend to be favoured by statisticians.

Risk versus Odds

Relative Risk and **Odds Ratios** are both methods of comparing the likelihood of an outcome occurring between two groups. The difference, and particularly the concept of odds ratios, are **commonly confused**. Relative risk tends

be much more intuitive than odds ratios. Imagine a trial has been performed, where group A was exposed group:

- In group A, the mortality was 50%
- In group B, the mortality was 25%

The **relative risk** is intuitive: $RR = \frac{\text{Risk of death in exposed}}{\text{Risk of death in unexposed}} = \frac{0.5}{0.25} = 2$

The **odds ratio** is not: $OR = \frac{\text{Odds of death in exposed}}{\text{Odds of death in unexposed}} = \frac{1/1}{1/3} = 3$

A RR of 2 is intuitive, but the OR of 3 is not. Now, imagine another trial where:

- In group A, the mortality was 90%
- In group B, the mortality was 10%

$$RR = \frac{\text{Risk of death in exposed}}{\text{Risk of death in unexposed}} = \frac{0.9}{0.1} = 9$$

$$OR = \frac{\text{Odds of death in exposed}}{\text{Odds of death in unexposed}} = \frac{9/1}{1/9} = 81$$

The relative risk is 9, but the OR is 81!

So why use odds ratios at all? Odds ratios are:

- Required when subjects are selected on the basis of outcome, rather than the basis of exposure
- Used by many statistical tests because the log odds ratio is normally distributed, which is a mathematically useful property

Note that relative risk has a weakness as well - it is dependent on how the question is framed. Using the first trial above, we calculated that RR for death was 2 and the OR was 3. Rather than calculating mortality, an alternative method could be to look at survival:

- In group A, the survival was 50%
- In group B, the survival was 75%

$$RR = \frac{\text{Risk of survival in exposed}}{\text{Risk of survival in unexposed}} = \frac{0.5}{0.75} = 0.6\bar{6}$$

$$OR = \frac{\text{Odds of survival in exposed}}{\text{Odds of survival in unexposed}} = \frac{1/1}{3/1} = 1/3$$

Note that the relative risk is not 0.5 (as you may initially assume). However, the odds ratio is simply the inverse of the previous value.

References

1. Myles PS, Gin T. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001.
2. Course notes from "Introduction to Biostats", University of Sydney, School of Public Health, circa 2013.
3. Simon S. Odds ratio vs. relative risk. "Steve's Attempt to Teach Statistics (StATS)". Children's Mercy Hospital, 2006.
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Significance Testing

Understand concept of significance and testing of significance

Significance testing is:

- The **process** of **determining whether a difference** between groups in a study **is due to a real difference, or chance alone**
- Usually performed using p-values
- Does not imply clinical significance

For a result to be statistically significant, there must be a 'real' difference between groups.

- This difference does not have to be clinically meaningful
e.g. A drug may reliably cause a 5mmHg decrease in SBP - this is unlikely to cause a meaningful drop in cardiovascular mortality but may be statistically significant.

This section assumes we are conducting null hypothesis significance testing. Bayesian analysis, despite its many benefits and similarity to clinical reasoning, has not gained the attention of the college.

p-values

The ***p*-value** is the **probability of obtaining a summary statistic** (e.g. a mean) **equal to or more extreme than the observed result, provided that the null hypothesis is true.**

Using p-values:

The *p*-value is commonly (mis)used in frequentist significance testing

- Prior to performing an experiment, a significance threshold (α) is selected
 - Traditionally 0.05 (5%) or (less-traditionally) 0.01 (1%)These values define the “false-positive rate”.
 - When multiple tests are being performed on one set of data, the chance of a false-positive will increase
To reduce the chance of a false positive occurring, the significance threshold for each test can be reduced. One method of this is the Bonferroni correction, where α is divided by the number of tests being performed.
- Then the experiment is performed, and a value for p is calculated
If $p < \alpha$, it suggests that the results are inconsistent with the null hypothesis (at that significance level), and it should be rejected.

Problems with p -values

p -values, when employed correctly, are useful. However, they do have several weaknesses:

- Assume the null hypothesis is true¹
- The p -value assumes that there is no real difference between groups.
 - This may not be the case
 - Not all hypotheses are created equal
There may be significant prior evidence supporting (or refuting) H_A - this will be ignored when interpreting a p -value.

¹It is a common misconception that the p -value estimates the chance that the result is true.

This is not the case.

The p -value measures **how inconsistent the observed results are with the null hypothesis.**

- Any study with significant results must therefore be interpreted in the context of:
 - Biological plausibility of those results
 - The previous evidence on the topic
 - A threshold of 0.05 is not always appropriate
The cost of being wrong must be included when interpreting a p -value. If this is a true result, what are the potential benefits? If this is a false positive, what are the potential harms?
 - Vulnerable to multiple comparisons
Conducting repeated analyses will eventually find a 'significant' result. At an α of 0.05, we would expect 1/20 analyses to be a false positive. Conducting 20 analyses would therefore generate one false positive result.
 - Do not quantify effect size
A significant p -value simply suggests a difference exists, it does not measure how big this difference is.
 - A result may be statistically significant but clinically unimportant, e.g. an antihypertensive medication causing a decrease in SBP by 2mmHg may be statistically significant, but clinically unimportant.
 - Related to sample size p -values are affected by sample size:
 - A large effect size may be hidden by an insignificant p -value if sample size is small
 - Similarly, a tiny effect size may be detected (i.e. a significant p -value) if sample size is large
 - Do not account for bias
Like other statistical test, the p -value cannot account for bias or confounding.
-

References

1. Wasserstein RL, Lazar NA. The ASA's Statement on p -Values: Context, Process, and Purpose. The American

Statistician. 2016 vol: 70 (2) pp: 129-133.

Drug Approval and Development

Describe the processes by which new drugs are approved for research and clinical use in Australia, and to outline the phases of human drug trials (Phase I-IV)

Drug Approval

The Therapeutic Goods Administration (TGA) approves medicine for both research and clinical use in Australia.

Research

Drug trials are approved for research purposes under two schemes:

1. Clinical Trials Exemption
Drugs must be evaluated by an expert committee to evaluate all aspects of pharmacology, toxicology, mutagenicity, teratogenicity, organ dysfunction, and other side-effects.
2. Clinical Trials Notification
A drug which has been approved in another nation with similarly stringent requirements (New Zealand,

Netherlands, UK, Sweden, US) may be used in a trial with oversight by a local ethics committee.

Clinical Use

The TGA classifies medicines into:

- **Registered Medicines**

Assessed by the TGA for quality, safety, and efficacy.

- All prescription (high-risk) medicines. Assessed on:

- **Quality**

- Composition of drug substance
 - Batch consistency
 - Stability data
 - Sterility data (if applicable)
 - Impurities

- **Non-clinical**

- Pharmacology data
 - Toxicology data

- **Clinical**

- Efficacy: results of clinical trials
 - Most OTC (low-risk) medicines
 - Some complementary medicines

- **Listed Medicines**

Assessed by the TGA for quality, safety, but not efficacy.

- Some OTC medicines
- Most complementary medicines

Phases of Drug Development

- **"Phase 0"**

- Pre-clinical R&D
 - *In vitro* and animal testing

- **Phase I**

- First administration in healthy human volunteers
 - Basic pharmacokinetic and toxicology data

- 20 - 100 human subjects
 - **Phase II**
 - Administration to select patient groups
 - Aim to establish dose-response curve
 - Evidence of efficacy and safety in the target population
Usually surrogate outcome measures.
 - **Phase III**
 - Full-scale evaluation of benefits, potential risks and costs analysis
 - 2000-3000 patients, usually treated in groups of several hundred for relatively short durations (3-6 months), regardless of the length of time the drug will be used in practice^[3]
 - May not reveal uncommon or long-term risks
 - **Phase IV**
 - Post-marketing surveillance
-

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1. PS Myles, T Gin. Statistical methods for anaesthesia and intensive care. 1st ed. Oxford: Butterworth-Heinemann, 2001.
2. Medicines and TGA classifications. Therapeutic Goods Administration. Available at: <https://www.tga.gov.au/medicines-and-tga-classifications>
3. Chris Anderson. Pharmaceutical Aspects and Drug Development. ICU Primary Prep.

Pharmacology

Pharmacokinetics

Additives

Describe the pharmaceuticals and formulation of drugs including ~~packaging, formulation, isomerism, compatibility~~ and excipients (additives) as they pertain to drugs

Additives are **components of a drug preparation** which **do not exert** the **pharmacological effect**. They include:

- **Preservatives**
 - Benzyl alcohol
 - Antimicrobial when > 2%
 - Can be used as a solvent when > 5%
 - Toxic
- **Antioxidants**
 - Sulfites
 - Hypersensitivity
 - Neurotoxic if given intrathecally
- **Solvents**
 - Water

Appropriate for dissolving polar molecules.
 - Non-aqueous solvents

Used to dissolve non-polar molecules, or to produce more stable preparations of semi-polar molecules. Examples include:

 - Propylene glycol
 - Hypotension
 - Arrhythmia

With rapid injection.
 - Pain on injection
 - Thrombophlebitis
 - Mannitol

- Diuresis
 - Soybean oil
 - Pain on injection
 - Allergy
 - **Emulsion**

Formed when drops of a liquid are dispersed through-out another liquid in which it is immiscible. Emulsions are:

 - Unstable
Emulsifiers are used to enhance stability.
 - Prone to contamination
Due to the water component.
 - Prone to rancidity
Due to the oil component.
 - **Buffers**

Maintain pH in a particular range in order to:

 - Maximise stability
Preserve shelf life.
 - Maintain solubility
 - Maximise preservative function
-

References

1. MacPherson RD. Pharmaceuticals for the anaesthetist. *Anaesthesia*. 2001 Oct;56(10):965-79.
2. Petkov V. *Essential Pharmacology For The ANZCA Primary Examination*. Vesselin Petkov. 2012.

Isomerism

Describe the pharmaceuticals and formulation of drugs including ~~packaging, formulation, isomerism, compatibility and excipients (additives)~~ as they pertain to drugs.

Describe and give examples of the clinical importance of isomerism

Isomerism:

- Describes groups of **compounds which have the same chemical formula but different chemical structures**
- Has relevance because **different isomers may have different** enzymatic and receptor affinities, altering their pharmacokinetic and pharmacodynamic **properties**

Types of Isomerism

Isomers can be divided into:

- Structural Isomers
- Stereoisomers

Structural Isomers

Have an **identical** chemical **formula** but **different arrangement** of atoms. Structural isomerism is subdivided into:

- **Static**

Further subdivided into:

- Chain isomer

The carbon skeleton varies, but position of functional groups is static.

- Position isomer

The carbon skeleton is static, but the position of functional groups varies.

- e.g. Isoflurane vs. enflurane

- **Dynamic** (also known as tautomer)

The molecule exists in a different molecular structures depending on the environment.

- e.g. Midazolam has pH dependent imidazole ring opening. When the pH is less than 4 the ring remains open, maintaining water solubility. Midazolam is supplied at pH of 3.5, and so is water soluble on injection but (due to its pKa of 6.5) becomes 89% unionised at physiological pH therefore able to cross lipid membranes.

Stereoisomers

Have the **same order of connections** between atoms, but a **different orientation of functional groups**. Stereoisomers are **not super-imposable**, meaning the different isomers can't be rotated so that they look the same. Stereoisomers are divided into:

- **Geometric Isomers**

Have a chemical structure (e.g. a carbon-carbon double-bond) prevents free rotation of groups, so different locations of chemical groups will create an isomer. Geometric isomers are known as **cis-** or **trans-** depending on whether the subgroups are

(Note the d- and l- nomenclature is *different* to D- and L-; the D-isomer refers to the molecule synthesised from (+)glyceraldehyde.)

on the same or opposite sides (respectively) of the chemical structure.

- e.g. Atracurium

- **Optical Isomers**

Optical isomers are **chiral**. This means they have **no plane of symmetry**. Optical isomers:

- Were initially named based on how they rotated under polarised light:

- **Dextrorotatory**

(d- or (+) isomers) molecules rotate **clockwise** under polarised light.

- **Levorotatory**

(l- or (-) isomers) molecules rotate **counter-clockwise** under polarised light.

- Unfortunately, some molecules were found to rotate in different directions depending on the temperature

Therefore, **another classification scheme (R/S)** is also used:

- Based on chemical structure
 - "Priority" is assigned to each atom in the structure
 - Highest priority is usually those with the highest molecular weight, but other rules exist for ambiguous or very large molecules
 - The molecule is arranged in space such that the lowest priority atom is facing "away"
 - An arrow is then drawn from the highest priority to the lower priority atoms:
 - If this arrow travels **clockwise** it is the **R (Rectus)** isomer
 - If this arrow travels **counter-clockwise** it is the **S (Sinister)** isomer

- Optical isomers are divided into:

- **Enantiomers**

Possess one chiral centre.

- e.g. levobupivacaine is less cardiotoxic than racemic bupivacaine.

- **Diastereoisomers**

Possess multiple chiral centres, and may have multiple stereoisomers. Since not all are

mirror images, these are not *enantiomers*.

- For a molecule with n chiral centres up to n^2 isomers are possible, though some of these may be duplicates.

Preparations

Drugs can be provided as:

- **Racemic solutions**

A **racemic** solution is one which where the different enantiomers are present in equal proportions.

- **Enantiopure preparations**

A drug produced with a single isomer, which may be **more efficacious** or **less toxic** (and definitely more expensive) than the racemic preparation.

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1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
2. CICM. The Mock Exam.
3. ChemGuide. Geometric isomerism
4. ChEBI. Misoprostol. European Molecular Biology Laboratory.
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Pharmacokinetic Terms

Explain the concept of pharmacokinetic modelling of single and multiple compartment models and define: Half life, Clearance, Zero and first order kinetics, Volume of distribution, Bio-availability, Area under the plasma concentration time curve, Extraction Ratio

Pharmacokinetics describes **what the body does to a drug**.

Volume of Distribution (V_D)

The volume of distribution is defined as the **theoretical volume into which an amount of drug would be distributed to produce the observed plasma concentration**.

$$V_d = \frac{\text{Total bioavailable drug}}{\text{Plasma concentration}}$$

Properties:

- Units are ml.kg^{-1}
- Describes what proportion of a drug is confined to plasma, and what proportion distributes to other tissues

It does not correspond to any particular volume, however a V_D of:

- Less than 40ml.kg^{-1} indicates a drug is confined to plasma

- Up to 200ml.kg^{-1} indicates a drug is confined to the ECF
- Up to 600ml.kg^{-1} indicates a drug is dissolved into the TBW
- Greater than 1L.kg^{-1} indicates a drug is highly protein bound or lipophilic
Agents which cross the blood brain barrier typically have a V_D of $1-2\text{L.kg}^{-1}$.
- Subtypes of the volume of distribution are used to describe drug distribution at different times or with different models
 - These include:
 - V_1
Volume of central compartment.
 - V_{Dss}
Volume of distribution at steady state.
 - V_{Dpe}
Volume of distribution at peak effect.
 - Which V_D to use depends on the pharmacological question
e.g. Intubating dose for opioid should use the V_{Dpe} as it will allow a target concentration to be selected for the time at which intubation will occur relative to drug administration
- Affected by changes in total-body water

Significance of the Volume of Distribution

Allows calculation of:

- **Expected drug concentrations** produced by a given dose
Both in plasma, as well as in target sites.
- **Loading doses**

Half-life

The half-life ($t_{1/2}$) is **the time it takes for a process to be 50% complete**¹.

- With respect to drug clearance, it is the time it takes for concentration (typically in plasma) to fall by 50%
- A **process** is considered to be **complete after 4-5 half-lives**
 - As the value decreases by 50% after each half-life, so **after 5 half-lives** concentration will be **3.125% of its starting value**
 - This also applies to wash in - it will take ~4-5 elimination half-lives of a drug for a constant-rate infusion to reach its final concentration

Various types of half-life are described:

- $t_{1/2\alpha}$ describes the rapidity of the distribution phase following drug administration
- $t_{1/2\beta}$ describes the rapidity of the *elimination* phase occurring after drug distribution equilibrium
This only evaluates clearance from plasma, and so is a composite of both excretion from the body (e.g. renal and hepatic clearance) and ongoing distribution to peripheral tissues.
 - The elimination half-life is generally not useful to predict drug offset, as this is affected by many

¹The formula for half-life can be derived from the equation for a wash-in exponential.

1. Wash in exponential is given by: $C = 1 - e^{-kt}$

2. $y = 0.5$ can then be substituted and the equation solved for t :

$$0.5 = 1 - e^{-kt}$$

$$-0.5 = e^{-kt}$$

$$\ln -0.5 = -kt$$

$$\ln 2 = kt$$

$$\frac{\ln 2}{k} = t$$

$$\frac{0.693}{k} = t$$

factors

However, it does set an **upper limit** on how long it will take plasma concentration to fall by 50%.

Half-life is mathematically related to many other key pharmacokinetic terms.

$$t_{1/2} = 0.693 \times \tau = \frac{0.693}{k} = \frac{0.693 \times V_D}{Cl}$$

Where:

- τ is the time constant
- k is the rate constant for elimination
- V_D is the volume of distribution
- Cl is the clearance

Significance of Half-Life

Half-life determines:

- The time taken:
 - To achieve **steady state**
Drugs with very long half-lives often require loading doses for this reason, as otherwise achieving an effective plasma concentration may take days-weeks.
 - For drug elimination
- **Dosing intervals**
If a drug is given:
 - Every half-life, the plasma concentration will fluctuate two-fold over the dosing interval
 - More frequently than one half-life, the peaks and troughs will be less than two-fold
 - Less frequently than one half-life, greater fluctuation will occur

Note that complete drug elimination is also another form of steady-state – both are achieved after 3–5× half-lives.

The significance of fluctuating drug levels is drug dependent. How much half-life affects dosing therefore depends on how much drug level affects:

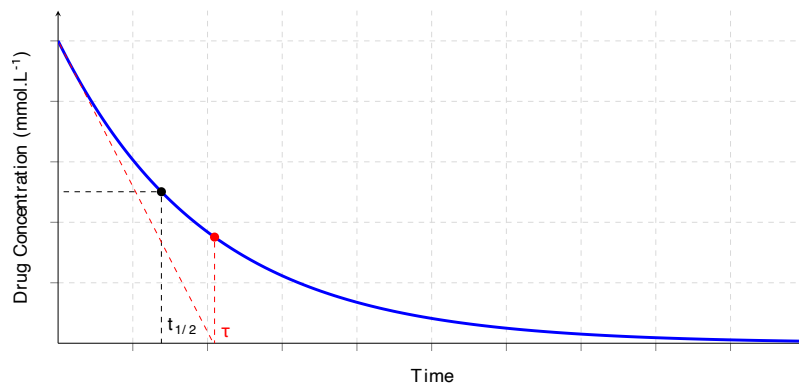
- Clinical efficacy
- Toxicity

Time Constant (τ)

The time constant is the **time taken for a process to complete if it continued at its initial rate of change**. Time constants are similar to the half-life, but are **better suited when modeling change in exponential processes**.

- Elimination will be virtually complete after three time constants
- A time constant is the **inverse of the rate constant for elimination**, i.e. $\tau = \frac{1}{k}$
- Illustration of the relationship between half-life and time constant:

Time constants are discussed in more detail under respiratory Chapter .



Clearance and Rate of Elimination

Clearance describes the elimination of drug from the body. Clearance is the **volume of plasma completely cleared of a drug per unit time** measured in $\text{ml} \cdot \text{min}^{-1}$. Clearance:

- Does not include redistribution
- Is calculated from the area under the concentration time curve: $Cl = \frac{\text{Dose}}{AUC}$
- Is the sum of drug clearances from each organ

The **rate of elimination** is the **amount of drug removed by the body** per unit time, and is:

- The product of the clearance and the current concentration: *Rate of Elimination* = $Cl \times C$, in $\text{mg} \cdot \text{min}^{-1}$
- *Not* the same as clearance
- Also **not** the same as the rate *constant* for elimination

Significance of Clearance

Clearance determines the **maintenance dose required** to achieve **steady-state**.

Total Clearance

Total clearance is the sum of clearances from individual organs, e.g.:

- $Cl_{total} = Cl_{renal} + Cl_{liver} + Cl_{lung} + Cl_{organ\ independent} + Cl_{other}$, where:
 - $Cl_{renal} = \frac{[U] \times V}{[P]}$, where:
 - $[U]$ is urine concentration in $\text{mmol} \cdot \text{L}^{-1}$, a function of glomerular filtration, reabsorption, and secretion.
 - V is the urine flow in $\text{ml} \cdot \text{min}^{-1}$
 - $[P]$ is the plasma concentration in $\text{mmol} \cdot \text{L}^{-1}$
 - $Cl_{hepatic} = HBF \times ER_{liver}$, where:
 - HBF is the hepatic blood flow in $\text{ml} \cdot \text{min}^{-1}$
 - ER is the extraction ratio (see below)

In a one compartment model, total clearance can be expressed as either:

- $Cl = k \cdot V_D$ in $\text{ml} \cdot \text{min}^{-1}$, or:
- $Cl = \frac{V_D}{\tau} = \frac{0.693 \times V_D}{t_{1/2}}$

This uses the property that $\tau = \frac{1}{k}$.

Generally, total clearance is equivalent to the sum of hepatic (via metabolism to inactive metabolite, or biliary secretion) and renal (intrinsic clearance and reabsorption) clearance is a constant.

Clearance Kinetics

Drug clearance can follow either first order or zero-order kinetics:

- **First-order Kinetics**

A constant **proportion** of the drug in the body is eliminated per unit time.

- Most drugs are eliminated by first order kinetics, as the capacity of the elimination system exceeds the concentration of drug

- **Zero-order kinetics**

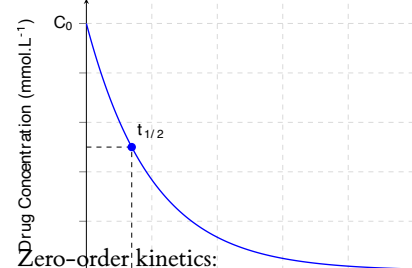
A **constant amount of drug** is eliminated per unit time, independent of how much drug is in the body.

- Occurs when there is saturation of enzyme systems

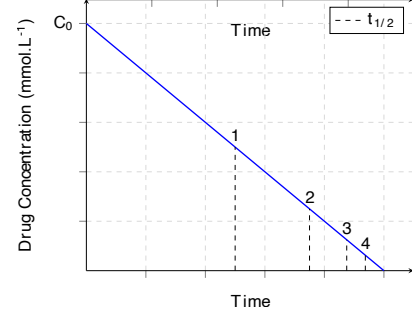
It is also known as **saturation kinetics** for this reason.

- e.g. Phenytoin follows first order kinetics at lower doses, but zero-order kinetics at doses within the therapeutic range
This is clinically relevant as the narrow therapeutic index means that toxic levels may occur rapidly with a small increase in dose.
- e.g. Ethanol also follows zero-order kinetics within the "therapeutic range", as it is a very weak (doses are in grams) positive allosteric modulator of the GABA_A receptor
- Zero-order kinetics is concerning as:
 - Plasma concentrations will **rapidly increase** with only modest dose increase
 - There is essentially **no steady state**: if drug input exceeds output, plasma levels will continue to rise

First-order kinetics:



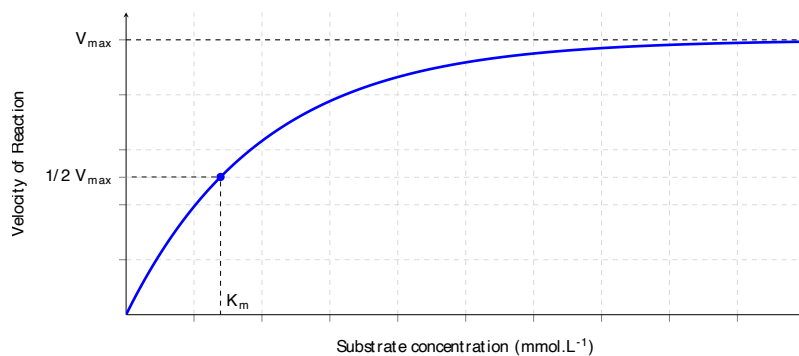
Zero-order kinetics:



Michaelis-Menten

The Michaelis-Menten equation describes the transition from first order to zero order kinetics as drug concentration increases:

- Metabolism increases proportionally with concentration as long as the concentration of drug leaving the organ of metabolism (e.g. in the hepatic vein) is less than half of the maximal concentration of drug that organ can metabolise
- This is usually $\sim 1/3^{\text{rd}}$ of the maximal rate of metabolism



References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
2. Plasma Volume. Mosby's Medical Dictionary, 8th edition. 2009.
3. Stanski RD, Shafer SL. The Biophase Concept and Intravenous Anesthesia.
4. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.

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Pharmacokinetic Models

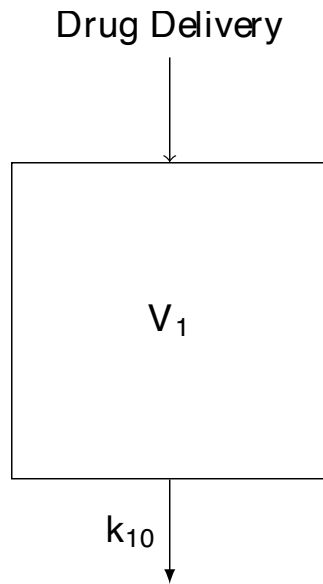
Explain the concept of pharmacokinetic modeling of single and multiple compartment models.

Explain the concept of pharmacokinetic modelling of single and multiple compartment models and define: Half life, Clearance, Zero and first order kinetics, Volume of distribution, Bioavailability, Area under the plasma concentration time curve, Extraction Ratio

Pharmacokinetic models are mathematical concepts used to predict plasma concentrations of drugs at different time points.

One-Compartment Models

The simplest model imagines the body a single, well-stirred compartment.



In a one compartment model, the concentration of a drug (C) at time t is given by the equation:

$$C = C_0 e^{-kt}$$

Where:

- C_0 is the concentration at time 0
As drug can only be eliminated from the compartment, this is also the peak concentration.
- k is the **rate constant for elimination**
This is the fraction of the V_d from which the drug is removed per unit time. The rate constant **determines** the **slope** of the curve.
 - A **high rate constant for elimination** results in a **steep curve** and therefore a **short time constant**

Steady State

At steady state, **input is equal to output**. Therefore concentration at steady state is:

- Proportional to the concentration of the infusion and infusion rate

- Inversely proportional to the clearance:

$$Input = Output$$

$$C_i \cdot I = C_{ss} \cdot Cl$$

$$C_{ss} = \frac{C_i \cdot I}{Cl}$$

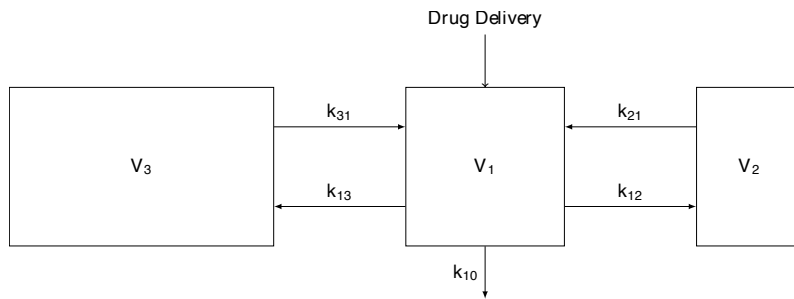
- **Concentration** of drug can therefore be **determined by the amount infused and the clearance**
- Note steady state requires peripheral compartments to be saturated, and so will only occur after an infusion of many hours

Multiple Compartment Models

Multi-compartment models imagine the body as a series of differently-sized containers, which distribute drug between them at different rates. Multi-compartment models:

- Are used because they tend to have a better fit with experimental data than single compartment models
Three-compartment models are the most common: additional compartments typically offer no extra fidelity but are mathematically more complex.
- Are usually conceptualised as a:
 - Plasma (or central) compartment
Drug is usually given into and eliminated from the central compartment.
 - A well-perfused (or vessel-rich group) compartment
 - A poorly-perfused (or vessel-poor group) compartment

This doesn't mean that they *should* be conceptualised this way - they are a mathematical technique used to calculate plasma concentration at a given time.



Plasma concentration in multi-compartment models is:

- Predicted through the net effect of several negative exponential equations
- Dependent on the relative effects of:
 - **Distribution**
Distribution describes the movement of drug from the central compartment (V_1) to the peripheral compartment(s).
 - Rapid fall in plasma concentration of a drug after administration is generally due to distribution
 - Distribution is an important method for drug offset in short-acting drugs.
 - **Redistribution**
Redistribution refers to the movement of drug from the peripheral compartment(s) back into plasma.
 - Drugs which have a large V_D in a peripheral compartment tend to **distribute quickly** along this concentration gradient, and **redistribute slowly** back into plasma
 - Drugs which tend to **distribute slowly** tend to **redistribute quickly** once administration has ceased
 - **Excretion**
Excretion is the removal of drug from the body.

Clearance in Multi-Compartment Models



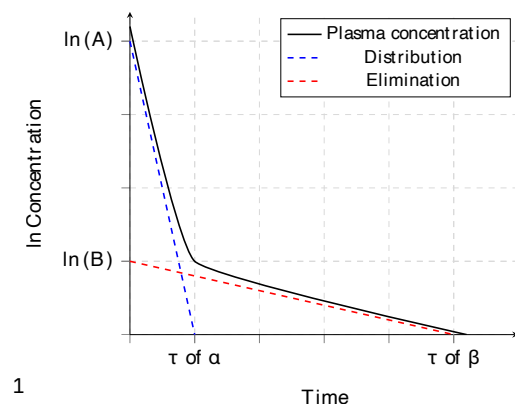
This will use two-compartment models as an example of clearance in multi-compartment models. The principles are the same for more complex models.

Removal of drug in two-compartment models is via:

- Distribution from the central to the peripheral compartment
- Elimination from the central compartment
- This produces a **bi-exponential** fall in plasma concentration

Consists of two phases:

- Phase α
Distribution phase: A rapid decline in plasma concentration due to distribution to peripheral tissues.
- Phase β
Elimination phase: Slow decline in plasma concentration due to:
 - Elimination from the body
 - Redistribution into plasma



¹Note that the distribution and elimination curves appear straight because the y-axis is log-transformed.

If plasma concentration was plotted on the y-axis, then each of these

This curve is given by the equation $C = Ae^{-\alpha t} + Be^{-\beta t}$, where:

- C is the concentration of drug in plasma
- A is the y-intercept of the distribution exponent
Used to calculate distribution half-life.
- B is the y-intercept of the elimination exponent
Used to calculate elimination half-life.
- α is the rate constant for distribution
The value of α is dependent on the ratio of rate constants for distribution and redistribution (i.e. $\frac{k_{12}}{k_{21}}$).
 - If distribution greatly exceeds redistribution, the gradient of α will be very steep and plasma concentration will fall rapidly after administration
- β is the rate constant for elimination

The Effect Site

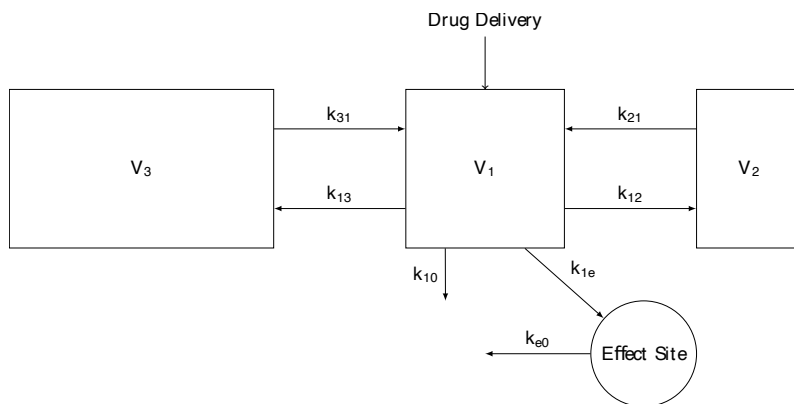
Pharmacokinetic models typically display the plasma concentration. Clinically however, we are interested in **drug concentrations** at the **site of action** (e.g. the brain); this is **known as** the **effect site** or **biophase**. Effect site concentration:

- This cannot be measured, and so is a calculated value
- Is given by C_e
- Will differ from plasma concentration (C_p) prior to reaching steady state
This delay between plasma and effect site concentrations is an example of **hysteresis**.
- Can be modeled as an additional compartment in three-compartment models
The effect site is modeled as a **compartment of negligible volume** contained **within V_1** , but with its own rate constants. Therefore:
 - Effect site volume changes as V_1 changes

Note neither the $t_{1/2ke0}$ or the k_{e0} are the time to peak effect.

curves would be a negative exponential (wash-out curve).

- The k_{1e} is the rate constant for drug diffusion from plasma into the effect site
- The k_{e0} is the rate constant for elimination of drug **from the effect site**
This is a theoretical elimination pathway - drug is not usually metabolised at the effect site.
- The $t_{1/2 ke0}$ **is the effect - site equilibration time**
It **describes how rapidly the effect site reaches equilibrium with plasma.**
 - A **high k_{e0}** (rapid drug flow) equals a **short $t_{1/2 ke0}$**
 - After one $t_{1/2 ke0}$, 50% of the final effect site concentration will be reached provided plasma concentration remains constant **A shorter $t_{1/2 ke0}$ indicates that the effect site concentration will reach equilibrium with plasma more rapidly, and therefore a more rapid clinical effect following administration is seen.**
- For an infusion run at constant plasma concentration the peak effect will be seen at 3 - 5x the $t_{1/2 ke0}$
- The time to peak effect is a function of both plasma kinetics and the $t_{1/2 ke0}$
e.g. Adenosine has such a short elimination $t_{1/2}$ the effect site concentration will reach its peak rapidly regardless of the k_{e0} .



Non-Compartmental Models

Compartment models are not appropriate for describing the behaviours of all drugs. Non-compartmental models are used when drug:

- Clearance is organ-independent
- Elimination does not occur solely from the central compartment

These models use **AUC**, which is calculated by measuring the plasma concentration of a drug at different time intervals, and **plotting the area under the curve (AUC)**. This can be used to:

- Determine **clearance**

$$Cl = \frac{Dose}{AUC}$$
- Determine **Bioavailability**
 Difference between the AUC of the same dose of drug administered IV and via another route.

References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
2. Plasma Volume. Mosby's Medical Dictionary, 8th edition. 2009.
3. Stanski RD, Shafer SL. The Biophase Concept and Intravenous Anesthesia.
4. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.

Absorption

Describe absorption and factors that will influence it with reference to clinically utilised sites of administration.

Absorption is dependent on the route of administration. Routes of administration are selected based on:

- Effect site of the drug
- Drug factors
 - Bioavailability
 - Available preparations
- Patient factors
 - Ability to take or absorb oral medications
 - Preference

Bioavailability

Bioavailability is the **proportion of drug given which reaches the systemic circulation unchanged**, compared to the IV form. Bioavailability is affected by:

- Formulation
- Physicochemical Interactions
 - Interactions with other drugs and food.
- Patient Factors
 - Malabsorption syndrome
 - Gastric stasis
- First-pass metabolism

Significance of Bioavailability

Bioavailability indicates how the dose-exposure relationship is affected by:

- Drug factors
 - Formulation
 - Solubility
- Patient factors
 - Comorbidities
 - Diet
 - Critical illness

Enteric Administration

Enteric administration is affected by:

- Gastric pH
- Gastric emptying and intestinal motility
- First pass metabolism

Gastric pH

Gastric pH affects absorption as it alters the proportion of drug that is in an unionised form, and therefore available to cross the gastric mucosa:

- ↓ Gastric pH ↑ absorption of alkaline drugs
- ↑ Gastric pH ↓ absorption of acidic drugs
- Gastric pH is:
 - Neutral at birth
 - Becomes briefly acidic (pH 1 - 3) a few hours after birth, but returns to neutral after 24 hours
 - Adult values for gastric acidity are reached at ~2 years

First-pass Metabolism

First-pass (pre-systemic) **metabolism** is the extent to which drug concentration is reduced after its first passage through an organ, prior to reaching the systemic circulation. First pass metabolism is:

- Typically used when referring to passage of orally-administered drugs through the liver
May also refer to metabolism by the:
 - Lungs
First pass of intravenously injected drugs prior to entering the arterial side of the circulation, e.g. fentanyl.
 - Vascular endothelium
- Relevant in:
 - Understanding differences between PO and IV dosing
 - Alternative routes of administration for drugs with low PO bioavailability
 - Delivery of prodrugs via PO mechanisms
Increases active drug concentration.
 - Understanding enzyme interactions
 - Understanding the effects of hepatic disease
 - Porto-systemic shunts decrease first pass metabolism
 - Altered bioavailability of drugs with high hepatic extraction ratios

References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.

2. Chong CA, Denny NM. Local anaesthetic and additive drugs.
3. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.
4. Starkey ES, Sammons HM. Practical pharmacokinetics: what do you really need to know? Arch Dis Child Educ Pract Ed. 2015 Feb;100(1):37–43.

Distribution

Describe factors influencing the distribution of drugs (for example, protein binding, lipid solubility, pH, pKa) and their alteration in physiological and pathological disturbance

Drug distribution is dependent on many factors, which can be subdivided¹ into:

- Diffusion characteristics
- Protein binding
- Regional blood flow

Diffusion Characteristics

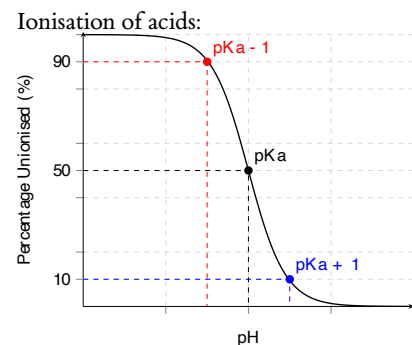
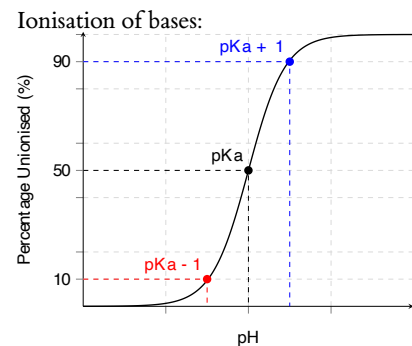
Diffusion of drugs to tissues is a function of the:

¹This approach is fundamentally an abstraction of Fick's Law that aligns neatly with both pharmacological properties and the principles of flashcard design.

If you get lost, remember Fick's Law:

$$\dot{V} = \frac{A \times D \times \Delta P}{T} \text{ Where:}$$

- A = Area for diffusion (regional blood flow)
- D = Diffusion constant (solubility, ionisation, molecular weight, pH, pKa)
- ΔP = Concentration gradient (dose, method, tissue mass, protein binding)
- T = Thickness of diffusion membrane



- **Concentration gradient**

Increasing the concentration of drug will increase the rate it distributes into tissues. This is affected by:

- Dose
- Method of administration
Bolus, infusion.
- Tissue mass
Lower mass results in a more rapid rise in concentration and a subsequent reduction in speed of distribution.

- **Molecular Weight**

Larger molecules are less able to cross cell membranes, and so a greater portion will remain in the compartment they are delivered to.

- **Lipid Solubility and Ionisation**

Ionised drugs are polar, and so are less lipid soluble.

- Ionisation is a function of:
 - pKa
The pKa is the pH at which a weak acid or weak base will be 50% ionised.
 - As solvent pH changes, the proportion of ionised vs. unionised drug will differ
 - How depends on whether the drug is an acid or base:
 - **B**ases are ionised **B**elow their pKa
 - **A**cids are ionised **A**bove their pKa
 - pH
In combination with pKa, affects the ionised portion.
- Unionised drugs:
 - Cross cell membranes more readily than the ionised form
 - Are typically hepatic metabolised
 - Are typically not renally eliminated
- Ionised drugs:
 - Are typically renally excreted without undergoing metabolism
 - Are poorly lipid soluble and do not cross cell membranes readily
 - May be ion trapped

This occurs when an unionised drug moves across a membrane and becomes ionised due to a change in pH. The now-insoluble drug is trapped in the new compartment. This is relevant in the:

- Placenta
Foetal pH is lower than maternal pH, which can trap basic drugs (e.g. LA, opioids) in foetus. This effect becomes more significant with a greater divergence of pH (e.g. in placental insufficiency).
- Kidney
Urinary alkalinisation is used to accelerate elimination of acidic drugs, as they become ionised and trapped in urine.

Protein Binding

Proteins and drugs may be bound together by weak bonds. These include ionic bonds, van der Waal's forces, and hydrogen bonds. Protein binding is important as only unbound drugs can:

- Cross cell membranes
- Interact with receptors
- Undergo metabolism
Reduced protein binding increases clearance of drugs with low **extraction ratios**.
- Be filtered by the glomerulus

Protein binding may occur in:

- Plasma
 - **Albumin**
Binds acid and neutral drugs.
 - High capacity
 - Two major binding sites (six total)
 - Site I (warfarin)
 - Site II (diazepam)

- **α 1 - acid glycoprotein**
 Binds basic drugs.
 - Single binding site
 - Low capacity
 Typically results in lower total binding (compared to albumin) of alkaline drugs, despite its increased affinity.
- Lipoprotein
 For lipid soluble drugs.
- Tissues
- Receptors

Protein binding is affected by:

- Affinity of drug for protein
 - Ionised drugs do not bind to protein pH.
 - Competition between drugs for binding sites
- Amount of protein
 - Disease
 Due to:
 - Hypoalbuminaemia
 Negative acute phase reactant.
 - Increased α 1 - acid glycoprotein
 Acute phase reactant.
 - Pregnancy
 - Decreased α 1 - acid glycoprotein
 - Competition
 Source of pharmacokinetic interactions.

Therefore, **highly tissue bound drugs**:

- Have a **long duration** of action
- Have a **high V_D** , prolonging their elimination
- May **build up in tissues**, leading to adverse effects
 e.g. Corneal deposition, lung fibrosis.

Protein binding *typically*:

- Correlates with lipid solubility
- Is important only when it is very high
- Results in a decreased V_{Dss} when plasma binding is high

- Results in an increased V_{Dss} when tissue binding is high
- Is important in duration of action as it also relates to affinity for *tissue* proteins

Regional Blood Flow

Regional blood flow affects concentration gradients between blood and tissue, and is affected by cardiac output. Regions include:

- Vessel Rich Group
 - Brain
 - Heart
 - Liver
 - Kidneys
 - Vessel Poor Group
 - Connective tissues:
 - Bones
 - Ligament
 - Teeth
 - Hair
 - Muscle groups
 - Fat
-

References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
2. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.

Metabolism

Describe the mechanisms of non-hepatic and hepatic metabolism of drugs including: Phase one and phase two reactions, Hepatic extraction ratio and its significance, First pass effect, enzyme induction and inhibition

Metabolism refers to the transformation of a drug to a new (often inactive) form by a living organism, usually via an enzymatic pathway. Metabolism may occur:

- Hepatically
- Non-hepatically
 - Organ-dependent
 - Organ-independent

Hepatic Metabolism

The principal organ of drug metabolism is the liver. Hepatic metabolism:

- Usually decreases the function of a drug, though:
 - **Prodrugs** have increased pharmacological activity after liver metabolism
 - Some drugs have active or toxic metabolites
- Can be divided into **two phases**:
 - Phase I
 - Phase II

Phase I

Phase one reactions:

- Occur in the **endoplasmic reticulum**
- Improve water solubility by exposing a functional chemical group
- Typically occur prior to phase II reactions for most drugs
- Include three key processes:
 - **Oxidation**
Loss of electrons.
 - Main phase I reaction
 - CYP450 driven
 - **Reduction**
Gain of electrons.
 - CYP450 driven
 - **Hydrolysis**
Addition of a water molecule, which may result in two new compounds.
 - Esterase driven
Therefore rapid, high capacity, organ-independent elimination.
 - Butylcholinesterase
 - Non-specific plasma cholinesterase
 - RBC esterase

The mnemonic OIL RIG: **O**xidation **I**s **L**oss, **R**eduction **I**s **G**ain can be used to remember the direction of electron transfer in oxidation and reduction reactions.

CYP450 System

The CYP450 family of enzymes are:

- A *superfamily* of enzymes vital in drug metabolism
Over 1000 enzymes, with ~50 functionally active.
- Named after the wavelength of light they absorb when reduced and combined with carbon monoxide
- Located in:
 - Liver
Endoplasmic reticulum of hepatocytes.
 - Lungs

- Kidney
- Gut
- Brain
- Classified by the degree of shared amino-acid sequence into:
 - Families
CYP1, CYP2, CYP3...
 - Subfamilies
CYP1A, CYP1B...
 - Isoforms
CYP1A1, CYP1A2...

Key CYP450 Enzymes

Table 1: Examples of Drugs Metabolised by Major CYP450 Isoenzymes

En- zyme	Substances
CYP2B6	• Propofol
CYP2C9	• Propofol
	• Parecoxib
CYP2C19	• Warfarin
	• Diazepam
	• Omeprazole
	• Clopidogrel
CYP2D6	• Phenytoin
	• Codeine
	• Metoprolol
CYP2E1	• Flecainide
	• Volatile anaesthetic agents
CYP3A4	• Paracetamol
	• Common benzodiazepines
	• Fentanyl
	• Alfentanil
	• Lignocaine
CYP3A5	• Vecuronium
	• Diazepam

CYP2D6:

- Important because genetic polymorphism leads to significant inter-patient variability
May result in significant over- or under-metabolism of drugs, and therefore significant inter-individual variability in response.
 - 5-10% of the population are poor metabolisers
 - 2-10% are intermediate metabolisers
 - 1-2% are ultra-rapid metabolisers
 - Bulk of the population (70-90%) are extensive metabolisers
- Clinical effect will depend on the type of drug
 - Pro-drugs
 - Extensive/ultra-rapid metabolisers will convert more drug to the active form, and see a greater effect
May lead to overdose.
 - Poor metabolisers will excrete more pro-drug prior to metabolism, and see a reduced clinical effect
 - Active drug
 - Extensive/ultra-rapid metabolisers will inactivate more drug, and see a reduced effect
 - Poor metabolisers will see a prolonged clinical effect
 - Clinical effect may be altered by enzyme interactions
e.g. Oxycodone use by an ultra-fast metaboliser, in combination with a CYP3A4 inhibitor (e.g. diltiazem) will result in a significant increase in the clinical effect of oxycodone.
- Drugs metabolised by CYP2D6 include:
 - Analgesics
 - Codeine (prodrug)
 - Oxycodone (metabolised to the significantly more potent oxymorphone)
 - Methadone
 - Tramadol (metabolised to form with greater MOP selectivity)

- Psychiatric drugs
 - SSRIs
 - TCAs
 - Haloperidol
- Cardiovascular drugs
 - Amiodarone
 - Flecainide
 - Mexiletine

Phase II

Phase II reactions:

- Involve conjugation with another compound, producing a highly polar metabolite
This increases water solubility and therefore renal elimination.
- Typically occur in the hepatic endoplasmic reticulum
- Include:
 - **Glucuronidation**
Addition of glucuronic acid.
 - **Sulfation**
Addition of a sulfa group.
 - **Acetylation**
Addition of an acetyl group.
 - Also occurs in the **lung** and **spleen**.
 - **Methylation**
Addition of a methyl group.

Extraction Ratio

The **extraction ratio** is the proportion of a drug that is cleared from circulation during each pass through the organ, typically the liver:

$$ER = \frac{\text{Drug Absorbed} - \text{Drug reaching systemic circulation}}{\text{Drug Absorbed}}$$

Extraction ratio is dependent on:

- Blood flow

- Hepatocyte uptake
- Enzyme capacity
Described by the **Michaelis Constant**: The concentration of a substrate which causes an enzyme to work at 50% of its maximum capacity.

Drugs can have either a high or low extraction ratio:

- **High extraction ratio**

These drugs have a rapid uptake and high capacity, so elimination is **perfusion dependent**.

- Free drug is rapidly removed from plasma, bound drug is released from plasma proteins and a concentration gradient is maintained
- Metabolism of drugs with a high extraction ratio is:
 - **Independent of protein binding**
 - **Dependent on liver flow**
Typically doubling liver blood flow will double hepatic clearance.
- There is a **high variability** in plasma concentration between individuals due to the variation in liver blood flow
- Drugs with high extraction ratios are generally independent of enzyme activity - decreasing enzyme activity from 99% to 95% has a minimal effect on hepatic clearance
 - The **key exception** is **first pass metabolism**, as the above change will result in a five-fold difference in dose reaching the systemic circulation
Therefore drugs with a high extraction ratio have low PO bioavailability.

- **Low extraction ratio**

Elimination is **capacity - dependent**.

- Amount of free drug available for metabolism is greatly affected by the degree of protein binding
- Metabolism is:
 - Largely **independent of flow**
Drugs have good PO bioavailability.
 - **Dependent on hepatocyte function and protein binding**

Table 2: Drug and Patient Factors Affecting Hepatic Metabolism

Drug Factors	Patient Factors
Lipid solubility	Age
Ionisation	Obesity
Protein binding	Pregnancy
Enzyme competition	Genetics: Slow vs. fast acetylators
	Hepatic flow/Extraction Ratio
	Enzyme Inhibition/Induction
	Hepatic disease
	Smoking, ETOH

Organ Independent Metabolism

Mechanisms of organ independent metabolism include:

- **Hofmann Degradation**

Spontaneous degradation or metabolism of substances occurring in plasma.

- e.g. Cisatracurium undergoes Hofmann degradation

- **Plasma Esterases**

Plasma esterases are non-microsomal enzymes which hydrolyse ester bonds. They:

- Are typically synthesised in the liver and erythrocytes
- Have a **high capacity**

This, combined with the organ-independent elimination, means drugs metabolised by plasma esterases have a reliable offset.

- e.g. Suxamethonium hydrolysis by plasma esterases.

References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
2. Essential pharmacology for the ANZCA primary exam
3. Birkett, DJ. Pharmacokinetics made easy 9: Non-linear pharmacokinetics. 1994. Australian Prescriber.

Elimination and Excretion

Describe the mechanisms of drug clearance and how physiological and pathological disturbance may affect these

Elimination is the **removal of drug from the plasma**, and includes distribution and metabolism, whilst **excretion** is the **removal of the drug from the body**. Excretion can occur via:

- Urine
- Bile
- Sweat
- Breast milk
- Tears
- Exhaled gas

Renal Excretion

Renal excretion can occur via **two** mechanisms:

- Glomerular filtration
Requires the drug is:
 - **Not protein bound**
Only free drug present in filtered plasma will be excreted.
 - Concentration of filtered drug will be the same as in *unfiltered* plasma

- **Highly protein bound** drugs are **poorly filtered**
There is only a weak concentration gradient favouring dissociation from plasma proteins.
- **Small**
 - Substances less than 7,000 Da are freely filtered
 - Substances greater than 70,000 Da are essentially impermeable
- **Hydrophilic/lipophobic**
Lipophilic drugs may be filtered at the glomerulus but will be freely reabsorbed during their passage down the tubule, such that only trivial amounts are eliminated in urine.
- Secreted in the tubules
 - **Active** process allows secretion **against concentration gradients**
 - Separate mechanisms for acidic and alkaline drugs
 - Saturatable process
Saturation may occur of a basic transporter whilst still allowing excretion of acidic drugs, and vice versa.

Renal elimination is **impaired** by tubular absorption, which occurs due to **passive diffusion** down a concentration gradient:

- Hydrophilic molecules can only be reabsorbed by a specialised transport mechanism
- Acidic drugs will become ionised in an alkaline urine (and vice versa), reducing their solubility
This is the physiological justification for urinary alkalisation.

Hepatic Excretion

Biliary elimination occurs for drugs unable to be filtered by the glomerulus. These are typically:

- Large
Greater than 30,000 dalton.
- Lipid soluble

Enterohepatic Recirculation

Drugs excreted in bile may:

- Be hydrolysed in the small bowel by bacteria and then reabsorbed
 - Then pass through the portal circulation and get metabolised again
 - This process may occur many times
-

References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.

Bolus and Infusion Kinetics

Explain the concepts of intravenous bolus and infusion kinetics. To describe the concepts of effect-site and context sensitive half time.

Explain and describe the clinical application of concepts related to intravenous and infusion kinetics including: Effect-site and effect-site equilibration time, Concept of context sensitive half time, Calculation of loading and maintenance dosage regimens

Continuous Infusions

Plasma concentrations of an IV infusion are influenced by both pharmacokinetics and the method of infusion. In absence of a loading dose, the **concentration of drug infused at a constant rate** increases in a **negative exponential fashion**:

- Plasma concentration initially rises rapidly
- **Distribution** into peripheral compartments is the **main method** for drugs to **leave plasma**

At the start of an infusion there is a large concentration gradient between plasma and peripheral compartments. As peripheral compartments fill the concentration gradient between plasma and compartments

falls, and redistribution becomes relatively less important.

- Excretion has a greater effect on plasma concentration in prolonged infusions
- **Steady state** is achieved when **concentrations in compartments are equal, and input is equivalent to clearance**

- Concentration at steady state is determined by the ratio of infusion rate to clearance:

$$C = \frac{\text{Infusion Rate}}{\text{Clearance}}$$

- Therefore, at steady state with drugs with 100% bioavailability:

$$\text{Infusion Rate} = C_{\text{target}} \cdot Cl$$

- For drugs given by a route with less than 100% bioavailability:

$$\text{Route Dosing Rate} = \frac{Cl \cdot C_{\text{target}}}{\text{Bioavailability}}$$

- If the dosing is given intermittently, then:

$$\text{Dosing Rate} = \text{Dose} \cdot \text{Dose Interval}$$

- **Volume of distribution at steady state** is termed **V_{Dss}**

This is the apparent volume into which a drug will disperse during a prolonged infusion, and is the sum of all compartment volumes in the model.

Continuous Rate Infusions with Bolus Dosing

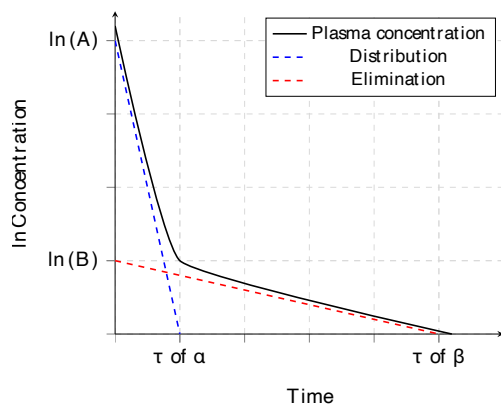
As seen, above starting an infusion at the rate required to maintain steady state is inefficient:

- For any desired plasma concentration, it will take three time constants (4 - 5 half-lives) for a continuous infusion to reach this concentration
 - If the half-life is long, then achieving a therapeutic level will take some time
- A bolus dose aimed to fill the V_D will allow steady-state to be reached immediately: $\text{Loading dose} = V_D \cdot C_{\text{target}}$

Stopping an Infusion

For a bi-exponential model (i.e. only one peripheral compartment), decline in plasma concentration can be modeled by the equation $C = Ae^{-\alpha t} + Be^{-\beta t}$. In this model:

- α is the time-constant for redistribution
- β is the time-constant for **terminal elimination** (Provided the infusion has reached steady-state).
- Neither α or β correspond to any individual rate constant



Factors affecting rate of offset of an infusion can be classified into pharmacokinetic, pharmacodynamic, and other drug factors:

- **Pharmacokinetic factors**
 - Distribution
 - V_D
High V_D will decrease clearance from central compartment. Factors affecting V_D include:
 - Ionisation
Ion trapping can cause drug to be sequestered.
 - Protein binding
 - Lipid solubility
Affected by body fat.

- Speed of distribution
 - CO
 - Affects organ blood flow.
 - **Redistribution**
 - During an infusion, peripheral compartments become saturated with drug. When an infusion ceases, drug is redistributed central compartment.
 - This is related to **context-sensitive half time** (see below)
 - Metabolism
 - Route of clearance
 - Organ-dependent
 - Organ failures
 - Extraction ratio
 - Organ blood flow
 - Organ-independent
 - Saturable kinetics
 - Zero-order kinetics.
 - Presence of active metabolites
 - Elimination
 - Route of excretion of active drug or active metabolites.
 - Organ failures
- **Pharmacodynamic Factors**
 - Age
 - Sensitivity
 - Dose required for effect and dose required for recovery.
 - Organ failures
 - Pregnancy
- **Other drug factors**
 - Pharmacokinetic interactions
 - Enzyme inhibition/induction
 - Pharmacodynamic interaction
 - Drug tolerances
 - Tachyphylaxis

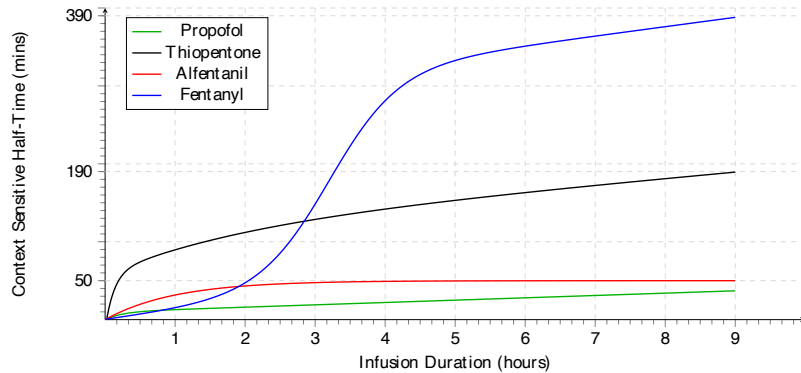
- Drug action
Drugs which alter gene or receptor expression, or bind irreversibly (e.g. clopidogrel) may show ongoing effects even after the drug has left the system.

Context-Sensitive Half-Time

Context-sensitive half time is the **time taken for plasma concentration to decrease by 50% after discontinuation of an infusion**. The CSHT is:

- A method to describe the variability in plasma concentrations after ceasing an infusion
The "**context**" is the **duration of infusion**.
- Used because terminal elimination half-life has little clinical utility for predicting drug offset
Half-lives are often misleading when discussing drug infusions.
- Dependent on both the:
 - **Duration of the infusion**
During an infusion, drugs distribute out of plasma into tissues. When the infusion ceases, drug is cleared from plasma and tissue drug redistributes back into plasma.
 - The longer an infusion, the more drug has distributed out of tissues, and the longer the redistribution phase
 - The longest context-sensitive half time occurs when an infusion is at steady-state
 - Extent of **Redistribution**
The maximal CSHT reached depends on the:
 - **V_{Dss}**
Drugs with a larger V_{Dss} have a longer CSHT, as only a small proportion of the drug in the body will be in plasma and able to be cleared.
 - **Rate constant for elimination**
Drugs with a smaller rate constant for elimination have a longer CSHT.

Drugs with a longer context-sensitive half-times will wear off less predictably.



Remifentanyl has little redistribution and a small V_D , and so has a very short context-insensitive half time. It wears off reliably and quickly following cessation of infusion.

Context-Sensitive Decrement Time

The context-sensitive decrement time:

- Describes the **time taken** for a **drug level** to **fall to a particular percentage of its starting value** after discontinuation of an infusion
- Is used because the half-time does not describe **mono-exponential decay**
i.e. The time taken for drug concentration to reach 25% of its starting value is not twice the context sensitive half-time.

The context-sensitive half-time could also be described as the 50% context-sensitive decrement time.

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1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.

2. Hill SA. Pharmacokinetics of Drug Infusions. *Contin Educ Anaesth Crit Care Pain* (2004) 4 (3): 76-80.

Drug Monitoring

Explain clinical drug monitoring with regard to peak and trough concentrations, minimum therapeutic concentration and toxicity

Drug monitoring describes the individualisation of dosing by maintaining plasma drug levels within a target range. Drug monitoring is indicated to:

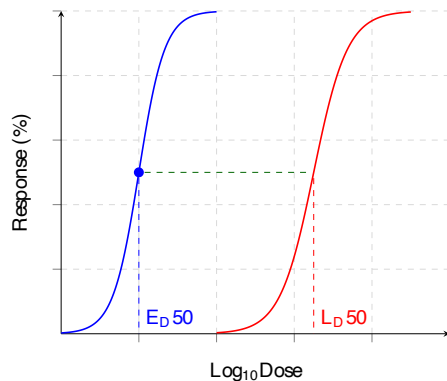
- Avoid **toxicity**
 - Adjust dosing for **efficacy**
 - Monitor **compliance**
 - Account for **high inter-patient variability** in response
- Variability can be:
- Pharmacokinetic
Adjusting drug dose by monitoring plasma levels reduces pharmacokinetic variability.
 - Pharmacodynamic
Drug dose is adjusted by evaluating the clinical effect.

Therapeutic Index

The **therapeutic index** is:

- $\text{Therapeutic Index} = \frac{LD_{50}}{ED_{50}}$, where:
 - **ED₅₀**
The dose required to produce the minimum clinical effect in 50% of the population.

- **LD₅₀**
The dose which is lethal in 50% of the population.
- An indication of how safe a medication is at the pre-scribed dose
 - A **higher** TI indicates a greater margin for safety
A drug generally has a good safety profile if it has a **TI >10**.
 - Drugs with a narrow TI have a small range between dose for efficacy and dose for harm



Narrow therapeutic index drugs are known as "poison pills" or "drugs with a narrow therapeutic range". The therapeutic index (TI) is the ratio of the minimum toxic concentration to the minimum effective concentration.

- TI is <2
- Ratio of the minimum toxic concentration and minimum therapeutic concentration is <2
- Safe use requires titration and monitoring

Indications

Drugs that typically require monitoring have a:

- Narrow target range
- Significant pharmacokinetic variability
- Relationship between the concentration in plasma and clinical effects
- Determined concentration range
- Validated monitoring assay

Drugs where the effect can be measured clinically (e.g. antihypertensives) tend to be adjusted based on observed effects. This is not possible when:

- The clinical response is the *absence* of a condition, e.g. antiepileptics
- The drug has a narrow therapeutic range

Table 1: Commonly Monitored Drugs in the ICU Setting

Drug	Therapeutic Range
Digoxin	0.8-2 microgram/L
Vancomycin	10-20 mg/L *
Tacrolimus	5-20 microgram/L
Serolimus	5-15 microgram/L
Phenytoin	10-20 mg/L

* Indicates trough level.

Timing of Samples

Timing depends on the reason for sampling:

- Time of **peak concentration** is mostly important for toxicity
This requires accounting for absorption and distribution.
 - e.g. Digoxin levels should be performed >6 hours following a dose to allow time for distribution to occur
 - If symptomatic, samples taken at this time may demonstrate toxic concentrations
- **Steady state** is optimal for **monitoring**
 - i.e. after 4-5 elimination half-lives
 - For drugs with very long half-lives (such as amiodarone), sampling tends to occur earlier to ensure toxic levels have not been reached, as steady state may take months to achieve
- **Trough levels** are best for drugs with **short half-lives**
These are pre-dose levels, and are best because they are usually the least variable point in the dosing interval.

Timing of sampling is less important for drugs with long half-lives.

Interpretation

Interpretation of drug levels is dependent on:

- **Timing** of sample
 - Duration of treatment at the current dose and **dosing** schedule
 - Individual characteristics that may affect the **pharmacokinetics**
 - Age
 - Physiology
 - Comorbidities (hepatic, renal, cardiac)
 - Drug interactions
 - Genetics
 - Environmental
 - **Protein binding**
 - Assays measure bound and unbound drug
Only unbound drug is pharmacologically active.
 - If binding is changed by disease or displacement by other drug, the proportion of unbound drug may change and targeted levels may need to be adjusted accordingly
 - **Active metabolites**
Active metabolites are not measured but will contribute to the response.
-

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2. Ghiculescu RA. Therapeutic drug monitoring, which drugs, why, when, and how to do it. Aust Prescr 2008;31:42-4.

3. Tamargo J, Le Heuzey JY, Mabo P. Narrow therapeutic index drugs: a clinical pharmacological consideration to flecainide. *Eur J Clin Pharmacol.* 2015;71(5):549–67.
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Routes of Administration

Describe absorption and factors that will influence it with reference to clinically utilised sites of administration.

Describe the pharmacokinetics of drugs in the epidural and subarachnoid space.

Intravenous

Intravenous administration is characterised by:

- Rapid Onset
 - 100% bioavailability
- Some drugs may still undergo metabolism in the pulmonary circulation, such as fentanyl, lignocaine, propofol, and catecholamines.

Oral

Oral absorption is through gut mucosa, through either:

- Transport mechanisms
- Diffusion of unionised (lipid soluble) molecules
 - Acidic drugs are absorbed more rapidly in the stomach Note that the small bowel absorbs both acid (despite being ionised) and alkaline drugs due to high surface area.
- Lowest bioavailability of any route due to:

- First-pass metabolism
 - Gut metabolism of drugs
 - Bacterial metabolism of drugs
- Drugs must be lipid soluble enough to cross cell-membranes and water soluble enough to cross interstitium

Factors affecting GIT Absorption:

- Drug Factors
 - Molecular Weight
 - Concentration Gradient
 - Lipid Solubility
 - pH and pKa
 - Pharmaceutical Preparation
 - Physiochemical Interactions
 - Food
 - Other drugs
- Patient Factors
 - GIT blood flow
 - Surface Area

Small bowel has the largest surface area of any GIT organ.
 - pH
 - Motility
 - Digestive Enzymes
 - GIT bacteria and subsequent metabolism
 - Disease
 - Critical Illness
 - Bowel Obstruction
 - Emesis/Diarrhoea

Epidural

Factors affecting epidural absorption include:

- Dose given
- Volume given

Increased volume increases area of subarachnoid that the drug is in contact with, increasing rate of diffusion.

- Rate of administration
May be via bolus or infusion.
- Solubility
Affected by:
 - pKa and pH
Determines unionised portion available to cross into CSF. Lignocaine has a more rapid epidural onset than bupivacaine as it has a pKa of 7.9 (compared to 8.4) and therefore a greater unionised portion at physiologic pH.
 - Protein binding
Determines free drug portion able to cross into CSF.
 - Lipid solubility
- CSF flow
Alters concentration gradient between epidural and subarachnoid space.
- Intrinsic vasoactive properties of the delivered drug
- Additives

Subarachnoid/Intrathecal

Intrathecal administration is characterised by:

- Very small dosing
- Minimal systemic spread

Extent of subarachnoid spread is dependent on:

- Volume and total dose
And concentration - these three factors are interdependent.
- Baricity
Iso-, hypo-, and hyperbaric preparations are available and affect direction of spread:
 - Hyperbaric solutions will sink with gravity
e.g. Heavy bupivacaine (0.5% bupivacaine with 8% dextrose)
 - Hypobaric solutions will rise against gravity

- Patient position
The combination of a supine or Trendelenburg patient position with high-specific gravity solutions increases cranial spread of the block.
- Intrinsic vasoactive properties of the delivered drug
- Additives

Inhalation

Inhalational administration characterised by:

- Rapid diffusion to circulation due to high surface area and no first-pass metabolism
- Variable systemic absorption; dependent on particle size
 - Large particles reach the bronchioles
 - <1 micron diameter particles may reach the alveolus

Transdermal

Transdermal absorption is determined by the **Fick Principle**:

- Concentration gradient
 - Amount of drug given
 - Amount of drug in skin
- Regional blood flow
 - Histamine release
- Surface Area
- Skin thickness
- Lipid solubility
 - pH of skin and emulsion
 - pKa of drug
- Molecular weight

Advantages:

- Convenient
- Painless
- No first pass metabolism
- Steady plasma concentration once established

Disadvantages:

- Slow onset
- Variable plasma concentration initially
- Overdose and abuse potentials
- Dose requirement
Large dose requirements cannot be effectively given transdermally.

Subcutaneous

- Absorption dependent on regional blood flow

Sublingual

- Rapid onset
- Bypass portal circulation (drains into SVC)

Rectal

- Variable absorption
 - Distal rectal absorption bypasses portal circulation
 - Proximal rectal absorption does not and may result in hepatic first pass metabolism
 - Small surface area for absorption

Intramuscular

- Bioavailability close to 1
 - Absorption dependent on regional blood flow
 - Potential local complications:
 - Abscess
 - Haematoma
-

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1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
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3. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.
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5. Factors influencing distribution of bupivacaine after epidural injection. Diaz Notes.

Target-Controlled Infusions

Describe the pharmacological principles of and sources of error with target controlled infusion

Target-Controlled Infusions describes the use of pharmacokinetic models (typically combined with microprocessor-controlled infusion pumps) to achieve a target concentration of intravenous drugs in a particular body compartment. TCI is related to, but distinct from, total intravenous anaesthesia (TIVA), which is the use of IV agents alone to achieve hypnosis, analgesia, and muscle relaxation.

Many TCI models, and hence electronically controlled pumps are not licensed or available inside the US.

Principles of TCI

Open-loop:

- Effects of drug are not measured (unlike with end-tidal gas monitoring), which introduces a vulnerability that can increase risk of awareness.
e.g. Compared to inhalational anaesthetics, where the loop is closed by using end-tidal drug monitoring.

Bolus, Elimination, and Transfer principle:

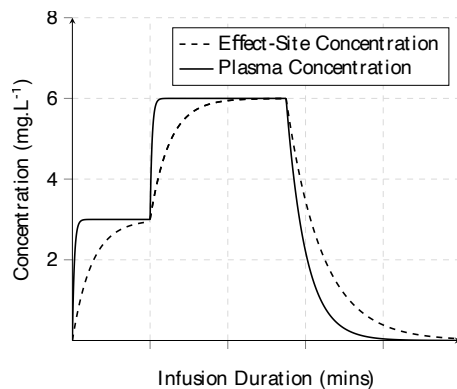
- A loading dose is given to saturate the volume of distribution to achieve target concentration
 $\text{Loading Dose} = V_D \times C_P$
- Infusion rate is then set to maintain a target plasma or effect site concentration:
 $\text{Maintenance Infusion Rate (MIR)} = Cl \times C_{P/e}$

- Rate compensates for:
 - Drug elimination
 - Drug distribution (transfer)
- Target C_P (or C_e) can be adjusted:
 - For a higher concentration:
 - A small bolus is given
 - Infusion rate is increased
 - For a lower concentration:
 - Infusion is paused until desired level is reached
 - Infusion rate restarts at a lower rate

Plasma Targeting C_P

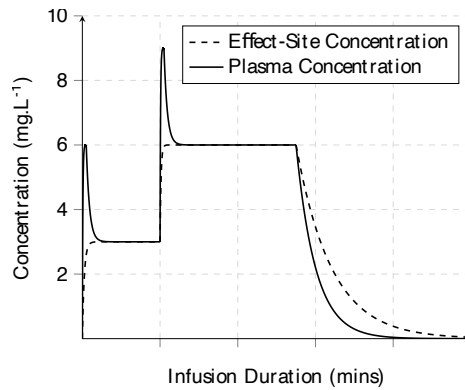
In plasma targeted infusions, C_P will not approximate C_e until steady state is reached. Therefore:

- Increase C_P during induction, so that C_e will rise more quickly
- C_P should be adjusted to the level of the surgical stimulus



Effect-Site Targeting

In C_e models, the "over-pressure" occurs automatically, so there is no requirement to increase target during induction.



TCI Models for Propofol

The Bristol Model:

- First pharmacokinetic model
- Based on three-compartment model of health patients
- Assumes:
 - Premedication with temazepam
 - Fentanyl $3\mu\text{g.kg}^{-1}$ on induction
 - Inhaled N_2O
 - A target plasma concentration (C_P) of $3\mu\text{g.ml}^{-1}$
- The model:
 - 1mg.kg^{-1} induction bolus
 - 10-8-6 maintenance:
 - $10\text{mg.kg}^{-1}.\text{hr}^{-1}$ for 10 minutes
 - $8\text{mg.kg}^{-1}.\text{hr}^{-1}$ for 10 minutes
 - $6\text{mg.kg}^{-1}.\text{hr}^{-1}$ thereafter

Marsh and Schnider Models:

- These are computer controlled models
 - Both were derived on very small groups of patients (18 and 24 respectively)
 - The models differ mostly in the first 10 minutes after induction, and progressively converge
- The initial behaviour of the model is key in deciding which model to apply to any particular patient.

Table 1: Comparison of the Marsh and Schnider Propofol TCI Models

Prop- erty Marsh	Schnider
Tar- Typically target plasma concentration, but can target effect site. Effect site targeting is usually done with the modified Marsh model, due to the large bolus dosing given by the standard Marsh model.	Typically effect site, but can target plasma concentration. Plasma targeting gives inconsistent results, as the fixed size of V_1 means any increase in desired plasma concentration results in the same size bolus being given, irrespective of patient parameters.
Re- TBW (overestimates required reduction (but not variable maintenance) in obese patients, consider using IBW), Age (but not used in calculation)	Age, height (to calculate lean body mass), TBW
Val- Variable compartment sizes but bigger V_1	Fixed V_1 (4.27L) and V_3 , variable V_2 and K_{eo}
Other- The 'modified Marsh' model uses a k_{eo} of $1.2\text{L}\cdot\text{min}^{-1}$ instead of $0.26\text{L}\cdot\text{min}^{-1}$, which decreases the C_p required to achieve the target C_e quickly. The modified Marsh is therefore preferable in patients at higher risk of overdose.	Limits BMI to < 42 for males and < 35 for females, to prevent absurd compartment sizes being calculated from the method used to calculate lean body mass
Over- Faster induction due to all larger V_1 , which results in a larger loading dose	Reduced rate of adverse events. Overall less propofol used.

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2. Naidoo D. Target Controlled Infusions. Department of Anaesthetics, University of Kwazulu-Natal. 2011.
3. Engbers FH, Sutcliffe N, Kenny G, Schraag S. Pharmacokinetic models for propofol: Defining and illuminating the devil in the detail. *Br J Anaesth*. 2010 Feb;104(2):261-2; author reply 262-4.
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Pharmacodynamics

Receptor Theory

To explain the concept of drug action with respect to: receptor theory

To define and explain dose-effect relationships of drugs, including dose-response curves with reference to: graded and quantal response, therapeutic index, potency and efficacy, competitive and non-competitive antagonists, partial agonists, mixed agonist-antagonists and inverse agonists

To explain the Law of Mass Action and describe affinity and dissociation constants

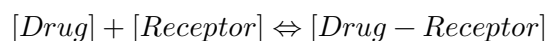
A **receptor** is a component of a cell which interacts with a drug and **initiates a sequence of events leading to an observed change in function**.

- Existence of receptors is inferred from the response of tissues to drugs, genome sequencing, and molecular biology.
- A drug binds to a receptor forming a **receptor-drug complex**, which initiates a cascade of events to exert a pharmacological effect.

Dissociation Constants

Interaction between a receptor and a drug is based upon the **law of mass action**, which states that the rate of a

chemical reaction is proportional to the masses of reacting substances. This can be expressed as:



The ratio of the **rate constant** for the forwards reaction (formation of the drug-receptor complex, $K_{\text{association}}$) and the backwards reaction (dissolution of the drug-receptor complex to individual drug and receptor components, $K_{\text{dissociation}}$) is the **dissociation constant**:

- This is the concentration of drug when 50% of receptors are occupied: $K_D = \frac{[D][R]}{[D-R]}$
- A **low** K_D value indicates that a lower concentration of drug is required to occupy 50% of the receptor, indicating that the drug has a **high affinity** for the receptor

Physiological factors which affect the dissociation constant are determined by the **Arrhenius equation**, $k = Ae^{\frac{-E_A}{RT}}$, where:

- A is a constant
- T is **temperature** in kelvin
- E_A is the activation energy required, which may be lowered by a **catalyst**
- R is the gas constant

Properties of Drugs in Receptor Theory

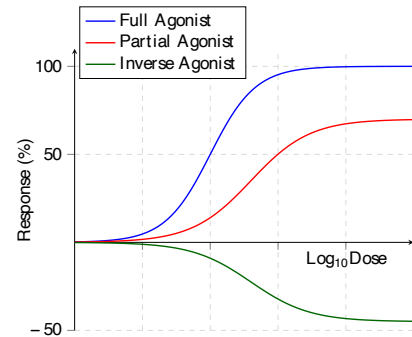
Key properties of drugs include:

- **Potency**
The **amount** of drug required to achieve a given effect.
◦ Typically expressed by the E_{D50}
- **Efficacy**
The **maximal** effect that a drug can generate.
- **Intrinsic activity** (activity)
The size of effect a drug has when bound, which is graded from 0 to 1.

Drug-Receptor Interactions

Drugs can be classified by the way they interact with receptors into:

- **Agonists**
 - Partial agonists
 - Inverse agonists
- **Antagonists**
 - Indirect antagonists
- **Mixed Agonist-Antagonists**
- **Allosteric Modulators**



Agonists

An agonist will generate a **maximal** response at the receptor site. An agonist has **high affinity** and an **activity of 1**. Agonists can be compared by:

- **Relative potency**
If two agonists are equally *efficacious*, the more *potent* one requires a **smaller dose** to achieve an effect.
- **Relative efficacy**
The agonist with greater relative efficacy can achieve a greater *maximal effect*.

Partial Agonist

A partial agonist:

- Generates a **submaximal** response at the receptor
- Has **high affinity** and an **activity between 0 and 1**
- Can act as an effective antagonist in the presence of a full agonist, as it will prevent maximal binding at a receptor, even with a high agonist concentration

Inverse Agonist

A drug which has a **negative activity** (between 0 and -1) producing the **opposite** response (compared to the endogenous agonist) at receptor. This occurs due to loss of **constitutive activity** at that receptor.

Antagonist

An antagonist produces no response at the receptor site, and prevents other ligands binding. Antagonists have **high affinity** and an **activity of 0**.

Antagonists with these properties are also known as **direct antagonists**, and can be either:

- **Competitive antagonists**

Displace other ligands from a binding site. Competitive antagonists can be:

- Reversible

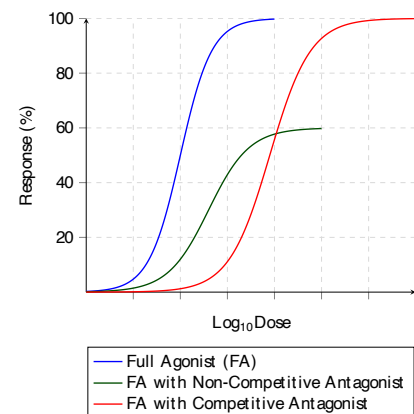
The effect **can be overridden** by increasing the dose of agonist.

- Irreversible

Drug **cannot be overridden** by increasing dose of agonist. Dose-response curve appears similar to that of the non-competitive antagonist.

- **Non-competitive antagonists**

Create a conformational change in the receptor. They **cannot be overridden** by increasing the dose of agonist.



Indirect Antagonist

Indirect antagonists reduce the clinical effect of a drug, but do so via means other than receptor interaction. They include:

- **Chemical antagonists**

Where the drug binds directly to another. Examples include protamine and heparin, and sugammadex and rocuronium.

- **Physiologic antagonists**

A countering effect is produced by agonism of other pathways.

Mixed Agonist–Antagonist

A drug which has may act as either agonist or antagonist, depending on the receptor.

Allosteric Modulator

A drug which binds to an allosteric site on the receptor and produces conformational change that alters the affinity of the receptor for the endogenous agonist.

Allosteric modulators can be:

- Positive
Increases affinity for endogenous agonist.
 - e.g. Benzodiazepines are positive allosteric modulators at the GABA_A receptor
- Negative
Decreases affinity for endogenous agonist.

References

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2. Encyclopaedia Britannica. Available at: <https://www.britannica.com/science/law-of-mass-action>

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Receptor Types

To explain receptor activity with regard to: ionic fluxes, second messengers and G proteins, nucleic acid synthesis, evidence for the presence of receptors, regulation of receptor number and activity, structural relationships.

A receptor is a protein, usually in the cellular membrane, to which a ligand may bind to generate a response.

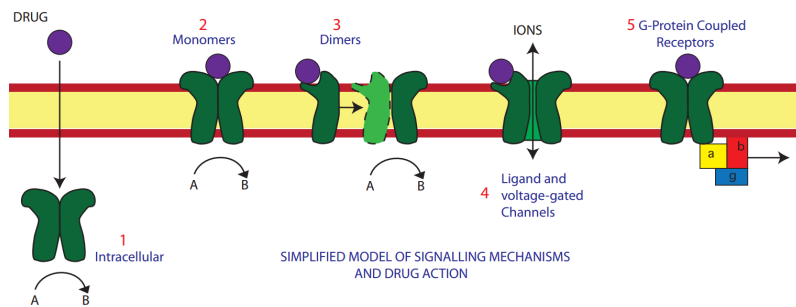


Figure 1: Receptor types and methods of signaling. Used without permission from icuprimaryprep.com

Key receptor types include: * Intracellular * Enzyme-linked
* Ion-channel * G-protein coupled

Intracellular Receptors

Intracellular receptors may be:

- **Cytoplasmic**
- **Intra-nuclear**

Intranuclear receptors are activated by lipid soluble molecules (such as steroids and thyroxine) to alter

DNA and RNA expression. This results in an alteration of production of cellular proteins, so the effects tend to be slow acting.

Enzyme-linked Receptors

Enzyme-linked receptors are activated by a ligand and cause enzymatic activity on the intracellular side. They can be either:

- **Monomers**
- **Dimers**

Where two proteins join, or dimerise, on binding of a ligand.

Ion-channel Receptors

Ion-channel (ionotropic) receptors create a channel through the membrane that allows electrolytes to flow down their electrical and concentration gradients. They can be either:

- Ligand-gated
- Voltage-gated

Ligand-gated Channels

Ligand-gated channels undergo a conformational change when a ligand is bound. There are three important families of ligand channels:

- Pentameric family
Consist of five membrane spanning subunits. Include:
 - Nicotinic ACh receptor
 - GABA_A receptor
 - 5-HT₃ receptor

- Inotropic glutamate receptors
Bind glutamate, a CNS excitatory neurotransmitter. Include:
 - NMDA receptor
High Ca^{2+} permeability
- Inotropic purinergic receptors
Form cationic channels that are permeable to Ca^{2+} , Na^+ , and K^+ , and activated by ATP

Voltage-gated Channels

Open when the **threshold voltage** is reached, facilitating electrical conduction in excitable tissues.

- In their normal physiological state, voltage gated channels do not generally behave as receptors for a ligand
However some drugs (e.g. local anaesthetics) bind to voltage gated channels to exert their effect.
- Have a common 4-subunit structure (each with 6 transmembrane segments) surrounding a central pore
This pore is selective for the particular ion, which include:
 - Na^+
 - Located in myocytes and neurons
 - Important in generating and transmitting an action potential by permitting sodium influx into cells
 - Inhibited by local anaesthetics, anti-epileptics, and some anti-arrhythmics
 - Ca^{2+}
Divided into subtypes, including:
 - L
Muscular contraction.
 - T
Cardiac pacemaker.
 - N/P/Q
Neurotransmitter release.

- K^+
Located in myocytes and important in repolarisation following an action potential.
- Undergo a conformational change when the threshold potential is reached
This is sensed by the **S4 helix**, which acts to open and close the channel.
- Exist in one of three functional states:
 - Resting
Pore is closed.
 - Active
Pore is open, and ions can pass.
 - Inactive
Transient refractory period where the pore is open, but ions cannot pass. This creates the **absolute refractory period** of a cell.

G-protein Coupled Receptors

G-proteins coupled (metabotropic) receptors are a group of **heterotrimeric** (containing three units; α , β , γ) proteins which bind GDP. When stimulated, the GDP is replaced by GTP and the α -GTP subunit dissociates to activate or inhibit an effector protein. The **effect depends** on the **type of α -subunit**:

- G_s proteins
Are **stimulatory**. These
 - Increase cAMP, leading to a biochemical effect
- G_i proteins
Are **inhibitory**. These:
 - Inhibit adenylyl cyclase, reducing cAMP
- G_q proteins
Have a variable effect, depending on the cell. These:
 - Activate phospholipase C
This affects the production of:
 - Inositol triphosphate (IP_3)
Stimulates Ca^{2+} from the SR, affecting enzy-

- matic function or causing membrane depolarisation.
 - Diacylglycerol (DAG)
Activates protein kinase C, which has cell-specific effects.
 - Activate intracellular **second messenger proteins** when stimulated
Second messenger systems:
 - Result in both transmission and **amplification** of a stimulus, as a single activated receptor can activate multiple proteins and each activated protein may activate several other intermediate proteins
This is known as a G-protein **cascade**.
-

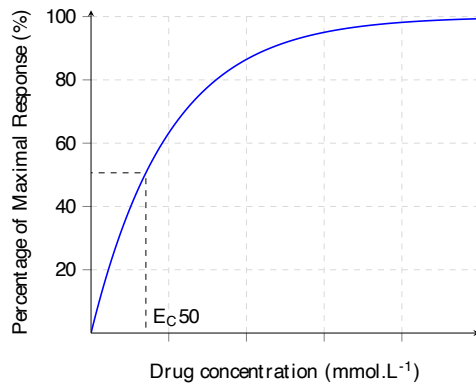
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1. Anderson, C. Pharmacodynamics 2. ICU Primary Prep.
2. Law of Mass Action. Encyclopaedia Britannica.
3. ANZCA August/September 2001
4. Catterall WA. Structure and Function of Voltage-Gated Ion Channels. Annu. Rev. Biochem. 1995. 64:493 - 531.

Dose-Response Curves

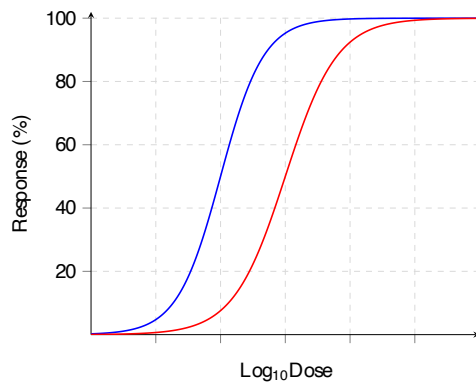
To define and explain dose-effect relationships of drugs, including dose-response curves with reference to: graded and quantal response.

The standard dose-response curve is a graph of concentration against the fraction of receptors occupied by a drug.



Log-Dose Response Curves

It is difficult to compare drugs using standard dose-response curves. Therefore, dose is commonly log-transformed to produce a log-dose response curve.



This curve:

- Compares log-dose versus clinical effect
- Demonstrates that the blue drug has greater **potency** than the red drug, though both are full agonists

Responses can be either **graded** or **quantal**:

- **Graded responses** demonstrate a **continuous increase** in effect with dose
 - E.g. Blood pressure and noradrenaline dose
- **Quantal responses** demonstrate a response once a certain proportion of receptors are occupied

Examples include:

- ED_{95}
Median dose of neuromuscular blocker required to produce a 95% loss of twitch height.
- MAC
Mean alveolar concentration of agent required to prevent movement in response to a surgical stimulus.

References

1. Anderson C. Pharmacodynamics 1. ICU Primary Prep. Available at: <https://icuprimaryprep.files.wordpress.com/2016/08/Pharmacodynamics-1.pdf>

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Mechanisms of Action

To explain the concept of drug action with respect to: receptor theory, enzyme interactions, and physicochemical interactions.

Drugs can act in four major ways:

- **Receptors**
 - GPCR
 - Intracellular
 - Cytoplasmic
 - Intranuclear

e.g. Steroids, which alter RNA expression.
- **Ion Channels**
- **Enzyme interaction**
 - Antagonism
 - False substrates
- **Physicochemical**
 - Osmotic
 - Acid-base
 - Chelation
 - Redox reactions

Receptors and ion channels are discussed in detail under Chapter .

Enzyme Interaction

An enzyme is a biological catalyst, increasing the speed of reaction. Drugs can interact with enzymes by:

- Antagonism

- False substrate

Enzyme Antagonism

Most drugs which interact with enzymes inhibit their activity. This results in:

- Increased concentration of enzymatic substrate
- Decreased concentration of the product of the reaction

Drugs can be competitive¹, non-competitive, or irreversible inhibitors² of enzymatic activity.

False Substrates

False substrates compete with the enzymatic binding site, and produce a product³.

Physicochemical

Drugs whose mechanism of action is due to their physicochemical properties, such as:

- Tonicity⁴
- Chemical reactions⁵
- Surface tension

¹Ramipril is a competitive inhibitor of angiotensin-converting enzyme.

²Aspirin is an irreversible inhibitor of cyclo-oxygenase.

³Methyldopa is a false substrate of the enzyme dopamine decarboxylase.

⁴Mannitol reduces ICP because it increases tonicity of the extracellular compartment (and is unable to cross the BBB), drawing free water from the intracellular compartment as a consequence.

⁵Aluminium hydroxide reacts with gastric acid to form aluminium chloride and water, reducing stomach pH.

References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
2. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.

Variability in Drug Response

Adverse Effects

Classify and describe adverse drug reactions.

Describe the mechanisms of adverse drug effects

An **adverse effect** is:

- A noxious or unintended effect associated with administration of a drug at the **normal dose** (i.e., not an overdose)
Occur:
 - Mainly in young and middle-aged individuals
 - Twice as common in women
 - May be exacerbated by asthma and pregnancy.
- Distinct from an **adverse event**, which is an untoward occurrence during treatment that does not necessarily have a causal relationship to drug administration

Adverse effects can be classified by mechanism as follows:

Type A Adverse Reactions

These are related to the pharmacological action of the drug. They are:

- Common
- Related to dose (dose-response relationship)

- Temporally associated with drug administration
- Reproducible
- Pharmacologically predictable based on understanding of the drug in question
 - e.g hypokalaemia secondary to diuretic use

They typically result in:

- Organ-selective injury
- More pronounced with long-term use and in risk groups:
 - Extremes of age
 - Pregnancy
 - Renal failure
- High morbidity but low mortality
Treatment is to decrease dose.

Type B Adverse Reactions

These are patient-specific or idiosyncratic reactions. They are:

- Rare
Potentially genetic, but poorly understood.
- Independent of dose
 - Occur with low doses
 - Do not have a dose-response relationship
- Not pharmacologically predictable
Important causes include:
 - Acetylator status
 - CYP450 variants
 - Receptor abnormalities
 - Enzyme alterations/deficiencies
 - e.g. Suxamethonium apnoea
- Not necessarily reproducible

They typically result in:

- Immuno-allergic reactions

- Pseudo-allergy
- Idiosyncratic reaction
- Low morbidity but high mortality
 - e.g. Stevens-Johnson Syndrome or anaphylaxis following penicillin administration

The treatment is to cease the medication.

Type C Adverse Reactions

These are 'statistical effects' associated with monitoring. They are:

- Typically an increased frequency of background disease that is detected due to increased screening
 - Atypical for a drug reaction and not pharmacological predictable
 - No identifiable temporal relationship
 - Not reproducible
-

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1. RHB Meyboom, M Lindquist, ACG Egberts. An ABC of Drug-Related Problems. *Drug Safety* 2000;22:415-23.
2. Pirmohamed M, Breckenridge AM, Kitteringham NR, Park BK. Adverse drug reactions. *BMJ*. 1998 Apr 25;316(7140):1295-8. Open Access Review.
3. Lazarou J, Pomeranz BH, Corey PN. Incidence of adverse drug reactions in hospitalized patients—a meta-analysis of prospective studies. *JAMA* 1998;279:1200-5.

Safety of Drugs in Pregnancy

The Therapeutic Goods Administration classifies drugs for suitability in pregnancy based on the potential of a drug to cause:

- Birth defects
- Detrimental effects at birth
- Problems in later life

The classification system is:

- Valid only for the dose and route of administration listed
 - Does not apply in overdose
- Not hierarchical
 - 'B' drugs are not safer than 'C' drugs

Categories

- **Category A**
Taken by *large number* of women without detrimental effects.
- **Category B**
Taken by a *limited number* of women without detrimental effect. Subclassified by results from animal studies into:
 - Category B1
Animal studies show no evidence of detrimental effect to the foetus.

- Category B2
Animal studies are inadequate or lacking, but available data shows no evidence of detrimental effect to the foetus.
 - Category B3
Animal studies show evidence of foetal damage, but the significance of this in humans is unknown.
 - **Category C**
 - Drugs which have caused (or are suspected to cause) detrimental foetal effects, but without malformations
 - These effects may be reversible
 - **Category D**
 - Drugs which have caused (or are suspected to cause) an increased incidence of foetal malformations or damage
 - May also have detrimental effects
 - **Category X**
 - Drugs which have a high risk of causing permanent damage
 - Should not be used in pregnancy, or when pregnancy is possible
-

References

1. Australian categorisation system for prescribing medicines in pregnancy. Therapeutic Goods Administration.

Drug Interactions

Classify and describe mechanisms of drug interaction.

Drug interactions occur “when the action of one drug modifies that of another”. Drug interactions may be:

- Physicochemical
- Pharmacokinetic
- Pharmacodynamic

Physicochemical

Physicochemical interactions occur because of an incompatibility between chemical structures.

Thiopentone and suxamethonium precipitate out of solution if prepared together or delivered together in the same line

Pharmacokinetic

Pharmacokinetic interactions can be sub-classified into those affecting:

- Absorption
- Distribution
- Metabolism
- Elimination.

Absorption

For oral medications, absorption may be affected by drugs which alter:

- Gastric pH
- Gastric emptying time

Metoclopramide is prokinetic and improves absorption of orally administered drugs

Distribution

Distribution may be affected by:

- Alteration of cardiac output¹
- Competition for plasma protein binding²
Loss of albumin and α_1 -acid glycoprotein.
- Displacement from tissue binding sites
This typically occurs due to alteration of metabolic capacity of one drug by the other.
- Chelation of drug from tissues
Chelating agents bind toxic elements and prevent tissue damage

Metabolism

Metabolism may be affected by changes to the CYP450 enzymes:

- Enzyme induction
 - Barbiturates
 - Phenytoin
 - Carbamazepine

¹ β -blockers reduce cardiac output and prolong the time to fasciculation of suxamethonium.

²Phenytoin is usually highly (90%) protein bound. A reduction in protein binding to 80% will double the free phenytoin level.

For drugs with first-order kinetics, metabolism will increase proportionally however phenytoin rapidly saturates the enzyme system, leading to zero-order kinetics and a high plasma level.

- Enzyme inhibition
 - Amiodarone
Amiodarone inhibits metabolism of S-warfarin by CYP2C9, enhancing its effect.
 - Diltiazem
 - Verapamil
 - Ciprofloxacin
 - Macrolides
 - Metronidazole
 - Grapefruit juice

Elimination

Renal elimination can be affected by:

- Changes in urinary pH
- Competition for active tubular transport mechanisms

Sodium bicarbonate increases urinary pH and enhances excretion of weak acids such as aspirin.

Pharmacodynamic

Pharmacodynamic interactions can be **direct**, due to interaction on the same receptor system; or **indirect**, when they act on different receptor system. These interactions can be classified as either:

- **Additive**
When the effects summate.
- **Antagonistic**
When the effects oppose each other.
- **Synergistic**
When the combined effect is greater than would be expected from summation alone.

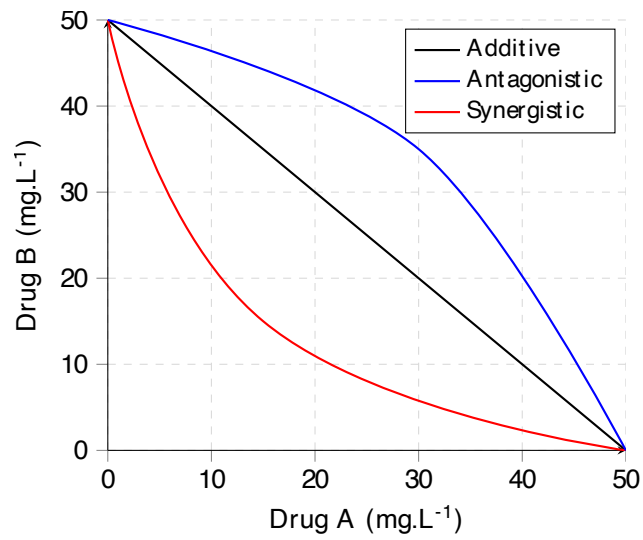
These three interactions can be graphically demonstrated using an **isobologram**, which draws a **line of equal activity** versus **concentration** of two drugs.

Examples:

Addition: Administering midazolam with propofol reduces the amount of propofol required to generate an effect.

Antagonism: Neostigmine indirectly antagonises the effect of NDMRs by increasing the level of ACh at the NMJ

Synergism: Co-administration of remifentanyl and propofol has a synergistic effect in maintenance of anaesthesia



References

1. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
2. Khan S, Stannard N, Greijn J. Precipitation of thiopental with muscle relaxants: a potential hazard. JRSM Short Reports. 2011;2(7):58.

Alterations to Drug Response

Define tachyphylaxis, tolerance, addiction, dependence and idiosyncrasy

Describe alterations to drug response due to physiological change with particular reference to the elderly

Describe alterations to drug response due to pathological disturbance with particular reference to cardiac, respiratory, renal and hepatic disease

There are four mechanisms which result in variable response to drug:

- Alteration in drug that reaches the receptor
This is typically due to pharmacokinetic factors.
- Relative difference in presence of exogenous and endogenous ligands
Antagonists will have a greater effect in the presence of high endogenous ligand concentration.
- Variation in receptor function and number
Up-regulation and down-regulation of receptors may occur as a consequence of prolonged stimulus.
- Alteration in function distal to the receptor

Key Terms

- **Tachyphylaxis** is the **rapid decrease in response** to repeated dosing over a **short time period**, usually due to depletion of transmitter
- **Desensitisation** is the **loss in response** over a **long time period**, usually due to change in receptor morphology or loss in receptor numbers
- **Withdrawal** is a pathological response when a drug is ceased
 - During administration receptors may be:
 - **Up-regulated** in the continued presence of an antagonist
 - **Down-regulated** in the continued presence of an agonist
 - Loss of receptor numbers may precipitate withdrawal when the agonist or antagonist is ceased
- **Addiction** is a behavioural pattern characterised by compulsive use and fixation on acquiring and using a drug
- **Idiosyncrasy** is an individual patient response to a drug
Typically mediated by a reactive metabolite rather than the drug itself.

Tolerance

Tolerance is the requirement for a **larger dose** to achieve the **same effect**, due to altered sensitivity of the receptors to the stimulant. Can be either:

- Pharmacokinetic tolerance
- Pharmacodynamic tolerance

Pharmacokinetic Tolerance

Occurs via:

- **Altered drug metabolism**

Metabolism may be **increased or decreased**:

- Enzymatic induction and increased drug metabolism
Increased hepatic enzyme pathway capacity increases metabolism and lowers plasma concentration.
- Decreased metabolism
Decreased metabolism of a prodrug can result in a reduced effect.

Pharmacodynamic Tolerance

Occurs via:

- **Change in receptor morphology**

Can occur with ion-channel receptors and GPCRs:

- Ion-channel receptors bind the ligand but do not open the channel
- GPCR become 'uncoupled' - phosphorylation of the receptor makes it unable to activate second messenger cascade, though it can still bind the ligand.

- **Receptor down-regulation**

Prolonged exposure to agonists causes transmembrane (typically hormone) receptors to become **internalised**. This occurs more slowly than uncoupling.

- **Receptor up-regulation**

Prolonged exposure to antagonists causes an up-regulation of receptor.

- Can lead to rebound effects when a drug is ceased (e.g. hypertension with cessation of clonidine)

- **Exhaustion of mediators**
Similar to tachyphylaxis - depletion of a mediating substance decreases the effect.
- **Physiological adaptation**
Actions of a drug may be countered by a compensatory homeostatic response.
- **Active removal of the drug from the cell**

Patient Factors Affecting Drug Response

Pharmacokinetics and pharmacodynamics are affected in:

- Pregnancy
- Extremes of age
 - Foetus
 - Neonate
 - Elderly
- Other

Pregnancy

Pharmacokinetic:

- **Absorption**
 - Decreased gastric emptying
 - Nausea and vomiting
 - Increased cardiac output
 - Increases IM and SC absorption
 - Volatiles:
 - Increased onset due to increased MV and reduced FRC
 - Decreased onset due to increased CO
- **Distribution**
 - Increased V_D due to:
 - Increased TBW
 - Increased plasma volume
 - Increased fat mass

- Decreased albumin and α_1 -glycoprotein
- **Metabolism**
 - No change to HBF
 - Progesterone induces enzymes
 - Oestrogen competes for enzymes
 - Decreased plasma cholinesterase activity
- **Elimination**
 - Increased RBF
 - Increased GFR

Pharmacodynamic:

- Decreased MAC
- Increased LA sensitivity due to decreased α_1 -glycoprotein

Foetus



Drugs that cross the placenta can be teratogenic to the foetus, besides exerting their usual pharmacological effects.

Pharmacokinetic factors predominantly **affect placental transfer**, and include:

- **Lipid solubility**
Lipid soluble drugs diffuse more rapidly.
- **Molecular size**
Drugs with a molecular weight >1000 dalton cross the placenta slowly.
- **Protein binding**
- **Placental transporters**
Some medications are actively removed from foetal circulation.
- **Placental metabolism**
The placenta can metabolise some medications, although in some cases results in toxic metabolites.

A teratogen is a drug which adversely affects foetal development. Teratogens have diverse mechanisms that are not well understood.

Maternal **pharmacodynamic** factors predominantly affect the uterus and breast; **major organ systems** are **not significantly affected**.

Neonates

At < 1 year of age, pharmacokinetics are significantly altered:

- **Absorption**

- Delayed gastric emptying, increasing absorption of drugs metabolised in the stomach
- Decreased secretion of pancreatic enzymes and bile salts impairs absorption of lipid soluble medications
- Smaller muscle mass and higher relative muscle blood flow increases IM onset
- Increased $V_A:FRC$ ratio increases onset of volatiles

- **Distribution**

- TBW is 70-75% (compared to 50-60% for an adult), and extracellular water is 40% (compared to 20%), which typically **increases V_D**
- Preterm infants have reduced body fat
- Greater proportion of cardiac output goes to head, increasing onset of centrally acting (e.g. anaesthetic) drugs
- Decreased albumin and α_1 -glycoprotein
- Immature BBB increases uptake of partially ionised drugs

- **Metabolism**

- Enzymatic capacity of all pathways is reduced, which prolongs elimination half-lives and reduces clearance.
 - Hepatically metabolised drugs must be dose adjusted accordingly
- The glucuronide pathway may not mature until age 4

- **Excretion**

- **GFR is proportionally lower** and dose not reach adult equivalence until 6-12 months
 - GFR is **further reduced** in **pre-term** infants
 - GFR is increased in **1-3 year** olds

Pharmacodynamic considerations:

- Smaller ACh reserves increase sensitivity to NMBs
- Increased MAC but more rapid onset
- NSAIDs cause closures of ductus arteriosus

Elderly



Though there is a linear decrease in functional capacity of major systems beginning at 45, alterations are predominantly a consequence of polypharmacy and subsequent drug interactions.

Pharmacokinetic:

- **Absorption**
 - Laxatives and prokinetic increase gastric emptying and reduce absorption of oral agents
- **Distribution**
 - There is a proportional increase in fat
 - There is a proportional decrease in:
 - Lean body mass
 - Total body water
 - Albumin
- **Metabolism**
 - ↓ Hepatic blood flow
 - ↓ Enzymatic activity
Phase I > Phase II.
- **Elimination**
 - Loss of nephron number with age reduces renal clearance

Pharmacodynamic:

- Increased sensitivity to sedatives, opioids, and hypnotics
- Decreased sensitivity to β -agonists and antagonists
- Decreased MAC
- Polypharmacy increases potential for drug interactions

Disease Factors Affecting Drug Response

Pharmacokinetics and pharmacodynamics are altered in disease states, including:

- Cardiac disease
- Hepatic disease
- Obesity
- Renal disease

Cardiac Disease

Pharmacokinetic:

- **Absorption**
 - Decreased cardiac output decreases PO absorption due to decreased gradient
- **Distribution**
 - Decreased CO prolongs arm-brain circulation time
 - Increased α_1 -glycoprotein increasing binding of basic drugs
 - Decreased V_D
- **Metabolism**
 - **Low-cardiac** output states reduce hepatic flow and will **reduce** metabolism of drugs with a **high extraction ratio**
 - **High-output** states have the opposite effect
- **Elimination**
 - Decreased renal blood flow

Hepatic Disease

Pharmacokinetic:

- **Absorption**
 - Porto-caval shunting
 - Decreased first pass metabolism.
- **Distribution**
 - Impaired synthetic function reduces plasma proteins and increases unbound fraction
 - Increased V_D due to fluid retention
 - Metabolic acidosis changes ionised fraction
- **Metabolism**
 - Impaired phase I and II reactions
 - Reduced plasma esterase levels
- **Elimination**
 - Reduced biliary excretion

Pharmacodynamics:

- Hepatic encephalopathy increases sensitivity to sedatives and hypnotics

Renal Disease

Pharmacokinetic:

- **Absorption**
 - Uraemia prolongs gastric emptying
- **Distribution**
 - Increased V_D due to fluid retention
 - Metabolic acidosis adjusts ionised fraction
- **Metabolism**
 - Buildup of toxic metabolites may inhibit drug transporters
 - Uraemic toxins inhibit enzymes and drug transporters
- **Elimination**

- Reduced clearance of active metabolites/active drug cleared renally

Obesity

Pharmacokinetic:

- **Absorption:**
 - Delayed gastric emptying
 - Decreased subcutaneous blood flow
 - Practical difficulty with IM administration
- **Distribution:**
 - Increased V_D of lipid soluble drugs
 - Dosing of lipid-soluble drugs by actual body weight
 - Dosing of water-soluble drugs by lean body weight
 - Increased CO
 - Increased α_1 -glycoprotein
 - Increased blood volume
 - Greater lipid binding to plasma proteins, increasing free drug fractions
- **Metabolism:**
 - Increased plasma and tissue esterase levels
 - Normal or increased hepatic enzymes
- **Elimination**
 - Increased renal clearance due to increased CO

Non-Specific Alterations to Drug Response

Pharmacokinetic:

- **Absorption:**
 - Site of administration
Drugs given centrally will act faster than those given into peripheral veins.
 - Rate of administration
Faster rate of administration will increase rate of onset.

Pharmacodynamic:

- Drug tolerance Increase requirement of drug.
 - e.g. induction anaesthetic agents in patients tolerant to CNS depressants.
 - Drug interaction
May be:
 - Synergistic
 - Additive
 - Antagonistic
-

References

1. Anderson C. Variability in Drug Response 1. ICU Primary Prep.
2. Rang HP, Dale MM, Ritter JM, Flower RJ. Rang and Dale's Pharmacology. Sixth Edition. Churchill Livingstone.
3. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.
4. CICM Examiner Report: Sep/Nov 2012
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Pharmacogenetics

Outline genetic variability

Develop an understanding of variations in individual drug responses together with clinical application of this knowledge

Genetic polymorphism occurs when several functionally distinct genes exist within a population. Genetic polymorphism is:

- Common
 - Important in determining an individual's susceptibility to adverse drug reactions
 - A goal of **personalised medicine**
Aims to adjust drug therapies for interpatient variability.
-

References

1. Rang HP, Dale MM, Ritter JM, Flower RJ. Rang and Dale's Pharmacology. Sixth Edition. Churchill Livingstone.
2. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.

Pharmacogenetic Disorders

Explain the mechanisms and significance of pharmacogenetic disorders (eg malignant hyperpyrexia, porphyria, atypical cholinesterase and disturbance of cytochrome function)

Key (examinable) disorders include:

- Pseudocholinesterase deficiency
- G6PD
- Malignant Hyperthermia
- Porphyrrias

Pseudocholinesterase Deficiency

A condition where plasma cholinesterase is unable to breakdown suxamethonium, prolonging its duration of action. This disease:

- May be congenital or acquired
 - Congenital is autosomal recessive
 - Has four alleles
 - Usual
 - Atypical (dibucaine-resistant)
 - Silent (absent)
 - Fluoride-resistance
 - Acquired is due to a loss of plasma cholinesterase
 - Pregnancy
 - Organ failure
 - Hepatic

- Renal
 - Cardiac
- Malnutrition
- Hyperthyroidism
- Burns
- Malignancy
- Drugs
 - OCP
 - Ketamine
 - Lignocaine and ester local anaesthetics
 - Metoclopramide
 - Lithium
- Has been traditionally measured using the **dibucaine number**

Dibucaine is:

- An amide local anaesthetic which **inhibits plasma cholinesterase**

Different forms are inhibited to different extents, with greater inhibition indicating a less severe mutation.

- Percentage inhibition correlates with different genotypes, e.g.:
 - Normal (Eu:Eu) has a dibucaine number of 80 (80% inhibited)
 - Dibucaine resistant (Ea:Ea) has a dibucaine number of 20 (20% inhibited)
- Note that acquired disease will have a normal dibucaine number, as the enzyme itself is working correctly, however does not exist in a large enough quantity to metabolise suxamethonium rapidly

G6PD

A common x-linked recessive condition that may cause haemolysis following administration of oxidative drugs. These include:

- Aspirin

- Sulfonamides
- Some antibiotics

Malignant Hyperthermia

Autosomal dominant deficiency in the skeletal muscle ryanodine receptor gene resulting in a defect of intracellular calcium regulation. This mutation:

- Causes massive calcium release from sarcoplasmic reticulum in the presence of volatile anaesthetic agents (and potentially suxamethonium)

Leads to:

- Increased muscle activity
- Rapid increase in body temperature and lactic acidosis
- High mortality from hyperthermia, hyperkalaemia/rhabdomyolysis, leading to ventricular arrhythmia and cardiac arrest
- Mutation present in 1:5,000 - 1:50,000
- Presents with:
 - Initially:
 - Tachycardia
 - Masseter spasm
 - Hypercapnea
 - Arrhythmia
 - Intermediate:
 - Hyperthermia
 - Sweating
 - Combined metabolic and respiratory acidosis
 - Hyperkalaemia
 - Muscle rigidity
 - Late:
 - Rhabdomyolysis
 - Myoglobinuria
 - Elevated CK
 - Coagulopathy
 - Cardiac arrest
- Management consists of:

- Cease administration of volatile
 - Start TIVA
- Give **dantrolene**
 - 2.5mg.kg⁻¹ increments up to 10mg.kg⁻¹
 - 20mg vials reconstituted with 60ml sterile water
 - 3g mannitol as additive
 - Highly alkaline
- Treat complications:
 - Hyperkalaemia
 - Hyperthermia
 - Acidosis
 - Arrhythmias
 - Renal failure

Porphyria

Autosomal dominant deficiency in the first step of haeme synthesis. These mutations:

- Result in a partial deficiency of enzymes
 - Lead to accumulation of porphyrin precursors
 - May be precipitated by many drugs:
 - Ketamine
 - Clonidine
 - Ketorolac
 - Diclofenac
 - Phenytoin
 - Erythromycin
 - Barbiturates
-

References

1. Rang HP, Dale MM, Ritter JM, Flower RJ. Rang and Dale's Pharmacology. Sixth Edition. Churchill Livingstone.
2. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.
3. Petkov V. Essential Pharmacology For The ANZCA Primary Examination. Vesselin Petkov. 2012.

Cellular Physiology

The Cell Membrane

Describe the cell membrane ~~and cellular organelles and their properties.~~

Outline basic cellular physiology in particular:
The structure of the cell membrane ~~and trans-membrane transport mechanisms,~~
~~The composition and regulation of intracellular fluid,~~
The generation of the trans-membrane potential,
~~Energy production by metabolic processes in cells~~

Cell membranes are the layer that separates the interior of a cell from the outside environment (extracellular space). They are:

- Formed by a **phospholipid bilayer**
- **Semi-permeable**
Leads to different ionic concentrations (and therefore electrical charge) on either side of the membrane.
 - Alteration in charge means the membrane acts as a **capacitor**, with most cells having a **resting potential 70-80mV lower** than extracellular fluid

Ion Permeability

At rest, the cell is:

- Permeable to **potassium**
Potassium flows out down its concentration gradient, which makes the resting potential becomes more negative.
 - This negative charge opposes the further movement of potassium and so an equilibrium is established between opposing electrical and chemical gradients
- Relatively **impermeable to other cations**
The membrane is not perfectly impermeable to sodium, and Na^+ will leak in down its concentration gradient.
 - The $3\text{Na}^+ - 2\text{K}^+$ ATPase pumps **three sodium ions** outside in exchange for **two potassium ions** in order to maintain these gradients
As there is an unequal exchange of charge, this pump is **electrogenic**.

Table 1: Typical Intracellular and Extracellular Ion Concentrations

Ion	[Intracellular]	[Extracellular]
Na^+	15	140
K^+	150	4.5
Cl^-	10	100

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Membrane Transport

Explain mechanisms of transport of substances across cell membranes, including an understanding of the Gibbs-Donnan effect.

Outline basic cellular physiology in particular: The structure of the cell membrane and trans-membrane transport mechanisms, ~~The composition and regulation of intracellular fluid, The generation of the trans-membrane potential, Energy production by metabolic processes in cells~~

Substances can cross cell membranes by **three** mechanisms:

- **Diffusion**
- **Active transport**
- **Exo- or endocytosis**

Diffusion

There are several types of diffusion:

- **Simple diffusion**
Molecules pass through the cell membrane or via a channel. This process is **passive**, and occurs down a concentration gradient.

- Only lipid soluble molecules (gases, steroids) can pass directly through the lipid bilayer without a specialised channel
- Voltage-gated and ligand-gated channels facilitate simple diffusion
- **Facilitated diffusion** (uniporters)
Molecules bind to a carrier protein, and move together through the lipid bilayer, before separating on the other side. Facilitated diffusion is **concentration gradient-dependent**, and **limited by** the amount of **carrier protein** available..

The rate and extent of diffusion is affected by:

- Hydrostatic pressure gradients
- Concentration gradients
- Electrical gradients

Active Transport

Substances that are moved against a concentration gradient require active transport, and requires energy in the form of ATP. Active transport mechanisms may be:

- **Primary** active transport
The substance itself is moved.
- **Secondary** active transport
The substance moves against a concentration gradient with another molecule that had a gradient established by active transport.
 - The "other molecule" is typically sodium
- **Co-transporters** (symporters)
Uses carrier proteins and moves two substances (e.g. sodium and an amino acid) across a membrane.
 - This process will be passive if the energy gained moving one substance down its concentration gradient is greater than the energy required to move the other substance up its concentration gradient

- **Counter-transporters** (antiporters)
Use carrier proteins and moves two substances in opposite directions across the membrane.
 - May be active or passive

Key transporters include:

- The **Na⁺ -K⁺ ATP-ase** pump
This moves three sodium ions out of a cell and two potassium ions in, cleaving one ATP in the process.
This pump has many functions:
 - Maintenance of cellular volume (which would otherwise burst from the influx of water with changing ECF tonicities) by net loss of osmoles
 - Maintenance of the potential difference across the membrane
 - Establishment of chemical gradients to be used in secondary active transport mechanism
 - e.g. Reabsorption of glucose in the kidney via the S-GLUT transporter

Exo- and Endocytosis

These processes describe the formation of a vesicle (typically from membrane phospholipid) to transport substances:

- **Exocytosis**
Vesicle containing a substance to be secreted fuses with the cell membrane when activated by calcium, depositing the substance outside the cell.
- **Endocytosis**
The cell membrane invaginates around the substance, absorbing the substance into the cell. A vesicle (or vacuole) may or may not be created. Endocytosis may be subdivided into:
 - Phagocytosis, where leukocytes engulf bacteria into a vacuole
 - Pinocytosis, where substances are endocytosed but not into a vacuole

Gibbs-Donnan Effect

The Gibbs-Donnan Effect describes the **tendency of permeable ions to distribute themselves such that the ratios of the concentrations are equal** when they are in the **presence of non-diffusible ions**.

Remember: all ions are charged.

The **consequence** is an uneven distribution of ions on either side of a semi-permeable membrane, and is important for:

- Maintenance of cell volume
Na⁺ acts as an effective osmole, reducing cellular swelling.
- Plasma oncotic pressure
Increased plasma ion concentration increases oncotic pressure.
- Resting Membrane Potential

It occurs when:

- A semi-permeable membrane (e.g. a cell membrane) separates two solutions
- At least one of those solutions contains a non-diffusible ion
The two main contributors to the Gibbs-Donnan effect in the body are sodium and protein, as the cell membrane is:
 - Impermeable to protein
Intracellular protein concentration is high.
 - Effectively impermeable to sodium
Due to the Na⁺-K⁺ ATP-ase pump.

The **distribution** of permeable ions will be **influenced by both**:

- **Valence**
- **Distribution of non-diffusible ions**
Such that at equilibrium **the products of the concentrations of paired ions on each side of the membrane will be equal**:

$$[Na^+]_A \times [Cl^-]_A = [Na^+]_B \times [Cl^-]_B$$

Gibbs–Donnan Equilibrium

Distribution of particles on either side of a membrane is a function of:

- Concentration gradients
- Electrostatic gradients
- Osmotic gradients

These gradients are dynamic and interdependent, and result in **no stable steady state**:

- As the distribution of charged ions changes, *tonicity* on either side of the cell membrane is altered
- This causes movement of water down the new osmotic gradient, which then alters *concentration* of substances on either side of the membrane
- Particles then move down their **concentration** gradient, and tonicity is altered
- etc

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Eaton DC, Pooler JP. Vander's Renal Physiology. 6th Ed (Revised). McGraw-Hill Education - Europe. 2004.

Organelles

Describe the cell membrane and cellular organelles and their properties

Outline basic cellular physiology in particular:
The structure of the cell membrane and trans-membrane transport mechanisms, The composition and regulation of intracellular fluid, The generation of the trans-membrane potential, Energy production by metabolic processes in cells

Organelles are specialised functional subunits within a cell, typically contained within their own lipid bilayer.

Key organelles include:

- Mitochondria
- Endoplasmic reticulum
- Golgi apparatus

Mitochondria

Mitochondria:

- Produce ATP via aerobic metabolism
Only method of aerobic metabolism in the body.
 - Mitochondria exist in greater numbers in more metabolically active cells

- Consist of two membranes (outer and inner), which create three spaces,
 - Cytoplasm
Outside the outer membrane.
 - Intermembrane space
Between the membranes.
 - Outer membrane separates mitochondria from cytoplasm, but contains pores allowing some substances (pyruvate, amino acids, fatty acids) to pass
 - Inner membrane:
 - Isolates the electron transport chain from the intermembrane (space between inner and outer membranes) space.
 - Proteins on the inner membrane conduct the redox reactions important for ATP production
 - Electron transport chain pumps hydrogen ions into the intermembrane space
 - Inner mitochondrial matrix
Contents important in many metabolic processes:
 - Citric acid cycle
 - Fatty acid metabolism
 - Urea cycle
 - Haeme synthesis
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Excitable Cells

Explain the basic electro-physiology of neural tissue, including conduction of nerve impulses and synaptic function.

Outline basic cellular physiology in particular:
The structure of the cell membrane and trans-membrane transport mechanisms, The composition and regulation of intracellular fluid, The generation of the trans-membrane potential, Energy production by metabolic processes in cells

Membrane Potential

At rest, membranes are:

- Permeable to potassium
- Impermeable to other cations

Generation of membrane potential:

- Intracellular potassium concentration is much higher than extracellular potassium concentration
Due to the action of the $\text{Na}^+ - \text{K}^+$ pump.
- As the membrane is permeable to potassium, potassium will attempt to diffuse down this gradient, generating a negative intracellular charge which opposes further diffusion

- At some point, an electrochemical equilibrium is reached between:
 - The concentration gradient dragging potassium out of the cell
 - Negative electrical charge pulling it in
- This equilibrium is the **resting membrane potential**
 - RMP is determined by:
 - Permeability of the membrane to different ions
 - Relative ionic concentrations on either side of the membrane
 - Impermeable ions do not contribute to the resting membrane potential

Altering membrane permeability causes a flow of ions and a change in voltage.

Nernst Equation

The potential difference generated by a **permeable** ion in electrochemical equilibrium when there are different concentrations on either side of the cell can be calculated via the **Nernst Equation**:

$$E (mV) = \frac{R.T}{z.F} \ln \frac{[ion]_{outside}}{[ion]_{inside}}$$

Where:

- E is the equilibrium potential for the ion
- R is the gas constant ($8.314 \text{ J.deg}^{-1}.\text{mol}^{-1}$)
- T is the temperature in Kelvin
- F is Faraday's Constant
- z is the ionic valency (e.g. +2 for Mg^{+2} , -1 for Cl^{-})

Goldman-Hodgkin-Katz Equation

The Nernst equation describes the equilibrium potential for a single ion, and assumes that the membrane is completely permeable to that ion.

However, calculation of membrane potential requires examining the effects of many different ions with different permeability. This can be performed with the **Goldman-Hodgkin-Katz equation**:

$$E \text{ (mV)} = \frac{R.T}{F} \ln \frac{P_K[K^+]_o + P_{Na}[Na^+]_o + P_{Cl}[Cl^-]_i}{P_K[K^+]_i + P_{Na}[Na^+]_i + P_{Cl}[Cl^-]_o}$$

Where:

- P_x is the permeability constant for the ion, x
If the membrane is impermeable to x , then $P_x = 0$.

Note that:

- This model does not consider valency
- The concentrations of negative ions are reversed relative to positive ions

Action Potential

Excitable cells can respond to a stimulus by a changing their membrane potential. This may be mediated:

- Chemically
e.g. ACh receptors causing Na^+ channels to open.
- Physically
Pressure receptors physically deforming and opening Na^+ channels.

Generating an Action Potential

- Stimulation increases Na^+ permeability, which increases (makes *less* negative) the membrane potential
- If several stimuli, or a large enough stimulus raises the membrane potential above the **threshold potential**, then an action potential will be generated.
- This process occurs due to **voltage-gated** (fast) **Na^+ channels**

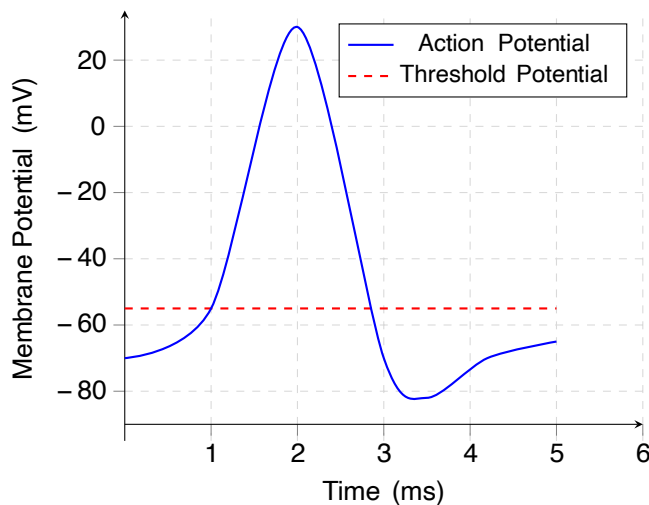
Voltage-gated channels:

- Open when **membrane potential** exceeds **threshold potential**

Threshold potential is typically -55mV .

- Generate the **all-or-nothing** response:
 - Stimuli below the threshold potential do not generate an action potential
 - Stimuli above threshold potential generate an action potential

The **size of the stimulus does not affect the magnitude of the action potential**, as this is determined by the fast sodium channels.



Key Players in the Action Potential

Fast Na^+ channels are responsible for depolarisation. They exist in three states:

- **Closed**
Impermeable to Na^+ .
- **Open**
Permeable to Na^+ . Occurs when the membrane potential reaches threshold potential.
 - Different voltage-gated channels may have slightly different opening (threshold) potentials
- **Inactivated**
Impermeable to Na^+ . Occurs shortly after the open state, and lasts until the membrane potential falls below -50mV .

Voltage-gated K^+ channels:

- Are vital for repolarisation
- **Open slowly** with depolarisation
This increases potassium permeability and reduces membrane potential.

Phases of the Action Potential

This describes the peripheral nerve action potential. The heart is covered under the cardiac action potential.

1. Rising Phase

A stimulus which rises above the threshold potential opens fast Na^+ channels, increasing Na^+ influx.

- Additional Na^+ has a positive feedback effect, causing additional Na^+ channels to open and further depolarisation
- This drives the membrane potential towards the Nernst equilibrium for Na^+

2. Peak Phase

Inactivation of fast-channels and delayed activation of K^+ channels slows depolarisation.

- Membrane potential **peaks** at **30mV**
- 3. **Falling Phase**

As potassium exits the cell, membrane potential continues to fall.

 - Voltage-gated K^+ channels start to close at - **50mV**
 - Inactivation of fast sodium channels defines the **absolute refractory period**

No Na^+ can be conducted, regardless of the intensity of the stimulus, and so an action potential cannot be generated

 - The absolute refractory period lasts **~1ms**
- 4. **Hyperpolarisation**

As potassium channels close slowly, the membrane potential slightly undershoots resting potential, causing slight hyperpolarisation of the cell.

 - This is the **relative refractory period**

A large enough stimulus may overcome the additional hyperpolarisation and generate a second action potential.

 - The relative refractory period lasts **10-15ms**
- 5. **Resting**

Cell is stable at resting membrane potential.

Propagation of the Action Potential

- An increase in Na^+ in one region will diffuse down the cell, raising the membrane potential above the resting potential in the adjacent membrane
- This causes local fast Na^+ channels to open, and the cell depolarises
- This results in a propagating wave of depolarisation and repolarisation
- Regions of a nerve cell covered by a myelin sheath do not have ion channels
- In these cells, propagation is **saltatory**

This describes the “jumping” of the action potential between gaps in the myelin sheath.

- These gaps are known as **nodes of Ranvier**
Ion channels generate an action potential at the nodes in the usual manner.
- Between nodes, conduction is via local electrical currents
- Myelination:
 - **Increases conducting velocity**
 - **Reduces energy expenditure**
Via reduction in total ion flux.

Classification of Nerve Fibres

Classified on their diameter and conduction velocity:

- Type A
Myelinated, 12-20 μm in diameter, conduct at 70-120 m.s^{-1} . Subdivided into:
 - A α
Motor fibres.
 - A β
Touch fibres.
 - A γ
Intrafusal (proprioceptive) muscle fibre.
 - A δ
Pain fibres.
 - Type B
Myelinated, < 3 μm , conduct at 4-30 m.s^{-1} . Innervate pre-ganglionic neurons.
 - Type C
Unmyelinated, 1 μm , conduct at 0.5-2 m.s^{-1} . Pain fibres.
-

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1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Fluid Compartments

To describe the composition and control of intracellular fluid ~and the mechanisms by which cells maintain their homeostasis and integrity~

Outline basic cellular physiology in particular:
~~The structure of the cell membrane and trans-~~
~~membrane transport mechanisms,~~ The composition and regulation of intracellular fluid,
~~The generation of the trans-membrane potential,~~ Energy production by metabolic processes in cells

On average, the human body is **~60% water**. Distribution of water content can be divided conceptually into:

- Intracellular fluid
- Extracellular fluid

Intracellular Fluid

Intracellular fluid:

- Composes 2/3rds of total body water.
- Not a contiguous fluid space
- Useful conceptually as the composition of cellular contents is relatively uniform:
 - Potassium is the dominant intracellular cation
 - Sodium concentrations are low.

- The dominant anion is protein
Chloride concentration is relatively low.
- Low in magnesium

Extracellular Fluid

Extracellular fluid:

- Composes the remaining 1/3rd of total body water
- Is subdivided into:
 - Intravascular fluid
 - Composes ~20% of ECF
 - Refers solely to plasma volume (as the volume of blood from cellular components is ICF)
 - Is vital for transporting nutrients, waste, and chemical messengers between the plasma and cells
 - Transcellular fluid
 - Composes ~7% of ECF
 - Includes the volume of CSF, urine, synovial fluid, gastric secretions, and aqueous humor
 - Interstitial fluid
 - Composes the bulk of ECF volume
 - Refers to the fluid that occupies the volume between cells

Variations

Actual total body water content varies predominantly with **fat content**. This leads to differences in TBW concentrations between:

- Neonates
~75-80%.
- Elderly
~50% by the age of 60, due to increased adiposity.

- Women
Typically ~55%.

Measuring Volumes of Fluid Compartments

All methods rely on the indicator-dilution method:

- A known amount (i.e. known volume of a known concentration) of indicator with affinity to a particular compartment is given and allowed to equilibrate
- The concentration of the indicator is then measured
- The difference between the measured concentration and the initial concentration is proportional to the volume of the compartment

Indicators used for calculation of:

- Plasma volume
A colloid that will be retained in the vascular compartment; e.g. radio-labeled albumin.
 - ECF volume
A substance which can enter the interstitium but not cells; e.g. thiosulfate.
 - Total body water
A substance which can enter all compartments freely; e.g. heavy water (2H_2O).
 - ICF volume
Can be measured by the difference between calculated ECF volume and TBW.
-

References

1. Brandis, K. Fluid Compartments. Anaesthesia MCQ.

2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Cell Homeostasis

To describe the ~~composition and control of intra-cellular fluid and the~~ mechanisms by which cells maintain their homeostasis and integrity

Cellular respiration describes the production of ATP through a series of redox reactions. Oxygen is used as the oxidising agent, whilst the catabolic fuel may be glucose, fat, or protein.

Cellular respiration can be broken down into:

- Glycolysis/Lipolysis/Proteolysis
- Citric Acid Cycle
- Electron Transport Chain

Glycolysis

Glycolysis, or the **Embden - Meyerhof pathway**, describes the production of pyruvate from glucose. Glycolysis:

- Occurs in the cytoplasm
- Begins with the **phosphorylation of glucose** to glucose-6-phosphate
- Produces:
 - 2 ATP
 - 2 Pyruvate
 - 2 NADH
- Note that **oxygen is not consumed and carbon dioxide is not produced**

- In aerobic conditions:
NADH exchanges electrons across the mitochondrial wall, regenerating NAD^+ and allowing glycolysis to continue
- In anaerobic conditions:
NAD⁺ is regenerated through the production of lactate
 - When aerobic conditions are restored, lactate can be oxidised back to pyruvate and enter the CAC
 - Transported to the liver and converted back to pyruvate (and enter the CAC), or produce glucose (**Cori cycle**)

Citric Acid Cycle/Kreb's Cycle

- Takes place in the mitochondria
- Complicated
- Can take many various substrates:
 - Acetyl CoA
Produced by β -oxidation of fatty acids and pyruvate.
 - Pyruvate
 - Ketoacids
- Does **not consume oxygen** but also **doesn't function under anaerobic conditions**, due to its requirement on fresh NAD^+ from the ETC
- Produces:
 - NADH
 - FADH_2
 - CO_2

Electron Transport Chain

- Final stage of carbohydrate, fat, and protein catabolism
- ETC consists of five protein complexes

- Electrons are passed along the chain and combine with oxygen, releasing energy which stimulates the movement of hydrogen ions
 - Each time a hydrogen ion crosses the mitochondrial matrix, an ATP is produced
 - This is called **coupled phosphorylation**
 - Uncoupled phosphorylation allows hydrogen ions to travel down their gradient without generating ATP, which produces excess heat instead
 - 36-38 ATP are produced by aerobic glycolysis
Sources disagree on exactly how much ATP is produced.
 - 2 from the Embden-Meyerhof pathway
 - 34-36 from the CAC and ETC
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Respiratory Physiology

Anatomy

Airway and Alveolar Anatomy

Describe the function and structure of the upper, lower airway and alveolus.

Describe the basic structural anatomy of the upper airway including the larynx

Outline the anatomy of the lower airways

The airway can be functionally divided into the:

- Upper airway
- Lower airway
- Alveolus

Upper Airway

The upper airway consists of the:

- **Mouth**
- **Nasal cavity**
 - Hairs filter large particles
 - Olfactory receptors detect harmful gases prior to inhalation
- **Pharyngeal structures**

- Pharynx
- Larynx

Nasal Cavity

The nasal cavity provides:

- Good **humidification** and **filtration** of inhaled particles because the septum and turbinates have:
 - High mucosal surface area
 - High mucosal blood flow
 - Generate turbulent flow
- High resistance to flow
At a **high minute ventilation**, oral breathing is favoured.

Pharyngeal Structures

- **Pharyngeal dilator muscles**
Including genioglossus and levator palati. Prevent pharyngeal collapse during negative-pressure ventilation and during sleep.
- **Larynx**
Important for **airway protection, speech, and effort closure**.
 - Prevents aspiration during swallowing by elevating the epiglottis and occluding of the aryepiglottic folds
 - **Phonation** is achieved by adjusting tension (and therefore resonance) of the vocal cords by action of the cricothyroid
 - During **inspiration**, **cricoarytenoid** muscles rotate the arytenoid cartilage and abduct the vocal cords to reduce resistance to airflow
 - During **expiration**, the **thyroarytenoid** muscles adduct the cords and increase resistance, providing **intrinsic PEEP**
 - 3-4 cmH₂O of PEEP is generated

- Maintains patency of small airways
Prevents alveolar collapse and therefore maintains FRC.
- **Effort closure** is tighter occlusion of the laryngeal inlet, in which the aryepiglottic muscles contract strongly to act as a sphincter, allowing the airway to withstand up to 120cmH₂O of pressure.

Lower Airway

The lower airway consists of the **tracheobronchial tree**, which is divided into two zones based on whether they contain alveoli and therefore are able to participate in gas exchange:

- The **conducting zone** is the first **16 divisions**
- The **respiratory zone** is the last 7 divisions

There are a total of 23 divisions in the lower airway.

Conducting Zone

The **first 16 divisions** constitute the **conducting zone**. Anatomically, the conducting zone consists of the:

- **Trachea**
 - Mean diameter of 1.8 cm and a length of 11cm
 - D-shaped cross section
Curved cartilages anteriorly and longitudinal muscle (trachealis) posteriorly. External pressure of 40cmH₂O is sufficient to occlude the extrathoracic trachea.
 - Flow is typically turbulent in the trachea and large airways
- **Bronchi**
 - Comprise the first four divisions of the trachea

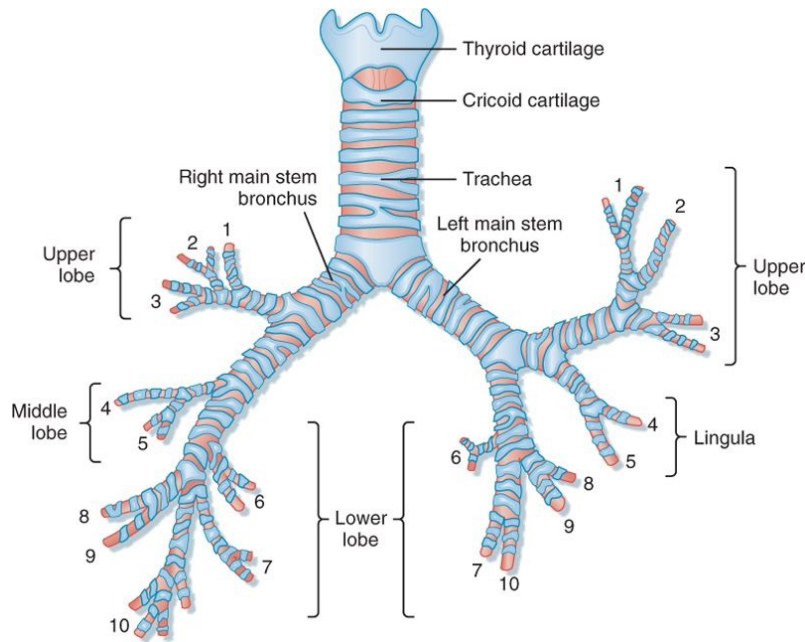
Segmental bronchi are those that show peribronchial cuffing and perihilar haze on x-ray in early pulmonary oedema.

- Right main bronchus is wider and deviates less from the axis of the trachea (the left main bronchus has a tighter turn over the heart), which is why foreign bodies will tend to the right side
- The two main bronchi divide into a total of 5 lobar bronchi, which in turn divide into a total of 18 segmental bronchi
 - Right main bronchi is **~2cm** long
 - Left main bronchi is **~5cm** long
- Cross-sectional area of the respiratory tract is lowest at the third division
 - These bronchi will collapse when intrathoracic pressure exceeds intraluminal pressure by $5\text{cmH}_2\text{O}$.
 - Segmental bronchi travel with branches of the pulmonary artery and lymphatics
 - Flow is typically transitional in the smaller bronchi and bronchioles
- Contain **goblet cells** in their walls which secrete **mucous**, which traps inhaled particles
- Contain **cilia**, move rhythmically to drive the **mucociliary elevator**, driving mucous up to the epiglottis, where it is then swallowed or expectorated
- **Bronchioles**
 - Embedded in the lung parenchyma
 - Do not have cartilage in their walls to maintain patency - are held open by lung volume
 - Resistance to flow tends to be negligible due to large cross sectional area, unless there is spasm of helical muscle bands in bronchial wall
- **Terminal bronchioles**
 - Flow may become laminar in the smallest bronchioles as flow decreases

Physiologically:

- Flow in the conducting zone during inspiration is fast and turbulent

- No gas exchange occurs in the conducting zone
The volume of the conducting zone therefore contributes to **anatomic dead space**.
- Blood supply to the conducting zone is via the **bronchial circulation**



Respiratory Zone

The **remaining 7 divisions** make up the **respiratory zone**.
This region:

- Makes up the majority of lung volume
All non-anatomical dead space volume is in the respiratory zone, and is $\sim 30\text{ml.kg}^{-1}$ (FRC) at rest
- Blood supply is via the pulmonary circulation
- Gas flow in the terminal respiratory zone is slow due to the exponential increase in cross-sectional area with each airway division
 - Diffusion is the predominant mechanism of gas movement

Alveolus

The alveolus is optimised for gas exchange:

- Spherical shape maximises surface area to volume ratio
- Total surface area of lung alveoli is 50-100m²
- Alveolar walls are extremely thin (0.2-0.3µm)
Consequently, they are fragile and can be damaged by increases in capillary pressure
- Alveolar walls contain a dense mesh of capillaries **7 to 10µm** thick, which is just large enough for an erythrocyte to pass through
The alveolar-capillary barrier consists of three layers:
 - Type I pneumocytes
 - Extracellular matrix
 - Pulmonary capillary endothelium

Alveoli are composed of three types of cells:

- **Type I pneumocytes**
Thin-walled epithelial cells optimised for gas exchange.
 - Form ~90% of the alveolar surface area
 - **Type II pneumocytes**
Specialised secretory cells.
 - Secrete surfactant
Alveoli are inherently unstable, and surface tension of alveolar fluid favours collapse of the alveoli. **Surfactant** reduces surface tension, allowing the alveoli to expand.
 - Form ~10% of alveolar surface area
 - **Alveolar macrophages**
Alveoli have no cilia - inhaled particles are phagocytosed by alveolar macrophages in alveolar septa and lung interstitium.
-

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1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.
2. West J. Respiratory Physiology: The Essentials. 9th Edition. Lippincott Williams and Wilkins. 2011.
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Chest Wall and Diaphragm

Describe the structure of the chest wall and diaphragm and to relate these to respiratory mechanics.

Discuss the structure of the chest wall and diaphragm and the implications for respiratory mechanics

Mechanics of breathing depend on the:

- Chest wall
 - **Ribs**
Slope antero-inferiorly, and are connected by the external, internal, and innermost intercostal muscles.
 - **Intercostal muscles**
 - External intercostals slope antero-inferiorly
 - Internal and innermost intercostals slope infero-posteriorly
- **Diaphragm**
Complex dome-shaped membranous structure, consisting of a central tendon and peripheral muscles
 - Performs the majority of inspiratory work of breathing
 - Able to dramatically increase intraabdominal pressure, so is essential in:
 - Coughing
 - Vomiting

- Sneezing
- Role in maintaining lower oesophageal sphincter tone
- It has three perforations:
 - **T8** for the **vena cava** (eight letters)
 - **T10** for the **oesophagus** (ten letters)
 - **T12** for the **aorta**, thoracic duct, and azygos vein

Inspiration

- During **inspiration**, the **diaphragm** and **external intercostal** muscles **contract**
 - Diaphragm pushes the intraabdominal contents down, increasing thoracic volume and generating a negative intrathoracic pressure
 - Diaphragm is supplied by the phrenic nerves from C3/4/5.
 - External intercostals pull the ribs antero-superiorly, which increases the cross-sectional area of the chest, further increasing thoracic volume (and negative pressure)
 - Intercostal muscles are supplied by intercostal nerves from the same spinal level
 - Paralysis of the external intercostals does not have a dramatic effect on inspiratory function provided the diaphragm is intact
- **Accessory muscles** include **sternocleidomastoid** and the **scalene**, which elevate the sternum and first two ribs respectively. They are active in hyperventilation.

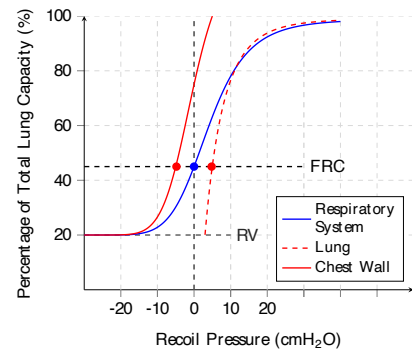
Expiration

- Expiration is passive during quiet breathing as elastic recoil of the lung will return them to FRC

- When minute ventilation is high, expiration becomes an active process:
 - **Abdominal wall** muscles (rectus abdominis, internal oblique, external oblique, transversus abdominis) contract, raising intraabdominal pressure and forcing the diaphragm up
 - **Internal** and **innermost** intercostals contract, pulling the ribs **downwards** and **inwards**, further decreasing thoracic volume

Respiratory Mechanics in Spinal Injury

- Paralysis of the abdominal wall muscles (e.g. spinal injury) has significant effect on respiratory mechanics
 - In the initial phases of injury, spinal shock results in a flaccid paralysis of the abdominal wall
 - Intraabdominal pressure is low, and so the diaphragm moves inferiorly
This results in a higher FRC but limits tidal volumes, as contraction of the diaphragm only increase thoracic volume by a small fraction.
 - Nursing in a supine position causes the abdominal contents to push the diaphragm superiorly, causing:
 - Lower FRC
 - Greater proportional expansion with respiration, improving tidal volumes
 - Once spastic paralysis ensues, the abdominal wall is rigid and the patient can be sat up



References

1. West J. Respiratory Physiology: The Essentials. 9th Edition. Lippincott Williams and Wilkins. 2011.

Variations in Upper Airway Anatomy

Understand the differences encountered in the upper airway for neonates, children and adults.

Neonates and Children

Changes are most obvious below 1 year of age, and typically resolve by ~8 years of age.

- Head and neck changes
 - Obligate nose breathers
Nasal obstruction may significantly impair respiration.
 - Proportionally **enlarged head** and occiput
Optimal intubating position is neutral rather than ramped.
 - Proportionally **short neck**
Favours airway obstruction when flexed.
- Laryngeal changes
 - Disproportionately **large tongue** that complicates laryngoscopy
 - **Epiglottis** is **u-shaped, longer**, and **stiffer**
 - **Larynx lies at C4** (rather than C6 in adults)
 - Narrowest part of the upper airway is **the transverse diameter of the vocal cords**

The cricoid used to be thought of the narrowest part of the child's airway, but this has been subsequently disproved.

- Intrathoracic changes
 - Intrathoracic trachea is also shorter
May be only 4cm long, so there is little margin for error in tube placement.
 - Left and right bronchi arise at similar angles, so endobronchial intubation may occur on either side
 - Airways themselves are narrower, and have a higher resistance to flow.
-

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1. Nickson, C. Paediatric Airway. LITFL.
2. Anderson, C. Anatomy of the Respiratory system.. ICU Primary Prep.
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
4. Tobias JD. Pediatric airway anatomy may not be what we thought: implications for clinical practice and the use of cuffed endotracheal tubes. Paediatr Anaesth. 2015 Jan;25(1):9-19.

Mechanics

Control of Breathing

Describe the control of breathing

Describe the neural and chemical control of ventilation via central and peripheral chemoreceptors and indicate how this is altered by anaesthesia and abnormal clinical states

Ventilation is controlled by a feedback loop involving:

- Inputs
- Integration and control centres
- Effectors

Inputs

Inputs to the respiratory centre comes from a number of sensors:

- Chemoreceptors
Chemoreceptors act synergistically. Chemoreceptors are divided into:
 - Peripheral
 - Central
- Mechanoreceptors
- Other effects

Peripheral Chemoreceptors

Peripheral chemoreceptors are divided into:

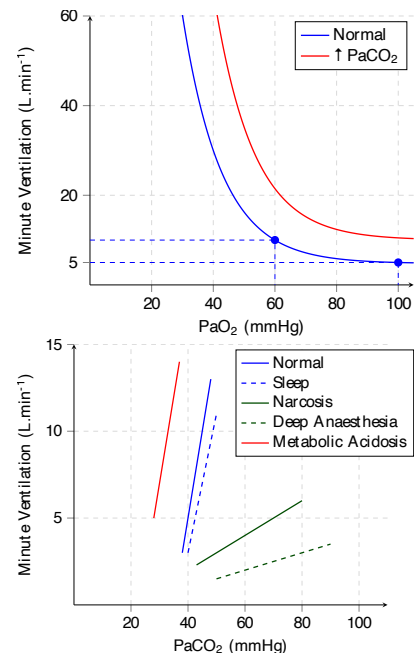
- The **carotid body**
Located at the bifurcation of the common carotid artery, and are innervated by the glossopharyngeal nerve (CN IX).
- The **aortic body**
Located in the aortic arch, and innervated by the vagus (CN X).

Peripheral chemoreceptors are stimulated by:

- **Low PaO_2**
Peripheral chemoreceptors are stimulated by **low O_2 tension**
- **High PaCO_2**
Peripheral receptors have a **rapid** ($\sim 1-3\text{s}$) but **weaker** ($\sim 20\%$ of response) to changes in CO_2 , compared to central chemoreceptors, which take $\sim 11\text{s}$ to respond
- **Acidaemia**
(Carotid bodies only)
- **Hypotension**

Central Chemoreceptors

- Central chemoreceptors are located on the **ventral medulla**, and are **stimulated** by a **fall in CSF pH**
 - H^+ and HCO_3^- are ionised, and cannot cross the BBB by diffusion
 - Because of this, central chemoreceptors **respond indirectly to changes in arterial PaCO_2**
 - Carbon dioxide is lipid soluble and freely diffuses into CSF
 - In CSF, carbon dioxide combines with water (catalysed by carbonic anhydrase) to form H^+ and HCO_3^-



- This gives the central chemoreceptors a number of special properties:
 - **Increased sensitivity**
Increased relative to plasma due to minimal buffering (as there is less protein in CSF)
 - **Respond to respiratory acidosis**
Fixed acid does not cross the blood brain barrier and so have a minimal response on CSF pH. **Cerebral hypoxia** increases **CSF lactate**, which will stimulate respiration.

Mechanism of CO₂ Retention

- Prolonged respiratory acidosis (i.e. prolonged CSF acidosis) stimulates active secretion of bicarbonate into the CSF
- When pH normalises, the stimulation of central chemoreceptors ceases
- Similarly, renal absorption of bicarbonate increases, which normalises arterial pH and reduces peripheral chemoreceptor stimulation

Mechanoreceptors

Stretch receptors in bronchial muscle are stimulated by overinflation, and stimulate the **apneustic centre** to reduce inspiratory volumes. This is the **Hering-Breuer reflex**.

Other Stimulants

Other inputs which stimulate respiration include:

- Juxta-capillary receptors (J-receptors)
Receptors in alveolar walls, potentially stimulated by oedema and emboli.
- Irritant receptors
Inhalation of noxious gases stimulates respiration.

- Pain receptors
- Thalamus
Increased core temperature stimulates respiration.
- Limbic system
Emotional responses.
- Cerebral cortex
Conscious control of breathing.
- Muscle spindles
Ventilatory response to exercise.

Integration and Control

The respiratory centre is located in the **medulla** and the **pons**. It consists of four groups:

- **Dorsal Respiratory Group (DRG)**
Controls the diaphragm, and is so only involved with inspiration.
 - **Ventral Respiratory Group (VRG)**
Controls the intercostal muscles, and so is involved in inspiration and expiration.
 - **Apneustic Centre**
Modulates DRG function to prevent over-expansion. Loss of this area causes **apneusis** - long, deep breaths.
 - **Pneumotaxic Centre**
Also modulates the DRG, increasing RR and decreasing V_T to maintain MV.
-

References

1. CICM February/April 2015
2. CICM March/May 2009
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

4. Brandis K. The Physiology Viva: Questions & Answers. 2003.

Intrapleural Pressure

Describe the inspiratory and expiratory process involving the chest wall, diaphragm, pleura and lung parenchyma

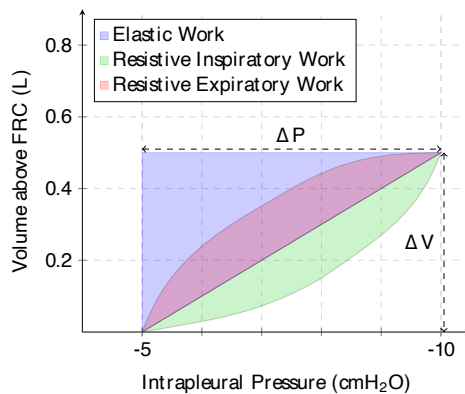
Explain the significance of the vertical gradient of pleural pressure and the effect of positioning

Explain the vertical gradient of pleural pressure and its significance

Intrapleural pressure (P_{pl}) is the pressure in airtight the space between the visceral and parietal pleura, or (physiologically) between the lungs and the chest wall. Intrapleural pressure:

- Is usually **negative**, typically **-5cmH₂O at rest**
 - Balance between the:
 - Outwards recoil of the chest wall
 - Inwards recoil of the lungs (P_{el})
- Varies with vertical distance in the lung
 - Gravity pulls the lung parenchyma inferiorly
 - Intrapleural pressure is therefore:
 - **More negative** in the **apex**
Typically **-10cmH₂O** at FRC
 - **Less negative** in the **base**
Typically **-3cmH₂O** at FRC
- This changes the degree of inflation at FRC
 - Apical alveoli are maximally inflated

- Basal alveoli are relatively deflated
- **Changes over the respiratory cycle**, which drives **change in lung volume** and therefore **breathing**
 - An airtight pleura is required, so that the trachea is the only route through which gas can enter or exit the lung
 - During inspiration, the pleural pressure changes evenly throughout the lung, however the basal alveoli are better ventilated because their compliance is increased (due to lower resting volume)



Inspiration

- Diaphragmatic and external intercostal/accessory muscle contraction causes an increase in the volume of the thorax
- Intrapleural pressure becomes more negative, typically to **-8-10cmH₂O**
- When $P_{pl} > P_{el}$, the lungs expands
- Alveolar pressure (P_A) becomes sub-atmospheric, and inspiration occurs
- At end inspiration:
 - $P_{pl} = P_{el}$
 - $P_A = P_{atmospheric}$

Expiration

- Muscular relaxation causes the chest wall to passively return to their resting position
 - Thoracic volume falls
 - P_{pl} falls to $-5\text{cmH}_2\text{O}$
 - The elastic recoil of the lung causes it to collapse until $P_A = P_{\text{atmospheric}}$
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Compliance

Define compliance (static, dynamic and specific), its measurement, and relate this to the elastic properties of the respiratory system.

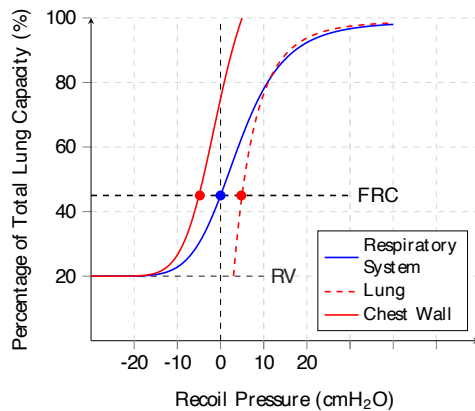
Compliance is the **change in volume** for a **given a change in pressure**. In the lung, compliance:

- Occurs due to the tendency of a tissue to resume its original position after removal of an applied force
- Is the inverse of **elastance**, which is the force at which the lung recoils for a given degree of distension
- If **decreased**, means the transpulmonary **pressure must change by a greater amount for a given volume**
This increases elastic work of breathing.

Units for compliance are $\text{ml.cmH}_2\text{O}^{-1}$

Compliance of the Respiratory System

- Compliance of the respiratory system is a function of both lung and chest wall compliance: $\frac{1}{C_T} = \frac{1}{C_L} + \frac{1}{C_W}$.



- The compliance curve is not linear as compliance varies with lung volume
- In the normal range however, (-5 to -10cmH₂O) compliance of the lung and chest wall **independently** is typically stated as $^{**}200\text{ml.cmH}^{-1}_{2}\text{O}^{**}$.
- **Compliance of the respiratory system** as a whole is therefore $^{**}100\text{ml.cmH}^{-1}_{2}\text{O}^{**}$

Measurement of Lung and Chest Wall Compliance

- Lung compliance is calculated from the alveolar-intrapleural pressure gradient
- Chest-wall compliance is calculated from the intrapleural-ambient pressure gradient
- Total compliance is calculated from the alveolar-ambient gradient
- Measuring ambient and alveolar pressure is straightforward, as is calculating compliance of the respiratory system
 - Alveolar pressure is measured by taking a plateau pressure
- Separating lung and chest wall compliance requires measurement of intrapleural pressure
This is performed by measuring oesophageal pressure (using a balloon) with an **open glottis**, as oesophageal pressure approximates intrapleural pressure.

- Measurement of compliance of each system individually determines what proportion of plateau pressure is distributed to each
 - If the lung is significantly less compliant than the chest wall, a greater pressure is required to distend the lung
 - Therefore, the alveolar-intrapleural gradient will be much greater than the intrapleural-ambient gradient
 - This can be expressed by the equation:

$$\Delta P_{Pl.} = P_{Plat.} \times \frac{C_L}{C_L + C_W}$$

Static Compliance

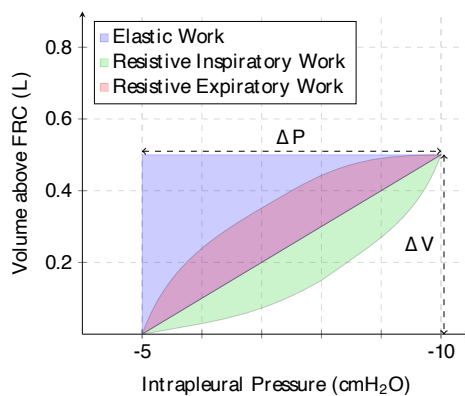
- **Static compliance** is the compliance of the system at a given volume when there is no flow
- Therefore there is no pressure component due to resistance
- A static compliance curve is made by measuring the pressure across a range of lung volumes, with patient taking incremental breaths
- Static compliance is a function of:
 - Elastic recoil of the lung
 - Surface tension of alveoli

Dynamic Compliance

Dynamic compliance is the compliance measured during respiration, using continuous pressure and volume measurements. Dynamic compliance:

- **Includes the pressure required to generate flow** by overcoming resistance forces
 - This means it is also a bit of misnomer

- **Dynamic compliance is always less than static compliance**, as there will always be a degree of airway resistance
- Dynamic compliance is a function of respiratory rate
In normal lungs at normal respiratory rates it approximates static compliance.
- Reduced in lung units with unequal time constants at high respiratory rates
 - Due to incomplete filling of alveoli - the portion of pressure that is used to overcome airways resistance is therefore proportionally greater



Specific Compliance

Specific compliance is the compliance per unit volume of lung. Specific compliance is:

- Expressed as: $C_S = \frac{C_{Tot}}{FRC}$
- Used to compare different lungs

Hysteresis

- In general, hysteresis refers to any process where **the future state of a system is dependent on its current and previous state**

- Specific to the lung, it means the compliance of the lung is different in inspiration and expiration
- There is hysteresis in both static and dynamic curves:
 - In dynamic compliance curves:
Airways resistance is a function of flow rate. Flow rate (therefore resistance) is maximal at the beginning of inspiration and end-expiration.
 - In static compliance curves:
There is no resistive component. Hysteresis is due to viscous resistance of surfactant and the lung.

Changes in Compliance

Respiratory system compliance can be affected by changes to either lung or chest wall compliance, and can be increased or decreased.

Increased Lung Compliance

- Normal ageing
- Asthma attack
- Emphysema

Decreased Lung Compliance

- Alterations in lung volume and consolidation
Compliance is reduced at **extremes of lung volume**.
It is **highest at FRC**.
 - Children
 - Pneumonectomy/lobectomy
 - Atelectasis/collapse
 - Pneumonia
 - ARDS
- Increased pulmonary blood volume/**venous congestion**

- APO
- Increased surface tension
 - **Reduced surfactant**
 - Hyaline Membrane Disease
- **Impaired parenchymal compliance**
 - Pulmonary fibrosis

Increased Chest Wall Compliance

- Collagen disorders

Decreased Chest Wall Compliance

- **Chest wall restriction**/structural abnormalities
 - Obesity
 - Spastic paralysis of chest wall musculature
 - Ossification of costal cartilages
 - Kyphosis/scoliosis
 - Scarring/constriction (e.g. circumferential burns)
- **Position**
 - Prone (60% reduced compliance)/supine
This is due to the effect of position on lung volume.

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. Kenny JE. Heart-Lung Interaction Lecture Series. From heart-lung.org.

Time Constants

Explain the concepts of time constants

Discuss 'fast' and 'slow' alveoli, including the concept of 'time constants'

The time-constant is used in respiratory physiology in:

- Timing inspiration and expiration
- Elimination of inhalational anaesthetics
- The change in PaO_2 and PaCO_2 after changes in ventilation

In ventilation:

- The time constant is affected by:
 - Compliance
 - Resistance
 - Inflation pressure

At a **constant inflation pressure**, the **time constant is equal to the product of resistance and compliance**, i.e. $\tau = C \times R$.

- For two lung units of **equal compliance and resistance**
 - Inflation will occur as per the exponential growth function
 - Time-constants of each lung unit will be equal
 - No redistribution of gas will occur at end-inspiration as the pressure and volume of each unit is the same

A Brief Refresher on Time Constants:

A time constant is **time** that a **process** would take to complete *if* its **initial rate of change** remained constant.

It is useful when modeling a process using exponential functions, as the **rate of change** of an **exponent** is **proportional to its current value**.

For a curve that **decreases over time**, the general case is:

$$y = y_0 e^{-kt}$$

Where:

- y_0 is the value of y at $t = 0$
- $-k$ is the rate constant
- t is time

A positive rate constant (k) indicates that the curve **increases** over time. k is the **reciprocal of the time constant**, τ

- For two lung units, where **one has half the compliance but twice the resistance**
 - The time constants are equal, therefore both reach peak filling at the same time
 - However, the poorly compliant unit will only reach half the volume
 - No redistribution of gas will occur at end-inspiration as the pressure and volume of each unit is the same
- For two lung units, where one has **twice the resistance** of the other
 - The time-constants are unequal
 - The resistant unit will fill at half the rate of the other
 - If inspiration is prolonged both will reach the same volume
 - If inspiration is halted early, and expiration prevented, there will be a pressure gradient between the units (as compliance is the same), and gas will redistribute from the low-resistant unit to the high-resistant unit
- For two lung units where one has **half the compliance**
 - The time constants are unequal
 - The poorly compliant unit will fill at half the rate of the other
 - If inspiration is prolonged they will both reach the same pressure
The volume in the poorly compliant unit will be half that of the more compliant unit.
 - During inspiration, the pressure rises more rapidly in the poorly compliant unit, and if inspiration is stopped and expiration prevented, this will result in redistribution into the more compliant unit until pressures are equal

In general:

- **Rate of filling is determined by time constants**
 - **High-resistance** lung units have **longer time constants** and take longer to fill

- **Final volume** (assuming an indefinite inspiration) **is a function of compliance**
 - Poorly compliant units with **empty and fill rapidly**
This creates the concept of **fast** and **slow alveoli**, depending on their time constants.
- At a sustained inflation pressure:
 - A low-resistance unit shows initial greater volume change but rapidly approaches equilibrium volume
 - A high-compliance unit takes a greater overall volume over a longer period
- At end-inspiration:
 - Pressure in units with a shorter time-constant rises more rapidly and if a breath is held will result in redistribution to those units with a longer time-constant.

Clinical Significance

If time-constants are equal:

- The pressure in each unit is identical throughout inspiration and distribution
Therefore, dynamic compliance will be independent of respiratory rate.

If time-constants are unequal:

- Long-time constant units may still be inhaling whilst the rest of the lung has stopped, or begun exhalation
This is called **pendelluft**.
- In pendelluft, distribution of inspired gas is dependent on respiratory rate
 - As respiratory rate increases, the proportion of the tidal volume that is delivered to the region with a long time-constant decreases
Fast alveoli are preferentially inflated, causing V/Q scatter or shunt in the unventilated slow alveoli.

- Dynamic compliance will decrease as respiratory rate rises and be markedly different from static compliance
-

References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.

Resistance

Explain the relationship between resistance and respiratory gas flow

Describe the factors affecting airway resistance, and its measurement

Explain the physics of gas flow and the significance of the relationship between resistance and flow in the respiratory tract

Resistance (measured in $\text{cmH}_2\text{O} \cdot \text{L}^{-1} \cdot \text{sec}^{-1}$) comprises the energy lost as frictional and inertial impedance to gas flow, where energy is lost as heat. Flow is a function of pressure gradient, resistance, and type of flow.

Types of Flow

Flow can be either laminar or turbulent. In laminar conditions flow is proportional to driving pressure, whilst in turbulent conditions flow is proportional to the square root of driving pressure.

Reynolds' Number

Type of flow can be predicted by **Reynolds's Number**, a dimensionless index:

$$\text{Reynolds' Number} = \frac{2r.d.v}{\eta}$$

Where:

- r = Radius
- d = Gas density
- v = Velocity
- η = Gas viscosity

A Reynolds' Number of <2000 suggests flow is predominantly laminar, whilst >4000 suggests flow is predominantly turbulent.

Laminar Flow

In laminar flow:

- Gas moves in a series of concentric cylinders which slide over one another
 - Gas in the centre moves **twice as fast** compared to the outside, where it is almost stationary
- Gas appears in cross-section as an **advancing cone**
Gas may reach the end of the tube when the volume of flow is less than the volume of the tube.
 - This is the mechanism of alveolar ventilation when tidal volumes are less than anatomical dead space volume

In a straight unbranched tube, flow can be quantified by the **Hagen - Poiseuille Equation**:

$$F = \frac{\Delta P \pi r^4}{8l.\eta}$$

Where:

- F = Flow
- ΔP = Driving pressure
- r = Radius

- l = Length
- η = Viscosity

However, as in laminar conditions flow is proportional to the driving pressure and inversely proportional to resistance, flow can be substituted and the equation solved for resistance:

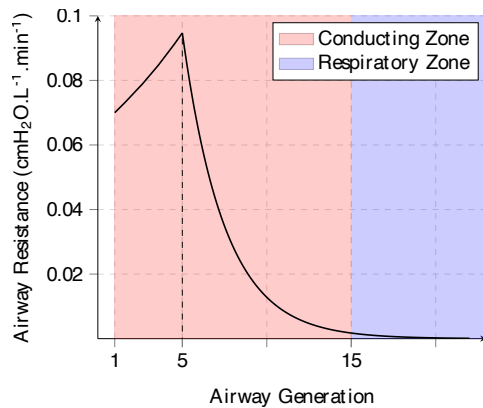
$$Resistance = \frac{8l.\eta}{\pi r^4}$$

This form of the equation can be used to illustrate the **factors affecting resistance**:

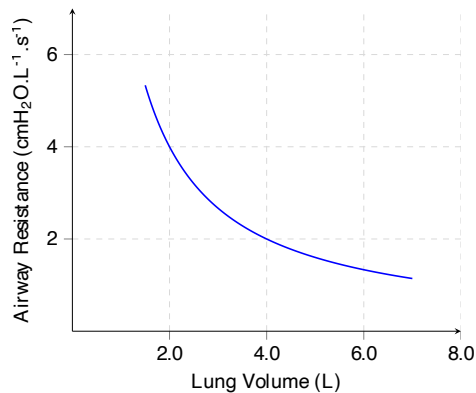
- **Length**
Fixed constant.
- **Viscosity**
Varies with the particular gas mixture being used.
- **Radius**
Main determinant. May be divided into:
 - **Extraluminal factors**
 - Compression:
 - Haemorrhage, tumour, dynamic hyperinflation, atelectasis compressing airways, etc.
 - Lung volume:
 - Airway radius increases when lung volume expands due to radial traction on airways (until dynamic hyperinflation occurs, at which point airways are compressed again)
 - **Luminal constriction**
Bronchospasm, bronchoconstriction.
 - **Intraluminal obstruction**
Sputum plugging, aspiration.

Note that airway resistance:

- **Peaks** at the 5th generation
- Rapidly **decreases** with each airway division thereafter
This is due to the total cross-sectional area increasing dramatically.



- Reduces with increasing lung volume, as radial tension distends airways, increasing their cross sectional area



Turbulent Flow

High flow rates and branching of airways disrupt disciplined laminar flow. Turbulent flow is:

- Dominant in the upper airway (where velocity is high)
- Dominant in early-generation airways due to regular branching, changes in diameter, and sharp angles
- Reduces after the **11th generation** bronchioles
- Proportional to the square root of the driving pressure
Therefore, **resistance is higher in turbulent flow** than in laminar flow.
 - Driving pressure is proportional to gas density, and independent of viscosity

Resistance in turbulent flow is managed by making flow less turbulent:

- Achieved by reducing Reynolds number
 - Helium mixtures reduce gas density
 - Of greater benefit in upper airway than lower airway disease.

Transitional Flow

Transitional flow is the change between laminar and turbulent flow. It occurs at branches and angles in the airways, seen in most of the bronchial tree.

References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.

Surfactant

Describe the properties, production and regulation of, surfactant and relate these to its role in influencing respiratory mechanics

Surface Tension

- Surface tension describes the tendency of a fluid to minimise its surface area
- It is related to the attraction between particles in the fluid relative to particles outside the fluid
- Surface tension is why:
 - Water scattered on a surface forms rounded droplets
 - Why multiple droplets will tend to coalesce into a single larger droplet
- This relationship is described by **Laplace's Law**
 $P = \frac{2T\Theta}{r}$, where:
 - P is pressure
 - T is wall thickness
 - Θ is surface tension
 - r is radius
- Alveoli obey Laplace's Law
- High surface tension causes three problems with alveoli
 - Compliance falls when the alveolus is empty
 As the radius falls, the pressure required to open

- it (at a given surface tension) will be increased. This increases work of breathing.
 - Smaller alveoli will preferentially empty into bigger alveoli
 - Smaller alveoli require greater transmural pressures to remain inflated. This causes smaller alveoli to empty into larger ones.
 - Fluid transudation
 - Surface tension draws fluid from interstitial spaces and contributes to pulmonary oedema.
- Overall, high surface tension is detrimental to the lungs

Surfactant

- Surfactant is a substance which substantially reduces work of breathing by reducing alveolar surface tension
 - Surfactant is produced by type II alveolar cells in response to lung inflation and respiration
 - It is composed of:
 - 85% phospholipid
 - 5% neutral lipid
 - 10% protein
 - Surfactant is **amphipathic**
 - Each component has a hydrophobic and hydrophilic end.
 - This causes the molecules to orient themselves along the air-liquid interface, disrupting the attractive bonds between water molecules
 - Surface tension is reduced in proportion to the concentration of molecules
 - The concentration of surfactant changes throughout the respiratory cycle
 - During expiration alveoli collapse
 - The decrease in alveolar radius is offset by the increase in surfactant concentration, so the fall in radius is mitigated by the drop in surface tension.
-

References

1. CCM September/November 2012
2. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.

Volumes and Capacities

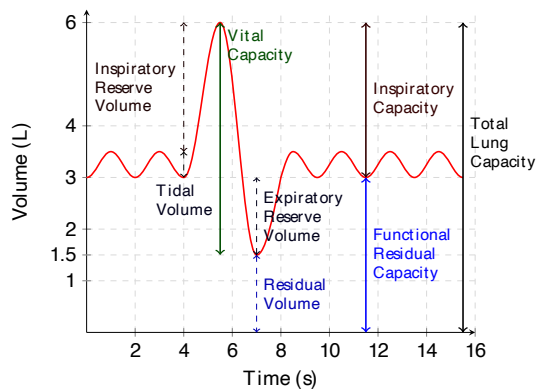
Explain the measurement of lung volumes and capacities, and factors that influence them

State the normal values of lung volumes and capacities

Describe closing capacity and its relationship to airway closure and explain its clinical significance and measurement

The lung has four volumes and four (main) capacities:

- A **volume** is measured directly
- A **capacity** is a **sum of volumes**



Volumes

- **Tidal volume** (V_T)
Volume of air during normal, quiet breathing.
 - Normal is **7ml.kg⁻¹**, or 500ml
- **Inspiratory reserve volume** (IRV)
Volume of air that can be inspired **above tidal volume**.
 - Normal is 45ml.kg⁻¹, or **2500ml**
- **Expiratory reserve volume** (ERV)
Volume of air that can be expired following tidal expiration.
 - Normal is 15ml.kg⁻¹, or **1500ml**
- **Residual volume** (RV)
Volume of air in the lungs following a maximal expiration.
 - Normal is 15-20ml.kg⁻¹, or **1500ml**

Capacities

- **Functional Residual Capacity** (FRC)
FRC = RV + ERV.
 - Normal is **30ml.kg⁻¹** or 3000ml
 - FRC decreases 20% when supine, and a further 20% under general anaesthesia
- **Vital Capacity** (VC)
VC = ERV + V_T + IRV.
 - Normal is 4500ml
- **Inspiratory Capacity** (IC)
IC = V_T + IRV.
 - Normal is 3000ml
- **Total Lung Capacity** (TLC)
TLC = RV + ERV + V_T + IRV.
 - Normal is 6000ml

Functional Residual Capacity

The FRC has many important physiological functions:

- **Gas exchange**

The FRC allows blood in the pulmonary circulation to become oxygenated throughout the respiratory cycle (if there was no FRC, then at expiration the lungs would be empty and no oxygenation would occur).

- **Oxygen Reserve**

FRC is the only clinically modifiable oxygen store in the body, and allows continual oxygenation of blood during apneic periods.

- **Minimise Work of Breathing**

Work of breathing is a function of lung resistance and compliance.

- The lung sits on the **steepest** part of the **compliance occurs** at FRC

Compliance is optimised as:

- Alveoli are open and minimally distended
 - Below FRC, some alveoli collapse and the volume of lung available to receive the tidal volume decreases
Re-expansion of collapsed alveoli requires more work than expanding open alveoli.
 - Above FRC, some alveoli will become overdistended and their compliance will fall

- **Airway resistance decreases** as airway radius increases as lung volume increases

- **Minimise RV Afterload**

PVR is minimal at FRC.

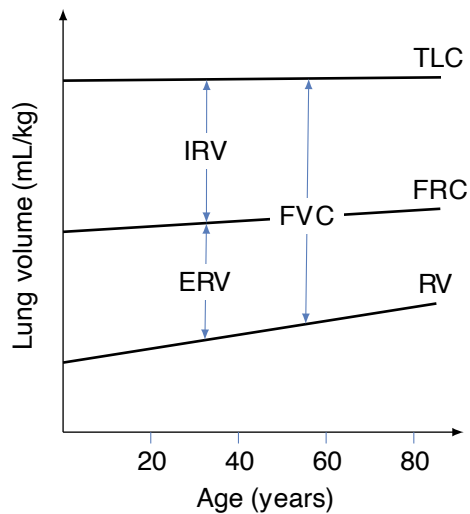
- Above FRC, compression of intra-alveolar vessels occurs and PVR increases
- Below FRC, extra-alveolar vessels collapse and PVR increases

- **Maintain lung volume above closing capacity**

If closing capacity (see below) exceeds FRC, then shunt will occur.

Factors affecting FRC:

- FRC is reduced by:
 - Supine positioning
Falls by ~20%.
 - Anaesthesia
Falls by ~20%.
 - Raised intra-abdominal pressure
 - Impaired lung and chest wall compliance
- FRC is increased by:
 - PEEP
 - Extrinsic
 - Intrinsic (gas trapping)
 - PEEP
 - Emphysema
 - Acute asthma
 - Age
May increase slightly.



Measurement of Lung Volumes and Capacities

- ERV, V_T , and IRV can all be measured directly using spirometry

- A **spirometer** is a flow meter
 - The patient exhales as fast as possible through the flow meter
 - A flow-time curve is produced
 - This curve can be integrated to find volume
- Any capacity which is a sum of these (IC, VC) can therefore be calculated
- RV **cannot** be measured by spirometry, as it can't be exhaled
 - Therefore FRC and TLC cannot be calculated
- RV can be measured using:
 - Gas dilution
 - Body plethysmography

Gas Dilution

- Gas dilution relies on two principles:
 - Conservation of Mass
 - Helium has poor solubility and will not diffuse into circulation
- **Limitations** of gas dilution:
 - Only gas communicating gas can be measured - will underestimate FRC in gas-trapping
- Method:
 - Patient takes several breaths from a gas mixture containing a known concentration of helium (giving time for equilibration)
 - The concentration of expired helium is then measured

From the law of conservation of mass:

$$C_1 V_1 = C_2 V_2$$

- V_2 is equal to the volume of the gas mixture the patient was breathing from (V_1) and the patient's FRC
- Therefore:

$$C_1 V_1 = C_2 (V_1 + FRC)$$

$$FRC = V_1 \frac{C_1 - C_2}{C_2}$$

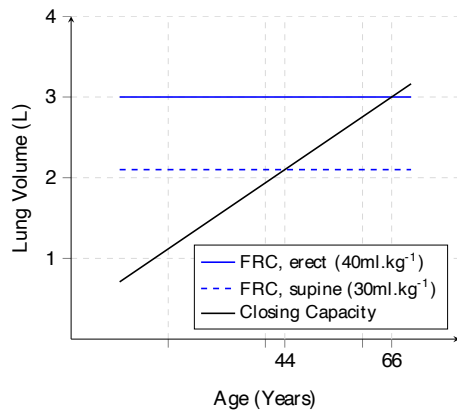
Body Plethysmography

- Body plethysmography relies on:
 - Boyle's law
Pressure and volume are inversely proportional at a constant temperature, i.e. ($P \times V = k$).
- Method:
 - Patient is placed in a closed box, with a mouth-piece that exits the box
 - The patient inhales against a closed mouthpiece:
 - When the patient inhales, the volume of gas in the box decreases (the patient takes up more space) and therefore the pressure increases
 - The change in volume of the box is given by:
 $P_1 V_1 = P_2 V_2$, where:
 - V_2 is the change in box volume, or $V_1 - \Delta V$
 - Therefore:
 $P_1 V_1 = P_2 (V_1 - \Delta V)$
As ΔV is the only unknown value, it can be calculated.
 - The change in volume of the lung must be the same as the volume of the box (ΔV)
 - In the case of the lung, the initial volume (V_1) is FRC
 - Therefore:
 $P_1 \times FRC = P_2 (FRC + \Delta V)$
 $FRC = \frac{P_2 \Delta V}{P_3 - P_4}$

Closing Capacity

- Closing capacity is **lung volume at which small airways begin to close**
Closing capacity is the sum of residual volume and closing volume.
 - Because dependent lung is compressed by gravity, dependent (typically basal) airways are of smaller calibre than non-dependent (typically apical) airways

- During expiration, these airways are compressed first
Alveoli connected to these airways are isolated, and **V/Q scatter or shunt** occurs.
- If **closing capacity exceeds FRC**, then airway closure occurs during normal tidal breathing
This occurs when:
 - FRC is decreased
 - CC is increased
- Increases with age
 - CC exceeds FRC in the supine patient at 44
 - CC exceeds FRC in the erect patient at 66
- This is clinically relevant during **preoxygenation**, as it will limit the denitrogenation that can occur



Measurement of Closing Capacity

Closing capacity is measured using Fowler's Method, and is covered under Dead Space.

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Spirometry

Describe the pressure and flow - volume relationships of the lung, chest wall and the total respiratory system

Describe the measurement and interpretation of pulmonary function tests, including diffusion capacity.

Pulmonary function tests are performed with a **spirometer**, which measures either volume or flow (integrated for time) to quantify lung function.

Basic spirometry can be used to quantify:

- Lung volumes and capacities
All except residual volume (and therefore FRC and TLC).
- Dynamic measurements
 - FEV₁
Volume of air forcibly exhaled in one second.
 - FVC
Forced vital capacity.
 - PEFR
Peak expiratory flow rate.
 - Flow - volume loop

Additional testing can be performed to measure:

- Residual volume
FRC and TLC can therefore be calculated.
- Diffusion capacity

Basic Spirometry

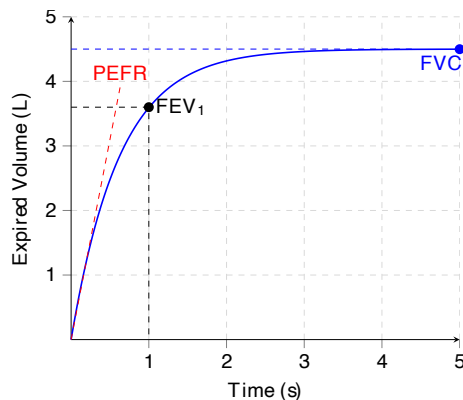
Basic spirometry includes:

- Forced spirometry

Patient forcibly exhales a vital capacity breath, producing an exponential (wash-in) curve. This calculates:

 - PEFR from the gradient at time 0 (assuming maximal effort)
 - FEV_1 is the volume expired in 1s
Normal is $> 80\%$ of predicted.
 - FVC is the total volume exhaled.
 - The FEV_1/FVC ratio
Normal is > 0.7 .
 - These values also quantify disease severity:
 - In obstructive airways disease:
 - $FEV_1 < 80\%$ predicted
 - FEV_1/FVC ratio < 0.7
 - Restrictive disease:
 - $FEV_1 < 80\%$ predicted
 - FVC $< 80\%$ predicted
 - FEV/FVC ratio > 0.7

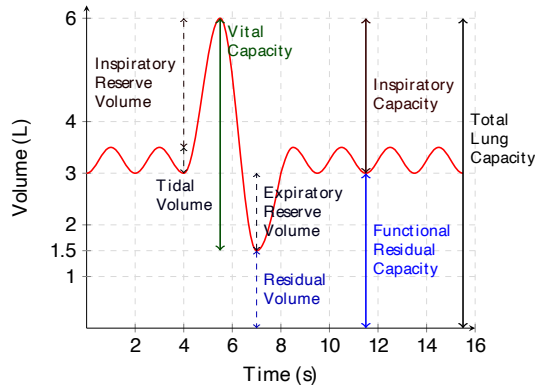
The ratio is normal as the FEV_1 and FVC fall proportionally.



- Volume - Time Graph (also known as a spirograph or spiogram)

Quantifies static lung volumes by having a patient perform:

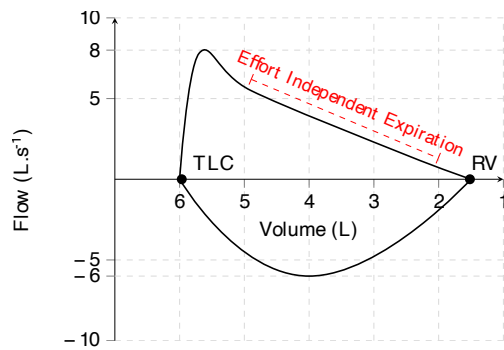
- Normal tidal breathing
- Vital capacity breath
- Vital capacity exhalation



Flow-Volume Loops

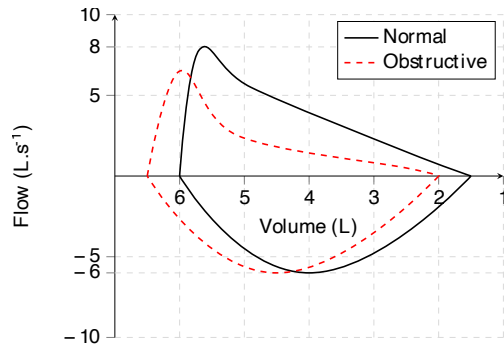
Normal:

- Peak expiratory flow of $\sim 8 \text{ L} \cdot \text{s}^{-1}$
Initial flow is highest as the increased lung volume **increases the calibre** of lung airways, reducing airways resistance.
 - This is called the **effort dependent** part of the curve
- Flow tails off later in expiration
 - Lungs collapse, and airway calibre falls
 - Small airways are compressed
Any increase in expiratory pressure will increase airway resistance proportionally.
 - This is called **dynamic airways compression**, and results in a uniform flow rate that is **independent of expiratory effort**
This is therefore labeled the **effort independent** part of the curve.



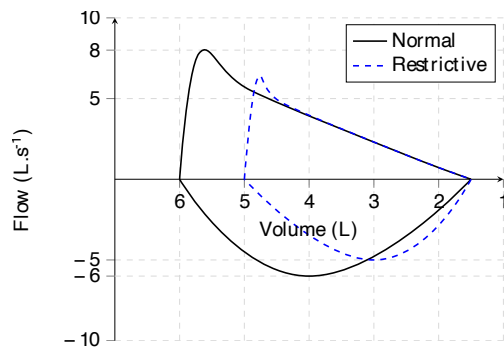
Obstructive lung disease:

- RV and TLC are increased due to gas trapping
- Peak flow is limited
- Effort-independent portion becomes concave



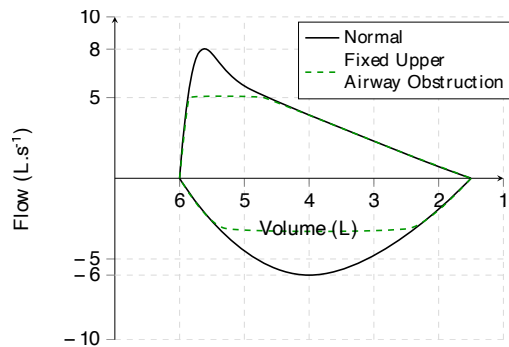
Restrictive lung disease:

- TLC is reduced, but residual volume is unchanged
- Peak flow may be reduced (as seen here)
However, this reduction is **proportional** to the decrease in volume, such that the FEV₁:FVC ratio is normal. If peak flow is preserved, the FEV₁:FVC ratio will be increased.
- Effort independent part is linear



Fixed upper airway obstruction:

- Describes an upper airway obstruction that does not change calibre during the respiratory cycle.
- Peak inspiratory and expiratory flow rates are limited by the stenosis

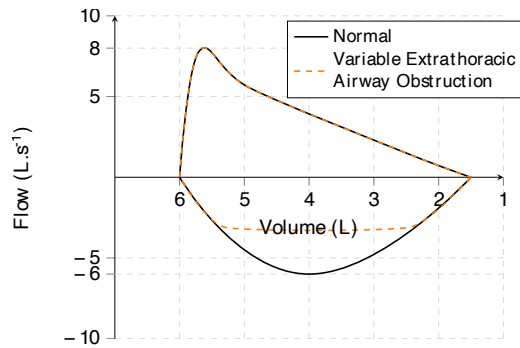


Variable *extrathoracic* obstruction:

- Variable as the obstruction changes during the respiratory cycle:
 - During (negative pressure) inspiration the lesion is pulled into trachea, reducing inspiratory flow
 - During expiration the lesion is pushed out of the trachea

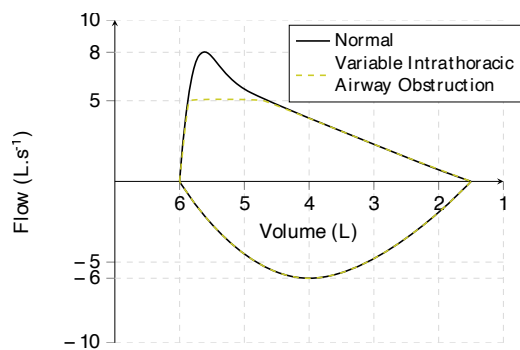
The way to remember this is an **extrathoracic** obstruction impedes **inspiration**

- The reverse effect occurs in positive pressure ventilation



Variable *intrathoracic* obstruction:

- The opposite to extrathoracic obstruction:
 - During inspiration the airway calibre increases and inspiratory flow is unimpeded
 - During expiration the airway calibre falls and expiratory flow is reduced



References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Work of Breathing

Describe the work of breathing and its components

Work of breathing is the energy used by the muscles for respiration. It is defined as: $Work = Pressure \times Volume$:

- This gives the work for a single respiratory cycle
Energy expenditure over time is better described as the "power of breathing", as $Power = \frac{Work}{Time}$ or **rate of work**, and is measured in Watts.
- It does not take into account **respiratory rate** or **flow rate**
These factors have a significant effect on energy requirement
 - Tidal breathing is efficient and uses < 2% of BMR
- The oxygen requirement of breathing at rest is ~2-5% of VO_2 , or ~3ml.min⁻¹

Determinants of Work of Breathing

Work of breathing is divided into:

- **Elastic work**
About 65% of total work, and is stored as elastic potential energy.
 - Elastic work describes the energy required to overcome elastic forces:
 - Lung elastic recoil

The derivation of $W = PV$ is as follows.

Work is classically defined as:

$$W = FD$$

Where:

- W is Work in Joules
- F is Force in Newtons
- D is Distance in Metres
- V is Volume in Litres

Pressure is defined as:

$$P = \frac{F}{A}$$

Where:

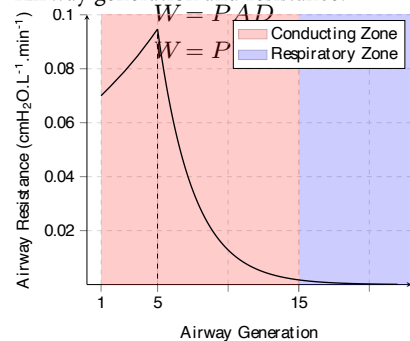
- P = Pressure in Pascals
- A = Area in metres squared

Rearranging:

$$F = PA$$

Substituting:

Airway generation and resistance:

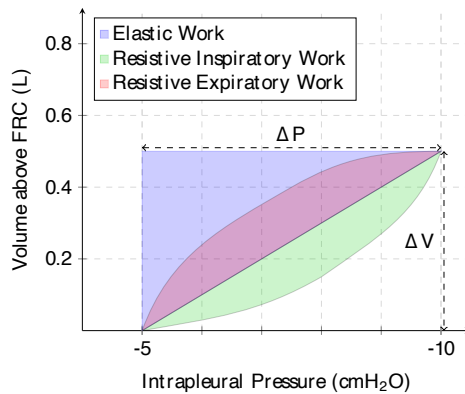


- Surface tension of alveoli
- Stored as elastic potential energy and reclaimed during passive exhalation
- **Resistive work**

About 35% of total work, and is lost as heat. This is due to the energy required to overcome frictional forces:
- Between tissues
 - Increased with increased interstitial lung tissue
- Between gas molecules
 - Increased at high flow rates
 - Increased with turbulent flow
 - High respiratory rates
 - Upper airway obstruction
 - Increased airway density
 - Hyperbaric
 - Diving
- Increased with decreased airway radius
 - Low lung volume
 - Inadequate PEEP
 - Decreased respiratory muscle tone
 - Bronchoconstriction
 - Dynamic airway compression
Effort-independent expiration.
 - Apparatus
 - Endotracheal Tube
 - HME filters
 - Airway division
 - Resistance peaks at the 3rd airway division (lobar bronchi)
 - Falls with increasing airway divisions due to increased cross-sectional area

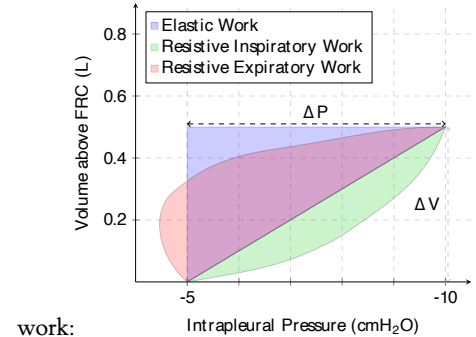
Graphing Work of Breathing

Work of breathing can be evaluated with a dynamic lung compliance curve:



- If there **were no resistive forces**, then this curve would be a straight line
 - The triangular area is the elastic work done
- The resistive work of breathing causes the deviation of the inspiratory and expiratory lines:
 - The area between the compliance line and the inspiratory line is additional **resistive inspiratory** work done
 - The area between the compliance line and expiratory line is additional **resistive expiratory** work done
 - This work is typically done by elastic recoil of the lungs
 - If this area falls within the area of elastic work of breathing, it is a purely passive process, using the stored elastic potential energy of inspiration
 - If part of this area falls outside the area of elastic work of breathing, it demonstrates additional **active work of expiration** which may occur in obstructive lung disease or when minute ventilation is high

Active expiratory



Minimising Work of Breathing

Work of breathing can be minimised by optimising the determinants:

- Elastic work
 - PEEP
Keep lung volume at FRC and maximise number of ventilated alveoli.
 - Positioning
Optimise lung volume.
 - Surfactant
Minimising surface tension.
 - Optimise respiratory rate
Elastic work of breathing typically decreases with increased respiratory rate.
 - Resistive work
 - Decrease respiratory rate
Respiratory rate is directly proportional to resistive work.
 - Increase laminar flow
Laminar flow is more efficient than turbulent flow.
Laminar flow can be increased by:
 - Reducing gas density
Heliox.
 - Increase Radius
 - Increase lung volume
 - Bronchodilators
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.

Diffusion of Gases

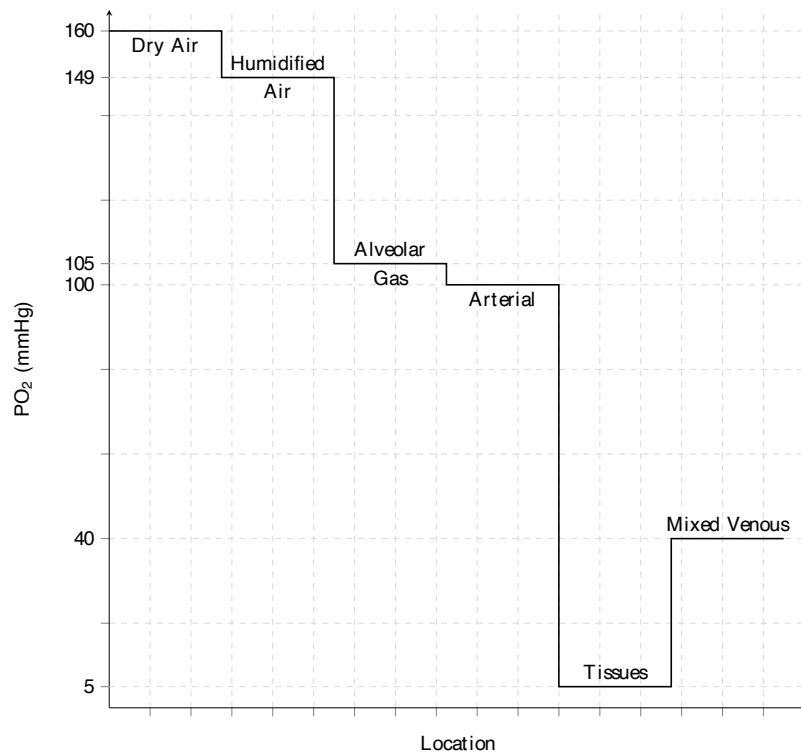
Oxygen Cascade

Describe and explain the oxygen cascade

The **oxygen cascade** describes the **passive transfer** of oxygen down a series of **partial pressure gradient** from air to mitochondria:

- Dry atmospheric gas
- Humidified tracheal gas
- Alveolar gas
- Arterial blood
- Mitochondria
- Venous blood

Partial pressure is the determinant of rate and extent of gas transfer, but **perfusion** depends on **oxygen content**



Atmospheric Gas

Atmospheric partial pressure of oxygen is a function of barometric pressure and the FiO_2 :

- $PO_2 = P_B \times FiO_2$, where:
 - P_B is 760mmHg
 - FiO_2 is 0.21
- Therefore, atmospheric PO_2 at sea level is **160mmHg**

Humidified Tracheal Gas

- Gas is humidified during inspiration
- Gas in the proximal trachea is **heated to 37°C** and has **100% relative humidity**

- The **saturated vapour pressure** of water at 37°C is **47mmHg**
- Therefore:
 - $PO_2 = (P_B - P_{SVP \text{ of Water}}) \times FiO_2$, where:
 - P_B and FiO_2 are as above
 - PO_2 is **149mmHg**

Alveolar Gas

Ideal alveolar PO_2 is calculated using the **alveolar gas equation**:

$$P_AO_2 = P_iO_2 - \frac{P_aCO_2}{R} + F$$

Where:

- P_AO_2 is the alveolar partial pressure of oxygen
- P_iO_2 is the inspired partial pressure of oxygen
- P_aCO_2 is the arterial partial pressure of carbon dioxide
- R is the respiratory quotient, where $R = \frac{\text{Volume of } CO_2 \text{ produced}}{\text{Volume of } O_2 \text{ consumed}}$
 - R is used in the alveolar gas equation to correct for the change in inspired relative to expired volume

As generally less CO_2 is produced than O_2 consumed, expired volumes are typically less than inspired volumes
 - R is dependent on the metabolic substrates used for metabolism:
 - Pure fat ≈ 0.7
 - Pure protein ≈ 0.9
 - Pure carbohydrate ≈ 1
 - The normal value for a Western diet is quoted as **0.8**
 - F is a correction factor, usually equal to $\sim 2\text{mmHg}$, and is given by $F = P_aCO_2 \cdot FiO_2 \cdot \frac{1-R}{R}$

Alveolar oxygen is therefore dependent on:

- PiO_2 , which is a function of:

- FiO_2
- Air pressure
- Alveolar ventilation

Essentially, $P_a O_2 \propto \frac{k}{V_A}$.

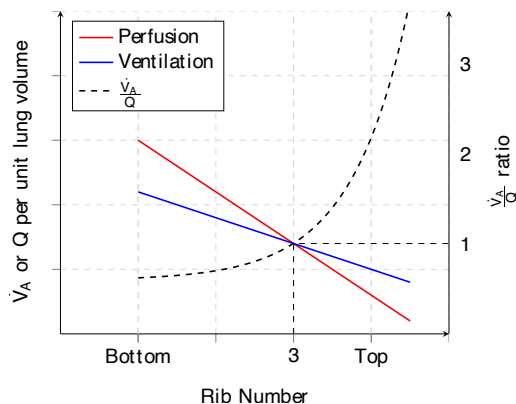
k here is abstracting out the CO_2 production, unit conversions, and the respiratory quotient. This also assumes steady state, and no A-a gradient.

Arterial Blood

The difference in partial pressure of oxygen between alveolar and arterial blood is called the **A-a gradient**:

$$A-a \text{ gradient} = PAO_2 - PaO_2 = PiO_2 - \frac{PaCO_2}{R} - PaO_2$$

- The normal A-a gradient is $\frac{Age}{4} + 4$
Therefore, the normal arterial PO_2 in a Standard Guyton Man is 100mmHg.
- The drop in PO_2 is due to:
 - Shunt/VQ scatter
A physiological right-to-left shunt is normal due to blood from the bronchial circulation and thebesian veins draining into the left heart.
 - Physiological V/Q mismatch



Mitochondria

- PO_2 varies with metabolic activity, but typically quoted as 5mmHg
- The **Pasteur point** is the partial pressure of oxygen at which oxidative phosphorylation ceases, and is ~1mmHg

Venous Blood

- PO_2 is greater than mitochondrial PO_2
Mixed venous blood typically quoted as 40mmHg.
 - Higher than mitochondria as not all arterial blood travels through capillary beds
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. Brandis K. The Physiology Viva: Questions & Answers. 2003.

Diffusion and Diffusing Capacity

Explain perfusion-limited and diffusion-limited transfer of gases

Define diffusing capacity and its measurement

Describe the physiological factors that alter diffusing capacity

The rate of diffusion of gases is given by **Fick's Law**:

$$\text{Rate of Diffusion} = \frac{\Delta P \times A \times s}{T \times \sqrt{MW}}$$

Where:

- ΔP is the pressure gradient across the membrane
- A is the area of the membrane
- s is the solubility of the substance
- T is the thickness of the membrane
- MW is the molecular weight of the substance

These can be grouped into:

- Pressure gradient
- Lung factors
 - Surface area
 - Thickness
- Substance factors

- Molecular weight
- Solubility

Pressure Gradient

In the lung, this is a function of:

- Partial pressure of the gas in the alveolus
This is affected by:
 - Atmospheric pressure
 - Ventilation
Alveolar hypoventilation will:
 - Increase PACO_2
 - Decrease PAO_2
- Partial pressure of the gas in blood
This is affected by:
- Solubility of the gas in blood
 CO_2 is ~20 times as soluble as O_2 in blood.
- Binding of gas to protein:
 - Particularly **haemoglobin**
Affects the rate of uptake of O_2 and CO , and is why calculated DL_{CO} is corrected for haemoglobin.
 - The shape of the oxy-haemoglobin dissociation curve allows a large volume of oxygen to be bound before PaO_2 begins to rise substantially.
 - Formation of carbamino compounds
 - Anaesthetic agents to plasma contents
e.g. albumin, cholesterol.

Lung Factors

The effective alveolar-capillary surface area is affected by:

- Parenchyma volume
 - Body size
 - Pathology
 - Many lung diseases will reduce surface area for gas exchange.
- V/Q mismatch
 - Both shunt and dead space reduce the surface area available for gas exchange.
- Pulmonary blood volume
 - Vascular distension and recruitment also affects surface area. Factors affecting pulmonary blood volume include:
 - Cardiac output
 - Increased recruitment of vasculature in high output states
 - Decreased recruitment and increased V/Q mismatch in shock states.
 - Posture
 - Increased surface area when supine relative to sitting or standing.

Increasing alveolar - capillary membrane thickness impedes gas exchange. Causes of this include:

- Pathology
 - e.g. Pulmonary oedema, interstitial lung disease.

Substance Factors

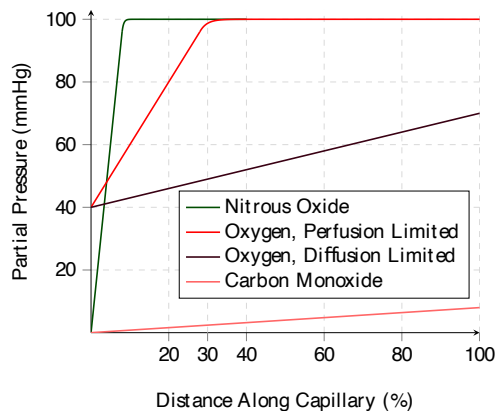
Include:

- Solubility
 - More soluble substances will diffuse more quickly.
- Molecular weight
 - Smaller substances will diffuse more quickly.

Diffusion and Perfusion Limitation

Limitation refers to what process limits gas uptake into blood:

- Gases which are **diffusion limited** fail to equilibrate, i.e. the partial pressure of a substance in the alveolus does not equal that in the pulmonary capillary
 - e.g. Carbon Monoxide
- Gases which are **perfusion limited** have equal alveolar and pulmonary capillary partial pressures, so the amount of gas *content* transferred is dependent on blood flow
 - e.g. Oxygen



Oxygen

- Oxygen diffusion takes **~0.25s**
- Pulmonary capillary transit time is **0.75s**
- Therefore, under normal conditions oxygen is a **perfusion limited** gas
- However, **oxygen may become diffusion limited** in certain circumstances:
 - **Alveolar-capillary barrier** disease
 - Decreases the rate of diffusion.
 - Decreased surface area
 - Increased thickness

- **High cardiac output**
Decreases pulmonary transit time.
- **Altitude**
Decreases PAO_2 .
- Reduced diffusion capacity leads to Type 1 respiratory failure as oxygen is affected to a greater extent than carbon dioxide

Carbon Dioxide

- Carbon dioxide is an exception to these categories and limited by *ventilation*, rather than diffusion or perfusion.
- This is because CO_2 is constantly being produced by the body, and needs to be *removed* - it therefore moves in the opposite direction to the other gases.
- There is a large amount of carbon dioxide in venous blood, present in various forms:
 - Dissolved in plasma (CO_2 is 20x more soluble in blood than oxygen)
 - Bicarbonate ions (part of the CO_2 and pH buffer system)
 - Carbamino compounds (largely bound to Hb for carriage to the alveolus)
- This means that although CO_2 readily diffuses into the alveolus, the partial pressure in the blood does not change because it is constantly being replenished both from the above stores, and ongoing production by cellular metabolism.
- If equilibrium is reached across the alveolar-capillary membrane, CO_2 transfer will stop *regardless* of speed of diffusion or ongoing perfusion.
- Therefore, the only way to ensure ongoing removal of CO_2 from the blood is to clear it from the alveolus i.e. by maintaining alveolar ventilation.

Other Gases

- **Carbon monoxide**

Diffusion limited due to:

- High affinity for haemoglobin
Continual uptake into Hb results in a low partial pressures in blood.

- **Nitrous oxide** Perfusion limited as equilibrium between alveolus and blood is rapidly reached as it is:

- Not bound to haemoglobin
- Relatively insoluble

Diffusing Capacity

- Measurement of the ability of the lung to transfer gases
- Measured as DL_{CO} or diffusing capacity of the lung for carbon monoxide
Carbon monoxide is used as it is a diffusion limited gas.
- Process:
 - Vital capacity breath of 0.3% CO
 - Held for 10s and exhaled
 - Inspired and expired CO are measured
 - Difference is the amount of CO which is now bound to Hb
 - DL_{CO} is corrected for:
 - Age
 - Sex
 - Hb
- DL_{CO} is decreased in:
 - Thickened alveolar - capillary barrier
 - Interstitial lung disease
 - Reduced surface area
 - Emphysema
 - PE
 - Lobectomy/pneumonectomy

- DL_{CO} is increased in:
 - Exercise
Recruitment and capillary distension.
 - Alveolar haemorrhage
Hb present within the lung binds CO.
 - Asthma (may be normal)
Potentially due to increased apical blood flow.
 - Obesity (may be normal)
Potentially due to increased cardiac output.
-

References

1. Brandis K. The Physiology Viva: Questions & Answers. 2003.
2. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.
3. ANZCA March/April 1999
4. Yartsev, A. Carbon Dioxide Storage and Transport.

V/Q Relationships

Basics of V/Q Matching

Optimal gas exchange occurs when regions of lung are ventilated in proportion to their perfusion, i.e. $V/Q = 1$.

Uneven distribution of ventilation and perfusion causes inefficient gas exchange:

- Excessive ventilation causes excessive work
- Inadequate ventilation causes inadequate gas exchange

Distribution of Ventilation

- The right lung is slightly better ventilated than the left
- In an erect patient the bases of the lung are better ventilated

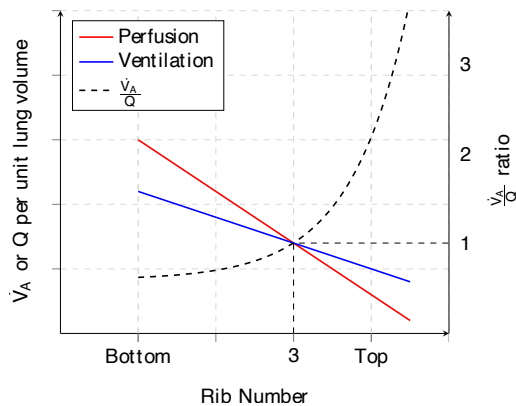
The weight of lung above compresses the lung below, improving the compliance of dependent lung whilst stretching the non-dependent lung.

 - This is only significant at low inspiratory flow rates
 - The V/Q ratio in the bases is ~ 0.6
 - The V/Q ratio in the apices is >3
- In the lateral position:
 - The dependent lung is better ventilated in a spontaneously breathing patient
 - The non-dependent lung is better ventilated in a ventilated patient

Distribution of Perfusion

- The pulmonary circulation is a low pressure circulation
 - Gravity therefore has a substantial effect on fluid pressure
 - Consequently, the distribution of blood throughout the lungs is uneven:
 - The **bases perfused better than the apices**
- This is affected by lung volume, with the effect:
- Becoming more pronounced at TLC (with apical perfusion falling precipitously)
 - Reversing slightly at RV

V/Q Ratios



- The global V/Q ratio for normal resting lung is **0.9**
- The global V/Q ratio improves to 1.0 during exercise

V/Q Mismatch and Etymology

- V/Q mismatch occurs when $V/Q \neq 1$:
 - **$V/Q > 1$** (Dead Space)
- Ventilation in excess of perfusion.
- However, pulmonary blood is passing ventilated alveoli and PaO_2 is normal

- **V/Q 0 to 1** (V/Q scatter)
Perfusion in excess of ventilation.
 - Increasing in PAO_2 will increase PaO_2
 - This is commonly referred to by the general term of V/Q mismatch
 - **V/Q = 0** (Shunt)
Mixed venous blood entering the systemic circulation without being oxygenated via passage through the lungs. PaO_2 falls.
-

References

1. West J. Respiratory Physiology: The Essentials. 9th Edition. Lippincott Williams and Wilkins. 2011.
2. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.

West's Zones

Describe West's zones of the lung and explain the mechanisms responsible for them

West's Zones describe the **changes** to **pulmonary blood flow** and **V/Q matching** that occur due to the **relative difference** in alveolar and pulmonary arterial pressure, and the subsequent **effects** on **pulmonary blood flow** and **V/Q matching**.

The lung is divided into **four zones**:

- **West Zone 1: $P_A > P_a > P_v$**

Alveolar pressure exceeds arterial pressure.

- The alveolus compresses the capillary, and no blood flow occurs
- This occurs when:
 - Alveolar pressure is high
 - PEEP
 - Arterial pressure is low
 - Shock
 - Hypovolaemia

As there is ventilation but no perfusion, **West Zone 1** can also be thought of as dead space.

- **West Zone 2: $P_a > P_A > P_v$**

Arterial pressure exceeds alveolar pressure, which exceeds venous pressure.

- Blood flow occurs **intermittently** during the cardiac cycle
- Alveolar pressure acts as a **Starling resistor**
Flow is proportional to the $P_a - P_A$ gradient.
 - When P_a falls below P_A (e.g. in diastole), then no blood flow will occur

- **West Zone 3: $P_a > P_v > P_A$**

Arterial pressure exceeds venous pressure which exceeds alveolar pressure.

- Blood flow occurs throughout the cardiac cycle
Flow is proportional to the $P_a - P_v$ gradient.
- For an accurate measure of PCWP, a PAC must be placed in West Zone 3 (so there is a continual column of blood)
- This tends to happen naturally as the majority of pulmonary flow is to this region

- **West Zone 4: $P_a > P_i > P_v > P_A$**

Interstitial pressure acts as a Starling resistor for pulmonary blood flow.

- It is seen when interstitial pressure is high (e.g due to pulmonary oedema).
-

References

1. Brandis K. The Physiology Viva: Questions & Answers. 2003.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Dead Space

Define dead space and its components. Explain how these may be measured and describe the physiological impact of increased dead space.

Discuss dead space, its measurement and apply the Bohr equation and alveolar gas equation

Dead space is the proportion of minute ventilation which does not participate in gas exchange.

Types of Dead Space

Dead space can be divided into:

- **Apparatus dead space**

Dead space from equipment, such as tubes ventilator circuitry. Some apparatus dead space may actually *reduce* total dead space, as an ETT bypasses the majority of anatomical dead space of the patient (nasopharynx).

- **Physiological dead space**

Dead space from the patient. Physiological dead space is divided into:

- **Anatomical dead space**

The volume of the conducting zone of the lung. Anatomical dead space is affected by:

Note that West Zone 1 (where $P_A > P_a > P_v$) physiology is increased dead space.

- Size and Age
3.3ml.kg⁻¹ in the infant, falls to 2.2ml.kg⁻¹ in the adult
- Posture
Decreases when supine.
- Position of the neck and jaw
Increased with neck extension.
- Lung volumes
Increases by ~20ml per litre of additional lung volume.
- Airway calibre
Bronchodilation increases airway diameter and therefore V_D.
- **Pathological/Alveolar Dead Space**
Dead space caused by disease. Causes of pathological dead space include:
 - Erect posture
 - Decreased pulmonary artery pressure/impaired pulmonary blood flow
 - Hypovolaemia
 - RV failure/Increased RV afterload:
 - HPV
 - MI
 - PE
 - Increased alveolar pressure
Increases West Zone 1 physiology.
 - PEEP
 - COAD

Calculation of Dead Space

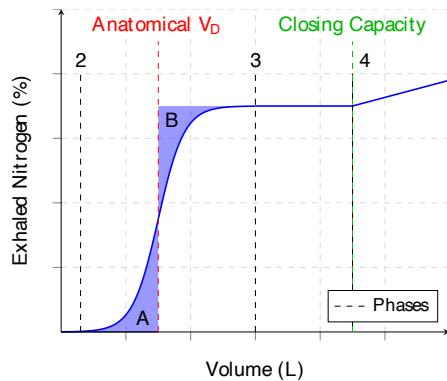
Two methods exist to allow dead space volumes to be calculated:

- Physiological dead space may be measured with **Bohr's method**
- Anatomical dead space may be measured by **Fowler's method**

- Pathological dead space may be calculated by subtracting anatomical dead space (Fowler's method) from physiological dead space (Bohr's Method)

Fowler's Method

Fowler's Method is a **single-breath nitrogen washout test**, used to calculate **anatomical dead space** and **closing volume**.



Method:

- At the end of a normal tidal breath (at FRC) a **vital-capacity breath of 100% oxygen** is taken
- The patient then exhales to RV
Expired nitrogen concentration and volume is measured.
- A plot of expired nitrogen concentration by volume is generated, producing a graph with four phases:
 - Phase 1 (Pure Dead Space)
Gas from the anatomical dead space is expired. This contains 100% oxygen - no nitrogen is present.
 - Phase 2
A mix of anatomical dead space and alveolar (lung units with short time constants) is expired. The midpoint of phase 2 (when area A = area B) is the volume of the **anatomical dead space**.

- Phase 3
Expired nitrogen reaches a plateau as just alveolar gas is exhaled (lung units with variable time constants).
- Phase 4
Sudden increase in nitrogen concentration.
 - Occurs because:
 - Basal alveoli are more compliant than apical alveoli
 - Therefore, during inspiration basal alveoli inflate more than apical alveoli
The single 100% oxygen breath therefore preferentially inflates the basal alveoli. At the end of the vital capacity breath, the oxygen concentration in basal alveoli is greater than that of apical alveoli.
 - In expiration, the process is reversed:
 - Basal alveoli preferentially exhale
 - At closing capacity, small basal airways close and now only apical alveoli (with a higher concentration of nitrogen) can exhale
 - Measured expired nitrogen concentration increases
 - Onset of phase 4 indicates the point at which **closing capacity** is reached
 - The volume exhaled between phase 4 commencing and reaching RV is the **closing volume**
This can be combined with a measurement of RV to quantify the closing capacity.

Bohr's Method

Physiological dead space is measured using the Bohr equation. This calculates dead space as a ratio, or as a proportion of tidal volume:

A **normal** value for physiological dead space during normal tidal breathing is 0.2–0.35

$$\frac{V_D}{V_T} = \frac{V_T - V_A}{V_T}$$

The Bohr equation is based on the principle that **all CO₂ exhaled must come from ventilated alveoli**.

$$\frac{V_D}{V_T} = \frac{PA_{CO_2} - \bar{P}E_{CO_2}}{PA_{CO_2}}$$

The PaCO₂-ETCO₂ difference is a consequence of dead space, as dead space gas dilutes alveolar gas

Note that:

- $\bar{P}E_{CO_2}$ is the mixed - expired carbon dioxide Partial pressure of CO₂ in an expired tidal breath.
- The Bohr equation requires alveolar PCO₂ to be measured
As this is impractical, the **Enghoff modification** is typically used, which assumes that PAO₂ ≈ PaO₂.
The equation then becomes:

$$\frac{V_D}{V_T} = \frac{Pa_{CO_2} - \bar{P}E_{CO_2}}{Pa_{CO_2}}$$

Physiological Consequences of Increased Dead Space

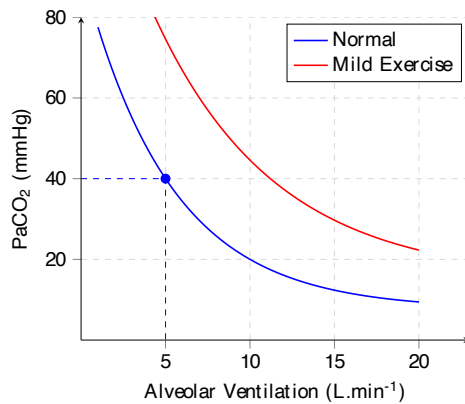
In dead space:

- The V/Q ratio approaches infinity as alveolar **perfusion** falls
- This results in a rise in PaCO₂
- In a spontaneously - ventilating individual, this stimulates the respiratory centre to increase minute ventilation to return alveolar ventilation (and therefore CO₂) to normal
- There is minimal effect on PaO₂, as in pure dead space all blood is passing through ventilated alveoli and therefore undergoes gas exchange

Relationship between Alveolar Ventilation and PaCO_2

Atmospheric air contains negligible CO_2 . As MV increases, PaCO_2 will fall, as will the gradient for further CO_2 diffusion. This can be expressed by the equation:

$$\text{PaCO}_2 \propto \frac{1}{\dot{V}_A}$$



Note that this graph:

- Describes the change in PaCO_2 for a change in alveolar ventilation
A doubling of alveolar ventilation will halve PaCO_2 .
- Does not describe the change in ventilatory drive for a given change in PaCO_2
This is covered under removal of CO_2 .

References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.
2. West J. Respiratory Physiology: The Essentials. 9th Edition. Lippincott Williams and Wilkins. 2011.

Shunt

Explain the concept of shunt, its physiological effects and its measurement.

Describe the shunt equation

Shunt is blood reaching the systemic circulation without being oxygenated via passage through the lungs.

Factors Contributing to Shunt

- Normal shunt
 - Anatomical shunt
 - Thebesian veins, which drain directly into the left cardiac chambers
 - Bronchial circulations, which drain into the pulmonary veins
 - Functional shunt

Blood draining through alveoli with a V/Q between 0 and 1.

 - This may not be true shunt, as blood may have some oxygen content but not be maximally oxygenated
- Pathological shunt

Pathological shunting can be anatomical (e.g congenital cardiac malformations), or physiological (e.g. pneumonia causing alveolar consolidation).

- Intra-cardiac
e.g. VSD
- Extra-cardiac
e.g. Pulmonary AVM, PDA

Calculation of Shunt

- Shunt cannot be directly measured
- This is because we cannot separate true shunt ($V = 0$) from V/Q scatter ($V/Q < 1$) when sampling blood entering the left heart
- **Venous admixture** is used instead
Venous admixture is **the amount of mixed venous blood that must be added to pulmonary end-capillary blood to give the observed arterial oxygen content**. Venous admixture:
 - Is a calculated, theoretical value
 - Assumes that alveoli have either complete shunt (no ventilation at all, i.e. $V/Q = 0$) or no shunt ($V/Q = 1$)
 - Is expressed as a ratio, or **shunt fraction**:

$$\frac{\dot{Q}_S}{\dot{Q}_T} = \frac{C_c O_2 - C_a O_2}{C_c O_2 - C_v O_2}, \text{ where:}$$
 - $\dot{Q}_S$ is Shunt blood flow
 - $\dot{Q}_T$ is Cardiac output
 - $C_c O_2$ is Pulmonary end-capillary oxygen content, assumed to have an oxygen tension equal to PAO_2 (with the corresponding oxygen saturation)
 - $C_a O_2$ is Arterial oxygen content
 - $C_v O_2$ is Mixed venous oxygen content

Physiological Consequences of Shunt

Effect on Carbon Dioxide

- No CO_2 can diffuse from shunted blood
- Therefore PaCO_2 might be expected to rise, however:
 - In a spontaneously breathing patient the increased PaCO_2 increases respiratory drive, and alveolar ventilation increases
 - Therefore, **shunt does not tend to increase PaCO_2** unless:
 - The shunt fraction is large and
 - The patient is unable to increase their alveolar ventilation to compensate
 - Additionally, the steepness of the CO_2 dissociation curve at the arterial point means that although CO_2 *content* increases, the increase in PaCO_2 is small

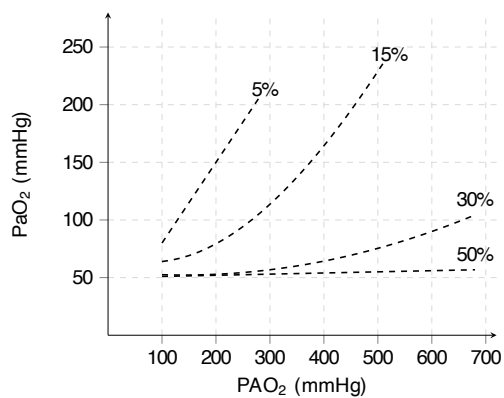
Effect on Oxygen

- **PaO_2 falls proportionally to shunt fraction**
- As shunted alveoli are perfused but not ventilated, **true shunt** is said to be **unresponsive to an increase in FiO_2**
 This is where technical definitions become important to avoid confusion.
 - For an alveolus with a V/Q between 0-1 (V/Q mismatch or V/Q scatter, but not true shunt):
 - There is perfusion, but *relatively less* ventilation
 - Therefore blood passing through this alveolus will be *partially* oxygenated
 - Increasing PAO_2 will improve oxygenation (assuming no diffusion limitation):
 - Administration of supplemental oxygen
 - Hyperventilation

- As per the alveolar gas equation
- For an alveoli with a V/Q of 0 (true shunt)
 - There is no ventilation. Regardless of the increase in PAO_2 , PaO_2 will not improve.

The Isoshunt Diagram

- Isoshunt diagram plots the relationship between FiO_2 and PaO_2 against a set of 'virtual shunt lines'
- These 'shunt fractions' are calculated from the above equation and so are actually V/Q admixture fractions



References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.
2. West J. Respiratory Physiology: The Essentials. 9th Edition. Lippincott Williams and Wilkins. 2011.
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Gas Transport

Oxygen Storage

Describe the oxygen and ~~carbon dioxide~~ stores in the body

The Standard Guyton Man (70kg) contains **~1.5L** of oxygen, split between:

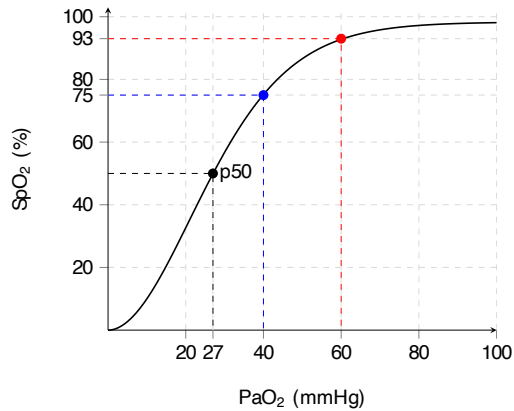
- ~850ml in blood
There is 20.4ml of oxygen per 100ml of blood, divided up as: * 20.1ml bound to haemoglobin * 0.3ml dissolved
- ~250ml bound to myoglobin
- ~450ml contained in FRC (21% of 2.4L)
This is why preoxygenation increases safe apnoea times, as the nitrogen washout increases the volume of oxygen stored.

Oxygen-Haemoglobin Dissociation Curve

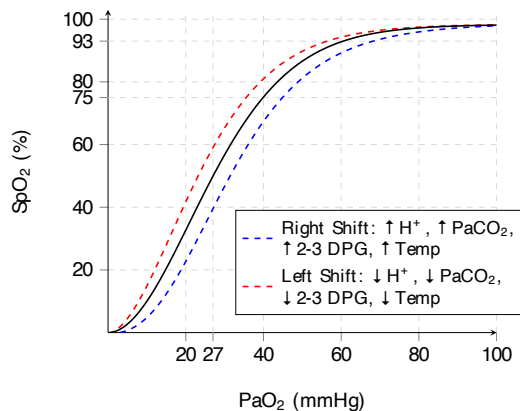
The sigmoid shape of the oxygen-haemoglobin dissociation curve offers many physiological advantages:

- Buffering in case of low PaO_2
The plateau allows oxygen **content** to remain high, even if the PaO_2 falls
- Maintenance of diffusion gradient to tissues
The steep section allows a large amount of oxygen to be delivered with only a small drop in PaO_2 , which

allows the rate of oxygen delivery to be maintained (as the blood-tissue partial pressure gradient is steep) with an increase in oxygen demand.



- The sigmoid shape exists due to **cooperative binding**. Each oxygen which binds to Hb causes conformational changes which allow it bind additional oxygen molecules more easily.
 - When the fourth oxygen molecule has bound, Hb is said to be in the **relaxed conformation** (R state)
 - When no oxygen is bound, Hb is said to be in the **tense state** (T state)
- The curve can be right or left-shifted by changes in temperature, pH, CO₂, and 2-3 DPG

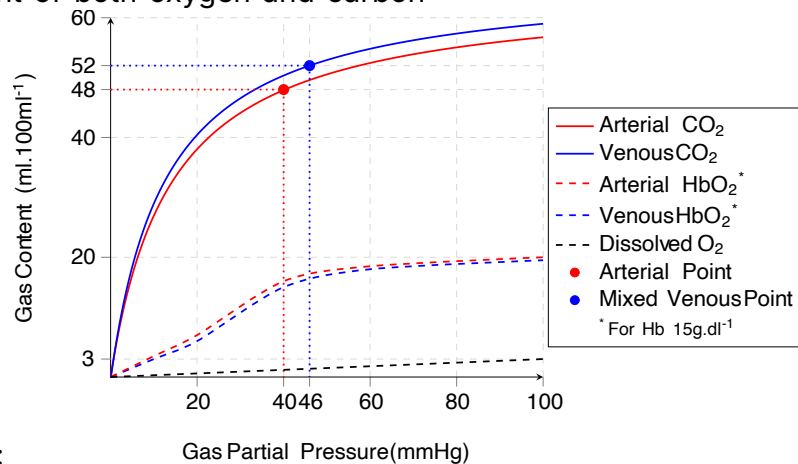


- Note that the **mixed venous point is not on the arterial curve** (unlike how it is displayed above), as the

venous dissociation curve is right-shifted relative to the arterial curve

Comparison Curve

Comparison of gas content of both oxygen and carbon



dioxide per 100mL of blood:

Haemoglobin Species

Haemoglobin is a four-tetramer molecule, and its species can be physiological or pathological:

- **Physiological**
 - **HbA**
 - Most common
 - 2 alpha and 2 beta subunits ($\alpha_2\beta_2$)
 - **HbA₂**
 - Less common
 - 2 alpha and 2 delta subunits ($\alpha_2\delta_2$)
 - **HbF**
 - Foetal Hb
 - Higher affinity for oxygen due to lack of 2,3-DPG
 - 2 alpha and 2 gamma subunits ($\alpha_2\gamma_2$)
- **Pathological**

◦ **HbS**

Sickle-cell disease.

- Abnormal beta subunit
- Unable to deform as they pass through capillaries
 - Increases blood viscosity, thrombus, and ischaemia through capillary occlusion
 - Often causes splenic infarction
 - Reduced red cell lifespan to 10-20 days

◦ **MetHb**

Methaemoglobinaemia.

- Ferrous iron (Fe^{2+}) is oxidised to ferric iron (Fe^{3+})
- Cannot bind oxygen, and left-shifts the oxyHb curve for normal Hb which reduces oxygen offloading at tissues
- Normally prevented by:
 - Glutathione in red cell reduces oxidising agents
 - Methaemoglobin reductase enzyme uses NADH to reduce MetHb
- Occurs due to:
 - Oxidising agents overwhelm capacity of glutathione system, e.g.:
 - SNP
 - NO
 - Amide local anaesthetics
 - Sulfonamides
 - Failure of the methaemoglobinaemia reductase enzyme:
 - G6PD

◦ **COHb**

Carboxyhaemoglobin.

- Haemoglobin binds carbon monoxide with greater affinity than oxygen

◦ **CyanoHb**

- Haemoglobin **irreversibly** binds cyanide molecules, causing a **functional anaemia**
- Cyanide inhibits cytochrome oxidase in the electron transport chain, preventing oxidative

phosphorylation occurring

Oxygen Saturation

Oxygen Saturation can be defined in two ways:

- **Functional Saturation**

$$SpO_2 (\%) = \frac{[HbO_2] \times 100}{[HbO_2] + [DeoxyHb]}$$

However, additional haemoglobin species exist in varying amounts, and this definition may deceptively imply good oxygen delivery when this is not the case.

- **Fractional Saturation**

$$SpO_2 (\%) = \frac{[HbO_2] \times 100}{[HbO_2] + [DeoxyHb] + [MetHb] + [COHb]}$$

Fractional saturation includes carboxy- and met-haemoglobin, and so is a more accurate estimator of oxygen saturation.

Note that pulse oximetry doesn't measure either of these and is dependent on the calibration, but will typically measure functional saturation

Myoglobin

Muscle is highly metabolically active and has a large O₂ demand. Myoglobin serves as an O₂ store for muscle. It is similar to Hb in that it is a large O₂-binding iron-containing protein myoglobin, and is different because it:

- Contains one globin chain and one haeme group (binding one O₂ molecule), and so **does not exhibit cooperative binding**
The myoglobin dissociation curve therefore has a **rapid upstroke** and an **early plateau**.
- Has a **P₅₀ of 2.7mmHg**
This allows it to take up oxygen from haemoglobin (as the partial pressure gradient favours diffusion into the cell), and unload it into the cell (so it can actually be used).
- Is found in skeletal and cardiac muscle

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. Brandis K. The Physiology Viva: Questions & Answers. 2003.

Carbon Dioxide

Describe the ~~oxygen~~ and carbon dioxide stores in the body

Describe the carbon dioxide carriage in blood including the Haldane effect and the chloride shift

Explain the carbon dioxide dissociation curve

Describe the movement of carbon dioxide from blood to the atmosphere

Discuss the carriage of carbon dioxide in blood, the carbon dioxide dissociation curve and their clinical significance and implications

Carbon dioxide is:

- Produced in the mitochondria during the citric-acid cycle as a product of aerobic metabolism
- Eliminated (and, at steady state, produced) at a rate of **200ml.min⁻¹**

Additionally, a large amount (~**120L**) is stored throughout the body, with 1.6L.kg⁻¹ - 1.8L.kg⁻¹ in relatively inaccessible compartments.

Carbon Dioxide in Blood

In blood, CO₂ is stored as:

- Bicarbonate (90%)
- Dissolved gas
- Carbamino compounds

Table 1: Forms of Carbon Dioxide in Arterial and Venous Blood

Form	Arterial Blood	Additional CO ₂ in venous blood
Bicarbonate	90%	60%
Dissolved	5%	10%
Carbamino compounds	5%	30%

Bicarbonate

- CO₂ diffuses freely into erythrocytes, where it can be catalysed by carbonic anhydrase to produce bicarbonate:



- To maintain bicarbonate production, the products (H⁺ and HCO₃⁻) are then removed:
 - H⁺ ions are **buffered to haemoglobin**

$$\text{HbO}_2 + \text{H}^+ \rightleftharpoons \text{HHb} + \text{O}_2$$
 - Intracellular **HCO₃⁻** is then exchanged with extracellular **Cl⁻** via the BAND3 membrane protein
 - This is called the Hamburger, or **Chloride Shift**
 - Chloride entering the cell draws water in along its osmotic gradient, **increasing the haematocrit of venous blood** relative to arterial blood

Dissolved Gas

- As per **Henry's Law**, the amount of carbon dioxide dissolved in blood is proportional to the P_{aCO_2}
- As carbon dioxide is 20x as soluble as oxygen in water, dissolved carbon dioxide contributes much greater proportion of carbon dioxide *content* than dissolved oxygen does to oxygen content

Carbamino Compounds

- CO_2 can bind directly to proteins (predominantly haemoglobin), which displaces a H^+ ion:

$$Hb + CO_2 \rightleftharpoons HbCOO^- + H^+$$
 - The H^+ ion is then buffered by another plasma protein (also predominantly haemoglobin)

$$Hb + H^+ \rightleftharpoons HHb$$

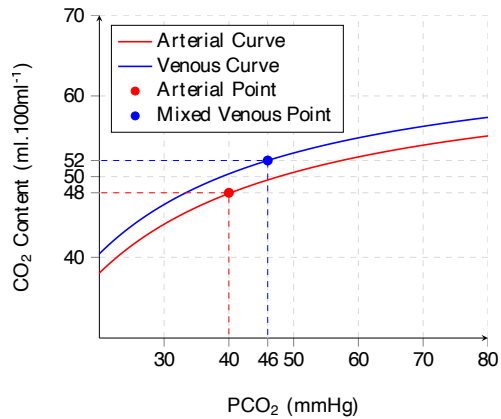
- Bound CO_2 does not contribute to the partial pressure gradient
- Carbamino compounds are only a small contributor to overall CO_2 carriage, but contribute about one third of the arterio-venous CO_2 difference due to the **Haldane effect**

The Haldane effect states that deoxyHb binds CO_2 more effectively than oxyHb. This is because:

- DeoxyHb is a **better buffer** of H^+
 pKa of deoxyHb is 8.2, compared to that of oxyHb which is 6.6.
 - Enhanced buffering contributes ~30% of the Haldane effect
- DeoxyHb forms carbamino compounds more easily Deoxy-Hb has 3.5x the affinity for CO_2 than Oxy-Hb.
 - This forms ~70% of the Haldane effect

CO₂ Dissociation Curve

This curve plots PCO₂ against blood CO₂ content in ml.100ml⁻¹.

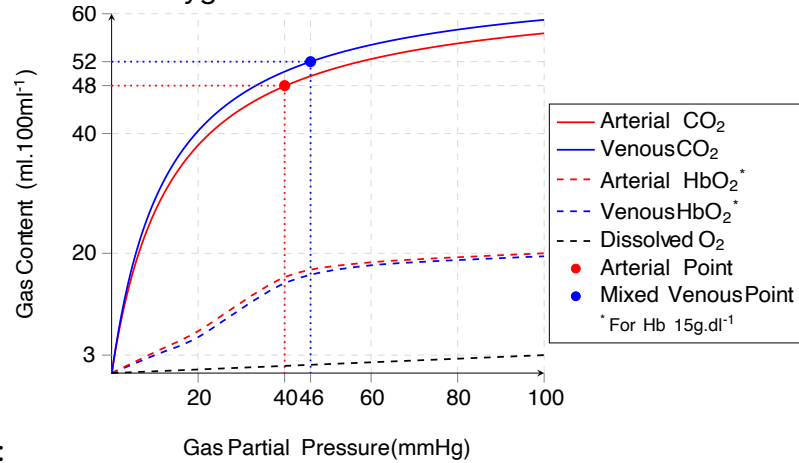


Key points:

- Mixed venous CO₂ content is 52ml.100ml⁻¹, at a PCO₂ of 46mmHg
- Arterial CO₂ content is 48ml.100ml⁻¹, at a PCO₂ of 40mmHg
- Approximately 50% of the arterial-mixed venous difference occurs due to the upwards shift of the curve, which is due to the Haldane effect
This is the mechanism for changes in PO₂ affecting the CO₂ dissociation curve.

Comparison Curve

Comparison of gas content of both oxygen and carbon



dioxide per 100mL of blood:

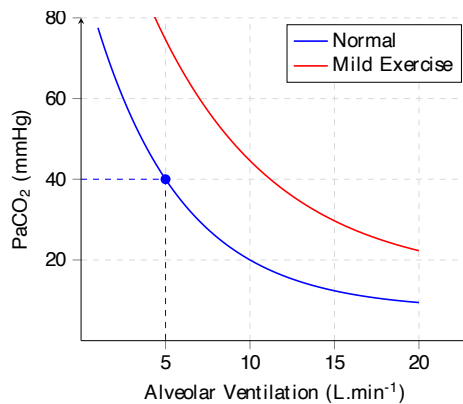
Removal of CO₂

CO₂ dissolves from pulmonary arterial blood into the alveolus down a concentration gradient. As inspired CO₂ is negligible, PACO₂ is a function of alveolar ventilation and CO₂ output, given by the equation:

$$PACO_2 = \frac{CO_2 \text{ output}}{\dot{V}_A}$$

Simplified, PaCO₂ is inversely proportional to alveolar ventilation:

$$PACO_2 \propto \frac{1}{\dot{V}_A}$$



Distribution of Carbon Dioxide

CO₂ in the body can be considered as a three-compartment model:

- Well-perfused (blood, brain, kidneys)
- Moderately-perfused (resting muscle)
- Poorly-perfused (bone, fat))
- Each of these tissues has a different time-constant, such that a mismatch of ventilation with metabolic activity may take **20-30 minutes** to equilibrate across compartments
- Therefore hypoventilation and hyperventilation have different effects on PCO₂:
 - **Hyperventilation** causes a rapid decrease in PCO₂ in blood, subsequent (slower) redistribution from peripheral compartments
 - **Hypoventilation** causes a rise in PaCO₂, the rate of which is determined both by production and distribution into plasma
 - With **no ventilation**, PCO₂ rises at **3-6 mmHg.min⁻¹**
 - Due to the Haldane effect the PaCO₂ will rapidly increase during passage through the pulmonary capillary (despite the fact that carbon dioxide *content* is

unchanged) as the proportion of OxyHb increases

- Therefore:
 - PaO_2 is more sensitive at detecting early hypoventilation provided PAO_2 is normal
 - Steady-state PCO_2 gives the best indication of adequacy of ventilation
 - In acute hypoventilation, produced CO_2 is preferentially stored in tissues, decreasing CO_2 elimination
 - In acute hyperventilation, CO_2 is mobilised from tissues resulting in increased CO_2 elimination

CO_2 Cascade

Table 2: Normal Partial Pressures Along the Carbon Dioxide Cascade

Region	Value (mmHg)
Mixed Venous	46
Alveolar	40
(Arterial)	40
Mixed-expired	27

- Venous CO_2 diffuses into the alveolus, reaching equilibrium with arterial PCO_2
- Alveolar CO_2 is then diluted by dead space gas, resulting in a lower $\text{ME}'\text{CO}_2$

References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.

2. FRCA: Anaesthesia Tutorial of the Week - Respiratory
Physiology

Applied Respiratory Physiology

Positive Pressure Ventilation

Describe the physiological consequences of intermittent positive pressure ventilation and positive end-expiratory pressure.

Physiological effects of positive pressure ventilation are mostly related to the **increase** in **mean airway pressure**. This increase is a function of:

- Ventilation mode
- Tidal volume and peak (and plateau) airway pressure
- Respiratory rate
- I:E ratio
- **PEEP**
 - PEEP has a much larger effect than the other factors.
 - PEEP is defined as a positive airway pressure at the end of expiration
 - PEEP is distinct from positive airway pressure (which is not confined to a phase of the respiratory cycle) and CPAP (which is a mode of ventilation)
 - iPEEP refers to **intrinsic PEEP**, auto PEEP or dynamic hyperinflation
 - iPEEP is PEEP generated by the patient, and occurs when expiration stops before the lung volume reaches FRC.
 - Application of external PEEP may limit the generation of iPEEP by maintaining airway patency in late expiration

Respiratory Effects

- **Decreased work of breathing**
 - Decreased VO_2
More important when work of breathing is high.
- **Alteration in anatomical/apparatus dead space**
 - Intubation typically reduces dead space, as the additional apparatus dead space is of smaller volume than the anatomical dead space it replaces
 - Non-invasive ventilation masks cause a large increase in dead space
- **Increased lung volume (and FRC, for PEEP) by an amount proportional to the compliance of the system**
 - Improves oxygenation via alveolar recruitment
 - Improves lung compliance via alveolar recruitment, reducing work of breathing
 - Elevated airway pressures may increase the proportion of West Zone 1 physiology and alveolar dead space
In healthy lungs an increase in the $\frac{V_D}{V_T}$ ratio is seen when PEEP exceeds 10-15cmH₂O.
- **Reduced airway resistance**
Airway resistance decreases as lung volume increases.

Cardiovascular Effects

- **Alteration in cardiac output**
 - PEEP and IPPV generally decrease CO via decreasing VR due to the increase in intrathoracic pressure.
Leads to reduction in RV filling pressure, LV filling, and CO.
 - This is the predominant reason why CO falls with the application of PEEP
In a well patient, CO falls by:

- 10% with IPPV and ZEEP
- 18% with IPPV and 9cmH₂O of PEEP
- 36% with IPPV and 16cmH₂O of PEEP
- These changes are:
 - More marked in hypovolaemia
Changes are reversed with volume expansion.
 - Less severe with poor lung compliance
Reduced compliance greatly reduces the effect of PEEP and IPPV on the vasculature, as the change in intrapleural pressure is reduced.
- LV preload may also be reduced due to increased RV afterload
 - Increased RV afterload may increase RV EDV, displacing the interventricular septum into the LV
 - The bulging septum decreases LVEDV, causing LV diastolic function and reduced LV filling
This is an example of **ventricular interdependence**.
- Reduced LV afterload due to reduced LV transmural pressure
In some cases, IPPV augments circulatory function by reducing LV afterload to a greater extent than preload.
 - Effects in a well patient are minimal, as PEEP is relatively small in magnitude compared to systemic arterial pressures
 - In patients generating highly negative intrathoracic pressures, the LV transmural pressure can increase markedly, increasing LV afterload and reducing cardiac output
- **Reduction in MAP**
MAP decreases as PEEP increases.
- **Changes to oxygen flux**
PEEP will tend to improve PO₂ whilst reducing CO.
- **Changes to pulmonary vascular resistance and RV afterload**

- If lung volume is lower than FRC, then PVR will reduce as PEEP stretches open extra-alveolar vessels
 - Alveolar recruitment will reduce hypoxic-pulmonary vasoconstriction, further reducing PVR
- If lung volume is higher than FRC, then PVR will increase as PEEP compresses alveolar vessels
- Therefore, PEEP has *variable effects* on RV after-load depending on how it changes lung volume with respect to FRC

End-Organ Effects

- **Reduced urine output** due to:
 - Reduced CO and renal blood flow
 - ADH release as a consequence of reduced atrial stretch and ANP release
May worsen oedema in patients with prolonged periods of ventilation.
 - **Reduced hepatic blood flow** due to:
 - Increased CVP and decreased CO lowering the pressure gradient for hepatic flow
 - May result in circulation only intermittently throughout the cardiac cycle
-

References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.
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7. Yartsev, A. [PEEP and Intrinsic PEEP}(<http://www.derangedphysiology.com/main/core-topics-intensive-care/mechanical-ventilation-0/Chapter%202.1.6/peep-and-intrinsic-peep>). Deranged Physiology.

Hypoxia

Explain the physiological effects of hyperoxia, **hypoxaemia**, hypercapnia, hypocapnia, and carbon monoxide poisoning.

- **Hypoxaemia** is a low partial pressure of oxygen in blood
- **Hypoxia** is an oxygen deficiency at the **tissues**, due to:
 - Impaired **oxygen delivery**
 - Impaired **oxygen extraction**

Oxygen delivery is given by the equation:

$$DO_2 = HR \times SV \times (1.34 \times [Hb] \times SpO_2 + PaO_2 \times 0.03)$$

Where:

- 1.34 is Hüfner's constant
This is the oxygen carrying capacity of haemoglobin, in ml.g^{-1} (of Hb).
 - The theoretical maximum is 1.39
 - *In vivo* it is 1.34 due to the effect of carboxyhaemoglobin and methaemoglobin compounds, which limit O_2 binding
- 0.03 is the solubility coefficient of O_2 in water at 37°C , in $\text{mls.L}^{-1}.\text{mmHg}^{-1}$
Can also be expressed as $0.003 \text{ mls.dL}^{-1}.\text{mmHg}^{-1}$ (mls per deciliter per mmHg). Different texts use different values, depending on whether haemoglobin is reported in g.L^{-1} or g.100ml^{-1} .

Classifications and Causes of Hypoxia

Hypoxia can be categorised into four types:

- Hypoxaemic hypoxia
- Anaemic hypoxia
- Ischaemic hypoxia
- Histotoxic hypoxia

Hypoxaemic Hypoxia

Hypoxic hypoxia, or **hypoxaemia**, is hypoxia due to low PaO_2 (and therefore low SpO_2), typically defined as a $\text{PaO}_2 < 60$.

Causes of hypoxaemia can be further classified based on their A-a gradient:

- Causes of hypoxaemia with a **normal A-a gradient**:
 - Low PiO_2
 - Decreased alveolar ventilation
- Causes of hypoxaemia with a **raised A-a gradient**:
 - Diffusion limitation
 - Shunt
 - (Increased oxygen extraction)

Low FiO_2

Hypoxaemia occurs at high altitudes when the PO_2 is decreased.

Decreased Alveolar Ventilation

A fall in alveolar ventilation ($\dot{V}_A = \text{Respiratory Rate} \times (V_T - V_D)$) causes a rise in PACO_2 , and therefore decreases PAO_2 . Decreased $\dot{V}_A$ can occur with:

- Respiratory centre depression:
 - Drugs

- Head injury (Raised ICP, closed head injury)
- Encephalopathy
- Fatigue
- Nerve dysfunction:
 - Spinal cord injury
 - GBS
 - MND
- NMJ dysfunction:
 - Paralysis
 - MG
- Muscular dysfunction:
 - Myopathy
 - Fatigue
 - Malnutrition
 - Dystrophy
- Chest wall abnormalities:
 - Kyphoscoliosis
 - Ankylosing Spondylitis
 - Pleural fibrosis

Diffusion Limitation

Impaired diffusion of O_2 across the membrane results in a lowered PaO_2 . Diffusion limitation occurs due to:

- Decreased alveolar surface area
- Increased alveolar capillary barrier thickness
 - Pulmonary fibrosis
 - ARDS

Shunt

Shunt occurs when **blood reaches the systemic circulation without being oxygenated via passage through the lung**.
With respect to shunt:

- As the alveolus is **perfused** but **not ventilated**, thus the **V/Q ratio is 0**
- Administration of 100% O_2 has less effect on PaO_2 as shunt fraction increases

Shunt physiology is explored in more detail under shunt

- Oxygen content of shunted alveoli is identical to mixed venous content
- Oxygen content of non-shunted alveoli does not increase appreciably at high partial pressures as haemoglobin is already fully saturated

Increased Oxygen Extraction

- Increased oxygen extraction (VO_2) will not typically cause hypoxia
- This is because:
 - Normal VO_2 is $250\text{ml}\cdot\text{min}^{-1}$
 - Normal DO_2 is $1\text{L}\cdot\text{min}^{-1}$
 - Maximal oxygen extraction ratio is ~70% (though it varies between organs)
Therefore VO_2 can increase until it reaches 70% of the DO_2 , a point called critical DO_2 .
- However, it may worsen hypoxia in the presence of a supply-side (DO_2) pathology

Anaemic Hypoxia

- Impaired oxygen delivery due to low Hb
- Typically asymptomatic at rest but limits exercise tolerance
- Compensation occurs by increasing levels of 2,3-DPG, causing a right-shift in the Hb- O_2 dissociation curve to favour oxygen off-loading at tissues

Carbon Monoxide Poisoning

- CO poisoning is classified as a subset of anaemic hypoxia as carboxyhaemoglobin **reduces** the **effective** amount of **haemoglobin** in solution
- CO has 210 times the affinity for Hb than O_2

- CO rapidly displaces O₂ from Hb and is liberated slowly
- CO poisoning causes headache and nausea, but no increased respiratory drive since the PaO₂ is unchanged

Ischaemic Hypoxia

- Ischaemic hypoxia is due to impaired cardiac output resulting in impaired oxygen delivery

Histotoxic Hypoxia

- Histotoxic hypoxia is due to impaired tissue oxidative processes, preventing utilisation of delivered oxygen
- Most common cause of histotoxic hypoxia is cyanide poisoning, which inhibits cytochrome oxidase and prevents oxidative phosphorylation
- Managed by using methylene blue or nitrites, which form methaemoglobin, in turn reacting with cyanide to form the non-toxic cyanmethaemoglobin

Effects of Hypoxia

- With a normal PaCO₂, PaO₂ must fall to 50mmHg before an increase in ventilation occurs
- With a rising PaCO₂, a fall in PaO₂ below 100mmHg will stimulate ventilation via action on carotid and aortic body chemoreceptors
 - The effects of each stimuli are synergistic, and greater than what is seen with either effect alone
- Prolonged hypoxaemia will also lead to cerebral acidosis (via anaerobic metabolism), which will stimulate central pH receptors and stimulate ventilation

Acid-Base Changes

- Hypoxia results in both fixed and volatile acid-base disturbances
- Anaerobic metabolism results in **lactate** production
- Production of fixed acid results in a **base deficit**, and a low bicarbonate
- Hypoxia and metabolic acidosis stimulate ventilation and hypocarbia

CO₂ retention

- In chronic hypercarbia the CSF pH normalises (as bicarbonate is secreted into CSF), with a raised CO₂
- Fall in PaO₂ becomes the predominant stimulus for ventilation

References

1. West J. Respiratory Physiology: The Essentials. 9th Edition. Lippincott Williams and Wilkins. 2011.
2. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
3. CICM July/September 2007
4. ICU Basic Book.

Hypo and Hypercapnea

Explain the physiological effects of hyperoxia, hypoxaemia, hypercapnia, hypocapnia, and carbon monoxide poisoning

Carbon dioxide is lipid soluble and can rapidly cross membranes, allowing it affect acid-base status in any compartment.

Hypercapnea

- **Respiratory Effects**

- Increased respiratory drive via chemoreceptor stimulation

- **CVS effects**

- Peripheral vasodilation
 - May cause tachycardia from sympathetic stimulation
- Pulmonary vasoconstriction
- Myocardial depression
Intracellular acidosis.
- Arrhythmogenic

- **CNS effects**

- Increased CBF
- Increased ICP secondary to increased CBF
- SNS activation

- CNS depression
When $\text{PaCO}_2 > 100\text{mmHg}$

Hypocapnea

- **Respiratory Effects**
 - Left-shift of oxyhaemoglobin dissociation curve
 - Respiratory depression
 - **CVS effects**
 - Myocardial depression
Intracellular alkalosis.
 - **CNS effects**
 - Decreased cerebral blood flow
 - **Electrolyte effects**
 - Decreased serum K^+
 - Decreased serum Ca^{2+}
Leads to paresthesias and twitches.
 - Ca^{2+} binds to H^+ binding site on albumin
-

References

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Position and Ventilation

Explain the effect of changes in posture on ventilatory function

Altered patient position can cause significant changes to V/Q matching. In general:

- The dependent lung has increased perfusion independent of anaesthesia, paralysis, or ventilation method
- The independent lung has progressively more ventilation through each of:
 - Spontaneous ventilation
 - Controlled ventilation
 - Thoracotomy

Factors Affecting Distribution of Blood Flow

Key factors:

- Gravity
- HPV
- Anatomy Blood flow is greater in central than peripheral portions.
- Airway pressures
Due to effects on:
 - Volume
 - Via:

- Altering vascular distension and subsequently PVR
- Alterations in extra-alveolar and intra-alveolar pressures at FRC may alter regional blood flow.
- Right ventricular afterload
 - Lung volume
- When both lungs are being ventilated, V/Q matching can be improved with selective application of PEEP to the dependent lung, which improves compliance

Factors Affecting Distribution of Ventilation

Key factors:

- Gravity

Intrapleural pressure is more negative in the elevated portions of the lung, due to the weight of the lung pulling down. This results in:

 - **Larger intra-alveolar volume** of the upper portion of the lung, and **reduced compliance**
 - **Smaller intra-alveolar volume** of the dependent portion of the lung, and **improved compliance**
 - Which results in the upper portion of the lung being higher on the compliance curve
 - Reducing the relative contribution of this portion of the lung to a tidal volume, and relatively reduced ventilation

These changes may be exaggerated in disease, e.g. COPD.

Lateral Decubitus

Effects depends on the method of ventilation:

- In the **spontaneously ventilating** patient:

- Dependent lung ventilation and perfusion improves by ~10%
Due to the effect of gravity.
- Dependent lung corresponds more to West Zone 3
- Non-dependent lung corresponds more to West Zone 2
- In the **anaesthetised patient**:
 - FRC ↓ by ~20% in both lungs due to reduced muscle tone
Each lung therefore changes in compliance:
 - Non-dependent lung falls to the steep portion of the curve, and compliance improves
 - The dependent lung falls from the steep position to the lower shallow segment, and compliance decreases
Ventilation is now directed preferentially to the **non-dependent lung**.
 - Perfusion is unchanged
- In the paralysed and **positively-pressure ventilated** patient:
 - FRC reduces further due to cephalad movement of:
 - Mediastinum
 - Abdominal organs
 - Ventilation is further directed to the non-dependent lung
 - Distribution of perfusion is unaffected, but the application of PEEP increases dead space and further worsens V/Q matching

Thoracotomy

Opening the non-dependent hemithorax causes:

- No change in perfusion pressure
Unless mediastinal structures move.

- Variable effects on the lung:
 - If lung isolation:
 - Lung will collapse under its own elastic recoil
Improved V/Q matching, as there is no ventilation, and perfusion will decrease as the lung collapses.
 - If no lung isolation:
 - Expansion of the lung, now unrestrained by the chest wall
Worse V/Q matching.
-

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1. Dunn, PF. Physiology of the Lateral Decubitus Position and One-Lung Ventilation. *Thoracic Anaesthesia*. Volume 38(1), Winter 2000, pp 25-53.
2. Graph from Benumof JL, ed. *Anesthesia for thoracic surgery*. 2nd ed. Philadelphia: WB Saunders Company, 1995.
3. ANZCA August/September 2015

Humidification

Define humidity and give an outline of the importance of humidification

Humidification:

- Describes the amount of water vapour present in air:
 - **Absolute Humidity** is the amount of water vapour in a given volume of air (g.m^{-3})
 - **Relative Humidity** is the ratio between the amount of water vapour in a sample of air (absolute humidity) and the amount of water required to fully saturate that sample at its current pressure and temperature
- Of inspired air is important to avoid drying out mucosa and sputum, which leads to tissue damage and failure of the mucociliary elevator
Optimal function requires a relative humidity of **greater than 75%**.

Moisture is the water produced by condensation when relative humidity exceeds 100%.

Mechanism

The nose is:

- Optimised for humidification
The **septum** and **turbinates** increase contact of air with mucosal surfaces by:
 - Increasing **surface area**
 - Generating **turbulent flow**

- The preferred orifice for breathing unless airways resistance becomes a significantly limiting factor
This is relevant in:
 - Airway obstruction (e.g. polyps)
 - At high minute ventilations ($> 35\text{L}\cdot^{-1}$)
- Humidifies inspired gas to 90%, compared to **60%** for the mouth

Method of humidification:

- Fluid lining the airway acts as a heat and moisture exchanger
- In inspiration:
 - Relatively dry air is evaporates water from the airway lining
 - Relative humidity is increased to 90% in the nasopharynx and 100% BTPS by the second generation of bronchi
This gives a water vapour pressure of **47mmHg** at BTPS, with an absolute humidity of **44g.m⁻³**.
- In expiration:
 - Air cools in the upper airway
As cooler air has a lower saturated vapour pressure, moisture condenses on the airway.
 - Moisture is reabsorbed
This reduces potential water losses from the airway from **300ml.day⁻¹** to **150ml.day⁻¹**.

References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.
2. WeatherFaq. Absolute and Relative Humidity.
3. CICM September/November 2012

Cough Reflex

Explain the pathways and importance of the cough reflex

Coughing:

- Is an **airway protection** reflex
- Involves deep inspiration followed by **forced expiration against a closed glottis**

The sudden opening of the cords causes a violent rush of air at $>900\text{km.h}^{-1}$, removing irritants and secretions from the airways.

Sensation

Vagus afferents have exquisitely sensitive light touch and corrosive chemical receptors in the larynx, carina, terminal bronchioles, and alveoli.

Integration

Vagal afferents synapse in the medulla, which coordinates the effector response.

Effector

A series of processes occur in three phases:

- Inspiratory phase
A close to vital capacity breath is taken.
 - Compressive phase
Effort closure of the epiglottis to seal the larynx, followed by a violent contraction of abdominal musculature and internal intercostals, causing a rapid rise in intrapleural pressure to $>100\text{mmHg}$.
 - Expulsive phase
Wide-opening of the cords and epiglottis, causing a violent expiration.
 - Compression of the lungs causes narrowing of the noncartilaginous airways and increases turbulent flow, removing adherent material from the tracheobronchial tree
-

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.

Non-Respiratory Functions

Outline the non-ventilatory functions of the lungs

The lungs are a unique organ as:

- The entire cardiac output passes through the pulmonary circulation
- They have a huge capillary bed which almost all of this blood volume
- They have a large interface with the external environment

Consequently they are adapted to a number of non-respiratory functions which include:

- Filtration
- Metabolism
- Blood reservoir
- Defence
- Drug delivery
- Endocrine

An alternative mnemonic: - **F**iltration

- Immune defence
- Blood reservoir
- Metabolism
- Taking up drugs
- (Inhalational Anaesthetics)
- Synthetic
- Endocrine

Filtration

The entire cardiac output passes through the 7µm pulmonary capillaries, which act as an effective sieve for particulate matter. This function may be impaired by intra-cardiac shunting (e.g. PFO) or pre-capillary anastomoses.

Complementing this role, the lungs are able to clear thrombi more rapidly than other organs as pulmonary endothelium has a high concentration of plasmin activator and heparin.

Metabolism

The pulmonary endothelium has a variety of effects on drugs and endogenous hormones:

Table 1: Pulmonary Endothelial Activation and Inactivation of Circulating Substances

Class	Activated	Inactivated
Amines		5-HT, Noradrenaline
Peptides	Angiotensin I (via ACE)	Bradykinin, ANP
Arachidonic acid derivatives	Arachidonic acid	Many prostaglandins
Other Drugs		Lignocaine, fentanyl

Blood Reservoir

The highly compliant pulmonary circulation contains a reservoir of ~500ml of blood which acts as a **volume reserve** for the LV.

Defence

The large surface area required for gas exchange leaves the lung vulnerable to invasion by airborne substances. This is attenuated by:

- Mucous

A mucous layer protects large airways, as large ($>8\mu\text{m}$) particles impact into the mucous.

 - Mucous is exocytosed by goblet cells in response to noxious stimuli including chemical irritation as well as inflammatory and neuronal stimulation
 - The efficacy of the mucous-cilia system is enhanced by bronchoconstriction, which reduces flow velocity and causes particulate matter to settle
- Cilia

Cilia are projections from epithelium which beat rhythmically at $\sim 12\text{Hz}$ to propel mucous out of the airway at a rate of $\sim 4\text{mm}\cdot\text{min}^{-1}$.

 - Ciliary function can be impeded by pollutants, smoke, and infection
 - Ciliary function is stimulated by anaesthetic agents
- Inhaled particles which reach the respiratory zone are not trapped by mucous, but instead phagocytosed by alveolar macrophages
- Bronchoconstriction reduces flow velocity and causes particulate particles to settle in the mucous

Drug Delivery

The same properties that optimise the lung for gas exchange optimise it for delivery of inhaled agents. Drugs absorbed in the pulmonary circulation are:

- Lipophilic
- Alkaline ($\text{pK}_a > 8$)

Endocrine

Important endocrine functions of the lung include:

- Release of inflammatory mediators such as histamine, endothelin, and eicosanoids
 - Release of nitric oxide to regulate smooth muscle
 - ACE metabolises angiotensin I to angiotensin II
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Altitude

Altitude causes a number of physiological effects, related to:

- Reduce atmospheric pressure
- Reduced temperature
- Reduced relative humidity
- Increased solar radiation

Pressure Effects

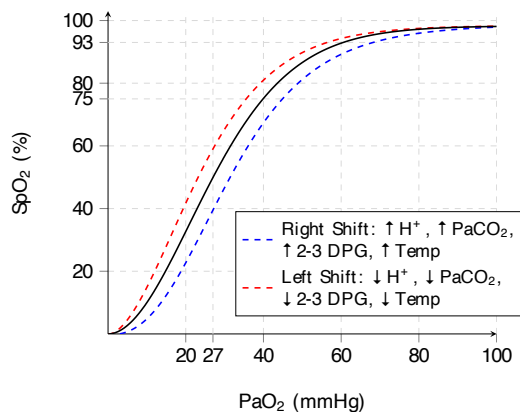
Reduced air pressure results in a proportional decrease in PO_2 :

- At 3,000m, alveolar PO_2 is 60mmHg
- At 5,400m, consciousness is lost in unacclimatised individuals
- At 10,400m, air pressure is 187mmHg
With 47mmHg of water vapour and an alveolar PCO_2 of 40, breathing 100% O_2 gives an alveolar PO_2 of 100mmHg.
- At 14,000m, consciousness is lost despite 100% O_2
- At 19,200m, the ambient pressure is so low that the boiling point of water is $37^\circ C$
This is the **Armstrong limit**.

Respiratory

- Fall in PaO_2 is compensated by increasing minute ventilation, which decreases $PACO_2$ and therefore increases PAO_2

- Limits of compensation are reached on 100% oxygen at 13,700m
- Effective compensation is limited by the respiratory alkalosis, this is known as the **braking effect**:
 - Peripheral chemoreceptors detect hypocapnea
 - Central chemoreceptors detect alkalosis
- The subsequent respiratory alkalosis generates a compensatory metabolic acidosis
This acidosis relaxes the braking effect and allows further hyperventilation, and is therefore an important part of acclimatisation.
- There is an **initial left-shift** of the oxygen-haemoglobin dissociation curve due to alkalosis
- This stimulates a compensatory increase in 2,3-DPG to right-shift the curve and improve oxygen offloading at the tissues



Cardiovascular

- PVR increases due to HPV
- Heart rate increases due to increased SNS outflow
- Stroke volume falls (cardiac output remains the same) due to decreased preload:
 - Plasma volume falls due to:
 - Pressure diuresis
 - Insensible losses from hyperventilation and reduce relative humidity
- Myocardial work increases

- Increased HR
- Increased viscosity of blood due to high haematocrit
- Increased RV afterload from high PVR
Increased pulmonary capillary hydrostatic pressures lead to fluid transudation and pulmonary oedema

Haematological

- Increased risk of thrombotic events due to increased haematocrit
 - Increased red cell mass due to EPO secretion
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Obesity

Discuss the effect of morbid obesity on ventilation

Obesity is a multisystem disorder defined by an elevated body mass index (BMI):

- Normal: BMI < 25
- Overweight: BMI 25 - 30
- Obese: BMI > 30
- Morbidly Obese:
 - Obesity related disease and a BMI > 35
 - BMI > 40

Effects of Obesity on the Respiratory System

Morbid obesity causes several changes to the respiratory system:

- **Airway**
 - Increased risk of OSA
 - Increased risk of GORD and aspiration
 - Increased risk of difficult bag-mask ventilation
 - Increased risk of difficulty laryngoscopy
- Changes to **respiratory pattern**
 - Increased minute ventilation
 - Secondary to increased VO_2 and VCO_2
Due to the increase in LBW and adiposity.

- Increased airway reactivity
Central adiposity increases circulating cytokines, including TNF- α , IL-6, leptin.
 - Changes to **volumes and capacities**
 - Reduced respiratory system compliance
 - Decreased chest wall compliance
Due to abdominal and chest wall fat.
 - Fat distribution may be more important than absolute BMI
 - Decreased lung compliance
Basal atelectasis due to abdominal compression and reduced respiratory compliance.
 - Decreased ERV and FRC
 - Note that RV is generally relatively unchanged
 - Increased airway resistance
Due to decreased airway radius at lower lung volumes.
 - Increased work of breathing
Due to reduced respiratory compliance and increased airway resistance.
 - Closing capacity encroaches on FRC
As FRC falls, closing capacity becomes closer to FRC.
 - If closing volume exceeds expiratory reserve volume, then small airways will collapse during normal tidal breathing, causing shunt
 - Changes to **blood gases**
 - Increased A-a gradient
Occurs when closing capacity exceeds FRC.
 - Changes to **pulmonary circulation**
 - PVR increases due to reduced FRC causing increased HPV
May lead to secondary PHTN and right heart dysfunction.
-

References

1. Alvarez A, Brodsky J, Lemmens H, Morton J. Morbid Obesity: Peri-operative Management. Cambridge: Cambridge University Press. 2010.
2. Lotia S, Bellamy MC. Anaesthesia and morbid obesity. Contin Educ Anaesth Crit Care Pain 2008; 8 (5): 151-156.

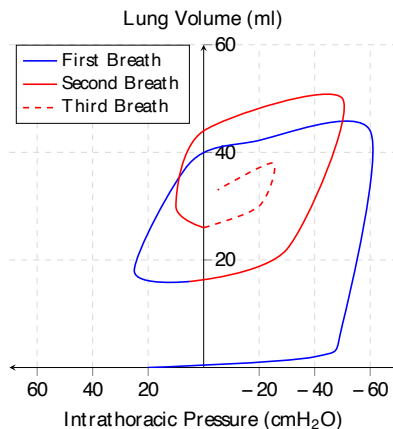
Neonates and Children

Transition at Birth

Transition from placental gas exchange to pulmonary gas exchange occurs within 20s after birth:

- Compression of the thorax through the vaginal canal expels foetal lung water
- Elastic recoil, combined with cooling of the skin and mechanical stimulation (which stimulate the respiratory centre), facilitate first breath
- The rapid drop in pulmonary vascular resistance with spontaneous breathing drives the changes in the cardiac circulation
- The **first three breaths** establish functional residual capacity

Large changes in intrathoracic pressure in the first three breaths pressure drive alveolar amniotic fluid into the circulation, and establish FRC.



Neonates and Children

- **Compliance**

Neonatal chest walls are highly compliant relative to their lungs (due to both a reduced lung compliance and increased chest wall compliance), as compared to adults where lung and chest compliance is equal. Therefore elastic work of breathing is largely determined by the lungs.

- **Oxygenation**

- O_2 consumption is $\sim 10 \text{ ml.kg}^{-1}.\text{min}^{-1}$ in neonates, and $6 \text{ ml.kg}^{-1}.\text{min}^{-1}$ in children
- There is a $\sim 10\%$ shunt after birth which contributes to a greater A-a gradient

- **Ventilation**

- Obligate nose breathers
- Increased CO_2 production due to higher metabolic rate
- Increased minute ventilation, which is due to increased respiratory rate (25-40 breaths per minute)

- **Neurological control of breathing**

Respiratory patterns change following birth, and complete change to adult respiratory patterns may take some weeks. Patterns include:

- **Periodic breathing** is a slowly oscillating respiratory rate and V_T
- **Periodic apnoea** is intermittent apnoea interspersed with normal breathing.

- **Volumes and capacities**

- Closing capacity is increased relative to adults, causing shunt
- Functional residual capacity is unchanged
- Tidal volume and dead space are unchanged

- **Laryngeal anatomy**

- Large head
- Large tongue
- Large, stiff, U-shaped epiglottis
- Elevated larynx
Glottis is at C-3C4 (C6 in adults).

- Upper airway is narrowest at the cricoid ring (rather than the glottis).
 - Trachea is shorter and narrower
4 - 5cm long, 6mm diameter in the neonate.
 - **Small airways**
 - Reduced bronchial smooth muscle so bronchospasm is uncommon
 - Bronchioles contribute 50% of airways resistance
Bronchiolitis much more distressing in neonates and children.
-

References

1. Lumb A. Nunn's Applied Respiratory Physiology. 7th Edition. Elsevier. 2010.
2. CICM March/May 2013

Cardiovascular Physiology

Structure and Function

Cardiac Anatomy

Describe the anatomy of the heart, ~~the peri-~~
~~cardium and coronary circulation~~

Echocardiographic Anatomy

The left ventricle is:

- Divided into four parts
From base to apex, in **equal thirds** along the long axis of the ventricle:
 - Basal
 - Mid-cavity
Identified by presence of the papillary muscles.
 - Apical
 - Apex
Tip of the ventricle, beyond where the cavity ends.
- Each part is divided into segments
Total of **seventeen segments** between:
 - 6 basal and mid-cavity segments
 - Inferior
Mid-cavity contains the postero-medial papillary muscle.
 - Inferoseptal
 - Inferolateral
 - Anterior
 - Antero-septal

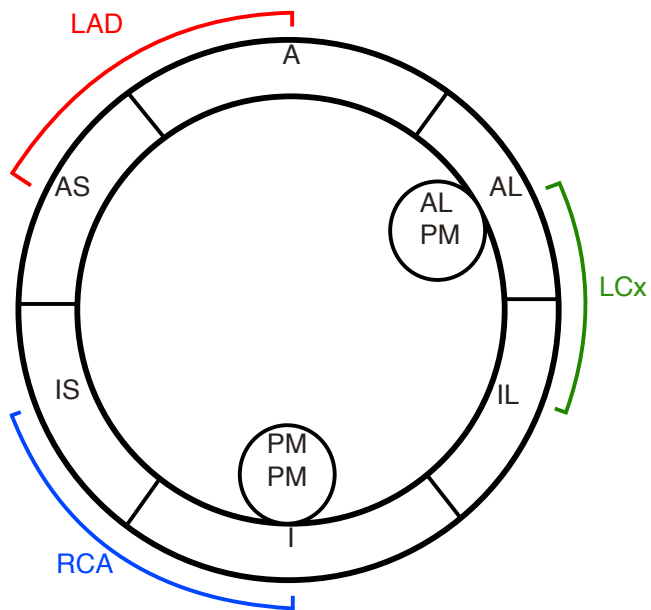
- Antero-lateral
Mid-cavity contains the anterolateral papillary muscle.
- 4 apical segments
 - Inferior
 - Anterior
 - Lateral
 - Septal
- Apical cap

Coronary Supply

The segments of the basal and mid-cavity parts are supplied by all three vessels¹:

¹There is both variability in coronary supply to the same wall of different sections of the LV, as well as inter-individual variation (although the LAD territory is relatively fixed).

Therefore, this is not an *entirely* accurate representation, but it is a good general approximation.



In the apical part, the:

- LAD
 - Supplies:
 - Anterior
 - Septal
- LCx
 - Supplies:
 - Lateral
- RCA
 - Supplies:
 - Inferior

The apical cap is supplied by the **LAD**.

References

1. Alfred Anaesthetic Department Primary Exam Tutorial Series
2. AHA 17 Segment Model. PMOD.

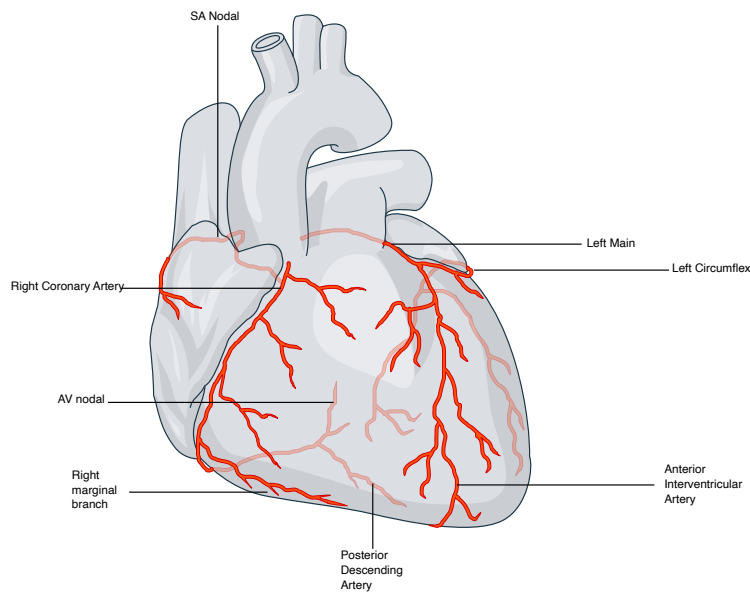
Coronary Circulation

Describe the anatomy of the heart, the pericardium and coronary circulation

Describe the anatomy of the heart including the coronary circulation and territories supplied.

Vascular Anatomy

Coronary Artery Anatomy



The **left main** coronary artery:

- Arises from the posterior aortic sinus superior to the left coronary cusp of the aortic valve
Eddy currents produced in the **Sinuses of Valsalva** (outpouchings of the aortic wall) prevent the valves occluding the os of the LM and RCA during systole, so they remain patent throughout the cardiac cycle.
- The **left main** is 5-10mm long, and bifurcates to form the LAD and LCx
- The **intermediate** artery is an anatomical variant that arises from the bifurcation, creating a trifurcation

The **LAD**:

- Courses along the **anterior interventricular groove** to the apex of the heart

The proximal LAD is defined as the area of LAD between the origin and D_1 , mid LAD is between D_1 and D_2 , etc.

Here, it anastomoses with the posterior descending artery from the RCA.

- **Supplies** the **anterolateral myocardium** and **anterior 2/3rds** of the interventricular **septum**
- **Branches** of the LAD include:
 - **Diagonal** vessels
Branches are named successively from proximal to distal (i.e. LADD₁, LADD₂, etc), and wrap around the anterior wall of the ventricle.
 - **Septal perforators**
Penetrate into the myocardium.

The **LCx**:

- Courses along the **left atrioventricular** groove between the LA and LV in the epicardial fat pad
- **Supplies** the **inferolateral wall** of the **LV**
- Gives off three **obtuse marginal** branches (OM₁, OM₂, etc) which follow the left margin of the heart
- Runs in close approximation with the coronary sinus for much of its course

The **RCA**:

- Arises from the anterior aortic sinus, superior to the right coronary cusp of the aortic valve
- Courses vertically downwards in the **right atrioventricular groove**
- Supplies the RA and RV
- RCA gives off:
 - **Acute marginal arteries**
 - **Posterior branch to the LV** (PLV)

Acute marginal arteries are almost never grafted.

The **posterior descending artery**:

- Arises from either the LCx or RCA
These vessels travel in opposite directions around the atrioventricular groove.
- Descends in the posterior interventricular groove before coursing along the base to anastomose with the LAD at the apex of the heart
- Is also known as the posterior interventricular artery

Coronary Dominance

Coronary dominance refers to which vessel gives rise to the PDA:

- In a **right-dominant** circulation the PDA is supplied by the **RCA**
- In a **left-dominant** circulation the PDA is supplied by the **LCx**

Additionally:

- The **SA node** is supplied by the **RCA in 60%** of individuals
- The **AV node** is supplied by the **RCA in 90%** of individuals

Venous Anatomy

- 85% of venous drainage occurs via the **coronary sinus**
 - Drains the **LV**
 - Formed from the cardiac veins:
 - The **great cardiac vein** runs with the LAD
 - The **middle cardiac vein** follows the PDA
 - The **small cardiac vein** runs with the RCA
 - The **oblique vein** follows the posterior part of the LA
- Most of the remainder is via **anterior cardiac veins** which drain directly into the RA
- A small proportion of blood from the heart is drained via the **thebesian veins** directly into four the cardiac chambers

Most into the right atrium, and least into the left ventricle. The portion of blood draining into the left side of the circulation contributes to **physiological shunt**.

Coronary Blood Flow

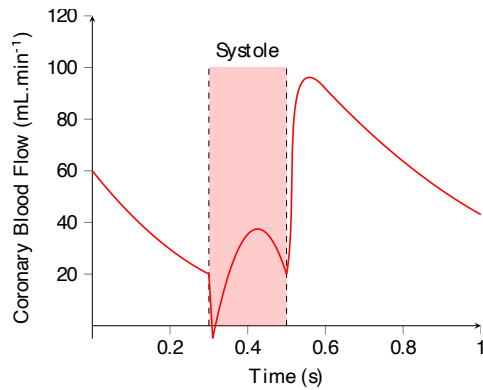
Coronary Blood Flow:

- Normal is **~250ml.min⁻¹** (~5% of resting CO)
- May increase 4x during strenuous exercise
Myocardial work may increase up to 9x, though as myocardial oxygen extraction is unchanged efficiency is actually improved during exercise.

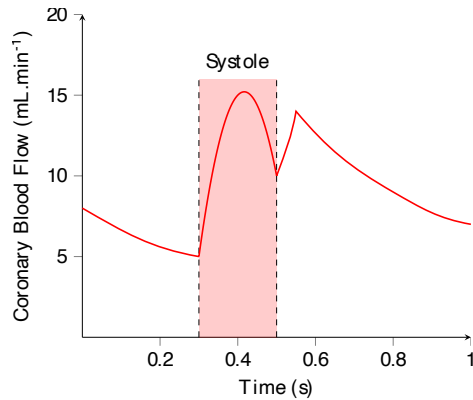
CBF is dependent on:

- Coronary vascular resistance
- Coronary perfusion pressure
The difference between aortic root pressure and the greater of RAP or intracavity pressure: i.e. $CBF = \frac{P_{Aorta} - P_{Cavity \text{ or } RAP}}{CVR}$.
 - Note that the pressure gradient is usually Aorta-Cavity rather than Aorta-RA
This is because the pressure in the ventricle acts as a Starling resistor - coronary flow is independent of RAP whilst $RAP < P_{Cavity}$
- Heart rate
LV CBF is affected in systole due to the changes in perfusion pressure, and compression of intramuscular vessels (causing an increase in CVR).
 - RV CBF is less affected, as the force of contraction is significantly smaller and a pressure gradient is maintained
 - Tachycardia reduces diastolic time and subsequently LV CBF

Left Ventricular coronary flow, demonstrating diastolic-dependent perfusion:



Right Ventricular coronary flow, demonstrating flow throughout the cardiac cycle:



Control of Coronary Blood Flow

CBF is controlled by:

- **Autonomic** mechanisms
 - **Direct** effects include:
 - Parasympathetic and sympathetic innervation of coronary vessels, with release of ACh or NA and A decreasing or increasing coronary blood flow
 - **Indirect** effects * Are more important than direct effects * Are related to autoregulation occurring with changing levels of myocardial work

in response to parasympathetic or sympathetic stimuli

- **Autoregulation**

- **Myogenic autoregulation**

This is common to many organ systems, and occurs within the coronaries.

- Increasing transmural pressure increases the leakiness of smooth muscle membranes, depolarising them
 - Resistance increases proportionally to pressure, such that flow remains constant

- **Metabolic autoregulation**

Anaerobic metabolism results in production of vasoactive mediators such as **lactate** and **adenosine**, which stimulate vasodilation and therefore increase flow (and oxygen delivery).

- This is the predominant means for autoregulation in the heart
 - Typical **myocardial oxygen extraction is 70%** and raising this further is difficult. Therefore, increasing oxygen supply requires an increase in blood flow.
-

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1. Hall, JE, and Guyton AC. Guyton and Hall Textbook of Medical Physiology. 11th Edition. Philadelphia, PA: Saunders Elsevier. 2011.
2. CICM July/September 2007
3. McMinn, RMH. Last's Anatomy: Regional and Applied. 9th Ed. Elsevier. 2003.
4. Coronary Artery Graph based on Coronary Arterial Circulation - es. 2/3/2013. (Image). By Addicted04 (Own work) CC BY 3.0, via Wikimedia Commons.

Cardiac Cycle

Describe the normal pressure and flow patterns (including velocity profiles) of the cardiac cycle with the electrical and ionic events

Discuss the physiological basis of electrical activity and its relationship to mechanical events including the:

- Physiological basis of the electrocardiograph in normal and common pathological states
- Factors that may influence cardiac electrical activity
- Correlation of the mechanical events of the cardiac cycle with the electrical and ionic events

The cardiac cycle:

- Describes sequence of events that occur in the heart over one beat
- Consists of **two phases** divided into **six stages**
- Typically is described as beginning in **late diastole** when the myocardium is relaxed and the ventricles are passively filling

Phases of the cardiac cycle:

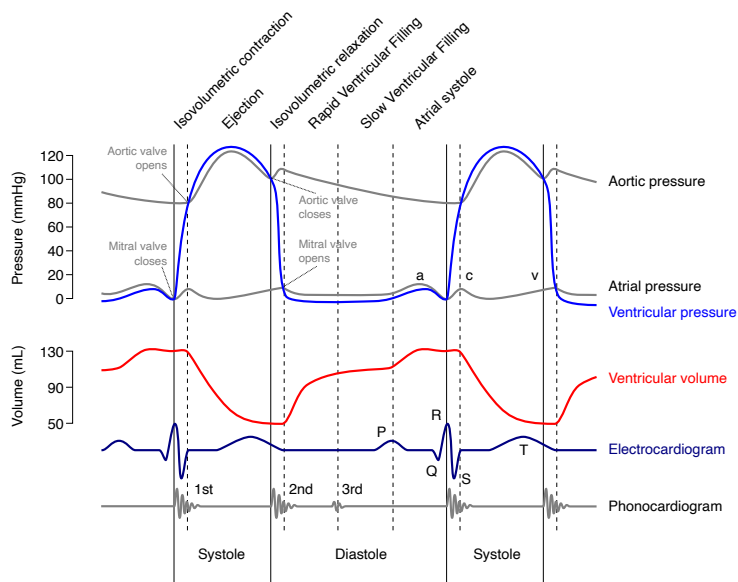
- Diastole
 - Isovolumetric Ventricular Relaxation

Slow ventricular filling is usually described as the beginning of the cycle.

- Rapid Ventricular Filling
- Slow Ventricular Filling (Diastasis)
- Atrial Contraction
- Systole
 - Isovolumetric Ventricular Contraction
 - Ejection

Phases of the Cardiac Cycle

Events during each phase of the cardiac cycle are represented on **Wigger's Diagram**:



Slow Ventricular Filling (Diastasis)

In slow ventricular filling:

- The AV valves are open and the semi-lunar valves are closed

- The ventricle is **relaxed completely** and fills slowly
The ventricles have been mostly filled during rapid ventricular filling and so the pressure gradient is reducing.
 - The pressure in each ventricle is almost zero
- **Arterial pressure is falling**, as it is end-diastole
- **CVP is slowly rising** as the ventricle and atria fill
This period occurs **after the y descent**.
- The **ECG** will show the beginnings of a **P-wave** at the end of this phase

Atrial Contraction

The atria contract, and remaining blood in the atria is ejected into the ventricle. This supplies **10%** of the ventricular **filling at rest**, but up to **40% in tachycardia**.

In atrial contraction:

- **Arterial pressure** is still **falling**
- The **CVP** waveform demonstrates the **a** wave as atrial contraction also causes blood to reflux into the SVC
- The **ECG** will show the **PR interval**

Isovolumetric Ventricular Contraction

Once the action potential passes through the AV node and bundle of His, ventricular contraction begins.

In isovolumetric contraction:

- Ventricular pressure rises, and the **AV valves close**
This gives rise to the first heart sound, S_1 .
 - As ventricular pressure is still less than systemic vascular pressure, the semilunar valves remain closed
- **Arterial pressure** is still **falling**

- The **CVP** waveform shows the **C** (closure) wave, as the tricuspid valve herniates back into the RA during ventricular contraction
There is a similar spike in LA pressure as the mitral valve also bulges back into the LA.
- The **ECG** will show the remainder of the QRS or the start of the QT interval
 - Atrial repolarisation occurs at this stage, but is typically masked by ventricular depolarisation

Ejection

When ventricular pressure exceeds arterial pressure, the semilunar valves open and ejection occurs. **Initial ejection is rapid**, but as ventricular pressure falls and systemic pressure rises the gradient falls ejection becomes slower.

During ejection:

- **Arterial pressure rises rapidly**, and is slightly less than ventricular pressure during this stage
- The **CVP** waveform shows the **x descent**, as the shortening RV pulls the RA down, rapidly lowering CVP
- The **ST segment** shows on the ECG as the ventricles are fully depolarised, though the **T wave** may appear in late ejection

Isovolumetric Relaxation

When contraction is complete, the ventricles begin to relax. **Inertia** means that ejection continues for a short time.

During isovolumetric relaxation:

- The **semilunar valves close**
This gives rise to the second heart sound, S_2 , and marks the beginning of isovolumetric relaxation.

- This occurs when ventricular pressure falls below vascular pressure
- **Arterial pressure begins to fall**, interrupted by the **di-crotic notch** which is a brief increase in arterial pressure as the semilunar valves close
- The **v wave** is visible on the CVP waveform
Due to atria filling against closed AV valves.
- The **end of the T wave** is visible on the ECG as ventricular repolarisation occurs

Rapid Ventricular Filling

Most of ventricular filling occurs in this phase. This is because in early ventricular diastole the ventricle is **still relaxing** and so a **pressure gradient is maintained** between the atria and ventricle.

During rapid ventricular filling:

- The **AV valves open** and ventricular filling occurs
This occurs when atrial pressure exceeds ventricular pressure.
 - **Arterial pressure is falling**
 - The **y descent** occurs when the AV valves open, causing a rapid drop in CVP as the ventricles fill
 - No electrical activity is produced - the ECG shows the TP interval
-

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1. Hall, JE, and Guyton AC. Guyton and Hall Textbook of Medical Physiology. 11th Edition. Philadelphia, PA: Saunders Elsevier. 2011.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

3. Wigger's Diagram (with some modifications) from Wigger's Diagram. 21/3/2012. (Image). By DanielChangMD (revised original work of DestinyQx); Redrawn as SVG by xavax. CC BY 3.0, via Wikimedia Commons.

Electrical Properties

Cardiac Action Potential

Explain the ionic basis of spontaneous electrical activity of cardiac muscle cells
Describe the normal and abnormal processes of cardiac excitation and electrical activity

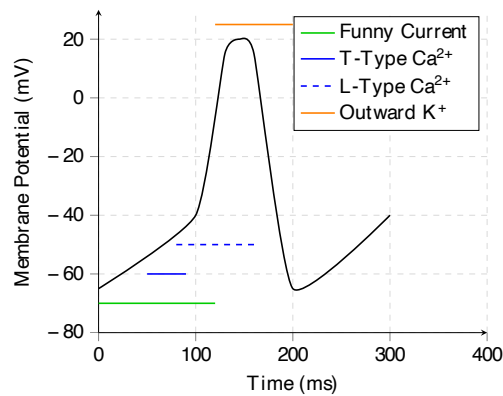
Discuss the physiological basis of electrical activity and its relationship to mechanical events including the:

- Ionic basis of automaticity the normal and abnormal processes of cardiac excitation

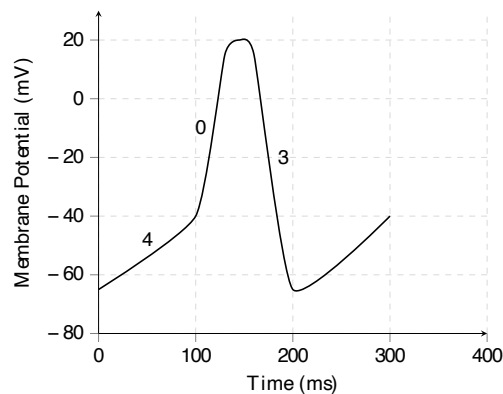
An action potential is a propagating change in the membrane potential of an excitable cell, used in cellular communication and to initiate intracellular processes. It is caused by altering the permeability of a membrane to different ions.

Pacemaker Potential

This pattern of electrical activity is seen in the SA and AV nodes. It **has no resting state**, and is continually depolarising.



Phases of the Pacemaker Potential



- **Phase 0**

Begins at the threshold potential of **-40mV**, with a peak membrane potential of **20mV**. Driven predominantly by the voltage-gated L-type (long-lasting) Ca^{2+} channels causing an influx of calcium ions.

- **Phase 3**

Repolarisation phase, which occurs as K^+ channels open and Ca^{2+} channels close. The nadir is called the **maximum diastolic potential** and is **-65mV**.

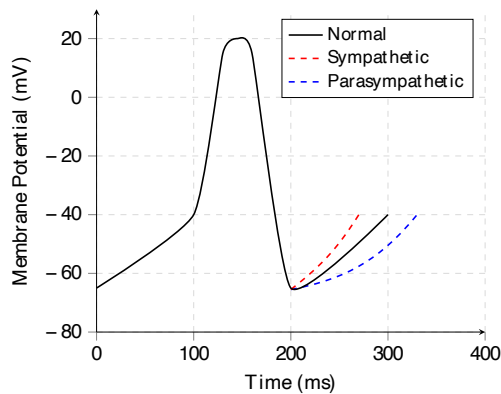
- **Phase 4**

Phase 4 consists of:

- The **funny current**

A steady influx of Na^+/K^+ which gradually depolarises the cell.

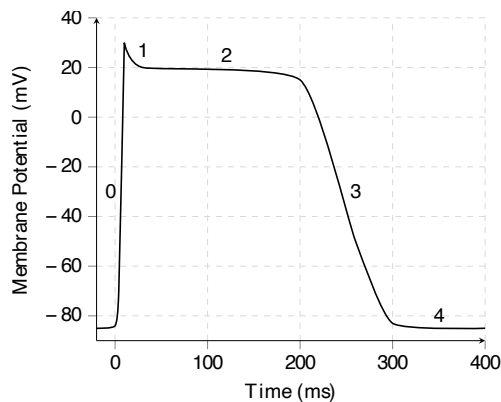
- Sympathetic stimulation increases the funny current, increasing the rate of depolarisation.
- Parasympathetic stimulation increases K^+ permeability, hyperpolarising the cell and flattens the gradient of phase 4.
- Calcium current
In phase 4, this is the transient calcium current, driven by T-type calcium channels. They open when the membrane potential reaches $\sim -50\text{mV}$, also causing depolarisation.



Ventricular Action Potential

To prevent tetanic contraction ventricular muscle has a long plateau prior to repolarisation, which lengthens the absolute refractory period to **250ms**. The relative refractory period is **50ms**.

(Tetanic ventricular contraction would be bad.)



Phases of the Ventricular Action Potential

- Phase 0: Depolarisation**
 At the threshold potential, voltage-gated **fast- Na^+** channels open briefly, causing depolarisation. The membrane potential peaks at **30mV**.
- Phase 1: Partial Repolarisation**
 The closure of Na^+ channels results in K^+ fleeing the cell down its electrochemical gradient, causing a slight drop in voltage called partial repolarisation.
- Phase 2: Plateau**
L-type Ca^{2+} channels open, causing a slow inward Ca^{2+} current which maintains depolarisation and facilitates muscle contraction.
- Phase 3: Repolarisation**
 Membrane permeability normalises, and outward potassium current returns the membrane potential to normal.
- Phase 4: Resting Potential**
 Membrane potential returns to its resting **-85mV**.

Propagation of the Cardiac Action Potential

Pacemaker cells:

- Are responsible for automaticity and rhythmicity of the heart
- The fastest pacemaker is the focus for myocardial conduction

This is typically the SA node.

- Should the SA node fail, the next fastest pacemaker will take over
- This provides an element of redundancy

Conduction pathway:

- **Atrial Conduction**

From the SA node, the impulse travels at $\sim 1\text{m.s}^{-1}$, depolarising the atria.

- Current travels down **Bachmann's Bundle**, which connects the right atrium to the left atrium

- **AV node**

The AV node is the only (normal) site of connection between the atria and ventricles. AV nodal cells:

- Transmits with a delay of 0.1s
This allows time for atrial contraction to finish before ventricular contraction begins.
- Have a prolonged refractory period and cannot conduct more than **220** impulses per minute
 - This period is prolonged by vagal stimulation, which increases potassium permeability and hyperpolarises the cell
 - Conversely, sympathetic stimulation increases calcium permeability and allows more rapid transmission
- Conducts via three pathways:
 - Bachmann Pathway
Also conducts to the LA.
 - Wenckebach pathway
 - Thorel pathway

- **Ventricular Conduction**

From the AV node, the signal propagates:

- Initially via the **Bundle of His** to the right and left bundles
- Secondly via the **Purkinje fibres** which conduct at $1 - 4\text{m.s}^{-1}$
Purkinje fibres have a long refractory period, and

spontaneously depolarise with an intrinsic rate of 30-40 bpm.

- Lastly, ventricular muscle is depolarised
Endocardium, papillary muscle and septum contract first, followed by apex, followed by the chambers.

Autonomic Control

• Parasympathetic Innervation

- SA node by the right vagus
There is continual PNS input ("Vagal tone") via inhibitory ACh GPCR, reducing the SA node from its intrinsic rate of 90-120bpm to a more sedate 60-100bpm.
- AV node by the left vagus
- The atria are innervated by parasympathetic neurons, whilst the ventricles are only minimally innervated
- PNS stimulation may have no direct effect on inotropy, instead acting indirectly via changes in chronotropy

• Sympathetic Innervation

- SNS activity causes release of noradrenaline (at post-ganglionic synapse) and adrenaline from adrenal medulla which stimulate cardiac β_1 receptors causing:
 - Positive chronotropy at the SA node
 - Positive inotropy at ventricular muscle
 - Positive lusitropy
 - Shorter action potential duration (due to opening of rectifying K^+ channels)
 - Increased AV conduction

Cardiac Transplant

The transplanted heart has no vagal/parasympathetic innervation but still expresses β_1 receptors, so it:

- Defaults to a resting heart rate of ~100bpm
 - Becomes highly preload dependent as it cannot respond quickly to changes in SVR
 - Not responsive to parasympatholytics (atropine, glycopyrrolate) or ephedrine (as this is indirectly-acting) to increase chronotropy - isoprenaline may be used
 - Gradual response to demands in exercise (lacks local SNS innervation, but will still respond to circulating catecholamines)
 - Increased sensitivity to catecholamines due to increased expression of β_1 receptors
-

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
3. Matsuura W, Sugimachi M, Kawada T, Sato T, Shishido T, Miyano H, Nakahara T, Ikeda Y, Alexander J Jr, Sunagawa K. Vagal stimulation decreases left ventricular contractility mainly through negative chronotropic effect. Am J Physiol. 1997 Aug;273.

Cardiac Output

Determinants of Cardiac Output

Define the components and determinants of cardiac output

Discuss the factors that determine and control cardiac output and the implications for clinical practice including:

- Preload, afterload and contractility
- The Frank-Starling mechanism
- Cardiac output and vascular function curves
- Pressure volume relationships in the heart

Cardiac output a function of Heart Rate (HR) and Stroke Volume (SV); $CO = HR \times SV$:

- **Heart rate** is fairly intuitive
- **Stroke volume** is defined as the difference between ESV and EDV, i.e. $SV = EDV - ESV$
Stroke volume is a function of three factors:
 - **Preload**
 - **Afterload**
 - **Contractility**

This section covers determinants of cardiac output from the perspective of the heart. The relationship to and importance of preload, afterload and contractility is covered in Chapter 10. Venous Return. For the purpose of the exam, it's good to have both a laboratory and a clinical definition. These definitions are those which have appeared in old examiner reports, or given to me by cardiac anaesthetists.

Preload

Preload is defined as the **myocardial sarcomere length just prior to contraction**:

- As this is not measurable without removing the heart and cutting it into tiny pieces, clinically it is usually **approximated by EDV** or, less appropriately, by EDP
 - EDV is typically calculated on echocardiography
 - EDP is typically measured using a CVC or PAC
 - CVP $\approx$ RVEDP
 - PCWP $\approx$ LVEDP

Determinants of Preload

Preload is a function of:

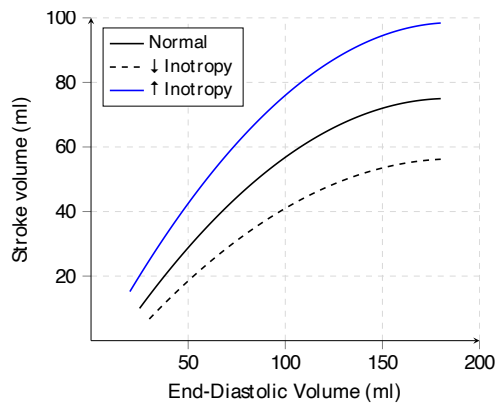
- **Venous Return**
 - Intrathoracic Pressure
 - MSFP
 - Venous compliance
A decrease in venous compliance will increase LVEDP.
 - Volume state
- **Ventricular compliance**
Reduced in diastolic dysfunction.
- **Pericardial compliance**
- **Valvular disease**
 - AV valve disease will **impair** preload
 - Semilunar valve disease will **increase** preload
- **Atrial kick**
- **Wall thickness**
Increased ventricular wall thickness decreases preload.
 - HOCM/Hypertrophy

Preload and the Respiratory Cycle

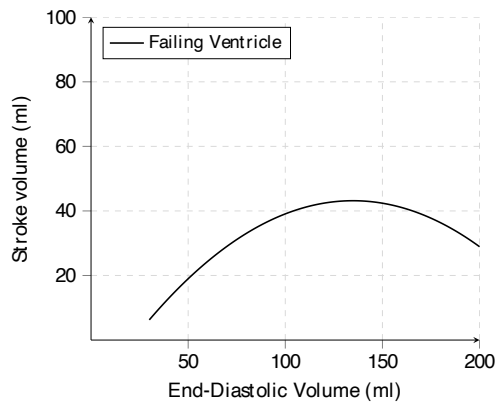
- Negative intrathoracic pressure causes RAP and PCWP to fall
- This increases RA filling, so and RVEDP and RVEDV increase relative to the pleural pressure (though absolute pressure is still low)
- LV effects are more variable
Negative intrapleural pressures:
 - Increase LV transmural pressure
This impairs ejection.
 - Cause bowing of the interventricular septum into the LV
This reduces LVEDV.

Frank-Starling Mechanism

- The **Frank-Starling Law of the Heart** states that the **strength of cardiac contraction is dependent on initial fibre length**
- At a cellular level, additional stretch increases:
 - The number of myofilament crossbridges that can interact
 - Myofilament Ca^{2+} sensitivity
- This law is represented by the **ventricular function curve**
Plot of preload against stroke volume (or cardiac output, assuming a constant heart rate).
 - Right shift of the curve demonstrates negative inotropy
 - Left shift of the curve demonstrates positive inotropy



The failing ventricle:



- In cardiac failure, the ventricle becomes overstretched. This reduces the number of overlapping crossbridges, reducing contractility.
- This is limited in the acute setting by constriction of the pericardium, which prevents excessive ventricular dilation.

Afterload

Afterload is the ventricular wall stress at the onset of systole. This is given by the **Law of Laplace** $\theta \propto \frac{P \times r}{T}$, where:

- θ is ventricular wall stress

- r is ventricular chamber radius
This is a proxy for ventricular size, or end-diastolic volume.
- P is ventricular transmural pressure
- T is ventricular wall thickness

End-Diastolic Volume

An increase in EDV increases ventricular radius and therefore wall tension.

Myocardial Wall Thickness

Increasing wall thickness (seen clinically as ventricular hypertrophy) *decreases* afterload by sharing wall tension (the product of pressure and radius) between a larger number of sarcomeres.

Ventricular Transmural Systolic Pressure

Transmural pressure is the difference between intrathoracic pressure and the ventricular cavity pressure during ejection. Transmural pressure is dependent on:

- **Intrathoracic Pressure**
Negative intrathoracic pressure will increase afterload, as the ventricle has to generate a greater change in pressure to achieve ejection.
 - PEEP reduces LV afterload
 - Negative-pressure ventilation with a high work of breathing increases afterload
This is why APO deteriorates - increased work of breathing increases LV afterload and worsens LV failure, increased pulmonary oedema, causing increased work of breathing...
- **Ventricular cavity pressure**
To facilitate ejection, the ventricle must overcome:

- **Outflow tract impedance**
 - Valvular disease
 - e.g. aortic stenosis
 - HOCM
- **Systemic arterial impedance/Aortic input impedance**
 Determined by resistance (SVR), reflected pressure waves, **inertia**, and **compliance**:
 - Determinants of **resistance** are stated in the Poiseuille Equation:

$$R = \frac{8\eta l}{\pi r^4}$$
 where:
 - η = **Viscosity**
 Affected by **haematocrit** (e.g. increased in polycythaemia)
 - l = Vessel length
 Essentially fixed.
 - r = Vessel **radius**
 - Greatest determinant
 - Function of **degree of vasoconstriction** of resistance vessels
 - Reflected pressure waves
 Reflection of pulse pressure waves from the distal circulation increases aortic pressure.
 - In normal circumstances this occurs during diastole (and so has less contribution to afterload)
 - Decreased compliance of the peripheral circulation increases the speed of propagation, so the pressure wave may return during systole and contribute to systolic blood pressure and afterload
 - **Inertia**
 - Given by the mass of blood in the column
 - Affected by **heart rate**
 - Arterial **compliance**
 Decreased arterial compliance increases afterload.
 - During ejection, the aorta and large arter-

- ies distend, reducing peak systolic pressure (impedance to further ejection)
- Decreased arterial compliance increases the change in pressure for any given volume, increasing afterload during ejection
 - Decreased arterial compliance increases the speed of propagation of **reflected pressures waves** returning to the aortic root
 - Wave arrival in diastole augments coronary blood flow
 - Wave arrival during systole further increases afterload
 - In diastole the arteries recoil and blood pressure and flow are maintained - the **Windkessel effect**.

Contractility

Contractility describes the **factors other than heart rate, preload, and afterload** that are **responsible for** changes in **myocardial performance**.

Determinants of Contractility

Contractility is primarily dependent on **intracellular Ca^{2+}** . Determinants include:

- Drugs
- Disease
 - Ischaemia
Reduced ATP production secondary to hypoxia, which impairs sarcoplasmic reticulum Ca^{2+} function. Further exacerbated by intracellular acidosis from anaerobic metabolism.

- Heart Failure
Impaired **contractility reserve**, i.e. minimal increase in contractility with sympathetic stimulation.
 - Reduced peak Ca^{2+} and sarcoplasmic reticulum uptake of Ca^{2+}
- Autonomic Tone
- Bowditch Effect
Contractility improves at faster heart rates. This is because the myocardium does not have time to remove calcium, so it accumulates intracellularly.
- Anrep Effect
Contractility increases as afterload increases.

Measuring Contractility

As with the other determinants of cardiac output, there has been some difficulty in developing measurable indices for contractility, as all are affected by preload or afterload to a greater or lesser degree. Those used include:

- **$\text{dP}/\text{dt}_{\text{max}}$ ($\frac{\Delta P}{\Delta T}$)**
The **rate of rise** of LVP, assuming a constant preload and afterload
 - This index is **preload dependent** but **afterload independent**
 - Typically, the $\text{dP}/\text{dt}_{\text{max}}$ in isovolumetric ventricular contraction is used
 - A greater rate of rise indicates a more forceful contraction
 - Measurement requires LV catheterisation
- **End-Systolic Pressure-Volume Relationship**
 - Uses the ventricular Pressure-Volume Relationship
 - Line plotted at the tangent to the curve from the **end-systolic point** (when isovolumetric ventricular relaxation begins)
 - The steeper the gradient the greater the contractility

- **Ejection Fraction**

Most common method used clinically is **ejection fraction**:

$$EF (\%) = \frac{SV}{EDV} \times 100 = \frac{EDV - ESV}{EDV} \times 100$$

Further Reading

- This site has a nice overview of wall tension, and the relationship of pressure to radius
 - This article discusses the wall stress definition for preload and afterload
 - Changes with ventilation are described with pretty graphs here
-

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1. Brandis K. The Physiology Viva: Questions & Answers. 2003.
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The use of wall stress for preload and afterload comes from this text, but is not used in the CICM texts.
4. Norton JM. Toward Consistent Definitions for Preload and Afterload. Advances in Physiology Education Mar 2001, 25 (1) 53–61.
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Venous Return

Define the components and determinants of cardiac output

The venous system has two key cardiovascular functions:

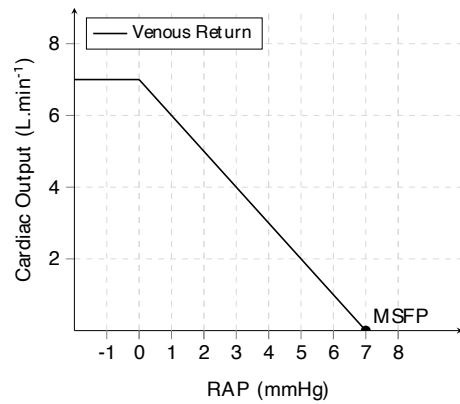
- **Blood reservoir**
Contains 65% of blood volume.
- **Conduit for return** of blood to the heart

Venous return is the rate at which blood is returned to the heart (in $\text{L}\cdot\text{min}^{-1}$). **At steady state**, venous return is equal to cardiac output, and can be expressed as $VR = \frac{MSFP - RAP}{RVR}$, where:

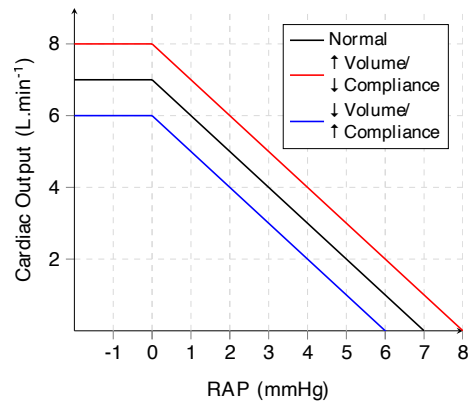
- VR is venous return
- $MSFP$ is the **mean systemic filling pressure**
This is the mean pressure of the circulation when there is no flow. It is an indicator of circulatory filling, and is a function of circulating volume and vascular compliance.
 - Normal mean systemic filling pressure is **~7mmHg**
- RAP is the right atrial pressure
An elevated RAP reduces venous return.
- RVR is the resistance to venous return

This relationship can be expressed graphically:

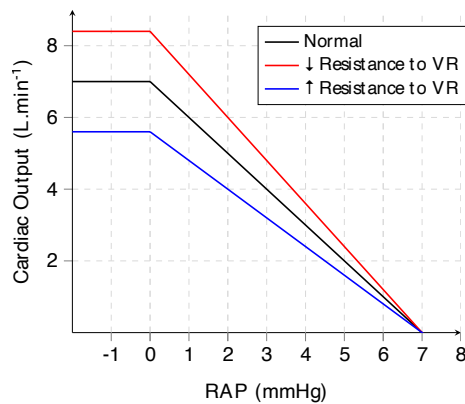
This section describes determinants of cardiac output from the perspective of venous return. The relationship to and importance of the determinants of cardiac output is covered in Chapter : Determinants of Cardiac Output.



- When venous return is 0, the measured right atrial pressure is an indication of mean systemic filling pressure



- Alterations to circulating volume and compliance affect both venous return and mean systemic filling pressure



- Alterations to the resistance to venous return affect venous return but mean systemic filling pressure is unchanged

Factors Affecting Venous Return

Venous return will be altered by any of the variables in the above equation:

- MSFP
 - Volume
e.g. Haemorrhage, resuscitation.
 - Compliance
- RAP
 - Respiratory pump
Negative intrathoracic pressure reduces RAP, improving venous return.
 - Positive pressure ventilation
 - Pericardial compliance
 - Constriction
 - Tamponade
- Resistance to Venous Return
 - Posture
 - Vascular compression
 - Obesity
 - Pregnancy

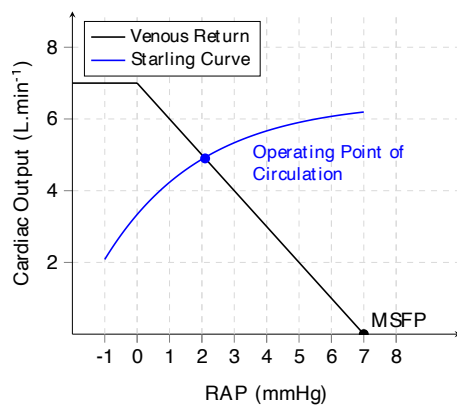
- Laparoscopy

Other factors affecting venous return include:

- Skeletal muscle pump
Contraction of leg muscles in combination with an intact venous system propels blood back towards the heart.

Interaction between Venous Return and Cardiac Function Curves

Guyton's curve can be superimposed on Starling's curve to examine the interaction between venous and cardiac function over a range of conditions:



References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. Brandis K. The Physiology Viva: Questions & Answers. 2003.

3. Henderson WR, Griesdale DE, Walley KR, Sheel AW. Clinical review: Guyton - the role of mean circulatory filling pressure and right atrial pressure in controlling cardiac output. *Critical Care*. 2010;14(6):243. doi:10.1186/cc9247.

Myocardial Oxygen Supply and Demand

Describe myocardial oxygen demand and supply, and the conditions that may alter each

Describe the factors determining myocardial oxygen supply and demand and their clinical implications

Myocardial:

- Oxygen **supply** is a function of **coronary blood flow**
- Oxygen **demand** is determined by **myocardial work**
- **Ischaemia** occurs when **demand exceeds supply**

Myocardial Oxygen Supply

Myocardial oxygen supply is dependent on three factors:

- Oxygen content of blood
- Oxygen extraction
- **Coronary artery flow**
Coronary flow is *the* determinant (and therefore the limiting factor) in normal physiology, as:
 - Oxygen content in individuals without pulmonary disease is maximal

- Resting myocardial oxygen extraction is near-maximal (~70%)
This high ER makes the heart **less tolerant of anaemia** than organs with a low ER.

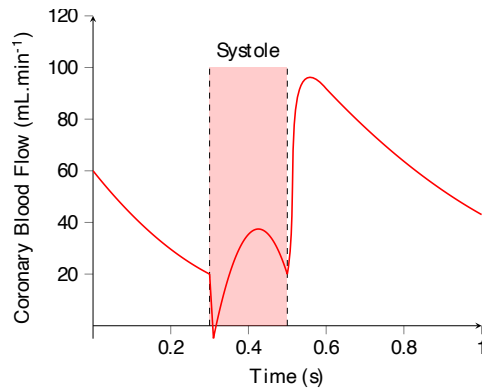
Coronary Blood Flow

Coronary blood flow is given by the equation:

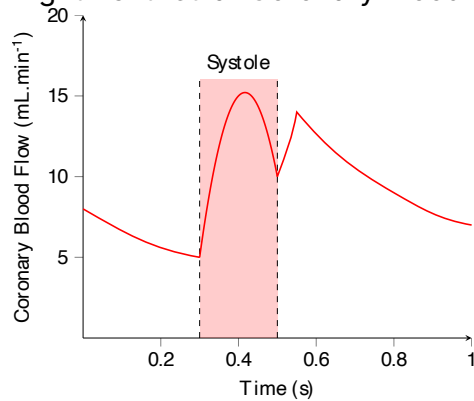
$$CBF = \frac{P_{Aortic\ Root} - P_{Cavity}}{Coronary\ Vascular\ Resistance}$$

Where:

- Aortic *root* pressure is the driving pressure for coronary flow
- Cavity (ventricular) pressure acts as a **Starling resistor** for coronary flow
 - Note that if RAP exceeds cavity pressure, RAP will be the pressure opposing coronary flow (due to downstream pressure at the coronary sinus)
- Note that **cavity and aortic root pressure change throughout the cardiac cycle**, therefore:
 - The flow to each ventricle is different during the cardiac cycle
 - The left ventricle is **best perfused in diastole**
Therefore **heart rate** is an important determinant of coronary blood flow, as **tachycardia will decrease coronary blood flow**
- Flow to each ventricle is a function of how relationships change over the cardiac cycle
 - Left Ventricular Coronary Blood Flow:



◦ Right Ventricular Coronary Blood Flow:



Myocardial Oxygen Demand

Normal myocardial oxygen consumption (MVO_2) is $21\text{--}27 \text{ mL} \cdot \text{min}^{-1}$. There are **five determinants**, which can be split into **three major** and **two minor** determinants.

Major Determinants

The **three major determinants** are:

- **Heart rate**

A change in heart rate will change the number of tension-generating cycles, causing a proportional change in MVO_2 .

- **Contractility**

Refers to the **rate** of tension development as well as its **magnitude**. Changing $\frac{\Delta P}{\Delta t}$ will change MVO_2 .

- **Ventricular wall tension**

Ventricular wall tension is **pressure work**, or the work done by the ventricle to **generate pressure but not to eject volume**.

- Wall tension is given by the **Law of LaPlace**

Wall Tension = $\frac{P \cdot r}{2}$, where:

- P = Pressure during contraction
- r = Radius

- Wall tension is therefore a function of:

- **Afterload**

Increasing afterload will increase the pressure during contraction.

- **Preload**

Increasing preload will increase radius, but to a **lesser extent than increasing afterload**.

- This is because *volume* and *radius* are *not directly proportional*

Minor Determinants

The **two minor determinants** of myocardial work are:

- **External work**

External work can also be thought of as **volume work**, or the energy expended to eject blood from the ventricle.

- This is encompassed by the area enclosed by the pressure-volume loop

- Conversely, internal work is defined as the work required to change the shape of the ventricle and prepare it for ejection

On the pressure-volume loop internal work is represented by a triangle between the point of 0 pressure and volume, the end systolic point, and the beginning of rapid ventricular filling.

- This is a minor determinant because the majority of ventricular work is generating the pressure required to eject blood, not actually move volume
- External work is of greater importance at high CO
- External work is used to calculate **cardiac efficiency**, given by the equation:

$$\text{Cardiac Efficiency} = \frac{\text{External Work}}{\text{Myocardial } O_2 \text{ Consumption}}$$

- **Basal oxygen consumption**

Basal oxygen consumption ($\sim 8 \text{ ml} \cdot \text{min}^{-1} \cdot 100 \text{ g}^{-1}$) comprises $\sim 25\%$ of MVO_2 .

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Pressure-Volume Relationships

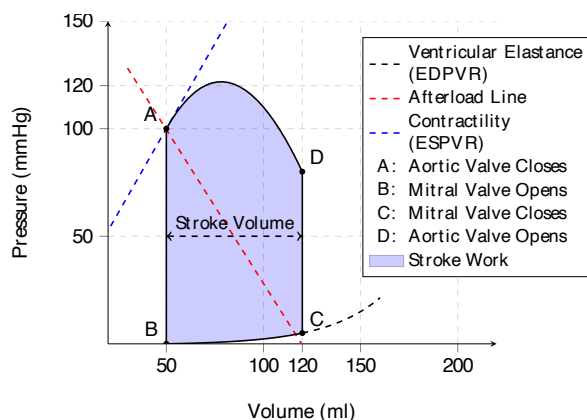
Describe the pressure-volume relationships of the ventricles and their clinical applications

Discuss the factors that determine and control cardiac output and the implications for clinical practice including:

- Pressure volume relationships in the heart

Pressure-volume loops are a graphical representation of the relationship between pressure and volume over the cardiac cycle. Time is **not directly** demonstrated, but the stages of the cardiac cycle can be inferred.

The **Left Ventricular P-V Loop** is the standard introduction to the topic:



- Stages of the cardiac cycle are indicated by the following intervals:
 - A-B is isovolumetric relaxation
Ventricular pressure is less than aortic pressure but greater than atrial pressure, so both mitral and aortic valves are closed.
 - B-C is rapid and slow ventricular filling, followed by atrial systole
Atrial systole is sometimes demonstrated by a sharp 'bump' towards C, as ventricular pressure will briefly rise out of proportion to ventricular volume
 - C-D is isovolumetric contraction
The ventricle contracts. As ventricular pressure is greater than atrial pressure but less than aortic pressure, the mitral valve closes (point C) and the aortic valve remains closed. Pressure increases without a change in volume.
 - D-A is ventricular ejection
When ventricular pressure exceeds aortic pressure, blood is ejected into the aorta and ventricular volume decreases.
- The slope of the line B-C gives the elastance of the ventricle
This is also known as the **End-Diastolic Pressure Volume Relationship (EDPVR)**, and is often (erroneously) referred to as ventricular compliance.
 - **Elastance** of the ventricle **increases as it is filled**
This is demonstrated by the dashed line.
 - The ventricle only overfills at high filling pressures
 - Increased elastance (such as in **diastolic dysfunction**) is demonstrated by an **increased slope** of this line, such that ventricular pressure will be higher at any given volume
Both ventricular and arterial elastance are low in normal circumstances (a state known as ventricular-arterial coupling), as this allows the ventricle to achieve a wide range of volume transfers in ejection with minimal change in filling

pressure.

- The horizontal distance between point B (ESV) and C (EDV) give the stroke volume

Ejection fraction can then be calculated.

- **Preload** is given by the EDV
- **Afterload** is:
 - *Technically given* by the pressure-volume relationship throughout the entirety of ejection i.e. the slope D-A.
 - This comes from La Place's law:

$$Afterload = \frac{P_{ejection} \times Radius_{ejection}}{2 \times Wall\ Thickness}$$
 - Usually *assumed to given* by the slope of a line drawn from the EDV (on the x-axis) to the **end-systolic point** (point A)

This is also known as the arterial elastance line.

- The gradient of the arterial elastance line can be worked out from the loops

$$E_A = \frac{Pressure_{End-Systole}}{Stroke\ Volume}$$
- This is different from the above formula because it only considers the pressure-volume relationship at end-systole, not throughout the entirety of ejection
- E_A is a good substitute for afterload because it is relatively independent of preload and contractility, and will vary with changes in afterload
 - i.e. For a given stroke volume, an increase in E_a leads to an increase in SBP. Similarly, if the ventricle is unable to maintain a given stroke volume as E_a increases, then SBP will fall.
- **Contractility** is given by the slope of the end-systolic pressure volume-relationship
 - Also known as elastance at end-systole, or E_{es} , and is given by the tangent to the curve at end-systole.
 - This measurement is not entirely independent of other factors, as it is influenced by afterload

Basic Pressure-Volume Loops

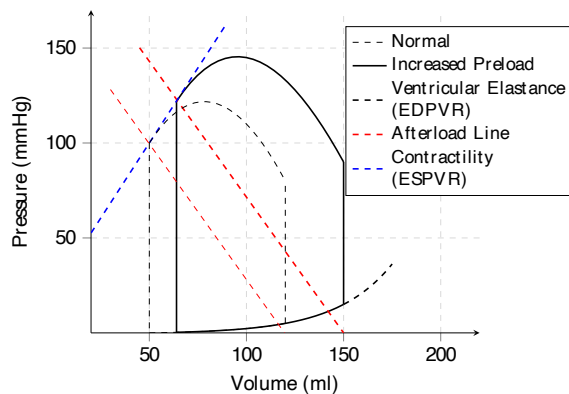
These loops:

- Show isolated changes to one factor only
- Are not accurate of real-world physiology

In reality:

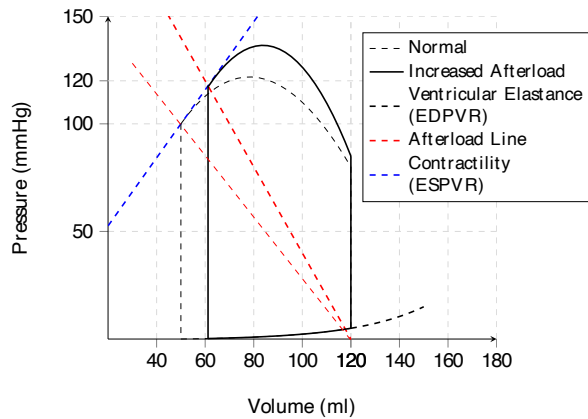
- Changing one factor will influence other factors
- These values change beat-to-beat

Increased Preload



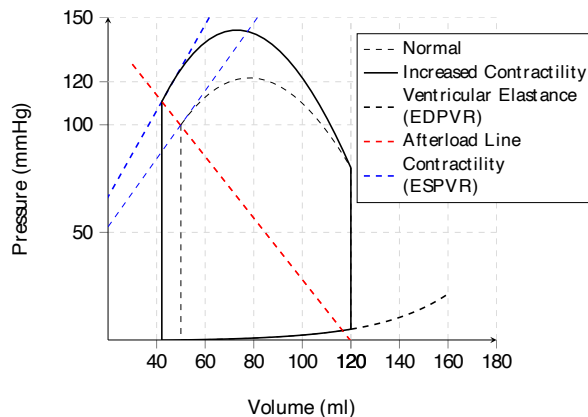
- EDV is increased, by definition
- The slope of the ESPVR remains unchanged (as contractility is unchanged)
- The slope of the afterload line (E_{ea}) is unchanged (as afterload is unchanged), but it is right-shifted due to the increased end-diastolic volume
- ESV is increased, though less than EDV, such that stroke volume increases

Increased Afterload



- EDV is unchanged (as preload is unchanged)
- The slope of the ESPVR remains unchanged (as contractility is unchanged)
- The slope of the afterload line (E_{ea}) has increased, but its x-intercept is unchanged
Note that the pressure-volume relationship throughout ejection is also steeper, and diastolic pressure has increased.
- ESV is increased, causing a reduction in stroke volume

Increased Contractility



- EDV is unchanged (as preload is unchanged)

- The slope and x-intercept of the afterload line (E_{ea}) is unchanged (as afterload is unchanged)
- The slope of the ESPVR has increased, though its x-intercept is the same
- ESV is decreased, causing an increase in stroke volume

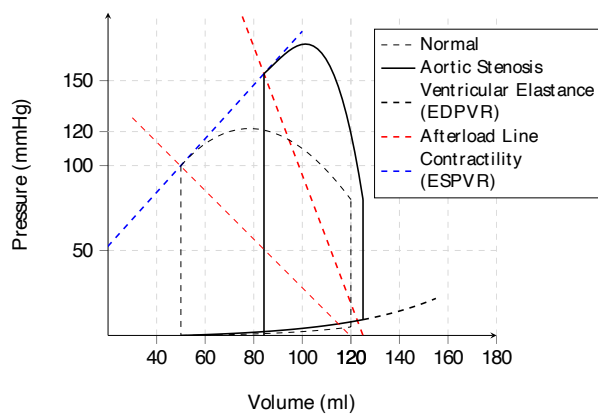
Advanced Pressure Volume Loops

The easiest way to approach more complicated pressure-volume loops is to address each of the basic factors before trying to draw the curve:

- How is preload changed?
- How is afterload changed?
- How is contractility changed?
- How are isovolumetric contraction and isovolumetric relaxation changed?

These show the loop for the primary physiological change, *without compensatory responses*:

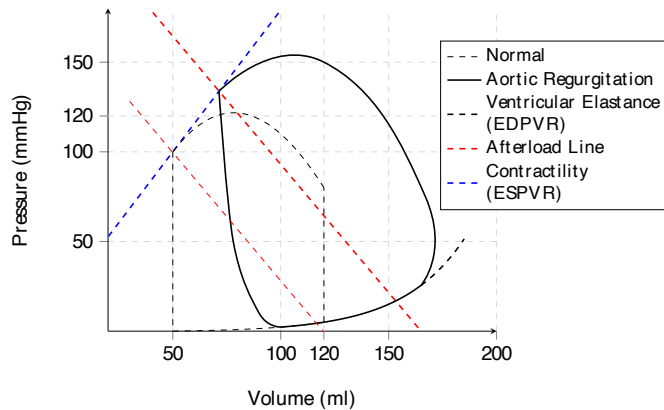
Aortic Stenosis



- Preload is increased due to the higher ESV, as the ventricle starts filling from a higher point

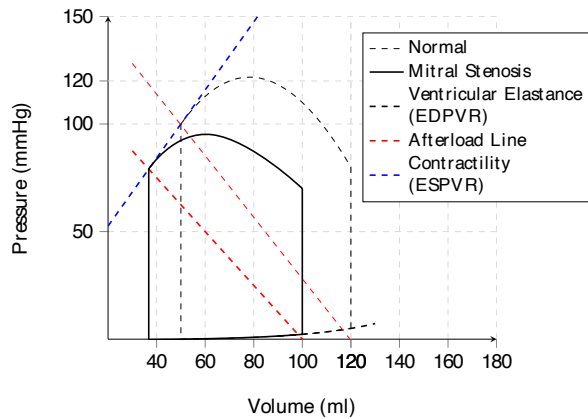
- Outflow tract impedance increases ventricular wall stress and therefore afterload
This leads to the decrease in stroke volume.
- Contractility is unchanged

Aortic Regurgitation



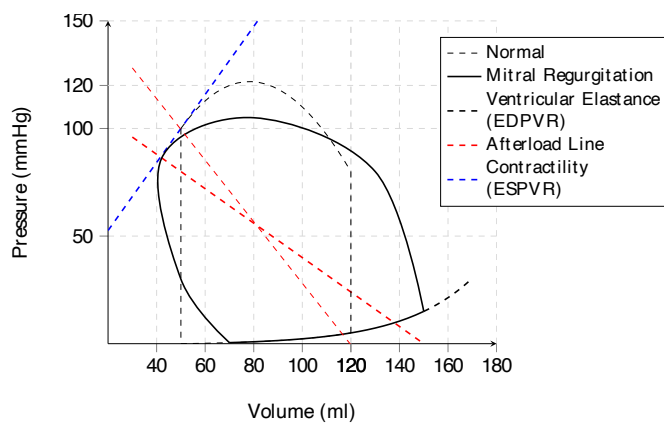
- Preload is dramatically increased as the ventricle fills from both the aorta and atria during diastole
- Afterload is increased due to the greater wall stress during ejection
- Contractility is unchanged
- There is no true isovolumetric relaxation, as the ventricle will begin to fill from the aorta at the completion of ejection
- Diastolic pressure is decreased and so the period of isovolumetric contraction is brief

Mitral Stenosis



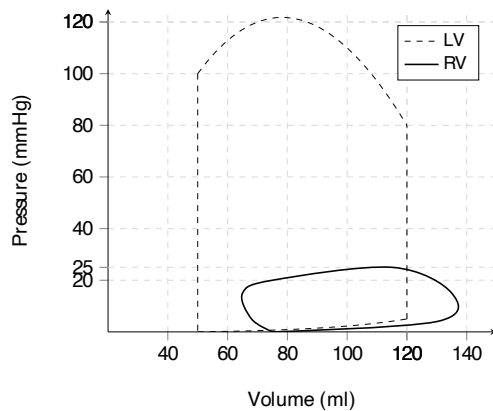
- Preload is reduced due to the increased gradient across the mitral valve
The effect of this is heart rate dependent, and will worsen as heart rate increases.
- Afterload is unchanged
Afterload may fall due to the reduction in ventricular wall stress.
- Contractility is unchanged
- ESV decreases (due to the reduced preload), though less than EDV, such that stroke volume is reduced

Mitral Regurgitation



- Preload is increased as the regurgitant volume increases left atrial pressure and therefore ventricular filling pressure
- Afterload is reduced as blood is ejected into the low-pressure atrial system
- Contractility is unchanged
- There is no true isovolumetric contraction phase as blood is ejected into the atria while ventricular pressure exceeds atrial pressure
- There is no true isovolumetric relaxation phase, as once atrial pressure exceeds ventricular pressure the ventricle will begin to fill
- *Apparent* stroke volume is increased due to the large difference between EDV and ESV, however *effective* stroke volume is reduced as only a portion of this is forward flow

Right Ventricular Pressure-Volume Loop



- The right ventricular curve is very different to the left ventricular curve
- RV preload is increased relative to LV preload
Note that stroke volume is the same (as both sides should have the same cardiac output).
- RV afterload is dramatically reduced due to the low-resistance pulmonary circulation

- Much of the RV ejection occurs after systolic pressure is reached
 - The right ventricle is very sensitive to changes in afterload
 - Contractility is reduced
Right heart contractility is partially dependent on co-ordinated contraction with the LV (particularly the septum), and therefore is decreased with LV systolic failure or conducting system disease (such as bundle branch block).
-

Further Reading

The Khan Academy series Changing the Pressure-Volume Loop is a fantastic introduction to the topic.

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3. Desai, R. Arterial elastance (Ea) and afterload. Khan Academy.
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Cardiac Reflexes

Describe the cardiac reflexes

Cardiac reflexes are fast-acting reflex loops between the CVS and CNS which contribute to the maintenance of cardiovascular haemostasis.

They include:

- **Baroreceptor reflex**
Aortic arch and carotid sinus reflexes.
- **Bainbridge reflex**
Atrial stretch receptor reflexes.
- **Chemoreceptor reflex**
Decreased $\text{PaO}_2 < 50\text{mmHg}$ or decreased pH sensed by peripheral chemoreceptors causes subsequent tachycardia and hypertension.
- **Cushing reflex**
Brainstem compression causes ischaemia of the vasomotor centre leading to Cushing's Triad:
 - Hypertension
May have a wide pulse pressure.
 - Bradycardia
Due to baroreceptor response from hypertension.
 - Irregular respirations
- **Bezold-Jarisch reflex**
Stimulation of C fibres of the vagus nerve in the cardiopulmonary region.
 - This causes:
 - Significant bradycardia
 - Hypotension
 - Apnoea, followed by rapid shallow breathing.

- These fibres can be stimulated by a number of substances, including:
 - Capsaicin
 - Serotonin
 - Those produced in myocardial ischaemia
 - **Oculocardiac reflex**

Pressure on the globe or traction on ocular muscles causes a decrease in heart rate. This is mediated by the:

 - Trigeminal nerve (afferent limb)
 - Vagus nerve (efferent limb)

Increased vagal tone reduces SA nodal activity.
-

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Peripheral Circulation

Starling Forces

Describe the essential features of the micro-circulation including fluid exchange (Starling forces) and control mechanisms present in the pre- and post-capillary sphincters

Interstitial fluid is an ultrafiltrate of plasma, with the net filtration pressure determined by the net effect of opposing hydrostatic and oncotic pressures:

$$NFP = P_{Capillary\ Hydrostatic} - P_{Interstitial\ Hydrostatic} - (P_{Capillary\ Oncotic} - P_{Interstitial\ Oncotic})$$

These four variables are known as **Starling's forces**.

Actual fluid movement is (of course) more complicated. **Hydrostatic pressure falls along the capillary**, and movement of solute and water are affected by other factors. Some of these are described by the:

- **Reflection coefficient (σ)**
This describes the fact that a small amount of protein leaks from the capillary, slightly increasing interstitial oncotic pressure and slightly decreasing capillary oncotic pressure. It is dependent on the interstitial protein content, and has a value between 0 and 1.
- **Filtration coefficient (K_f)**
Encompasses membrane permeability (to water) and membrane surface area. Varies between tissues:

The **Starling Equation** becomes:

$$\text{Net Fluid Movement} = K_f(P_{CH} - P_{IH} - \theta(P_{CO} - P_{IO}))$$

Table 1: Typical Values for Starling Pressures (mmHg)

	Arteriolar End	Venous End
Capillary Hydrostatic Pressure	25	10
Interstitial Hydrostatic Pressure	-6	-6
Capillary Oncotic Pressure	25	25
Interstitial Oncotic Pressure	5	5

Organ-Specific Values

In the glomerulus:

- Reflection coefficient is close to 1 due to the impermeability of the glomerulus to protein
- K_f is high due to both high permeability and a large surface area.
- Hydrostatic pressure is high
- Glomerular oncotic pressure is essentially 0

In the liver:

- Reflection coefficient is close to 0 in hepatic sinusoids as they are very permeable to protein

In the lungs:

- Reflection coefficient of ~0.5 in the lungs due to significant leak of protein
 - Protein leak decreases as interstitial oncotic pressure rises, limiting further oedema formation

- The oncotic pressure gradient is small, and favours reabsorption
- Hydrostatic pressure gradient is small, but favours extravasation of fluid
 - Interstitial hydrostatic pressure becomes more negative closer to the hilum, drawing fluid into the pulmonary lymphatics

Causes of Oedema

Oedema can be localised or generalised, and in both cases caused by:

- **Increased filtration pressure** Occurs when capillary hydrostatic pressure exceeds interstitial hydrostatic pressure. Causes:
 - Increased Venous pressure
This includes an increase in CVP:
 - CCF
 - TR
 - Increased venoconstriction
 - Increased MSFP
 - Impaired venous return
 - Obstruction
 - Respiratory muscle pump
 - Skeletal muscle pump
 - Positioning
- **Decreased oncotic pressure gradient**
 - Decreased plasma protein
 - Hepatic failure
 - Critical Illness
 - Increased interstitial oncotic pressure
 - Mannitol/starch extravasation
- **Increased capillary permeability**
 - Inflammatory proteins
 - Substance P
 - Histamine
 - Kinins
- **Inadequate Lymph Flow**

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1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. Brandis, K. Starling's Hypothesis. Anaesthesia MCQ.
3. ANZCA August/September 2001

Variations in Blood Pressure

Describe the physiological factors that may contribute to pulse variations in blood pressure

Blood pressure is not uniform throughout the circulation. Ventricular ejection generates two waves:

- A **blood flow wave**
Travels at $\sim 20\text{cm.s}^{-1}$.
- An **arterial pressure wave**
Distends the elastic walls of the large arteries during systole, which then recoil during diastole to facilitate continual blood flow.
 - This wave travels at 4m.s^{-1}
 - **This is** what is **felt** when **pulses** are **palpated**, and what is **seen** on the **arterial line** waveform

This recoil is known as the **Windkessel effect**.

Key pressures measured are:

- **Systolic blood pressure**
Maximal pressure generated during ejection.
 - Determined by:
 - Stroke volume
 - Systolic time
 - Arterial compliance
 - Reflected pressure wave
 - Relevant for:
 - Bleeding
- Clot disruption
- Aneurysmal wall pressure

- **Diastolic pressure**

Pressure exerted by the circulation upon the aortic valve.

- Determined by:
 - Circulatory compliance
 - Circulating volume
 - Aortic valve (in)competence
- Relevant for:
 - Coronary perfusion

- **Mean arterial pressure**

Average pressure in the circulation throughout the cardiac cycle, as measured by the **area under the curve** of the arterial line waveform.

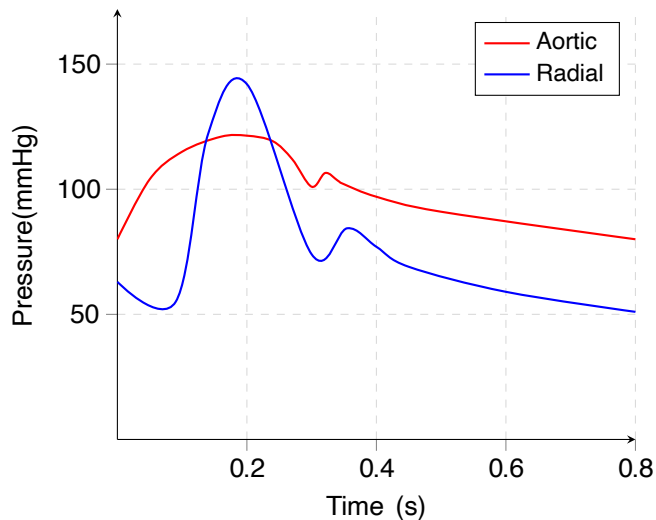
- Determined by:
 - Systolic blood pressure
 - Diastolic blood pressure
 - **Heart rate**
Increasing HR will tend to increase MAP, as overall systolic time (and therefore time spent at higher pressure) is increased.
 - Shape of the arterial waveform/diastolic runoff
The slow decrease in pressure after peak systolic pressure represent elastic recoil of large arteries, increasing the pressure driving blood into the peripheral circulation. A longer diastolic runoff period leads to a larger area under the curve, and a higher MAP.
- Relevant for:
 - Organ perfusion

Changes by Site of Measurement

Measured pressure changes predictably at more distal sites:

- All gradients are increased
Arterial upstroke and falloff are both steeper.

- The SBP increases
- DBP decreases
- **MAP is constant**
- The dicrotic notch occurs later and becomes less sharp
This occurs due to reflections in arterial pressure waves.



Respiratory Variation

Ventilation causes variation in peak systolic pressure due to dynamic changes in cardiac loading conditions:

- **Negative pressure respiration** (i.e. regular breathing) generates a negative intrathoracic pressure during inspiration
This:
 - **Augments right** ventricular function
 - Increased VR
Via negative intrathoracic pressure.
 - Increases RVEDV
RV moves up the Starling curve.
 - **Impedes left** ventricular function
 - Pooling of blood in the pulmonary circulation

- Decreased LVEDV
LV filling restricted by increased RV EDV; this is an example of **intraventricular interdependence**.
- Increased afterload
Negative intrathoracic pressure increases LV transmural pressure, increasing wall tension, and therefore afterload.
- Decrease in SV and SBP
- **Positive pressure ventilation** causes *generally* opposite effects Increased intrathoracic pressure during inspiration:
 - Decreases VR but increases LV filling via compression of the pulmonary circulation
 - Note that LV afterload is reduced throughout the respiratory cycle with the application of PEEP
- When this change is **>10mmHg**, it is known as **pulsus paradoxus**
- The magnitude of this effect varies with:
 - Magnitude of intrathoracic pressure change
Large changes in intrathoracic pressure cause correspondingly larger changes in ventricular filling.
 - Other factors affecting cardiovascular function
 - Preload
 - Volume state
 - Compliance
 - Pericardial compliance
 - Constriction
 - Tamponade
 - Cardiac compliance
 - Diastolic dysfunction
 - Afterload
 - PE
 - Raised intrathoracic pressure
 - PEEP
 - Tension PTHx
- These differences can be measured:
 - Qualitatively
By looking at respiratory swing on an arterial line or plethysmograph; or by palpation.

- Quantitatively
Using **pulse pressure** or **stroke volume variation**.

Pulse Pressure Variation

Describes the variation in pulse pressure over the course of a respiratory cycle. Pulse pressure variation is:

- Mathematically defined as $PPV = \frac{PP_{max} - PP_{min}}{PP_{mean}} \times 100$,
where:
 - PPV is pulse pressure variation
 - PP is Pulse Pressure
- Calculated as a **percent**
- Used as an indicator of **fluid responsiveness**
 - Patients higher on the **Frank-Starling curve** will have less change in stroke volume with an increase in preload, and therefore:
 - Reduced PPV
 - Be less fluid responsive
 - A PPV of **>12%** suggests volume responsiveness
- Reliant on several assumptions:
 - Regular sinus rhythm
Irregular heart rates (particularly AF) lead to significant alterations in ventricular filling and therefore pulse pressure, independent of the respiratory cycle.
 - Controlled mechanical ventilation
No spontaneous efforts.
 - Adequate tidal volumes
Must be $>8\text{mL.kg}^{-1}$.
 - Normal chest wall compliance
Requires a closed chest.

Note that this does not necessarily mean a fluid responsive patient needs fluid, or that fluid will not be harmful.

Stroke Volume Variation

SVV is:

- Alternately defined as:
 - The percent change in stroke volume during in-spiration and expiration over the previous **20 seconds**
 - Variation of beat-to-beat SV from the **mean value** over the previous 20 seconds
$$SVV = \frac{SV_{max} - SV_{min}}{SV_{mean}} \times 100$$
- Calculated by specialised devices from an invasive arterial waveform
Calculation incorporates:
 - Pulse pressure
 - Vascular compliance
Estimated from nomograms based on patient age, gender, height, and weight.
 - Vascular resistance
Estimated from arterial waveform shape.
- An alternative to PPV in measuring fluid responsiveness
Relies on similar principles.
- Probably less specific but more sensitive than PPV for identifying fluid responders

Circulatory Factors

Changes in circulatory function:

- **Inotropy**
The rate of systolic upstroke is related to $\frac{\Delta P}{\Delta t}$, and therefore contractility.
- **SVR**
The gradient between the peak systolic pressure and the dicrotic notch gives an indication of SVR. E.g., a steep downstroke suggests a low SVR, as the pressure in the circulation rapidly falls when ejection ceases.
- **Preload**
A beat-to-beat variation is seen with the respiratory cycle, due to the change in preload occurring with changes in intrathoracic pressure.

Pathological Changes

Some pathological causes include:

- **Aortic Stenosis**

Causes a reduction in:

- Pulse pressure
Due to reduced stroke volume.
- Gradient of upstroke
Due to reduced stroke volume.

- **Aortic Regurgitation**

- Wide pulse pressure

Combination of:

- Increased SBP due to the increased force of ejection due to increased preload (Starlings Law), which occurs due to high ESV
 - Decreased DBP due to part of the stroke volume flowing back into the ventricle through the incompetent valve
-

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Pulmonary Circulation

Outline the anatomy of the pulmonary and bronchial circulations

Describe the physiological features of the pulmonary circulation and its resistance

Understand the differences between the pulmonary and systemic circulation

The pulmonary circulation is:

- A **low-pressure, high-flow**, high-pulsatility circulation
- Supplied by the pulmonary trunk (pressure 25/8 mmHg), driven by the RV (pressure 25/0 mmHg)
 - Arteries and veins run with the bronchi as far as the terminal bronchioles, dividing at the same points
 - Beyond this, they form a capillary bed so thin it is essentially sheet of flowing blood punctuated by alveoli

The **bronchial circulation**:

- Arises from the systemic circulation, and supplies blood to the conducting zone of the lung
- A third drains back to the systemic circulation
- The remainder drains into the pulmonary vessels - this is a **physiologic shunt**
 - Supply to tumours is predominantly from the bronchial circulation (rather than the pulmonary circulation) as these vessels respond to angiogenic factors.

Differences between Pulmonary and Systemic Circulations

Blood Pressure

Normal pulmonary arterial pressure is 25/8 mmHg (MAP 15 mmHg) compared to 120/80 mmHg (MAP 100 mmHg) in the systemic circulation¹. This is because the systemic circulation must:

- **Regulate flow** to different organs at different times
It therefore contains resistance vessels which allow it to allocate cardiac output accordingly.
- **Maintain flow** to organs far above the heart

Conversely, the pulmonary circulation must:

- **Accept the entirety of cardiac output**, with little capacity to regulate flow (hypoxic vasoconstriction being the exception)
- **Minimise extravasation** of fluid
 - As per Starling's Law, fluid movement out of the capillary is given by the difference in hydrostatic gradients and oncotic gradients
 - The net oncotic gradient is small (but favours reabsorption), however the pulmonary interstitium has **no hydrostatic pressure**
 - Increased pulmonary capillary pressure therefore causes extravasation of large volumes of fluid
- Consequently, pulmonary vessels are thin walled and contain minimal smooth muscle

¹According to Ohm's Law, pressure is a function of both flow and resistance:

$$\frac{\Delta P = Resistance}{Flow} =$$

As the LV CO = RV CO in normal circumstances, the difference between pulmonary and systemic pressures is due to the relative difference between PVR and SVR.

Similarly, an increase (or decrease) in CO will also alter systemic and pulmonary blood pressures, assuming a constant resistance.

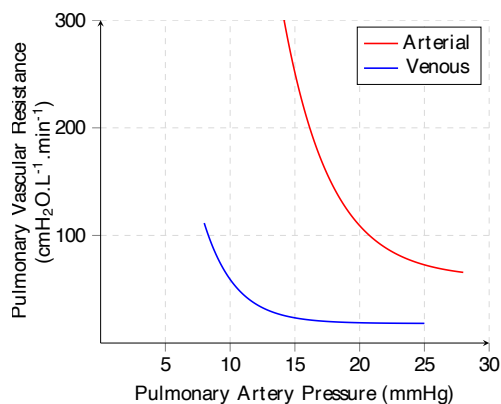
- This makes the pulmonary circulation **highly compliant** - the volume of blood is able to change substantially with minimal change in pressure

Pulmonary Vascular Resistance

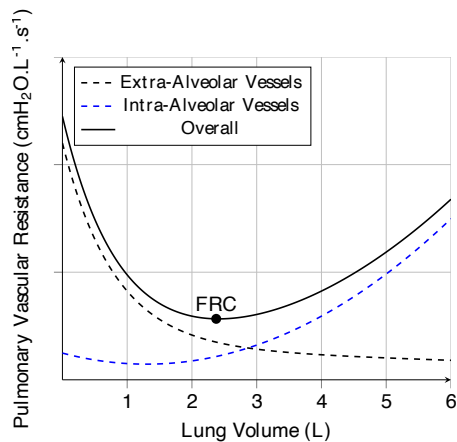
Normal PVR is $\sim 1/10^{\text{th}}$ that of the systemic circulation, and is a function of:

- **Pulmonary Artery Pressure**
Increased PAP causes a decrease in PVR. This occurs because:
 - Previously closed pulmonary capillaries are re-recruited when their **critical opening pressure** is reached
This is more important when MPAP is low.
 - Vessels distend at higher pressures
This is more important when MPAP is high.

This is because the pressure drop across the pulmonary circulation is 10mmHg (MPAP - LAP), $\sim 1/10^{\text{th}}$ that of the systemic circulation, and flow is the same.



- **Lung volume** Lung volume has a variable effect on PVR.
 - At large lung volumes:
 - Resistance in large extra-alveolar vessels decreases as the vessels are pulled opening by distension of elastic tissues
 - Resistance in small intra-alveolar vessels increases as they are compressed by the high lung volumes
 - At small lung volumes, the reverse occurs



• Hypoxic Pulmonary Vasoconstriction

Low PAO_2 causes a vasoconstriction in the vessels supplying that alveolus, increasing PVR and directing blood to better ventilated alveoli.

- Low alveolar PO_2 is the primary determinant
- Low mixed venous PO_2 also contributes
- Constriction begins when $\text{P}_\text{A}\text{O}_2$ falls below 100mmHg, and becomes dramatic below 70mmHg
- This is important in:
 - Foetal circulation
 - Alveolar consolidation
 - Pneumonia
 - Cardiogenic pulmonary oedema
- HPV is attenuated by:
 - Elevated LAP
Greater than 25mmHg.
 - High CO

• Minor factors which affect PVR:

- **Increase PVR:**
 - Hypercarbia

- Hypothermia
- Acidaemia
- Pain
- **Decrease** PVR:
 - Bronchodilators
 - Volatiles

Response to Substances

Oxygen:

- The pulmonary circulation constricts when PO_2 falls, whilst the systemic circulation dilates

Carbon Dioxide:

- The pulmonary circulation constricts when PCO_2 rises, whilst the systemic circulation dilates

Distribution of Pulmonary Flow

Gravity has a significant effect on pulmonary blood flow:

- In the upright lung, flow decreases almost linearly with height
 - In the supine lung, flow to posterior regions exceeds that of anterior regions
- This occurs due to the low driving pressure of the pulmonary circulation, which means gravity has a much more significant effect on pulmonary blood flow than systemic blood flow.

Due to difference in lung size:

- The **right lung** receives **~55%** of cardiac output
- The **left lung** receives **~45%** of cardiac output

West's Zones

The lung is divided into four zones, based on the relationship between alveolar and vascular pressures:

- **West's Zone 1**

In West's Zone 1, $P_A > P_a > P_v$.

- This should not occur in normal conditions, because a normal pulmonary artery pressure is normally (just) sufficient
This is because in the upright lung, the hydrostatic pressure difference will be about 30cmH₂O.
- However, if alveolar pressure is raised (e.g. IPPV), or arterial pressure falls (shock), there may be a region where **alveolar pressure exceeds arterial pressure**

- **West's Zone 2**

In West's Zone 2, $P_a > P_A > P_v$.

- Here, flow is determined by the arterial-alveolar pressure gradient rather than the arterial-venous gradient
Alveolar pressure acts as a **Starling Resistor**, where **flow is independent of downstream pressure**.

- **West's Zone 3**

Occurs when alveolar pressure falls below venous pressure, i.e. $P_a > P_v > P_A$. Flow is dependent on the arterial-venous pressure gradient. Capillary pressure increases along their length, increasing transmural pressure and mean width.

- **West's Zone 4**

Occurs at low lung volumes, as extra-alveolar vessels collapse and shunt occurs. The interstitium is acting as a Starling resistor, which can be expressed as: **$P_a > P_{int} > P_v > P_A$** .

Hypoxic Pulmonary Vasoconstriction

As discussed above, HPV allows redirection of blood flow from poorly ventilated regions of the lung, and so improve V/Q matching. HPV is relevant in disease states, as well as specific physiologic circumstances:

- At high altitude, the PAO_2 is globally reduced, leading to high pulmonary artery pressures
- In utero, PAO_2 is negligible, and PVR is therefore very high

This diverts blood from the pulmonary circulation into the left side of the heart via the foramen ovale. When the first breath is taken, pulmonary vessels dilate and the right-to-left shunt is reversed.

References

1. Dunn, PF. Physiology of the Lateral Decubitus Position and One-Lung Ventilation. Thoracic Anaesthesia. Volume 38(1), Winter 2000, pp 25-53.
2. West J. Respiratory Physiology: The Essentials. 9th Edition. Lippincott Williams and Wilkins. 2011.
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Cerebral Blood Flow

Describe the distribution of blood volume and flow in the various regional circulations and explain the factors that influence them, including autoregulation. These include, but not limited to, the cerebral and spinal cord, ~~hepatic and splanchnic, coronary, renal and utero-placental~~ circulations

With respect to cerebral blood flow:

- Normal CBF is **$\sim 750 \text{ ml} \cdot \text{min}^{-1}$** or $\sim 15\%$ of resting cardiac output
Note that the brain makes up only $\sim 2\%$ of body weight.
 - A relatively high blood flow is required due to the high **cerebral metabolic rate for oxygen** (CMRO_2) of **$50 \text{ ml} \cdot \text{min}^{-1}$**
- The brain is sensitive to interruptions in flow as it has:
 - A high metabolic rate
 - No capacity to store energy substrates

The factors affecting cerebral blood flow can be classified by the factors in the **Hagan - Poiseuille Equation**:

$$CBF \approx \frac{\Delta P \pi R^4}{8 \eta L}$$

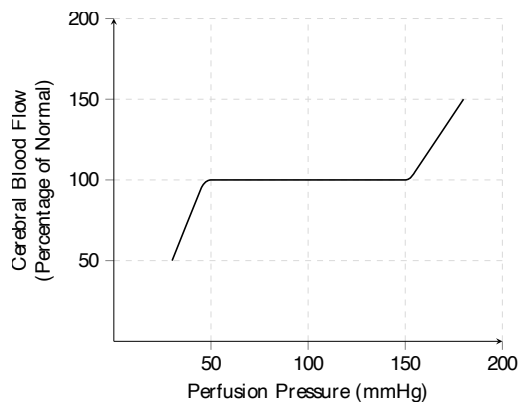
Where:

- ΔP is the pressure difference driving flow, i.e. CPP
 - R is the radius of the blood vessels
 - η is the blood viscosity
- These are also called rheologic factors.

- L is the length of the tube, a fixed quantity

Factors Affecting Perfusion Pressure

- Cerebral Perfusion Pressure is the difference between mean arterial pressure and intracranial pressure: $CPP = MAP - ICP$
- A **normal CPP** is **~80mmHg**
- In normal individuals, CBF is classically thought to be autoregulated over a **CPP range of 60-160mmHg**
 - This occurs by **myogenic means**, similar to the kidney
 - In normal circumstances, this is dependent on MAP (i.e., with a normal ICP < 10mmHg, CBF is regulated over a MAP range of 50-150mmHg).
- Note that more **recent evidence** would suggest that CBF is autoregulated over a much narrower range of perfusion pressures, and has a greater capacity to buffer an increased rather than decreased perfusion pressure



- At the lower limit, the reduced perfusion pressure means flow cannot be maintained even with maximal vasodilation
- At the upper limit, the high perfusion pressure overcomes maximal vasoconstriction

- Additionally, the increased CBF may result in damage to the blood-brain barrier
- The curve is **left-shifted in neonates and children** (due to lower normal MAP)
- The curve is **right-shifted in chronic hypertension**
- The curve is probably inaccurate in the pathological conditions where it would otherwise be useful, such as malignancy, subarachnoid haemorrhage, CVA, or TBI
 - This may be due to damage to either the feedback mechanisms, or the effectors (vasculature)
 - **Flow may become pressure-dependent**, and small changes in MAP can have large changes in CBF

Factors Affecting Vessel Radius

Vasodilation and constriction affect both cerebral blood flow and ICP, as vasodilatation increases cerebral blood volume and therefore may increase ICP through the Monroe-Kellie doctrine.

Vessel calibre is affected primarily by five factors:

- Cerebral metabolism
- **PaCO₂**
- **PaO₂**
- Neurohormonal factors
- Temperature

Cerebral Metabolism

Cerebral metabolism (typically given by the cerebral metabolic requirement for oxygen, CMRO₂) has a linear association with cerebral blood flow - this is known as **flow-metabolism coupling**. This is controlled locally through the release of vasoactive mediators, such as H⁺,

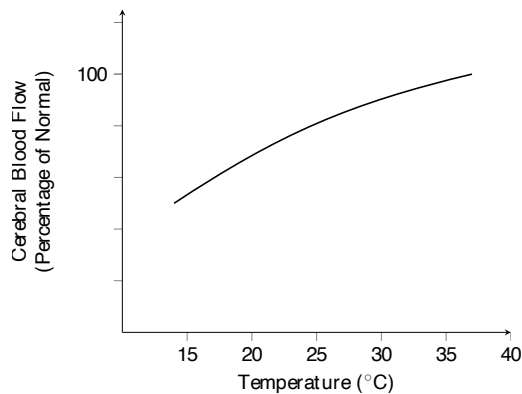
adenosine, and NO. Determinants of cerebral metabolism include:

- **Drugs**

Cerebral metabolism may be decreased by use of drugs such as benzodiazepines, barbiturates, and propofol.

- **Temperature**

CMRO₂ decreases linearly by ~7% per degree centigrade, allowing prolonged periods of reduced CBF without ischaemic complications.



PaCO₂

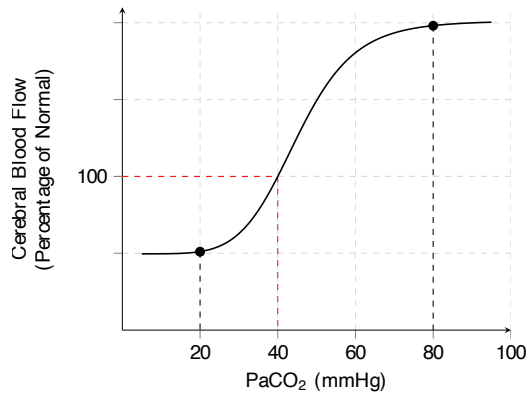
Carbon dioxide acts as a cerebral vasodilator.

- Between **20mmHg and 80mmHg**, CBF is almost linear
 - Above 80mmHg, the circulation is maximally dilated
 - Below 20mmHg, the circulation is maximally constricted

Additionally, the alkalosis causes a left-shift of the oxyhaemoglobin curve. This reduces off-loading of oxygen, causing hypoxia and subsequent vasodilation.

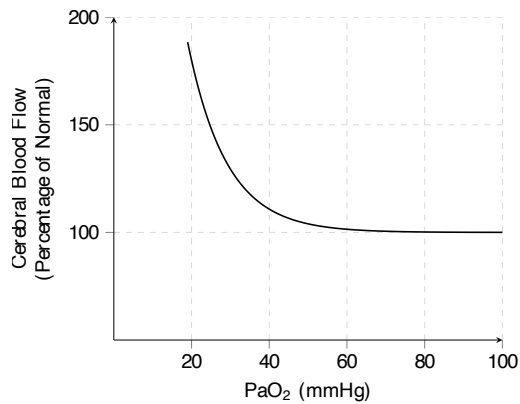
- There is a right-shift in chronic hypercapnea. The mechanism of action is complex, but involves local increase in H⁺ ions.

- Changes to CBF with CO_2 are dependent on current arteriolar tone - vasodilatory effects of CO_2 are significantly reduced when the perfusing pressure is low.



PaO₂

- CBF increases rapidly when PaO₂ falls below 60mmHg so that cerebral oxygen delivery is maintained
Hypoxia causes a release of adenosine and reduced calcium uptake, with subsequent vasodilation



Neurohormonal

- Autonomic control of cerebrovascular tone is limited, though is responsible for the right-shift in the autoregulation curve with sustained hypertension

Factors Affecting Blood Viscosity

- Blood viscosity is dependent on haematocrit
 - Reduced haematocrit is associated with increased CBF, but reduced O₂-carrying capacity
The optimal haematocrit is ~0.3-0.35, which provides the best balance between reduction of viscosity to improve cerebral blood flow, without reducing DO₂.
-

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2. Hill L, Gwinnutt C. Cerebral Blood Flow and Intracranial Pressure. FRCA Website.
3. Willie CK, Tzeng YC, Fisher JA, Ainslie PN. Integrative regulation of human brain blood flow. J Physiol. 2014 Mar 1;592(5):841-59.
4. Muizelaar JP. CBF and management of the head-injured patient. In: Narayan RK, Wilberger JE, Povlishock JT, eds. Neurotrauma. New York: McGraw-Hill, 1996:553-561.

Hepatic Blood Flow

Describe the distribution of blood volume and flow in the various regional circulations and explain the factors that influence them, including autoregulation. These include, but not limited to, ~~the cerebral and spinal cord~~, hepatic and splanchnic, ~~coronary, renal and utero-placental~~ circulations

The liver serves as a blood reservoir (30ml per 100g, half of which may be mobilised in hypovolaemia), and receives **25%** of cardiac output from a unique dual blood supply:

- **Hepatic arterial system**
 - Supplies about **one-third** of blood, but 40-50% of O_2
 - **SpO_2 of ~98%** (as would be expected)
 - High-pressure, high-resistance, high-flow system (average velocity 18cm.s^{-1})
 - Capacity to autoregulate
- **Portal venous system**
 - Supplies the remaining **two-thirds** of blood
 - **SpO_2 of portal venous blood varies** depending on gut activity:
 - In the **resting gut**, SpO_2 is **~85%**
 - In the **active gut**, SpO_2 is **~75%**
 - Low-resistance, low-pressure, low-velocity system (average flow 9cm.s^{-1})
 - No capacity to autoregulate

Regulation of Flow

As with other organs, blood flow is autoregulated via intrinsic and extrinsic mechanisms, and may be affected by external factors.

Intrinsic Autoregulation

- Myogenic autoregulation
- Hepatic arterial buffer response
This is also known as the "hepatic artery-portal venous semi-reciprocal interrelationship".
 - Hepatic arterial resistance is proportional to portal venous blood flow, such that a reduction in portal venous flow causes a decrease in hepatic arterial resistance and increases hepatic arterial flow
This is probably mediated by adenosine.

Extrinsic Autoregulation

- Autonomic Nervous System
Both the hepatic and portal vasculature have sympathetic innervation:
 - The hepatic artery has dopamine receptors, as well as β - and α -adrenoreceptors
 - The portal vein has only α -adrenoreceptors
Activation of these receptors causes vasoconstriction, reducing the compliance of the hepatic vasculature and mobilising up to **250ml** of blood in times of sympathetic stress.
- Endocrine and hormonal effects
A number of substances affect portal flow:

Table 1: Regulators of Hepatic and Portal Blood Flow

Regulator	Portal Vein Effect	Hepatic Artery Effect	Overall Effect on Flow
Adrenaline	Constriction	Constriction (α), then dilation (β)	Reduced
Glucagon	Dilation	-	Increased
Secretin	-	Dilation	Increased
An-giotensin II	Constriction	Constriction	Reduced
Vaso-pressin	Constriction	Constriction	Reduced
PCO₂	Constriction	-	Reduced

External Factors

Flow in the hepatic vein is dependent on venous return:

- Increased venous return (e.g. negative-intrathoracic pressure) increases hepatic flow
- Decreased venous return (e.g. positive-pressure ventilation, tamponade, haemorrhage), reduces hepatic flow, and in extreme cases flow may only occur intermittently throughout the cardiac cycle
- **Exercise** reduces both portal vein and hepatic arterial flow

Microvasculature

Hepatic arterioles and portal venules form the **hepatic triad** with a bile canaliculi. Hepatic arterioles and venules anastomose to form **sinusoids**, which create a specialised low-pressure (~2mmHg) capillary system which drains into the central veins of the hepatic acinus.

This arrangement:

- **Optimises hepatic O₂ extraction**
Increased hepatic O₂ demand is met by increasing O₂ extraction, rather than by increasing flow (as occurs in the heart).
 - **Prevents shunting and retrograde flow**
-

References

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4. Kogire M, Inoue K, Sumi S, Doi R, Yun M, Kaji H, Tobe T. Effects of gastric inhibitory polypeptide and glucagon on portal venous and hepatic arterial flow in conscious dogs. Dig Dis Sci. 1992 Nov;37(11):1666-70.

Circulatory Control

Baroreceptors

Describe the function of baroreceptors and to relate this knowledge to common clinical situations.

Baroreceptors are **stretch receptors** which monitor changes in arterial pressure. **Arterial pressure** is monitored by receptors in the:

- **Aortic arch**
 - Innervated by CNX
- **Carotid sinus**
 - Small dilation of the ICA at the level of the bifurcation.
 - Innervated by CNIX
 - Remember the carotid *sinus* is a baroreceptor, the carotid *body* is a chemoreceptor

Venous pressure is monitored by **low-pressure** stretch receptors:

- Respond to increased venous return
- Are inhibited by positive pressure ventilation
- Act by stretch and typically described as **volume receptors**
- Are located in the:
 - **Atrial walls**
 - **SVC and IVC**
 - **Pulmonary circulation**

Baroreceptor Control

Afferent fibres from CNIX and CNX travel to the NTS in the **medulla**. Effector neurons from the RVLM are GABAergic and therefore inhibitory, i.e. increased baroreceptor discharge *reduces* tonic sympathetic tone and increases vagal tone.

Increased baroreceptor activity therefore **results in**:

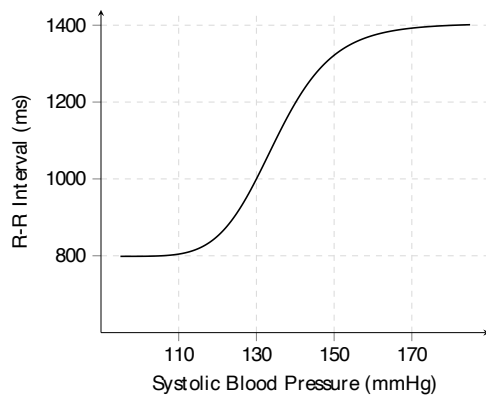
- Arterial and venous vasodilation
- Hypotension
- Bradycardia
- Decreased cardiac output
- Decreased respiratory rate

Conversely, **increased** activity of **low-pressure** stretch receptors results in an **increase** rather than a decrease in **heart rate**.

Baroreceptor Activity

Baroreceptors are:

- **More sensitive to pulsatile pressure** than constant pressure
A decrease in pulse pressure without a change in MAP will decrease baroreceptor firing.
- Active throughout the cardiac cycle
Rapid compensatory responses are vital in the short-term control of blood pressure, e.g. with posture.
- Active over the range from 50mmHg to 200mmHg



- This curve is left-shifted in children and neonates, and right-shifted in chronic hypertension, though this is reversible

Hormonal control

Activation of atrial/ventricular stretch receptors stimulates ANP/BNP release respectively, which act to reduce blood pressure in the following ways:

- Increased GFR
Act to constrict the efferent arteriole and dilates of the afferent arteriole. This subsequently inhibits renin secretion through increased hydrostatic pressure at the JGA and increased Na^+ and Cl^- delivery to the macula densa.
 - Decreased aldosterone
Via inhibition of aldosterone secretion.
 - Vasodilation
Causes vasodilation of peripheral smooth muscle.
-

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. CICM September/November 2014
3. ANZCA July/August 2000

Valsalva Manoeuvre

Explain the response of the circulation to situations such as changes in posture haemorrhage, hypovolaemia, anaemia, intermittent positive pressure ventilation, positive end-expiratory pressure, and the Valsalva manoeuvre.

A Valsalva is forced expiration against a closed glottis. This can be achieved by increasing P_{AW} to 40mmHg for 15 seconds. This increase in intrathoracic pressure alters many haemodynamic parameters.

Phases

A Valsalva manoeuvre consists of four phases:

Phase I

- P_{AW} is increased to 40cmH₂O, with a corresponding increase in $P_{Thoracic}$
- SBP and DBP increase due to:
 - Compression of the aorta
 - Increased LV preload due to ejection of blood in the pulmonary vasculature

Phase II

- VR falls due to increased P_{Thoracic}
- CO falls due to decreased VR
- SBP and DBP fall due to decreased CO
- Baroreceptor activity is $\downarrow$ by the fall in BP, and SNS outflow increases, causing:
 - Increased HR
 - Increased SVR
 - BP therefore starts to recover late in Phase II

Phase III

- The Valsalva ceases, and P_{AW} returns to 0cmH₂O
- PVR rapidly drops as alveolar vessels re-expand
- SBP and DBP rapidly fall due to:
 - Decreased PVR causing decreased LV preload
 - Loss of high intrathoracic pressure compressing the aorta

Phase IV

- VR normalises
- CO normalises due to normal VR and PVR
- SBP and DBP transiently increase due to a normal CO entering a baroreceptor-driven high-SVR vascular bed
- Baroreceptors respond to high SBP and DBP by increasing vagal tone:
 - HR falls (reflex bradycardia)
 - BP falls

Abnormal Responses

Abnormal responses occur in cardiac failure and autonomic neuropathy.

Cardiac Failure

In CCF a square-wave pattern is produced:

- Increasing P_{AW} resulting in a sustained **increase** in SBP and DBP
- There is a slight decrease in SBP and DBP for the few seconds in phase III when airway pressure is released

This square-wave pattern appears to be due to the increased circulating volume, as this difference resolves in venesectioned cardiac patients, and is demonstrated in normal individuals who are transfused to a high circulating volume.

Autonomic Neuropathy

Baroreceptor response to the Valsalva is minimal in both phase II and IV:

- In phase II, there is no compensatory increase in sympathetic outflow, so BP continues to fall until P_{AW} returns to 0mmHg
- In phase IV, there is no compensatory increase in vagal tone and so BP returns to normal without overshooting

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Judson WE, Hatcher JD, Wilkins RW. Blood Pressure Responses to the Valsalva Maneuver in Cardiac Patients with and without Congestive Failure. Circulation. 1955;11:889-899.

Cardiovascular Changes with Obesity

Describe the cardiovascular changes that occur with morbid obesity

Obesity is a multisystem disorder defined by an elevated body mass index (BMI):

- Normal: BMI < 25
- Overweight: BMI 25 - 30
- Obese: BMI > 30
- Morbidly Obese:
 - Obesity related disease and a BMI > 35
 - BMI > 40

Characteristics of obesity include:

- Complex genetic and environmental causes
- Increased caloric intake
- Increased metabolic rate (normal for BSA)

The effect of obesity on the cardiovascular system is complex, and can be classified into:

- Hormonal
- Cardiac

Hormonal Changes

Abdominal visceral fat is responsible for secreting a large number of hormones which affect cardiovascular param-

ters:

- Increased leptin
Contributes to cardiac remodelling and LVH.
- Angiotensinogen
Leads to systemic hypertension and LV remodelling.
 - Small amounts are produced in adipocytes, which increases as fat volume increases
- Plasminogen activator inhibitor - 1
Reduces fibrinolysis and predisposes to VTE.
- Inflammatory adipokines
Impair endothelial function, leading to increased SVR.
- Catecholamines
Increased contractility, SVR, and worsen endothelial function.
 - Released with:
 - Hypoxia
 - Hypercapnea
 - Negative intrathoracic pressure
 - Fragmented sleep
Due to OSA.

Cardiovascular Changes

These can be subdivided into:

- Changes in **key cardiovascular parameters**
 - Increased VO_2
Due to increased LBM and fat mass.
 - Increased Blood Volume
Due to increased angiotensin II and aldosterone.
 - Increased Stroke Volume
Due to:
 - Increased preload (major factor)
 - Increased contractility (minor factor)
Due to increased circulating adrenal hormones.
 - Increased Cardiac Output
To maintain DO_2 .

- Initially with preserved ejection fraction
- **Myocardial changes**
 - Diastolic dysfunction
Due to myocardial fibrosis impairing relaxation.
 - Fatty infiltration of myocardium and conducting system
 - Predisposes to arrhythmias
Risk is worsened by change in myocardial architecture, hypoxia, and increased circulating catecholamines.
 - Biventricular hypertrophy as a response to increased afterload
 - LV afterload increased due to systemic hypertension
LVH is much more common than RVH.
 - Eccentric hypertrophy due to volume overload
 - Concentric hypertrophy due to pressure overload or hormonal changes
 - RV hypertrophy due to:
 - LV diastolic failure
 - Increased PVR
 - Hypoxia
Due to:
 - Effects of OSA
 - Increased shunt through collapsed lung bases
 - Acidosis

References

1. Alvarez A, Brodsky J, Lemmens H, Morton J. Morbid Obesity: Peri-operative Management. Cambridge: Cambridge University Press. 2010.
2. Lotia S, Bellamy MC. Anaesthesia and morbid obesity. Contin Educ Anaesth Crit Care Pain 2008; 8 (5): 151-156.

Cardiovascular Effects of Ageing

Describe the cardiovascular changes that occur with ageing.

CVS effects of ageing can be divided into:

- Cardiac
- Vascular
- Autonomic

Cardiac Changes

- Decreased maximum heart rate
Due to fibrosis of the SA node causing reduced pace-maker cell number and function, and reduction in catecholamine receptor density.
 - $\text{Maximum heart rate} \approx 220 - \text{Age}$
- Decreased inotropy
Minor.
- Increased reliance on atrial kick
Reduced ventricular compliance increases the reliance on atrial kick to achieve adequate preload.
- Decreased diastolic compliance
 - Due to hypertrophy from increased afterload
- Decreased receptor density and number

Vascular Changes

- Reduced compliance
Due to loss of elastic tissue in the large arteries.
- Increased SVR
Reduced compliance results in increased vascular resistance.
- Reduced endothelial cell function (decreased NO)
Impairs the ability of the vascular tree to adapt to changes in pressure/volume leading to:
 - Elevated SBP
 - Reduced DBP
Reduced elastic recoil causes diastolic run off and a fall in diastolic blood pressure.
- Reduced catecholamine receptor density
Reduced responsiveness to (and increased number of) circulating catecholamines.

Autonomic Changes

- Impaired autonomic function
Due to decreased catecholamine responsiveness.
- Impaired baroreceptor response
- Decreased exercise tolerance
Reliance on preload to maintain cardiac output.

References

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2. Cheitlin MD. Cardiovascular physiology - changes with ageing. Am J Geriatr Cardiol. 2003 Jan-Feb;12(1):9-13.

Adrenoreceptors

Understand the pharmacology of adrenoreceptor blocking drugs.

Adrenoreceptors are classified by their varying sensitivity to different catecholamines. Additionally:

- All adrenoreceptors are G protein-coupled receptors
 - Each receptor contains seven transmembrane α -helical subunits, three extracellular loops, and three intracellular loops
- Alpha receptors have different subunits and mechanisms of action
- **All beta receptors** are:
 - G_s coupled
 - Activate adenylate cyclase increasing cAMP, leading to increased Na^+/K^+ ATPase activity and hyperpolarisation

This covers the pharmacology of adrenoreceptors. The production and metabolism of endogenous catecholamines is covered under Chapter . Detailed information on specific sympathomimetic agents, including structure-activity relationships, is covered under Chapter .

Adrenoreceptor Subtypes

α_1 - receptors:

- Are present in smooth muscle
Agonism causes vasoconstriction, relaxation of GIT muscle (via presynaptic receptors), and contraction of GU muscle.
- They are:
 - G_q coupled

- Phospholipase C activated increases IP_3 , increase calcium

α_2 - receptors:

- Are present in the CNS, arterioles, pancreas
Agonism causes sedation, analgesia, vasodilatation, and inhibition of insulin release.
- They are:
 - G_i coupled
 - Inhibits adenylate cyclase, decreasing cAMP

β_1 - receptors:

- Are present in cardiac muscle and the JGA
 - Cardiac agonism increases inotropy, chronotropy, and dromotropy
 - JGA agonism increases renin release
- Increase in cAMP increases intracellular calcium

β_2 - receptors:

- Are present in skeletal vascular and bronchial smooth muscle, the liver, and on cell membranes
- Agonism causes:
 - Vasodilation and bronchodilation
 - Hepatic glycogenolysis
 - Increases activity of the $Na^+ - K^+$ ATPase pump, increasing intracellular potassium
- Increase in cAMP increases Na^+ / K^+ ATPase activity and hyperpolarisation

β_3 - receptors:

- Are present in fat
Agonism causes lipolysis and thermogenesis.
-

References

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2. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
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Renal System

Renal Physiology

Functional Anatomy and Control of Renal Blood Flow

Describe the functional anatomy of the kidneys and renal blood flow.

Functional Anatomy

The functional unit of the kidney is the **nephron**. Nephrons:

- Are composed of the glomerulus, proximal tubule, loop of Henle, distal tubule, and collecting duct
- Are divided by their location into:
 - **Superficial cortical** nephrons
Have **short loops** of Henle.
 - **Juxtamedullary** nephrons
Have **long loops** of Henle, and the efferent arteriole forms the vasa recta for the kidney.
 - **Mid-cortical nephrons**
May have either long or short loops.

Control of Renal Blood flow

The kidneys:

- Receive **22% of cardiac output** at rest
- Extract only **10% of delivered O_2**
- Have a **high renal blood flow** exceeds that required for metabolism
High flow is instead needed **to produce** the large volume of **glomerular filtrate** ($125\text{ml}\cdot\text{min}^{-1}$) required for excretion of waste.

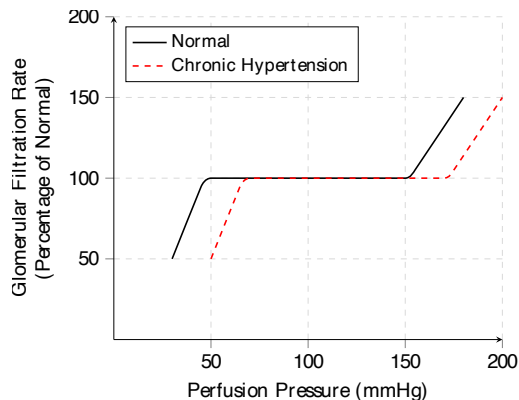
Autoregulation

Renal blood flow is **autoregulated** over a wide range of mean arterial pressures (60–160 mmHg) via:

- **Myogenic** autoregulation
- **Tubuloglomerular** feedback

Myogenic autoregulation:

- Describes the intrinsic constriction of the afferent arteriole in response to an increased transmural pressure
- This increases vascular resistance in proportion to the increase in pressure, keeping flow constant



Tubuloglomerular feedback is more complicated, and describes the constriction or dilation of the afferent arteriole in response to adenosine or NO (respectively) release from the macula densa:

- The macula densa lies in the wall of the ascending limb of the loop of Henle
- It detects change in tubular flow rate (probably via changing Na^+ flux across its membrane)
 - Increased flow in the loop indicates an increased perfusion pressure, prompting release of adenosine and constriction of the afferent arteriole
 - Decreased flow indicates a decreased perfusion pressure, reducing adenosine release and prompting the release of NO and renin, which causes the afferent arteriole to dilate

Notably, flow to **juxtamedullary** nephrons is **not autoregulated**. High blood pressure increases juxtamedullary flow, increasing GFR and impairing renal concentration, resulting in a **pressure diuresis**.

Neuronal Control

The kidneys are innervated by noradrenergic sympathetic nerves, which causes:

- Afferent and efferent arteriolar constriction
This increases capillary hydrostatic pressure (increasing filtration) and also increases capillary oncotic pressure (decreasing filtration).
 - This leads to an overall slight reduction in GFR

Hormonal Control

Renin:

- Is released from the juxtaglomerular apparatus by β_1 stimulation
- Catalyses the production of angiotensin I from circulating angiotensinogen
Angiotensin I is then converted into Angiotensin II by circulating ACE.

The actions of the RAAS are described in more detail in the endocrine functions of the kidney.

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
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Glomerular Filtration and Tubular Function

Describe glomerular filtration and tubular function.

Glomerulus

- The **glomerulus** is a set of capillaries which invaginate **Bowman's capsule**
- Fluid filters out of the capillary bed into Bowman's space based on **Starling forces**:
 - Membrane permeability
 - Hydrostatic pressure gradients
 - Oncotic pressure gradient
 - Reflection coefficient

Glomerular Filtration Rate

Glomerular Filtration Rate is:

- The volume of *plasma* filtered by the glomerulus each minute
Normal renal *blood* flow is 1.1 L.min^{-1} , however renal *plasma* flow is less (600 ml.min^{-1} for a normal haematocrit). Therefore, the normal filtration fraction (proportion of renal blood flow which is filtered) is ~20%.

- Typically **125ml.min⁻¹**

- Decreases with age (partially due to loss of nephron number)

GFR can be expressed as the **product** of **Net Filtration Pressure** and the combination of membrane permeability and membrane surface area, designated **K_f** (the filtration coefficient):

$$GFR = NFP \times K_f$$

- **Net Filtration Pressure** is given by opposing **Starling Forces** across the glomerular membrane:

$$NFP = P_{\text{Glomerular Hydrostatic}} - P_{\text{Bowman's Hydrostatic}} - P_{\text{Glomerular Oncotic}}$$

As protein is not filtered in normal states, the oncotic pressure in Bowman's Space is usually assumed to be 0mmHg.

- The average capillary NFP is ~17mmHg
- **Hydrostatic pressure**
Determined by renal blood flow and the relative constriction of the **afferent** and **efferent** arterioles. Hydrostatic pressure **decreases along the capillary**, and is affected by:
 - MAP
 - Catecholamines
 - Local autoregulation
 - Myogenic
 - Tubuloglomerular Feedback
 - Hormones
 - Angiotensin II constricts the efferent arteriole more than the afferent arteriole, causing an increase in renal resistance with only a small decrease in GFR.
 - Prostaglandin E2 dilates the afferent arteriole, increasing GFR
- **Osmotic pressure**
Increases along the capillary, as protein free-fluid is filtered leaving a higher concentration of protein within the capillary. This change in capillary oncotic pressure is proportional to the fil-

tration fraction - a greater filtration fraction will cause a higher oncotic pressure of fluid in the capillary.

- **Membrane permeability**

Overall permeability is:

- A function of:
 - Membrane permeability, in turn affected by:
 - **Capillary endothelium**
 - **Basement membrane**
Negatively charged molecules have **reduced filtration** as the basement membrane is also negatively charged which opposes movement out of the capillary.
 - **Foot processes** of podocytes
Molecules **less** than **7000 Dalton** are **freely filtered**, whilst larger molecules are filtered less.
 - Membrane Surface Area
Typically very high for water and solutes.
 - Affected by:
 - Glomerulonephritis
 - Change in basement membrane or podocyte foot processes
 - Angiotensin II causing contraction of mesangial cells

Tubular Function

Proximal Tubule

The proximal tubule reabsorbs 60% of glomerular filtrate. It **reabsorbs basically everything**, including protein, and secretes H^+ , organic ions (such as uric acid and salicylates), ammonium, and up to 60% of filtered urea load.

Loop of Henle

The loop of Henle consists of a **thin descending limb** and a **thick ascending limb**:

- The descending limb reabsorbs water only
- The thick ascending limb:
 - Reabsorbs common ions (Na^+ , K^+ , Cl^-) and HCO_3^-
 - Excretes H^+
- The function of the loop is to concentrate urine in states of water deprivation
This is done via the countercurrent mechanism.

Countercurrent Multiplier

The countercurrent concentrating system is:

- Formed from the loop of Henle and collecting ducts
- Driven entirely by the removal of NaCl from the ascending limb
- Most easily understood in stages:
 - NaCl is actively transported out of the thick ascending limb, increasing interstitial osmolality *at that level*
 - Increased interstitial osmolality results in water reabsorption from the descending limb, increasing tubular osmolality *at that level*
 - This more concentrated tubular fluid then flows to a deeper, more concentrated level, and more water is reabsorbed
 - The effect is progressive concentration of tubular and interstitial fluid, but with a low and stable energy cost as the relative gradients that each transport pump works against is small
 - The end result is a dilute urine leaving the ascending limb, but a highly concentrated medullary interstitium

Countercurrent Exchange

The **vasa recta** are peritubular capillaries that:

- Surround the loop of Henle of juxtamedullary nephrons
- Follow the loop into the medulla
- Have typically low blood flow

This prevents “washout” of the countercurrent multiplier, as the slow blood flow allows solute concentrations to equalise at each level of the loop.

- In hypovolaemic situations, renal blood flow falls and vasa recta flow decreases, further reducing washout
- When renal blood flow is high, vasa recta flow increases

This washes out part of the medullary concentration gradient and reduces the concentrating ability of the kidney.

Distal tubules

Fluid entering the distal tubule has about **one-third** the **osmolality** of plasma. The distal tubule:

- Reabsorbs: Na^+ , Cl^- , HCO_3^- , Ca^{2+}
- Secretes: K^+ , H^+

Collecting Ducts

- The collecting ducts lie in the interstitium (concentrated by the loop of Henle)
- In the absence of aquaporins, the collecting ducts are **impermeable to water**
 - Osmolality can fall as low as **50 mmol.L⁻¹** due to continued reabsorption of solute
 - In the presence of aquaporins, water flows down the osmotic gradient into the concentrated interstitium, resulting in a highly concentrated urine

- ADH also **increases** collecting duct permeability of **urea**
 - Urea moves via solvent drag with water
-

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2. Greger R, Windhorst U. Comprehensive Human Physiology: From Cellular Mechanisms to Integration. Springer-Verlag Berlin Heidelberg. 1996.
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Handling of Organic Substances

Describe the role of the kidney in the handling of glucose, nitrogenous products and drugs

Broadly speaking, the kidney:

- Reabsorbs important substances
- Filters and secretes waste products

Methods of Reabsorption

Reabsorption from tubule to blood can occur via two mechanisms:

- **Transcellular reabsorption**
Substance is absorbed into tubular epithelium and then secreted into blood. This is typically achieved by symporters, which rely on the low intracellular sodium concentration to move substances out of the tubule against their concentration gradient.
- **Paracellular reabsorption**
Substance passes through the matrix of tight junctions between epithelial cells.

Rate Limitation

There are **two** functional upper limits on the rate of reabsorption of substances from the tubule:

- **Tubular Maximum ($T_{\max}$) Limited**

Saturation of transporters occur, so a further increase in solute concentration does not increase the rate of substance reabsorption. The maximum solute concentration for a $T_{\max}$ system is a function of the transporter.

- **Gradient Limited**

Leaks in the tight junctions will result in solute moving from the interstitium back into the tubule if the tubular concentration falls too low. The maximum solute concentration for a gradient limited system is related to the permeability of the tight junctions.

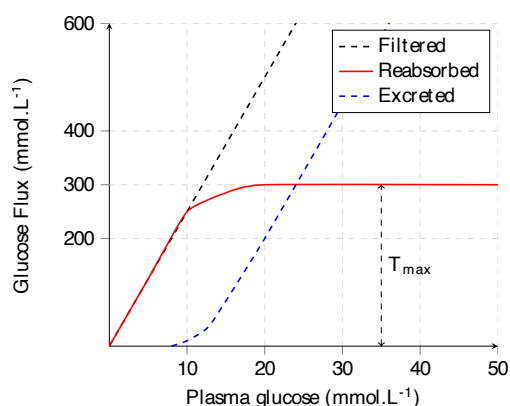
Glucose

Glucose is:

- Freely filtered at the glomerulus
- Completely reabsorbed via the **transcellular route** in the proximal convoluted tubule under normal circumstances
- Actively transported via the SGLUT (Sodium-dependent Glucose symporter) transmembrane protein
 - Secondary active transport (down the established Sodium gradient)
 - There are two subtypes of the SGLUT protein:
 - Low-affinity, high-capacity
Rapidly reabsorbs glucose, but is ineffective when glucose concentration is low. It is located early in the PCT, and reabsorbs ~90% of filtered glucose.
 - High-affinity, low-capacity
Slowly reabsorbs glucose, but remains

effective even when glucose concentration is low. It is located late in the PCT, where glucose concentration is lower (having already been reabsorbed by the high-capacity transporter), and reabsorbs ~10% of filtered glucose.

- As GFR increases, glucose filtration and therefore glucose absorption increase
- As glucose is co-transported with Na^+ , absorption of Na^+ and H_2O also increase
- This phenomenon is known as **glomerulo-tubular balance**
- Glucose reabsorption is a $T_{\max}$ system, and is overwhelmed when filtered glucose exceeds $300\text{mg}\cdot\text{min}^{-1}$ or $16\text{mmol}\cdot\text{min}^{-1}$
 - This typically occurs when plasma (and therefore filtered) glucose concentrations exceed **$12\text{mmol}\cdot\text{L}^{-1}$**



Consequences of Glycosuria

Glycosuria occurs when filtered glucose exceeds the capacity of the PCT to reabsorb it, and causes:

- Increased urine volume:
 - Glucose acts as an osmotic diuretic by:
 - Reducing Na^+ reabsorption in the PCT
 - As some glucose is not absorbed, the sodium

that would normally be reabsorbed with (tubuloglomerular balance) is remaining in the tubule.

- Reducing water and salt reabsorption in the Loop of Henle

Due to high tubular flow rates.

- Impairs the formation of the medullary concentration gradient, limiting concentrating capacity
- Stimulates ADH release
- Electrolyte derangements
 - Hypokalaemia due to:
 - Reduced K^+ reabsorption due to high tubular flow rates
 - Aldosterone release due to hypovolaemia, increasing Na^+ reabsorption and K^+ secretion
 - ADH release in response to hypovolaemia
- Loss of substrate for ATP generation
- Increase risk of urinary infections

Nitrogenous Products

- Amino acids are reabsorbed by amino-acid transporters
These are not (entirely) selective, and reabsorb several structurally similar amino acids.
 - These shared pathways create competition for binding sites between amino acids
 - Excess of one substance will lead to both excretion of this substance in urine, as well as inappropriate excretion of related substances
- Larger proteins (such as albumin) are in fact filtered at the glomerulus (though in very small amounts)
Reuptake occurs in several stages:
 - Endocytosis at the luminal membrane
This is an energy-dependent process, requiring protein to bind to membrane receptors.
 - Degradation of protein into individual amino acids

- Reuptake across the basolateral membrane
- Smaller proteins and peptides (e.g. insulin, angiotensin II) are completely filtered
 - Catabolism occurs in the tubular lumen by membrane-surface peptidases
 - Amino acids are reabsorbed by standard amino-acid transporters

Urea

Urea is a small, water soluble molecule produced in the liver from ammonia as a method for eliminating nitrogenous waste.

Urea excretion is complex, as it has an important role in the counter current multiplier. This means that in the short term (hours to days) elimination may not match production, although over weeks they will be equal. Urea is:

- Freely filtered
- ~50% of filtered load is reabsorbed in the PCT by solvent drag (with water reabsorption)
Urea concentration is slightly increased as more water is reabsorbed than urea.
- The urea reabsorbed in the PCT is then secreted into the Loop of Henle via UT uniporters
 - Luminal concentration of urea is much higher in the ascending limb due to the absorption of water
- ~50% is reabsorbed (again) in the medullary collecting ducts
Here, urine becomes so concentrated that luminal concentration of urea exceeds medullary concentration.
 - Overall, 50% of filtered load is excreted

pH Dependent Drug Reabsorption

- Many substances, such as drugs, are weak acids or bases
 - Reabsorption of these substances is pH dependent
 - **Weak acids** are proportionally **more ionised at a pH above their pKa**
 - **Weak bases** are proportionally **more ionised at a pH below their pKa**
 - Unionised substances are lipid soluble, and able to diffuse into tubular cells down concentration gradients
 - Ionised substances are trapped within the lumen
-

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1. Eaton DC, Pooler JP. Vander's Renal Physiology. 6th Ed (Revised). McGraw-Hill Education - Europe. 2004.

Measurement of GFR

Describe the principles of measurement of glomerular filtration rate and renal blood flow

Renal clearance of a substance quantifies the effectiveness of kidneys in excreting substances. The definition of clearance is the *volume* (typically of plasma) cleared of a drug per unit time. Renal clearance can therefore be expressed as:

$$Cl = \frac{U_C \cdot U_Q}{P_C}, \text{ where:}$$

- Cl = Clearance
- U_C = Urine concentration
- U_Q = Urine flow rate
- P_C = Plasma concentration

Clearance and GFR

As the elimination of most substances is dependent on glomerular filtration, clearance of a substance can be used to estimate GFR. Methods include:

- **Inulin**

Inulin is a naturally occurring polysaccharide.

- Inulin clearance accurately measures GFR as it is:
 - Freely filtered by the glomerulus
 - Not secreted at the tubules
 - Not reabsorbed

- However, inulin is not produced by the body and so must be given by IV infusion
This limits its clinical utility.

- **Creatinine**

Creatinine is a byproduct of muscle catabolism.

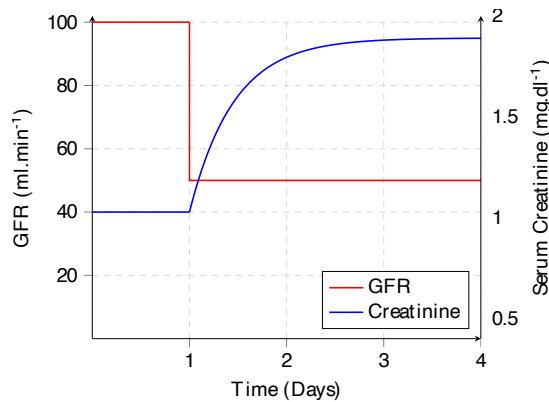
- Creatinine is used clinically to measure renal function because it is:
 - Produced at a relatively constant rate
Factors affecting creatinine production include:
 - Race
 - Muscle mass
 - Age
 - Sex
 - Diet
 - Not metabolised
 - Freely filtered by the glomerulus
 - Minimally secreted
As GFR falls the *proportion* of creatinine secreted by renal tubules increases, so plasma creatinine will overestimate GFR when GFR is low.
 - Not reabsorbed
- GFR can be approximated by creatinine clearance
This is given by the equation: $GFR \approx Cl_{Cr} = \frac{U_{[Cr]} \cdot U_Q}{P_{[Cr]}}$

Serum Creatinine

This formula demonstrates that **GFR is inversely proportional to serum creatinine** concentration¹

¹The relationship between GFR and creatinine concentration only holds when both creatinine production and glomerular filtration are at steady-state. A **sudden drop in glomerular filtration** (e.g. aortic cross-clamp) **will not** result in an **immediate rise in creatinine** concentration.

During acute reductions in GFR, serum creatinine will underestimate GFR until a new steady state is reached.



Estimating Creatinine Clearance

Using the above formula requires measurement of urine volume. This is:

- Typically performed by taking a 24 hour urine collection
- Tedious, and so creatinine clearance is often estimated

A common method is the **Cockcroft - Gault formula**, which has a correlation of ~0.83 with creatinine clearance:

$$Cl = \frac{1.2 \times (140 - A) \times W \times S}{Cr}, \text{ where:}$$

- Cl = Clearance
- A = Age
- S = Sex coefficient (Male = 1, Female = 0.85)
- Cr = Creatinine in $\mu\text{mol.L}^{-1}$

Alternative formulas are **MDRD** and **CKD - EPI**. These equations have two advantages of Cockcroft - Gault:

- They are better predictors of GFR
- They do not require weight, and so can be calculated by the laboratory automatically. Other required data (e.g. age) can be taken from hospital records.

These estimates have similar weaknesses to the above:

- Dependent on serum creatinine, which can be highly variable. Formulas are derived from average values of dependent variables, and so will be unreliable at extremes of:
 - Age
 - Muscle mass
 - Critically ill
 - Malignancy
 - Diet
-

References

1. Hall, JE, and Guyton AC. Guyton and Hall Textbook of Medical Physiology. 11th Edition. Philadelphia, PA: Saunders Elsevier. 2011.
2. Cockcroft DW, Gault MH. Prediction of Creatinine Clearance from Serum Creatinine. Nephron 1976;16:31-41
3. Levey AS, Stevens LA, Schmid CH, Zhang YL, Castro AF 3rd, Feldman HI, Kusek JW, Eggers P, Van Lente F, Greene T, Coresh J; CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration). A new equation to estimate glomerular filtration rate. Ann Intern Med. 2009 May 5;150(9):604-12.
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Endocrine Functions of the Kidney

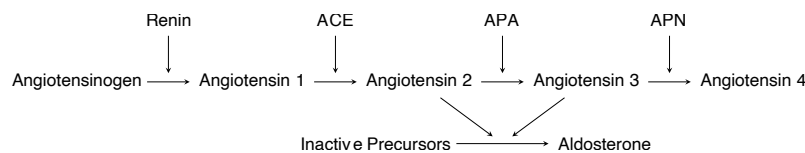
Outline the endocrine functions of the kidney

The kidney is involved in a number of endocrine processes and produces or metabolises a number of hormones:

- RAAS
- Vitamin D
- EPO
- Prostaglandins

Renin–Angiotensin–Aldosterone System

The RAAS is a signaling pathway involved in blood pressure control. It involves a number of hormones:



- **Angiotensinogen** is produced by the liver in response to:
 - Glucocorticoids
 - Thyroid hormones
 - Oestrogens
 - Angiotensin II
 - Various inflammatory proteins

- **Renin** is a protease produced by the kidneys in response to β_1 stimulation or hypotension, and exists to **cleave angiotensinogen to angiotensin I**
- **ACE** cleaves angiotensin I to angiotensin II, and also cleaves bradykinin into inactive metabolites
- **Angiotensin II** increases blood pressure via a number of mechanisms:
 - Stimulates **aldosterone release** from the adrenal cortex, increasing sodium and water retention
 - **Vasoconstriction of efferent greater than the afferent arterioles**
Results in slight decrease in GFR at a lower perfusion pressure, but increases filtration fraction.
 - NB: Different sources quote different changes (increase or decrease) in GFR
The final effect may vary depending on the contribution of other autoregulatory processes.
 - **Reduces K_f** through constriction of glomerular mesangial cells
 - Increased SNS activity and **central and peripheral vasoconstriction**
 - **Increases thirst** via hypothalamic stimulation
 - Stimulates **ADH release**, reducing renal water excretion
 - Stimulates release of angiotensinogen
- **Aldosterone** acts on the distal convoluted tubule to:
 - Increase reabsorption of Na^+ and water
 - Increase elimination of K^+ and H^+

Vitamin D

Vitamin D has a complex metabolic pathway which meanders through a number of organ systems:

- Vitamin D_3 may be absorbed in diet or produced in skin by the action of UV light on 7-dehydrocholesterol

- Vitamin D₃ is then hydrolysed in the liver by CYP450 enzymes to form 25-hydroxycholecalciferol (25-OHD₃)
- 25-OHD₃ is then converted in the proximal tubule to calcitriol - the active form

Erythropoietin

Erythropoiesis is stimulated by EPO release:

- In adults, EPO is released from the:
 - Peritubular capillary fibroblasts (85%)
 - Liver (15%)
- EPO is released in response to:
 - Hypoxia
 - Hypotension
 - Low Hct
- Erythropoiesis is inhibited by:
 - High red cell volume
 - Renal failure

Production of EPO is decreased in renal failure, which is why patients with end-stage renal disease require exogenous EPO.

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Acid-Base Balance

Describe the role of the kidneys in the maintenance of acid/base balance

Acids produced by the body can be:

- **Volatile** (CO_2)
 - Body produces and eliminates $\sim 13 - 20 \text{ mol} \cdot \text{day}^{-1}$
 - Removed by the lungs
- **Fixed** (everything else)
 - Include lactate, sulphate, phosphate, and ketones
 - Body produces and eliminates $10 \text{ mmol} \cdot \text{kg}^{-1} \cdot \text{day}^{-1}$
 - Eliminated by the kidney

Mechanisms for elimination of acid include:

 - Reabsorption of HCO_3^-
This is equivalent to the removal of the same amount of H^+ .
 - As there is usually a net production of acid, under normal circumstances **all filtered HCO_3^- is reabsorbed**
 - Note that removal of an acid load is associated with greater HCO_3^- generation and reabsorption, not increased H^+ secretion
 - Bound to filtered buffers
 - As ammonium
 - The rate and extent of these reactions is dependent on ECF pH and ion concentrations, which gives the kidney control over ion concentrations
 - Urinary pH can fall as low as ~ 4.4 , before the active transport of H^+ is inhibited

Bicarbonate and the Kidney

Buffer systems minimise changes in pH until the kidney can eliminate excess hydrogen.

Bicarbonate is the predominant ECF buffer system (see Acid-Base physiology for more on buffers). By adjusting the level of HCO_3^- the kidney is able to adjust pH, as per the Henderson-Hasselbalch equation:

$$\begin{aligned} \text{pH} &= \text{pK}_a + \log \frac{[\text{HCO}_3^-]}{[\text{CO}_2]} \\ &= 6.1 + \log \frac{24}{1.2} \\ &= 7.4 \end{aligned}$$

Where:

- $\text{pK}_a = 6.1$, the pK_a of HCO_3^-
- $[\text{HCO}_3^-] = 24$, the normal $[\text{HCO}_3^-]$ in mmol.L^{-1}
- $[\text{CO}_2] = 1.2$, the normal $[\text{CO}_2]$ in mmol.L^{-1}

Bicarbonate is:

- Freely filtered
4320 mmol.day^{-1} of HCO_3^- is filtered ($24 \text{mmol.L}^{-1} \times 180 \text{ L.day}^{-1}$, normal range is 4-5 mol.day^{-1})
- Reabsorbed in the PCT (90%), thick ascending limb, DCT, and CT
Adjusting rate of absorption allows correction of an acidosis or alkalosis. All **HCO_3^- reabsorption is equivalent to a loss of H^+ .**

Reabsorption of Bicarbonate

Reabsorption of bicarbonate involves several steps:

- H^+ is secreted into the lumen in one of three ways:
 - Primary H^+ ATPase in the PCT and DCT
 - $\text{H}^+ - \text{Na}^+$ antiporter in the PCT and ascending limb
 - $\text{H}^+ - \text{K}^+$ ATPase in the CT

- Secreted H^+ combines with filtered HCO_3^- to form CO_2 and H_2O
- CO_2 and H_2O diffuse into the tubular cell
- CO_2 and H_2O are converted back into HCO_3^- and H^+ in the tubular cell
- HCO_3^- is reabsorbed into the capillary via the $HCO_3^- - Cl^-$ antiporter, and the H^+ ion is available to be secreted into the tubule (in exchange for K^+ in the collecting ducts and Na^+ in the proximal tubule)

This complicated process allows HCO_3^- to be moved from the tubule to the tubular cell and then to the capillary. **There is no elimination of H^+** by this method - the purpose of H^+ secretion is to facilitate the **reabsorption of HCO_3^-** into the tubular cell.

Ammonia

Glutamine provides a mechanism for elimination of a large number of H^+ ions:

- This is important in:
 - Elimination of excess metabolic acid
 - Renal compensation for acidosis
- This occurs via:
 - Filtered glutamine is absorbed into **proximal tubular cells** and metabolised to NH_4^+ (ammonium) and HCO_3^-
 - HCO_3^- diffuses into blood, and the NH_4^+ is secreted into the tubule via the $NH_4^+ - Na^+$ antiporter and eliminated in urine
 - The $NH_3 + H^+ \rightleftharpoons NH_4^+$ reaction has a pKa of 9.2 meaning:
 - Ammonia cannot act as an effective urinary buffer
 - Ammonia is not a **titratable acid**, as it will not release H^+ ions as urinary pH increases
 This means filtered ammonia does not contribute to the lower limit of urinary pH (4.4),

which is why it is so important in the renal correction of severe metabolic acidosis.

Bound to Filtered Buffers

Secreted H^+ may also combine with a filtered buffer (e.g. PO_4^{3-}). **These H^+ ions are not reabsorbed.** About **36mmol** of H^+ is eliminated with filtered PO_4^{3-} each day, with each PO_4^{3-} binding **two** H^+ ions.

References

1. CICM Sep/Nov 2014
2. ANZCA Feb/April 2012
3. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
4. Acid-Base Online Tutorial, University of Connecticut
5. Brandis, K. Renal Regulation of Acid-Base Balance, in 'Acid-base physiology'.

Dialysis

Dialysis is the **separation of particles in a liquid based on their ability to pass through a membrane.**

Indications

Failure of normal renal functions, i.e.:

- **Acid**
- **Electrolyte derangement**
Particularly hyperkalaemia.
- **Intoxications**
- **Overload**
- **Uraemia**

Physical Mechanisms

Principles underlying the methods of dialysis include:

- **Diffusion**
Diffusion is the **spontaneous movement of substances from a higher concentration to a lower concentration**, where rate of movement is proportional to the concentration gradient (as per Fick's Law).
- **Ultrafiltration**
Movement of water, as determined by Starling's Forces.

- **Convection**

Movement of solute together with a solvent, due to the frictional forces between pulling dissolved substances along. Also known as bulk flow or **solvent drag**. Responsible for clearance of many small molecules during haemofiltration.

Implementation

- **Haemodialysis**

Uses diffusion.

- Blood is pumped through an extracorporeal circuit that contains a dialyser.
- Dialysate flow is countercurrent, which maximises the gradient for diffusion.
- Solutes move across a membrane between blood and dialysate, as per Fick's Law:
 - Concentration gradient between blood and dialysate
 - Flow rate of blood and dialysate
 - Solubility of the solute
 - Mass
 - Charge
 - Protein binding
 - Dialysis membrane permeability
 - Thickness
 - Porosity
 - Surface area

- **Haemofiltration**

Uses ultrafiltration.

- Both a positive hydrostatic pressure in blood and a negative hydrostatic pressure in dialysate is generated, causing ultrafiltration and removal of solutes via solvent drag.
- Elimination via bulk flow is **independent of solute concentration gradients across the membrane**.

- Transport is dependent on Starling Forces:
 - The transmembrane pressure generated
This is a function of:
 - Blood flow to the membrane
Determines hydrostatic pressure.
 - Oncotic pressure gradient
 - Porosity of the membrane
- Additionally, a high filtration fraction will cause excessive haemoconcentration, and clotting of the filter
- The filtered fluid (ultrafiltrate) is discarded, and replaced with another fluid depending on the desired fluid balance.

Differences

- Renal Replacement Therapy (RTT) can be via:
 - Peritoneal dialysis (PD)
 - Intermittent haemodialysis (IHD)
IHD causes greater cardiovascular instability compared to CRRT as the fluid and electrolyte shifts occur more rapidly.
 - Continuous Renal Replacement Therapy (CRRT)
 - Continuous Veno-Venous Haemofiltration (CVVH)
 - Continuous Veno-Venous Haemodiafiltration (CVVHDF)

Method chosen depends desired effect:

- Small molecules (<500 Da) and electrolytes can be removed by filtration or dialysis
- Medium-sized molecules (500-5000 Da) are best removed by filtration
- Low molecular weight proteins (5000-50000 Da) are removed by filtration
This includes removal of inflammatory proteins, which may be beneficial in sepsis.
- Water is best removed by filtration

Pharmacokinetics of RRT

Pharmacokinetics are unpredictable, but are broadly affected by:

- **Drug factors**
 - **Free drug in plasma**

Drugs with a small proportion of free drug in plasma are (unsurprisingly) poorly removed by RRT (but **may be removed via plasmapheresis**). These include:

 - Highly (> 80%) protein bound substances
Examples included phenytoin, warfarin, and many antibiotics.
 - Not that this **may not apply in overdose**
Once protein binding sites are saturated, both free drug fraction and efficacy of dialysis is increased.
 - Drugs with a V_D greater than 1L.kg^{-1}
 - **Size/Molecular Weight**
 - Small molecules (< 500 Da) are more easily cleared by diffusive methods of RTT
 - Molecules > 15kDa are poorly dialysed
This includes proteins, heparins, and monoclonal antibodies.
 - **Volume of distribution**

Drugs with high volumes of distribution are poorly dialysed, as removal of drug from plasma only removes a small proportion of total-body drug content.
- **Dialysis factors**
 - **Dose/Flow rates**

Reduced flow rates will reduce clearance.

 - Conventional high-flux haemodialysis has more rapid clearance compared to lower-flux haemoperfusion or CRRT
 - **Membrane permeability**
 - **Timing**

Drugs given between IHD or SLED sessions will not be cleared until the next session.s

- **Patient factors**

- **Residual renal function**

Patients residual GFR will also affect pharmacokinetics.

Table 1: An Incomplete List of Drugs Removed and Not Removed by Renal Replacement Therapy

Drugs Removed on RRT	Drugs not removed on RRT
Barbiturates	Digoxin
Lithium	TCAs
Aspirin	Phenytoin
Sotalol/Atenolol	Other
	beta-blockers
Theophylline	Gliclazide
Ethylene Glycol	Benzodiazepines
Methanol	Warfarin
Aminoglycosides, metronidazole, carbapenems, cephalosporins, penicillins	Macrolides, quinolones

References

1. Johnson CA, Simmons WD. Dialysis of Drugs. Nephrology Pharmacy Associates.

Fluids and Electrolytes

Sodium and Water

Describe the function, distribution, regulation and physiological importance of sodium, ~~chloride, potassium, magnesium, calcium and phosphate~~ ions

Normal total body Na^+ is 60mmol.kg^{-1} , 70% of which is exchangeable. Total body Na^+ is distributed as:

- 50% in ECF
Sodium is the dominant extracellular cation.
 - Typical ECF $[\text{Na}^+]$ of 140mmol.L^{-1} .
- 45% in bone
- 5% in ICF
A minor intracellular cation.
 - ICF $[\text{Na}^+]$ varies with cell type, but is typically $12\text{--}20\text{mmol.L}^{-1}$.
 - Concentration is kept low by the action of the $2\text{Na}^+ - 3\text{K}^+$ ATPase exchange pump and the low permeability of the cellular membrane to Na^+

Function of Sodium

- **Regulation of ECF volume**
Principal ECF cation. Changes in sodium levels cause compensatory fluid shifts. Loss of sodium *content* will result in hypotension/hypovolaemia, with consequent baroreceptor stimulation and activation of the RAAS.

Baroreceptors will activate with a **7-10% change in volume**.

- **Osmolarity**

Changes in sodium *concentration* affect osmoreceptors and will affect ADH and thirst mechanisms. Osmoreceptors will activate with a **1-2% change in osmolality**.

- **Acid-Base balance**

$\text{Na}^+ - \text{H}^+$ exchange pumps in the kidney are stimulated in acidosis.

- **Resting Membrane Potential**

Alterations in sodium concentration will affect intracellular potassium to a similar degree, which will alter the RMP.

Regulation of Sodium and Water

Regulation of any system is typically a balance between **input** and **output**:

- Sodium intake is **essentially unregulated**
- Therefore, sodium concentration is a function of:
 - Sodium elimination
 - Sodium reabsorption
 - **Water homeostasis**
Control of total body water is a major mechanism to regulate sodium concentration.

Sodium Elimination

Sodium is eliminated in:

- Sweat and GIT
Obligatory and not amenable to regulation.
 - Acclimatisation to hot environments improves the efficiency of sweating by reducing its tonicity, reducing sodium loss
- GIT

- Urine
 - Adjust renal elimination is the main mechanism to regulate sodium concentration
- Can be performed in two ways:
- **Changes in GFR**
Changes in GFR due to hyper or hypovolaemia will (indirectly) adjust sodium elimination. Increased plasma volume increases GFR, and vice versa.
 - **Changes in sodium reabsorption**
This is the main mechanism for controlling sodium in euvolaemia, and is mediated primarily by **aldosterone**.

Sodium Reabsorption

Given that:

- Normal glomerular filtrate is $\sim 180\text{L}\cdot\text{day}^{-1}$
- The dominant osmole in glomerular filtrate is sodium
- Normal urine output is $\sim 1.5\text{L}\cdot\text{day}^{-1}$

The majority of filtered sodium must be reabsorbed. This is called **bulk reabsorption** and occurs in the PCT and LOH:

- 60% of total reabsorption is by the $\text{Na}^+ - \text{K}^+$ ATPase pump in the PCT
- 30% of total reabsorption is by the $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ co-transporter in the LOH

The remaining 10% of sodium reabsorption occurs in the DCT and CT. As it is under the influence of aldosterone, it is the component which is important in regulation. Aldosterone increases Na^+ reabsorption by increasing the number or activity of these pumps:

- $\text{Na}^+ - \text{Cl}^-$ pumps in the DCT
- $\text{Na}^+ - \text{K}^+$ ATPase pumps in **principal cells** of the DCT
- $\text{Na}^+ - \text{H}^+$ pumps in **intercalated cells** of the CT

Water Homeostasis

Body water homeostasis involves:

- Sensors
Osmoreceptors present in the:
 - Macula densa
 - Circumventricular organs
Subfornical organ and the vascular organ of the lamina terminalis.
 - Change in cellular volume secondary to changes in osmolality alter hormone secretion.
 - Effectors
Predominantly hormonal:
 - ADH
 - RAAS
 - Natriuretic peptides
-

References

1. Brandis K. The Physiology Viva: Questions & Answers. 2003.
2. CICM September/November 2014
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
4. National Research Council. Recommended Dietary Allowances. 10th Ed. 1989. National Academies Press.

Potassium

Describe the function, distribution, regulation and physiological importance of ~~sodium, chloride, potassium, magnesium, calcium and phosphate~~ ions.

Potassium is the major intracellular cation, with 90% of total body potassium present in the ICF. A further 8% is sequestered in bone, with 2% present in the ECF:

- Normal ECF concentration is $3.5 - 5 \text{ mmol.L}^{-1}$
- Normal ICF concentration is $\sim 150 \text{ mmol.L}^{-1}$

Function and Dysfunction

Potassium is important for:

- Regulation of intracellular pH
- Control of intracellular volume
- DNA and protein synthesis
- Enzymatic function
- Resting membrane potential

The **resting membrane potential** is determined by the ratio of intracellular:extracellular potassium, as per the Nernst equation:

- Small changes in extracellular ion concentration produce large changes in voltage
This has significant effect on excitable tissues.

- Rapid changes in potassium concentration cause symptoms at lower levels than chronic changes
Symptoms are related to the change in action potential generation.

Hyperkalaemia

Hyperkalaemia causes:

- The resting membrane potential to become less negative
As per the Nernst equation.
- This results in the resting membrane potential being closer to the threshold potential, increasing irritability
- Several symptoms, including:
 - Weakness
 - Paralysis
 - Parasthesias
 - ECG findings are those of **prolonged depolarisation** and **rapid repolarisation**:

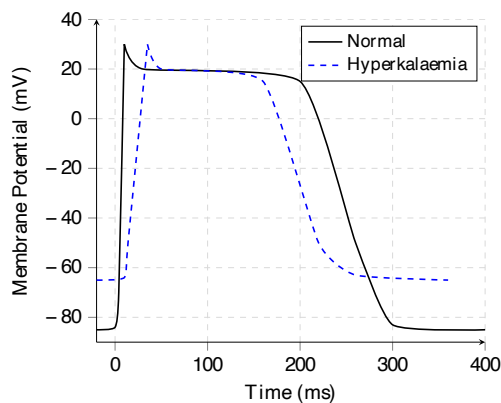


Table 1: ECG Findings at Increasing Serum Potassium Concentrations

Serum [K ⁺] (mmol/L)	ECG Findings
5.5–6.5	Tall tented T waves

Serum [K ⁺] (mmol/L)	ECG Findings
6.5 - 7.5	Loss of P wave, lengthening PR interval
7.5 - 8.5	Widening QRS
>8.5	Sine-wave QRS

Hypokalaemia

Hypokalaemia:

- Causes the resting membrane potential to become more-negative
This makes it more difficult for a stimulus to reach the threshold potential, and therefore it is harder to generate and propagate action potential.
- ECG findings are those of **rapid depolarisation** and **prolonged repolarisation**, and include:
 - Prolonged PR
 - Long QT
 - Flat T waves or TWI
 - U waves
 - ST depression
 - Severe hypokalaemia may result in:
 - Frequent supraventricular and ventricular ectopics
 - Supraventricular arrhythmias
 - Ventricular arrhythmias

Regulation

Serum potassium is dependent on intake, sequestration, and elimination.

Intake

Dietary intake may be highly variable. Potassium is completely absorbed from the upper GI tract.

Sequestration

Several factors affect potassium sequestration:

- **Insulin** and **β_2 -agonism** results in increase activity of the $\text{Na}^+ - \text{K}^+$ ATPase pump, shifting potassium into cells following a meal and during exercise
- **Acidosis** causes an **extracellular** shift of potassium, as hydrogen ions are exchanged for potassium ions. The reverse occurs in alkalosis.
- **Cell lysis** may release a large amount of potassium into circulation and cause significant hyperkalaemia if a large number of cells are destroyed
- **Aldosterone** increases uptake of potassium into cells

Elimination

Elimination of potassium occurs via the kidneys, and is dependent on production of large volumes of glomerular filtrate and secretion by the **distal convoluted tubule** and **collecting duct**.

In normal conditions:

- The PCT and ascending limb reabsorb the majority of absorbed potassium
This is essentially fixed.
 - PCT absorbs ~55%
 - Ascending limb absorbs ~30%
- The principal cells of the DCT and collecting duct secrete potassium
Altering potassium secretion is the main method by which the kidney regulates serum potassium.

- The collecting duct has a much greater role than the DCT
- With normal dietary intake, more potassium is secreted than reabsorbed
This changes in conditions of potassium depletion.

Control of Tubular Secretion

Tubular potassium secretion is mainly a function of:

- **Plasma $[K^+]$**
Increased plasma $[K^+]$ stimulates the $Na^+ - K^+$ ATPase pump in the principal cells, and also stimulates aldosterone release from the adrenal cortex.
- **Tubular flow rate**
Movement of potassium out of principal cells occurs down a passive concentration gradient. Increasing tubular flow rate increases the concentration gradient for potassium
- **Aldosterone**
Aldosterone increases production of the $Na^+ - K^+$ ATPase pump, which increases potassium secretion and uptake into cells.

Minor contributors include:

- Sodium and water content
 - High sodium content inhibits aldosterone release, reducing potassium elimination
 - High water content inhibits ADH excretion and reduces secretion of potassium, however high water content also increases flow through the renal tubule, which indirectly increases tubular secretion of potassium.
- Alkalosis
Alkalosis increases elimination of potassium as the $Na^+ - K^+$ ATPase pump is stimulated by low H^+ ion concentration.

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Hall, JE, and Guyton AC. Guyton and Hall Textbook of Medical Physiology. 11th Edition. Philadelphia, PA: Saunders Elsevier. 2011.
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4. Parham WA, Mehdirad AA, Biermann KM, Fredman CS. Hyperkalemia Revisited. Texas Heart Institute Journal. 2006;33(1):40-47.

Acid-Base

Principles of Acid-Base Physiology

Explain the principles underlying acid-base chemistry

There have been several different theories of acid-base chemistry. The one most relevant for the primary exam is the **Brønsted-Lowry definition**, which defines:

- An **acid** as a **proton donor**
- A **base** as a **proton acceptor**

pH

- Stands for the **power of hydrogen**
- Is a measure of **hydrogen ion activity** in a solution
- Activity can be approximated by concentration
 - Therefore, pH can be expressed as a function of hydrogen ion concentration: $pH = -\log_{10}[H^+]$
Using pH rather than concentration makes it easier to compare different solutions.

pKa

- Strong acids (and bases) dissociate completely in solution

- Weak acids (and bases) only partially dissociate
They have a dissociated state (A^-) and an undissociated state (HA)

- The ratio of concentrations on each side can be used to calculate the **acid dissociation constant, K_a**

$$K_a = \frac{[A^-] \times [H^+]}{[HA]}$$

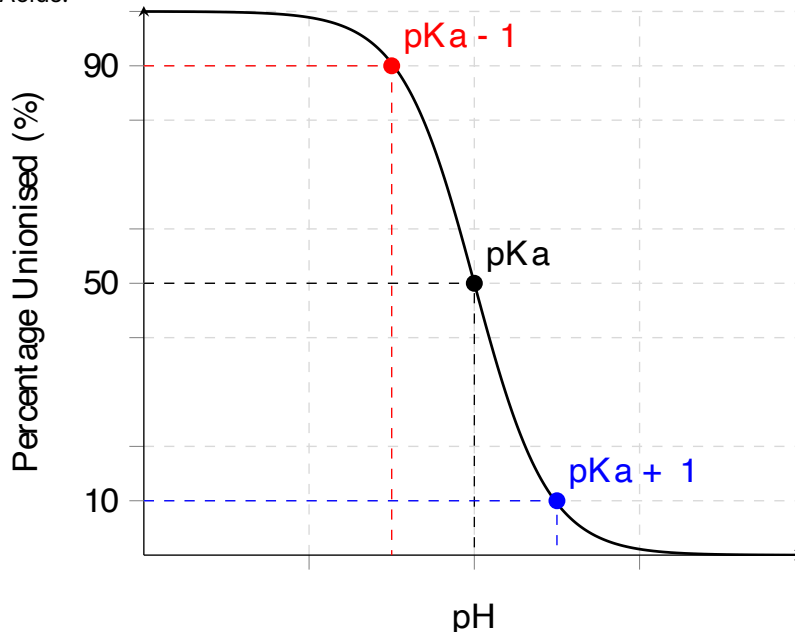
This equation describes the strength of an acid by indicating how readily the acid gives up its hydrogen.

- Similar to pH, this value is often log transformed to **pKa** produce an index, which allows easy comparison of different substances:

$$pK_a = -\log_{10} K_a$$

- pKa has several useful properties:
 - An acid of base will be 50% ionised when the pH of its solution equals its pKa
 - A**cids are more ionised **a**bove their pKa¹
 - B**ases are more ionised **b**elow their pKa²
 - An increase in pH of 1 above the pKa will result in that substance being either 90% (for an acid) or 10% (for a base) ionised

¹Acids:



²Bases:

Systemic Effects of Acid-Base Disorders

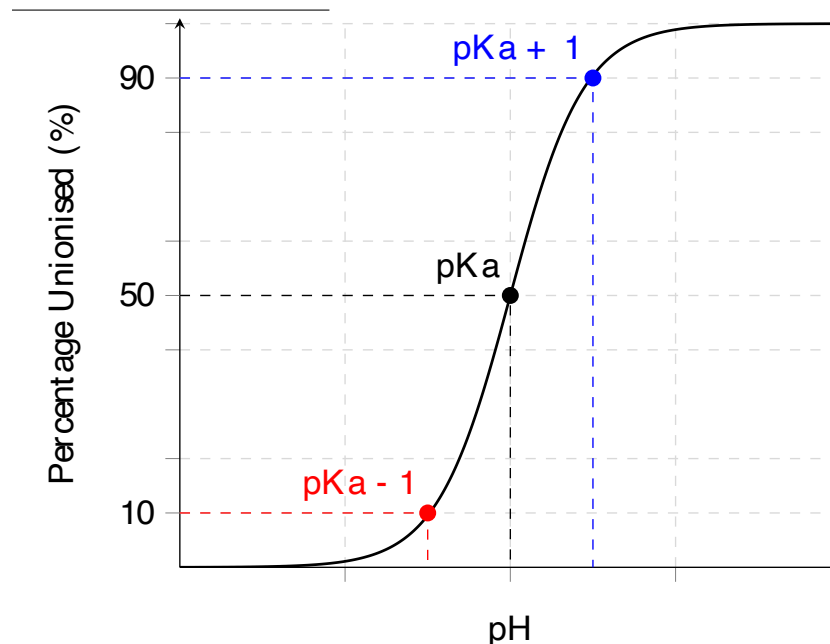
pH disturbance affects many organ systems:

- **Respiratory**

- Increased $\dot{V}_A$
Peripheral and central chemoreceptors increase ventilation in response to a fall in pH.
- Oxyhaemoglobin-Dissociation Curve
Right-shifted by a fall in pH.
- Bronchoconstriction
Hypercapnea causes parasympathetically-mediated bronchoconstriction.

- **Cardiovascular**

- Inotropy
Inotropy falls in acidosis due to a direct myocardial depressant effect. May be offset by increased SNS tone in low-grade acidosis. Alkalosis may increase inotropy by increasing responsiveness to circulating catecholamines.
- Decreased response to catecholamines
When $\text{pH} < 7.2$.



- Arrhythmias
Secondary to altered SNS tone and electrolytes.
- Vasodilation
Directly due to hypercapnea.
- **CNS**
H⁺ ions cannot cross the BBB, however CO₂ can.
- **Fluid and Electrolyte**
 - Plasma **K⁺ increases** by** 0.6mmol.L⁻¹ for every 0.1 unit fall** in **pH**
This is due to impairment of the Na⁺/K⁺ -ATPase
 - **H⁺ ions bind** to the **same site on albumin as calcium**, so ionised calcium will **increase**
- **MSK**
 - Bones
Chronic metabolic acidosis consumes bone phosphate to buffer H⁺ ions, causing osteoporosis.
- **Cellular**
 - Enzyme function
Denaturation and functional impairment.
 - Molecular ionisation
Change in ionisation may change a molecules ability to cross cell membranes (e.g. reducing dose of thiopentone in acidosis), or affect their function
 - Resting membrane potential
Change in ion permeability will alter RMP, and therefore how easy it is to generate an action potential.

Change with Temperature

pH is temperature dependent:

- **pH increases by 0.015** for every **1°C fall** in **temperature**
Due to decreased ionic dissociation of water.
- **Gas solubility** almost always** increases** when **temperature falls**

Dissolving is typically (not always) an **exothermic** reaction. As the kinetic energy content of a molecule falls, its ability to dissociate from solution decreases.

- As CO_2 dissolves, PaCO_2 falls
- As blood gas machines operate at 37°C , a measurement error will occur if a patient is not close to 37°C
 - A **hypothermic** patient will have a **higher pH** and **CO_2** than measured

There are two common methods for **managing** pH of significantly hypothermic patients (e.g., those on CPB): **pH-stat** and **alpha-stat**.

pH-stat

- CO_2 is added to the circuit so that pH and PaCO_2 are normal when corrected for temperature
- This theoretically improves oxygen delivery by preventing the left-shift in the oxyhaemoglobin dissociation curve
- The increased CO_2 also causes cerebral vasodilation, which:
 - Increases speed and uniformity of cerebral cooling
 - Increases risk of cerebral embolic events

alpha-stat

- pH and CO_2 values are maintained at 'normal for 37°C '
 - Measured values will be different, as:
 - pH will be increased
 - CO_2 will be decreased
 - Cellular autoregulation is preserved
 - Unlike pH-stat, this does not cause cerebral vasodilation
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. ANZCA July/August 1999
3. Chemlab. Solubility. Florida State University.

Compensation

Explain the principles underlying acid-base chemistry

Metabolic Acidosis

Compensation to metabolic acidosis includes:

- Buffering
 - Occurs over minutes to hours. Includes:
 - ECF buffers
 - Bicarbonate
 - Plasma proteins
 - Albumin
 - ICF buffers
 - Include phosphate, proteins
 - Leads to hyperkalaemia due to H^+/K^+ exchange
 - K^+ increases by 0.6mmol.L^{-1} per 0.1 unit fall in pH.
 - Bone
 - Exchange of Na^+ and Ca^{2+} in bone.
 - Leads to demineralisation and release of alkaline compounds
- Respiratory compensation
 - Occurs in minutes.
 - Rapid response
 - Cannot compensate completely

- Renal compensation
Occurs over days to weeks. Includes:
 - Elimination of H^+ bound to filtered buffers
 - Include ammonium, phosphate
 - Reabsorption of bicarbonate
 - Active secretion of H^+ in the DCT/CT
Under control of aldosterone.
-

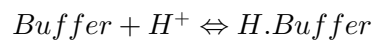
References

1. Diaz, A. Describe how the body handles metabolic acidosis. Primary SAQs.

Buffers

Describe the chemistry of buffer mechanisms and explain their relevant roles in the body

A **buffer** is a solution which consists of a **weak acid and its conjugate base**, that can **resist a change in pH** when a stronger acid or base is added.



Buffering:

- Is a key part of acid-base homeostasis
- Allows compensation for large changes in acid or alkali load with minimal change in hydrogen ion concentration
 - In one experiment, dogs were infused with $14,000,000 \text{ nmol.L}^{-1}$ of H^+ , with a corresponding rise in H^+ of only 36 nmol.L^{-1}

Efficacy of a buffer system is determined by:

- **pKa** of the buffer
80% of buffering occurs within 1 pH unit of the pKa of the system.
- pH of the solution
- **Amount of buffer**
- Whether it is an **open or closed system**
An open buffer system can have the amount of chemical at one (or both) ends adjusted by physiological means.

- This alters the concentration of reactants at either end of the equation, thus altering the speed of the reaction via the Law of Mass Action

Buffer Systems

Important buffer systems include:

- Bicarbonate buffer system
- Protein buffer system
 - Haemoglobin buffer system
- Phosphate buffer system

All buffer systems are in equilibrium with the same amount of H^+ . This is known as the **isohydric principle**.

Bicarbonate Buffer System

The bicarbonate buffer system is:

- The most important ECF buffer system
 - Bicarbonate is formed in the erythrocyte and then secreted into plasma
 - Bicarbonate diffuses into the interstitium and is also the dominant fluid buffer in interstitial space
- Formed in the erythrocyte
- A **buffer pair** consisting of bicarbonate and carbonic acid
 Carbonic acid is exceedingly short lived in any environment even remotely compatible with life and it rapidly dissociates to HCO_3^- and H^+ .

Hydrogen ions are consumed or released by the following reaction:

- $H_2CO_3 \rightleftharpoons HCO_3^- + H^+ \rightleftharpoons CO_3^{2-} + 2H^+$
- Carbonic anhydrase (present in erythrocytes) is an enzyme which allows rapid conversion of H_2O and CO_2 to H_2CO_3 (and back again)
- Each stage of the reaction has an individual pKa:

- As the pKa of the $H_2CO_3 \rightleftharpoons HCO_3^- + H^+$ system is 6.1, these substances predominate at physiological pH
- The pKa for the second stage of the reaction is 9.3 and so essentially no CO_3^{2-} exists in blood Clinically this reaction can be ignored.
- In clinical conditions, the reaction becomes:

$$H_2O + CO_2 \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+$$
 - Addition of a strong acid drives the above reaction to the left, forming (briefly) H_2CO_3 before it dissociates to CO_2 and H_2O
 - CO_2 is then able to be exhaled, which prevents equilibration and allows the system to buffer more acid

Bicarbonate is an effective buffer because it is:

- Present in large amounts
 - **Open at both ends**
 - CO_2 can be adjusted by changing ventilation
 - Bicarbonate can be adjusted by changing renal elimination
 - This prevents the bicarbonate buffer system from equilibrating and allows it to resist large changes in pH despite its low pKa
- However, because it relies heavily on changes in pulmonary ventilation it is **unable to effectively buffer respiratory acid-base disturbances**.

Protein Buffer System

- All proteins contain potential buffer groups
 However, the useful one at physiological pH is the **imidazole groups** of the **histidine residues**.
- Extracellularly, proteins have a small contribution which is entirely due to their low pKa
- Intracellularly proteins have a much greater contribution because:

- Intracellular protein concentration is much greater than extracellular concentration
- Intracellular pH is much lower (~6.8) and closer to their pKa

Haemoglobin Buffer System

Haemoglobin is:

- A protein buffer system
- Quantitatively the most important non-bicarbonate buffer system of blood

This is because haemoglobin:

- Exists in greater amounts than plasma proteins (150g.L^{-1} compared to 70g.L^{-1})
- Each molecule contains 38 histidine residues
This results in 1g of Hb ~3x the buffering capacity of 1g of plasma protein.

In the cell:

- Haemoglobin exists as a weak acid (HHb) as well as its potassium salt (KHb)
- In acidosis:
 - Additional H^+ ions are bound to Hb molecules
 - HCO_3^- diffuses down its concentration gradient into plasma
Electroneutrality is maintained through the inwards movement of Cl^- .
 - Dissolved CO_2 will also form carbamino compounds by binding to the terminal amino groups
- The **pKa** of Hb is **variable** depending on whether it has bound oxygen:
 - **Deoxyhaemoglobin** has a **pKa of 8.2**
Because of its higher pKa, deoxyhaemoglobin will more readily accept H^+ ions which makes it a better buffer of acidic solutions.
 - **Oxyhaemoglobin** has a **pKa of 6.6**
 - Both are essentially equidistant from normal pH, and are equally effective buffers

- Quantitatively, **per mmol of oxyhaemoglobin reduced, ~0.7mmol of H⁺ can be buffered**
Therefore 0.7mmol of CO₂ can enter blood **without a change in pH**.
- This is the mechanism behind the **Haldane effect**, and why venous blood is only slightly more acidic than arterial blood

Phosphate Buffer System

Phosphoric acid is:

- Tribasic and can therefore potentially donate three hydrogen ions
- However, only one of these reactions is relevant at physiological pH, with a pKa of 6.8:
$$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$$
- The quantitative effect is low despite the optimal pKa due to the low plasma concentration of phosphate
- At higher concentrations, such as intracellularly and in urine, it is a significant contributor
- In prolonged acidosis, CaPO₄ can be mobilised from bones and can be considered as an alkali reserve

Further Reading

1. Alex Yartsev offers an excellent discussion on buffering in his excellent trademark prose at Deranged Physiology
2. Brandis's anaesthesia MCQ is required reading

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Nervous System

Blood-Brain Barrier

Discuss the significance of the blood brain barrier

The blood-brain barrier is a **physiological barrier** which prevents substances in the ECF of the body moving freely into the ECF of the brain. The functions of the BBB are:

- Maintain a stable extracellular milieu
Optimises neuronal function by preventing fluctuations in plasma K^+ , Na^+ , and H^+ affecting cerebral cells.
- Protection of the brain
Isolates the brain from toxins.
- Protection of the body
Isolates the rest of the body from CNS neurotransmitters.

Anatomy

The BBB occurs in three layers:

- Capillary endothelial cells
Joined with **tight junctions**, preventing free movement of solvent and solute.
 - Substances must move through capillary endothelium to reach the brain
Capillary endothelial cells contain high numbers of mitochondria, due to the higher energy cost of the active transport mechanisms.
- Basement membrane

- **Astrocytes**
Glial cell which extends foot processes around the basement membrane, and reduce permeability of endothelial cells.

Due to their function, several important CNS structures must exist outside of the BBB. These are known as the **circumventricular organs**, and include:

- **Sensing structures**
 - Chemoreceptor trigger zone (Area Postrema)
Identifies toxins in the systemic circulation, triggering vomiting.
 - Hypothalamus
Osmoreceptors detect systemic osmolarity.
 - Subfornical organ
Role in CVS and fluid balance.
 - Organum vasculosum
- **Secreting structures**
 - Pituitary
Secretes hormones.
 - Pineal gland
Secretes melatonin.
 - Choroid plexus
Produces CSF via secretion and ultrafiltration of plasma.

Movement of Substances

Substances can move via:

- **Diffusion**
For lipid soluble molecules only; e.g:
 - CO₂
 - O₂
- **Facilitated diffusion**
For movement of larger/less soluble molecules down their concentration gradient, e.g:
 - Glucose
 - Water

- **Active transport**

Responsible for movement of most small ions; e.g:

- Na^+
- Cl^-
- K^+
- Mg^{2+}
- Ca^{2+}

Other substances are specifically excluded:

- **Catecholamines**

Metabolised by MAO in capillary endothelium, preventing their action as CNS neurotransmitters.

- **Amino acids**

Prevent action as neurotransmitters.

- **Ammonia**

Metabolised in astrocytes to glutamine, limiting its neurotoxic effects.

References

1. Lawther BK, Kumar S, Krovvidi H. Blood-brain barrier. Continuing Education in Anaesthesia Critical Care & Pain, Volume 11, Issue 4, 1 August 2011, Pages 128-132.
2. Brandis K. The Physiology Viva: Questions & Answers. 2003.

Cerebrospinal Fluid

Describe the physiology of cerebrospinal fluid

Describe the dynamics and metabolism of cerebrospinal fluid

CSF is a **transcellular** fluid in the ventricles and subarachnoid space. ~150ml (2ml/kg) of CSF exist in a normal individual, divided evenly between the head and spinal column.

Functions

- **Mechanical Protection**

Due to its low specific gravity, CSF reduces the effective weight of the brain (by a factor of 30) and therefore reduces trauma caused by the acceleration and deceleration of the brain.

- **Buffering of ICP**

CSF can be displaced to the spinal subarachnoid and have its rate of reabsorption increased in order to offset an increase in ICP by another space-occupying lesion.

- **Stable Extracellular Environment**

Neurons are sensitive to ionic changes in the extracellular environment. Ionic concentrations in CSF are

tightly controlled, which ensures stable neuronal activity. Additionally, toxins are actively removed from CSF.

- **pH Regulation**

pH of extracellular fluid is important in the control of respiration, and is also tightly regulated.

- **Nutrition**

Supply of O_2 and simple sugars and amino acids, and removal of CO_2 occurs in CSF.

Formation

CSF is produced in the **choroid plexus** (70%) and brain **capillary endothelial cells** (30%) at a rate of $0.4 \text{ ml} \cdot \text{min}^{-1}$ ($500 \text{ ml} \cdot \text{day}^{-1}$). It is produced by a combination of **ultrafiltration** and **secretion** from plasma:

- Na^+ is actively transported
Drives flow of Cl^- ions and water.
- Glucose is transported via facilitated diffusion down its concentration gradient

Factors Affecting Formation

Formation is relatively constant within normal parameters (altering the rate of absorption is the predominant means to control pressure), though it is reduced by:

- **Decreased Choroidal Blood Flow**
CPP $< 70 \text{ mmHg}$ reduces CSF formation.

Contents

In summary:

- $[Na^+]$ is unchanged
- $[Mg^{2+}]$ and $[Cl^-]$ are increased
- Concentrations of everything else is less

Table 1: Composition of Cerebrospinal Fluid Relative to Plasma

Content	Relative Change	CSF
Na ⁺	-	140 mmol.L ⁻¹
Cl ⁻	↑	124 mmol.L ⁻¹
K ⁺	↓	2.9 mmol.L ⁻¹
Gluc	↓	3.7 mmol.L ⁻¹
pH	↓	7.33
PCO ₂	↑	50mmHg
Protein	↓	Variable*
Ca ²⁺	↓	1.12 mmol.L ⁻¹
Mg ²⁺	↑	1.2 mmol.L ⁻¹

Relative to plasma CSF [protein] is variable:

- Highest in the lumbar sac
- Lowest in the ventricles
- Always lower than plasma [protein]

Due to the low [protein], CSF is a **poor buffer solution**, which increases its sensitivity to derangements in respiratory acid-base status.

Circulation

CSF flow is driven by respiratory oscillations, arterial pulsations, and ongoing production in the choroidal plexus:

- **Production** in the **choroidal plexus** in the **lateral ventricles**
- To the **third ventricle** via the **Foramen of Munro**
- To the **fourth ventricle** via the **Aqueduct of Sylvius**
- To the **cisterna magna** via the two lateral **Foramina of Luschka** and the midline **Foramen of Magendie**
- It may now pass either:
 - **Cranially**, to the **basilar cisterns** and via the **Sylvian fissure** to the **cortical regions**
 - **Caudally**, to the **spinal subarachnoid space** via the **central canal**

Reabsorption

Reabsorption of CSF:

- Occurs in the **arachnoid villi**, which are located in the **dural walls** of the **sagittal** and **sigmoid sinuses**
 - 85% of reabsorption occurs in intracranial arachnoid villi
 - Remainder by spinal arachnoid villi
 - Is predominantly via pinocytosis and opening of extracellular fluid spaces
 - Is **pressure-dependent**
 - Reabsorption occurs when the CSF pressure is **1.5mmHg greater** than venous pressure
Typically an ICP < 7mmHg results in minimal CSF reabsorption. Above this, CSF absorption increases in a linear fashion up to 22.5mmHg.
-

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Hall, JE, and Guyton AC. Guyton and Hall Textbook of Medical Physiology. 11th Edition. Philadelphia, PA: Saunders Elsevier. 2011.

Spinal Cord Anatomy

Describe the major sensory and motor pathways (including anatomy)

Spinal Cord Anatomy

The spinal cord in transverse section consists of a central section of grey matter containing neuronal cell bodies and synapses, and a peripheral section of white matter containing myelinated ascending and descending pathways. Important pathways are:

- **Corticospinal tract**
Motor function. Crosses at the brain stem.
- **Dorsal column**
Light touch and proprioception. Crosses at the brain stem.
- **Spinothalamic tract**
Pain and temperature. Crosses within two vertebral segments.
- **Spinocerebellar tract**
Unconscious proprioception. Does not cross.

Spinal Cord Blood Supply

Supply is predominantly from **three** major arteries:

- **One** anterior (median) spinal artery
 - Located along anterior midline of cord
 - Arises from the vertebral arteries caudal to the basilar
 - A branch from each vertebral join between C1 and C6 and form the single vessel.
 - Supplies **75%** of arterial blood to the spinal cord.
 - Reinforced by supply from:
 - Unpaired radicular arteries
 - Usually **5-10** total but only one in midthoracic regions
 - Originate from posterior intercostal arteries
 - Artery of Adamkiewicz
 - Variable origin between T7 and L4
 - Left sided in the majority of patients
 - Principle supply to the lower two-thirds of the cord **Anterior** cord **supply** is much **more vulnerable** than the posterior portion.
- **Two** posterior (lateral) spinal arteries
 - Fed by smaller radicular arteries at almost every spinal levels
- An **arteriolar plexus** is formed by both anterior and posterior arteries, and forms a fine pial plexus known as the **vasa corona**
This plexus is not able to compensate for a loss of supply to either anterior or posterior vessels.

Additional perfusion to the cervical region from:

- Posterior inferior cerebellar arteries
- Segmental branches

Spinal Cord Syndromes

Lesions to certain anatomical regions of the spinal cord produce a particular constellations of findings.

Complete Transection

A complete transection results in loss of movement and sensation below the level of the lesion. Initially, paralysis is flaccid (and other signs, such as priapism, may be absent in this 'spinal shock' phase) becomes spastic after a few weeks. Bowel and bladder function is lost.

Lesions above T10 will result in impaired cough in the initial stage as the abdominal wall is unable to contract (intercostal muscle function may be impaired as well, but this is of less importance clinically).

Central Cord Syndrome

Central cord syndrome results in a flacid paralysis and loss of sensation of the **upper limbs greater than the lower limbs**.

Anterior Cord Syndrome

Anterior cord syndrome spares the dorsal columns only, therefore motor function and pain and temperature sensation are affected below the level of the lesion.

Brown-Sequard Syndrome

Hemisection of the cord results in:

- **Ipsilateral** loss of **motor function** below the level of the lesion
- **Ipsilateral** loss of **light touch and proprioception** below the level of the lesion
- **Contralateral** loss of **pain and temperature** sensation below the level of the lesion
- **Ipsilateral** loss of **pain and temperature** sensation **at the level** of the lesion

Cauda Equina

Cauda Equina syndrome results from compression of lumbosacral nerve roots below the level of the conus medullaris. It may produce a combination of UMN and LMN signs:

- Radiculopathy
 - Sacral sensory loss
 - Asymmetric LMN weakness and atrophy
 - Erectile dysfunction and inability to ejaculate
 - Urinary retention and overflow incontinence
 - Constipation and overflow incontinence
-

References

1. Goldber S. Clinical Neuroanatomy Made Ridiculously Simple. 3rd Ed. Medmaster. 2005.
2. McMinn, RMH. Last's Anatomy: Regional and Applied. 9th Ed. Elsevier. 2003.

Intracranial Pressure

Explain the control of intra-cranial pressure

Outline the factors determining intracranial pressure and discuss its regulation

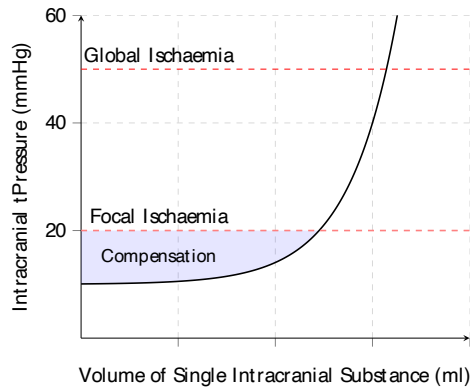
Normal ICP is $\leq 13\text{mmHg}$, with some rhythmic variation occurring on the transduced waveform:

- **P1** is the first peak, and represents arterial pulsation
- **P2** is the second peak, and represents intracranial compliance
If $P2 > P1$, this is suggestive of poor intracranial compliance
- **P3** is the third peak, and is a dicrotic wave representing valve closure

In addition, a second set of Lundberg waves are described:

- **A waves** are **pathological**, and consist of square-wave plateaus up to 50mmHg lasting **5-20 minutes**. They are suggestive of **herniation**, and are always pathological.
- **B waves** are variable spikes in ICP at 30-120 second intervals, suggestive of cerebral **vasospasm**
- **C waves** are oscillations that occur 4-8 times per minute, and are a benign phenomena occurring with respiratory and blood pressure variations

Raised intracranial pressure may cause focal ischaemia when ICP >20mmHg, and global ischaemia when the ICP >50mmHg:



Monroe Kellie Doctrine

This states that:

- The skull is a rigid container of a fixed volume, containing approximately 8 parts brain, 1 part blood, and 1 part CSF
- As it has negligible elastance, any **increase in volume** of one substance must **be met with a decrease in volume** of another or a **rise in ICP**¹

Physiological Responses to an Increase in ICP

- Displacement of CSF into the spinal subarachnoid space
- Compression of vascular bed
- Increased CSF reabsorption

¹Elastance is the *technically* correct term when discussing ICP, as we are discussing a change in *pressure* for a given change in volume.

Compliance is a change in *volume* for a given change in pressure.

- The Cushing reflex may occur in brainstem herniation. Occurs as a reflex SNS response to medullary ischaemia, and describes a **triad** of:
 - Hypertension
To improve CPP.
 - Bradycardia
Due to a baroreceptor response.
 - Irregular respiration
Due to respiratory centre dysfunction.

Physiological Basis of Treatment

Treatment can be classified as per the Monroe Kellie doctrine:

Brain

- **Osmotic agents** such as mannitol and hypertonic saline
Increase plasma osmolality and expand blood volume, creating an osmotic gradient between brain parenchyma and blood with a resulting reduction in brain oedema and ICP.
- Timely **evacuation** of mass lesions and intracranial haemorrhage

CSF

- **External Ventricular Drain**
Facilitates removal of CSF.

Blood

- Reducing cerebral metabolic rate
Results in reduced blood flow due to **flow-metabolism coupling**. May be achieved with:
 - **CNS depressants** such as propofol, benzodiazepines, or barbiturates
Have several beneficial effects:
 - Depress cerebral metabolism which reduces oxygen requirements
 - Reduce **seizure** risk, which is detrimental because it greatly increases cerebral O₂ demand and impairs venous return
 - Improves ventilator dyssynchrony, limiting coughing and bearing down, and subsequent rises in ICP
 - **Avoid hyperthermia**
Causes a reduction in cerebral metabolism and risk of seizures.
 - **Prevention of hypoxia or hypercapnea**
Hypoxia and hypercapnea both cause vasodilatation, with a subsequent increase in cerebral blood volume, blood flow, and ICP.
 - Aim **low-normal CO₂**
Causes vasoconstriction and a subsequent reduction in cerebral blood flow and blood volume. This leads to:
 - Reduction in ICP
 - Reduction in cerebral oxygen delivery
Consequently, a low-normal ETCO₂ target is used to avoid tissue hypoxia.

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Intraocular Pressure

Normal intraocular pressure is ~**15mmHg**, with a range of 12 - 20mmHg. Regulation of intraocular pressure is important for:

- Vision
Sustained high (>25mmHg) can lead to blindness due to compression of axons of the optic nerve and the optic artery at the optic disc.

Determinants of Intraocular Pressure

As the globe has typically poor compliance, a small increase in volume can cause a large increase in intraocular pressure. Factors affecting volume include:

- Volume of **aqueous humor**
Aqueous humor is a clear fluid that fills the anterior and posterior chambers of the eye, and provides avascular tissues with nutrients and oxygen whilst still allowing light to pass freely between the lens and retina. Volume of aqueous humor is a function of:
 - Production
Aqueous humor is produced by secretion and filtration from capillaries in the ciliary body in the posterior chamber, and circulates through into the anterior chamber.
 - Production is accelerated by β_2 agonism
 - Production is inhibited by α_2 agonism
 - Carbonic anhydrase inhibitors decrease aqueous humor production probably by decreasing sodium secretion into the eye

- Reabsorption
Aqueous humor is reabsorbed into venous blood in the **canal of Schlemm**.
 - The **trabeculae meshwork** is the main source of resistance to reabsorption
If this is blocked, a significant reduction in reabsorption can occur and IOP will increase.
 - Reabsorption is affected by:
 - Haemorrhage
Blocks trabecular meshwork.
 - Muscarinic antagonism
Dilates pupil, which brings the iris closer to canal and decreases absorption.
 - α_1 agonism
Dilates the pupil, decreasing absorption.
 - $\text{PGF}_{2\alpha}$
Relaxes ciliary muscle, increasing absorption.
- Volume of **blood** within the globe
Affected by:
 - MAP
 - Venous obstruction
- **External factors**
Other factors affecting volume or compliance of the globe:
 - Extraocular muscle tension
 - Extraocular compression

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Sleep

Describe the physiology of sleep

Sleep is a naturally occurring state of unconsciousness from which one can be aroused by an external stimulus.

Sleep is important in:

- Homeostasis of many organ systems
- Memory formation
- Preservation of cognitive function

Stages of Sleep

Stages of sleep are classified into REM or non-REM sleep based on EEG changes. Periods of REM sleep alternate with non-REM sleep during the night, with an average of 4-5 cycles of REM sleep per night.

- **REM sleep**

Characterised by EEG activity resembling that of awake individuals. REM sleep:

- Lasts for 5-30 minutes
Event frequency decreases with age.
- In REM sleep:
 - Irregular eye movements
 - Dreaming occurs
 - Irregular HR and RR
 - Muscle contraction occurs (but muscle tone is decreased)

- **Non-REM sleep**

Deep sleep, characterised by depression of HR, SVR, BP, RR, and metabolic rate (~0.9 METs) It is divided into four stages on EEG:

- Stage 1: 4-6Hz θ waves replace α -waves
Dosing, easily roused.
- Stage 2: Similar to stage 1 with occasional high frequency 50 μ V bursts (sleep spindles)
- Stage 3: 1-2Hz high-voltage δ waves appear
- Stage 4: Large δ waves become synchronised
Deep sleep.

Respiratory Effects

GABAergic neurons depress the respiratory centre, leading to respiratory depression:

- Decreased MV
 - Decreased V_T
Greatest decrease occurs during REM sleep, where it falls by ~25%.
 - Unchanged RR
- Increased PaCO_2
- Decreased PO_2
More pronounced in elderly.
- Collapse of airway soft tissue
Due to reduced tonic activity of pharyngeal muscles.

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Pain

Describe the physiology of pain, including peripheral nociception, conduction, mediators and pathways, spinal cord modulation, central processing of pain, changes in the older patient.

Describe the pain and sensory pathways

Key Definitions

- **Pain**

Pain is an “unpleasant sensory or emotional experience associated with actual or potential tissue damage, or described in such terms.” Pain can be broadly classified into:

- Aetiology

- **Nociceptive pain**

- Stimulation of nociceptors by noxious stimuli.

- **Visceral pain**

- **Neuropathic pain**

- Nervous system dysfunction.

- Duration

- **Acute pain**

- Pain due to symptoms of current pathology.

- **Chronic pain**

- Pain occurring after the pathological process has resolved.

- **Hyperalgesia**

Increased response to a normally painful stimulus.

- Primary hyperalgesia

- Local reduction in pain threshold.

- Secondary hyperalgesia

- Hyperalgesia away from the site of injury due to alteration in spinal cord signaling.

- **Allodynia**

Painful response to a normally painless stimuli. Occurs due to pathological synapse between second-order neurones in the spinal cord.

- **Anaesthesia dolorosa**

Pain in an area which is anaesthetised.

Peripheral Nociception

Nociceptors are receptors which respond to a noxious stimulus. Nociceptors:

- Can be **stimulated** or **sensitised** by:
 - Chemical signals
See table.
 - Mechanical signals
 - Shear stress
 - Thermal signals
 - Hot nociceptors activate above 43°C
 - Cold nociceptors activate below 26°C
- Stimulation initiates a nervous impulse
- Sensitisation increases a receptors sensitivity to a stimulating mediator

Key chemical stimulating and sensitising mediators include:

Table 1: Chemical Mediators That Stimulate or Sensitise Nociceptors

Stimulating Mediators	Sensitising Mediators
H ⁺	Prostaglandins
K ⁺	Leukotrienes
ACh	Substance P
Histamine	Neurokinin A
5-HT	Calcitonin GRP
Bradykinin	

Nociception

Impulses are conducted by two types of **primary afferent fibres**:

- A δ fibres:
 - Small (~2-5 μ m diameter)
 - **Myelinated**

- Conduct sharp pain at up to 40m.s^{-1}
- Mediate initial reflex responses to acute pain
- Synapse in laminae I in the dorsal horn
Substance P is the neurotransmitter at the NK1 receptor.
- C fibres:
 - **<2 μm** diameter
 - **Unmyelinated**
 - Conduct dull pain at 2m.s^{-1}
 - Synapse in laminae II in the dorsal horn
Substance P is the neurotransmitter at the NK1 receptor.

Pain Pathways and Site of Action of Analgesics

The response to a painful stimulus requires a cascade of processes:

- **Activation of nociceptors**
Membrane depolarisation in response to stimulus. If the stimulus is great enough to reach the threshold potential, an action potential is generated.
 - **NSAIDS** reduce nociceptor mediated inflammation
 - **Opiates** act on peripheral MOP receptors
 - **Local anaesthetics** prevent signal propagation
- **Synapse in the dorsal horn**
Input from both A δ and C fibres, and descending interneurons.
 - Descending inhibitory input reduces nociceptive transmission
Basis of "gate control" theory. Descending input increased with:
 - Touch
A β 'touch' fibres stimulate inhibitory interneurons in the dorsal horn, 'closing the

gate' by increasing **descending inhibition** and prevent signals from peripheral C fibres from rising to the thalamus.

- Arousal
- Opioid receptors
Opioids act pre-synaptically to reduce Substance P and glutamine release.
- α_2 receptors
Clonidine, tricyclic antidepressants, noradrenaline-reuptake inhibitors, and endogenous catecholamines.
- **Gabapentin and pregabalin** inhibit presynaptic neurotransmitter release

- **Wide dynamic range** neurones
Receive afferent input from chemical, thermal, and mechanoreceptors.

- Typically more difficult to stimulate
- Important in **wind-up**
Mediated by NMDA agonism.
 - **Ketamine** reduces windup and central sensitisation
- Lead to secondary hyperalgesia
- Lead to allodynia
Via additional synapses to sensory neurones in lamina III and IV.

- Interneuron synapses with a **second-order neurones** fibre

These secondary afferents:

- Cross within 1-2 vertebral segments and ascends in the spinothalamic tract
- Receives input from descending fibres
- **Opioids** act post-synaptically to hyperpolarise second-order neurones

- **Reflex arc**

- **Higher centres**

Pain perception occurs in the somatosensory cortex.

Neuropathic Pain

Pain due to a lesion of the somatosensory system, rather than a stimulus itself. Neuropathic pain is divided into:

- Central neuropathic pain
From CNS injury, e.g. spinal cord injury, CVA, multiple sclerosis.
- Peripheral neuropathic pain
Damage from:
 - Diabetes
Ischaemia of Schwann cells causes demyelination, causing the exposed axon to generate action potentials inappropriately.
 - Trauma
Transected axons may regrow with endings that spontaneously fire or that have altered threshold potentials.

Mechanisms of Neuropathic Pain

- Neuroma
Healing of damaged nerves leads to neuroma formation. Neuromas:
 - Are more sensitive to painful stimuli
 - Cause spontaneous pain
 - May sprout and innervate local tissues
Movement of these tissues may lead to pain.
- Windup
- Phantom limb pain
Neurons damaged in removal of a limb develop additional synapses, leading to phantom sensations.

Features of Neuropathic Pain

Neuropathic pain is associated with:

- Injury or disease that causes nerve injury

- Burning or electrical quality
- Reduced or absent sensation
- Poor response to typical analgesia

Chronic Regional Pain Syndrome

Damage to the SNS can lead to abnormalities in autonomic function:

- Change in temperature due to vasomotor dysfunction
- Altered sweating
- Reduced hair growth
- Osteoporosis
- Hyperalgesia and allodynia

Pain in the Elderly

Nervous System Changes:

- **Peripheral Nervous System**
 - Nerve deterioration
 - Decreased myelination
 - Decreased conduction velocity
 - Reduced range and speed of ANS responses
 - Increased resting sympathetic tone
- **Central Nervous System**
 - Decreased pain perception
 - Increased sensitivity to anaesthetic and analgesics
Reach ceiling effects more rapidly.
 - Degeneration of myelin
Subsequent cognitive dysfunction due to neuronal circuit dysfunction.
 - Generalised atrophy
 - Decreased neurotransmitter production

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Autonomic Nervous System

Describe the autonomic nervous system, including anatomy, receptors, subtypes and transmitters (including their synthesis, release and fate).

Describe the anatomy of the autonomic nervous system.

Describe the autonomic nervous system and its physiological roles including:

- Autonomic receptors and cellular effects of receptor activation
- Autonomic transmitters, their synthesis, release and fate

The ANS is the section of the nervous system which regulates involuntary and visceral functions. These include:

- Haemodynamics
- Digestion
- Urination and defecation
- Thermoregulation
- Sexual function

The autonomic nervous system can be divided into:

- **Central ANS**
Control occurs in the **hypothalamus, brainstem, and spinal cord.**

- **Peripheral ANS**

Divided anatomically and functionally into the:

- **Sympathetic** nervous system
- **Parasympathetic** nervous system

Central Control

The hypothalamus controls autonomic functions by neural and endocrine mechanisms. It is subdivided anatomically into four regions:

- **Anterior hypothalamus**
Controls the PNS and thermoregulation. It also **releases ADH** in response to increased plasma osmolarity, and **oxytocin**.
- **Medial hypothalamus**
Inhibits appetite in response to increase in blood glucose.
- **Lateral hypothalamus**
Contains the thirst centre and drive to seek food.
- **Posterior hypothalamus**
Controls vasomotor centres, modulating sympathetic vasoconstriction, as well as positive and negative inotropy and chronotropy. Also modulates wakefulness in response to sympathetic stimuli.

Signals from the hypothalamus have a tonic output to:

- All smooth muscle
- Heart
- Exocrine organs
- Endocrine organs
- GIT
- GU

Central Anatomy

In the grey matter of the spinal cord, efferent nerves synapse with two other nerves connected in series. This maintains tonic autonomic outflow.

Efferent nerves exit the spinal root anteriorly, and form the **ventral root**.

Conversely, **afferent** nerves enter posteriorly, forming the **dorsal root** and then **dorsal root ganglion**, before synapsing in the spinal cord.

Sympathetic Nervous System

The sympathetic nervous system optimises the body for short-term survival.

Sympathetic innervation is from the **sympathetic trunks**. These:

- Are a paired bundle of sympathetic neurons which run lateral to the vertebral bodies from **T1 to L2**

The trunk is subdivided into four parts:

- The **cervical part** innervates the head, neck, and part of the thorax
- The **thoracic part** is further subdivided into:
 - Upper thoracic from T1-T5, which innervates the aorta, heart, and lungs
 - Lower thoracic from T6-T12, which innervates the foregut and midgut
- The **lumbar part** forms the coeliac plexus
- The **pelvic part** innervate the pelvic visceral and lower limb vasculature
- Contain the sympathetic ganglion, which is a synapse between the:
 - **Short pre-ganglionic fibre**
Cell body is located in the lateral horn of the spinal cord, and connects to the sympathetic ganglion.

- Releases ACh to stimulate the post-ganglionic fibre.
- **Long post-ganglionic fibre**
Cell body is located in the sympathetic ganglion, and stimulates the effect site.
 - Has a **nicotinic** ACh receptor
 - Releases NA at the effect site
 - Sensitivity (for ACh) and activity (for NA release) is modulated by a number of other substances:
 - Enkephalin
 - Neuropeptide Y
 - Dopamine
 - Adrenaline
 - Prostaglandin
 - GABA
 - Neurotensin

There are three exceptions to the above structure:

- The **adrenal gland** is a **modified sympathetic ganglion**. It is:
 - Directly innervated by preganglionic neurons releasing ACh
- Sweat glands have muscarinic receptors, and are stimulated by ACh rather than noradrenaline
- Skeletal muscle arterioles also have muscarinic ACh receptors, and are stimulated by ACh

Effect

Sympathetic stimulation has a number of effects by either direct neural innervation or adrenaline release. They are consistent with a 'fight or flight' response, and optimise the body for short-term stress conditions.

Table 1: Effects of Sympathetic Nervous System Activation by Organ System

Effector Organ	Sympathetic Innervation	Response
Eye	Cervical	Pupillary dilatation
Lungs	Thoracic	Bronchodilation
Heart	Thoracic	↑↑↑ Chronotropy, ↑↑↑ inotropy, ↑↑↑ lusitropy, ↑↑ dromotropy
Vasculature	Sacral	Constriction
MSK	Sacral	Sweating, contraction, lipolysis
Endocrine	Lower thoracic	Adrenaline and noradrenaline release
GIT	Thoracic, lumbar	Decreased salivation and GIT motility, increased sphincter tone, gluconeogenesis
GU	Pelvic	Detrusor relaxation, sphincter contraction, ↑ uterine tone

Parasympathetic Nervous System

Parasympathetic innervation arises from the:

- **Cranial nerves**

From CN III, VII, IX, and (mostly) X.

- The vagus is the major cranial parasympathetic, innervating the:
 - Heart via the cardiac plexus
 - The SA node is innervated by the right vagus
 - The AV node is innervated by the left vagus
- The ventricles are also sparsely innervated from the left vagus.

- Lungs via the pulmonary plexus
- Stomach, liver, spleen, and pancreas, and gut proximal to the splenic flexure via the gastric plexus.
- **Hypogastric plexus**
Arises from S2-S4, and innervates the bladder, uterus, and gut distal to the splenic flexure.

The parasympathetic nervous system ganglia sit close to the target organ. This means that the:

- Pre-ganglionic fibre is long
 - Preganglionic cell body sits within the brainstem (cranial nerves) or sacral grey matter (hypogastric plexus)
 - Releases ACh to stimulate the post-ganglionic neurone at a **nicotinic ACh** receptor
- Post-ganglionic fibre is short
 - Releases ACh to stimulate the target organ at a **muscarinic ACh** receptor

Table 2: Effects of Parasympathetic Nervous System Activation by Organ

Effector Organ	Parasympathetic Innervation	Response
CNS	CN III via the Edinger-Westphal nucleus CN VII	Pupillary constriction (CN III) Lacrimation (CN VII)
Lungs	CN X	Bronchoconstriction

Effector Organ	Parasympathetic Innervation	Response
Heart	CN X	↓↓↓ Chronotropy ↓↓↓ Dromotropy ↓ Inotropy ↓ Lusitropy
GIT	CN VII (submaxillary and mandibular salivary glands) CN IX (parotid gland) CN X (stomach to proximal two-thirds of the transverse colon) Hypogastric plexus (distal one-third of the transverse colon to rectum)	Salivation ↓ Sphincter tone ↑ Motility
GU	Hypogastric plexus	Detrusor contraction Erection

The decrease in inotropy and lusitropy from parasympathetic tone is greater in the atria than the ventricles.

Ganglion Blockade

Blockade of the ganglion (at the nicotinic ACh receptor) blocks transmission and reduces sympathetic and parasympathetic impulse transmission. Clinical effect of ganglion blockade depends on which part of the ANS is dominant in that organ system:

- SNS dominant organ systems
Effective sympatholysis:
 - Vasculature
Vasodilation, hypotension.
 - Sweat glands
Anhydrosis.

- PNS dominant organ systems
Effective parasympatholysis:
 - Heart
Tachycardia.
 - Iris
Mydriasis.
 - GIT
Decreased tone.
 - Bladder
Urinary retention.
 - Salivary
Reduced secretions.

Enteric Plexus

The **enteric plexus** is a system of autonomic nerves in the GIT which is free of CNS control. It consists of sensory and integrative neurons as well as excitatory and inhibitory motor neurons which generate coordinated muscular activity.

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Acetylcholine Receptors

Understanding of the pharmacology of anti-cholinesterase drugs.

Describe the adverse effects of anti-cholinesterase agents.

Acetylcholine is a neurotransmitter vital for normal function of:

- CNS
- ANS
- Muscle contraction

This covers the pharmacology of acetylcholine receptors and the production and metabolism of ACh. Detailed information on specific agents is in the pharmacopeia.

Synthesis, Release, and Metabolism

ACh is produced in the nerve cytoplasm by **acetyltransferase** from:

- Choline
From diet and recycled ACh.
- Acetyl-coenzyme A
Produced in the inner mitochondrial matrix.

Once synthesised, ACh is then packaged into vesicles (each containing ~10,000 ACh molecules), which are **released in response to calcium** influx occurring at the culmination of an action potential.

Acetylcholine is metabolised by acetylcholinesterase on the post-junctional membrane. AChE:

- Has two binding sites:
 - Anionic binding site
Binds the positively charged quaternary ammonium moiety.
 - Esteratic binding site
Binds the ester group of ACh.
- Once bound, ACh is acetylated
- Acetylated-ACh is then hydrolysed to produce **acetic acid**

ACh Receptor Subtypes

There are two types of ACh receptor:

- **Nicotinic ACh receptors**
 - Inotropic
Linked to an ion channel.
 - Non-specific - may allow Na^+ , K^+ , or Ca^{2+} to cross
 - Consists of five subunits:
 - Two α
Bind ACh.
 - One β
 - One δ
 - One γ
 - Located in:
 - Post-synaptic NMJ
 - Preganglionic autonomic nervous system
Antagonism causes **ganglion blockade**.
 - Brain
 - Known as nicotinic because nicotine agonises this receptor
 - Activation:
 - 2 ACh molecules must bind to activate the receptor

- Once bound, receptor undergoes a conformational change which opens the central ion pore
Permeability to Na^+ (and to a lesser extent, K^+ and Ca^{2+}) increases, leading to depolarisation

- **Muscarinic ACh receptors**

- Metabotropic
G-protein coupled.
 - Known as muscarinic because muscarine also agonises this receptor
 - Subdivided into:
 - M_1 (Gq)
Secretory glands and CNS.
 - M_2 (Gi)
Heart.
 - M_3 Gq
Bronchial and arteriolar smooth muscle.
 - M_4 (Gi) and M_5 (Gq)
CNS.
-

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Endocrine System

Hormones

A **hormone** is a chemical messenger produced by a **duct-less gland** which has its action at a distant target cell via a specific receptor. Hormones can be classified by composition into:

- **Lipid hormones**

Further divided into:

- **Steroids**

Steroids are synthesised from cholesterol, and are released as they are produced (they are not stored). They are highly lipid soluble and act on cytoplasmic and intra-nuclear receptors.

- Aldosterone
- Testosterone
- Oestrogen
- Cortisol

- **Eicosanoids**

Eicosanoids are formed from cell membrane phospholipid.

- Prostaglandins
- Thromboxanes
- Leukotrienes

- **Peptide hormones**

Peptide hormones are stored in granules and released by exocytosis. Further divided into:

- **Short-chain**

- Insulin
- ADH
- Oxytocin
- ACTH

- **Long-chain**

- GH

- Prolactin
 - **Glycopeptides**
Proteins with carbohydrate groups.
 - LH
 - FSH
 - TSH
 - **Monoamine derivatives**
Derived from a single amino acid. Further divided into:
 - Serotonin
 - Thyroxine
 - Catecholamines
Stored in granules and act at membrane receptors.
 - Adrenaline
 - Noradrenaline
-

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Insulin, Glucagon, and Somatostatin

Describe the physiology of insulin, glucagon and somatostatin.

Insulin

Insulin is a **polypeptide** hormone, and is:

- Synthesised from **proinsulin** in the rough endoplasmic reticulum of **B cells** in the **Islets of Langerhans**
- Excreted via exocytosis in response to an increase in intracellular Ca^{2+}
- **Minimally protein bound** with a **tiny volume of distribution**
 V_D 0.075 L.kg^{-1} , increased to 0.146 L.kg^{-1} in diabetics.
- **Metabolised** in liver, muscle, and kidney **by glutathione insulin transhydrogenase**, with renal elimination of inactive metabolites
Circulatory **half-life** of **~5min**.

Actions of Insulin

Insulin binds to a specific insulin receptor (a membrane-spanning protein composed of α and β subunits) on the cell

membrane. The complex is internalised, and its effects are mediated by tyrosine kinase.

Table 1: Actions of Insulin

	Seconds	Minutes	Hours
Muscle	↑ Glucose uptake Active transport via GLUT ₄ . ↑ Amino acid ↑ Ketone ↑ K ⁺ uptake	↑ Anabolism ↑ Catabolism	
Fat	↓ Glucose uptake ↓ Amino acid uptake K ⁺ uptake	↑ Glycerol phosphate synthesis	↑ Fatty acid synthesis
Liver		↓ Gluconeogenesis ↓ Ketogenesis ↑ Glycogen synthesis ↑ Glycolysis ↑ Protein synthesis ↑ Lipid synthesis	
General			↑ Cell growth

Note that cells do not require insulin to take up glucose. GLUT₁ receptors allow free entry of glucose into cells via facilitated diffusion; i.e. **Metabolism drives glucose uptake**.

Active transport of insulin in fat and muscle leads to triglyceride and glycogen production, respectively. Incidentally, the **hyperglycaemia** seen in **diabetes** is due **uninhibited hepatic gluconeogenesis**, and the acidosis due to uninhibited hepatic ketogenesis.

Glucose Tolerance

Hyperglycaemia occurs in diabetes due to **decreased peripheral utilisation** as glucose uptake is reduced due to absence of or resistance to insulin. In addition, the suppressive effect of insulin on hepatic gluconeogenesis is absent or reduced.

Glucagon

Glucagon is a **polypeptide** hormone, and is:

- Synthesised in the **A cells** of the pancreas
- Has a circulating **half-life** of **~5min**
- Metabolised predominantly in the liver
Secreted directly into the portal vein, and **undergoes first-pass metabolism** resulting in low circulating levels.

Table 2: Physiological Effects of Glucagon

System	Effect
Liver	Glycogenolysis
	Gluconeogenesis
	Glucose release
	Ketone formation
CVS	Inotropy
Fat	Lipolysis
	↑ Metabolic rate
Metabolic	GH release
	Somatostatin release
	Insulin release

Secretion of glucagon is influenced by a number of factors:

Table 3: Factors Affecting Glucagon Secretion

Stimulate Release	Inhibit Release
Hypoglycaemia and starvation	Somatostatin
Amino acids	Secretin
Physiological stressors:	Free Fatty Acids
Exercise	
Infection	
β -agonists	α -agonists
Cortisol	Insulin
ACh	Ketones
Theophylline	GABA

Somatostatin

Somatostatin is a **polypeptide** hormone that:

- Inhibits secretion pancreatic polypeptides including insulin and glucagon
- May function as a neurotransmitter in the CNS

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. Peck TE, Hill SA. Pharmacology for Anaesthesia and Intensive Care. 4th Ed. Cambridge University Press. 2014.

Control of Blood Glucose

Explain the control of blood glucose.

Normal blood glucose in the non-diabetic is $4-6 \text{ mmol.L}^{-1}$, though will rise after consumption of carbohydrate. Glucose levels are influenced by the:

- Liver
 - Insulin and glucagon act on the liver to continually adjust the relative rates of glycogenolysis and glycogenesis, allowing it to function as an effective **buffer** of **blood glucose**.
 - Hepatic disease significantly limits the efficacy of this system, and results in a widely-fluctuating blood glucose level
- Kidney
 - A transient glycosuria may be seen as hyperglycaemia decreases renal absorption of glucose

Table 1: Physiological Responses to Hypoglycaemia

BSL (mmol.L^{-1})	Symptoms	Endocrine Response
4.6		↓ Insulin secretion
3.8	Autonomic dysfunction	↑ Glucagon, adrenaline, and GH secretion
2.8	CNS dysfunction	
2.2	Lethargy, Coma	
1.7	Convulsions	
0.6	Permanent brain damage, Death	

Regulation

Mechanisms of glucose regulation can be divided into:

- Hormonal
 - Short-term
Regulation via secretion or inhibition of insulin and glucagon from the pancreatic islets.
 - Long-term
Regulation via both neuronal (SNS activation) and hormonal (cortisol, GH) mechanisms.
- Neuronal

Hormonal: Short Term

Glucose levels are sensed directly in the pancreas and will result in insulin release when the **BGL is >5.6 mmol.L⁻¹**.

Pancreatic B cells respond directly to glucose by secreting insulin in a biphasic fashion:

- An initial, rapid increase in release
 - **Glucose enters** via the GLUT-2 transporter, and is converted to pyruvate which enters **the citric acid cycle** and produces ATP
 - ATP **inhibits** ATP-sensitive **K⁺ channels**, reducing K⁺ efflux and **causing depolarisation**
 - Depolarisation causes **Ca²⁺ release**, resulting in **exocytosis** of insulin granules
- A prolonged, slow increase in release
 - Glutamate is produced as a by-product of the citric acid cycle
 - Glutamate stimulates maturation of other insulin granules
 - Release of these granules causes the second phase of insulin release

Conversely, a low glucose level stimulates secretion of glucagon. This is typically less important than the effect of

insulin unless in situations of starvation or severe physiological stress.

Hormonal: Long Term

Sustained hypoglycaemia increases fat utilisation and decreases glucose utilisation (limiting further drops in blood glucose), via stimulating release of:

- GH
- Cortisol

Neuronal

Hypoglycaemia directly stimulates the hypothalamus, causing:

- Increased SNS tone
Adrenaline release in turn stimulates hepatic glucose release.
-

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
2. Hall, JE, and Guyton AC. Guyton and Hall Textbook of Medical Physiology. 11th Edition. Philadelphia, PA: Saunders Elsevier. 2011.

Pituitary Hormones

Describe the control, secretions and functions of the pituitary and the hypothalamus

The pituitary gland secretes **eight hormones**, and is divided into **two lobes**:

- **Anterior pituitary**
Secretes **six** hormones, all in response to a hypothalamic endocrine stimulus.
- **Posterior pituitary**

Anterior Pituitary Hormones

Hormones from the anterior pituitary are further divided into:

- **Stimulating** hormones
Act at another gland:
 - **ACTH**
Short-chain peptide that stimulates cortisol release from the zona fasciculata. Release is stimulated by CRH, and inhibited by cortisol.
 - **TSH**
Glycoprotein that stimulates synthesis and release of T_3 and T_4 . Release is stimulated by TRH, and inhibited by T_3 .
 - **FSH**
Glycoprotein gonadotropin. Release is stimulated

by GnRH, and inhibited by circulating sex steroids.
Has different effects depending on sex:

- Females: Stimulates oestrogen synthesis and ovarian follicle development.
- Males: Stimulates sperm maturation.

◦ **LH**

Glycoprotein gonadotropin with different effects depending on sex:

- Females: Rapid increase stimulates ovulation and corpus luteum development.
- Males: Stimulates testosterone synthesis.

• **Direct acting hormones:**

◦ **GH**

Long-chain peptide released in a pulsatile fashion. Release is stimulated by GHRH and is typically high with exercise, hypoglycaemia, and stress. Release is inhibited by somatostatin and IGF-1. GH has generally anabolic effects:

- Directly stimulates lipolysis, increasing circulating FFA
- Indirectly stimulates IGF-1 release, promoting cell growth and development

◦ **Prolactin**

Long-chain peptide which promotes breast development during gestation, and lactation after delivery.

Posterior Pituitary Hormones

Secretes **two** structurally similar short-chain peptide hormones:

• **Anti-Diuretic Hormone (ADH)**

Released in response to a change in osmolality detected by osmoreceptors in the circumventricular organs. ADH:

- Release is:
 - ↓ When osmolality is $<275 \text{ mOsm.L}^{-1}$

- ↑ When osmolality is $>290 \text{ mOsm.L}^{-1}$
 - Effective at:
 - V_1 receptors in vascular smooth muscle, causing vasoconstriction
Effect predominates in muscle, skin, and fat; and may relatively spare coronary, cerebral, and mesenteric circulations.
 - V_2 receptors in the:
 - Kidney, where it has two actions:
 - ↑ Aquaporin expression, leading to ↑ water reabsorption
 - ↑ Urea transport in the collecting duct, leading to ↑ medullary concentrating
 - Endothelium to increase vWF and factor VIII release
 - V_3 receptors in the pituitary to stimulate ACTH release
 - **Oxytocin**
Role in:
 - Uterine contraction
 - Let-down reflex
Stimulates milk release on suckling.
 - Psychological
Pair bonding.
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. Nickson, C. Vasopressin. LITFL.

Hypothalamic Hormones

Describe the control, secretions and functions of the ~~pituitary and the~~ hypothalamus

The hypothalamus is a circumventricular organ that regulates a large number of autonomic processes:

- **Thermoregulatory**
Integrates thermoreceptor input and controls activity of heat loss and heat gain mechanisms.
- **Satiety**
Feelings of hunger are modulated by glucose, CCK, glucagon, and leptin.
- **Water balance**
 - Contains osmoreceptors which control ADH release from the posterior pituitary
 - Angiotensin II stimulates thirst and ADH release via the subfornical organ and organum vasculosum
- **Circadian rhythms**
Balance between anterior and posterior hypothalamic stimulation controls sleep-wake cycle.
- **Pituitary control**
 - Anterior pituitary by hormone secretion into the long portal vein. Secreted hormones include:
 - GnRH, stimulates FSH and LH release
 - CRH, stimulates ACTH release
 - GHRH, stimulates GH release
 - TRH, stimulates TSH release
 - Somatostatin, inhibits GH and TSH release
 - Dopamine, inhibits prolactin release

- Posterior pituitary by neuronal innervation
- **Behaviour**
Punishment and reward centres.
- **Sexual function**

Hypothalamic-Pituitary Axis

The **hypothalamic-pituitary axis** describes the complex feedback loops between these endocrine organs:

- **Short-loop feedback** describes negative feedback from the pituitary on the hypothalamus, e.g. GH inhibiting GHRH release
 - **Long-loop feedback** describes negative feedback from a pituitary target gland (i.e. thyroid, adrenal, gonads) on the hypothalamus, e.g. cortisol inhibiting CRH (as well as ACTH) release.
 - These axes are also named with target gland, e.g. hypothalamic-pituitary-adrenal axis
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
2. Nickson, C. Vasopressin. LITFL.

Thyroid

Describe the control, secretions and functions of the thyroid.

The thyroid gland:

- Produces and secretes two hormones in response to TSH:
 - T_4 (thyroxine, 93%)
 - T_3 (tri-iodothyronine, 7%)
- Secretions are controlled via a negative-feedback loop on the hypothalamic-pituitary-thyroid axis
Increased TSH results in:
 - Increased iodine uptake
 - Increased iodination to form T_4 and T_3
 - Increased proteolysis of thyroglobulin, which releases T_4 and T_3
- Secretions are decreased with decreased iodine uptake
 - Perchlorate
Blocks Na^+/I^- symporter.
 - Wolff-Chaikoff effect
A reduction in thyroid hormone production due to a high circulating [iodide].

Synthesis

Thyroid hormones are:

In brief:

- Iodide is taken into the thyroid follicles by secondary active transport, and oxidised to iodine
- Thyroglobulin is synthesised in the follicle, and excreted into the follicular cavity
- Iodine is secreted into the follicular cavity, where it combines with thyroglobulin to produce T_4 and T_3

- **Synthesised in follicles**

A follicle is formed of a single layer of cuboidal epithelium around a central lumen (follicular cavity) containing thyroglobulin.

- **Iodide** is transported into follicular cells via a **secondary active transport** mechanism Na^+/I^- co-transporter.
- Iodide is then **oxidised to iodine**
- Thyroglobulin is synthesised in the endoplasmic reticulum of the follicular cell and excreted into the follicular cavity
- Iodine is excreted into the follicular cavity using a chloride exchange pump
- In the follicular cavity:
 - **Thyroid peroxidase** catalyses the iodination of thyroglobulin, forming mono-iodotyrosine and di-iodotyrosine
 - These are subsequently **oxidised**, forming T_3 and T_4 respectively

Secretion and Metabolism

Thyroid hormones are:

- Secreted in vesicles via endocytosis into the surrounding capillaries
 - Colloid enters thyroid cell via pinocytosis at the apical membrane
 - Vesicles then fuse with lysosomes
 - Thyroid hormone cleaved from thyroglobulin by proteases
 - Free T_3 and T_4 diffuse through the base of the thyroid cell into surrounding capillaries
- Highly protein bound to albumin and thyroxine-binding globulin
 - T_4 has a $t_{1/2}$ of 7 days
 - T_3 has a $t_{1/2}$ of 24 hours
 - Both are deiodinated in the liver, kidney, and muscle

- 55% of T_4 will be first deiodinated to T_3

Physiological Effects

Thyroid hormones:

- Act on thyroid receptors in the **cell nucleus**
Increasing gene transcription, protein synthesis, and mitochondria size and number.
- **T_3 is 3-5x more active than T_4**

Table 1: Effects of Thyroid Hormone

System	Effect
Resp	↑ MV due to ↑ CO_2 production ↑ HR
CVS	↑ Inotropy ↑ CO ↓ SVR
CNS	↑ Excitability - Seizures, tremor.
MSK	↑ Osteoblastic activity
GU	- Impotence (men) - Oligomenorrhoea (women)
GIT	↑ GIT motility ↑ BMR - Up to 100%.
Metabolic	↑ Carbohydrate metabolism ↑ Glucose uptake ↑ Glycolysis ↑ Gluconeogenesis
	↑ Lipolysis ↑ Non-shivering thermogenesis ↓ Plasma cholesterol ↓ Plasma phospholipids ↓ Triglycerides

System	Effect
	↑ Protein metabolism ↑ Anabolism at physiological levels ↑ Catabolism at high levels

Effects of thyroid hormone are predominantly metabolic.

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. CICM September/November 2008

Adrenal Hormones

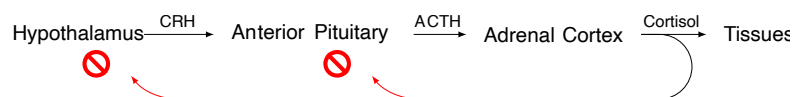
Describe the control, secretions and functions of renal and adrenal hormones

The adrenal glands are paired triangular glands at the superior pole of the kidney. The gland can be divided into:

- Adrenal cortex
 - Consists of three layers which produce **steroid hormones** (A good mnemonic is GFR for the layers, and ACT(H) for hormones)
 - Zona Glomerulosa
 - Predominantly produces mineralocorticoids (aldosterone).
 - Zona Fasciculata
 - Predominantly produces glucocorticoids (cortisol).
 - Zona Reticularis
 - Predominantly produces sex steroids (testosterone).
- Adrenal medulla
 - Produces catecholamines.

This covers the production of adrenal hormones. Information specific to catecholamines receptor function can be found under Chapter , whilst detailed information on specific agents, including structure-activity relationships of catecholamines, is covered under Chapter .

Steroid Hormones



Mineralocorticoids

Aldosterone is the key mineralocorticoid hormone, accounting for 95% of mineralocorticoid activity:

- Release is stimulated by:
 - ↑ Serum K^+
 - ↑ Angiotensin II
 - Hypovolaemia
 - ↓ Osmolarity
 - ↑ ACTH
 - ↓ Serum pH
- Acts to increase sodium and water retention (and removal of potassium), via:
 - ↑ Expression and activation of Na^+/K^+ pumps on the basolateral membrane of DCT and CT cells, causing increased Na^+ (and water) reabsorption and K^+ elimination
 - Stimulation of the Na^+/H^+ pump in intercalated cells on the DCT

Glucocorticoids

Cortisol (hydrocortisone) is the **primary endogenous glucocorticoid**, accounting for 95% of effect. Cortisol is:

- Produced at $\sim 15-30\text{mg}\cdot\text{day}^{-1}$
- Released in response to ACTH

ACTH is released in response to CRH, which is:

 - Released in response to stress
 - Modulated by circadian rhythms, and demonstrates diurnal variation:
 - CRH peaks just before waking
 - CRH troughs during sleep

Cortisol has effects on many organ systems, and in **physiological** amounts cause:

- CVS
 - ↑ Sensitivity to catecholamines

Metabolic effects of cortisol are essentially anti-insulin effects.

- ↑ Fluid retention
- Metabolic
 - Gluconeogenesis
To provide substrates, it also stimulates:
 - Proteolysis
 - Lipolysis
 - ↓ Glucose uptake

Catecholamines

Naturally occurring catecholamines include:

- Adrenaline
- Noradrenaline
- Dopamine

Synthesis of catecholamines:

- Occurs in the **adrenal medulla**
This is a **modified sympathetic ganglion** composed of **chromaffin cells**.
- Depends on **ACh release** by the presynaptic neuron
Unlike many other hormones, catecholamine secretion does not exhibit a negative-feedback loop.
- Occurs as follows:
 - **Tyrosine** is concentrated in the adrenal medulla
 - Tyrosine is hydroxylated to DOPA by **tyrosine hydroxylase**
This is the **rate-limiting** step, and is probably the best enzyme to remember.
 - DOPA is decarboxylated to dopamine
 - Dopamine is converted to noradrenaline
 - Noradrenaline is converted to adrenaline by **PNMT** (Phenylethanolamine N-methyltransferase)
This may only occur in the adrenal medulla.

Plasma half-lives of noradrenaline and adrenaline are small as a consequence of their metabolism and elimination.

- Extraneuronal uptake in the lungs, liver, kidney, and GIT
 - Neuronal uptake by sympathetic nerve endings
 - Inactivation by MAO in nerve cytoplasm
 - Inactivation by COMT in the liver and kidney
-

References

1. Brandis K. The Physiology Viva: Questions & Answers. 2003.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Calcium Homeostasis

Describe the function, distribution, regulation and physiological importance of sodium, chloride, potassium, magnesium, calcium and phosphate ions

Describe the control of plasma calcium.

Calcium is a bivalent cation. Almost all (99%) of calcium is located in bone, with the remainder in plasma and soft tissues.

Normal plasma levels are $2.2-2.55 \text{ mmol.L}^{-1}$, which (in plasma) may be:

- Ionised (free) calcium (50%)
Normal range $1.1 \text{ to } 1.3 \text{ mmol}^{-1}$.
- Bound to albumin (40%)
- As calcium compounds (10%)

Functions of Calcium

Physiological roles of calcium include:

- **Cell signaling**
Calcium has a number of roles in cell signaling:
 - Affects cell sodium permeability and therefore the RMP of excitable cells

- Calcium triggers exocytosis of neurotransmitter vesicles
- Calcium is an important second messenger for some G proteins
- **Bone**
Calcium has two functions in bone:
 - Physical structure
 - Alkali reserve
Calcium phosphate can be mobilised to buffer acidosis.
- **Enzymatic cofactor**
Calcium is an important cofactor in enzymatic pathways, including the coagulation cascade. Clinical hypocalcaemia does not cause coagulopathy however, as calcium levels low enough to prevent coagulation are not compatible with life.

Regulation of Calcium

Calcium is regulated to maintain a stable **ionised** calcium level. Three hormones are involved in the regulation of calcium:

- **Parathyroid Hormone**
Protein hormone secreted by the four parathyroid glands, located on the posterior surface of the thyroid, in response to a ↓ in iCa^{2+} levels, and acts to ↑ **plasma calcium**:
 - ↑ Calcium reabsorption in the PCT and late DCT
 - ↑ Osteoclastic activity in bone
 - ↑ Vitamin D activation in the intestine, which in turn increases intestinal absorption of dietary calcium
- **Vitamin D/Calcitriol**
Once converted to calcitriol in the kidney (via stimulation from PTH), vitamin D acts to:
 - ↑ Calcium reabsorption from kidney and gut
 - ↑ Bone calcification

- **Calcitonin**

Peptide hormone secreted by the C cells of the thyroid gland, in response to a rise in iCa^{2+} greater than 2.4mmol.L^{-1} . Calcitonin acts to:

- ↓ Absorption of calcium from gut and kidney
 - ↓ Osteoclastic activity of bone
-

References

1. Brandis K. The Physiology Viva: Questions & Answers. 2003.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Histamine

Describe the physiology of histamine and serotonin.

Histamine is an endogenous amine produced by decarboxylation of **histidine**. Histamine is:

- Present in all tissues
Particularly abundant in those exposed to the outside environment:
 - Lungs
 - Gut
 - Skin (lungs, gut, skin)
- Produced in and released by:
 - Mast cells
Released by exocytosis during inflammatory and allergic reactions.
 - Basophils
 - Histaminocytes in the stomach
 - Histaminergic neurons in the CNS
- Metabolised by:
 - Histaminase
 - Imidazole N-methyltransferase

Histamine Receptors and Effects

Histamine acts on:

- H₁ receptors
Gq receptor involved broadly in inflammation and vasodilation.

- H₂ receptors Gs receptor involved in gastric acid secretion.
- H₃ receptors
Gi presynaptic receptor in the CNS.
- H₄ receptors
Gi receptor located in bone marrow and other solid haematological organs (spleen, liver, thymus).

Table 1: Physiological Effects of Histamine

System	H ₁	H ₂	H ₃	H ₄
Resp	Bronchoconstriction	Bronchodilation		
	↑ Vasoconstriction	↑ HR		
CVS	(endothelial effect) Coronary vasoconstriction ↓ AV nodal conduction	↑ Inotropy Coronary vasodilation ↑ Capillary permeability		
CNS			Presynaptic inhibition of neurotransmission	
MSK	Weal due to local vasodilation			

System	H ₁	H ₂	H ₃	H ₄
	Itch ↑ Noci- ception ↑ Peri- stalsis	↑ Gastric acid secretion		
GIT				
Haeme				Alter IL-16 release

References

1. Parsons ME, Ganellin CR. Histamine and its receptors. British Journal of Pharmacology. 2006;147(Suppl 1):S127-S135. doi:10.1038/sj.bjp.0706440.
2. Rang HP, Dale MM, Ritter JM, Flower RJ. Rang and Dale's Pharmacology. 6th Ed. Churchill Livingstone.
3. Bowen R. Histamine. Vivo. Colorado State.

Prostanoids

Prostanoids are a diverse family of eicosanoids (20-carbon molecules), produced from arachidonic acid, and include:

- Thromboxane
- Prostacyclin
- Prostaglandins

Synthesis

Arachidonic acid is converted into:

- Leukotrienes by LOX
- Cyclic endoperoxidases by COX enzymes
These undergo further metabolism to produce:
 - Thromboxanes
 - Thromboxane A₂
 - Prostacyclins
 - PGI₂
 - Prostaglandins
 - PGE₂
 - EP₁
 - EP₂
 - EP₃
 - PGF_{2α}
 - PGD₂

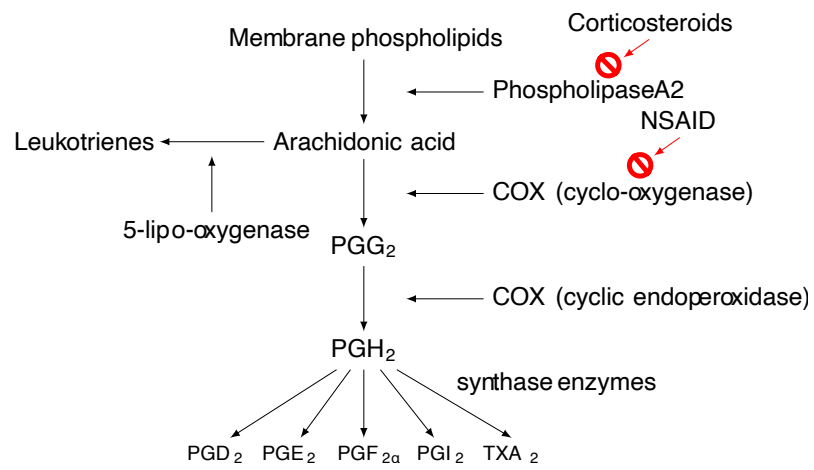


Table 1: Physiological Effects of Prostanoids

Receptor	Receptor	Respiratory	Vascular	GIT	GU	Other
Thromboxane A₂	Gq		Vasoconstriction			Platelet aggregation
PGI₂	Gs	Bronchodilation	Vasodilation (renal and pulmonary)			
PGE₂ EP₁	Gq	Bronchoconstriction		↑ Contraction	Renal vasodilation	
PGE₂ EP₂	Gs	Bronchodilation	Closure of ductus arteriosus	↓ Contraction	Renal vasodilation	
PGE₂ EP₃	Gi			Gastric mucous production GIT contraction	Uterine contraction	
PGF_{2α}	Gq	Bronchoconstriction	Vasoconstriction		Uterine contraction	
PGD₂	Gs				Renal vasodilation	Promotes sleep

References

1. Ricciotti E, FitzGerald GA. Prostaglandins and Inflammation. *Arteriosclerosis, thrombosis, and vascular biology*. 2011;31(5):986-1000. doi:10.1161/ATVBAHA.110.207449.

Musculoskeletal

Skeletal Muscle

Describe the anatomy and physiology of skeletal ~~, smooth, and cardiac~~ muscle

Describe the mechanism of excitation-contraction coupling

Describe the relationship between muscle length and tension

Outline the physiology of skeletal muscle including mechanism of excitation contraction coupling

Functions of skeletal muscle include:

- Facilitate movement
- Posture
Via tonic contraction of antagonistic muscle groups.
- Soft tissue support
Abdominal wall and pelvic floor support viscera.
- Voluntary sphincter control
- Heat production

Structure and Contents

Skeletal muscle consists of long tubular cells, known as **muscle fibres**, which run the length of the muscle. Muscle fibres consist of:

- **Connective tissue** layer
- **Skeletal muscle cells**
Contain **sarcomeres**, which are the **functional unit** of muscle.

Muscle Connective Tissue

Muscle fibres are surrounded by **three layers** of connective tissue:

- **Endomysium**
Thin layer which surrounds each muscle fibre.
- **Perimysium**
Surrounds bundles of muscle fibres.
- **Epimysium**
Thick layer which surrounds an entire muscle.

These layers of connective tissue join at the end of a muscle to form a tendon or aponeurosis.

Muscle Cells

Skeletal muscle cells:

- Are under voluntary control from the somatic nervous system via α -motor fibres
 α -motor fibres may control multiple myofibres, forming a **motor unit**.
- Are 10-100 μ m in diameter
- Contain several hundred peripheral nuclei
- Contain multiple mitochondria
Determines the fibre type:
 - **Slow oxidative fibres** (red fibres)
 - Contain multiple mitochondria
 - Produce sustained contraction
 - Resistant to fatigue
 - **Fast glycolytic fibres** (white fibres)

- Contain low numbers of mitochondria
- Contain large amounts of glycogen
- Produce strong contractions
- Are more easily fatigued
- Contain **sarcoplasmic reticulum**
- Contain large amounts of **glycogen**
~200g total.
- Contain myoglobin
- Appear striated microscopically due to the arrangement of myofibrils
 - Myofibrils are multiple myofilaments arranged in parallel
 - Myofilaments are formed from multiple sarcomeres arranged in series

Sarcomere

The sarcomere is the functional contractile unit of muscle, and contains two main proteins:

- Myosin (**thick**) filaments
Myosin is a large protein with two heads, which bind actin and ATP. The myosin head flexes on its neck during contraction.
- Actin (**thin**) filaments
Actin is a smaller protein than myosin, and potentiates the ATPase of myosin. Actin filaments have a groove which contains another protein called **tropomyosin**, to which **troponin** attaches to.
 - Troponin has three subunits:
 - Troponin T
Binds troponin to tropomyosin.
 - Troponin I
Prevents myosin binding to actin by physically obstructing the binding site.
 - Troponin C
Binds Ca^{2+} which initiates contraction.

These proteins are arranged to form three bands and two lines:

The average sarcomere length is 2.5 μ m.

- A-band
The myosin filaments.
- H-band
The section of myosin filaments not overlapping with actin filaments.
- I-band
The section of actin filaments not overlapping with myosin filaments.
- Z-line
Each end of the sarcomere. Actin from adjacent sarcomeres are connected at the Z line.
- M-line
Band of connections between myosin filaments.

Excitation-Contraction Coupling

Muscle contraction requires coordinating electrical (signaling) events with mechanical events. This occurs as follows:

- **ACh stimulates nicotinic receptors**
 $\uparrow$ Na^+ and K^+ conductance of the end-plate, generating an **end-plate potential**.
- **Muscle fibres** undergo successive **depolarisation**, generating an action potential along T tubules
These deliver the action potential deep into the cell, close to the sarcoplasmic reticulum.
- **Ca^{2+} is released from sarcoplasmic reticulum**
 - **Initial release** occurs via the **Dihydropyridine receptor**
 - Specialised voltage-gated L-type Ca^{2+} channel, activated by T-tubular depolarisation

- Responsible for a small amount of Ca^{2+} transport
- **Secondary** release occurs via the **Ryanodine receptor**
 - Second Ca^{2+} channel which is attached to, and activated by, the dihydropyridine receptor
 - Causes a much larger release of Ca^{2+}
Intracellular Ca^{2+} $\sim 2000\times$.
- Ca^{2+} binds to **troponin C**, which:
 - Weakens the troponin I - actin link
 - Uncovers myosin-binding sites on actin
- **Cross-linkages** form **between actin and myosin**, which releases ADP
- ADP triggers a **power stroke**, which is a process of attachment, pulling, and detachment
Each cycle shortens the sarcomere by **$\sim 10\text{nm}$** :
 - The myosin head rotates on its 'neck', moving to a new actin binding site
 - ATP binds to the (now free) binding site on the myosin
 - ATP is hydrolysed to ADP, in the process "re-cocking" the myosin head

This process causes the thick and thin filaments to slide on each other, with the myosin heads pulling the actin filaments to the centre of the sarcomere. Therefore, over the course of a power stroke:

 - The A-band is unchanged
 - The H-band shortens
 - The I-band shortens

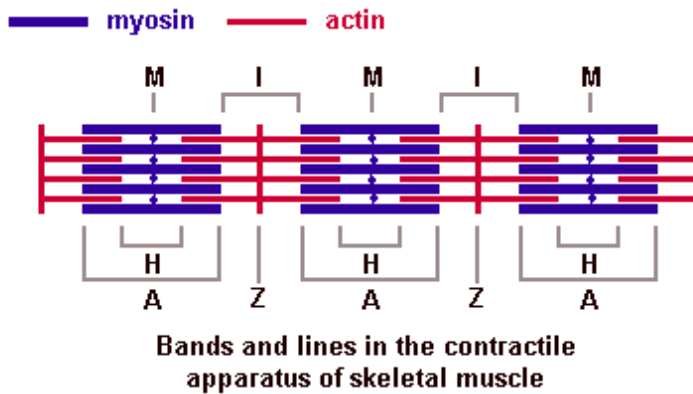


Figure 1: Sarcomere Contraction

Muscular Relaxation:

Power strokes continue as long as there is ATP and Ca^{2+} available. Therefore, in order to relax:

- Ca^{2+} is pumped back into the sarcoplasmic reticulum. This is an ATP-dependent process, and is why muscle relaxation is active.
- The $\downarrow$ intracellular Ca^{2+} causes troponin to release Ca^{2+} .
- Myosin binding sites on the actin are occluded by troponin, and no further contraction occurs.

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.
3. Slomianka, L. Muscle. University of Western Australia - School of Anatomy and Human Biology. The animated muscle contraction image is from this source.

Skeletal Muscle Innervation

Explain the concept of motor units

Describe the monosynaptic stretch reflex, single twitch, and tetanus

Motor Units

A motor unit consists of an α -motor neuron and the group of muscle cells that it innervates:

- An action potential in this neuron will cause contraction of all the myocytes in the unit
- Large muscles have many myocytes per unit
- Small, precise muscles (e.g. extraocular) have few myocytes per unit

Force of Contraction

Muscle tension is dependent on three factors:

- **Initial myocyte fibre length**
Optimal stretch maximises the number of overlapping actin and myosin filaments.
- **Number of contracting myocytes**
Recruitment of additional motor units increases the force of contraction.

- **Frequency of Action Potentials**

High frequency action potentials cause accumulation of calcium in the cytoplasm (the Bowditch or Treppe effect), increasing force of contraction.

- As the absolute refractory period of skeletal muscle is shorter than cardiac muscle, **tetany**, or sustained muscle contraction, can occur

Proprioception

Proprioception is the ability of the body to determine its position in space. There are two key proprioceptive sensors:

- Muscle spindles
- Golgi tendon organs

Muscle Spindles

Muscle spindles sense **changes in muscle length**. They:

- Are a specialised muscle fibre, known as **intrafusal fibres**
- Run **parallel** to myocytes (also known as **extrafusal fibres**)
- Consist of two elements:
 - Central, non-contractile portion which senses tension
 - Contractile endsThis allows the muscle spindle to adjust its length with its muscle, so that a constant tension in the non-contractile portion can be maintained over a range of muscle lengths.
- **Have both afferent and efferent innervation:**
 - Afferent type Ia fibres adjust their electrical output to signal both current fibre length and rate of change
 - Afferent type II fibres only signal fibre length

- Efferent γ neurons innervate the contractile elements

Golgi Tendon Organs

Golgi tendon organs are stretch receptors located between muscle and tendon. They:

- Run **in series** to myocytes
- Sense stretch
- Cause reflexive muscle **relaxation**, intended to prevent muscle damage

Voluntary muscle contraction results in contraction of both motor units (α 1 neurons) and intrafusal fibres (α -motor neurons).

Tonic innervation of α -motor neurons increases muscle tone by stretching the non-contractile portions, increasing Ia firing and subsequent α -motor unit firing.

Reflexes

A reflex is an **involuntary, predictable movement in response to a stimulus**. There are two types:

- **Monosynaptic**
Motor neuron synapses directly with the sensory neuron. Monosynaptic reflexes:
 - Are **rapid**
 - Generate **simple** responses
 - Have **five components**:
 - Sensory receptor
Typically muscle spindles.
 - Afferent neuron
Type Ia afferents relay signal from muscle spindle to ventral horn via the dorsal root.
 - Synapse between afferent and efferent neuron
In the ventral horn
 - Efferent neuron
 α -motor neuron travels from the ventral horn and innervates the motor unit.
 - Effector muscle
Innervated motor unit contracts in response.

- **Polysynaptic**

Motor neuron is separated from the sensory neuron by one or more interneurons in the dorsal horn.

Polysynaptic reflexes:

- Can be **modulated**
Interneuron activity can adjust reflex response.
- **Slower**
- More **complex**
e.g. Withdrawal of a limb from a hot object.

Twitch and Tetany

- A **twitch** is the response of a muscle to a **single stimulus** (action potential)
- A **tetanic** contraction describes the sustained contraction produced by **repetitive stimulation** before relaxation can occur
 - This stimulation must be causing above a **critical frequency**, which is dependent on the action potential duration for a cell
 - Repetitive stimulation causes repeated SR depolarisation, leading to sustained high intracellular Ca^{2+} levels as Ca^{2+} entry exceeds Ca^{2+} exit
 - Force from tetanic contraction is up to 4x greater than that of a twitch

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Nutrition and Metabolism

Basal Metabolic Rate

Define basal metabolic rate and outline the factors that influence it

Outline the factors that influence metabolic rate

Basal Metabolic Rate is the **energy output required to sustain life at rest**.

- 'Resting' is defined as an individual who is:
 - Fasted for 12 hours
 - In a comfortable external environment
 - At mental and physical rest
- Normal values are:
 - $100\text{W}\cdot\text{day}^{-1}$
 - $70\text{kcal}\cdot\text{hr}^{-1}$

Factors Affecting Metabolic Rate

Metabolic rate is the **actual energy consumption** of an individual, and is greater than BMR due to a number of factors.

Table 1: Factors Affecting Metabolic Rate

Factor	Effect	Detail
Age	BMR ↓ as age ↑	Neonates have a BMR twice that of an adult Children have an ↑ BMR relative to that of an adult BMR ↓ by 2% for each decade of life
Body Composition	Lean muscle has a greater energy requirement than fat	Higher body fat percentage results in a lower BMR Females have a lower BMR for this reason alone
Diet	The specific dynamic action of food describes the ↑ metabolic rate in metabolic rate due to the energy required to assimilate nutrients	Generally ~10% ↑ in BMR Protein > carbohydrate > fat Specific Dynamic Action for each macromolecule is not related to the respiratory quotient for that food type
Exercise	Skeletal muscle is the largest and most variable source of energy consumption	
Environment	Cooler environments ↑ BMR	

Factor	Effect	Detail
	Temperate environments ↓	
	BMR up to 10%	
Physiological states	Pregnancy ↑ BMR up to 20% in 2nd and 3rd trimester	
	Lactation ↑ BMR	
	Catecholamines ↑ metabolic rate	
	Corticosteroids ↑ metabolic rate	
Disease states	Malignancy ↑ metabolic rate	
	Sepsis ↑ metabolic rate	
	Hyperthyroidism ↑ metabolic rate	

Men and women have equivalent BMR when adjusting for when adjusted for body lean mass.

Measurement of BMR using Indirect Calorimetry

BMR is measured using **indirect calorimetry**, which calculates heat production via measurement of VO_2 and VCO_2 .

A number of methods exist to measure BMR. Which used depends on:

- Whether the patient is intubated
i.e. All inspired and expired gases can be measured.
- Whether they require supplementary oxygen

When measuring BMR:

- Patients should be relaxed and fasted
- FiO_2 needs to be calculated (or taken from the ventilator settings), and $\text{E}_\text{T}\text{CO}_2$ and $\text{E}_\text{T}\text{O}_2$ must be measured
- Steady-state should be achieved across a five minute period
 - The average MVO_2 and MVCO_2 changes by <10%
 - The respiratory quotient ($RQ = \frac{\text{VCO}_2}{\text{VO}_2}$) change by <5%
 - This ratio will vary depending on the substances metabolised:
 - Carbohydrates = 1
 - Protein ≈ 0.8
 - Fat ≈ 0.7

A variety of equations can be used to calculate Resting Energy Expenditure. One is the **Weir equation**:

$REE = 1.44 \times (3.94 \times \text{VO}_2 + 1.1 \times \text{VCO}_2)$, where:

- REE is in kCal.day^{-1}
- VO_2 is in L.min^{-1}
- VCO_2 is in L.min^{-1}

The **respiratory quotient** is the value of $\frac{\text{VCO}_2}{\text{VO}_2}$ at steady-state, whilst the **respiratory exchange ratio** is affected by metabolic rate.

Errors in Indirect Calorimetry

- Air leaks and measurement errors
- Measures consumption (rather than requirements)
- Point estimate of a dynamic process

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Fat Metabolism

Describe the physiology of fat, carbohydrate and protein metabolism

Digestion

Triglycerides are the main constituent of body fat in animals and vegetables, and therefore in dietary fat. They consist of **three fatty acid** molecules joined by a **glycerol molecule**.

As fats are not water soluble, they tend to clump together in chyme and are hard to digest due to the low surface area:volume ratio. **Emulsification** speeds up the digestive process, and occurs via the action of:

- Bile salts
Many bile salts have a hydrophobic and a hydrophilic end, which give a detergent action. Bile salts bound to fatty acids form a **mixed micelle** which can be further digested by enzymes or directly absorbed.
- Partially digested fats
- Mechanical action of the stomach

Once emulsified, triglycerides can be hydrolysed by **lipase** into fatty acids and monoacylglycerol.

Absorption

Absorption occurs in a number of stages:

- Mixed micelles, free fatty acids, monoacylglycerol, and cholesterol are absorbed via facilitated diffusion into the enterocyte
- From the enterocyte:
 - Short-chain fatty acids (those with < 12 carbon atoms) enter the portal vein and travel directly to the liver
 - Long-chain fatty acids are re-esterified and packaged with a layer of **protein** and **cholesterol** to form a **chylomicron**
Re-esterification maintains the concentration gradient for diffusion of fatty acids, allowing further uptake to occur.
 - Chylomicrons are ejected from the cell into the lymphatics and travel to the systemic circulation
 - Chylomicrons are removed from circulation by **lipoprotein lipase**
Lipoprotein lipase is found on capillary endothelium and bound to albumin.
 - Lipoprotein lipase breaks down triglyceride in chylomicrons and VLDL to free fatty acids and glycerol
This reaction uses **heparin** as a **cofactor**.
 - Free fatty acids and glycerol are then free to enter adipose tissue

Storage

Fat is stored as **triglycerides**, and forms the bulk of energy storage of the body.

Triglycerides are synthesised by the liver:

- When insulin levels are high and glycogen stores are full
- From excess carbohydrate and amino acids
These are converted to fatty acids and glycerol, and then esterified to form triglyceride. This is known as lipogenesis.

Metabolism

Free fatty acids can be:

- Absorbed by adipocytes for storage
 - **β -oxidised to acetyl CoA** in the liver
Acetyl CoA can then enter the citric acid cycle to produce ATP.
-

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Carbohydrate Metabolism

Describe the physiology of fat, carbohydrate and protein metabolism

Storage

Carbohydrates are stored in liver and muscle as glucose polymers known as **glycogen**. Glycogen:

- Production is stimulated by insulin, which is released as plasma glucose levels rise following carbohydrate ingestion
When plasma glucose levels fall, the release of glucagon and adrenaline stimulates glycogenolysis.
- Storage occurs in the liver and skeletal muscle
 - The liver contains ~100g of glycogen
This can maintain plasma glucose for ~24 hours.
 - Skeletal muscle contains ~200g of glycogen
This cannot be released into circulation, and is for use only by the muscle.

Glycolysis

Glycolysis:

- Also known as the **Embden-Meyerhof pathway**

- Describes the process of converting glucose into pyruvate
- Occurs in the cytoplasm
- Produces 2 ATP
- Does not consume oxygen or produce carbon dioxide
Glycolysis can therefore occur under both aerobic and anaerobic conditions.

Gluconeogenesis

Gluconeogenesis is the process that consumes ATP to produce glucose from other molecules. Gluconeogenesis is:

- Stimulated by glucagon
 - Inhibited by biguanides (metformin)
 - Fueled by:
 - Lactate
 - Pyruvate
 - Glycerol
 - Amino acids
 - CAC-intermediates
-

References

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Protein Metabolism

Describe the physiology of fat, carbohydrate and protein metabolism

Essential amino acids cannot be produced by transamination - they must be supplied in the diet.

Metabolism

Protein **catabolism** involves the **deamination** of amino acids. Deamination can occur in one of two ways:

- **Oxidative deamination**

Hepatic deamination, removing the amino group to create a ketoacid and ammonia. Ammonia produced in the liver enters the urea cycle and becomes urea, which requires **3 ATP**.

- **Transamination**

Amino group is transferred by aminotransferases to another amino acid or a ketoacid to produce:

- Keto acids, which:
 - Enter the citric acid cycle and produce ATP
 - Get converted to glucose or fatty acids
- Amino groups
 - Enter the urea cycle and become urea

Ammonia can also be produced in the kidney by the deamination of glutamate in the kidney. In this instance:

- It is eliminated directly in urine as ammonium
- Does not enter the urea cycle

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1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Fasting

Describe the physiological consequences of fasting and starvation

Fasting

Fasting is the metabolic state achieved after complete digestion and absorption of a meal.

Fasting can be divided into:

- **Early fasting**

Less than 24 hours.

- Plasma glucose falls due to consumption

Leads to hormonal changes:

- Insulin release decreases, causing:

- Liver

- Decreased glycogenesis
- Increased gluconeogenesis

- Muscle

- Decreased glucose utilisation
- Decreased glycogenesis
- Decreased protein synthesis

- Fat

- Decreased lipogenesis

Due to:

- Decreased glucose uptake
- Decreased TG uptake

- Increased lipolysis
 - Adrenaline release increases, causing:
 - Decreased insulin release
 - Increased lipolysis
 - Increased muscle FFA use
 - Increased hepatic glycogenolysis and gluconeogenesis
 - Glucagon release increases
- Cellular metabolism alters:
 - Decreased glucose uptake by non-obligate glucose consumers
e.g. Muscle.
 - Increased FFA and ketone body use
 β -oxidation of FFAs to meet ATP requirements, leading to formation of ketone body.
- **Sustained fasting**
Greater than 24 hours. See starvation below.

Starvation

Starvation is the failure to absorb sufficient calories to sustain normal body function, requiring the body to survive on endogenous stores.

- **Days:**
 - Energy is conserved through reduction in movement
 - Hormonal changes
 - Increased gluconeogenesis, using glycerol, lactate, and amino acids
 - Insulin concentrations fall further
 - Cortisol levels increase
 - Glucagon levels peak at 4 days
 - Metabolic changes
 - Glucose use continues to fall, and FFA use increases
 - Further fall in muscle protein synthesis
- **Weeks:**

- Tissues adapt to metabolise ketones (with plasma levels rising up to 7 mmol.L^{-1}), and gluconeogenesis falls
 - The brain still requires 100g of glucose per day
- BMR falls
- All but life-saving movement ceases
- Death typically occurs after 30-60 days, when muscle catabolism weakens the respiratory muscles such that secretions can no longer be cleared, and pneumonia occurs

Refeeding Syndrome

Refeeding syndrome is a deranged metabolic state that occurs with feeding after a period of prolonged fasting, typically >5 days.

There are three pathogenic mechanisms:

- A large spike in insulin causes increased cellular uptake (and low plasma levels) of:
 - Glucose
 - Magnesium
 - Phosphate
 - Potassium
- Sodium and water retention occurs, which may precipitate cardiac failure
- $\uparrow \text{CO}_2$ production increases minute ventilation and work of exhausted respiratory muscles

Management of refeeding syndrome is by slow institution of feeding and aggressive electrolyte management.

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Anaerobic Metabolism

Describe anaerobic metabolism and ketone production

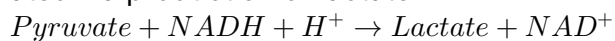
Lactate

Glycolysis (also known as the **Embden-Meyerhof** pathway):

- Describes the conversion of glucose to pyruvate (and two ATP)
- Does not consume O_2 or produce CO_2
Therefore it occurs in both anaerobic and aerobic conditions.
- Consumes two NAD^+ and produces two NADH

In anaerobic conditions (in the erythrocyte, and in the setting of cellular hypoxia):

- There is no oxygen available to allow further ATP production via the electron transport chain
There is also no regeneration of NAD^+ in the ETC.
- In order for glycolysis to continue, NAD^+ is regenerated via production of lactate



About 1400mmol (20mmol/kg) of lactate is produced per day. Lactate is either:

- Oxidised in the cell
This requires restoration of NAD^+ , e.g. resolution of cellular hypoxia.

- Circulated to the liver
Lactate is then:
 - Oxidised to pyruvate
 - Converted to glucose via the **Cori cycle**

Ketones

Ketones:

- β -oxidation of fatty acids in the liver produces acetyl-CoA
 - Acetyl-CoA usually enters the citric acid cycle to produce ATP
 - When large amounts of acetyl CoA are produced, they may instead condense to form **acetoacetate**, which can then be reduced to **β -hydroxybutyrate**
These substances are known as ketones
 - Ketones can only be produced by the liver, and only used as a substrate by the kidney, as well as skeletal and cardiac muscle
 - Production of ketones is accelerated by glucagon and adrenaline
-

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
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Body Temperature

Define heat and temperature

Outline the mechanisms for heat transfer between the body and its environment

Explain the mechanisms by which normal body temperature is maintained and regulated

Explain the physiological responses to hypothermia and hyperthermia.

Define the thermoneutral zone

Describe the mechanisms by which normal body temperature is maintained.

Outline temperature regulation in the neonate and the physiological responses to lowered and raised environmental temperature

The body is able to maintain a relatively constant core temperature under a wide range of environmental conditions by **balancing heat loss and heat production**, predominantly through behavioural mechanisms and skin.

The **thermoneutral zone** is the range across which the basal rate heat production (and oxygen consumption) is balanced by the rate of heat loss:

- For an adult it is **typically 27-31°C**
- In neonates it is higher, **typically 32-34°C**.

Principles

Net flux of heat is determined by the balance of metabolic heat production and the contribution of four mechanisms of heat loss:

- **Radiation**
- **Conduction**
- **Convection**
- **Evaporation**

Radiative

Radiative heat exchange:

- Describes the loss of heat through EMR by all objects above 0°K
 - Radiative heat loss is proportional to temperature
 - Radiative heat loss does not require a transfer medium
- Makes up ~45% of heat loss under thermoneutral conditions.
- Depends on the temperature differential between an individual and their environment
 - A cold environment (e.g. operating theatre) causes a large radiant heat lossThe heat loss from the patient is greater than the heat gain from the surrounding environment.

Conduction

Conduction is the **transfer of heat (as kinetic energy) by direct contact** from a higher temperature object to the lower temperature one. Conduction:

- Requires physical contact between bodies to conduct heat
 - Solids conduct heat better than gases
 - There is no conduction in a vacuum
- Heat loss via conduction is minimal in air but is a major cause of heat loss in immersion
 - As arteries and veins typically run next to each other, arterial heat tends to be transferred to the (cooler) veins, limiting further heat loss
This is similar to counter-current exchange in the kidney.
- As fat is a poorer conductor of heat than muscle, increased body fat will slow heat loss by conduction

Convection

Convection is loss of heat by conduction by a moving object. Convection is:

- The predominant mechanism of heat loss in the naked human
Effects are greater effects at higher wind speeds.

Evaporation

Evaporative losses describe the loss of heat energy due to the latent heat of vapourisation of water. Evaporation:

- Of 100ml of water will reduce body temperature by ~1°C

Temperature Sensation and Regulation

Temperature sensors are **central** and **peripheral**, whilst **regulation** occurs centrally. The **inter-threshold range** is:

- The range of core temperatures not triggering a response.
- Normally is 0.2 to 0.4°C.
- Widened under anaesthesia to ~4°C

Central Sensation

Central temperature sensors exist in the:

- Abdominal viscera
- Spinal cord
- Hypothalamus
 - **Anterior hypothalamus** is the most important central thermoreceptor, and responds to both increased and decreased temperatures by **altering their rate of depolarisation**, eliciting an array of neuronal and hormonal responses.
- Brainstem

Peripheral Sensation

Peripheral temperature sensors are:

- **Free nerve endings**
- Extremely sensitive
 - Alter their rates of firing by orders of magnitude in response to temperature change.
- Divided into:
 - **Cold receptors**
 - Lie beneath the epidermis, and are excited by cooling (inhibited by warming), active from 10-40°C, with a static maxima at 25°C.

- **Warm receptors**

Lie deep to the dermis, are excited by warming (and inhibited by cooling), active from 30-50°C, with a static maxima at 44°C.

Regulation

Temperature sensation runs from cutaneous receptors via the spinothalamic tracts and medulla to the hypothalamus. Cortical input is received via the thalamocortical relay, whilst primitive responses are effected via the midbrain.

Effector Responses

Table 1: Thermoregulatory Responses to Heat and Cold

	↑ Heat loss	↓ Heat loss/↑ Heat gain
CNS	Remove clothing Sprawl ↓ Activity	Huddle Seek shelter Add clothing
Cardio-vascular	↑ Peripheral vasodilation and AV shunting ↑ CO to improve flow to cutaneous tissues	Vasoconstriction Peripheral circulatory shut down
Musculo-cutaneous	Sweating	Piloerection Shivering
Metabolic		↑ BMR Non-shivering thermogenesis

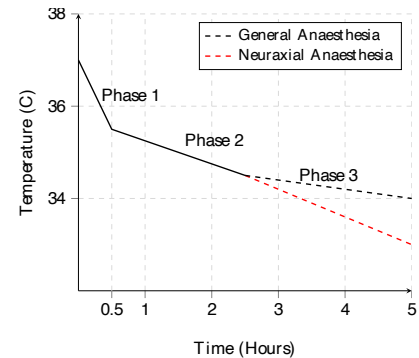
- **Vascular changes** are the least metabolically costly and can result in dramatic increases (up to 60% of cardiac output) in skin blood flow

- When environmental temperature exceeds body temperature, conduction and convection result in heat gain - **evaporative cooling via sweating** is the only way to reduce body temperature
- Efficacy of sweating is related to **relative humidity**
- Piloerection (hair standing on end) traps a layer of warm air close to the body to act as an insulator
This is of more importance in other primates than in man, as they have enough body hair to make it effective.
- ↑ BMR and 'waste' heat production is essential to maintain temperature in cold environments. This can be through:
 - Shivering
The simultaneous contraction of agonistic and antagonistic muscles.
 - Non-shivering thermogenesis:
 - Hormonal
Levels of thyroid hormone and adrenaline increase, raising metabolic rate in all cells
 - Brown fat
Brown fat produces heat through **uncoupled oxidative phosphorylation**, which uses the electron transport chain to produce heat rather than ATP. Brown fat is:
 - A vital mechanism for heat production in the **neonate** (they have an **immature shivering response**), and forms ~5% of neonatal mass
 - Located in:
 - Neck
 - Supraclavicular
 - Interscapular
 - Suprarenal
 - Sympathetically innervated
Contains large numbers of β_3 receptors

Effect of Anaesthesia

General anaesthesia causes a 1-3°C drop in core body temperature, which occurs in three phases:

- Rapid reduction
Core temperature falls by 1-1.5°C in the first 30 minutes.
 - Predominantly due to vasodilation, which is due to:
 - Reduction in SVR, with generalised vasodilation and increased skin blood flow
Heat redistribution is the major initial factor (rather than heat loss), as vasodilation leads to increased heat content of peripheries.
 - Impairs thermoregulatory vasoconstrictive responses
Inter-threshold range is widened to 4°C (up from 0.4°C)
- Gradual reduction
Further drop in core temperature of 1°C over following 2-3 hours.
 - Due to heat loss exceeding heat production
Non-shivering thermogenesis is the only response available to paralysed, anaesthetised patient.
- Plateau
Once core body temperature falls far enough, thermoregulatory responses are activated and further heat loss is attenuated by increased metabolic heat production.



With **neuraxial anaesthesia**:

- Hypothermia is less extreme as thermoregulation is only affected in areas covered by the blockade
- Plateau does not occur as vasoconstrictive responses are inhibited by the blockade

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Immunology

Inflammation

Describe the factors involved in the process of inflammation and the immune response, including innate and acquired immunity

Inflammation is a non-specific response triggered by a pathogen or tissue injury, which aims to limit further tissue damage.

Inflammation is classically characterised by:

- **Pain**
- **Heat**
- **Redness**
- **Swelling**
- **Loss of function**

This is a consequence of:

- **Vasodilation**
↑ Blood flow to area, which increases supply of immune cells and resources for cellular repair.
- **↑ Vascular permeability**
↑ Extravasation of protein and immune cells.
- **Migration of phagocytes**
Remove pathogens and cellular debris.

Process of Inflammation

- Tissue damage

- Trauma causes mechanical disruption of vasculature and mast cell degranulation, causing local inflammation and activation of haemostatic mechanisms
 - Infection stimulates degranulation of local macrophages, releasing inflammatory cytokines and triggering mast cell degranulation
 - Local inflammatory response
 - Histamine causes arteriolar and post-capillary venule dilatation and subsequent extravasation
 - Release of chemotactic molecules attracts circulating inflammatory cells
 - Systemic inflammatory response

Severe inflammation may lead to cytokines in the systemic circulation, causing:

 - Fever
 - Neutrophil recruitment from bone marrow
 - Release of acute-phase proteins from liver
-

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1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Innate Immunity

Describe the factors involved in the process of inflammation and the immune response, including innate and acquired immunity

The innate immune system consists of a composite of protective mechanisms that typically form the first line of defence against pathogens. Innate immunity is characterised by:

- **Immediacy**
- **Non-specific** response
- **Not modified** by repeat exposures

Components

The innate immune system is composed from **three components**:

- Physicochemical barriers
- Humoral mechanisms
- Cellular Mechanisms

Physicochemical Barriers

These include:

- Skin
- Mucous membranes

- Mucous
- Mucociliary elevator
- Gastric acid
- Urination
Optimised by high flow rates and low residual bladder volumes.

Innate Humoral Immunity

Humoral mechanisms describes the role of inflammatory proteins in innate immunity, and are divided into:

- **Complement**

The complement system is a complex group of about **25 plasma proteins** important in both innate and adaptive immunity.

- The complement system is activated by:
 - Antigen-antibody complexes
The 'classical pathway.'
 - Substances in the bacteria cell wall
The 'alternative pathway.'
- Complement has a number of inflammatory functions:
 - Destruction of bacteria
Several complement proteins come together to form a **membrane attack complex**, which creates large pores in cell membranes, causing water to diffuse in and bacteria to burst.
 - Opsonisation of bacteria
Bound complement acts as a binding site for phagocytes.
 - Activation of monocytes and phagocytes
 - Chemotaxis
Attracts leucocytes.
 - Mast cell degranulation
Augments inflammation.

- **Acute-Phase Proteins**

Inflammatory proteins with a number of effects:

- Opsonisation
- Inflammatory mediators
Increase blood flow and delivery of inflammatory cells via three mechanisms:
 - Dilatation and increased capillary permeability
 - Endothelial activation increasing leukocyte adhesion
 - Attraction of neutrophils and monocytes
- **Proteolytic enzymes**
Bactericidal enzymes located in saliva, tears, respiratory mucous, and neutrophils.

Innate Cellular Immunity

Cellular components of the innate immune system include:

- **Mast cells**
Exist in loose connective tissue and mucosa, and contain many intracellular granules of heparin and histamine.
- **Leukocytes**
 - Neutrophils (60% of all leukocytes)
Phagocytose bacteria and fungi (15 - 20 per neutrophil). This process consists of a number of steps:
 - Exit circulation by **marginating** along capillary border when activated
 - Migrate via **chemotaxis** towards the tissue insult
 - **Phagocytose opsonised** bacteria and fungi
 - Kill organisms with a **respiratory burst**:
A granule containing hydrogen peroxide, hydroxyl and oxygen radicals fuses with the target cell membrane, destroying both the target and the neutrophil.

- Monocytes
Become **macrophages** when they leave circulation and enter tissue. Macrophages have a lifespan of 2-4 months, and can phagocytose up to 100 bacteria before it dies. Functions include:
 - Phagocytosis and destruction of pathogen
Especially intracellular pathogens (*listeria*, *mycobacteria*), parasites, and fungi.
 - Breakdown of damaged body cells
 - Present antigen to T-helper cells
 - Secretion of inflammatory mediators
 - Eosinophils
Kill multicellular parasites.
 - Basophils
Contain heparin and histamine.
 - Lymphocyte
Subtype of leukocyte important in adaptive immunity. Include:
 - **Natural Killer cells**
Active against viral and tumour cells.
 - B cells
 - T cells
-

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Adaptive Immunity

Describe the factors involved in the process of inflammation and the immune response, including innate and acquired immunity

The adaptive immune system responds to an exposure, demonstrating **specificity** and **memory**, with improved efficacy on repeat exposure.

Adaptive immunity is divided into:

- **Active** immunity
 - Primary immune response generated by exposure to antigen.
 - Infection
 - Vaccination
 - An inactive (but still foreign and therefore antigenic) protein component of a pathogen is given to the patient, resulting in an immune response.
 - Subsequent exposure to the whole pathogen triggers a secondary immune response.
- **Passive** immunity
 - Preformed antibody is given to the patient. This will provide treatment/coverage for the life of the antibody, but immunity will be lost when the antibody breaks down or supplies are exhausted.
 - Transplacental
 - Colostrum
 - Administration of serum

Components

The active immune system is composed of **two components**:

- **Cellular**
Predominantly T lymphocytes.
- **Humoral**
Including complement and antibody.

Adaptive Cellular Immunity

Lymphocytes are divided into two types:

- **B lymphocytes**
Are produced in the bone marrow, and migrate to lymphoid (nodes, spleen, MALT) where they are renamed **plasma cells** and produce antibody. Functions include:
 - Production of antibody against specific antigens
 - Presentation of antigen to T-cells to activate them
 - Proliferation to form memory cells
- **T lymphocytes**
Are produced in the bone marrow and migrate to the thymus where they mature. T cells which express antibody to host protein apoptose, resulting in only 2% of immature T cells surviving. Mature T cells then spread to lymphoid tissue. There are five types of T cells, of which two are most important:
 - Helper T-cells
2/3rds of T-cells are helper cells, are identified by their **CD4** membrane protein. Functions include:
 - Cytokine production
 - B lymphocyte stimulation
 - Macrophage activation
 - Cytotoxic T-cells
Are identified by their **CD8** membrane protein. Functions include:

- Destruction of virally infected and tumour cells
All cells **express proteins** that they are producing **on membrane MHC I molecules**, for inspection by immune cells. Infected or tumour cells will express foreign proteins, and cause activation of cytotoxic T cells:
 - **Induce apoptosis** in the target cell
 - **Rapid division** of cytotoxic T cell, which then inspects other cells for infection
- Transformation to memory cells

Adaptive Humoral Immunity

Antibodies Y-shaped immunoglobulins which:

- Are produced in response to a pathogen
- Are specific to that pathogen

Antibody functions include:

- **Opsonisation**
- **Agglutination**
Each antibody can bind multiple pathogens, increasing target size for leukocytes.
- **Inactivation of pathogen**
Antibody binding may disable the pathogen.
- **Activation of complement**
Antibody-antigen complexes cause complement activation.

Adaptive Immune Response

The response of the adaptive immune system to a pathogen is divided into two phases:

- **Primary immune response**
Initial response to a novel pathogen, characterised by the formation of memory cells.
- **Secondary immune response**
Response to a previously encountered pathogen, characterised by an aggressive and targeted response.

Primary Immune Response

The process of invasion of a new pathogen to antibody production takes ~5 days, and occurs in a number of steps:

- APC phagocytose a pathogen
APCs include macrophages and dendritic cells.
- APC express antigen (bits of pathogen) on cell surface
- APC travel to lymphoid tissue and present it to B and T cells
- When an APC finds a B and T cell with a reciprocal antibody:
 - T helper cell becomes activated by APC
 - T helper cell rapidly proliferates ('clonal expansion')
 - Proportion become memory cells
 - B cells are activated by both the **APC and a T-helper cell** (requires both)
 - B cells rapidly proliferate
 - Proportion become memory cells
 - Proportion become **plasma cells**
 - Plasma cells produce antibody at a rate of 2000 molecules per *second*, which overrides normal cellular homeostasis, causing death within a week.
 - Antibody produced in a primary immune response is IgM, with some IgG produced later on.

Secondary Immune Response

Repeat invasion by the same pathogen is met with a much more rapid and aggressive immune response:

- APCs phagocytose a pathogen
 - APCs express and present antigen
 - Memory T and Memory B cells formed during the primary response are activated, and begin rapidly dividing and producing antibody
-

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Hypersensitivity

Explain the immunological basis and pathophysiological effects of hypersensitivity, including anaphylaxis.

Describe the immunology and pathophysiology of hypersensitivity reactions

Describe the immunology and pathophysiology of anaphylaxis

Hypersensitivity reactions are:

- **Exaggerated immune responses** that cause host injury.
- Classified by mechanism using the **Gel and Coombs system**
Common used but fails to classify more complex disease states.

Table 1: Gell and Coombs Classification of Hypersensitivity Reactions

Type	Timing	Mediator	Patho-physiology	Disease example(s)
Type I - Immediate hypersensitivity	Seconds to minutes	IgE	Basophil and mast cell degranulation	Anaphylaxis (systemic), Atopy (local)
Type II - Cellular hypersensitivity	5-8 hours	IgM, IgG	Antibody binding to cell surface antigen, resulting in cell death via complement membrane attack complexes, or phagocytosis by macrophages	Transfusion reactions, hyperacute allograft rejection

Type	Timing	Mediator	Patho-physiology	Disease example(s)
Type III - Immune-complex deposition	2-8 hours	IgM, IgG, IgA	Tissue deposition of Ab-Ag complexes. Accumulation of PMNs, macrophages, and complement.	SLE, necrotising vasculitis, post-Strep GN
Type IV - Delayed hypersensitivity	24-72 hours	T-cell	T-cell induced mononuclear cell accumulation. Release of monokines and lymphokines.	TB, Wegener's Granulomatosis, Granulomatous vasculitis

Type I Hypersensitivity

Mechanism of Type I hypersensitivity reactions:

- Antigen stimulates a B lymphocyte to produce a specific IgE against it
- This IgE then binds to Fc receptors on mast cells, sensitising them to this exposure
- On re-exposure the antigen binds (**cross-links**) IgE on mast cells, causing degranulation:

- Histamine, leukotrienes, and prostaglandins are released
- This may cause local or systemic effects, depending on method of exposure:
 - A systemic reaction is called **anaphylaxis**, and manifests as a combination of:
 - Hypotension
 - Bronchospasm
 - Laryngeal oedema
 - Rashes
 - Local reactions depend on the route of exposure, and include
 - Asthma
Inhaled.
 - Allergic rhinitis
Nasopharyngeal mucosa.

Type II Hypersensitivity

Mechanism of Type II hypersensitivity reactions:

- Antibodies bind to cell surface antigen
- Antibody - Antigen complex **activates complement**
- Complement generates an inflammatory response
- Cell death occurs via:
 - Complement membrane attack complex
 - Phagocytosis

Type III Hypersensitivity

Mechanism of Type III hypersensitivity reactions:

- Immune-complex reaction where Ab-Ag complexes are formed and deposited in tissues

Non-immune anaphylaxis (also known as anaphylactoid) reactions are characterised by a immediate generalised reaction clinically indistinguishable from true anaphylaxis, but the immune nature is unknown, or not due to a type I hypersensitivity reaction.

The clinical picture of type II hypersensitivity reactions depends on affected organs. Examples include:

- Hyperacute allograft rejection
- Transfusion reactions and haemolytic disease of the newborn
- Goodpasture's syndrome
- Autoimmune cytopaenias
- Myasthenia Gravis

- Subsequent complement activation causes inflammation and neutrophils activation, leading to tissue damage
- There are two subtypes of type III reactions:
 - Formation of complexes in circulation and subsequent deposition in tissues * e.g. Serum sickness
 - Formation of complexes in tissues
Small amounts are typically removed by the reticuloendothelial system, but in this case there are too many, or they are too small, to be cleared effectively.

Examples of type III hypersensitivity reactions include:

- The Arthus reaction
Form of localised vasculitis.

Type IV Hypersensitivity

Mechanism of Type IV hypersensitivity reactions:

- Antigen is presented to T lymphocytes which proliferate and become sensitised
- T-cells then release cytokines, attracting macrophages and leading to local inflammation
- During prolonged exposure, macrophages may fuse to form giant cells and form a granuloma.

Examples of type IV hypersensitivity reactions include:

- TB
- Granulomatous vasculitis

References

1. CICM July/September 2007
2. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Microbiology

Bacteria

Describe the classification of bacterium.

Bacteria are **prokaryotic** organisms, and are generally classified by the combination of:

- **Gram stain**

Gram staining:

- Separates bacteria according to their **cell wall composition**
- Cannot be used on organisms that lack a cell wall, such as mycoplasma
- Occurs applying a **crystal violet** stain followed by an **iodine** solution to the slide, which is then **washed with a solvent**
 - • **Gram +ve** organisms will **retain the stain** due to their **thick peptidoglycan cell wall**, whilst gram negative organisms become colourless
 - A **safranin pink stain** is then applied, which stains the **gram -ve** bacteria **pink**

- **Shape**

Bacteria can also be classified by shape into:

- **Cocci**
Appear round on microscopy.
- **Rods**

Table 1: Examples of Clinically Relevant Bacteria

Exam- ples	Gram Positive	Gram Negative
Cocci	<i>Staphylococcus Aureus</i> <i>Streptococcus Pneumoniae</i>	<i>N. Meningitidis</i> <i>N. Gonorrhoea</i>
Rods	<i>Listeria monocytogenes</i> <i>Clostridium difficile</i>	<i>Escherichia Coli</i> <i>Pseudomonas aeruginosa</i>

Bacterial Subclassification

Additional testing can be done to further classify bacteria:

- **Catalase** testing is performed on **Gram positive cocci**
Hydrogen peroxide is added to a bacterial sample, and in the presence of catalase will produce **oxygen**.
 - Catalase positive indicates Staphylococci
 - Catalase negative indicates Streptococci (and enterococci)
- **Coagulase** testing is performed on Staphylococcal species
Coagulase is an enzyme which cleaves fibrinogen to fibrin. The staphylococcal colony is added to rabbit plasma and incubated. In the presence of coagulase, fibrin is formed.
 - Coagulase positive strongly suggests *S. Aureus*
 - Coagulase negative examples include *S. epidermidis* or *S. saprophyticus*
- **Haemolytic** testing is performed on Streptococcal species
Bacterial colonies are added to blood agar, and the colour change (due to haemolysis) is noted.

Most clinically relevant bacteria can be classified in this fashion, but it does not capture spirochetes, mycoplasmas, chlamydias, and other less commonly encountered organisms.

A more complete classification uses six properties:

1. Cell Wall Structure
Includes:
 - Flexible (e.g. Spirochetes)
 - Rigid
 - Non-existent
e.g. *Mycoplasma spp.*
2. Morphology
 - Unicellular
 - Filamentous
3. Growth Location
 - Extracellular
 - Obligate intracellular parasites
e.g. *Chlamydia spp.*
4. Gram Stain
 - Gram positive
 - Gram negative
5. Shape
 - Cocci
 - Rods
6. O₂ tolerance
 - Aerobes
 - Anaerobes
e.g. *Clostridium spp.*

- **α haemolytic** organisms produce dark green agar, as methaemoglobin is produced by hydrogen peroxide produced by these organisms. Examples include:
 - *Strep. pneumoniae*
 - *Strep. viridans*
 - **β-haemolytic** organisms produce yellow agar from complete haemolysis. Examples include:
 - *Strep. pyogenes*
 - *Strep. agalactiae*
 - **γ-haemolytic** organisms leave the agar unchanged. Examples include:
 - *E. faecalis*
 - *E. faecium*
-

References

1. Harvey RA, Cornelissen CN, Fisher BD. Lippincott Illustrated Reviews: Microbiology (Lippincott Illustrated Reviews Series). 3rd Ed. LWW.
2. CICM September/November 2008

Viruses

Broadly outline the classification of viruses and fungi.

Viruses:

- Consist of molecules of either DNA or RNA shielded in a protein coat
- Require the use of host cell structures for reproduction and are therefore **obligate intracellular parasites**
- Can be classified by **five** properties:
 1. **DNA/RNA**
DNA viruses replicate in the cell nucleus using a host polymerase.
 2. Double-**stranded** or single-stranded
 - *Most* DNA viruses are double-stranded (ds-DNA)
 - *Most* RNA viruses are single-stranded (ss-RNA)
 3. Negative-**sense** or positive-sense (**single-stranded viruses only**)
 - Positive-sense genomes may be translated directly into mRNA
 - Negative-sense genomes require an RNA-dependent RNA polymerase to translate them to a positive-sense strand prior to translation.
 4. **Capsid Symmetry**
The protein coat may be either icosahedral or helical
 5. **Enveloped** or non-enveloped
In addition to a protein coat, viruses may have

a lipid membrane (acquired from the host cell)
around their protein coat.

References

1. Harvey RA, Cornelissen CN, Fisher BD. Lippincott Illustrated Reviews: Microbiology (Lippincott Illustrated Reviews Series). 3rd Ed. LWW.
2. CICM September/November 2008

Eukaryotes

Broadly outline the classification of viruses and fungi.

Eukaryotic organisms include fungi, protozoa, and helminths, as well as plants and animals. They differ from prokaryotic organisms in a number of ways:

Table 1: Comparison of Prokaryotic and Eukaryotic Cells

Property	Prokaryotes	Eukaryotes
Chromosomes	Single, circular	Multiple
Nucleus and Organelles	None	Membrane bound nucleus and organelles
Cell wall	Usually	In plants
Ribosome	70S	80S in cell, 70S in organelles
Size	0.2 - 2mm	10 - 100mm

Clinically relevant eukaryotes include:

- **Fungi**
Typically feed on dead/decomposing/the immuno-compromised, and produce spores. Fungi are sub-classified into:
 - **Yeasts**
Yeasts are unicellular. They are divided into:
 - Candida
 - Albicans

- Non-albicans Historically more treatment resistant.
 - Cryptococcus
 - **Moulds**
Moulds are filamentous. Include:
 - Aspergillus
 - Penicillium
 - **Dimorphous**
Have characteristics of both yeasts and moulds.
Include:
 - Histoplasma
 - **Protozoa**
Parasitic single-celled eukaryotes. Protozoa can be:
 - Intracellular or extracellular
 - **Helminths**
Parasitic multi-celled eukaryotes. Helminths can be:
 - Intracellular or extracellular
 - Divided into:
 - Tapeworms (cestodes)
 - Flukes (trematodes)
 - Roundworms (nematodes)
-

References

1. Harvey RA, Cornelissen CN, Fisher BD. Lippincott Illustrated Reviews: Microbiology (Lippincott Illustrated Reviews Series). 3rd Ed. LWW.
2. CICM September/November 2008

Antimicrobial Resistance

Describe the principles of anti-microbial resistance.

Resistance occurs when the **maximal level** of the agent tolerated is **insufficient** to **inhibit growth**, and may be produced by two broad mechanisms:

- **Genetic Alteration**
 - **Spontaneously**, through mutation and subsequent natural selection of resistant organisms
 - Transfer of resistance genes from organism to organism via **plasmids**
- **Protein Expression**
 - ↑ Or ↓ expression of proteins with subsequent change in efficacy of antimicrobials.

Mechanisms

Specific mechanisms of resistance (which may be genetic alterations or changes in protein expression) include:

- **Prevent access to target**
 - Can occur via:
 - ↓ Membrane permeability
 - Narrowing of porin channels
e.g. Streptococcal resistance to penicillins typically occurs by reducing access to PBPs.

- Loss of non-essential transporter channels
e.g. Anaerobes have no oxygen-transport channel which prevents penetration by aminoglycosides.
 - Active efflux of agent
↑ Efficiency or expression of efflux pumps. This results in the antimicrobial agent being:
 - Removed from cell
 - Trapped between cell wall layers
e.g. Glycopeptide resistance in VRSA.
 - **Alter antibiotic target site**
 - Changes in binding site protein will increase resistance to agents with low affinity
 - Over-expression of target protein
 - Synthesis of target-protecting proteins
 - **Modification or Inactivation of Drug**
 - Metabolism of drug
e.g. β -lactamases hydrolyse β -lactam rings
 - **Modification of Metabolic Pathways**
 - Development of metabolic pathways to bypass site of action of antibiotic
e.g. Resistance to Trimethoprim-Sulfamethoxazole by allowing bacteria to synthesise or absorb folic acid.
-

References

1. Harvey RA, Cornelissen CN, Fisher BD. Lippincott Illustrated Reviews: Microbiology (Lippincott Illustrated Reviews Series). 3rd Ed. LWW.
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3. Blair JM, Webber MA, Baylay AJ, Ogbolu DO, Piddock LJ. Molecular mechanisms of antibiotic resistance. Nat Rev Microbiol. 2015 Jan;13(1):42-51. doi: 10.1038/nrmicro3380.
4. Microraio - Mechanisms of Antimicrobial Resistance.

Obstetrics

Respiratory Changes of Pregnancy

Explain the physiological changes during pregnancy and parturition.

Describe the physiological changes and their implications for anaesthesia that occur during pregnancy, labour and delivery, in particular the respiratory, cardiovascular, haematological and gastrointestinal changes.

Respiratory changes in pregnancy are a function of:

- Anatomical compression of the chest
- $\uparrow$ VO_2 and VCO_2

Anatomical Changes

Key anatomical changes include:

- Superior displacement of the diaphragm by **~4cm**
- $\uparrow$ AP and transverse diameter of the chest wall (~2-3cm)
- Large airway dilation, $\downarrow$ airway resistance by ~35%

Volumes and Capacities

From conception until term:

- V_T increases by 40%
- Inspiratory capacity increases by 10%
- Expiratory capacity decreases by 30%
- Total lung capacity decreases by 5%
- Vital capacity is unchanged

From ~20 weeks until term:

- ERV decreases
- RV decreases
- FRC decreases
 - By 20% erect
 - By 30% supine

Ventilation

Progesterone stimulates respiratory centres, shifting the O_2 and CO_2 response curves to the left which causes hyperventilation and a **respiratory alkalosis**. From conception until term:

- MV $\uparrow$ by 50%
 - 10% $\uparrow$ in RR
 - 40% $\uparrow$ in V_T
- $PCO_2 \downarrow$ to ~26-32mmHg, with a compensatory drop in plasma $[HCO_3^-]$ to 18-21mmol.L⁻¹

Labour and Postpartum

During labour:

- MV $\uparrow$ 70% due to pain and $\uparrow O_2$ demand

- This causes hypocapnea, so cessation of uterine contractions (and the associated pain and oxygen demand) are followed by a hypoventilatory period producing desaturation

FRC and RV return to normal within 48 hours of delivery.

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Cardiovascular Changes of Pregnancy

Explain the physiological changes during pregnancy and parturition

Describe the physiological changes and their implications for anaesthesia that occur during pregnancy, labour and delivery, in particular the respiratory, cardiovascular, haematological and gastrointestinal changes.

Pregnancy is a time of ↑↑ metabolic demand, which is reflected in the cardiovascular adaptations. These include:

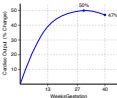
- ↑ Intravascular volume
Occurs via two mechanisms:
 - ↑ **Oestrogen** causes an ↑ **plasma volume**
 - This ↓ capillary oncotic pressure, predisposing to oedema
This may be exacerbated by the gravid uterus compressing the IVC, especially near-term
 - ↑ **EPO** causes an ↑ **red cell volume**
- ↑ Venous return
Due to ↑ intravascular volume and MSFP.
 - The gravid uterus may compress the IVC and impair VR, hence pregnant women are positioned with a left lateral tilt to displace the uterus off the IVC

- $\uparrow$ VR causes an $\uparrow$ in CO, with both an $\uparrow$ in HR and SV, as well a $\downarrow$ in SVR
- $\downarrow$ SVR results in $\downarrow$ SBP, DBP and MAP, which occurs despite the $\uparrow$ in CO

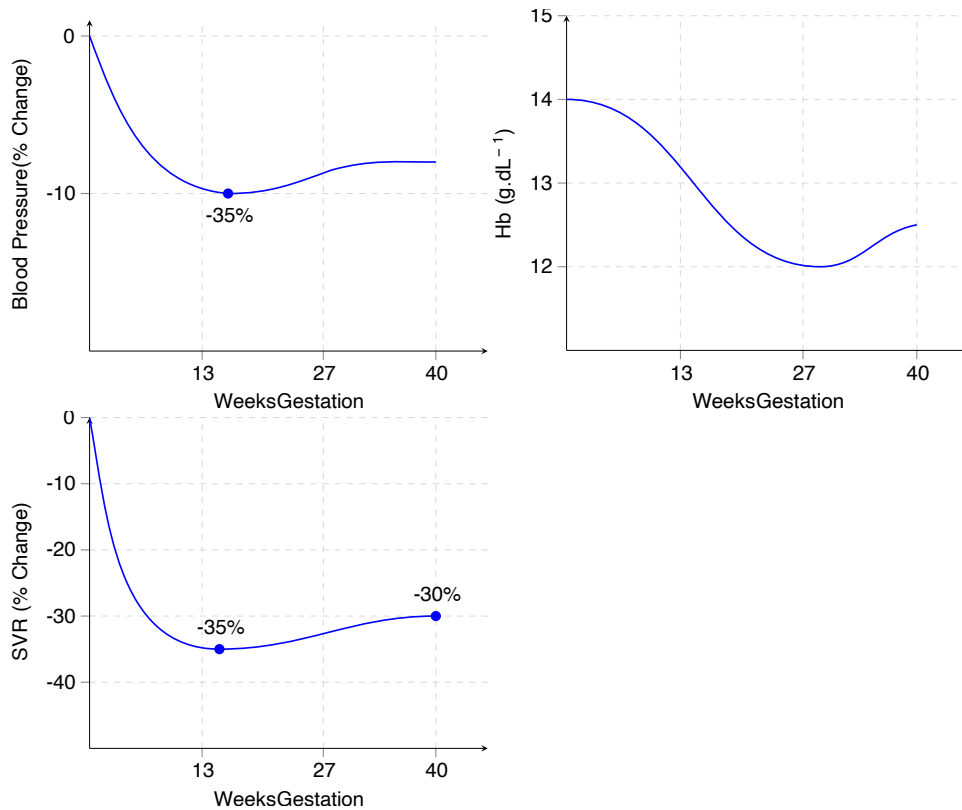
Magnitude of Changes by Trimester

Table 1: Cardiovascular Changes During Pregnancy by Trimester

Parameter	Direction	1 st Trimester	2 nd Trimester	3 rd Trimester	Notes
Plasma volume	↑	35%	45%	50%	Peaks between 32 - 36th week, ↓ slightly thereafter ↑ Less than the ↑ in plasma volume, resulting in the fall in haematocrit to 33%
Blood volume	↑	5%	15%	20%	
HR	↑	15%	18%	25%	↑ Progressively throughout
SV	↑	20%	25%	30%	↑ Progressively throughout

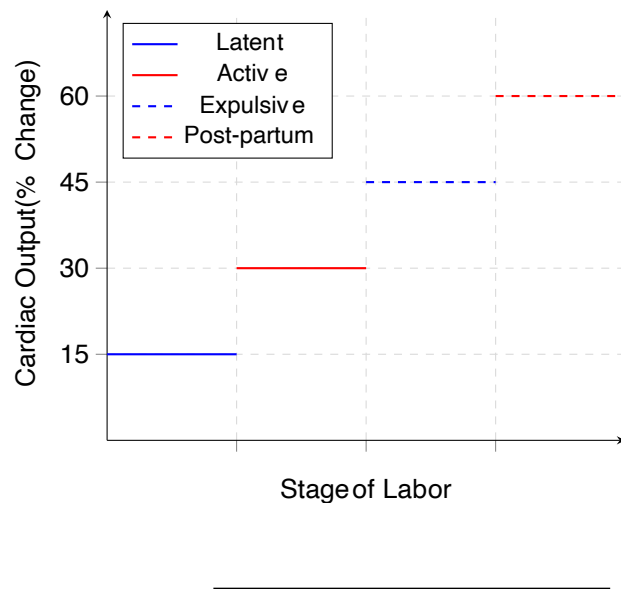
Parameter	Direction	1 st Trimester	2 nd Trimester	3 rd Trimester		Notes
CO	↑	20%	40%	45%		↑ Throughout, and dramatically in labour

Other Changes



Changes During Labour

- Uterine contraction boluses ~300ml of blood into the maternal circulation
Causes an $\uparrow$ in CO by up to 30% during the active phase and 45% during ejection.
 - Associated with corresponding increase in SBP and DBP by 10-20mmHg
- Post-partum CO is up to 80% of pre-labour values due to autotransfusion, and returns to normal within 2 weeks of delivery



References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Foetal Circulation

Describe the transition from foetal to neonatal circulation and the establishment of ventilation.

In Utero

The foetal circulation has a number of structural differences to the adult circulation:

- Two **umbilical arteries**
Return deoxygenated blood to the placenta.
 - PO_2 of 18mmHg (SpO_2 45%)
 - Over 50% of the combined output of both foetal ventricles enters the placenta
- One **umbilical vein**
Supplies oxygenated blood to the foetus.
 - Has a PaO_2 of 28mmHg (SpO_2 70%)
 - 60% of blood from the umbilical vein enters the IVC
 - 40% of blood enters the liver
- Two ducts:
 - **Ductus venosus**
Shunts blood from the umbilical vein to the IVC.
 - **Ductus arteriosus**
Shunts blood from the pulmonary trunk to the descending aorta.
- A **foramen ovale**
Shunts blood from the right atrium to the left atrium.

- **Immature myocardium**

Foetal myocardium does not obey Starlings Law, and does not adjust contractility for any given preload.

Therefore:

- SV is fixed
- CO is HR dependent

Normal HR at term is 110-160 bpm.

These structural difference alter the pathway of blood circulation:

- Oxygenated blood returns via the umbilical vein
 - 40% flows to the liver
 - 60% is returned to the IVC
- Oxygenated blood in the IVC is directed via the Eustachian valve through the foramen ovale
- Blood returning from the SVC is directed into the RV, and then into the descending aorta by the ductus arteriosus
 - ~10% of RV output flows through the pulmonary circulation

This arrangement has several features:

- Blood with the most oxygen is delivered to the arch vessels to supply the brain
- Blood with the least oxygen is delivered to the umbilical arteries for gas exchange
- Both the RV and the LV eject into systemic circulations, and are of similar size and wall thickness

Changes at Birth

Several changes happen at birth:

- Placental circulation is lost
There is a transition from circulation in parallel to circulation in series.
- An **FRC is established**
Reversal of hypoxic pulmonary vasoconstriction results in a rapid drop in PVR.

- The cord is clamped (or spasms, eventually)
The systemic vascular bed volume falls, and SVR increases due to the loss of the low-resistance placental circulation.

This results in:

- A ↓ in RV afterload, secondary to the ↓ in PVR
RAP falls due to the loss of hypoxic pulmonary vasoconstriction.
- An ↑ LV afterload, secondary to the ↑ in SVR
LAP rises as the LV moves up the Starling curve.
- Closure of the foramen ovale when LAP > RAP
A degree (~10%) of residual shunt remains. This shunt is generally bidirectional, and:
 - Left-to-right shunt is generally un concerning
 - Right-to-left shunt has usually only minor effects on systemic SpO₂
Will be increased with ↑ PaCO₂, excessive PEEP, ↓ pH.
 - Beware **embolic material**
Don't forget the bubbles.
- **Flow reversal** in the ductus arteriosus due to ↑ left-sided afterload
There is progressive closure of the ductus over hours to days, under the influence of prostaglandins and oxygenated blood flowing through the duct.
- Progressive fibrosis of the ductus venosus
Over a period of days to weeks.

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

3. Brandis K. The Physiology Viva: Questions & Answers. 2003.

The Placenta

Outline the functions of the placenta, and determinants of placental blood flow.

Describe the utero-placental circulation and the principles of placental physiology as related to placental gas exchange and regulation of placental blood flow.

The placenta is an organ of maternal and foetal origin which supports the developing foetus, and:

- Develops simultaneously from the foetus and the mother
 - From the uterine wall, the mother produces **blood sinuses** around the trophoblastic cords
 - These in turn send out **placental villi**
 - This creates a sinus of maternal blood invaginated by multiple foetal villi
 - Foetal villi are supplied by **two umbilical arteries** and a **single umbilical vein**
 - Maternal sinuses are filled from the **uterine arteries**
 - The maternal sinuses are supplied by **spiral arteries**
- Has **three broad functions**:
 - Interface between foetus and mother for exchange of **gases and nutrients**
 - **Immunological barrier**
 - **Endocrine**

Nutrient and Waste Exchange Functions

The primary purpose of the placenta is **diffusion of nutrients and oxygen**, and **removal of waste**.

As with the lung, diffusion is dependent on Fick's Principle, i.e.:

$$\dot{V} = \frac{A \times D \times \Delta P}{T}, \text{ where:}$$

- $\dot{V}$ is the flow of substance across the membrane
- A is the area of the membrane
- D is the diffusion constant for the substance, where:
 - $D \propto \frac{\text{Lipid Solubility}}{\sqrt{\text{Molecular Weight}}}$
 - Molecules < 600 Da in size more readily diffuse down concentration gradients
- ΔP is the concentration difference across the membrane
 - Maternal placental flow is $\sim 600 \text{ mL} \cdot \text{min}^{-1}$ at term - double that of foetal flow - which improves diffusion by increasing the concentration gradient for solutes
- T is the thickness of the membrane

Oxygen Transfer

At the end of pregnancy, the **PO₂ for foetal blood**:

- Entering the placenta via the umbilical artery is **18mmHg** (SpO₂ 45%)
- Leaving the placenta via umbilical vein is **28mmHg** (SpO₂ 70%)

The foetus is able to have adequate delivery of O₂ despite the low PO₂ for four reasons:

- **High Cardiac Index**
 ↑ Cardiac output increases DO₂.

The structure of the placental membrane has a significant effect on permeability. Consequently, the mature placenta is optimised for the determinants of Fick's law:

- Thin membrane - improved permeability
 Placental membrane permeability is small in early-to-mid pregnancy, reaching maximum at ~34 weeks
- Surface area of 14m²
- Weight of ~500g
- Blood flow of 600mL.min⁻¹ at term
 Flow is ↓ during contractions due to ↑ uterine pressure, and also with α-adrenergic stimulation.

• Foetal Hb

Contains **two gamma subunits** instead of beta subunits. These prevent the binding of 2,3-DPG, which result in a **left-shifted Oxy-Haemoglobin dissociation curve**, favouring oxygen loading at a low PaO_2 .

• Foetal [Hb] is 50% greater than maternal [Hb]

• The **Double Bohr effect**:

◦ The Bohr effect states that:

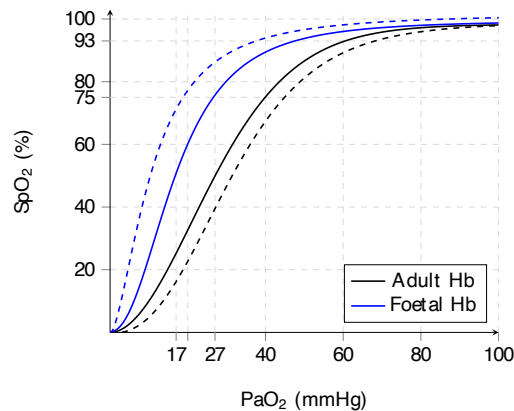
- An $\uparrow$ in PaCO_2 **right-shifts** the oxy-haemoglobin dissociation curve
- Conversely, the affinity of Hb for O_2 $\uparrow$ in alkalaemia

◦ The double Bohr effect describes the Bohr effect happening in *opposite* directions in the foetal and maternal circulations, the result of which favours transfer of O_2 to the foetus

◦ In the placenta, foetal CO_2 diffuses into maternal blood down its concentration gradient

This makes **foetal blood relatively alkaline, and maternal blood relatively acidic**. Therefore:

- O_2 unloading of maternal blood is favoured
- O_2 loading of foetal blood is favoured



Although *maternal* blood is relatively alkaline due to hyperventilation, *placental* blood is relatively acidic as it receives the entirety of foetal CO_2 production.

Carbon Dioxide Transfer

CO_2 is extremely lipid soluble, and so passes easily across membranes. Foetal PaCO_2 is $\sim 50\text{mmHg}$, and intervillous

PCO₂ is ~37mmHg. CO₂ offloading is favoured in the foetus by:

- A high Foetal [Hb] increases the amount of CO₂ that can be carried as carbaminohaemoglobin
- The **Double Haldane** effect:
 - The Haldane effect states that deoxygenated Hb binds CO₂ with more affinity than oxygenated Hb
 - The double Haldane effect describes this happening in opposite directions in the maternal and foetal circulations, favouring CO₂ transfer to the mother:
 - As maternal blood releases O₂, this favours maternal loading of CO₂ without an increase in maternal PCO₂ (Haldane effect)
 - The release of CO₂ from the foetal Hb favours O₂ loading, which in turn favours further maternal O₂ release.

Nutrient Diffusion

In late pregnancy, foetal caloric requirements are high (approximately the same as the mother). **Facilitated diffusion** of **glucose** via carrier molecules occurs in trophoblasts.

Active transport occurs for amino acids, Ca²⁺, Fe, folate, and vitamins A and C. Other transporters actively remove substances from foetal circulation.

Immunological Function

The placenta is selectively **permeable to IgG** via pinocytosis, which allows maternal antibodies to provide passive immunity to the foetus.

Endocrine Function

The placenta synthesises:

- β HCG
 - hPL
 - Oestriol
 - Progesterone
-

References

1. Hall, JE, and Guyton AC. Guyton and Hall Textbook of Medical Physiology. 11th Edition. Philadelphia, PA: Saunders Elsevier. 2011.
2. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Gastrointestinal System

Oesophagus

Describe the control of gastrointestinal motility, including (oesophageal) sphincter function.

Explain the factors preventing reflux of gastric contents into the oesophagus.

The oesophagus is a muscular tube connecting the pharynx to the stomach. The oesophagus has:

- **Skeletal muscle** in its **upper third**
- **Smooth muscle** in its **lower third**

Lower Oesophageal Sphincter

The lower oesophageal sphincter is:

- The most distal **2-4cm** of the oesophagus
- Macroscopically indistinguishable from the rest of the oesophagus
Microscopically, it has a higher concentration of nerve cells and is able to constrict at a higher pressure.
- Tonically innervated by the vagus
- Important in the prevention of reflux
Competency of the LoS is required to prevent reflux
 - **Barrier pressure is** the pressure difference between the pressure at the lower oesophageal sphincter and the pressure in the stomach, and

is typically **~15-25mmHg**

Barrier pressure is affected by:

- Changes in lower oesophageal sphincter pressure
 - Swallowing

Barrier pressure decreases during swallowing, and transiently increases immediately afterwards.
 - Anatomical
 - Age

Sphincter tone is decreased in neonates and the elderly.
 - Diaphragm

An external sphincter is formed by the diaphragmatic crura, and exerts a **pinch-cock** action on the oesophagus.
 - Stomach

A fold in the stomach wall just distal to the GOJ creates a **flap valve**, which occludes the GOJ when gastric pressure rises.
 - Oesophagus

The oesophagus enters the stomach at an oblique angle, limiting retrograde flow.
 - Hormonal
 - Gastrin, motilin, α -agonism **increase** LoS tone
 - Progesterone, glucagon, vasoactive intestinal peptide (VIP) **decrease** LoS tone
 - Drugs
 - ETOH, IV and volatile anaesthetic agents, and anticholinergics **decrease** LoS tone
 - Suxamethonium, metoclopramide, and anticholinesterases **increase** LoS tone
- Changes in gastric pressure
 - Raised intraabdominal pressure
 - Obesity

- Pregnancy
 - Disease
 - Hiatus hernia
 - GOJ moves into the thorax, causing:
 - Loss of pinch-cock action
 - Negative intrathoracic pressure reduces LoS pressure and therefore barrier pressure
-

The "pinch-cock" action refers to the external compression of the oesophagus from contraction of the diaphragm, which contributes to the action of the lower oesophageal sphincter.

The name is derived from the clamps used to compress tubing in laboratory equipment.

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
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3. Kahrilas PJ, Pandolfino JE. Hiatus hernia. GI Motility online. 2006.
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Gastric Secretions

Describe the composition, volumes, and regulation of gastrointestinal secretions.

Explain the composition of gastric fluid.

The gastrointestinal tract produces a number of substances, release of which occurs in **three phases**:

- **Cephalic**
Thought/sight/taste/smell of food, resulting in vagal-mediated stimulus to release gastrin. Accounts for ~30% of production.
- **Gastric**
Stretch of the stomach stimulates HCl secretion and gastrin release. Accounts for ~50% of production.
- **Intestinal**
A drop in pH of the proximal duodenum releases **secretin** to stimulate the exocrine pancreas.

Composition of Secretions

Table 1: Composition of Gastrointestinal Secretions

Saliva	Gastric	Pancreas	Small Bowel
H ₂ O (98%)	Digestive proteins:	H ₂ O	Endocrine:
Digestive proteins:	HCl	HCO ₃ ⁻	Secretin
Amylase	Gastrin	Digestive proteins:	Somatostatin
Lipase	Pepsin	Lipase	
Mucin	Intrinsic Factor	Amylase	
Hapto-corrin	Mucosal protection:	Trypsinogen	
Binds Vitamin B12.			
Immuno-logical proteins:	Mucous		
Lysozyme	HCO ₃ ⁻		
Lactoferrin			
IgA			

Salivary Secretions

Approximately **1L** of saliva is produced by the parotid, sub-mandibular, and sublingual glands each day.

Saliva has four main functions:

- **Lubrication**

- Mucin

- **Digestion**

- Amylase
- Lipase
Particularly important in neonates who produce little pancreatic lipase.

- **Neutralisation of acid**

For protection prior to vomiting.

- **Antibacterial**

Gastric Secretions

The stomach produces **~2L** of secretions **per day**. Key gastric secretions include:

- **Acid secretion**

Parietal cells contain an $H^+ - K^+$ exchange pump.

- H^+ is produced by carbonic anhydrase on CO_2 and water, with 'waste' HCO_3^- removed from the cell in exchange for Cl^- .
 - High levels of acid production result in large amounts of bicarbonate being secreted into blood
 - This creates an **alkaline tide** as portal venous pH increases dramatically
 - Respiratory quotient of the stomach may become negative due to **consumption of CO_2**
- This pump is activated in response to increased levels of intracellular Ca^{2+} from stimulation by:
 - **ACh**
 - **Histamine (H_2)**
 - **Gastrin**
- Inhibited by:
 - Low gastric pH
 - Somatostatin

- **Gastrin**

Gastrin is a peptide family secreted from antral **G cells**.

- Secretion is stimulated by:

- Neural (vagal) stimulation in the cephalic phase of digestion
Main mechanism.
- Protein and amino acids in the stomach
- Drugs
 - Alcohol
 - Caffeine
- Secretion is inhibited by:
 - Low pH
 - Secretin
 - Glucagon
- Gastrin has a number of pro-digestive effects:
 - Stimulates gastric acid secretion
 - Stimulates pancreatic secretion
 - Stimulates biliary secretion
 - Increases gastric and intestinal motility
- **Pepsinogens**
Chief cells secrete pepsinogen which is released by ACh or β stimulation. Pepsinogen is cleaved by hydrochloric acid in the stomach to become pepsin. Pepsin breaks down protein.
- **Intrinsic Factor**
Parietal cells produce intrinsic factor, which forms a complex with B₁₂ which facilitates its later absorption in the terminal ileum.
- **Mucous**
Neck cells produce mucopolysaccharide, glycoprotein, and HCO₃⁻ in response to stimulus by prostaglandins, which protects mucosa and lubricates food.

Pancreatic Secretions

Exocrine pancreatic secretions are produced by the **acinar** and **ductal cells**, at the rate of **1.5L per day**. Pancreatic secretions:

- Are released by an ↑ in:

- CCK
- Secretin
- ACh
- Via vagal stimulation.
- Consist of:
 - HCO_3^-
 - To alkalinise gastric contents.
 - Pancreatic bicarbonate production lowers venous pH, and neutralises the alkaline tide of the stomach.
 - Water
 - Enzymes
 - Trypsinogen
 - Activated to **trypsin** by duodenal enteropeptidases, and important in proteolysis.
 - Amylase
 - Hydrolysis of glycogen, starch, and complex carbohydrate.
 - Lipase
 - Hydrolysis of dietary triglycerides.

Small Bowel Secretions

- **Cholecystokinin (CCK)**
 - Peptide family secreted by intestinal enteroendocrine cells (I cells) in the mucosa of the duodenum and jejunum. Cholecystokinin:
 - Regulates satiety
 - Regulates leptin release from fat
 - Stimulates secretions from the gallbladder and duodenum
 - **Secretin**
 - Stimulates pancreatic release, and is released by the proximal duodenum in response to ↓ pH.
 - **Motilin**
 - Stimulates the migrating motor complex, and is released **cyclically** from M cells in the small bowel.
-

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Control of Gastric Emptying

Describe the control of gastrointestinal motility, including (oesophageal) sphincter function.

Gastric emptying is a neurally and hormonally mediated process which aims to present food to the small bowel in a controlled manner. Different drugs, hormones, and physiological states can either encourage or inhibit gastric emptying.

Determinants of Gastric Emptying

Rate of gastric emptying is a function of:

- **Antral pressure**
Primary determinant (pyloric resistance tends to be low), and is affected by:
 - Stomach
 - Duodenum
 - Systemic factors
 - Drugs
- **Pyloric resistance**

Stomach Factors

- Gastric **distension**
Vagal excitation from gastric stretch causes release of **gastrin**, increasing peristaltic frequency.
- Composition of chyme:
 - **Liquids** empty **faster** than **solids**
 - Liquids have a half-time of ~20 minutes, and empty in an exponential fashion
 - Solids have a half-time of ~2 hours, with a dwell time of ~30 minutes, and empty in a linear fashion
 - Protein independently stimulates gastrin release

Duodenal Factors

The duodenum has hormonal mechanisms which have a **negative feedback** on gastric emptying. These include:

- **Duodenal distension**
- **Hypoosmolar and hyperosmolar** chyme
- **Acidic** chyme
In response to acid the duodenum releases **secretin** and **somatostatin**:
 - Secretin directly inhibits gastric smooth muscle
 - Somatostatin inhibits gastrin release
- **Fat and protein**
Fat and protein breakdown products stimulate release of **cholecystokinin**, which **inhibits gastrin**.

Carbohydrate-rich meals empty faster than protein, which empty faster than fat.

Systemic Factors

- **Motilin**
Released by the small bowel and enhances the strength of the **migrating motor complex**, a peristaltic

wave of contraction through the whole GIT which occurs every 60-90 minutes.

- **Sympathetic** input Occurs via the coeliac plexus, and **inhibits** gastric emptying.
- **Parasympathetic** input
Enhances gastric motility.
- **Pregnancy**
Has a number of effects on gastric emptying:
 - Progesterone relaxes smooth muscle and inhibits gastric smooth muscle response to ACh and gastrin, as well as creating incompetence of the LoS leading to GORD
 - Gastrin production increases
Some gastrin is produced by the placenta.
 - Gastric acid production is increased during the third trimester

Drug Factors

Drugs which ↑ gastric emptying include:

- Metoclopramide
- Erythromycin

Drugs which ↓ gastric emptying include:

- Opioids
 - Alcohol
 - Anticholinergic agents
-

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Swallowing

Describe the control of gastrointestinal motility, including (oesophageal) sphincter function.

Explain the physiology of swallowing.

Swallowing is divided into **three phases**:

- **Oral Phase**

- Voluntary
- Food is pushed against hard palate by tongue

- **Pharyngeal Phase**

- Involuntary
Coordinated by medulla.
- Closure of nasopharynx
- Adduction of vocal cords
- Hyoid elevation and deflection of epiglottis
- Pharyngeal contraction
Propels food bolus towards oesophagus

- **Oesophageal phase**

- Involuntary
- Closure of UoS
Resting **barrier pressure 100mmHg**.
- Relaxation of LoS
Resting **barrier pressure 20mmHg**, which is a balance between:
 - LoS pressure (30mmHg)

- Antral pressure (10mmHg)
 - Oesophageal peristalsis
-

Impairment of any of these processes ↑
risk of aspiration. These include:

- Obtundation
Reduced cough reflex.
- Muscular weakness
- Impaired medullary
coordination

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Vomiting

Describe the control of gastrointestinal motility, including (oesophageal) sphincter function.

Explain the physiology of nausea and vomiting.

Vomiting is the active, forceful expulsion of gastric contents from the stomach. Vomiting is:

- Distinct from regurgitation, which is a passive process
- A mechanism to expel toxic substances from the GIT

Stimulation

Stimulants for vomiting are divided into:

- **Central stimuli**

Central stimuli may act:

- Directly on the vomiting centre

Include:

- Emotion
- Pain
- Olfactory
- Visual

- Via the chemoreceptor trigger zone

The CTZ is part of the **area postrema**, and is located outside of the blood-brain barrier. Therefore, the CTZ can be stimulated by circulating

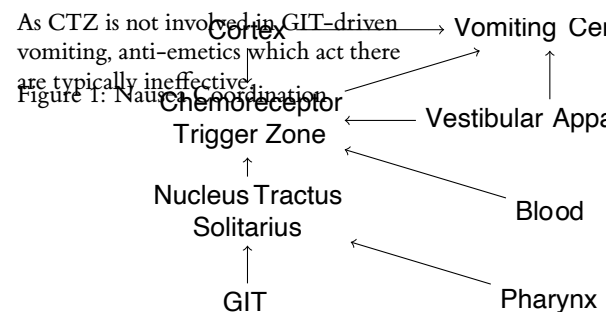
substances that have not crossed the BBB. Stimuli include:

- Vestibular stimuli, acting via:
 - H_1 receptors
 - ACh receptors
- Drugs/Toxins acting on:
 - 5-HT₃
 - D₂
 - μ -opioid receptors
- **GIT stimulation**
 GIT stimuli travel SNS and PNS afferents to the vomiting centre, and include **distension** and **toxins**. Neurotransmitters include:
 - 5-HT₃ in mucosal stretch receptors
 - ACh in NTS afferents
 - H_1 in NTS afferents

Postoperative Nausea and Vomiting

Central structures involved include:

- Chemoreceptor trigger zone
- NTS
- Multiple pathways exist (similar to those described above), and neurotransmitters involved include:
 - 5-HT₃
 - D₂
 - NK₁
 - H_1
 - mACh
- Risk factors
 - Patient factors
 - Female
 - Non-Smoker
 - Young age
 - History of PONV or motion sickness
 - Anaesthetic factors
 - Volatile use



- Nitrous oxide use
Relative risk of 1.4.
- Opioid use
- Anaesthesia duration
- Surgical factors
 - Gynaecological surgery
Likely not an independent risk factor, and simply confounded by female gender.
 - Strabismus surgery in children

Process of Vomiting

Vomiting consists of a set of processes coordinated by the **vomit centre** in the medulla oblongata, and is divided into three phases:

- **Pre-ejection phase**
 - Prodromal nausea
 - Salivation
 - Retrograde intestinal contraction which forces intestinal contents into the stomach
 - **Retching Phase**
 - Deep inspiration and breath-holding to splint the chest
 - Epiglottic closure
 - Elevation of the soft palate (prevents nasal soiling)
 - **Expulsive phase**
 - Relaxation of oesophageal sphincters
 - Pyloric contraction
 - Violent contraction of the diaphragm and abdominal muscles
-

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4. Pierre S, Whelan R. Nausea and vomiting after surgery. Continuing Education in Anaesthesia Critical Care & Pain, Volume 13, Issue 1, 1 February 2013, Pages 28–32.

Functions of the Liver

Describe the functions of the liver.

Outline the functions of the liver.

Key liver functions include:

- Storage
- Synthesis
- Metabolic
- Immune
- Acid-base

Storage

The liver is important in storage and release of:

- Carbohydrates as glycogen
The adult liver stores ~100g of glycogen.
- Fat as triglycerides
- All fat-soluble vitamins (A, D, E, K)
- Many water soluble vitamins including folic acid and B₁₂
- Iron
- Copper

Synthetic

The liver is critical for synthesis of:

- **Bile**
- Plasma proteins:
 - Clotting factors
 - Albumin production
120-300mg.kg⁻¹ of albumin is produced per day, dependent on nutritional status, plasma oncotic pressure, and endocrine function.

Metabolic

Metabolic functions include:

- Carbohydrate
- Fat
- Protein
- Bilirubin metabolism
- Drugs and Toxins

Carbohydrate Metabolism

The liver is critical for the production and storage of glucose:

- Monosaccharides and disaccharides passively diffuse into hepatocytes
Gradient is maintained by converting glucose to glucose-6-phosphate which is used to produce glycogen. This maintains the gradient for diffusion.
- Glycogen is either synthesised (glycogenesis) or broken down (glycogenolysis) depending on plasma glucose and insulin:
 - Increased blood glucose stimulates insulin release, increasing the formation of glycogen through activation of glycogen synthetase

- Decreased blood glucose stimulates glycogenolysis and gluconeogenesis from amino acids.

Lipid Metabolism

The liver is involved in both the production and consumption of fat, including:

- Storage of triglycerides
- Hydrolysis of fat to glycerol and fatty acids, which is used for ATP production

Protein Metabolism

Amino acids are absorbed from blood to be used for gluconeogenesis and for protein synthesis. In order to produce substrates for the CAC, Amino acids may be:

- Transaminated
- Deaminated
- Decarboxylated

The nitrogenous scrap of these reactions is urea, which is produced in several stages:

- A variety of metabolic processes convert amino acids to **glutamate**
- Glutamate is converted to **ammonia** by glutamate dehydrogenase
- Ammonia then enters the urea cycle to produce (surprisingly) urea, at the cost of **3 ATP**
 - A normal diet of 100g protein per day produces ~30g of urea, and 1000mmol of hydrogen ions

Endocrine

The liver:

- Produces angiotensinogen
- Produces IGF - 1
- Converts T4 to T3

Immune Function

The liver contains **Kupffer** cells, which are the tissue macrophages of the hepatic reticuloendothelial system. Kupffer cells phagocytose harmful substances including:

- Endotoxins
- Bacteria
- Viruses
- Immune complexes
- Thrombin
- Fibrin complexes
- Tumour cells

Acid-Base Balance

The liver may produce or consume large numbers of hydrogen ions:

- Carbon dioxide production
 - Metabolism of organic acid anions
 - Lactate
 - Ketones
 - Amino acids
 - Ammonium
 - Production of plasma proteins
Notably albumin
-

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
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Liver Function Assessment

Describe the laboratory assessment of liver function.

Describe the laboratory assessment of liver function and hepatic failure.

Synthetic Function

Measures of synthetic function include:

- **Albumin**

Main plasma protein.

- Normal range 28-58g.L⁻¹
- Half-life ~20 days
- Important in:
 - Maintenance of plasma oncotic pressure
 - Binding
 - Calcium
 - Drugs
- ↓ In liver dysfunction and malnutrition

- **Coagulation Assays**

Clotting factors are produced by the liver. Hepatic impairment may result in reduced production and abnormality of clotting assays, although functional clotting function may be normal (as pro-coagulant proteins are affected to a similar extent).

- INR
Test of the extrinsic pathway.
- APTT
Test of intrinsic pathway.

Metabolic Function

Transaminases are released when liver parenchyma is damaged, and are used to evaluate metabolic function:

- ALT
Normal range $< 54 \text{ U.L}^{-1}$.
- AST
Normal range $< 35 \text{ U.L}^{-1}$.

Obstructive Tests

- ALP (Alkaline Phosphatase)
Enzyme involved in dephosphorylation of many compounds. ALP is found in all cells, but particularly in the liver, bile duct, bone, kidney, and placenta.
 - Normal range is $30 - 120 \text{ U.L}^{-1}$
 - May be $\uparrow$ in isolation in bone disease
 - GGT
Enzyme found in biliary duct.
 - Normal range:
 - Males: $11 - 50 \text{ U.L}^{-1}$
 - Females: $7 - 30 \text{ U.L}^{-1}$
 - Bilirubin
Byproduct of haemoglobin metabolism. May be measured as total, or as conjugated and unconjugated bilirubin.
-

References

1. Diaz, A. Outline the clinical laboratory assessment of liver function. Primary SAQs.

Bile

Describe the physiology of bile.

Bile is a dark green solution produced by the liver to facilitate absorption of fat and fat-soluble vitamins through emulsification. Bile is:

- Produced by the liver at the rate of 1L per day
- Concentrated in the gallbladder
- Formed from:
 - Water
 - Protein
 - Bilirubin
 - Bile salts

The sodium and potassium salts of bile acids.

Bile acids:

 - Are produced from cholesterol
 - Are amphipathic, and act as emulsifiers of lipid

Break up large fat globules into smaller micelles, which can then be absorbed.
 - Major bile acids include:
 - Cholic acid
 - Chenodeoxycholic acid
 - Are absorbed in the terminal ileum, and re-cycled by the portal circulation
 - Lipids
 - Electrolytes

Fat-soluble vitamins include Vitamin A, D, E, and K.

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2. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.

Haematology

Erythrocytes

Outline the physiological production of blood and its constituents

Describe the physiological consequences of acute and chronic anaemia, ~~including iron deficiency.~~

Erythrocytes:

- Are **7.5µm** in **diameter**
- Are 2µm thick
- Have a lifespan of **120 days**
- Have:
 - No nucleus
Maximises cell volume available for Hb.
 - No mitochondria
Cannot perform aerobic metabolism - all ATP is generated via glycolysis.
 - No ribosomes
Incapable of producing protein
- Have a **biconcave disc** shape
This maximises surface area (optimising gas transfer) and makes the cells flexible enough to pass through capillary beds (which are narrower than the cell).
- Are important in:
 - Delivering O₂ to the tissues and delivering CO₂ to the lungs
 - Acid-Base balance

- Metabolism of some drugs
- Carry ~29pg of haemoglobin
- Comprise 40-50% of blood volume

Production

Erythrocytes have a **myeloid progenitor** which differentiates into the myeloid line. EPO stimulates myeloid progenitor cells to **differentiate** and then **proliferate**:

- Proerythroblasts begin synthesis of Hb, with ongoing production occurring until the cell is mature
- Further differentiation results in successive **loss of organelles, increasing Hb content**
- The loss of **ribosomes** and **nucleus** of the reticulocyte are the final stage of erythropoiesis
- The entire process takes ~7-10 days

EPO is covered in detail under Chapter .

Function

- Gas Carriage
- Acid-Base Buffering
 - Production of HCO_3^-
 - Binding of H^+ to Hb
- Metabolism

Esterases (and other -ases) in erythrocytes metabolise many drugs, including:

 - Remifentanyl
 - SNP

Reacts with Hb to form NO, CN, and Met-Hb.
 - Esmolol

Elimination

Old red cells are removed from circulation via:

- **Phagocytosis** by macrophages in:
 - Spleen
Major mechanism.
 - Liver
 - Bone marrow
- **Haemolysis**
~10% of red cell breakdown occurs in circulation, where the Hb dimers are then bound to **haptoglobin** by haemopexin.
 - This is important to prevent glomerular filtration of haeme, and loss of iron

Haemoglobin Metabolism

Haemoglobin is broken down into:

- **Globin**
Broken down into constituent amino acids.
- **Iron**
Re-enters haemoglobin synthetic pathway.
- **Haeme**
Complex metabolic pathway, notable as it is the only metabolic process that produces **carbon monoxide**:
 - Metabolised to **biliverdin** by splenic macrophages in the reticuloendothelial system of the **spleen**
 - Circulating erythrocytes are phagocytosed by splenic macrophages
 - Haptoglobin binds circulating Hb, the Hb-Haptoglobin complex is then phagocytosed by splenic macrophages
 - **Biliverdin** is reduced to **unconjugated bilirubin**
This is fat soluble, and binds to albumin.
 - **Unconjugated bilirubin** is conjugated in the liver to **conjugated bilirubin**

- **Conjugated bilirubin** is **secreted in bile** by active transport
This is impaired during hepatic disease, leading to increased bilirubin levels in plasma.
- Secreted **conjugated bilirubin** is metabolised to **urobilinogen** by gut bacteria
- **Urobilinogen** may have a number of fates:
 - Enterohepatic recirculation and elimination in bile (again)
 - Further metabolism by gut bacteria to **stercobilinogen** and then to **stercobilin**
 - Enterohepatic recirculation and urinary excretion, where it is oxidised to **urobilin**

Table 1: Alterations to Bilirubin Metabolism in Disease

	Blood	Urine	Faeces
Prehep- atic disease	↑ Unconjugated bilirubin	↑ Urobilinogen Bilirubin not present	Normal
Intrahep- atic disease	↑ Conjugated bilirubin ↑ Unconjugated bilirubin	Bilirubin present	Acholic
Posthep- atic disease	↑ Conjugated bilirubin	↓ Urobilinogen Bilirubin present	Pale

Acholic stools describes a pale or clay-coloured stool that occurs due to ↓ urobilinogen excretion in bile.

References

1. Barrett KE, Barman SM, Boitano S, Brooks HL. Ganong's Review of Medical Physiology. 24th Ed. McGraw Hill. 2012.

Iron Homeostasis

Describe the normal nutritional requirements, including vitamins and trace elements.

Describe the physiological consequences of ~~acute and chronic anaemia, including iron deficiency.~~

Approximately 3-5g of iron is found in the body as:

- **Oxygen-carrying globin molecules**
Haemoglobin (~70%) and myoglobin (~5%).
- **Catalyst** for biological reactions (~25%)
Catalase, peroxidase, and cytochromes all require iron.

Absorption

Dietary iron comes in two forms:

- Haeme groups
Directly absorbed via specialised transport proteins.
- Dietary iron salts
 - Ferrous (Fe^{2+}) iron is soluble, and is absorbed via facilitated diffusion across the enterocyte membrane
 - Reduced acidity of the stomach will reduce the absorption of ferrous iron

- Ferric (Fe^{3+}) iron precipitates when $\text{pH} > 3$, and so cannot be absorbed independently by the small bowel.
 - A pathway may exist for absorption of ferric iron from soluble chelates

Once in the enterocyte, iron can be:

- Stored, bound to **ferritin**
- Transported via **ferroportin** out of the enterocyte, where it is then oxidised to ferrous iron and bound to **transferrin**

Regulation

- Excretion is uncontrolled
 - Regulation of iron levels is only by absorption
 - **Hepcidin** is a liver protein which inhibits the action of ferroportin
 - High hepcidin prevents iron transport from the enterocyte
 - Hepcidin is deficient in haemochromatosis
-

References

1. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Platelets

Outline the physiological production of blood and its constituents.

Describe the process and regulation of haemostasis, coagulation and fibrinolysis.

Describe the physiology of haemostasis, including the role of platelets.

Platelets are anuclear circulating cell fragments, that perform a vital role in haemostasis by forming a **platelet plug**.
Platelets:

- Have a lifespan of **7 - 10 days**
- Are removed by the reticuloendothelial system in the spleen and liver

Production

Platelets:

- Bud from megakaryocytes
 - As the megakaryocyte cell volume increases, the cell membrane invaginates and small platelets bud off
 - The time from stem cell to platelet is **~10 days**, and is stimulated by **thrombopoietin**

- New platelets are held in the spleen for 36 hours until they mature

Contents

- **α -granules**
Contain fibronectin, fibrinogen, vWF, PDGF, and Thrombospondin, platelet factor 4.
- **δ -granules**
Contain 5-HT, ATP, ADP, and Ca^{2+} .
- **Contractile proteins**
Facilitate platelet deformation when activated.

Activation

Platelets are activated by exposure to:

- Collagen
Exposed by damaged endothelium.
- Adrenaline
- ADP
- Thrombin

Platelet activation results in a series of events:

- Exocytosis of granules
- Activation of membrane **phospholipase A_2** to form **thromboxane A_2**
- Deformation from a disc to a sphere with long projections
- Promotion of the coagulation cascade
- Change in glycoprotein (GP) expression by the action of ADP:
ADP antagonists (e.g. clopidogrel) prevent expression of the GPIIb/IIIa complex.
 - GP Ib/IIb/IIIa facilitate platelet attachment to vWF
vWF also binds to sub-endothelial connective tissue.

- GP IIb/IIIa are also receptors for fibrinogen, which encourages platelet aggregation
-

References

1. Kam P, Power I. Principles of Physiology for the Anaesthetist. 3rd Ed. Hodder Education. 2012.
2. Krafts K. Clot or Bleed: A Painless Guide for People Who Hate Coag. Pathology Student.

Blood Groups

Explain the major blood groups and the principles of cross matching.

Describe blood groups and the physiological basis of transfusion reactions.

Blood groups refer to the expression of surface antigens by red blood cells, as well as any antibody in plasma. Blood groups can be divided into three types:

- ABO
- Rhesus
- Other antibodies These are additional antibodies that a patient may express in plasma, and include Kell, Lewis, Duffy, etc.

ABO

The ABO blood group is:

- A complex carbohydrate-based antigens series
These may be either A or B antigen, and patients may express one, both, or neither, giving four blood groups (A, B, AB, O).
- Expressed on the **H-antigen stem** of RBCs, and on the surface of tissue cells.

- The **Bombay Blood Group** (or **hh** or **Oh** group) describes individuals who do not express the H antigen

These individuals:

- Don't express A- or B-antigen (as there is no H-antigen stem) and are 'universal donors'
- Express H-antibody
Can only receive blood from other individuals with the Bombay phenotype
- Individuals express IgM antibody to foreign blood groups
This develops within 6 months of birth, likely due to environmental exposure to similar antigens.
- Associated with a severe hypersensitivity reaction if an ABO-mismatch occurs

Rhesus

The Rhesus blood group is the next most important group after ABO. The Rhesus system:

- Consists of ~50 different antigens, the most important of which is **D**
Rhesus status is therefore expressed as positive (D - 85% of the population) or negative (anything-but-D).
- Rhesus antibody does not naturally occur in Rh(-) individuals
 - This is relevant in Rhesus disease
A Rh(-) mother exposed to Rh(+) blood will develop Anti-D antibody, which can cross placenta and induce abortion in a future Rh(+) foetus. This can occur with:
 - Incompatible transfusion
 - Foetal-maternal haemorrhage

Table 1: Antigen and Anti-body by Blood Group

Group	Red Blood Cell Antigen	Plasma Anti-body
A	A-antigen	B-antibody
B	B-antigen	A-antibody
O	-	A-antibody B-antibody
AB	A-antigen B-antigen	-

Compatibility Testing

Donor blood must be tested with recipient blood to avoid a transfusion reaction. This involves three steps:

- **Blood Typing (ABO/Rh)**

- Patient's blood is mixed with samples of plasma known to contain Anti-A, Anti-B, or Anti-AB antibodies.
- Agglutination occurs when antibodies in the plasma react with the antigen on the RBC surface, and indicates a mismatch

- **Antibody Screen**

For minor (non-ABO/Rh) group antibodies.

- Testing is similar to ABO screening, but now patient *plasma* is mixed with several samples of red cells which contain a known non-ABO/Rh antigen (e.g. Kell, Duffy), and monitored for agglutination

- **Cross-match**

Two methods:

- **Electronic cross-match**

Used if the antibody screen is negative and the patient has had no previous blood transfusion, as it is quicker and cheaper.

- **Serological cross match**

Indicated if the antibody screen is positive, or the patient has had previous exposure to blood (prior transfusion, pregnancy), as this gives the patient a risk of exposure to non-ABO or Rh antigens.

- Performed by mixing patient plasma with a small sample of donor RBC and monitoring for agglutination
- If there is no agglutination, the transfusion can proceed with the tested specimen **only**

Indirect Coomb's Test

Indirect Coomb's testing identifies IgG antibody in host *plasma* which would haemolyse transfused red cells, and involves:

- Incubating
Binds IgG Ab to antigen on RBC membrane.
- Washing
Removes serum and unbound IgG.
- Testing with an antibody to IgG, known as antiglobulin serum.
 - A positive test will cause clumping of red cells, as each antiglobulin serum will bind two IgG molecules, which have in turn been bound to red cells
 - A negative test will cause no agglutination, as the IgG has not been bound to red cells
 - If negative, the antiglobulin serum is re-used on a control sample to ensure that it is not a false negative

The indirect Coomb's test is now deprecated as it provides negligible additional safety over the above processes.

References

1. Blood Service. Classification & Incidence of Adverse Events. Australian Red Cross.
2. National Blood Authority. Patient Blood Management Guidelines. Australian Red Cross.
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Blood Product Storage

Understand the pharmacology of blood and its components, including individual factor replacement.

Describe the changes that occur during blood storage and their clinical implications.

Outline the changes that occur in stored blood.

Fractionation of whole blood into components ↑ efficiency of a blood bank, and ↑ shelf life of the components.

Blood can be stored for up to 35 days, which corresponds to 70% survival

Red Blood Cells

Packed red blood cells are:

- Collected in an aseptic fashion
- Kept at ~4°C
 - Balance between freezing and being too warm.
 - ↓ Cellular metabolic requirement
 - Inhibits bacterial growth
- Stored in special solutions:
 - SAGM is currently used by the Australian Red Cross:
 - Saline

- Adenine
Substrate for ATP synthesis
- Glucose
Substrate for RBC glycolysis
- Mannitol
- CPDA1 (citrate-phosphate-dextrose-adenine)
was traditionally used
 - Citrate binds calcium, preventing clotting
 - Phosphate acts as a buffer and phosphate source for metabolism
 - Dextrose
 - Adenine

Storage Lesion

Stored blood decays over time. The changes that occur in stored blood are known as **storage lesions**. This includes:

- Loss of 2,3 DPG
Less of a factor in CPDA1 blood.
- Haemolysis
- Hyperkalaemia
Typically not clinically relevant as potassium is taken up into red cells when metabolism resumes.
- Acidaemia
- Hyponatraemia
Not clinically significant.

Platelets

Platelets require particular storage conditions to remain functional:

- Temperature ~22°C
Below this, platelets deform and become non-functional.

- Gas exchange
Platelets are stored in a bag which allows gas exchange to occur, minimising lactic acid and carbon dioxide production.
- Agitation
Platelets are stored on an agitator which prevents clotting and ensures the platelets are well mixed, which maximises the diffusion gradient for gas exchange.
- pH control
pH is kept between 6.2 to 7.8 to prevent degranulation.

As platelets do not contain antigen, there is not a *strict* requirement for platelets to be type matched. However:

- Rh(+) platelets should be avoided in Rh(-) patients
The small amount of contaminating red cells may precipitate rhesus disease.
- Plasma incompatibility should be avoided as this may lead to haemolysis of recipient red cells
 - Children are at greater risk due to their proportionally smaller blood volume

Fresh Frozen Plasma

Fresh Frozen Plasma is:

- Prepared either via:
 - Separation from whole blood
 - Apheresis
Removal of a large volume (typically 800ml) of plasma from a single patient, with return of red cells to the donor.
- Frozen after collection and re-thawed in a water bath prior to use

Cryoprecipitate

Cryoprecipitate is prepared by removing the precipitate from FFP which forms at 1–6°C. Cryoprecipitate contains predominantly:

- Fibrinogen
- Fibronectin
- vWF
- Factor VIII
- Factor XIII

Whole Blood

Stored whole blood undergoes additional changes:

- White cells become nonfunctional within 4–6 hours of collection, though antigenic properties remain
 - Platelets become non-functional within 48 hours of storage at 4°C
 - Factor levels decrease significantly after 21 days
-

References

1. Blood Service. Classification & Incidence of Adverse Events. Australian Red Cross.
2. National Blood Authority. Patient Blood Management Guidelines. Australian Red Cross.
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Transfusion Reactions

Understand the adverse consequences of blood transfusion, including that of massive blood transfusion and storage lesions.

Describe blood groups and the physiological basis of transfusion reactions.

A transfusion reaction is an adverse event associated with the transfusion of whole blood or one of its components, and can be classified as either:

- Acute (< 24 hours) or delayed (> 24 hours)
- Immunological or non-immunological

Table 1: Acute Immunological Transfusion Reactions

Re- ac- tion	Inci- dence	Mechanism
ABO Mis- match	1:40,000	ABO incompatibility causing rapid intravascular haemolysis, which may cause chest pain, jaundice, shock, and DIC. RhD-reactions tend to cause extravascular haemolysis.
Haemolytic (acute)	1:16,000 (1:1.8 million fatal)	Immunological destruction of transfused cells (Type II hypersensitivity). Presents with fever, tachycardia, pain, progressing to distributive shock

Re- ac- tion	Inci- dence	Mechanism
Febrile, ~1:100 non- haemolytic		Cytokine release from stored cells causing a mild inflammatory reaction, with temperature rising to $\geq 38^{\circ}\text{C}$ or $\geq 1^{\circ}\text{C}$ above baseline (if $>37^{\circ}\text{C}$). Benign - but requires exclusion of a haemolytic reaction.
Ur- ticaria	1:100	Hypersensitivity to plasma proteins in the transfused unit
Ana- phy- laxis	1:20,000	Type I hypersensitivity reaction to plasma protein in transfused unit
TRALI	Vari- able	Donor plasma HLA activates recipient pulmonary neutrophils, causing fever, shock, and non-cardiogenic pulmonary oedema

Table 2: Acute non-Immunological Transfusion Reactions

Reaction	Inci- dence	Mechanism
Massive Transfu- sion Compli- cations	Variable	See below
Non- immune mediated haemoly- sis	Rare	Due to physicochemical damage to RBCs (freezing, device malfunction). May lead to haemoglobinuria, haemoglobinaemia, tachycardia and fevers.
Sepsis	1:75,000 1:500,000 (RBC)	Contamination during collection or (platelets), processing. Most common organisms are those which use iron as a nutrient and reproduce at low temperatures, e.g. <i>Yersinia Pestis</i> .

Reaction	Incidence	Mechanism
Transfusion Related Circulatory Overload (TACO)	<1:100	Rapid increase in intravascular volume in patients with poor circulatory compliance or chronic anaemia. May result in pulmonary oedema and be confused with TRALI.

Table 3: Delayed Immunological Transfusion Reactions

Reaction	Incidence	Mechanism
Delayed haemolytic transfusion reaction	1:2,500	Development of sensitisation with the reaction occurring 2 - 14 days after a single exposure. Typically Kidd, Duffy, Kell antibodies.
Post-transfusion Purpura	Rare	Alloimmunisation to Human Platelet Antigen causing sudden self-limiting thrombocytopenia
TA-GVHD	Rare	Transfused lymphocytes recognise host HLA as positive causing marrow aplasia, with mortality >90%
Alloimmunisation	1:100 (RBC antigens), 1:10 (HLA antigens)	Previous sensitisation leading to antibody production on re-exposure.
Transfusion related Immune Modulation	Not known	Transient immunosuppression following transfusion potentially due to cytokine release from leukocytes

Table 4: Delayed non-Immunological Transfusion Reactions

Re- ac- tion	Incidence	Mechanism
Iron Over- load	Chelation after 10-20 units, organ dysfunction 50-100 units	Each unit of PRBC contains ~250mg of iron, whilst average excretion is 1mg.day ⁻¹ .

Complications of Massive Transfusion

A **massive transfusion** is one where:

- Greater than **one-half** of **circulating volume** in **4 hours**
- Whole **circulating volume** in **24 hours**

Table 5: Complications of Massive Transfusion

Compli- cation	Mechanism
Air embolism	Inadvertent infusion
Hy- pothemia	Cooled products
Hypocal- caemia	Consumption with coagulopathy and bound to citrate added to transfused units
Hypo- magne- saemia	Bound to citrate in transfused units
Citrate toxicity	Citrate is added to stored units as an anticoagulant
Lactic acidosis	Hyperlactataemia due to anaerobic metabolism in stored units
Hyper- kalaemia	Potassium migrates from stored erythrocytes into plasma whilst in storage

Risk of complication from a massive transfusion is influenced by:

- Number of units
- Rate of transfusion
- Patient factors

References

1. Blood Service. Classification & Incidence of Adverse Events. Australian Red Cross.
2. National Blood Authority. Patient Blood Management Guidelines. Australian Red Cross.
3. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.

Haemostasis

Describe the process and regulation of haemostasis, coagulation and fibrinolysis.

Describe the physiology of haemostasis, including coagulation and the role of platelets.

Haemostasis describes the physiological processes that occur to stop bleeding. It involves three processes:

- **Vessel constriction**
↓ Flow, which limits further haemorrhage and reduces the shear stresses which break up forming clot
- **Platelet plug formation** or Primary Haemostasis
Platelets adhere to the damaged vessel wall and aggregate
- **Fibrin formation** or Secondary Haemostasis
Fibrin is formed from fibrinogen (via the coagulation cascade), which stabilises the platelet plug

Primary Haemostasis

Following a vascular injury, the exposure of subendothelial proteins stimulates platelets to form an occlusive plug via several processes:

- **Adhesion**

- Exposed collagen binds to GPIa receptor on platelets
- vWF also binds to platelets
- **Activation**
Metabolic activation of the platelet to facilitate clotting.
 - Metabolic activation is stimulated by:
 - Collagen
 - Adrenaline
 - ADP
 - Thrombin
 - Transforms the platelet from a disc to a sphere with long projections
Initiated by $\uparrow$ Phospholipase A_2 and Phospholipase C, causing $\uparrow$ platelet intracellular Ca^{2+} .
 - Activated platelets release ADP and thromboxane A_2 from their **alpha granules** and **dense bodies**
This further amplifies platelet aggregation and adhesion.
- **Aggregation**
With other platelets - held together by fibrin - forming a plug.
- **Contraction**
After some time platelets contract, retracting the clot and sealing the wall.

Secondary Haemostasis

The **coagulation cascade** is an amplification mechanism which activates clotting factors in order to produce fibrin. Participating factors in the coagulation cascade can be either **enzymes** or **cofactors**:

- Enzymes circulate in their inactive form, and become active (e.g. VII $\Rightarrow$ VIIa) when hydrolysed by their precursor factor
- Cofactors amplify the cascade

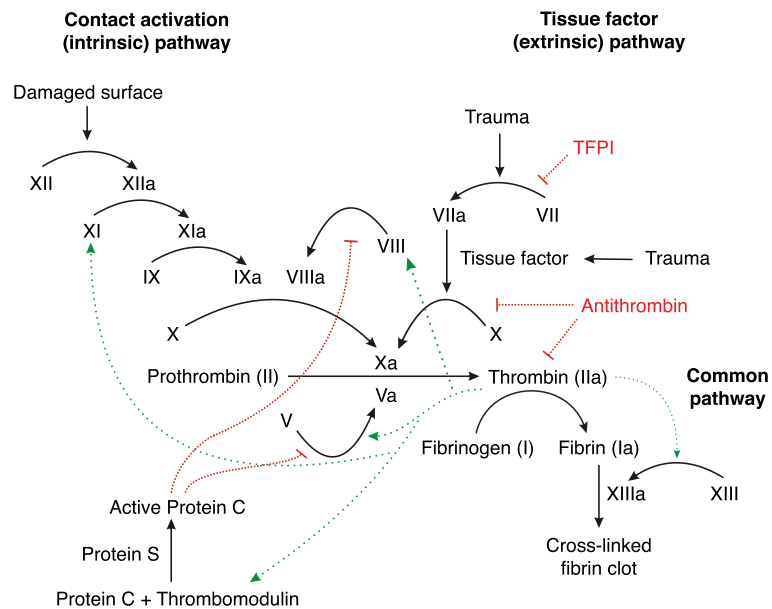


Figure 1: The Clotting Cascade

The clotting cascade is divided into the **intrinsic pathway** and **extrinsic pathway**, which join to form the **common pathway**.

Extrinsic Pathway

The extrinsic pathway contains two factors, and the process of activation **occurs in seconds**:

- **Tissue Factor**
Membrane protein on sub-endothelial cells, which is exposed when the vessel is damaged (it is found in a few other places as well). Tissue factor binds to factor VII to form VIIa, and thus activates the extrinsic pathway.
- **Factor VII**

In vitro, the intrinsic and extrinsic pathways operate separately. This is an artifact of lab measurement – *in vivo* the pathways are co-dependent.

Intrinsic Pathway

The intrinsic pathway is activated **over minutes**, and contains:

- Contact factors
 - Only important *in vitro* when conducting lab testing - deficiency of these factors does not cause a coagulopathy.
 - HMWK
 - HMWK activates factor XII.
 - Factor XII
 - Factor XIIa activates factor IX, as does thrombin.
- Factor XI
- Factor IX
- Factor VIII
 - Factor VIII circulates in a complex with vWF, preventing it from degradation. When activated by thrombin, it acts as a cofactor for factor IXa to activate factor X.

The intrinsic pathway is activated by:

- Thrombin
 - Main activator of the intrinsic pathway *in vivo*.
- Collagen
- Glass
 - In vitro*.

Common Pathway

The common pathway contains:

- Factor X
- Factor V
 - Cofactor (similar to factor VIII), which when activated by thrombin allows factor Xa to convert prothrombin into thrombin.
- Factor II (prothrombin)
 - Has several key roles:
 - Cleaves fibrinogen to fibrin

- Activates factor XIII
Factor XIIIa stabilises clot by forming cross-bridges between fibrin in a platelet plug.
- Amplification of the clotting cascade by activating factors V and VIII
- Activates protein C
Thrombin binds with thrombomodulin to form a complex which inhibits coagulation.
- Factor I (fibrinogen)

The Cell-Based Model of Coagulation

The cascade model (above) accurately describes the process of clotting *in vitro*, but not *in vivo*. An alternative model, the cell-based model, offers an alternative perspective that emphasises the central role of the platelet:

- **Initiation phase**
Coagulation begins with tissue factor being exposed, which also activates platelets.
- **Amplification phase**
A positive feedback loop occurs:
 - Production of Xa causes production of thrombin (IIa), **priming** the system
 - Thrombin then activates factors V, VIII, and IX, accelerating Xa production and further thrombin generation
- **Propagation phase**
Platelets bind activated clotting factors, causing high rates of thrombin formation around them.

References

1. Krafts K. Clot or Bleed: A Painless Guide for People Who Hate Coag. Pathology Student.

2. Chambers D, Huang C, Matthews G. Basic Physiology for Anaesthetists. Cambridge University Press. 2015.
3. Clotting Cascade 22/4/2007. (Image). By Joe D (Own work). CC BY 3.0, via Wikimedia Commons.

Haemostatic Regulation

Describe the mechanisms of preventing thrombosis including endothelial factors and natural anticoagulants.

Describe the process and regulation of haemostasis, coagulation and fibrinolysis.

Describe the physiological mechanisms of limiting and preventing thrombosis.

Haemostasis must be controlled to prevent rampant clotting of the vascular tree. This involves both:

- Endothelial factors
- Proteins

Endothelial Regulation

Intact endothelium and the glycocalyx prevent clotting in a number of ways:

- **Minimise stasis**
 - High blood flow
Especially where flow is turbulent (large arteries).
 - Maximise laminar flow
Glycocalyx smooths flow.

- **Inhibition of platelet adhesion and activation**
NO, prostacyclin, and ectonucleotidases (which de-grade ADP) inhibit platelet activation.
- **Membrane-bound anticoagulant proteins**
 - **Heparan** (not heparin)
Activates antithrombin III.
 - **Thrombomodulin**
Binds thrombin, preventing cleavage of fibrinogen to fibrin. The thrombin-thrombomodulin complex activates protein C (which in turn inactivates factors Va and VIIIa).
- **Prevent exposure of procoagulant protein**
 - Collagen
 - vWF
 - Tissue Factor
- **tPA secretion** (see 'Clot Lysis')

Clot Regulation

- Effect of blood flow
 - Dilutes clotting factors
Activated clotting factors are washed away and metabolised by the RES.
 - Laminar flow
Causes **axial streaming** of platelets, minimising endothelial contact and chance of activation.
- Activation of anticoagulant factors
 - **Tissue Factor Pathway Inhibitor**
Inhibits VIIa, antagonising the action of tissue factor
 - **Antithrombin III**
Inhibits the **serine proteases**, i.e. the non-cofactor factors in all three pathways - IIa, VIIa, IXa, Xa, XIa, XIIa.

- **Protein C**
Inactivates protein Va and VIIIa, and is activated by thrombin.
- **Protein S**
Cofactor which helps protein C.

Clot Lysis

Clot breakdown is performed by:

- **Tissue Plasminogen Activator (tPA)**
Binds to fibrin, and then cleaves plasminogen to plasmin. This keeps the plasmin formation in the vicinity of the clot, limiting its systemic spread of.
 - **Plasmin** cleaves fibrin into **fibrin degradation products**
FDPs conveniently inhibit further thrombin and fibrin formation.
-

References

1. Krafts K. Clot or Bleed: A Painless Guide for People Who Hate Coag. Pathology Student.
2. Van Hinsbergh VWM. Endothelium—role in regulation of coagulation and inflammation. Seminars in Immunopathology. 2012;34(1):93-106.

Coagulation Testing

Outline the methods for assessing coagulation (including TEG and ROTEM).

Outline the methods for assessing platelet function and fibrinolysis.

Outline the methods for assessing coagulation, platelet function and fibrinolysis.

All **coagulation tests measure how long it takes to make fibrin**. However, each test evaluates different parts of the coagulation cascade, and so helps to localise *where* a coagulopathy may be occurring.

In these tests:

- Citrate is added to blood sample
Binds calcium and prevents clotting.
- Sample is centrifuged
- Plasma is decanted
- Calcium (to replace the calcium lost by binding to citrate) and a reagent is added
- Time taken to clot is measured

Prothrombin Time/INR

The prothrombin time measures the **extrinsic pathway**. Any disorder of the extrinsic or common pathways will prolong

the PT, i.e. deficiency or inhibition of:

- Factor VII
- Factor X
- Factor II (prothrombin)
- Factor V
- Factor I (fibrinogen)

The extrinsic pathway is so named because a substance *extrinsic* to the sample (namely, Tissue Factor) must be added to initiate clotting.

However, as the PT varies significantly between different laboratories, the INR is used to allow values to be compared.

Although warfarin affects factors in all three pathways, its clinical effects are measured using INR. This is because:

- Factor VII has the shortest half-life of the clotting factors affected by warfarin
Therefore so its levels will fall the quickest.
- Therefore a fall in Factor VII levels is the earliest indication of changes in coagulation status due to warfarin
- As factor VII is only in the extrinsic pathway, the PT and INR are the only tests which can evaluate factor VII function

(Activated) Partial Thromboplastin Time

The partial thromboplastin time measures the **intrinsic pathway**, which begins produce fibrin when activated by the addition of phospholipid to the sample (as phospholipid is contained in platelets, it is not technically "extrinsic").

The *activated* partial thromboplastin time is the same test, except an activating agent is added to speed up the reaction. Any disorder of the intrinsic or common pathways will prolong the APTT, i.e. deficiency or inhibition of:

- Factor XI
- Factor IX
- Factor VIII
- Factor X

Anti-phospholipid antibodies prolongs the APTT by binding the added phospholipid.

- Factor V
- Factor II (prothrombin)
- Factor I (fibrinogen)

Heparin affects both sides of the pathway (IIa, IXa, Xa, XIa) however typically affects intrinsic factors more than extrinsic factors.

Activated Clotting Time

Activated Clotting Time is used for the dosing and reversal of heparin in cardiopulmonary bypass and other extracorporeal circuits. Testing involves:

- Adding fresh whole blood to a tube with an activator (e.g. glass beads) to stimulate the intrinsic pathway
- Measuring the time taken until clot formation occurs
Different activators will have different normal ranges as well as target ranges for the circuit in use.

Platelet Function Tests

Platelet function testing:

- Evaluates how well platelets aggregate in response to endogenous stimulators of platelet aggregation
These include:
 - ADP
 - Collagen
 - Arachidonic acid
 - Adrenaline
 - Involves adding aggregating agent to a tube of platelets, and measuring the change in turbidity
Different patterns of response (or non-response) can be diagnostic of different platelet function disorders.
-

References

1. Krafts K. Clot or Bleed: A Painless Guide for People Who Hate Coag. Pathology Student.
2. Activated Clotting Time - Practical Haemostasis.

Viscoelastic Testing

Outline the methods for assessing coagulation (including TEG and ROTEM).

Outline the methods for assessing coagulation, platelet function and fibrinolysis.

Viscoelastic coagulation testing, also known as point-of-care coagulation testing provides information on all phases of clotting, and:

- Involves testing of **whole blood**
Traditional testing uses plasma only.
- Includes the **cell-based model** of coagulation
- **May better represent actual clotting function** compared with traditional coagulation factor testing.
- Includes:
 - Thromboelastography
 - Rotational thromboelastometry

Table 1: Advantages and Disadvantages of Viscoelastic Testing

Advantages	Disadvantages
Rapid compared with traditional testing	Still measures coagulation in artificial conditions

Advantages	Disadvantages
Uses whole blood, providing a more complete picture of plasma-RBC-platelet interaction	Does not measure contribution of endothelium and therefore conditions affecting platelet adhesion (e.g. von Willebrand's disease)
Real-time display of clot evolution	Harder to institute QA outside of laboratory
Reduces non-evidence-based transfusion	Measurement methodology is not yet standardised between institutions
Predictive of post-operative hypercoagulable states	Baseline measurement does not predict post-operative bleeding
Very sensitive to heparin effect	Does not measure effect of hypothermia Requires training and competency of non-lab staff More expensive than traditional testing

Thromboelastography

Continuous measurement and display of viscoelastic properties of a blood sample from initial fibrin formation to clot retraction, and ultimately fibrinolysis. TEG involves:

- A known volume (typically 0.36ml) of whole blood added to activators in two disposable cuvettes (cups) heated to 37°C
 - Contact activators (such as kaolin) are added to the blood to accelerate clotting
 - A heparinase cuvette is also commonly used so clotting function can be measured during full anticoagulation (e.g. CPB)

- Pin attached to torsion wire immersed into blood Tor-sion on the pin is converted (by a transducer) into a TEG tracing.
- Cuvette rotates through $4^{\circ}45'$ in alternate directions Each rotation takes 10s.
- Pin initially remains stationary as it rotates through the un-clotted blood
This is represented by a straight line on the tracing.
- **As blood clots, cup rotation exerts torque on the pin**
- The stronger the blood clot, the greater the torque exerted on the pin

Rotational Thromboelastometry

ROTEM is a modified version of TEG:

- A pin fixed to a steel axis is rotated in blood via movement of a spring
The cuvette remains stationary.
- Two samples are used:
 - Tissue factor is added to measure the extrinsic pathway (known as the ENTEM cuvette)
 - Contact activator is added to measure the intrinsic pathway (INTEM cuvette)
- Impedance to rotation is detected by an optical system:
 - LED
 - Mirror on the steel axis
 - Electronic camera
- **Uses different reference ranges and nomenclature to TEG**

Interpretation

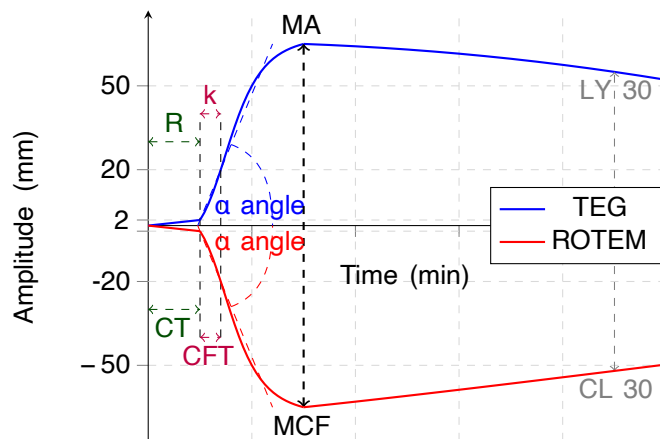


Figure 1: Thromboelastography

Note that reference ranges are not included, as these vary depending on the:

- Technique (TEG/ROTEM) used
- Activator used
- Adjuvants added
e.g. Citrated vs. recalcified samples.

Table 2: Interpretation of Viscoelastic Methods

Parameter (TEG)	Parameter (ROTEM)	Definition	Relevance
R (reaction) time	CT (clotting time)	Time until 2mm amplitude	Time until initial fibrin formation, dependent on plasma concentration of clotting factors
K time	CFT (clot formation time)	Time for amplitude to increase from 2 - 20mm	Measurement of clot kinetics (clot amplification), dependent on fibrinogen
α angle	α angle	Angle between the tangent to the tracing at 2mm and the midline	Rapidity of fibrin formation and cross-linking. Alternate measure of clot kinetics, dependent on fibrinogen
MA (maximum amplitude)	MCF (maximum clot thickness)	Greatest amplitude	Indicates point of maximal clot strength, dependent predominantly on platelets (80%) and fibrinogen (20%) , binding via GPIIb/IIIa . Treatment with platelets or DDAVP .
CL 30 (clot lysis 30)	LY 30	Percent decrease in amplitude 30 minutes after MA	Clot stability, dependent on fibrinolysis . Reduced CL 30 can be treated with an antifibrinolytic , such as TXA

References

1. Krafts K. Clot or Bleed: A Painless Guide for People Who Hate Coag. Pathology Student.
2. Activated Clotting Time - Practical Haemostasis.
3. Srivastava A, Kelleher A. Point-of-care coagulation testing. Contin Educ Anaesth Crit Care Pain. 2013;13(1):12-16.

Physics and Measurement

System of Units

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular, SI units.

The International System of Units (**SI**, or *Système International d'Unités*), is a set of measurement standards which defines (almost) all standards in terms of uniform natural phenomena, and form the base of the metric system.

Base Units

There are **seven base SI units**, which are defined by their mutual independence.

Table 1: Base SI Units

Quantity	Unit	Abbreviation	Definition
Time	Second	s	Duration of 9,192,631,770 periods of the radiation corresponding to the transition between two hyperfine levels of the ground state of an atom of Cs-133
Length	Metre	m	Distance that light travels in a vacuum in $1/299,792,458^{\text{th}}$ of a second
Current	Ampere	A	The constant current that would produce a force of 2×10^{-7} Newton between two conductors of infinite length and negligible cross section in a vacuum
Temperature	Kelvin	°K	$1/273.16^{\text{th}}$ of the triple point of water. The triple point is the temperature at which a substance exists in equilibrium in all three phases (solid, liquid, gas).
Amount	Mole	mol	The amount of substance which contains as many elementary entities as in 0.012kg of Carbon 12

Quantity	Unit	Abbreviation	Definition
Luminous Intensity	Candella	cd	Luminous intensity of a source which emits monochromatic radiation at 540×10^{12} Hz at radiant intensity of 1/683 watts per steradian
Mass	Kilogram	kg	Weight of the International Prototype Kilogram (IPK)

Derived Units

Derived SI units are defined by a combination of base units.

Table 2: Derived SI Units

Quantity	Unit	Abbrevi- ation	Conver- sion to Base SI Units	Definition
Area	Square metre	m^2	m^2	
Velocity	Metre per Second	$m.s^{-1}$	$\frac{m}{s}$	
Acceler- ation	Metre per Second per Second	$m.s^{-2}$	$\frac{m}{s^2}$	
Force	Newton	N	$N = \frac{kg.m}{s^2}$	Force required to accel- erate 1kg at $1m.s^{-2}$
Pressure	Pascal	Pa	$Pa = \frac{N}{m^2} = \frac{kg}{m.s^2}$	Force per area
En- ergy/Work/Quan- tity of Heat	Joule	J	$J = \frac{N.m}{s^2} = \frac{kg.m^2}{s^2}$	Energy con- verted when 1N is applied to 1kg over 1m
Dose Equiva- lence	Sievert	Sv	$Sv = \frac{J}{kg} = \frac{m^2}{s^2}$	Radiation dose per mass

Quantity	Unit	Abbrevi- ation	Conver- sion to Base SI Units	Definition
Power	Watt	W	$W = \frac{J}{s} = \frac{kg \cdot m^2}{s^3}$	Rate of energy conver- sion per second
Electro- motive Force	Volt	V	$V = \frac{W}{A} = \frac{kg \cdot m^2}{A \cdot s^3}$	Measure of electrical potential energy

References

1. Physical Measurement Laboratory. National Institute of Standards and Technology. Including the various subpages.

Electrical Safety

Explain the electrical concepts of current, potential difference, resistance, impedance, inductance, capacitance, frequency, and amplitude as they relate to biological signals and biomedical apparatus.

Describe basic physics applicable to anaesthesia, in particular electrical concepts, current, potential difference, resistance, impedance, inductance and capacitance.

Key electrical concepts include:

- **Charge**
The property of a subatomic particle which causes it to experience a force when close to other charged particles. Charge is measured in coulombs (C).
- **Current**
The flow of electrons through a conductor. Current is measured in amps (A).
- **Voltage**
The strength of the force that causes movement of electrons.
 - By tradition, voltages are quoted relative to **ground** (or **earth**)
 - If a potential difference exists, a current will flow from that object to the earth via the path of least resistance

- If this path contains a person, an electrical injury may result.

- **Resistance**

Describes to what extent a substance reduces the flow of electrons through it. Resistance is measured in ohms (Ω).

- Substances with **high resistance** are **insulators**
- Substances with **low resistance** are **conductors**

- **Inductance**

Property of a conductor by which a change in current induces an electromotive force in the conductor, and any nearby conductors

- **Capacitance**

Ability of an object to store electrical charge. Measured in Farads (F), where one farad is when one volt across the capacitor stores one coulomb of charge.

- A capacitor is an electrical component consisting of two conductors separated by an insulator (called a dielectric)
- When a **direct current** flows, electrons (a negative charge) build up on one of these conductors (called a plate), whilst an electron deficit (positive charge) occurs on the other plate
 - Current will flow until the build up of charge is equal to the voltage of the power source
 - Current can be **rapidly discharged** when the circuit is changed
- An **alternating current** can flow freely across a capacitor, and causes no buildup of charge

- **Impedance**

Describes to what extent the flow of **alternating current** is reduced when passing through a substance.

- Impedance can be conceptualised as 'resistance for AC circuits', and is a combination of **resistance** and **reactance**.
- Reactance is a function of two things:
 - Induction of voltage in conductors by the alternating magnetic field of AC flow
 - Capacitance induced by voltages between these conductors

Electrical Injury

Potential electrical injuries include:

- **Ventricular Fibrillation** secondary to **electrical shock**
Likelihood is a function of:
 - Current density
 - FrequencyLowest current density required is at **50Hz**.
- **Burns**
Function of current density. Burns typically occur at the entry and exit point as this is where current density is highest.
- **Tetanic Contraction**
Flexors are stronger than extensors, which may maintain grip on live wire. Death may result from either VF or asphyxiation from sustained respiratory muscle contraction.

Electrical Shock

Electrical shocks are divided into two types, based on their ability to induce VF:

- **Microshock**
Current required to induce VF when applied directly to myocardium.
 - Typical current is **0.05-0.1mA**
 - This requires **skin breach**
Potential causes:
 - Guidewire
 - Pacing lead
 - Column of conducting fluid
 - CVC
 - PICC
- **Macroshock**
Current required to induce VF from surface contact.
 - Typical current is **100mA**

- This is much higher than microshock because most of this current is not going to the ventricle, and so the total current must be greater to achieve sufficient **current density** in the myocardium to induce VF

Principles of Electrical Safety

Power points contain three terminals:

- **Active**
240V in Australia and New Zealand.
- **Neutral**
0V, relative to ground.
- **Earth**
Direct pathway into ground.

An electrical circuit is completed between an appliance and the power station by returning current to the station via the earth. This is an **earth referenced power supply**.

Electrical Dangers

Electrical risks include:

- **Shorting**
Active wire connected to some other part of the equipment (e.g. case), resulting in electrical current traveling through components that it was not designed to.
 - Principle of earth wire, which provides path of least resistance for current to travel should a short occur
 - Earthing ↓ risk of electrocution if a person becomes part of the electrical circuit - ideally electrical current should flow down the circuit

A variety of other detrimental effects can be seen at lower currents.

Table 1: Adverse Effects of Electrocution

Current (mA)	Effect
1	Tingling
5	Pain
8	Burns
15	Skeletal muscle tetany
50	Skeletal muscle paralysis & respiratory arrest

Note that measuring voltage for AC current is not intuitive, as the voltage will be negative half the time. The **root mean square** (RMS) is used instead - each value for the voltage is squared (giving a positive number), and then divided by the number of samples to give an average.

- **Fire**

Current that exceeds the capability of the conductor results in $\uparrow$ heat generation, equipment damage, and possible fire.

- Principle of fuses which trigger when current drain is excessive

Methods of Electrical Safety

Design features to improve electrical safety include:

- **Insulation**

Conductors are coated by a high-resistance substance, $\downarrow$ risk of shorting and heat generation.

- **High-resistance wires**

Reduce risk of electrocution via devices that *require* an electrical connection to a patient, e.g. EEG or BIS.

- High resistance wires

- **Fuses**

Safety devices which cease all current flow when current exceeds a certain threshold (typically 20A).

- If there is a fault which greatly lowers resistance (i.e. insulation breaks, causing a device to become live and drain via the earth wire), a high current will flow and the fuse will be triggered
- Activation of a fuse requires:
 - Equipment malfunction that causes a high current flow
 - The fuse to work correctly
- After a fuse has activated, electricity will be unable to flow and so the device will be inoperable. This presents its own safety concerns for critical equipment.

- **Residual Current Device**

An RCD measures the current difference between the active and neutral lines, and therefore detects a 'leak' of current.

- In an non-fault situation, these will be equal

- In a fault situation, current will be being delivered by the active line but not returned via the neutral. The 'current gap' indicates current is instead flowing to the ground via the equipment/patient/staff.
- An RCD will disconnect power within 10ms if there is a >10mA difference between the active and neutral lines
- Activation of an RCD requires:
 - Equipment malfunction causing current to leak
 - A single fault will turn off the circuit
- A working RCD will shut off power to the device, and so introduces similar risks to fuses
- **Line Isolation Supply, with a line isolation monitor**
 - A line isolated supply is a 'transformer' with an equal number of windings, such that the:
 - Voltage produced is the same on each side of the 'transformer'
 - **Powerpoint is not physically connected to the supply**
This creates an **earth-referenced, floating power supply**.
 - A line isolation monitor
 - A line isolation monitor continuously checks the impedance between the:
 - Active wire and ground
 - Neutral wire and ground
 - Continuously evaluates the impedance between the:
 - Active wire and ground
 - Neutral wire and ground
 - In a no-fault situation, both impedances should be the same, and *close* to infinite. Impedance won't be *absolutely* infinite as there will always be a small current leak from devices, and therefore always some impedance to flow.
 - Measures the **prospective hazard current**,

which states how much current *could* flow, *if* a fault completed the circuit

- In a no-fault situation, the impedances are very high (close to infinite), and so the prospective hazard current is very low
- In a single-fault situation, the calculated impedance for the affected line will be significantly lower, and therefore the prospective hazard current will increase
- An alarm will sound when the prospective hazard current exceeds 20mA
- This is called a **prospective hazard current**
 - In a single-fault situation, the calculated impedance for the affected line will be significantly lower, and therefore the prospective hazard current will increase
- Activation of the line isolation supply:
 - Requires **two** separate faults
 - Active wire must be connected to ground
 - Neutral wire must be connected to ground
 - Occurs when the circuit: active wire - ground - neutral wire is created
- The advantage of a line isolation supply is that:
 - A single fault is not dangerous
 - A single fault will not result in a power loss Important for vital equipment.
- The disadvantage of a line isolation supply is that:
 - ≥ 2 faults will not result in a power loss, and therefore are dangerous

- **Equipotential earthing**

This is the **only method which prevents microshock.**

- Ultra-low resistance earth cables are attached to electrical devices and the patients bed
- These cables are then attached to special wall earth connectors
- This ensures all equipment is referenced to a common ground, minimising the risk of leakage currents between devices and the patient

by creating a low-resistance pathway that does not include the patient

Classification of Electrically Safe Equipment

These classifications are designed to limit macroshock:

- **Class I: Earthed**

Any part that can contact the user is earthed to ground.

- If a fault develops such that parts of the device that the user can touch are live, then there is a risk of shock
- If the case is earthed, the path of least resistance should be via the earth wire
This will cause a large current to flow, and should blow a fuse, ceasing current flow.

- **Class II: Double-insulated**

All parts of the device that the user can touch have two layers of insulation around them, reducing the chance of the device becoming live.

- **Class III: Low-voltage**

Device operates at less than 40V DC/24V AC, limiting the severity of shock a device can deliver.

Classification of Electrically Safe Areas

- **B areas**

Protection against macroshock. Options include:

- Residual Current Devices
- Line Isolation Supply

- **BF areas**

Cardiac (microshock) protection.

- Requires equipotential Earthing

- **Z areas**

No particular protections.

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1. Electricity and Electrical Hazards.
2. Alfred Anaesthesia Primary Exam Tutorial Program
3. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

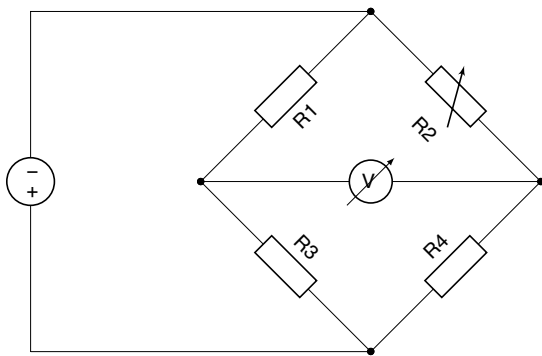
Wheatstone Bridge

The Wheatstone bridge is an electrical device used to accurately measure very small changes in electrical resistance.

The Wheatstone bridge is:

- Historically used in many other medical devices (e.g. invasive pressure monitoring)
- A device with **infinite gain**
- A **null deflection galvanometer**
- **Not an amplifier**
As it does not increase current amplitude.

Mechanism



The Wheatstone bridge consists of:

- Battery
- Four resistors
 - R_1 and R_3 are **known** and **fixed**
 - R_2 is **known** and **adjustable**

- R_4 is **unknown**
- Galvanometer

The Wheatstone bridge relies on the ratio of resistances between the known ($\frac{R_2}{R_1}$) and unknown ($\frac{R_4}{R_3}$) legs:

- When $\frac{R_2}{R_1} = \frac{R_4}{R_3}$, equal current flows down either limb and there is **no current flow** across the galvanometer. At this point the bridge is said to be balanced.
 - The equation can then be re-arranged to solve for R_4 :

$$R_4 = \frac{R_2}{R_1} \cdot R_3$$
 - Very small changes in R_4 lead to a current flow across the bridge
 - R_2 is then be adjusted until the bridge is balanced; at which point the value of R_4 is determined
-

References

1. Alfred Anaesthetic Department Primary Exam Tutorial Series

Neuromuscular Monitoring

Describe the monitoring of neuromuscular blockade.

Describe the concept of depth of neuromuscular blockade and explain the use of neuromuscular monitoring.

Describe the clinical features and management of inadequate reversal of neuromuscular blockade

The degree of neuromuscular blockade achieved using neuromuscular blocking drugs can be assessed:

- **Clinically**

Crude compared to electrical assessment. Described methods include:

- Sustained head lift >5 seconds
Suggests <30% blockade.
- $V_T > 10 \text{ ml.kg}^{-1}$
- Tongue protrusion

- **Electrically**

Using a nerve stimulator. Can be:

- Visual/tactile
Monitoring of twitch height by anaesthetist. Inaccurate.
- Electrical
Monitoring of twitch height by a device:
 - Accelerometer
Acceleration is proportional to force for

any given mass ($F = ma$), therefore an accelerometer taped to the thumb can be used to assess force of contraction.

- Mechanical force transducers
Muscle tension is measured using a strain gauge. Requires calibrating prior to drug administration.
- Electromyography
EMG response is measured using electrodes over the muscle. The AUC of the response curve can be used to calculate degree of blockade.

Nerve Stimulator

Depth of neuromuscular blockade can be assessed using a **nerve stimulator**, which:

- Consists of two electrodes, a power supply, and a controller
- Produces a monophasic, **square wave** at **constant current**, lasting no more than **0.3ms**
- Generates a **supra-maximal stimulus**
Ensures every nerve fibre is depolarised, which means a consistently reproducible response will be generated. A supra-maximal stimulus is 25% greater than the maximum required to depolarise all nerve fibres.
- Allows assessment of different muscle groups
Not all muscle groups are affected equally by neuromuscular blockade.
 - Typically **smaller muscle** groups are **more sensitive**
 - The **positive** (red) lead is placed **proximally**
 - Ulnar nerve
Electrodes are placed along the ulnar border of the wrist at the flexor crease, and thumb adduction is assessed.
 - Facial nerve
The positive electrode is placed at the outer can-

thus, and the negative electrode is placed anterior to the tragus. Eyebrow twitching is assessed.

- Posterior tibial nerve
Electrodes are placed posterior to the medial malleolus, and plantar flexion is assessed.

Stimulation Patterns

There are five common stimulation patterns:

- **Train of Four**

Four single twitches (0.1ms) delivered at 2Hz (i.e. 1.5s for all 4).

- Number of observed twitches gives an indication of receptor occupancy
With increasing blockade, the amplitude and number of observed twitches decreases.
 - **Fade** is the reduction of twitch height with repeated stimuli during a partial neuromuscular block
Occurs due to the effect of non-depolarising agents on the *presynaptic* membrane, reducing ACh production.
 - Number of observed twitches depends on the degree of blockade:
 - No twitches $\approx$ 100% blockade
 - One twitch $\approx$ 90% blockade
 - Two twitches $\approx$ 80% blockade
 - Three twitches $\approx$ 75% blockade
Reversal agents should not be given with a ToF count < 3 .
 - Four twitches $\approx$ $< 75\%$ blockade
- The ratio of the amplitude of T_1 to T_4 (ToF ratio) can also be used as a measure of blockade:
 - ToF ratio $> 90\%$ is adequate for extubation
 - ToF ratio $> 70\%$ suggests adequate respiratory function
- Should not be repeated faster than every 10s

- **Tetanic stimulation**

High frequency (50-200Hz) supramaximal stimulus for 5 seconds.

- Normal muscle will exhibit tetanic contraction
- Partially paralysed muscle exhibits fade
Degree of fade is proportional to degree of block-ade, and is very sensitive.

- **Post-tetanic count (PTC)**

Used in deep blockade when there is no response to ToF. A tetanic stimulus is given, followed 3s later by single twitches at 1Hz.

- No response may be seen in very deep blockade
- However, twitches may be seen prior to the re-turn of a ToF response.

This is called **post-tetanic facilitation**, and occurs due to the tetanic stimulus mobilising ACh vesicles into the pre-junctional area.

- Typically, a ToF of 1 will occur when the PTC ≈ 9
- Should not be repeated faster than every 6 minutes
Due to residual post-tetanic potentiation.

- **Double burst**

Two 0.2ms 50Hz (tetanic) stimuli are applied 750ms apart.

- Two identical contractions occur in normal muscle
- Amplitude of the second burst is reduced in partially paralysed muscle
DB ratio is similar to the ToF ratio, but is easier to assess clinically.
- A ratio > 0.9 is required for adequate reversal

- **Single twitch**

A single stimulus lasting ~ 0.2 ms is applied.

- $> 75\%$ blockade causes a depressed response
 - A twitch must be assessed prior to blockade so a baseline can be established
-

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1. Leslie RA, Johnson EK, Goodwin APL. Dr Podcast Scripts for the Primary FRCA. Cambridge University Press. 2011.
2. Saenz, AD. Peripheral Nerve Stimulator - Train of Four Monitoring. 2015. Medscape.
3. McGrath CD, Hunter JM. Monitoring of neuromuscular block. Continuing Education in Anaesthesia Critical Care & Pain, Volume 6, Issue 1, 1 February 2006, Pages 7–12.

Pressure Transduction

Describe the invasive and non-invasive measurement of blood pressure, including limitations, potential sources of error and the need for calibration.

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular measurement of volumes, flows, and pressures, including transducers and the measurement of blood pressure

A **transducer** converts one form of energy to another. Pressure transducers convert a pressure signal to an electrical signal, and require:

- Several components
 - Catheter
 - Tubing
 - Stopcock
 - Flush
 - Transducer
- Calibration

Required in order to read accurately. Two forms of calibration are performed:

 - Static calibration
 - Dynamic calibration

Pressure exerted by a static fluid is due to the **weight** of the fluid, and is a function of:

- Fluid **density** (in kg.L^{-1})
- **Acceleration** (effect of gravity, in m.s^{-2})
- The **height** of the fluid column

Note that this does not require the mass or the volume of liquid to be known, and is why pressure is often measured in height-substance units (e.g. mmHg, cmH_2O), as acceleration is assumed to be constant.

Static Calibration

Static calibration calibrates to a known zero, and involves:

- Leveling the transducer (typically to the level of the phlebostatic axis at the right atrium, or the external auditory meatus)
A change in transducer level will change the blood pressure due to the change in hydrostatic pressure (in cmH₂O).
- **Zeroing** the transducer
 - Opening the transducer to air
 - Zeroing the transducer on the monitor
A change in measured pressure when the transducer is open to air is due to **drift**, an artifactual measurement error due to damage to the cable, transducer, or monitor.

Dynamic Calibration

Dynamic calibration ensures the operating characteristics of the system (or **dynamic response**) are accurate. Dynamic response is a function of:

- **Damping**
How rapidly an oscillating system will come to rest.
 - Damping is quantified by the **damping coefficient** or **damping ratio**
 - Describes to what extent the magnitude of an oscillation falls with each successive oscillation
 - Calculated from the ratio of the amplitudes of successive oscillations in a convoluted fashion:

$$D = \sqrt{\frac{(\ln \frac{D_2}{D_1})^2}{\pi^2 + (\ln \frac{D_2}{D_1})^2}}, \text{ where:}$$

- **Resonant Frequency**

How rapidly a system will oscillate when disturbed and left alone.

- When damping is low, it will be close to the **natural frequency** (or undamped resonant frequency)

Resonance

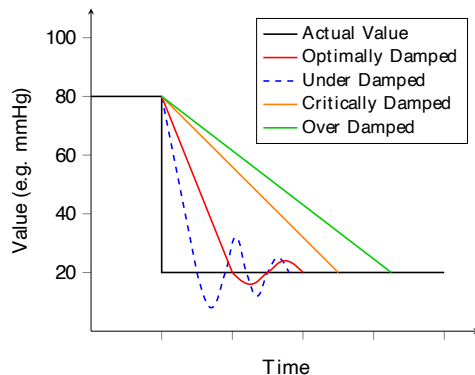
Resonance describes the progressive amplification of small oscillations or disturbances that occurs when the natural frequency of the system or its components aligns with the frequency of pressure fluctuations *within* the system. Resonance:

- Occurs if high frequency components of the pressure waveform approach the natural frequency of the system
- Results in a distorted output signal and an **overshoot in systolic pressure**

Damping and frequency is the frequency at which a system oscillates when disturbed and left alone but it is conceptually and mathematically easier to use damping and resonance, as they are both **easily measured** and **accurate** in describing the observed dynamic response.

Damping

A pressure transduction system should be adequately damped:



- An **optimally** damped waveform has a damping of **0.64**. It demonstrates:

- A rapid return to baseline following a **step-change**, with **one overshoot and one undershoot**
- Accuracy over a wide range of frequency responses
- A **critically damped** waveform has a damping coefficient of **1**. It demonstrates:
 - The most rapid return to baseline possible following a step-change **without overshooting**
- An **over-damped** waveform has a damping coefficient of **>1**. It demonstrates:
 - A slow return to baseline following a step-change with no oscillations
 - Slurred upstroke
 - Absent dicrotic notch
- Loss of fine detail
- An **under-damped** waveform has a damping coefficient close to 0 (e.g. **0.03**). It demonstrates:
 - A very rapid return to baseline following a step-change with several oscillations
 - Systolic pressure overshoot
 - Artifactual bumps

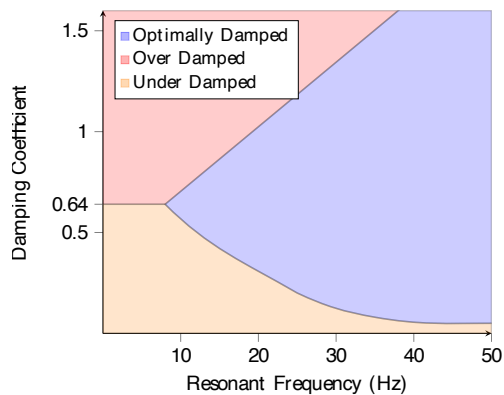


Figure 1: Optimally Damped Waveform

Dynamic Response

Dynamic response refers to how accurately the transducer can track and respond to changes in pressure over time, without distorting the signal. The dynamic response:

- Necessary to faithfully reproduce a signal is dependent on the nature of the pressure wave to be measured
- Must be greater to accurately reproduce an arterial waveform compared to a venous waveform

Principles of Waveform Analysis

An arterial pressure waveform is a periodic (repeating) complex wave. With respect to complex waves:

- Complex waveforms can be represented mathematically by **Fourier analysis**
- Fourier analysis involves expressing a complex waves as the sum of many simple sine waves of varying frequencies and amplitudes
- The frequency of the arterial wave (i.e., the pulse rate) is known as the **fundamental frequency**
- The sine waves used to reproduce it must have a frequency that is a *multiple* (or **harmonic**) of the fundamental frequency
 - Increasing the number of harmonics allows better reproduction of high-frequency components, such as a steep systolic upstroke
- Accurate reproduction of an **arterial** waveform requires up to **10 harmonics** - or **10 times the pulse rate**
- An arterial pressure transducer should therefore have a dynamic response of **30Hz**
 - This allows accurate reproduction of blood pressure in heart rates up to 180bpm (180 bpm = 3Hz, 3Hz x 10 = 30Hz)

Testing Dynamic Response

Dynamic response can be tested by inducing a **step-change** in the system, which allows calculation of both the

natural frequency and the damping coefficient. Clinically, this is performed by doing a **fast-flush test**.

- Fast flush valve is opened during diastolic runoff period (minimises systemic interference)
- The pressure wave produced indicates the natural frequency and damping coefficient of the system:
 - The **distance between** successive oscillations should be identical and equal to the natural frequency of the system
 - The **ratio of amplitudes** of successive oscillations gives the damping coefficient

Optimising Dynamic Response

The lower the natural frequency of a monitoring system, the smaller the range of damping coefficients which can accurately reproduce a measured pressure wave. Therefore, the optimal dynamic response is seen when the natural frequency is as **high as possible**. This is achieved when the tubing is:

- Short
- Wide
- Stiff
- Free of air

Introducing an air bubble will increase damping (generally good, since most systems are under-damped), however it will lower the natural frequency and is detrimental overall.

References

1. Brandis K. The Physiology Viva: Questions & Answers. 2003.
2. Alfred Anaesthetic Department Primary Exam Program

3. Miller, RD. Clinical Measurement of Natural Frequency and Damping Coefficient. In: Anesthesia. 5th Ed. Churchill Livingstone.

Non-Invasive Blood Pressure

Describe the ~~invasive and non-invasive measurement of blood pressure and cardiac output~~ including calibration, sources of errors and limitations

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular measurement of blood pressure.

Non-invasive blood pressure measurements is performed with either a:

- **Device for Indirect Non-invasive Automatic Mean Arterial Pressure (DINAMAP)**
Automatic blood pressure cuff.
- Von Recklinghausen's oscillotonometer
"Manual" blood pressure cuff.
 - Uses two cuffs, and therefore two tubes

DINAMAP

Components:

- One cuff
Performs both arterial occlusion and measurement.

- Tubing
- Device for inflating the occlusive cuff and gradually deflating it
- Pressure transducer
- Display

Method:

- Cuff is inflated above SBP
- Cuff deflates at a rate of $2-3\text{mmHg.s}^{-1}$
 - When cuff pressure equals:
 - SBP
Turbulent flow occurs past the cuff, creating pressure oscillations. The pressure at which these are **first detected** is the **SBP**.
 - MAP
The pressure at which **amplitude** of oscillations is maximal.
- DBP is calculated from MAP and SBP

Cons:

- Requires an appropriately sized cuff
 - Cuff should be ~20% greater than arm diameter.
 - Cuffs that are too small will over-read
 - Cuffs that are too wide will under-read
- Requires a regular rhythm
- Inaccurate at extremes of blood pressure
- Inaccurate when used more frequently than once per minute
- Inaccurate when the vessel is incompressible
 - Heavily calcified vessels
 - When applied to forearm/foreleg
- May cause neuropraxia

Von Recklinghausen's Oscillotonometer

Components:

- Two cuffs
 - Occlusive cuff

- Measurement cuff
- Tubing
- Device for inflating the occlusive cuff and gradually deflating it
- Aneroid barometer for transducing pressure
- Display

Process:

- Cuff is inflated until the radial pressure is no longer palpable
This approximates SBP.
- Cuff is deflated, and re-inflated to 20mmHg above the estimated SBP
- Cuff is deflated at a rate of 2-3mmHg.s⁻¹ whilst auscultating the brachial artery

When cuff pressure equals:

- SBP
Turbulent flow occurs past the cuff, turbulent flow causes the first of the **Korotkoff sounds** (clear tapping pulsations) to be heard.
- DBP
The cuff no longer compresses the vessel at all, so no turbulent flow occurs and nothing is auscultated.

References

1. ANZCA July/August 2000
2. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.
3. Leslie RA, Johnson EK, Goodwin APL. Dr Podcast Scripts for the Primary FRCA. Cambridge University Press. 2011.

Cardiac Output Measurement

Describe the methods of measurement of cardiac output including limitations, potential sources of error, the need for calibration and the values obtained.

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular measurement of cardiac output.

Cardiac output can be measured:

- Invasively
 - Pulmonary Artery Catheter
 - Thermodilution
 - Fick Principle
 - TOE
 - Arterial waveform analysis
 - PiCCO
 - Vigileo
- Non-invasively
 - TTE
 - MRI
 - Thoracic impedance

Pulse contour analysis is covered in detail under Chapter .

Thermodilution

Thermodilution:

- Remains the gold standard of cardiac output measurement
- Requires a pulmonary artery catheter
Various different designs exist. For CO measurement, they require:
 - A proximal port at the RA/SVC
 - A temperature probe at the tip
Typically a silicon oxide thermistor.
 - A balloon at the tip
To float it into position.
 - A distal (PA) port is required for measuring PAP and the PCWP, but is not required for CO calculation

Cardiac Output Measurement by Thermodilution

- A known volume of (typically dextrose) at a known temperature (classically cooled, but this is not required) is injected into the proximal port
- The temperature of blood is measured at the tip of the PAC
This produces a temperature-time curve.
- The area under the curve can be used to calculate cardiac output, as per the modified Stewart-Hamilton Equation:

$$Q = \frac{V(T_B - T_I)k_1k_2}{\int_{t_1}^{t_2} \Delta T dt}, \text{ where:}$$
 - Q = Cardiac output
 - V = Volume of injectate
 - T_B = Temperature of blood
 - T_I = Temperature of injectate
 - k_1 = Density constant
Relates to the specific heat and specific gravity of both injectate and blood.

- k_2 = Computation constant
Accounts for catheter dead space and heat exchange during injection.
- $\int_{t_1}^{t_2} \Delta T dt$ = Area under the change in temperature-time curve

Errors in Thermodilution

Sources of error include:

- Natural variability
Cardiac output varies up to 10% with changes in intrathoracic pressure during respiration. Therefore:
 - A mean of 3-5 measurements should be taken
 - Measurements should be taken at end-expiration
- Incorrect volume of injectate
 - Too much underestimates CO
 - Too little overestimates CO
- Warm fluid
The closer the temperature of injectate is to blood, the greater degree of error introduced to the measurement.
 - Colder injectate is more accurate, but carries the risk of inducing bradyarrhythmias
- Poorly positioned PAC
The PAC must be positioned in West's Zone 3 for blood flow to occur past the tip, and for the measured temperature to be accurate.
- Tricuspid regurgitation or Pulmonary regurgitation
Results in retrograde ejection of injectate back past the valve, resulting in a prolonged decay time of the thermodilution curve.
 - This *usually* results in **underestimation of CO**
 - May result in overestimation depending on the severity of regurgitation and the true CO
- Shunts
Thermodilution measures RV CO only. LV CO and RV CO are likely to not be equal in the presence of intra- or extra-cardiac shunts. This can be corrected for

Any factor that spuriously reduces the area under the thermodilution curve will result in an overestimation of CO.

by calculating shunt fraction, but is not performed by bedside CO computers.

- Arrhythmia
 - Fluctuation in BP
- Rapid changes in temperature result in unreliable measurements; classically seen in the few minutes after weaning from cardiopulmonary bypass when the core is rewarming.

Fick Principle

Cardiac Output can also be measured using the **Fick Principle**, which **states** that the **blood flow to an organ is equal** to the **uptake of a tracer substance divided by the arterio-venous concentration difference** of tracer substance.

In this case:

- The tracer substance is oxygen
 - The 'organ' is the whole body
 - This produces the equation: $CO = \frac{VO_2}{C_a - C_v}$, where:
 - CO is Cardiac Output
 - VO_2 is the patients oxygen consumption
Typically estimated as $3.5 \text{ ml.kg}^{-1}.\text{min}^{-1}$
 - C_a is arterial oxygen content
 - C_v is mixed venous oxygen content
 - Accurate measurement relies upon:
 - Mixed venous blood sampled from the pulmonary artery
 - Arterial blood sampled from a peripheral arterial line
-

References

1. Moise, S. F., Sinclair, C. J. and Scott, D. H. T. (2002), Pulmonary artery blood temperature and the measurement of cardiac output by thermodilution. *Anaesthesia*, 57: 562–566.
2. Nishikawa, T. & Dohi, S. Errors in the measurement of cardiac output by thermodilution. *Can J Anaesth* (1993) 40: 142.

Pressure Waveform Analysis

Describe the invasive and non-invasive measurement of ~~blood pressure and~~ cardiac output including calibration, sources of errors and limitations.

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular measurement of cardiac output.

Analysis of arterial pulse contour is:

- **Real-time** and **continuous**
- Used to **estimate** cardiac output
Less accurate but also less invasive (e.g. thermodilution) or technically demanding (e.g. echocardiography) than other methods.
 - Therefore also calculate (and often display) stroke volume variation and pulse pressure variation

Principles

All models recognise that the **amplitude of the systolic upstroke** is:

- Directly proportional to **stroke volume**

- Inversely proportional to **arterial compliance**

Other principles used by some (but not all) devices include:

- **Three - element Windkessel model**
Characterises the arterial tree as having three major features:
 - Aortic Impedance
 - Arterial Compliance
Predicted using patient characteristics.
 - Systemic Vascular Resistance
- **Conservation of Mass**

Devices

Devices can be **classified based on** whether they are:

- Calibrated/Uncalibrated
 - **Calibrated**
Initial estimation is refined using a dilution technique.
 - Dilutions may be by:
 - Thermodilution
 - Cold saline injected into SVC
Using an IJV or SCV CVC.
 - Temperature changed measured at the femoral artery
 - Lithium dilution
 - Small amounts of lithium chloride injected into a central vein
 - Change in lithium concentration measured in radial artery
 - CO by calculated Stewart-Hamilton equation
 - Periodically recalibrated to correct for drift
 - **Uncalibrated**
Not corrected for a measured 'true' cardiac output.

- Inaccurate for short term changes in arterial properties
- Not validated in:
 - Shock
 - ARDS
 - Hepatic surgery
Due to changes in arterial tone.
 - Cardiac surgery
- Invasive/Non-invasive
 - Invasive
Rely on a (usually femoral) arterial catheter.
 - Non-invasive
Rely on the **volume clamp** method:
 - Inflatable cuff wrapped around finger
 - Plethysmograph estimates blood volume in the digital arteries
 - Cuff inflates and deflates throughout the cardiac cycle, keeping the volume of the arteries constant
Arterial pressure is proportional to cuff pressure.
 - **Inaccurate** in:
 - Peripheral oedema
 - Vasoconstricted states

Common Devices in Use

- PiCCO/VolumeView/FloTrac
 - Calibrated
 - Invasive
 - 3-element Windkessel
 - Mechanism:
 - Calculates area under systolic part of the arterial curve
 - Divides calculated area by aortic compliance
Compliance estimated by proprietary algo-

- rithm each time the device is calibrated.
 - SVR is continuously estimated from calculated CO and measured BP
 - LiDCO
 - Calibrated
 - Invasive
 - Conservation of mass
 - Compliance inferred from biometric data
 - ClearSight/CNAP
 - Uncalibrated
 - Non-invasive
 - T-Line
 - Calibrated
 - Proprietary, *non-validated* auto-calibrating algorithm.
 - Non-invasive
 - Uses radial applanation tonometry
-

References

1. Jozwiak M, Monnet X, Teboul J-L. Pressure Waveform Analysis. Anesth Analg. 2017.
2. Francis, SE. Continuous Estimation of Cardiac Output and Arterial Resistance from Arterial Blood Pressure using a Third-Order Windkessel Model. MIT. 2007.

Pulse Oximetry

Describe the principles of pulse and tissue oximetry, co-oximetry including calibration, sources of errors and limitations.

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error, in particular oximetry.

Pulse oximetry relies on several principles:

- Oxygenated and deoxygenated haemoglobin **absorb light of different wavelengths** to different extents
 - Light of **660nm** and **940nm** is used
 - Deoxyhaemoglobin has a greater absorbance of red (660nm) light than oxyhaemoglobin
 - Oxyhaemoglobin has a greater absorbance of infrared (940nm) light than deoxyhaemoglobin
 - The relative absorbance of each wavelength of light allows determination of the proportions of oxygenated and deoxygenated haemoglobin
- The **Beer-Lambert Law(s)**:
Absorption of light passing through a substance is directly proportional to both the distance it travels through the substance and the concentration of attenuating species within the substance. It is a composite of:
 - **Beer's Law**

Absorption of light is proportional to the concentration of "attenuating species"

- **Lambert's Law**

Absorption is proportional to the thickness of the solution, or more precisely, that each layer of equal thickness absorbs an equal proportion of radiation that passes through it

- **Blood flow is pulsatile**

Allows absorption from blood to be separated from absorption from other tissues.

Method

A pulse oximeter consists of:

- Two diodes of the desired wavelengths
- Photocell
- Microprocessor

Process:

- During pulsatile flow, the expansion and contraction of the blood vessels alters the distance and haemoglobin concentrations, changing the absorption spectra of blood (as per the Beer-Lambert Law).
- Non-pulsatile elements are due to tissues and venous blood
- These are subtracted from the total, leaving the pulsatile element which represents the arterial component
- The **ratio of absorbance** of the **pulsatile** elements and the **non-pulsatile elements** is called **R**, and is calculated as:

$$R = \frac{Pulsatile_{660}/Non-Pulsatile_{660}}{Pulsatile_{940}/Non-Pulsatile_{940}}$$

- R is compared with a set of standardised values to deliver a calculated SpO₂
 - An R of 1 gives an SpO₂ of 85%
 - An R of 0.4 gives an SpO₂ of 100%
 - An R of 2 gives an SpO₂ of 50%

The Isobestic Point

- The **isobestic point** is the wavelength at which light is absorbed equally by both haemoglobin species
- Light absorption is therefore **independent** of saturation, and is instead a function of **haemoglobin concentration**
- This can be used to correct for haemoglobin concentration
- There are two isobestic points for oxygenated and deoxygenated haemoglobin, at **590nm** and **805nm**

Limitations

- Requires detectable **pulsatile flow**
 - Limited by poor peripheral perfusion (shock, hypotension, hypothermia) and non-pulsatile flow (ECMO, CPB)
 - Body movements confound readings (shivering, seizing)
- **Low saturations**
Inaccurate below 70%, and completely unreliable below 50%.
- **Venous pulsation**
Detected as pulsatile flow, and erroneously interpreted by the microprocessor as arterial flow.
- Confounded by **ambient light**
The diodes are cycled at several hundred times per second which allows the detector to compensate for the effect of ambient light (the values when the diodes are off give the effect of ambient light).
- Absorption spectra confounded by:
 - **Haemoglobinopathies**
 - Carboxyhaemoglobin causes the pulse oximeter to read artificially high due to as it also absorbs 660nm light
 - Methaemoglobinaemia causes the SpO₂ to trend towards 85%, as though it absorbs

660nm light is also absorbs 940nm light to a greater degree

- **Dyes**

- Methylene blue will cause the SpO₂ to read < 65% for several minutes
 - Indocyanine green will also cause a decreased SpO₂
-

References

1. Davis PD, Kenny D. Basic Physics and Measurement in Anaesthesia. 5th Ed. Elsevier. 2003.
2. Mardirossian G, Schneider RE. Limitations of Pulse Oximetry. Anesth Prog 39:194-196 1992.
3. CICA March/May 2014
4. Tremper KK, Barker SJ. Pulse oximetry. Anesthesiology. 1989 Jan;70(1):98-108.
5. Williams GW, Williams ES. Basic Anaesthesiology Examination Review. Oxford University Press. 2016.

Oxygen Analysis

Describe the principles of measuring oxygen concentration.

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular gas analysis, including capnography.

As oxygen is a molecule containing two similar atoms, its partial pressure cannot be determined using infrared techniques (unlike CO_2). Oxygen content of a gas is instead determined using either:

- Paramagnetic analyses
- Fuel Cells

Paramagnetic Analysis

Principles of paramagnetic analysis:

- Oxygen is paramagnetic
This means it is attracted by magnetic fields, but does not propagate the field.
 - This is because its two unpaired valent electrons have the same spin
- Many other gases weakly repelled by magnetic fields (diamagnetic)

- The attraction of a gas mixture to a magnetic field is therefore proportional to its oxygen content
- Many different methods exist which use this property to determine oxygen content

Pressure Method

- Gas being tested flows into a tube
- A reference gas flows into a parallel tube
- Both gases then pass through:
 - Flow restrictors
 - Alternating magnetic field
Field is turned on and off at ~100Hz.
- The gases combine in the magnetic field
- The greater the oxygen content of the gas, the more it will move into the magnetic field
This movement creates a negative pressure behind the gas.
- The pressure difference between the tested gas and the reference gas is proportional to the oxygen content of the test gas.

Temperature Method

Used in many modern devices. In this method:

- Gas flows through a magnetic field, causing the particles to align
This changes the thermal conductivity of the oxygen molecules.
- The change in thermal conductivity of the gas mixture is proportional to the oxygen content
- This is detected by measuring current passing through a heated wire

Pros

- Accurate
- Rapid response time
Modern analysers can identify breath-to-breath variation in FiO_2 .
- Does not require regular calibration

Cons

- Water vapour reduces accuracy
- Interference from other paramagnetic gases
 - Nitric oxide
Effect is minimal as nitric is delivered in far smaller volumes than oxygen, and is only weakly paramagnetic.

Fuel Cells

Fuel cells rely on reduction of oxygen to measure oxygen partial pressure. They consist of:

- Oxygen permeable membrane
- KOH solution
This contains:
 - Lead anode
Lead is consumed as the fuel cell operates.
 - Gold cathode

Method

- Oxygen diffuses across the membrane into the potassium hydroxide solution
- At the cathode:
$$\text{O}_2 + 4e^- + 2\text{H}_2\text{O} \rightarrow 4\text{OH}^-$$

- At the anode:
$$\text{Pb} + 2\text{OH}^- \rightarrow \text{PbO} + \text{H}_2\text{O} + 2\text{e}^-$$
- The oxygen consumption is proportional to the current generated, which is measured with an ammeter

Pros

- No power required
- Small
- Accurate

Cons

- Will accumulate nitrogen in the presence of N_2O
Results in an under-reading of PO_2 .
 - Must be replaced after 6-12 months
 - Requires regular two-point calibration
21% and 100% oxygen are used.
 - Relatively slow response time compared to paramagnetic analysers
~20s.
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

End-Tidal Gas Analysis

Describe the principles of capnography, including calibration, sources of errors limitations.

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular gas analysis, including capnography.

Principles

Several mechanisms for $E_T\text{CO}_2$ measurement exist:

- Infrared Spectroscopy
- Colourimetric Methods
- Rayman Scattering
- Gas Chromatography

Infrared Spectroscopy

Infrared spectroscopy relies on the fact that:

- Gases with two or more different atoms will **absorb infrared radiation**

- **Different gases absorb different wavelengths** to different degrees
- Measuring the absorbed wavelengths and comparing with the **likely composition** of a mixture, a system can be designed using a specific wavelength to measure gas concentrations and avoid interference

End-tidal gas analysis using infrared light is used in the measurement of:

- CO₂
Capnography is the continuous measurement and graphical display of the partial pressure of CO₂ in expired gas. This is the most common method to measure E_TCO₂.
- Anaesthetic agents

Measurement of CO₂

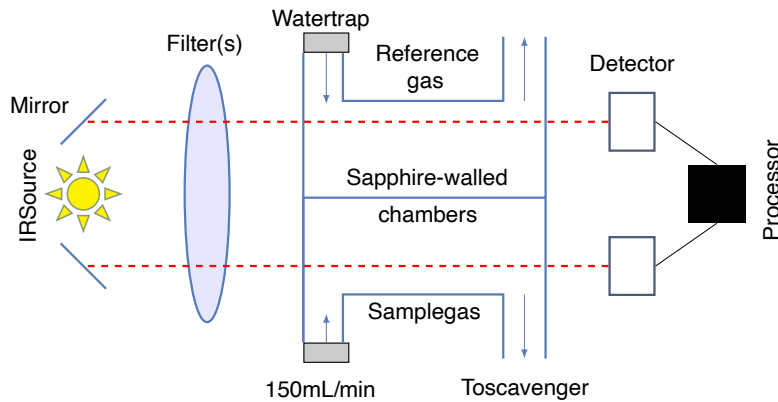
Components:

- Sapphire sampling chamber containing gas sample
 - CO₂ absorbs infrared radiation at a peak wavelength of 4.28µm
 - The sapphire lens only allows 4.28µm light through
- Emitter
- Detector
- Microprocessor
- Display

Method:

- Light is emitted and passes through the sampling chamber
A lens is used to focus emitted light.
- Levels of radiation are measured on the other side of the chamber

- Levels correspond to the amount of gas present in the sample
- The less radiation that reaches the detector, the more gas there is in the sample absorbing it



Equipment Errors

Errors can be classified into:

- Specific to **technique**
 - The **collision broadening effect**
Intermolecular forces vary depending on their proximity to other molecules in the gas mixture. A change in intermolecular forces may alter their bond-energy and the frequencies at which they absorb radiation. It can be overcome by:
 - Correcting for the presence of other gases
 - Manually adjusting the obtained values
 - Crossover with other gas mixtures
CO₂ and N₂O have similar absorbance spectra, and may lead to error when a device is not designed to measure both wavelengths.
- Failure of **equipment**
These can be overcome by use of **double-beam cap-nometer**. This uses a reference chamber which contains CO₂-free air, and the same emitter-detector system. All absorption from this system must occur

due to artefact (as no CO_2 is present). The artefactual component is then subtracted from the value detected in the main chamber. This corrects for:

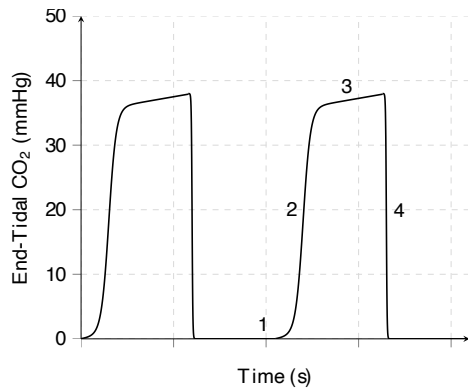
- Variable amount of infrared radiation released
- Variable sensitivity of the detector
- Variable efficacy of the crystal window and lens system

- Relating to **type of capnometer** used

$E_T\text{CO}_2$ may be either **side-stream** or **in-line**.

- Side-stream CO_2 involves a length of narrow tubing drawing gas from the expiratory limb of the breathing circuit (typically from the HME filter) to the capnograph
 - Side-stream requires a flow of 150 ml.min^{-1}
 - Has a (pretty insignificant) delay ($<1\text{s}$) in measurement
 - May be blocked by water vapour, and require use of a water trap to remove condensation
- In-line systems have a sampling chamber attached in-line with the ETT
 - The sampling chamber slightly increases the dead-space of the circuit
May be relevant in children or very difficult to ventilate patients.
 - Adds weight to patient end of the breathing circuit
 - Require heating to 41°C to avoid condensation

Normal $E_T\text{CO}_2$ Waveform



The normal trace consists of four components:

1. The **baseline**

This consists of:

- Inspiratory time
- Early **dead-space exhalation**

This is the period immediately before phase 2, where some gas with a PCO_2 of **0** is exhaled.

2. **Alveolar exhalation**, where PCO_2 **rises rapidly**

3. **Alveolar plateau**, where PCO_2 **flattens**

The highest-point of this curve is labeled $E_T\text{CO}_2$.

4. **Inspiration**, where PCO_2 returns to **0**

Waveform Variations

Airway Obstruction:

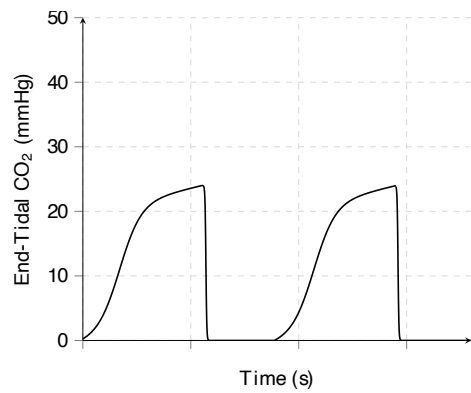


Figure 1: Airway Obstruction

- Occurs due to uneven emptying of alveoli with different time-constants

Hyperventilation:

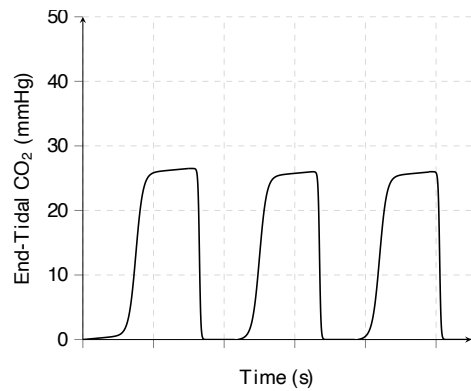


Figure 2: Hyperventilation

- Lower $E_T\text{CO}_2$ with shorter baseline
- Plateau phase may not occur at very high respiratory rates

Rebreathing:

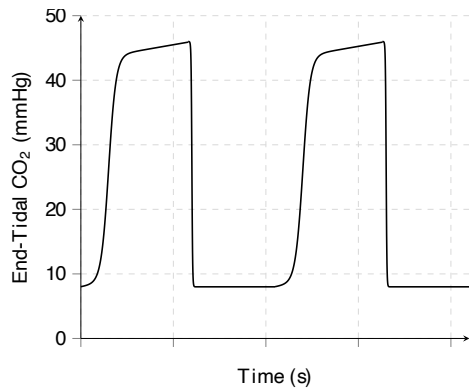


Figure 3: Rebreathing

- Baseline increases as inspired CO_2 is measured from gas analyser

Changes in $E_T\text{CO}_2$

Normal $E_T\text{CO}_2$ is 32-42 mmHg, whilst normal PaCO_2 is 35-45 mmHg.

High $E_T\text{CO}_2$

This may be from:

- ↓ Ventilation
 - ↓ RR
 - ↓ V_T
 - ↑ V_D and therefore a greater $V_D:V_T$ ratio
- ↑ Production of CO_2
 - ↑ Metabolic rate
 - Sepsis
 - Tourniquet release
 - ROSC following arrest
- ↑ Inspired CO_2
 - Rebreathing (i.e. equipment/ventilator malfunction)
 - External source of added CO_2

Low $E_T\text{CO}_2$

Rapid loss of $E_T\text{CO}_2$:

- Failure of ventilation
 - Circuit disconnect
 - Airway obstruction
 - Bronchospasm
- Failure of circulation
 - Cardiac arrest
 - Shock

Gradual loss of $E_T\text{CO}_2$:

- $\uparrow V_A$ (i.e. increased MV)
- $\downarrow \text{CO}_2$ production
 - Hypometabolic state
 - Hypothermia
- $\uparrow V_D$, i.e. V/Q mismatch
 - $\uparrow$ West Zone I physiology:
 - Hypotension
 - $\uparrow$ RV Afterload:
 - PE
 - High PEEP
- Sampling error
 - Air entrainment into the sample chamber
 - Inadequate V_T

Discrepancy between $E_T\text{CO}_2$, PaCO_2 , and PaCO_2

- The normal gradient between PaCO_2 and $E_T\text{CO}_2$ is 0-5 mmHg
- Healthy and awake individuals should have essentially no (<1ml) alveolar dead space, and so essentially no gradient
- This gradient is $\uparrow$ in patients with V/Q mismatch
 $E_T\text{CO}_2$ will underestimate arterial CO_2 as gas from un-perfused alveoli (with negligible CO_2) will dilute CO_2 expired gas.

Colourimetric Methods

Litmus paper which changes colour when exposed to hydrogen ions (produced by CO₂) can be used to confirm endo-tracheal intubation, though they may generate false-positive results due to gastric pH.

References

1. Cross ME, Plunkett EVE. Physics, Pharmacology, and Physiology for Anaesthetists: Key Concepts for the FRCA. 2nd Ed. Cambridge University Press. 2014.
2. Davis PD, Kenny D. Basic Physics and Measurement in Anaesthesia. 5th Ed. Elsevier. 2003.
3. Leslie RA, Johnson EK, Goodwin APL. Dr Podcast Scripts for the Primary FRCA. Cambridge University Press. 2011.

Blood Gas Measurement



Note that the revised ANZCA curriculum “makes clear that the physics of Clark, Severinghaus and pH electrodes is NOT examinable”, and both curricula have pivoted, rightly, to focus on the **clinical interpretation** of blood gases.

This section is left here for posterity.

Describe the methods of measurement of **pH** in blood.

Blood gas machines directly measure three variables and calculate the remainder. Measured variables are:

- PO_2
- CO_2
- pH

Calculated variables include:

- Bicarbonate
Using the pH, CO_2 and the Henderson-Hasselbalch equation.
- Base Excess
Calculated using the Henderson-Hasselbalch and Siggaard-Anderson equation. Can be expressed in two ways:
 - Base Excess
The amount of alkali that must be added to the

sample to return it to a normal pH, at a temperature of 37°C and a PaCO₂ of 40mmHg.

- Standardised Base Excess
As base excess, but calculated for blood with a Hb concentration of 50g.L⁻¹. This is thought to better represent the ECF as a whole.

Oxygen Tension

Oxygen tension is measured with a **Clarke electrode**. This consists of:

- A chamber for the blood sample
- A chamber containing a potassium chloride solution, which:
 - Is separated from the blood chamber by an oxygen-permeable membrane. This prevents blood being in direct contact with the cathode, which would lead to protein deposition on the cathode and incorrect measurement.
 - Contains a platinum cathode
 - Contains a silver/silver Chloride anode
- A battery applying 0.6V across the electrodes

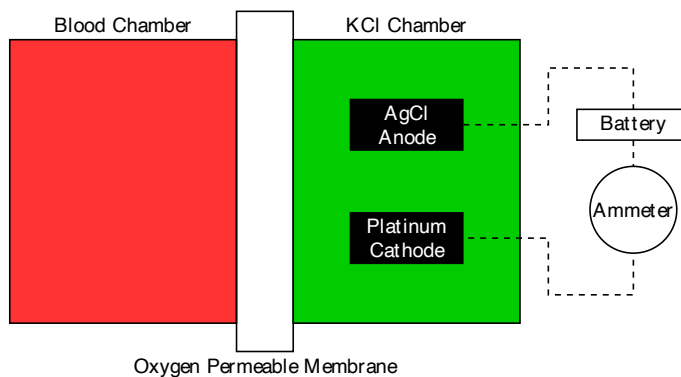


Figure 1: Clark Electrode

Method

- A voltage of **0.6V** is applied across the electrodes, causing the silver to react with chloride in the solution to produce electrons:

$$Ag + Cl^- \Rightarrow AgCl + e^-$$
 - This potential difference is required to start the reaction
 - 0.6V is chosen because it is enough to start the reaction but will have minimal effect on measured current flow
- At the cathode, oxygen combines with electrons and water to produce hydroxyl ions: $O_2 + 4e^- + 2H_2O \Rightarrow 4OH^-$
- For each oxygen molecule present at the cathode, four electrons can be consumed
- Increasing the oxygen available at the cathode $\uparrow$ the number of electrons consumed, and therefore $\uparrow$ **current flow**
 - Oxygen will move from the sample chamber to the measuring chamber according to its partial pressure
- Measured current flow is therefore proportional to oxygen tension in blood

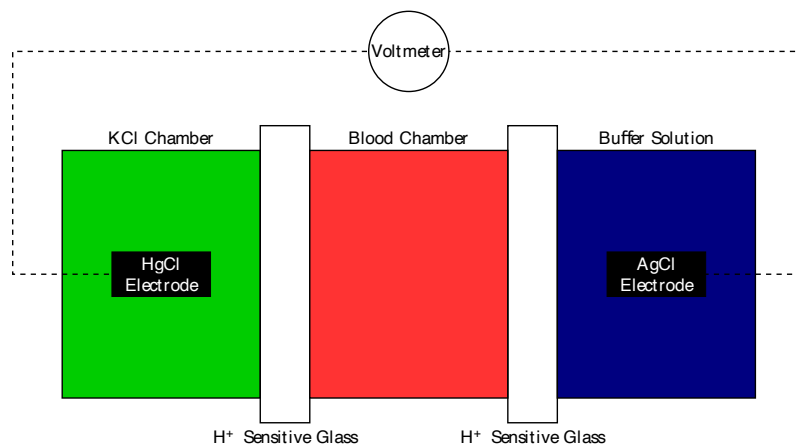
Calibration, Limitations, and Accuracy

- Calibration is performed with standard gas mixtures
Requires regular two-point calibration.
- Cathode must be kept clean from protein and not damaged
- Cathode must be kept at 37°C
- May read falsely high with halothane

pH Measurement

pH is a measure of the hydrogen ion concentration^[1] in solution, and is defined as the negative logarithm to the base 10 of the $[H^+]$:

- $pH = -\log_{10}[H^+]$
- A pH of **7.4** is a $[H^+]$ of **40nmol.L⁻¹** at 37°C
 - A change in a pH unit of 1 is equivalent to a 10-fold change in the $[H^+]$
 - A change in pH of 0.3 is equal to doubling or halving the $[H^+]$



Technically pH is defined as the **activity** of H^+ in a solution. Clinically, activity is identical to concentration, so in medicine these definitions are functionally the same.

The pH electrode consists of:

- A chamber for the blood sample
- A measuring chamber, separated from the sample by H^+ - permeable glass, which contains:
 - A buffer solution
 - A silver/silver chloride measuring electrode
- A reference chamber, also separated from the chamber by H^+ - permeable glass, which contains:
 - A KCl solution
 - Has no buffering properties.
 - A mercury/mercury chloride reference electrode

Method

- Relies on the principle that two solutions with different H^+ activities will develop a potential difference between them (proportional to the concentration gradient)
- H^+ passes through the glass along a concentration gradient:
 - A variable potential difference is generated in the measuring chamber, as H^+ ions are buffered and the concentration gradient is maintained
 - A constant potential difference is generated in the reference chamber, as there is no buffer of H^+ ions in the KCl solution
- Once H^+ has equilibrated between blood and the KCl solution, the potential difference between the measuring and reference electrodes is proportional to the H^+ concentration in blood

Calibration, Limitations, and Accuracy

- Calibration is performed with two phosphate buffer solutions containing two different (known) $[H^+]$
- Must be kept at $37^\circ C$
 - Hypothermia $\uparrow$ solubility of CO_2** and therefore *lowers* $PaCO_2$
 - A reduced partial pressure of CO_2 is required to keep the same number of molecules dissolved (as per Henry's Law)
 - Therefore, as blood cools its pH will increase
- Electrodes must be kept clean from protein and not damaged

Carbon Dioxide Tension

Carbon dioxide tension is measured with a **Severinghaus electrode**, which is based on the pH electrode, as PaCO_2 is related to $[\text{H}^+]$. The Severinghaus electrode consists of:

- A chamber for the blood sample, separated from the bicarbonate chamber by a CO_2 permeable membrane
- A chamber containing bicarbonate solution in a nylon mesh, and separated from both the measuring and reference chambers by H^+ -permeable glass
- A measuring chamber containing:
 - A buffer solution
 - A silver/silver chloride measuring electrode
- A reference chamber containing:
 - A KCl solution
 - A mercury/mercury chloride reference electrode

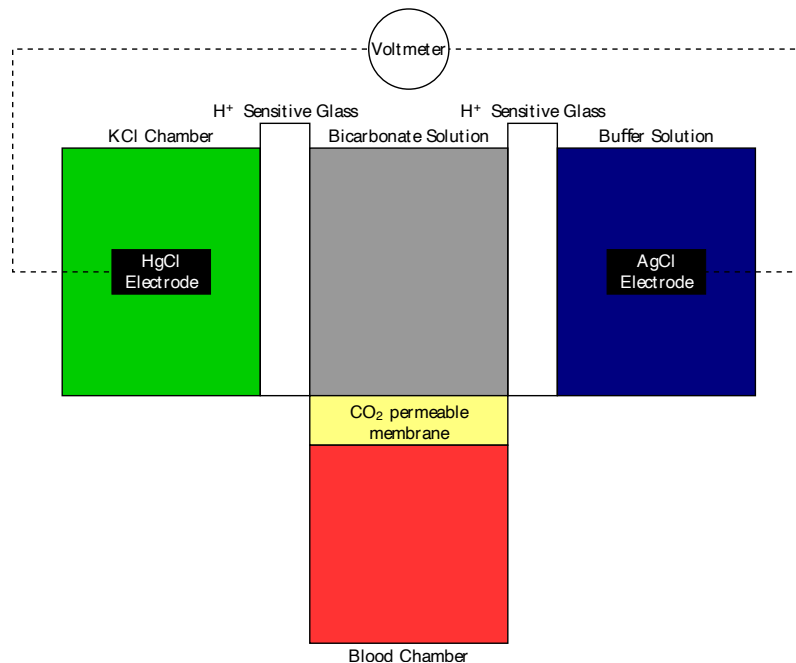


Figure 3: Severinghaus Electrode

Method

- CO₂ diffuses from blood into the bicarbonate chamber
- CO₂ reacts with water in the bicarbonate chamber to produce H⁺ ions
- From here, the process is identical to the pH electrode, except bicarbonate takes the place of blood:
 - H⁺ ions diffuse into the reference chamber until the H⁺ ion concentration has equilibrated
 - H⁺ ions continually diffuse into the measuring chamber (as they are buffered)
- This establishes a constant pH gradient
This gradient is proportional the H⁺ ion concentration in the bicarbonate chamber, which is proportional to the CO₂ content of blood.

Calibration, Limitations, and Accuracy

- Calibration is performed with solutions of known CO₂ concentration
- Must be kept at 37°C
Hypothermia ↓ solubility of CO₂ and therefore ↓ pH
- Electrodes must be kept clean from protein and not damaged
- Slow response time relative to pH electrode due to time taken for CO₂ to diffuse and react
This can be accelerated with carbonic anhydrase

References

1. Leslie RA, Johnson EK, Goodwin APL. Dr Podcast Scripts for the Primary FRCA. Cambridge University Press. 2011.

2. FRCA. Measurement of pO₂, pCO₂, pH, pulse oximetry and capnography.
3. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Gas Flow

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular measurement of volumes, flows, and pressures, including transducers.

Describe basic physics applicable to anaesthesia in particular behaviour of fluids (gases and liquids).

Flow can be described the motion of the fluid as either:

- **Laminar** flow
Fluid moving in a steady manner without turbulence.
- **Turbulent** flow
Irregular fluid movement in radial, axial, and circumferential axes.
 - Laminar flow is more efficient than turbulent flow, as it requires a smaller pressure gradient to generate the same flow
 - For two fluids moving at the same speed, the velocity of individual particles in laminar flow will be both higher and lower
- **Transitional** flow
Mixture of laminar and turbulent flow. Transitional flow is typically turbulent in the centre, and laminar at the edges.

Flowmeters

Devices used to measure gas flow include:

- Variable-Orifice Flowmeters
- Fixed-Orifice Flowmeters
Also known as pneumotachographs.
- Hot wire flowmeter

Note orifice based flowmeters rely on the **Hagan-Poiseuille Equation**:

$$Q = \frac{\Delta P r^4 \pi}{8 \eta l}, \text{ where:}$$

- Viscosity (η) and length (l) are fixed by both devices
- Fixed orifice flowmeters also fix radius (r), such that the change in pressure must therefore be proportional to flow:
 $Q = \Delta P \pi k$, where k is a constant
- Variable orifice flowmeters also fix pressure (ΔP), such that flow can be calculated from the radius:
 $Q = r^4 \pi k$

Variable Orifice Flowmeters

Constant pressure, variable orifice flowmeters are found on wall and cylinder gases. They consist of:

- An inverse conical tube (i.e. narrower at the bottom, and wider at the top)
- A needle valve
- A bobbin
May have a groove which causes the bobbin to spin, confirming it is not stuck.

Method:

- Gas flows from the bottom to the top of the tube
- The bobbin obstructs flow
Therefore there is a **pressure difference** across it.
 - Remember: $Pressure = \frac{Force}{Area} = \frac{Mass \times Acceleration}{Area}$
 - **At equilibrium**, the pressure exerted by the bobbin on the flow of gas ($P = \frac{Gravity \times Bobbin\ Mass}{Bobbin\ Cross-Sectional\ Area}$) is equal to the pressure exerted by the gas on the bobbin
- As flow is increased, the bobbin is pushed further up the flowmeter due to the increased pressure
- The bobbin will reach a new equilibrium position when the orifice of the flowmeter has become wide enough for the pressure on the bobbin to equal the pressure of gravity
- Flowmeters are calibrated for individual gases as:
 - Laminar (typically low flows) flow is proportional to viscosity

- Turbulent (typically high flows) flow is proportional to density

Pros:

- Cheap
- No additional power supply required
- Accurate

Readings may be altered by:

- Change in temperature affects viscosity and density of gas
- Change in pressure affects density of gas

Cons:

- Must be vertical
- Bobbins can become stuck

Fixed Orifice Flowmeters

Constant orifice, variable pressure flowmeter. Several different designs exist, and include:

- **Fleisch pneumotachograph**
 - Consists of several fine bore parallel tubes placed in the gas circuit
Decreased radius and increased resistance reduces gas flow velocity, improving laminar flow.
 - A **differential pressure transducer** is placed at either end of the tubes
 - The pressure drop across the tubing is directly proportional to flow
- **Pitot tubes**
 - Consists of two tubes placed into the gas circuit:
 - One faces into the gas flow
 - The other faces away from the gas flow
 - The pressure difference between tubes is proportional to flow

Pros:

- Accurate
- Continual measurement
- Allow calculation of volumes

$$Volume (L) = Flow (L.s^{-1}) \times Time (s)$$

Cons:

- ↑ Resistance
- ↑ Dead space
- Require laminar flow

Inaccurate when:

- Flows are higher than what the system is designed for
- Alteration in gas density
 - Change in gas mixture
- Alteration in gas temperature

Hot Wire Flowmeter

Components:

- Two fine platinum wires in the gas circuit
 - One heated to 180°C at $OL.min^{-1}$
 - One at 0°C
- Ammeter

Method:

- As gas flows, the wire cools
- Rate of heat dissipation is proportional to gas flow
- The amount of current required to return the wire to 180 is measured, and is proportional to flow

Pros:

- Accurate
- Fast

Cons:

- Fragile

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Principles of Ultrasound

Outline the physical principles of ultrasound including transducer properties, image resolution and the Doppler Effect.

Describe basic physics applicable to anaesthesia in particular principles of ultrasound imaging and use of doppler.

Ultrasound is an imaging technique where high-frequency sound waves (2-15MHz) are used to generate an image.

An ultrasound wave is produced by a probe using the **piezoelectric effect**:

- Certain crystalline structures will vibrate at a particular frequency when a certain voltage is applied across them
The conversion of electrical energy to kinetic energy is how the ultrasound probe creates an ultrasound wave.
- Similarly, they can generate a voltage when a vibration is induced in them
This is how the probe interprets reflected waves.

Basic Terms

- **Spatial resolution**

How close two separate objects can be to each other and still be distinguishable. It is divided into:

- **Axial resolution**, how far apart two objects can be when one is above the other (in the direction of the beam)
- **Lateral resolution**, how far apart two objects can be when side-by-side
- Contrast resolution is how similar two objects can appear (in echogenic appearance) and still be distinguishable
- **Higher frequency** settings offer greater spatial resolution but decreased penetration
- **Lower frequency** settings offer reduced spatial resolution but increased penetration
They are used for visualising deep structures.

Effect of Tissues on Ultrasound

At tissue interfaces, the wave may be:

- **Absorbed**

Sound is lost as heat, and increases with decreased water content of tissues.

- **Reflected**

Sound bounces back from the tissue interface, and returns to the probe.

- Reflection is dependent on the:
 - **Difference in sound conduction** between the two tissues
 - **Angle of incidence** (close to 90° improves reflection)
 - **Smoothness** of the tissue plane
- The amplitude of sound returning to the probe determines echogenicity, or how white the object will be displayed

- The time taken for the sound to return determines depth
 - The time taken for a wave to return is proportional to **twice** the distance of the object from the probe
 - Depth can be calculated using $d = vt$, where:
 - d is Depth
 - v is the speed of sound in tissue, and is assumed to be 1540 ms^{-1}
 - t is Time
- **Transmitted**
Sound passes through the tissue, and may be reflected or absorbed at deeper tissues.
- **Scattered**
Sound is reflected from tissue but is not received by the probe.
- **Attenuated**
Attenuation describes the loss of sound wave with increasing depth, and is a function of the above factors.
 - Attenuation is managed by increasing the **gain**
Gain refers to amplification of returned signal.
 - **Time-gain compensation** refers to amplification of signals which have taken longer to return, which amplifies signals returned from deep tissues

A combination of these effects occur; i.e. some ultrasound energy will be absorbed, some portion reflected, and some transmitted further through the tissue, etc.

Modes

Ultrasound modes include:

- **B-Mode** (brightness mode)
The standard 2D ultrasound mode, and plots the measured amplitude of reflected ultrasound waves by the calculated depth from which they were reflected.
- **M-Mode** (movement mode)
Selects a single vertical section of the image and displays changes over time (i.e. depth on the y-axis, and time on the x-axis).

Doppler Effect

The doppler effect is the change in observed frequency when a wave is reflected off (or emitted from) a moving object, relative to the position of the receiver. In medical ultrasound, this is the change in frequency of sound reflected from a moving tissue (e.g. an erythrocyte). It is given by the equation:

$$V = \frac{\Delta F s}{2F_0 \cos \theta}, \text{ where:}$$

V = Velocity of object

F = Frequency shift

s = Speed of sound (in blood)

F_0 = Frequency of the emitted sound

θ = Angle between the sound wave and the object

Reflected frequencies are **higher towards the probe** and **lower away**.

Calculation of Cardiac Output

- Heart rate is measured
- Stroke volume is calculated by:
 - Measuring the **cross-sectional area** of the **left ventricular outflow tract**
Obtained by measuring the diameter using ultrasound.
 - Measuring the **stroke distance**
Obtained via integrating the velocity-time waveform for time across the left ventricular outflow tract (LVOT VTI).
 - The integral of flow (m.s^{-1} and time (s)) for time (s), produces a distance (m)
 - Multiplying the LVOT cross-sectional area (m^2) by the stroke distance (m), produces a volume (m^3)
This is the stroke volume.

Remember, $CO = HR \times SV$.

References

1. Cross ME, Plunkett EVE. Physics, Pharmacology, and Physiology for Anaesthetists: Key Concepts for the FRCA. 2nd Ed. Cambridge University Press. 2014.
2. CICM July/September 2007.

Temperature

Describe the measurement of body temperature.

Describe the methods of measurement applicable to anaesthesia, including clinical utility, complications and sources of error in particular measurement of temperature.

Temperature is average kinetic energy of molecules within a material, and is measured in degrees. Temperature is distinct from **heat**, which describes the transfer of thermal energy from one body to another body, and is measured Joules. The two are related by the **specific heat capacity**, which describes how much energy (J) must be applied to a body to raise its temperature by 1°K , without a change in state.

Temperature Measurement

Temperature is measured by a number of methods:

- Liquid expansion thermometry
- Electrical thermometry

Liquid Expansion Thermometry

This is used in mercury thermometers. These consist of:

- A graduated **evacuated capillary** of negligible volume, attached to
- A mercury reservoir, of much greater volume, separated by
- A constriction ring
Prevents travel of mercury up the capillary by gravity.

Method:

- When heated, the kinetic energy of the mercury increases and it expands, forcing it up the capillary
As the thermal expansion coefficient for all liquids is very small, the capillary must be of a very small volume to create a usable device.
- The speed that this occurs is related to the **time-constant** of the system
This is typically 30 seconds. Measurement therefore takes ~4 time-constants, or 2 minutes.

Pros:

- Easy to use
- Accurate
- Reusable
- Sterilisable
- Cheap

Cons:

- Slow response
Only accurate once it has reached thermal equilibrium.
- Glass can break
May cause release of mercury or alcohol.
- Inaccurate at:
 - Low temperatures with mercury
Freezes at -38.8°C .
 - High temperatures with alcohol
Boils at 78.5°C .

Electrical Thermometry

Electrical methods include:

- **Resistance thermometer**

Platinum wire increases electrical resistance with increasing temperature.

- Therefore the voltage drop across the wire will correspond to the temperature of the wire
- Change in resistance is linear across the temperature range
- However, these are expensive.

- **Thermistor**

Metal (e.g. SiO_2) semiconductor which changes its resistance in a **predictably non-linear fashion** (run-away exponent) with temperature.

- Can be manufactured so that change is linear over the clinical range
- Much cheaper than wire resistance methods
- The degree of voltage drop is usually very small, however this can be amplified using a **Wheatstone bridge**
- e.g. Pulmonary artery catheter, oesophageal probe

- **Thermocouple**

At the junction of two dissimilar metals, a potential difference will be produced proportional to their temperature. This is known as the Seebeck effect.

- Non-linear (wash in exponent)
 - Degrade over time
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

2. Alfred Anaesthetic Department Primary Exam Tutorial Series

Humidity

Define humidification, outline the mechanisms of humidification and its importance.

Describe basic physics applicable to anaesthesia in particular principles of humidification and use of humidifiers.

Humidity refers to the water content of air, and may be either **absolute** or **relative**:

- **Absolute Humidity** is the mass of water vapour in a volume of gas
- **Relative Humidity** measures the percentage saturation of gas at *current temperature*, or more formally:

$$\text{Relative Humidity} = \frac{\text{mass of water vapour in volume of air}}{\text{mass of water vapour if air was fully saturated}} = \frac{\text{absolute humidity}}{\text{mass of water if air was fully saturated}}$$

Measurement of Humidity

Humidity can be measured by a number of methods:

- **Hair Hygrometer**
Hair (actual hair) changes elasticity depending on the humidity of air. Changes in elasticity can be related to changes in humidity.

- **Wet and Dry Bulb**

This system measures both temperature and relative humidity.

- Two thermometers are used
 - One is wrapped in a wick, which is attached to a water reservoir
This is the wet thermometer.
 - The dry thermometer gives a measurement of surrounding air temperature
- The wet thermometer is cooled due to evaporative cooling from the wick
High energy water molecules become vapour, leaving only low energy molecules behind.
- The **temperature difference** between the thermometers is a function of:
 - Latent heat of vapourisation of water
 - How much evaporative cooling is occurring
This is function of humidity.
 - At 100% relative humidity, no evaporative cooling will take place and the temperatures will be equal
 - As humidity decreases, evaporative cooling will cool the wet thermometer, and the temperature difference allows humidity to be determined

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.
2. Alfred Anaesthetic Department Primary Exam Tutorial Series

Humidifiers

Define humidification, outline the mechanisms of humidification and its importance.

Describe basic physics applicable to anaesthesia in particular principles of humidification and use of humidifiers.

Humidifiers add water vapour to inspired gas, taking the place of normal body mechanisms which are bypassed or impeded by invasive and non-invasive ventilation. Maintaining adequate humidity of inspired gas is important in:

- Reducing metabolic load
Humidification of inspired gas accounts for ~15% of basal heat expenditure.
- Maintaining function of the mucociliary elevator
Inspiration of dry gas increases viscosity of mucous.
- Reducing water loss
Water will be absorbed from mucosa to humidify gas.

Humidifiers can be classified into:

- Passive humidifiers
- Active humidifiers

Passive Humidifiers

Passive humidifiers:

- Do not require power
- Do not require water

The **Heat and Moisture Exchange (HME) filter** is the **classic passive humidifier**:

- Placed between the patient and the patient Y-piece
- Consists of:
 - A moisture exchange layer
Pleated, hygroscopically coated foam or paper.
 - Expired gas cools as it passes, condensing onto the foam, with condensation promoted by hygroscopic coating (usually this is NaCl)
 - The latent heat of vapourisation results in a decreased temperature of expired gas
 - A filter layer
Typically a electrostatic or hydrophobic material.
- Expired gas is cooled and dried
- Inspired gas is then heated and humidified
- An HME takes up to 20 minutes to be fully effective, and can achieve a relative humidity up to 70%
- Efficacy depends upon the patient's core temperature and the condition of the airway

Pros:

- Cheap
- Lightweight
- Straightforward
- May contain anti-bacterial filter

Cons:

- May be blocked with vomit and secretions
- Increase airway resistance
- Increase dead space
- Not as effective as powered active systems
- Only last 24 hours
- Takes 15-20 minutes to become fully effective

Active Humidifiers

Active Humidifiers:

- Require either:
 - Power
Unpowered humidifiers are typically less effective, and only operate well at lower flow rates.
 - Water
 - (Or both)
- Consist of:
 - A water bath
Typically sterile water.
 - A heating element
To heat the water bath.
 - A gas pipe
Inspired gases are bubbled through the water bath to humidify them.
 - A water trap
To trap condensed water. Should be changed regularly to minimise infection risk.

Pros:

- Greater humidification
- Appropriate for long-term ventilation

Cons:

- Bulky
- Expensive
- Require power
- Infection risk from water bath

References

1. McNulty G, Eyre L. Humidification in anaesthesia and critical care. Continuing Education in Anaesthesia Critical Care & Pain, Volume 15, Issue 3, 1 June 2015

Electrocardiography

Describe the principles behind the electrocardiogram ECG.

Discuss the physiological basis of electrical activity and its relationship to mechanical events
Physiological basis of the electrocardiograph in normal and common pathological states.

The ECG is a graphical representation of the electrical activity of the heart, as measured by the sum of electrical vectors at the patients skin.

Components

An ECG consists of:

- Electrodes
Disposable, sticky components which act as conductors due to a silver/silver chloride coating. To reduce electrode impedance, skin should be:
 - Hairless
 - Dry
 - Clean
- Cables
Shielded to prevent currents being induced and electrocuting the patient.
- Processor

- Monitor

Leads

ECG **leads** are created by taking the **potential difference** between two electrodes, which varies by **0.5 - 2mV** through the cardiac cycle as myocardium depolarises. ECG leads are divided into:

- Limb leads
Potential difference between limb electrodes:
 - I: RA to LA
 - II: RA to LL
 - III: LA to LL
- Augmented leads
Potential difference between the average of the limb leads (called the **indifferent electrode**) and each individual limb lead.
 - Augmented leads are of much lower voltage and must be amplified
 - Three augmented leads exist (one for each limb electrode)
- Precordial leads
Potential difference between the indifferent electrode and one of the six additional electrodes placed on the chest wall.

The relationship between electrodes and leads is described with **Einthoven's Triangle**:

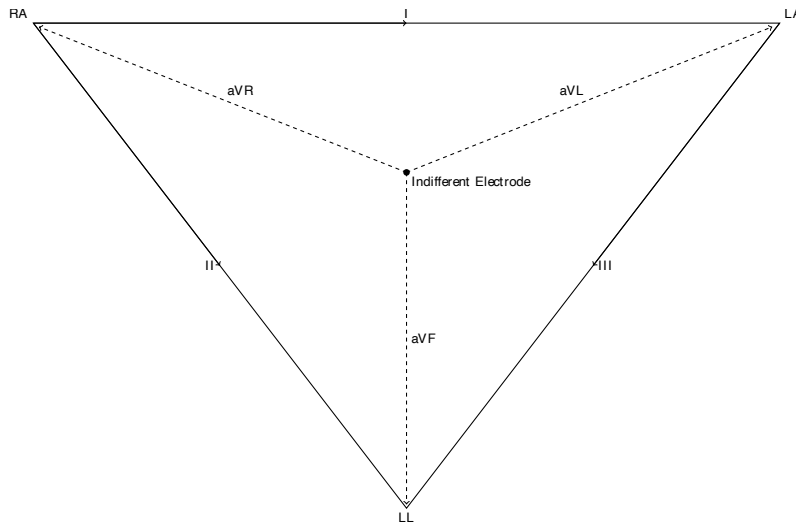


Figure 1: Einthoven's Triangle

Method

- As the myocardial membrane potential changes across the cardiac cycle, a potential difference can be measured at the skin.
 - A depolarisation wave traveling towards the positive electrode (or a repolarisation wave traveling away) will cause an upward deflection in the ECG
- These potential differences are very small, and therefore need to be:
 - Distinguished from background interference

Several techniques exist:

 - **Common mode rejection**

Identical electrical activity occurring in multiple electrodes is likely due to interference rather than cardiac activity, and is removed from the measured signal.

 - A ground electrode is typically used for this purpose
 - ECG modes

ECGs can be set to varying levels of sensitivity.

- Diagnostic mode
Responds to higher range of frequencies, but is at greater risk of interference.
- Monitor mode
ECG responds to a lower range of frequencies, reducing interference but also resolution. This is common on 3-lead ECG.
- High input impedance
Minimises signal loss.
- Amplified
Frequencies in the desired signal range are amplified.

Sources of Error

- Improve signal detection
 - Good adherence
 - Optimal skin contact
Ensure dry and hairless.
- Minimise external electrostatic forces
 - Earthed
 - Diathermy
 - Shivering

Risks

- ECG electrodes can act as an exit electrode for surgical diathermy
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.
2. CICM February/April 2016

Supplemental Oxygen

Describe different systems to deliver supplemental oxygen and the advantages and disadvantages of these systems.

Devices for delivery of oxygen can be classified into:

- Variable performance devices
- Fixed performance devices

Variable Performance Devices

Variable performance devices:

- Do not deliver a fixed FiO_2
 - This is because respiratory flow is non-uniform. Although minute ventilation may be $5-6\text{L}\cdot\text{min}^{-1}$, peak inspiratory flows are substantially higher.
 - Delivered FiO_2 is dependent on oxygen flow and inspiratory flow
 - $\uparrow$ Oxygen flow rate will increase FiO_2 , but the effect will vary depending on the device (volume, seal) and the patient
- Include:
 - Nasal Cannulae
 - Prongs delivering gas at $1-4\text{L}\cdot\text{min}^{-1}$.
 - Higher flows may dry mucosa, and lead to epistaxis

- Nasopharynx acts as an oxygen reservoir, somewhat increasing FiO_2
- Well tolerated
 - Allow eating, drinking, and talking
- Hudson Mask

Simple unsealed mask, allowing gas flow up to $15\text{L}\cdot\text{min}^{-1}$.

 - Cheap
 - Less well tolerated
 - Rebreathing may occur
- Non-Rebreather Mask

Modified version of the Hudson mask, containing a reservoir bag.

 - Reservoir bag is filled during expiration
 - Gas is drawn from the reservoir bag during inspiration, increasing FiO_2

Some air is entrained from around the mask and so FiO_2 is < 1 .

Fixed Performance Devices

Fixed performance devices:

- Theoretically deliver a fixed FiO_2

These are usually flow limited as well, and so FiO_2 may decrease at higher inspiratory flows.
- Include:
 - Venturi

Consists of a cone through which oxygen flows. Apertures on the side of the cone entrain room air.

 - Air is entrained via:
 - Frictional drag of molecules
 - The venturi effect (though this is controversial)

The widening of the cone leads to an increase in fluid velocity and therefore a decrease in pressure, as per the Bernoulli principle.

- Entrained air is proportional to flow rate, so the ratio of oxygen to air is constant for any given aperture size
This is known as the **entrainment ratio**.
 - Will deliver the specified FiO_2 provided oxygen flow is above the minimum rate
Therefore become variable performance devices when inspiratory flow greatly exceeds oxygen flow.
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Bispectral Index

Explain the principles involved in the electronic monitoring of depth of sedation and anaesthesia, including the use of EEG analysis.

Bispectral Index (BIS) is a proprietary signal-processed EMG and EEG monitor used to estimate depth of anaesthesia.

The BIS outputs four values:

- **BIS**
Dimensionless index between 0 and 100 where:
 - 0 represents cortical electrical silence
 - 85 - 100 represents normal awake cortical activity
 - 40 - 60 is consistent with general anaesthesia
- **Signal Quality Index (SQI)**
Dimensionless index between 0 and 100 which gives an indication of the accuracy of the BIS value.
- **Electromyography**
Gives an indication of the influence of muscle activity on BIS values.
- **Suppression Ratio (SR)**
Percentage of previous 63 seconds where EEG is isoelectric.

Method

Proprietary, but involves:

- Multivariate logistic regression of EEG features that correlate with clinical levels of sedation
- Initial validation on a cohort of healthy volunteers, not undergoing surgery
- Use of four frontotemporal EEG monitors

Analytic techniques:

- **Compressed Spectral Array**
 - The signal over a short period (e.g. 5-10 seconds) of EEG recordings are analysed together. Each period is known as an epoch.
 - A Fourier transformation is performed. This breaks the EEG signal down into the sine waves used to produce it.
 - A histogram of each frequency is plotted
 - As anaesthesia deepens, lower frequencies begin to dominate
 - The **spectral edge frequency** is the frequency greater than 95% of the frequencies in the compressed spectral array. It is an indicator of anaesthetic depth, but not of drug concentration.
- **Coherence**

Under anaesthesia, the electrical activity in different sections of the brain falls out of phase.

Pros

- Reduced anaesthetic awareness in high risk patient groups
Trauma, GA caesarian section, cardiac surgery.
- Non-invasive
- Use appears to result in reduced anaesthetic use and more rapid emergence

Cons

- Proprietary algorithm
 - Expensive
 - May be inaccurate with:
 - Hypothermia
 - Hypercarbia
 - Hypoxia
 - Muscle relaxants
BIS may fall inappropriately.
 - Non-GABAergic agents (e.g. ketamine, nitrous oxide)
May not fall appropriately.
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Medical Gas Supply

Describe the supply of medical gases (bulk supply and cylinder) and features to ensure supply safety including pressure valves and regulators and connection systems.

This section will focus on oxygen, covering:

- Production
- Storage
- Safety of gas supplies

Production

Oxygen production can be performed using:

- Fractional distillation
For bulk supply.
- Oxygen concentration

Fractional Distillation

Oxygen is produced on the industrial scale by fractional distillation of atmospheric air. This process:

- Relies on the fact that **different gases** have **different boiling points**
By liquefying air and then heating it gradually, each gas can be removed separately as it boils.

- Occurs in stages:
 - Filtration of atmospheric air
Removes dust and other contaminants.
 - **Compression** of filtered air to **6 atm**, followed by **cooling** to below ambient temperature
Water vapour condenses and is removed.
 - Cooled and compressed air passed through a **zeolite sieve** which removes CO₂
 - Gas is then allowed to re-expand
As it does so it loses heat energy as per **Gay-Lussac's Law**, and liquefies.
 - Air must be cooled below the boiling point of the desired gases
This requires getting gases very cold, and so the process may be mechanically assisted using a turbine, and/or a heat exchanger. Key boiling points (at 1 atm):
 - Nitrogen: 77°K
 - Oxygen: 90°K
 - Helium: 4°KHelium *can* be produced by fractional distillation, but liquefying it is understandably difficult given the very, very low boiling point. Helium can also be mined, as helium produced by alpha decay of radioactive materials may be trapped in gas pockets under the earth.
 - Liquid air is then **fractionally distilled**
Temperature of liquid air is raised slowly.
 - As the boiling point of each gas is reached (e.g. 77°K for nitrogen), that gas will begin to vapourise from the liquid, and can be collected
 - The remaining liquid can then be further heated, until the boiling point for the next gas is reached
 - This process can be repeated until all the desired gases have been separated

Oxygen Concentrator

Oxygen concentrators:

- Produce up to 95% oxygen from air by removing nitrogen
- Built using two **zeolite lattices**
 - Pressurised air is filtered through one lattice
 - Nitrogen and water vapour are retained in the lattice
 - Oxygen and **argon** are concentrated
Produces a 95% oxygen/5% argon mixture.
 - The unused column is heated to release the bound nitrogen and water

Pros of oxygen concentrators:

- Cheap
- Reliable
- Avoid need for oxygen delivery infrastructure

Cons of oxygen concentrators:

- Result in an accumulation of argon when used at low flows on a circle system
- Require continuous power
- Fire and explosion risk

Storage

Methods for gas storage include:

- Gas cylinders
Ideal for small quantities of a variety of different gases.
- Cylinder manifolds
Efficient methods to provide piped supply of a variety of gases.
- Vacuum insulated evaporator
Method to store large quantities of oxygen.

Medical Gas Cylinders

Gas cylinders are:

- Made from chromium molybdenum or aluminium
- Used as:
 - Backup for a piped supply
 - When a piped supply is not available (transports)
 - When the gas is uncommonly used (e.g. nitric oxide)
- The common cylinder used in hospital is **CD**
This contains **460 L** of oxygen at 15°C and 137 bar.
- Cylinders are not completely filled, to reduce risk of overpressure and explosions if the temperature rises
 - The **filling ratio** is the weight of liquid in a full cylinder compared to the weight of water that would completely fill the cylinder
 - In cool climates, the filling ratio is ~0.75
 - In warmer climates, the filling ratio is reduced to ~0.67
- Cylinders are tested for safety every 5-10 years
Tests include:
 - Endoscopic examination
 - Tensile tests1% of cylinders are destroyed to perform testing on the metal.

Pros of gas cylinders:

- Portable
- Reusable

Cons of gas cylinders:

- Heavy
- Limited supply

Cylinder Manifolds

Cylinder manifolds are formed of sets of large gas cylinders used in parallel. * All cylinders in a group are used together

* When the pressure falls below a set level, a pressure valve

will switch and gas will be drawn from another cylinder group * The first (now empty) cylinder group is exchanged for full cylinders

Pros of cylinder manifolds:

- Cheap
- Useful as a backup supply

Cons of cylinder manifolds:

- Less capacity than a VIE
- Fire and explosion risk

Vacuum Insulated Evaporator

The VIE:

- Stores **liquid oxygen**
It is **vacuum insulated** as it must keep oxygen below its critical temperature (-119°C). The VIE typically stores oxygen between -160°C and -180°C , and at 700kPa.
 - The gas is stored below its critical temperature and above its boiling point
 - The amount of oxygen remaining is calculated from its mass
- Does not require active cooling
Instead it is cooled by:
 - Insulation
 - Evaporation
Heat entering the VIE causes liquid oxygen to evaporate. Oxygen vapour is drawn off the VIE to the pipeline supply, so the VIE remains cool and at a steady pressure provided oxygen is being drawn from it.
- Has a pressure relief valve to prevent explosions if oxygen is not being used
- Has an evaporator to evaporate large volumes of oxygen rapidly if demand is high
 - This is simply an uninsulated pipe exposed to the outside temperature

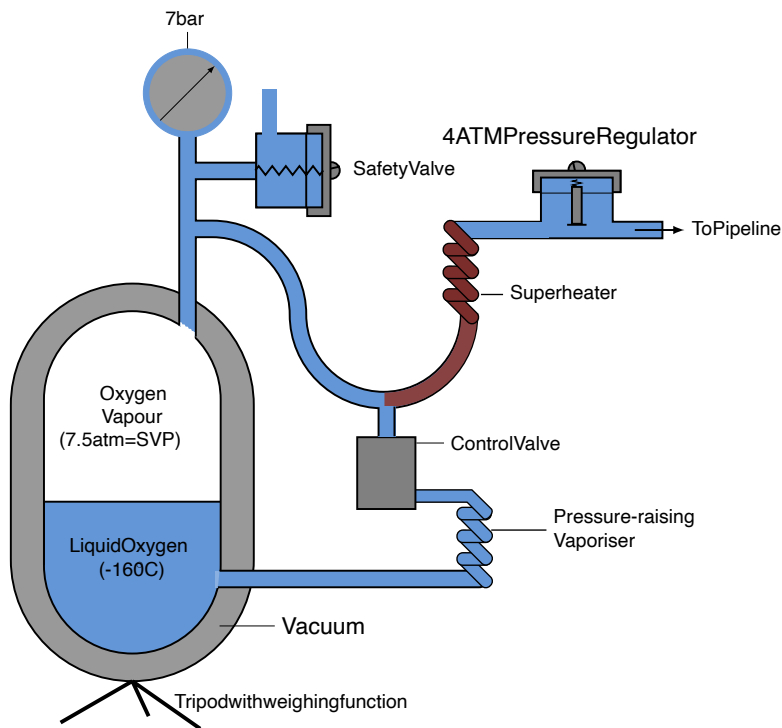


Figure 1: Vacuum Insulated Evaporator

Pros of the VIE:

- Cheapest option for oxygen delivery and storage
 - Storing oxygen as a liquid is much more efficient than as a gas
 - Does not require power

Cons of the VIE:

- Set-up costs are expensive
- Requires a back-up setup
- Will waste large volumes of oxygen if not being used continuously
- Fire and explosion risk

Safety in Medical Gas Supply

Many systems exist to ensure safety:

- Colour coding of cylinders and hoses
 - Oxygen is white
 - Nitrogen is black
 - Air is black with white shoulders
 - Nitrous oxide is blue
 - Helium is brown
 - Heliox is brown with white shoulders
 - Carbon dioxide is grey-green
- Labeling of connections
- The **pin index system**
Used to prevent the wrong gas yoke being connected to a cylinder.
 - Pins protrude from the back of the yoke
 - Holes exist on the valve block
 - Pins and holes must line up for the cylinder to be connected
 - There are six positions, divided into two groups of three
Common combinations include:
 - Oxygen: 2 - 5
 - Air: 1 - 5
 - Nitrous oxide: 3 - 5
- **Sleeve Index System**
Used in Australia when connecting pipeline gases.
 - Wall block contains a sleeve when prevents fitting the incorrect gas hose to the wall
 - Screw thread is identical in all cases
- **Non-Interchangeable Screw Thread (NIST)**
Used (but not in Australia) when connecting pipeline gases.
 - NIST connectors have a probe and a nut
 - Probe diameter is gas-specific, preventing the wrong gas from being connected
- **Testing**
 - Must demonstrate
 - Correct oxygen concentrations
 - Absence of contamination
 - Delivery of adequate pressure when several other systems on the same pipeline are in use

- Testing must be performed twice on a new installation:
 - First by engineers
 - Second by a medical officerIn theatres, this should be the director of the anaesthetic department or their delegate, who should hold fellowship of ANZCA.
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.
2. ANZCA August/September 2016
3. The Essential Chemical Industry Online - Oxygen, Nitrogen, and the rare gases.
4. Thompson, C. The Anaesthetic Machine - Gas Supplies. University of Sydney.

Vapourisers

Describe the principles and safe operation of vapourisers.

Delivery of gas that is fully saturated with anaesthetic agent would result in lethal doses being administered. The use of a vapouriser allows a safe dose of anaesthetic agent to be given. Vapourisers can be divided into:

- Variable bypass vapourisers
 - Air that is fully saturated with gas is mixed with a 'bypass' stream of gas, diluting the delivered concentration. Further subdivided into:
 - Plenum
 - Requires supra-atmospheric pressure to operate.
 - More accurate
 - Draw-over
 - Driven by the patients inspiratory effort.
 - Portable
- Measured flow vapourisers

Variable Bypass Vapouriser

Variable bypass vapourisers aim to deliver the same concentration of anaesthetic agent over a range of flows. They achieve this by:

- Flow management

- Baffles and wicks increase the surface area of the liquid/gas interface, increasing the rate of vapourisation.
 - Excessively high flow rates may result in gas not being fully saturated with agent when it exits the vapouriser stream
 - These are less effective in draw-over vapourisers, as resistance must be minimised
- Temperature management

The SVP of volatile agents increases non-linearly as temperature increases. Temperature changes:

 - Occur through:
 - Changes in ambient temperature
 - Loss through latent heat of vapourisation
Liquid agent from the vapouriser will cool over the course of an anaesthetic.
 - Are managed with:
 - Temperature stabilisation
Use of materials with both a high thermal conductivity and specific heat capacity, allowing the vapourising chamber to buffer changes in surrounding temperature.
 - Temperature compensation
Adjusts flow into either the vapourising chamber or bypass chamber to account for changes in environmental temperature.
Methods include:
 - Bimetallic strip
Metal strip which bends in response to environmental temperature, adjusting the amount of gas entering the vapourising chamber.
 - Aneroid bellows
Connect to a cone in the opening of the bypass chamber. As temperature decreases, the bellows contract and the cone partially obstructs the bypass channel.

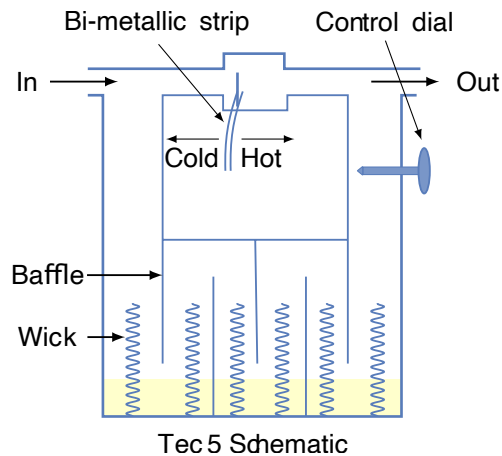


Figure 1: Tec 5 Plenum Vapouriser

Comparison of Plenum and Draw-Over Vapourisers

Plenum vapourisers are:

- More accurate
 - Designed to deliver accurate agent concentrations over a wide range ($0.25 - 15 \text{ L} \cdot \text{min}^{-1}$) of flow rates
 - Below $250 \text{ ml} \cdot \text{min}^{-1}$ the resistance of the flow splitting valve becomes more significant, causing the amount of gas in the bypass stream to be higher than intended
 - Above $15 \text{ L} \cdot \text{min}^{-1}$ gas may not be fully saturated
- Heavier
 - Typically built of metals such as copper to maximise thermal stability.
- High internal resistance
 - Must be used out-of-circle
 - Must be used with positive-pressure

Draw-Over Vapourisers are:

- Less accurate
 - Less use of baffles and wicks to minimise inspiratory resistance
- Less thermally stable

- Oxford Minature Vapouriser does not have a bimetallic strip
- Oxford Minature Vapouriser uses glycol as a thermal buffer

Measured Flow Vapourisers

Measured flow vapourisers have a separate stream of agent-saturated gas that is added to the gas flow. This requires the device to:

- Measure fresh gas flow rate
- Adjust vapour-gas flow rate so the desired concentration is delivered

This system is used for the delivery of desflurane, as desflurane:

- Has a very high SVP
Requires high bypass flow rate to dilute to a clinically useful concentration.
- Has a low boiling point
Intermittently boils at room temperature, which will cause large fluctuations in delivery:
 - Excessive agent delivery during boiling
This will lead to cooling due to the latent heat of vapourisation.
 - Cooled desflurane will have a much lower saturated vapour pressure
Significant under-delivery will then occur.

The Tec6 vapouriser:

- Heats desflurane to 39°C
SVP of desflurane at this temperature is 1500mmHg.
- Gaseous desflurane is then added to the fresh gas flow
The amount added depends on:
 - Desired concentration

- Fresh gas flow rate
As flow increases the resistance to flow of desflurane vapour decreases.

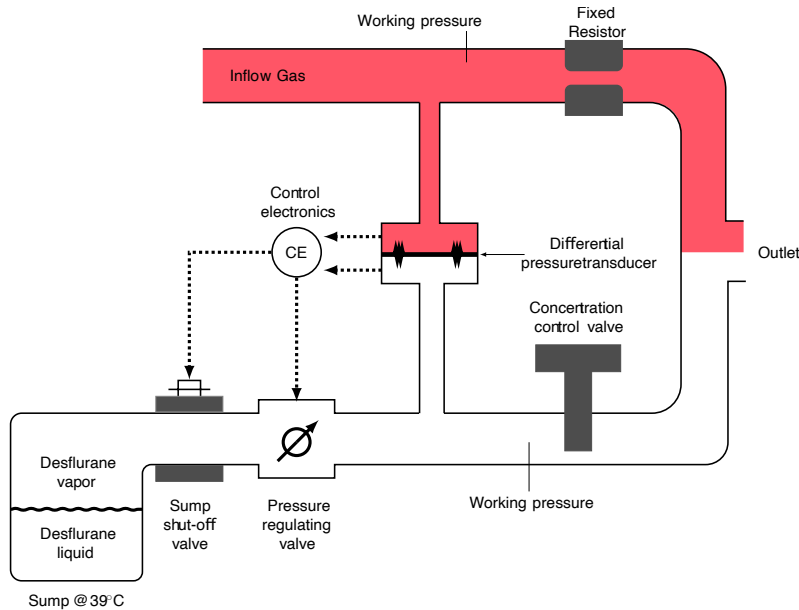


Figure 2: Tec 6 Vapouriser

General Safety Features of Vapourisers

Safety features of vapourisers include:

- **Agent specificity:**
 - Key indexed filling
 - Pin indexed safety system connectors
 - Colour coding of unit and agent containers
- **Single agent administration:**
 - Interlock mechanism
Prevents multiple vapourisers being turned on.
 - Single cartridge slot (Aladdin system)
- **Tipping and overfilling:**
 - Long vapourisation chamber inflow
 - Heavy construction
 - Transport modes
 - Side filling and overflow ports

- **Anti-pumping:**
 - Check valves and long vapourisation chamber inflow prevent entrainment of vapouriser gas in the inflow of the bypass channel
- **Agent depletion:**
 - Filling gauges
 - Low pressure alarms (Tec 6)

Other Factors Affecting Vapourisers

Carrier Gas Composition:

- Nitrous oxide and air are more viscous than oxygen
- This leads to decreased flow through the vapourising chamber when FiO_2 is low
This effect is not clinically significant.

Altitude:

- Clinical effect of volatile agent is a function of their partial pressure in tissues
- As SVP is independent of atmospheric pressure, this is unchanged at altitude
- A vapouriser set at 2% will deliver 4% gas at 0.5atm pressure, however as the atmospheric pressure is reduced the same partial pressure of vapour is delivered
- The delivered concentration of an agent at altitude is given by the equation:

$$\text{Delivered Concentration (\%)} = \text{Intended Concentration} \times \frac{P_{cal}}{P_{alt}}, \text{ where:}$$

- *Actual Delivered Concentration* is the concentration of agent in the gas delivered to the patient
The actual delivered concentration must be multiplied by the atmospheric pressure to find the partial pressure of agent delivered to the patient.
- *Intended Concentration* is the concentration dialed up on the vapouriser
- P_{cal} is the atmospheric pressure where the vapouriser was calibrated

- P_{alt} is the atmospheric pressure where the vapouriser is being used
-

References

1. Miller RD, Eriksson LI, Fleisher LA, Weiner-Kronish JP, Cohen NH, Young WL. Miller's Anaesthesia. 8th Ed (Revised). Elsevier Health Sciences.
2. Boumphrey S, Marshall N. Understanding vapourizers. Continuing Education in Anaesthesia Critical Care & Pain. Volume 11, Issue 6, 1 December 2011, Pages 199–203,

Breathing Systems

Describe and classify breathing systems used in anaesthesia. Evaluate their clinical utility and hazards associated with their use.

Classifications:

- **Open**

Anaesthetic gases not confined to the circuit.

- Limited current practical application
- Expensive, environmental contamination.
- e.g. Ether masks

- **Non-rebreathing**

No expired gas is re-inspired; requires a one-way valve.

- Limited practical application
- Requires a low-resistance draw-over vapouriser
- e.g. Tri-service apparatus
 - Robust
 - Inexpensive

- **Rebreathing systems**

Expired gas is re-inspired.

- Absorption systems
- Requires method for CO₂ absorption.
 - Circle
 - Common anaesthetic circuit, covered in detail under. Can be:
 - Vapouriser Out-of-Circuit
 - Common system, covered in detail under Chapter .
 - $[Alv_{agent}] \propto \frac{FGF}{MV}$

This provides a general overview of anaesthetic breathing systems. The circle system in particular is covered under Chapter , and the bag-valve-mask is covered under Chapter .

- Vapouriser in-circuit
Uncommon system.
 - $[Alv_{agent}] \propto \frac{MV}{FGF}$
 - In a spontaneous ventilation mode, the patient will increase agent concentration as minute ventilation $\uparrow$. This means that as surgical stimulus $\uparrow$, depth of anaesthesia also $\uparrow$.
- Waters
 - Mapleson B or C with a CO_2 absorption canister between bag and FGF.
- Non-absorption Rebreathing expired gas is part of circuit design.
 - Mapleson Systems

Mapleson System

The Mapleson System of circuits are a family of breathing circuits that allow both spontaneous and controlled ventilation:

- Rebreathing of expired gas **does not necessarily equate to CO_2 retention**, provided the FGF is above a certain multiple (circuit dependent) of the patients MV
 - $PaCO_2$ is a function of FGF and CO_2 production **only**
Increasing MV without increasing FGF will result in re-breathing of CO_2 and unchanged $PaCO_2$.
- In any spontaneous ventilation mode, patients will hyperventilate if FGF is inadequate

Mapleson A

The Mapleson A is efficient for spontaneous breathing, as expired gas is quickly vented via the APL, but inefficient for controlled ventilation due to the high FGF requirements.

Setup:

- APL valve close to mask
- Tubing between bag and mask
- FGF close to bag

Flow requirements:

- Spontaneous ventilation: $0.7 \times MV$
APL valve is set low. Initial exhalation (which is mostly **dead space**, and not containing CO_2) will fill bag until bag pressure exceeds APL valve opening pressure. Provided the APL valve is set *low*, the majority of CO_2 containing exhalation will exit through the APL valve, and FGF required to clear CO_2 from the circuit is low.
- Controlled ventilation: $> 3 \times MV$
APL valve is set high. More of the exhalation will fill the bag, and so a greater FGF is required to prevent re-breathing.

Mapleson B and C

Neither variation has a use case that another configuration doesn't perform better. Setup:

- APL valve and FGF are situated close to the mask
- Mapleson B has long tubing between the mask and bag
- Mapleson C has short tubing between the mask and bag

Flow requirements:

- Spontaneous and controlled ventilation are similar, at $3 \times MV$

Mapleson D

The Mapleson D circuit is a variation on the Mapleson A, with reversal of the FGF and APL. This circuit is much more

efficient for controlled breathing, but inefficient for spontaneous breathing. generally the best circuit to maximise efficiency across both spontaneous and controlled ventilation.

Setup:

- FGF is close to mask
 - Valve is close to bag
 - Tubing between FGF and APL valve
- Co-axial versions exist, but are functionally similar.

Flow requirements:

- Spontaneous ventilation: $2 \times MV$
 - Controlled ventilation: $0.8 - 1 \times MV$
- Best circuit for controlled ventilation.

Mapleson E

Also known as Ayre's T-piece, and is similar to a T-piece used to transport spontaneously breathing patients with a laryngeal mask.

Setup:

- T-shaped circuit with no valve or bag

Flow requirements:

- Only viable for spontaneous breathing.

Mapleson F

Also known as the "Jackson-Rees modification to the Ayre's T-piece," and very useful in paediatrics for both spontaneous and controlled ventilation.

Setup:

- Bag (with hole) added to the stem of the T of a Mapleson E
Allows monitoring of ventilation, and occluding the hole of the bag allows controlled ventilation.
- Functionally identical to a Mapleson D, with an operator-controlled APL valve

Flow requirements are identical to the Mapleson D:

- Spontaneous ventilation: $2 \times MV$
- Controlled ventilation: $0.8 - 1 \times MV$

References

1. Westhorpe, R. Paediatric Breathing Systems. RCH Anaesthetic Tutorial Program. 2019.

Circle System

Describe and classify breathing systems used in anaesthesia. Evaluate their clinical utility and hazards associated with their use.

The circle breathing system is a highly efficient system which:

- Has several key advantages
 - Preserves anaesthetic gases making volatile anaesthesia cost-effective
 - Preserves medical gases (oxygen) which is useful in resource-limited settings (e.g. prehospital)
 - Preserves heat and moisture
 - Reduces fire risk
 - Particularly with older agents.
- Requires **re-breathing** of expired gases
CO₂ is actively removed.
- Is a closed-circuit system
 - The only gases which must be replaced are those:
 - Consumed by the patient
 - Oxygen
 - Absorbed and metabolised volatile agents
 - Lost via leak

This covers the circle breathing system. A general overview of anaesthetic breathing systems is covered under Chapter .

Components

A circle circuit consists of:

- A Y-piece, connecting the circuit to the patient
- Expiratory and inspiratory valves, ensuring unidirectional flow
- A means of generating pressure

In most systems this consists of both a ventilator and a reservoir bag with APL valve attached, with a bag/vent switch to swap between circuits.

 - These are typically placed on the expiratory limb so that gas can be removed via scavenging prior to passage through soda lime
 - This reduces soda lime consumption, as some CO₂ will be scavenged.
- Soda lime

To absorb CO₂.
- Fresh gas flow
 - Includes oxygen, air and nitrous oxide
 - Oxygen enters the back-bar last
 - When the vapouriser is out-of-circuit, all fresh gas flow will pass through the vapouriser prior to entering the circle
- A separate high-pressure high-flow oxygen flush, which bypasses the vapouriser

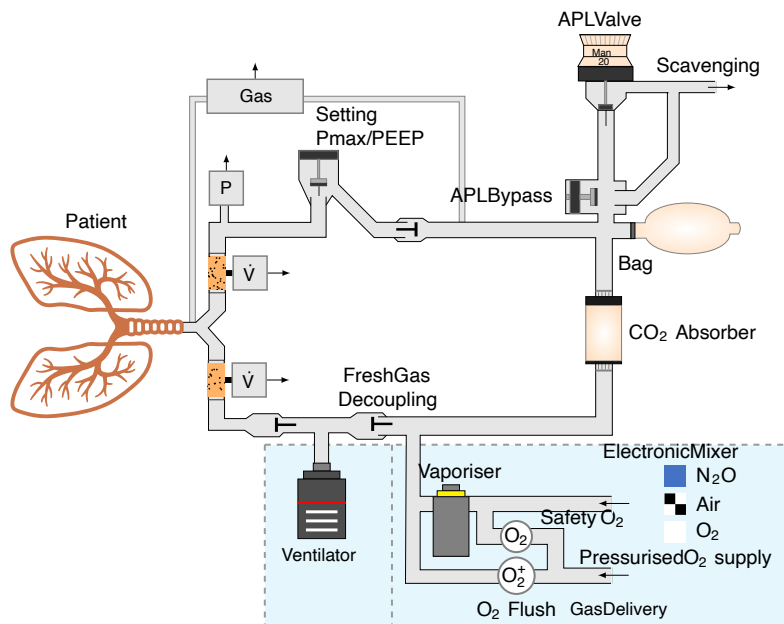


Figure 1: Circle System

Soda Lime

Soda lime:

- Consists of granules of:
 - 81% $\text{Ca}(\text{OH})_2$
 - 4% NaOH
 - 15% H_2O
 - Silicates
 - Hardens granules.
 - pH indicator
 - Visual representation of uptake of CO_2 by soda-lime.
 - Phenolphthalein
 - Red to white.
 - Ethyl violet
 - White to purple.
- Granules are 4-8 mesh in size
 - Will pass through a mesh with 4 holes per square inch, but not 8

- Balance between surface area (speed/efficacy of reaction) and resistance to flow
- Absorbs CO_2 by following reaction:

$$\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$$

$$\text{H}_2\text{CO}_3 + 2\text{NaOH} \rightarrow \text{Na}_2\text{CO}_3 + 2\text{H}_2\text{O} + \text{Heat}$$

$$\text{Na}_2\text{CO}_3 + \text{Ca(OH)}_2 \rightarrow \text{CaCO}_3 + 2\text{NaOH} + \text{Heat}$$
 - This ↑ the pH of the soda lime, causing the pH indicator to change colour
- 100g of soda lime can absorb ~26L of CO_2

Pros

- Cheaper to operate
- Conserves gases, heat, and moisture
- Low dead space
- Reduced greenhouse effects

Cons

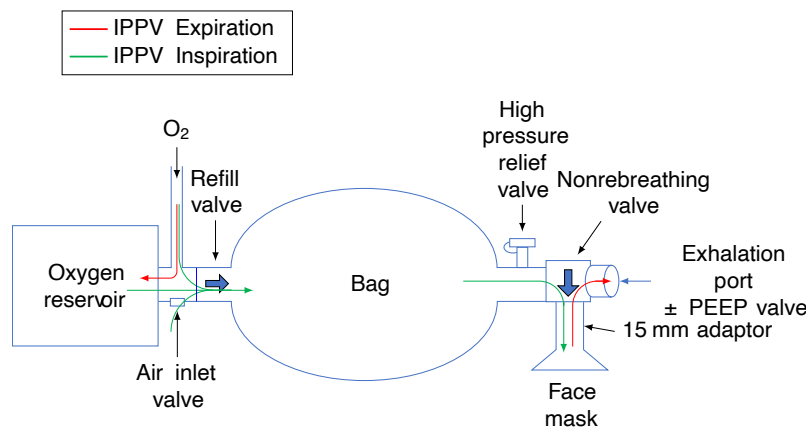
- Gas mixture settings are not delivered to the patient
Settings affect the fresh gas flow mixture, whilst the patient respire gas from the circuit. These are not identical, especially at low flows.
 - Nitrogen may build up in the circuit during low-flow anaesthesia, and potentially lead to delivery of a hypoxic gas mixture
 - Less portable than open-circuit systems
 - Increased circuit resistance
 - Requires soda lime, which can be toxic
 - Produces Compound A-E from sevoflurane
 - Produces carbon monoxide from desflurane, isoflurane, and enflurane
 - Dangerous if aspirated
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Bag-Valve-Mask

Describe and classify breathing systems used in anaesthesia. Evaluate their clinical utility and hazards associated with their use.



This covers the bag-valve-mask
 Figure 11.3 A general overview of
 anaesthetic breathing systems is covered
 under Chapter .

Components

Essential components:

- Self-inflating bag
 - In various sizes:
 - Infant: 240mL
 - Child: 500mL
 - Adult: 1600mL
- Oxygen source

- Air inlet valve
Draws room air into the bag if the oxygen reservoir is empty and the bag is still inflating.
- One-way valves
Sit between the self-inflating bag and the:
 - Gas source
Refill valve prevents reflux of gas into the reservoir instead of into the patient.
 - Exhalation port Non-rebreathing valve prevents expiration into the self-inflating bag, and forces expiration through the exhalation port (or PEEP valve, if present).
- 15mm adaptor
For mask or airway device.
- High pressure relief valve
Pop-off valve prevents Pip rising excessively (usually at $40-65\text{cmH}_2\text{O}$, depending on model).

Optional components:

- PEEP valve
Partially occludes exhalation port to provide PEEP. May be built-in or sold separately.
- Oxygen reservoir
Must be at least as large as the bag. Allows buffering of gas flow into the self-inflating bag (as peak flow will be less than that supplied from the oxygen source), and allows **100% O₂** to be delivered provided the **minute ventilation** is **less** the **O₂ flow rate**.

Method

Steps:

- Attach to supplemental oxygen
- Attach to mask or airway device
- Hand ventilate

Pros

Logistics:

- Cheap
- Effective
- Widely available
- Portable
- Requires no power

Function:

- Can deliver close to 100% oxygen
- Minimal dead space
- Pop-off valve reduces risk of barotrauma
- Highly compatible
15mm adaptor compatible with masks, LMA, ETT, and other in-line airway add-ons (ETCO₂, bronchoscopic Y-piece, etc.).

Cons

Logistics:

- Mis-assembly
Reusable BVM systems can occasionally have their valves inserted backwards, giving risk of barotrauma and failure of oxygen delivery.

Function:

- Less effective for spontaneous ventilation
 - Delivered FiO₂ may vary substantially between devices
 - Requires sufficient ventilatory effort to open one-way valves
Failure will result in air entrainment rather than oxygen.
- Lower FiO₂ delivered at high MV
- Bulky
- Easy to hyperventilate

- Unable to gauge lung compliance
Risk of barotrauma.
 - Difficulty assessing delivered tidal volumes
-

Scavenging

Outline the hazards of anaesthetic gas pollution and the methods of scavenging anaesthetic gases.

Scavenging is the removal and safe disposal of waste anaesthesia gases from the breathing circuit to avoid contamination of the theatre environment. Scavenging is important as continuous exposure of staff to anaesthetic gases has been implicated in:

- Cognitive impairment
- Spontaneous abortion
- Infertility
- Haematological malignancy

Methods of Scavenging

A scavenging system consists of:

- Gas collection assembly
 - Connects to the APL valve and ventilator relief valve
Collects gas vented from the circuit.
 - Uses a 30mm connector
Prevents accidental connection to the breathing system.
- Transfer tubing

- Scavenging interface

The structure of the scavenging interface depends on the type of scavenging system.

- Open interface

Active scavenging systems use a pump to generate a pressure gradient drawing gas to the disposal assembly. The scavenging interface is **open** to air to prevent the negative pressure being transmitted to the patient.

- Closed interface

Passive scavenging systems use a series of positive and negative pressure relief valves.

- When gas pressure in the collection assembly exceeds **5cmH₂O**, the positive relief valve opens and gas enters a reservoir bag
 - When gas pressure in the disposal assembly falls below **0.5cmH₂O**, the negative relief valve opens and gas enters the disposal assembly

- More transfer tubing

- Disposal assembly

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Diathermy

Outline the principles of surgical diathermy, its safe use and the potential hazards.

Diathermy is the use of an electrical current to cut tissue and coagulate blood via localised heating. Diathermy:

- Uses **high frequency, alternating current** passing between two electrodes
Frequencies between 300kHz and 2MHz are used, which have a negligible risk of inducing arrhythmia.
- Heat energy produced is proportional to electrical power dissipated (I^2R)
- Relies on the principle of current density

$$\text{Current Density} = \frac{\text{Current}}{\text{Area}}$$
 - A high current density at the electrode causes tissue damage
 - A low current density (e.g. at the plate of a unipolar electrode) causes heating without damage

Diathermy Types

Diathermy can be either:

- **Unipolar**
Consists of a probe containing one electrode, and a large plate (placed elsewhere on the patient) containing the other probe.

- **Bipolar**

Consists of a pair of forceps with each point containing a separate electrode. Minimises the current passing between probes, and is used when using diathermy on electrically sensitive tissues (e.g. brain).

Diathermy Modes

Diathermy modes include:

- Cutting
Low-voltage mode producing a high current in the shape of a continuous sine wave.
- Coagulate
High-voltage mode producing a damped sine wave response.
- Blended
Mixture of cutting and coagulate on different tissues.

Risks

- Burns
From incorrectly applied unipolar plate.
 - Electrocution
May injure patient, staff, or damage equipment and implants.
 - Electrical Interference
May inhibit pacing in certain pacemakers, or trigger ICDs.
 - Smoke production
Respiratory irritant, dissemination of viral particles, and may be carcinogenic.
 - Tissue dissemination
Potential source of metastatic seeding.
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Lasers



This is no longer examinable content for the ANZCA Primary Exam. This section is left here for posterity.

A laser is a device for light **amplification** by **stimulated emission** of **radiation**. Laser light is:

- Non-divergent
All photons move in parallel.
- Coherent
All photons are in phase.
- Monochromatic
All photons have the same wavelength.

Lasers are used clinically for:

- Precise incisions
Destruction of cells by localised vapourisation of water.
- Destruction of chemicals
Tattoos, oncological drugs.
- Tissue destruction without heating
Ophthalmology.

Principles

Method:

- An energy source is passed through a **lasing medium**, housed in a **resonator** made of mirrors

- As the lasing medium is excited, electrons enter a higher energy level
When more than 50% of electrons are at a higher energy level, **population inversion** has occurred.
- As electrons fall back to their resting state, they release a photon
 - A spontaneous emission occurs when an electron enters its resting state spontaneously
 - A stimulated emission occurs when an electron enters its resting state after being struck by a photon released from a spontaneous emission
 - Stimulated emissions result in amplification of light release
- The mirrors in the resonating chamber ensure most light is reflected back into the chamber, causing more stimulated emissions
- The exit from the chamber can be adjusted so only certain polarities of light are emitted
- A lens may be used to focus the laser beam
- Lasers may be:
 - Pulse wave
Uses short bursts of laser light to minimise collateral damage.
 - Continuous wave
May lead to excessive heating.

Pros

- Precise surgery and haemostasis

Cons

- Require multiple safety precautions
 - Laser safety officer
 - Eye protection
 - Warning signs on doors

- Cover theatre windows
 - Non-combustible drapes
 - Matte finish on equipment to minimise chance of reflection
 - Additional risks in airway surgery
 - Use lowest FiO₂ possible
 - Avoid N₂O
 - Consider use of heliox
 - Use specialised laser tubesNormal PVC ETTs are combustible.
-

References

1. Aston D, Rivers A, Dharmadasa A. Equipment in Anaesthesia and Intensive Care: A complete guide for the FRCA. Scion Publishing Ltd. 2014.

Procedural Anatomy

Internal Jugular Vein

Describe the anatomy relevant to central venous access (including ~~femoral~~, internal jugular, ~~external jugular~~, ~~subclavian~~ and ~~peripheral~~ veins).

Describe the anatomy and anatomical relations of the great veins relevant to performing central venous cannulation, including the ultrasound anatomy.

The internal jugular vein:

- Originates at the **jugular bulb**
This is a dilatation formed by the confluence of the inferior petrosal sinus and the sigmoid sinus.
- Exits the skull via the **jugular foramen**
- Descends laterally to the internal carotid (and later the common carotid) in the **carotid sheath**
- Terminates behind the sternal end of the clavicle, where it joins with the **subclavian vein** to form the **brachiocephalic vein**

Borders

- Anteriorly by SCM
- Posteriorly by the lateral mass of C1, scalene muscles, and lung pleura
- Medially by the internal carotid

Relationships

- Vagus nerve lies behind/between the carotid and IJV
- Cervical sympathetic plexus lies posterior to the carotid sheath
- Deep cervical lymph nodes lie close to the vein
- External jugular crosses the sternomastoid belly of SCM, running posteriorly and more superficial to the IJV, later perforating deep fascia to drain into the sub-clavian vein
- Pleura rises above the clavicle, and is close to the vein at its termination
- Thoracic duct passes lateral to the confluence of the left IJV and SCV, and may be injured during **left** IJV cannulation
 - The right lymphatic duct may be injured during right IJV cannulation, but due to its smaller size this is less common

Surface Anatomy

Identify the triangle formed by the two heads of SCM and the clavicle. Palpate the artery, and ensure the site of entry is **lateral** to the carotid. Aim:

- Caudally, at a 30 angle to the frontal plane
- Parallel to the sagittal plane
- Towards the ipsilateral nipple

Ultrasound Anatomy

Identify the vein deep to SCM, noting that it is (unlike the adjacent ICA):

- Non-pulsatile
- Thin walled
- Compressible

Approaches

- Anterior
At the medial border of SCM, 3-4cm above the clavicle. Requires retraction of the carotid medially.
 - Central approach
At the apex of the triangle formed by each muscle belly of SCM and the clavicle.
 - Posterior approach
At the posterior edge of SCM, just superior to where the EJV crosses the sternomastoid.
-

References

1. McMinn, RMH. Last's Anatomy: Regional and Applied. 9th Ed. Elsevier. 2003.
2. <http://radiopaedia.org/articles/internal-jugular-vein>
3. <http://www.frca.co.uk/article.aspx?articleid=100030>
4. Internal jugular vein catheterisation: Posterior and Central Approach

Subclavian Vein

Describe the anatomy relevant to central venous access (including femoral, internal jugular, external jugular, subclavian and peripheral veins)

Describe the anatomy and anatomical relations of the great veins relevant to performing central venous cannulation, including the ultrasound anatomy.

The subclavian vein:

- Is a continuation of the axillary vein as it crosses the **upper surface** of the **first rib**
- Travels posterior to the clavicle, separated from the subclavian artery by the anterior scalene
- **Joins** with the **internal jugular vein** to form the **brachiocephalic vein**

Borders

- Anteriorly by the clavicle, subclavius muscle, and pectoralis major
- Posteriorly by anterior scalene muscle and subclavian artery
- Inferiorly by first rib and lung apex
- Superiorly by skin, subcutaneous tissue, and platysma
- Medially by the brachiocephalic vein

- Laterally by the axillary vein

Surface Anatomy

The needle is placed in the deltopectoral groove, inferior and lateral to the middle third of the clavicle. The needle is inserted at a shallow angle, passing under the middle third of the clavicle aiming at the sternal notch.

Ultrasound Anatomy

Identify:

- The vein deep to subclavius
- The artery, superior to the vein
- The pleura, deep to the vascular structures

Approaches

- Infraclavicular
 - Sternal notch identified with index finger
 - Thumb placed two-third laterally along the clavicle
 - Needle advanced through the skin beneath the thumb and angled towards the sternal notch, into the clavicle
 - Needle 'walked off' the deep aspect of the clavicle, with continual gentle suction used to identify the vein
 - This maintains a flat trajectory - it is important to avoid a steep needle angle
 - The needle can be displaced below the clavicle with pressure on the overlying tissues, which limits the steepness of the angle

- Needle trajectory should not be adjusted whilst in the deep tissues
- Supraclavicular

Generally, landmarks are identified with the contralateral hand – if placing a right-sided subclavian line, identify the structures with your left hand.

References

1. McMinn, RMH. Last's Anatomy: Regional and Applied. 9th Ed. Elsevier. 2003.

Intercostal Catheter

Describe the anatomy relevant to the insertion of an intercostal catheter

An intercostal catheter drains the intrapleural space.

Surface Anatomy

An ICC should be placed in the **safe triangle**:

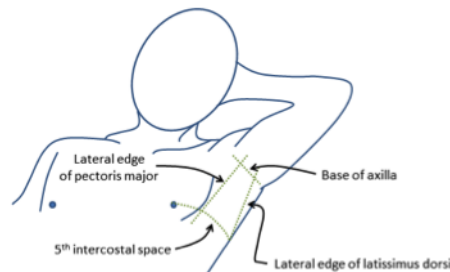


Figure 1: Safe Triangle

This is bordered:

- **Anteriorly** by pectoralis major
- **Posteriorly** by latissimus dorsi
Too far posterior will injure the long thoracic nerve.
- **Superiorly** by the base of the axilla
- **Inferiorly** by the 5th intercostal space
Too far inferiorly risks placement in the liver or spleen.

Layers of Dissection

- Skin
 - Subcutaneous tissue
 - External intercostal
 - Internal and innermost intercostal muscles
- Note the **neurovascular bundle** which sits on the inferior aspect of the ribs, therefore aim to place the ICC at the bottom of the intercostal space - "above the rib below".
- Parietal pleura
-

References

1. LITFL - Chest Drain

Antecubital Fossa

Describe the anatomy relevant to central venous access (including femoral, internal jugular, external jugular, subclavian and peripheral veins)

Describe the anatomy (including ultrasound anatomy) peripheral venous system relevant to performing intravenous cannulation and PICC line insertion.

The antecubital fossa is a **triangular space** on the anterior aspect of the forearm.

Borders

The triangular borders are formed:

- Medially by pronator teres
- Laterally by brachioradialis
- Superiorly by an imaginary line between the medial and lateral epicondyles
- The roof of the fossa is formed by subcutaneous tissue
- The floor is formed by brachialis and supinator

Contents

From medial to lateral:

- Median nerve
 - Brachial artery
 - Biceps tendon and aponeurosis
 - Radial and posterior interosseous nerves
 - Veins
 - Basilic vein
 - Cephalic vein
 - Venous variations:
 - A median cubital vein connecting the basilic and cephalic veins
 - A median vein of the forearm, which divides into a median basilic and median cephalic vein which drain into the basilic and cubital veins
-

References

1. FRCA. The Cubital Fossa.

Tracheostomy

Describe the anatomy relevant to the performance of a naso, or endo, tracheal intubation, a cricothyroidotomy or tracheostomy.

Trachea

The trachea is fibrocartilagenous tube which:

- Extends from the larynx superiorly to the Plane of Louis inferiorly
- Terminates by division into the right and left mainstem bronchi
- Runs at 15 degrees parallel to the surface of the neck, such that the distal trachea is deeper than the proximal trachea
- Has a D-shaped cross section
 - Anterior wall is formed by 18-22 incomplete cartilaginous rings which maintain tracheal patency
 - Posterior wall of the trachea is spanned by longitudinal smooth muscle known as trachealis
- Is typically:
 - **10cm long**
 - **2.3cm wide**
 - **1.8cm in AP diameter**

Relationships

- Lateral to the trachea are the:
 - Carotid sheaths
Contains the carotid artery, internal jugular vein, and vagal nerves.
 - Thyroid lobes (and inferior thyroid arteries)
 - Recurrent laryngeal nerves.
- Inferior to the thyroid isthmus lies the thyroid veins
- Posterior to the trachea are the:
 - Oesophagus
 - Vertebral column

Surface Anatomy

Midline neck structures are relevant surface anatomy:

- Laryngeal structures
Including: Hyoid, thyroid cartilage, cricothyroid membrane, cricoid cartilage.
- Sternal notch
- Thyroid lobes
Lie lateral to trachea.

Layers of Dissection

- Skin
 - Subcutaneous fat
 - Superficial and Deep Pretracheal fascia
 - Tracheal wall
 - Ideally between 1st and 2nd rings
-

References

1. McMinn, RMH. Last's Anatomy: Regional and Applied. 9th Ed. Elsevier. 2003.

Neuraxial Anatomy

Describe the anatomy relevant to the performance of a lumbar puncture.

Describe the anatomy of the vertebral column spinal cord and meninges relevant to the performance of central neuraxial block with appropriate surface markings.

The spinal **epidural space** is:

- A space lying within the vertebral column, that contains:
 - Nerve roots
 - Vessels
Vascular density is greater anteriorly.
 - Fat
- A **negative pressure space**
The epidural space is continuous with the paravertebral space, which is affected by changes in intrapleural pressure. As **intrapleural pressure** is usually **negative**, the epidural space is also a negative pressure space. This property facilitates the **hanging drop** technique of epidural placement. **Key exceptions** (and relative contraindications to hanging drop) include:
 - Positive pressure ventilation
 - Contractions in labour
Epidural veins are transfused by a contracting uterus, and pressure increases as they distend.

- Valsalva manoeuvre

The **intrathecal space** is:

- Deep to the epidural space
- Contains the spinal cord and CSF

Borders

The spinal **epidural space** is bordered:

- Anteriorly by the posterior longitudinal ligament
- Posteriorly by the ligamentum flavum
- Superiorly by the endosteal layer of dura
Prevents passage of contents through the foramen magnum.
- Inferiorly by the sacrococcygeal membrane at the sacral hiatus
- Laterally by the pedicles and intervertebral foraminae

Surface Anatomy

Identify:

- Spinous processes
Generally palpable in midline.
 - C2 is the first palpable process
 - C7 is the most prominent process
 - T7 corresponds with the tip of the scapula when the arms are by the patient's side
 - The line drawn across the iliac crest corresponds with the L4/5 interspace in most patients
This is known as **Tuffier's Line**.
- Gluteal crease
Identify midline in patients with impalpable spinous processes, who:
 - Are sitting flat

- Do not have scoliosis

Layers

Tissues passed through, from superficial to deep, include:

- Skin
 - Subcutaneous tissue
 - Supraspinous ligament
One continuous ligament.
 - Interspinous ligament
One ligament between each spinous process.
 - Ligamentum flavum
This yellow ligament is:
 - Thicker and harder than other ligaments
 - Paired at each intervertebral level
 - 5 - 6mm thick in the adult, though it is thinner with:
 - Youth
 - ↑ Level
 - Body habitus
 - Epidural space
 - Dura
 - Subarachnoid space
 - Spinal cord
Ends at L2 in adults and L3 in children.
-

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Pharmacopoeia



Toxicology

Tricyclic Antidepressant Overdose

Tricyclic antidepressants are weak bases typically used for depression and as an adjunct for analgesia. They have a complex mechanism of action, competitively inhibiting noradrenaline and serotonin reuptake, and also blocking muscarinic receptors, histaminergic receptors, α -adrenoreceptors, GABA-A receptors, and fast sodium channels.

Toxicity

In overdose, toxicity is predominantly due to cardiac and central effects, though there are effects on most of the major organ systems.

Cardiac toxicity

Cardiac toxicity is due to antagonism of α -adrenoreceptors use-dependent blockade of fast sodium channels.

α -antagonism results in vasodilatation and subsequent **hypotension**. Hypotension may also be due to myocardial depression from sodium channel blockade.

Blockade of fast sodium channels occurs in the His-Purkinje system, as well as the atrial and ventricular myocardium. This results in **decreased myocardial impulse conduction**. They block channels in the inactivated state, resulting in a **use-dependent blockade** such that

the **effect is greater at faster heart rates**. This results in an increased depolarisation and repolarisation time. ECG findings are consistent with this and are essentially pathognomonic:

- Widened QRS
- Right axis deviation of the terminal QRS
- ≥ 3 mm terminal R wave in aVR.

Additional ECG findings include:

- Tachycardia
- Any degree of heart block
- Ventricular arrhythmias

Central toxicity

Central toxicity is predominantly due to anticholinergic effects, though antihistaminic effects contribute.

Anti-cholinergic effects tend to occur prior to cardiac effects, and include:

- Confusion
- Agitation
- Seizures
- Pupillary dilatation and blurred vision

Antihistaminic effects include obtundation.

Management

Standard management of poisoning applies. TCAs are not dialysable and as they are weak bases are not amenable to urinary alkalinisation.

Cardiac toxicity

NaHCO₃ and hyperventilation to a pH >7.5 is used to manage cardiac toxicity. There are a number of proposed mechanisms of action for the benefit of alkalinisation:

- Plasma **alkalosis** results in **less ionised drug** and **increases distribution** into tissues
- Plasma **alkalosis increases protein binding** of drug
- Intracellular alkalosis results in less bound intracellular drug, favouring its movement out of cells
- Extracellular alkalosis results in reduced H⁺/K⁺ exchange, increasing intracellular potassium and **hy-polarising the cell**.
- In addition to the alkalinising effects, sodium load from the NaHCO₃ improves the sodium concentration gradient into cells

α-adrenoreceptor antagonism can be countered with use of an α-agonist such as noradrenaline.

Arrhythmias should be managed with drugs that do not prolong the action potential - so amiodarone and beta-blockers are contraindicated. Initial management should be using NaHCO₃, though MgSO₄ and lignocaine can be considered in refractory cases.

Central toxicity

Seizures should be managed with benzodiazepines, phenytoin, propofol, and phenobarbital. Avoid agents which result in QRS prolongation.

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Organophosphate Poisoning

Organophosphates are substances that bind **irreversibly** to **acetylcholinesterase**, causing cholinergic excess. Examples include fertilisers and sarin gas.

Toxicity

Effects (as expected) are signs of muscarinic and nicotinic over-activation. This can be remembered by 'BLUDGES' for the muscarinic effects:

- **B**radycardia (and subsequent hypotension)
- **L**acrimation
- **U**rination
- **D**efecation
- **G**IT upset
- **E**mesis
- **S**weating and **S**alivation

and 'M' for the nicotinic effects:

- **M**uscular spasm

Management

Management is aimed at reducing ACh burden:

- **Atropine**
Competitive antagonist of ACh at the muscarinic receptor.

- Atropine is preferred over glycopyrrolate as it will cross the blood brain barrier and treat central ACh toxicity

- **Pralidoxime**

Reactivates acetylcholinesterase by luring the organophosphate away from the enzyme with a tantalising oxime group.

- Pralidoxime must be used within the first few hours of poisoning
After which the organophosphate-enzyme group 'ages' and is no longer susceptible.
 - Does not cross the blood-brain barrier and so cannot treat central effects
-

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Toxic Alcohols

Alcohols include:

- **Ethanol**
- **Methanol**
- **Ethylene Glycol**

In toxicity:

- All present with symptoms of alcohol intoxication
- All contribute to the osmolar gap
- Different toxicities occur due to the different metabolites

Ethanol

Ethanol is a weak alcohol with a complicated mechanism of action similar to volatile anaesthetic agents:

- Enhanced GABA-mediated inhibition
This is **reversible with flumazenil**.
- Inhibition of Ca^{2+} entry
- Inhibition of NMDA function
- Inhibition of adenosine transport

Table 1: Ethanol

	Ethanol
Class	Weak alcohol
Dose	
Standard drink	One unit is ~8g/10ml of pure ethanol
Toxic alcohol	0.6g/kg, targeting a BAC of 0.1-0.15%

Ethanol	
Pharmacokinetics	
Absorption	Rapid oral absorption
Mechanism	
Target	GABA - A receptor NMDA receptor Ca ²⁺ channel Adenosine transporter
Action	↑ GABA-mediated inhibition (reversible with flumazenil). Inhibition of Ca ²⁺ entry. Inhibition of NMDA function. Inhibition of adenosine transport.

Methanol

- Metabolised by alcohol dehydrogenase to formaldehyde and then formic acid
- Formic acid is neurotoxic
Damages retina and the optic nerve.

Ethylene Glycol

- Metabolised by alcohol dehydrogenase to glycoaldehyde, and (via several intermediate steps) to oxalic acid
- Oxalic acid binds calcium, which causes:
 - Hypocalcaemia
 - Long QT
 - Acute renal failure

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Naloxone

Pure MOP antagonist used for:

- Treatment of opioid overdose
 - Reducing constipation
- In combination with PO oxycodone.

Table 1: Naloxone

Naloxone	
Class	Opioid antagonist
Chemistry	Synthetic phenanthrene
Dose	
Reversal	0.1 - 0.4mg IV Q5 minutes
Infusion	0.5mg/kg/hr
Pharmaceutics	
Presentation	Clear colourless solution at 400 mcg/mL
Solubility	pKa 7.9; highly lipid soluble
Pharmacokinetics	
Absorption	Approximately 2% oral bioavailability due to extensive first-pass metabolism
Distribution	pKa: 7.9
	Vd: 2 L/kg
	Protein binding: 50%

Naloxone	
Metabolism	Half-life: 30-80 minutes Duration: Approximately 30-40 minutes Pathway: Rapid hepatic glucuronidation Metabolites: Naloxone-3-glucuronide (inactive)
Elimination	Renal: Renal elimination of inactive metabolite
Mechanism	
Target	μ -opioid receptor (MOP)
Action	Competitive antagonist at MOP receptors, displacing opioids from their receptors and reversing their effects
Pregnancy	
Code	B1
Pharmacodynamics	
Respiratory	Reversal of opioid-induced respiratory depression ($\uparrow$ RR, $\uparrow$ VT)
CVS	$\uparrow$ SVR $\uparrow$ BP Arrhythmia due to $\uparrow$ SNS tone
CNS	$\downarrow$ Analgesia $\downarrow$ Sedation $\downarrow$ Miosis Antanalgesic in opioid-naïve patients Precipitation of opioid withdrawal Hyperalgesia Seizures
Gastrointestinal	Vomiting Diarrhoea
Considerations	Duration of action (~30-40 minutes) is shorter than many opioids — risk of re-narcosis requiring subsequent doses or infusion

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Flumazenil

Competitive antagonist and inverse agonist of the benzodiazepine receptor.

Table 1: Flumazenil

	Flumazenil
Aliases	Anexate
Class	Benzodiazepine-site antagonist
Chemistry	Imidazobenzodiazepine
Dose	
Sedation	0.2mg IV over 15sec, then 0.1mg Q1min; maximum 1mg
Overdose	0.3mg IV, then 0.1mg Q1min; maximum 2mg; infusion 0.1 - 0.4mg/h if resedated
Pharmaceutics	
Presentation	0.5mg/5mL solution for IV injection
Compatibility	Compatible with 5% glucose and 0.9% NaCl; complete infusion within 24h of preparation
Solubility	Weak lipophilic base
Pharmacokinetics	
Absorption	IV; onset 1 - 2min
Distribution	Vd: 0.95L/kg (Vss)
	Protein binding: 50%

Flumazenil	
Metabolism	Half-life: 53min (40-80min) Duration: Shorter than most benzodiazepines Pathway: Almost complete hepatic metabolism Metabolites: Inactive carboxylic acid and conjugate
Elimination	Renal: Urine 90-95%, faeces 5-10% (radioactivity); negligible unchanged drug
Mechanism	
Target	Benzodiazepine site on GABAA receptors
Action	Competitive antagonism → reverses benzodiazepine and Z-drug CNS effects
Contraindications	
	Benzodiazepine treatment of status epilepticus or raised ICP Severe tricyclic or tetracyclic antidepressant toxicity
Pregnancy	
Code	B3
Pharmacodynamics	
Respiratory	Reversal of benzodiazepine respiratory depression is unpredictable
CVS	Dysrhythmias after mixed tricyclic overdose
CNS	Rapid reversal of sedation Seizures and acute benzodiazepine withdrawal
Gastrointestinal	Nausea and vomiting

Flumazenil	
Considerations	Resedation is common Inpatient use under continued observation only; not outpatient/short-stay or diagnostic use Prolonged ICU-sedation reversal not recommended Hepatic impairment ↓ clearance and prolongs half-life Avoid rapid injection in benzodiazepine dependence

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Respiratory Pharmacology

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Oxygen

Table 1: Oxygen

Oxygen	
Aliases	Medicinal oxygen
Class	Essential medical gas
Chemistry	Diatomic elemental gas
Dose	
Oxygen (normobaric)	FiO ₂ 0.21 - 1.0; avoid >0.6 when possible
Pharmaceutics	
Presentation	≥99.5% compressed colourless gas
Compatibility	Strongly supports combustion; avoid oils, grease, smoking and naked flames
Solubility	Poor water solubility
Pharmacokinetics	
Absorption	Rapid alveolar diffusion down its partial-pressure gradient
Distribution	V _d : Haemoglobin-bound and dissolved in plasma
	Protein binding: Hb oxygen capacity ~1.34mL/g
Metabolism	Duration: Minutes
	Pathway: Mitochondrial electron transport and aerobic metabolism
	Metabolites: Carbon dioxide and water (products of aerobic metabolism)

Oxygen	
Elimination	Renal: CO ₂ exhaled; bicarbonate eliminated in urine
Mechanism	
Target	Cytochrome c oxidase
Action	Substrate for aerobic metabolism/ oxidative phosphorylation → ATP production
Pregnancy	
Code	A
Justification	Clinically required normobaric use at lowest concentration/shortest duration; category does not apply to hyperbaric treatment
Pharmacodynamics	
Respiratory	↑ Alveolar and arterial PO ₂ Small initial ↓ respiratory drive; absorption atelectasis CO ₂ retention in susceptible patients (mainly V/Q mismatch and Haldane effect) Prolonged exposure at PiO ₂ >0.6bar → pulmonary toxicity risk
CVS	Corrects hypoxic cardiovascular dysfunction Hyperoxia: ↑ SVR; ↓ CO and PVR/PAP; coronary vasoconstriction
CNS	Hyperbaric toxicity, usually >1.6bar → neurological symptoms and seizures Retinopathy in premature neonates
Metabolic	Reactive oxygen species → lipid peroxidation and cell injury
Haematological	Haemolysis with oxygen toxicity

	Oxygen
Considerations	Circulating O₂ reserve ≤ 4 min at basal consumption (healthy individuals) Use the lowest FiO₂ that corrects hypoxaemia Bleomycin exposure: use $<30\%$ where possible High FiO₂ may worsen injury from amiodarone, nitrofurantoin or paraquat Fire risk

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Helium

Helium is an inert gas which is used to reduce the specific gravity of inhaled gas mixtures. It is typically provided as a 0.79/0.21 Helium-Oxygen (Heliox) mixture (though other dilutions exist).

Table 1: Comparison of Helium, Heliox, and Oxygen

	Helium	Heliox (0.79/0.21)	Oxygen
Specific Gravity (as compared to air)	0.18	0.34	1.09

Reduced specific gravity results in a proportional reduction in Reynolds Number, improving laminar flow within the airways.

Table 2: Helium

	Helium
Aliases	Heliox
Class	Inert medical gas
Chemistry	Inert element (He)
Dose	
Airway obstruction	72% helium/28% O2 inhaled
Pharmaceutics	
Presentation	Heliox28 compressed premixed gas

Helium	
Compatibility	Heliox28: dedicated regulator; non-flammable but supports combustion
Solubility	Low tissue solubility at atmospheric pressure
Pharmacokinetics	
Absorption	Not significantly absorbed systemically at atmospheric pressure
Distribution	Vd: Proportional to tissue solubility and partial pressure
Metabolism	Pathway: Not metabolised
Elimination	Renal: Exhaled unchanged
Mechanism	
Target	Airway gas flow
Action	Low gas density → ↓ Reynolds number and turbulence → ↑ laminar flow through obstructed airways
Contraindications	
	Pneumothorax Subcutaneous emphysema
Pharmacodynamics	
Respiratory	↓ Work of breathing, PaCO ₂ and dyspnoea; ↑ PEFR Dry secretions and atelectasis
CNS	High-pressure neurological syndrome at >16atm
Metabolic	Hypothermia with prolonged inhalation
Considerations	Monitor SpO ₂ and add O ₂ if required
	↑ Vocal pitch
	Long-term safety is unstudied Avoid oils, grease, smoking and naked flames

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Anti-Asthma Drugs

Describe the pharmacology of anti-asthma drugs.

- **Oxygen**
Increases FiO_2 and improves saturation.
- **Heliox**
Reduces specific gravity of inhaled gas mixtures, improving laminar flow.
- **β_2 -agonists**
Acts on a G-protein coupled receptor to $\uparrow$ cellular levels of adenylyl cyclase, $\uparrow$ cAMP, which results in smooth muscle relaxation and bronchodilatation.
- **Corticosteroids**
Glucocorticoids are steroid hormones that bind to specific intracellular receptors and translocate into the nucleus, where they regulate gene expression in a tissue-specific manner. They are used in asthma as they cause:
 - Bronchodilatation by increasing bronchial smooth muscle response to circulating catecholamines
 - Decreased airway oedema by decreasing inflammatory responses and transudate production
- **Muscarinic antagonists**
Anti-muscarinics are synthetic quaternary ammonium compounds which competitively inhibit M3 muscarinic receptors on bronchial smooth muscle, antagonising the bronchoconstrictor action of vagal impulses.
- **Methylxanthines**
Methylxanthines are phosphodiesterase inhibitors, reducing levels of cAMP hydrolysis and increased in-

tracellular levels of cAMP (via a different mechanism, so they are synergistic with β_2 agonists) and causing smooth muscle relaxation.

- **Ketamine**

Increases sympathetic outflow and relaxes bronchial smooth muscle.

- **Volatile Anaesthetic Agents**

Volatile anaesthetic agents reduces bronchial smooth muscle constriction where this is preexisting (such as asthma).

- **Leukotriene Antagonists**

Selectively inhibits the cysteinyl leukotriene receptor, increased activity of which is involved in airway oedema and bronchial smooth muscle constriction.

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Beta Agonists

This covers the inhaled β -agonists used for bronchodilation. Information on catecholamines and sympathomimetics with activity on β -receptors is covered under adrenergic vasoactives.

Table 1: Pharmacodynamic Effects of Beta Agonists

Pharmacodynamic Effects	β -agonists
Resp	Bronchodilatation, $\downarrow$ HPV causing $\uparrow$ shunt and potential $\downarrow$ PaO ₂ if O ₂ is not co-administered.
CVS	$\uparrow$ HR (β_1 with higher doses), $\downarrow$ BP (β_2 with lower doses)
GU	Tocolytic.
Metabolic	Hypokalaemia from β_2 stimulation of Na ⁺ /K ⁺ ATPase, hyperglycaemia.
Other	Potentiates non-depolarising muscle relaxants

Table 2: Comparison of Beta-Agonists

	Salbutamol	Salmeterol
Aliases	Albuterol Ventolin	Serevent
Class	Short-acting selective β_2 agonist	Long-acting selective β_2 partial agonist

	Salbutamol	Salmeterol
Chemistry	Synthetic sympathomimetic amine	Synthetic sympathomimetic amine; xinafoate racemate
Dose		
MDI	100-200microg inhaled	
Nebulised	5mg inhaled Q4-6H	
Infusion (IV)	0.5microg/kg/min	
Maintenance		50microg inhaled BID; up to 100microg BID in severe asthma
Pharmaceutics		
Presentation	MDI 100microg/puff; nebules 2.5mg/2.5mL or 5mg/2.5mL; injection 5mg/5mL	50microg dry-powder inhalation
Compatibility	Nebules must not be injected; dilute IV concentrate before bolus injection	Accuhaler contains lactose and milk protein
Solubility	Salbutamol sulfate is water soluble	
Pharmacokinetics		
Absorption	~10% of inhaled dose deposited in lung; swallowed fraction undergoes first-pass metabolism	Local pulmonary action; plasma concentration $\leq 200\text{pg/mL}$ after inhalation
Distribution	pKa: 9.3 and 10.3 Vd: 1.3L/kg Protein binding: Low	Protein binding: 95-98%

	Salbutamol	Salmeterol
Metabolism	Half-life: 2.7 - 5h (inhaled/PO) Duration: Inhaled onset 5 - 15min; peak 60 - 90min; duration 3 - 6h Pathway: Hepatic sulfation, oxidative deamination and glucuronidation; intestinal first-pass sulfation; not metabolised in lung Metabolites: Inactive salbutamol 4'-O-sulfate	Half-life: Plasma radioactivity ~67h after an oral radio-labelled dose Duration: ~12h Pathway: Extensive hepatic CYP3A4-mediated aliphatic hydroxylation Metabolites: α-Hydroxysalmeterol
Elimination	Renal: ~72% within 24h as unchanged drug and metabolites; small faecal fraction	Renal: Predominantly faecal; lesser urinary excretion
Mechanism		
Target	β_2 adrenoceptors	β_2 adrenoceptors
Action	Gs activation $\rightarrow$ $\uparrow$ adenylyl cyclase $\rightarrow$ $\uparrow$ cAMP $\rightarrow$ bronchial smooth-muscle relaxation	Prolonged Gs activation $\rightarrow$ $\uparrow$ cAMP $\rightarrow$ bronchial smooth-muscle relaxation

	Salbutamol	Salmeterol
Contraindications	Non-IV formulations: not for arresting uncomplicated premature labour or threatened abortion IV: antepartum haemorrhage	Severe milk-protein allergy
Pregnancy		
Code	A	B3
Justification		Asthma - control benefit outweighs potential adverse pregnancy outcome
Pharmacodynamics		
Respiratory	Bronchodilation ↓ HPV → ↑ shunt and possible ↓ PaO ₂ without supplemental O ₂	Bronchodilation; onset 10-30min, peak 3-4h ↓ HPV → ↑ shunt and possible ↓ PaO ₂ without supplemental O ₂
CVS	↑ HR at high doses; ↓ MAP from peripheral vasodilation	↑ HR at high doses; ↓ MAP from peripheral vasodilation at lower doses
CNS	Insomnia	
Endocrine	Hyperinsulinaemia and hyperglycaemia	
Musculoskeletal	Tremor, muscle cramps; potentiates non-depolarising NMBs	Tremor; potentiates non-depolarising NMBs
Metabolic	Hypokalaemia; hyperglycaemia Lactic acidosis with high-dose IV/nebulised therapy	Hypokalaemia and hyperglycaemia

	Salbutamol	Salmeterol
Genitourinary	Uterine relaxation	Uterine relaxation
Considerations	Increasing use may indicate worsening asthma β blockers antagonise bronchodilation Hypokalaemia may be potentiated by xanthines, corticosteroids and diuretics	Not for acute bronchospasm Use with inhaled corticosteroid in asthma Strong CYP3A4 inhibitors $\uparrow$ exposure and may prolong QTc

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Antimuscarinics (Respiratory)

Antimuscarinics with predominantly cardiac effects are covered at Antimuscarinics (Cardiac), whilst atropine is covered separately.

These agents competitively antagonise ACh at M₃ receptors in bronchial smooth muscle, preventing parasympathetic mediated bronchoconstriction.

Table 1: Respiratory Antimuscarinics

	Ipratropium	Tiotropium
Aliases	Atrovent	Spiriva
Class	Short-acting inhaled muscarinic antagonist	Long-acting inhaled muscarinic antagonist
Chemistry	Quaternary ammonium atropine derivative	Non-chiral quaternary ammonium compound
Dose		
Bronchospasm	250-500microg nebulised QID	
Maintenance		5microg inhaled daily (2 puffs)
Pharmaceutics		
Presentation	250microg/mL or 500microg/mL inhalation solution	2.5microg/puff solution for Respimat inhaler
Compatibility	Dilute to 2-3mL with 0.9% NaCl; protect UDV from light	Use only with the Respimat inhaler

	Ipratropium	Tiotropium
Solubility	Water soluble; low lipid solubility	Sparingly water soluble
Pharmacokinetics		
Absorption	<1% of inhaled dose absorbed through bronchial mucosa; oral absorption contributes ~2% of systemic levels	~33% systemic bioavailability after inhalation (young healthy volunteers); 2-3% PO; peak 5-7min
Distribution	pKa: 15.3	Vd: 32L/kg
	Vd: 2.4-4.6L/kg	Protein binding: 72%
	Protein binding: <20%	
Metabolism	Half-life: 1.5-3h; IV terminal ~1.6h	Half-life: 27-45h (effective; inhaled COPD/asthma)
	Duration: Up to 6h	Duration: ≥24h
	Pathway: ~60% of systemic dose metabolised, probably hepatic	Pathway: Minimal; mainly non-enzymatic ester cleavage
	Metabolites: Inactive urinary metabolites	Metabolites: N-Methylscopine (inactive at muscarinic receptors), Dithienylglycolic acid (inactive at muscarinic receptors)

	Ipratropium	Tiotropium
Elimination	Renal: ~40% of systemic dose unchanged; biliary/faecal radioactivity <10% after IV dosing	Renal: 74% unchanged after IV dosing (young healthy volunteers); 18.6% of inhaled dose in COPD, with most remainder unabsorbed in faeces
Mechanism		
Target	Muscarinic receptors (non-selective; M3-mediated bronchodilation)	Muscarinic receptors (M3-mediated bronchodilation)
Action	Competitive antagonism → ↓ vagal bronchial smooth-muscle contraction	Competitive antagonism with slow M3 dissociation → prolonged bronchodilation
Pregnancy		
Code	B1	B1
Pharmacodynamics		
Respiratory	Bronchodilation without impaired mucociliary clearance Onset 3-5min; peak 1.5-2h	Bronchodilation
CVS	Tachycardia	
CNS	Minimal CNS penetration	Minimal CNS penetration
Genitourinary	Urinary retention	Urinary retention
Gastrointestinal	Dry mouth; ↓ GI secretions at large doses	Dry mouth, constipation and ileus

	Ipratropium	Tiotropium
Considerations	Eye exposure may cause mydriasis or acute angle-closure glaucoma Caution with prostatic hyperplasia or bladder-neck obstruction Additive bronchodilation with β_2 agonists or xanthines	Maintenance only; not for acute bronchospasm Caution with narrow-angle glaucoma or urinary outflow obstruction Moderate-severe renal impairment $\uparrow$ exposure

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Phosphodiesterase Inhibitors / Methylxanthines

Methylxanthines non-selectively inhibit phosphodiesterase, which results in reduced levels of cAMP hydrolysis and therefore increased intracellular cAMP, and subsequent smooth muscle relaxation. This effect is synergistic with β_2 agonists, which also increase cAMP by increasing production.

Table 1: Theophylline

Theophylline	
Aliases	Nuelin
Class	Methylxanthine bronchodilator
Chemistry	1,3-Dimethylxanthine
Dose	
Maintenance (SR)	200-300mg PO Q12H
Pharmaceutics	
Presentation	SR tablets 200mg, 250mg or 300mg; syrup 26.67mg/5mL
Compatibility	SR tablets may be halved; do not chew or crush
Pharmacokinetics	
Absorption	~100% PO bioavailability (SR); peak 4-6h
Distribution	Vd: 0.5L/kg
	Protein binding: 50-70% (adults); ~40% in neonates/cirrhosis

	Theophylline
Metabolism	Half-life: 3-12h (healthy adults) Pathway: Hepatic metabolism, mainly CYP1A2; low extraction ratio Metabolites: 1,3-Dimethyluric acid, 3-Methylxanthine (active), Caffeine (active; preterm neonates), 1-Methyluric acid
Elimination	Renal: Metabolites; ~10% unchanged
Mechanism	
Target	Phosphodiesterase Adenosine receptors
Action	Proposed non-selective PDE inhibition → ↓ cAMP hydrolysis → bronchial smooth-muscle relaxation; adenosine-receptor antagonism
Pregnancy	
Code	A
Pharmacodynamics	
Respiratory	Bronchodilation; ↑ diaphragmatic contractility
CVS	↑ Inotropy and chronotropy; tachyarrhythmias
CNS	CNS stimulation; insomnia, tremor and seizures
Metabolic	Hypokalaemia
Renal	Diuresis and natriuresis
Gastrointestinal	↑ Gastric secretion; nausea and vomiting

	Theophylline
Considerations	Narrow therapeutic index; target 10-20microg/mL SR preparations are unsuitable for acute asthma Smoking ↑ clearance; hepatic dysfunction, heart failure, hypoxia and acute illness ↓ clearance Numerous CYP interactions; avoid concurrent xanthines

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Leukotriene Antagonists

Selectively inhibit the cysteinyl leukotriene receptor, increased activity of which is involved in airway oedema and bronchial smooth muscle constriction.

Table 1: Montelukast

Montelukast	
Aliases	Singulair
Class	Leukotriene receptor antagonist
Chemistry	Montelukast sodium
Dose	
Maintenance (≥15y)	10mg PO daily; evening for asthma
Pharmaceutics	
Presentation	10mg film-coated tablet; 4mg or 5mg chewable tablet; 4mg granules (not supplied in Australia per PI)
Compatibility	10mg tablet is not bioequivalent to two 5mg chewable tablets
Solubility	Montelukast sodium is water soluble
Pharmacokinetics	
Absorption	64% PO bioavailability; peak 3h (fasted adults, 10mg film-coated tablet)
Distribution	Vd: 8-11L (Vss)
	Protein binding: >99%

	Montelukast
Metabolism	Half-life: 2.7 - 5.5h (healthy young adults)
	Duration: ~24h
	Pathway: Extensive hepatic CYP2C8; lesser CYP3A4 and CYP2C9
Elimination	Renal: Predominantly biliary; oral radiolabel recovery 86% in faeces over 5 days, <0.2% in urine
Mechanism	
Target	CysLT1 receptors
Action	Selective antagonism → blocks cysteinyl-leukotriene-mediated bronchoconstriction and airway inflammation
Pregnancy	
Code	B1
Pharmacodynamics	
Respiratory	Bronchodilation within 2h ↓ Airway oedema, mucus and bronchial smooth-muscle constriction
CNS	Neuropsychiatric effects, including depression and suicidality
Considerations	Not for acute bronchospasm Stop if neuropsychiatric symptoms occur Additive bronchodilation with β agonists

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Corticosteroids

Glucocorticoids are endogenous (hydrocortisone) and synthetic (prednisolone, methylprednisolone, dexamethasone) steroid hormones with **metabolic, anti-inflammatory, and immunosuppressive** effects. They bind to specific intracellular receptors and translocate into the nucleus, where they regulate gene expression in a tissue-specific manner.

Corticosteroids have multiple indications including:

- Replacement in adrenal suppression or other cortisol-deficient states
- Autoimmune disorders
- Anaphylaxis and atopic disorders, including asthma
- Hypercalcaemia
- Chemotherapy
- Immunosuppression following transplantation

Table 1: Systemic Effects of Corticosteroids

System	Effect
Resp	↑ Bronchial smooth muscle response to circulating catecholamines, ↓ airway oedema
CVS	↑ Inotropy, ↑ vascular smooth muscle response to circulating catecholamines (↑ receptor expression), ↑ BP secondary to mineralocorticoid effects
CNS	Mood changes, sleep disturbance, psychosis

System	Effect
MSK	Atrophy, thinning of skin
Renal	Glycosuria, Na ⁺ and fluid retention (mineralocorticoid effect), hypokalaemia
GIT	Gastric ulceration
Metabolic	↑ Gluconeogenesis, diabetes, ↑ protein catabolism, fat redistribution, adrenal suppression (negative feedback on ACTH), ↑ lipolytic response to circulating catecholamines
Immune	↓ Transudate production, ↓ production of inflammatory mediators, ↓ macrophage function, ↓ transport of lymphocytes, ↓ T-cell function, ↓ antibody production, ↑ susceptibility to infection,
Toxic Effects	Relative steroid deficiency in adrenal suppressed individuals with infection or surgery

Table 2: Comparison of Corticosteroids

	Hydrocortisone	Prednisolone	Methylprednisolone	Dexamethasone
Aliases	Solu-Cortef	Panafcortelone	Solu-Medrol	
Class	Short-acting glucocorticoid; mineralocorticoid	Intermediate-acting glucocorticoid	Intermediate-acting glucocorticoid	Long-acting glucocorticoid
Chemistry	Endogenous corticosteroid; sodium succinate ester for injection	Synthetic corticosteroid	Synthetic corticosteroid; sodium succinate ester for injection	Fluorinated synthetic corticosteroid
Dose				
Shock	100mg IV Q8H			2-6mg/kg IV adjunct for haemorrhagic/traumatic/surgical shock; repeat after 2-6h if persistent; usually $\leq 48-72$ h
Inflammation		Initially 20-40mg PO daily; up to 80mg/day		
ARDS			1-2mg/kg/day IV	
Life-threatening inflammation			30mg/kg IV adjunct over ≥ 30 min; repeat Q4-6H for ≤ 48 h	
Cerebral oedema				10mg IV, then 4mg IM Q6H
Airway oedema				8-16mg IV 1h before extubation
Nausea				4-8mg IV
Pharmaceutics				

	Hydrocortisone	Prednisolone	Methylprednisolone	Dexamethasone
Presentation	100mg, 250mg and 500mg powder for IV/IM injection	1mg, 5mg and 25mg tablets	40mg, 125mg, 500mg, 1g and 2g powder for IV/IM injection	4mg/mL dexamethasone phosphate solution for injection
Compatibility	Compatible with 0.9% NaCl and 5% glucose		Dilute in 0.9% NaCl or 5% glucose; do not mix directly with other drugs	Compatible with 0.9% NaCl and 5% glucose; protect from light and freezing
Solubility	Lipid soluble; water-soluble sodium succinate ester	Poorly water soluble; highly lipid soluble	Lipid soluble; water-soluble sodium succinate ester	Lipid soluble; water-soluble phosphate salt
Pharmacokinetics				
Absorption	97% PO bioavailability; rapid IM absorption	70% PO bioavailability; peak 1-2h	80% PO bioavailability	70-100% PO bioavailability; rapid IM absorption
Distribution	pKa: 12.59	pKa: 12.59	pKa: 12.61	pKa: 12.42
	Vd: 0.5L/kg	Vd: 0.64L/kg	Vd: 1.38L/kg	Vd: 0.65L/kg
	Protein binding: 90%	Protein binding: 65-91%; concentration-dependent	Protein binding: 77%	Protein binding: 65%

	Hydrocortisone	Prednisolone	Methylprednisolone	Dexamethasone
Metabolism	Half-life: ~60min; up to 2h at high doses Duration: Biological activity 8-12h Pathway: Mainly hepatic; 11β-HSD2 and CYP3A4 Metabolites: Inactive conjugates	Half-life: 2-3h Duration: Intermediate acting Pathway: Hepatic and renal conjugation; interconversion with prednisone Metabolites: Prednisone, Inactive conjugates	Half-life: 2-3h Duration: Biological activity 12-36h Pathway: Extensive hepatic metabolism Metabolites: 20α-hydroxymethylprednisolone, 20β-hydroxymethylprednisolone	Half-life: Plasma 4-5h Duration: Biological activity 36-72h Pathway: Extensive hepatic metabolism Metabolites: Inactive glucuronide and sulfate conjugates
Elimination	Renal: Minimal active drug; IV dose excretion nearly complete within 12h	Renal: Free and conjugated metabolites; appreciable unchanged drug	Renal: Inactive conjugates; ~1% unchanged	Renal: Inactive metabolites; small amount unchanged
Mechanism				
Target	Glucocorticoid receptor Mineralocorticoid receptor	Glucocorticoid receptor	Glucocorticoid receptor	Glucocorticoid receptor
Action	Receptor binding $\rightarrow$ genomic and non-genomic regulation of inflammatory, immune, metabolic and electrolyte pathways	Receptor binding $\rightarrow$ genomic and non-genomic regulation of inflammatory, immune and metabolic pathways	Receptor binding $\rightarrow$ genomic and non-genomic regulation of inflammatory, immune and metabolic pathways	Receptor binding $\rightarrow$ genomic and non-genomic regulation of inflammatory, immune and metabolic pathways

	Hydrocortisone	Prednisolone	Methylprednisolone	Dexamethasone
Contraindications	Systemic fungal infection Intrathecal or epidural administration Live vaccines with immunosuppressive dosing	Uncontrolled infection	Systemic fungal infection Cow's milk/dairy hypersensitivity (40mg ACT-O-VIAL) Intrathecal, epidural or local injection Live vaccines with immunosuppressive dosing	Systemic fungal infection; other systemic infection without specific anti-infective therapy Live vaccines with immunosuppressive dosing Myasthenia gravis Peptic ulcer, osteoporosis or psychosis
Pregnancy Code	A	A	A	A
Pharmacodynamics				
Respiratory	↓ Airway oedema; ↑ response to catecholamines	↓ Airway oedema; ↑ response to catecholamines	↓ Airway and alveolar inflammation; ↑ response to catecholamines	↓ Airway oedema; ↑ response to catecholamines
CVS	↑ Inotropy, vascular response and MAP	↑ Inotropy, vascular response and MAP	↑ Inotropy and vascular response to catecholamines	↑ Inotropy and vascular response to catecholamines
CNS	Insomnia, mood disturbance and psychosis	Insomnia, mood disturbance and psychosis	Insomnia, mood disturbance and psychosis	↓ Vasogenic cerebral oedema Insomnia, mood disturbance and psychosis
Endocrine	HPA-axis suppression and Cushingoid state	HPA-axis suppression and Cushingoid state	HPA-axis suppression and Cushingoid state	HPA-axis suppression and Cushingoid state

	Hydrocortisone	Prednisolone	Methylprednisolone	Dexamethasone
Musculoskeletal	Proximal myopathy, osteoporosis and skin thinning	Proximal myopathy, osteoporosis and skin thinning	Proximal myopathy, osteoporosis and skin thinning	Proximal myopathy, osteoporosis and skin thinning
Metabolic	↑ Gluconeogenesis, protein catabolism and lipolysis Hyperglycaemia, hypernatraemia and hypokalaemia	↑ Gluconeogenesis, protein catabolism and lipolysis Hyperglycaemia, Na ⁺ retention and hypokalaemia	↑ Gluconeogenesis, protein catabolism and lipolysis Hyperglycaemia; weak mineralocorticoid effect	↑ Gluconeogenesis, protein catabolism and lipolysis Hyperglycaemia; minimal mineralocorticoid effect
Renal	Na ⁺ and water retention; K ⁺ loss	Na ⁺ and water retention; K ⁺ loss	Na ⁺ and water retention; K ⁺ loss	
Gastrointestinal	Peptic ulceration and pancreatitis	Peptic ulceration and pancreatitis	Peptic ulceration and pancreatitis	Peptic ulceration and pancreatitis
Haematological	Neutrophilia, lymphopenia and impaired immune function	Neutrophilia, lymphopenia and impaired immune function	Neutrophilia, lymphopenia and impaired immune function	Neutrophilia, lymphopenia and impaired immune function

	Hydrocortisone	Prednisolone	Methylprednisolone	Dexamethasone
Considerations	Greater mineralocorticoid activity than prednisolone, methylprednisolone or dexamethasone Infection risk; taper after prolonged therapy to avoid adrenal crisis Hepatic impairment may ↑ effect; CYP3A4 inducers ↓ and inhibitors ↑ exposure	Avoid live vaccines during corticosteroid therapy (replacement dosing excepted) 25mg prednisolone ≈ 100mg hydrocortisone Infection risk; taper after prolonged therapy to avoid adrenal crisis Food affects initial absorption, not overall bioavailability	ARDS: start within 2wk; later initiation may ↑ mortality 4mg methylprednisolone ≈ 5mg prednisolone ≈ 20mg hydrocortisone Doses >250mg over ≥30min; rapid high-dose IV use may cause arrhythmia or cardiac arrest Infection risk; taper after prolonged therapy to avoid adrenal crisis	~25 times hydrocortisone glucocorticoid potency; minimal mineralocorticoid activity Infection risk; taper after prolonged therapy to avoid adrenal crisis CYP3A4 inducers ↓ and inhibitors ↑ exposure

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Pulmonary Vasodilators

Table 1: Pulmonary Vasodilators

	Nitric oxide	Iloprost	Sildenafil
Aliases	NO iNO	Ventavis	Viagra Revatio
Class	Pulmonary vasodilator	Pulmonary vasodilator	Pulmonary vasodilator
Chemistry	Free-radical inorganic gas (NO)	Synthetic prostacyclin analogue (eicosanoid)	Pyrazolopyrimidine
Dose			
PPHN	20ppm inhaled; ↓ to 5ppm within 4-24h; max 20ppm		
Cardiac surgery	10-20ppm inhaled; ↓ to 5ppm		
ARDS/RVF/PHTN	1-40ppm inhaled		
PAH		2.5-5µg neb 6-9 times/day	20mg PO TID
RV failure		20µg neb Q2h for ≤48h; then Q6h on days 3-4	
PAH (unable PO)			10mg IV TID
perioperative RVF			25-50mg PO/NG 3-4 times/day
Erectile dysfunction			25-100mg PO PRN 1h before; max OD
Pharmaceutics			

	Nitric oxide	Iloprost	Sildenafil
Presentation	Aluminium cylinders; 800 or 4,880ppm NO in N ₂ (IN-Omax); 100ppm also described in QMD/textbook	10µg/mL nebuliser solution; 20µg/2mL ampoule	20mg, 25mg, 50mg or 100mg tablets; 0.8mg/mL IV solution; pharmacopoeia also describes buccal spray
Compatibility	O ₂ → NO ₂ ; monitor NO ₂ <0.5ppm	Wellington Aeroneb protocol adds 3mL 0.9% NaCl; do not mix with other nebulised medicines	
Solubility	Water → HNO ₃ (pKa -1.3)		Sildenafil citrate 2.6mg/mL in water at 25°C
Pharmacokinetics			
Absorption	Rapid pulmonary and systemic absorption after inhalation	Rapid pulmonary absorption; undetectable 30-60 minutes after inhalation	41% PO bioavailability; peak 30-120 minutes; high-fat meal delays peak 60 minutes and ↓ C _{max} 29%
Distribution	pKa: -1.3 (HNO ₃)	Vd: 0.6-0.8L/kg (IV)	pKa: 8.7
	Vd: Unmeasurable	Protein binding: ~60%	Vd: 105L (~1.5L/kg)
	Protein binding: Avid haemoglobin binding		Protein binding: 96%

	Nitric oxide	Iloprost	Sildenafil
Metabolism	Half-life: <5 seconds Duration: Seconds Pathway: OxyHb → methaemoglobin + nitrate; lung water → nitrite → methaemoglobin; deoxyHb → nitrosyl-Hb Metabolites: Methaemoglobin, Nitrite, Nitrate	Half-life: 5-25 minutes (inhaled) Duration: 1-2 hours Pathway: Extensive β-oxidation of the carboxyl side chain Metabolites: Tetranor-ilo-prost (inactive)	Half-life: 3-5 hours; active metabolite ~4 hours Duration: Vasodilation maximal ~1-2 hours post-dose Pathway: Hepatic; CYP3A4 (primary) and CYP2C9 Metabolites: N-desmethyl sildenafil (active; ~50% potency of parent)
Elimination	Renal: Nitrate >70% excreted in urine; methHb metabolised back to Hb within hours by endogenous reductases	Renal: Metabolites; 68% urine and 12% faeces after IV dose	Renal: Metabolites; ~80% faeces and ~13% urine
Mechanism			
Target	Soluble guanylate cyclase (sGC)	Prostacyclin receptor (IP receptor)	Phosphodiesterase 5 (PDE5)
Action	Activates sGC → ↑ cGMP → ↓ intracellular Ca ²⁺ → vasodilation; inhalation targets ventilated alveoli and Hb limits systemic effect	IP receptor agonist → ↑ cAMP → pulmonary vasodilation and ↓ platelet aggregation	Inhibits PDE5-mediated cGMP hydrolysis → ↑ cGMP → pulmonary > systemic vasodilation

	Nitric oxide	Iloprost	Sildenafil
Contraindications	Pulmonary veno-occlusive disease (may precipitate pulmonary oedema)	High haemorrhage risk Severe CAD, unstable angina or recent MI Decompensated HF or severe arrhythmia Pulmonary veno-occlusive disease Significant valvular disease with myocardial dysfunction	Nitrates or NO donors Riociguat Potent CYP3A4 inhibitors (REVATIO) Severe hepatic impairment or BP <90/50mmHg Recent stroke/MI or hereditary retinal disease
Pregnancy Code	B2	B3	B1
Pharmacodynamics Respiratory	Inhibits HPV in ventilated alveoli → ↑ perfusion to ventilated units → ↑ V/Q matching No effect on unventilated (shunt) units	Pulmonary vasodilation Bronchospasm and cough	
CVS	↓ PVR ↓ PAP ↑ Capillary permeability Minimal systemic effect (inactivated by Hb before reaching systemic circulation)	↓ PVR ↓ PAP ↑ CO and SvO ₂ Minor ↓ SVR and MAP	↓ PVR ↓ SVR ↓ BP Reflex ↑ HR
CNS	↑ CBF (cerebral vasodilation)		Headache Visual disturbances (PDE6 inhibition in retina)
Genitourinary			Priapism

	Nitric oxide	Iloprost	Sildenafil
Haematological	Methaemoglobinaemia; neonatal MetHb >2.5% → ↓ dose Inhibition of platelet aggregation Thrombocytopenia (~10% of patients)	Inhibition of platelet aggregation	
Considerations	Abrupt cessation → rebound PHTN/hypoxaemia; wean stepwise Monitor inspired NO/NO ₂ , FiO ₂ and MetHb Sildenafil/other cGMP vasodilators → additive effect; caution Pulmonary oedema → consider PVOD/LV dysfunction	Short effect → 6-9 inhalations/day Pulmonary oedema → consider PVOD and stop Hepatic/severe renal impairment: initially Q3-4h	Inhaled NO → additive pulmonary vasodilation; use with caution

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Cardiovascular Pharmacology

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Inotropes

Understand the detailed pharmacology of inotropes and vasopressors

Inotropes are agents which alter myocardial contractility.

- Positive inotropes increase contractility
- Negative inotropes decrease contractility

Classes of Positive Inotrope

Table 1: Classification of Positive Inotropes

Classes	Class I: Increase Intracellular Calcium	Class II: Calcium Sensitisers	Class III: Metabolic/Endocrine
Examples	Adrenaline, milrinone, glucagon, digoxin	Levosimendan	T3, Insulin
General Mechanism of Action	Increase intracellular Ca^{2+} by a variety of different pathways	Increase sensitivity of actomyosin to Ca^{2+}	Variable. T3 potentiates the effect (or increases expression of) cardiac β_1 receptors

Detailed profiles of the adrenergic and non-adrenergic vasoactives are provided on the Adrenergic Vasoactives and Non-Adrenergic Vasoactives pages.

Table 2: Other Positive Inotropes

	Glucagon	Triiodothyronine
Aliases	GlucaGen	Liothyronine T3
Class	Pancreatic hormone analogue	Thyroid hormone
Chemistry	29-amino-acid peptide	Iodinated tyrosine derivative
Dose		
BB/CCB overdose (IV)	5mg; repeat once after 5min, then 2-5mg/hour if responsive	
Hypoglycaemia (IV/IM/SC)	1mg	
Myxoedema coma (IV)		25-50µg initial; subsequent doses Q4-12H; ≥65µg/day initially
Pharmaceutics		
Presentation	White 1mg powder; reconstitutes to clear colourless 1mg/mL solution	10µg/mL solution in 1mL amber vial
Compatibility	D5W, water for injection	
Solubility	pKa 7.1; poorly water-soluble at high concentrations	pKa 8.4; reluctantly water-soluble
Pharmacokinetics		
Absorption	No oral bioavailability; SC effect within ~20 minutes	97% oral bioavailability; IV metabolic response within 2-4 hours

	Glucagon	Triiodothyronine
Distribution	pKa: 7.1	pKa: 8.4
	Vd: 0.25L/kg	Vd: 1.8L/kg
	Protein binding: Minimal	Protein binding: Lower affinity than T4
Metabolism	Half-life: 20 - 30 minutes	Half-life: 6 - 10 hours
	Pathway: ~30% hepatic, ~30% renal; remainder degraded by the reticuloendothelial system	Duration: Maximum IV response within 2 days
		Pathway: Tissue deiodination; hep- atic metabolism with enterohep- atic recirculation in overdose
		Metabolites: In- active iodinated compounds
Elimination	Renal: Minimal un- changed drug	Renal: Substantially excreted
Mechanism		
Target	Glucagon receptor	Nuclear thyroid hormone recep- tors Cytoplasmic and transmembrane thyroid hormone receptors

	Glucagon	Triiodothyronine
Action	Gs/Gq agonism → ↑ cAMP → glycogenolysis, gluconeogenesis and β-receptor-in- dependent cardiac stimulation	Receptor binding → altered gene transcription and protein synthesis; non-genomic acute effects
Contraindications	Phaeochromocytoma Insulinoma Glucagonoma	Uncorrected adrenal cortical insufficiency Untreated thyrotoxicosis Artificial rewarming in myxoedema coma
Pregnancy		
Code	B3	A
Pharmacodynamics		
Respiratory		↑ Rate and depth of respiration
CVS	↑ HR ↑ Inotropy ↑ CO ↑ Myocardial O ₂ consumption	↑ HR ↑ Inotropy ↑ CO ↓ SVR
CNS		↑ Cerebration
Endocrine	↑ Blood glucose ↑ Insulin release	
Musculoskeletal		↑ Protein synthesis, blood flow and O ₂ consumption
Metabolic	↑ Glycogenolysis and gluconeogenesis ↑ β-oxidation and ketosis ↓ K ⁺ at high doses	↑ BMR and O ₂ consumption ↑ Thermogenesis
Renal		↑ RBF and clearance

	Glucagon	Triiodothyronine
Gastroin- testinal	↓ Tone and motility N/V	↑ Motility ↑ Appetite
Considera- tions	Hepatic glycogen required for hyper- glycaemic effect BB/CCB overdose use is second-line with limited clinical evidence No overdose re- sponse after 10mg → abandon treat- ment Use immediately after reconstitution	Give with gluco- corticoids for myx- oedema coma Cardiac disease → initial 10-20µg IV ↑ Catecholamine and anticoagulant effects Monitor T3 and TSH

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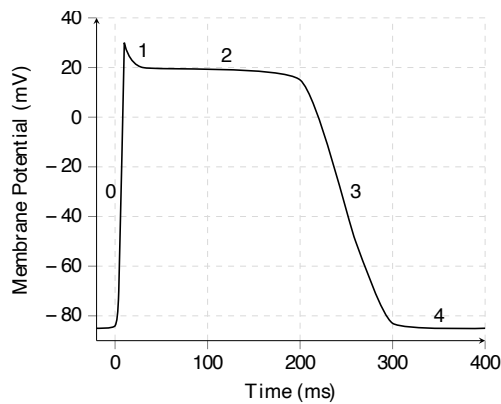
Antiarrhythmics

Understand the pharmacology of antiarrhythmic drugs

Antiarrhythmic drugs are typically classified using the **Vaughan Williams** classification system, which divides drugs into four classes based on their effect on the cardiac action potential. Many drugs will act via multiple mechanisms.

- **Class I:** Block voltage-gated Na^+ channels
 - Class Ia: Intermediate dissociation
 - Class Ib: Fast dissociation
 - Class Ic: Slow dissociation
- **Class II:** β -Blockers
- **Class III:** Prolong the action potential (Usually via K^+ channel blockade)
- **Class IV:** Ca^{2+} antagonists

This classification is notably incomplete, as some drugs (such as amiodarone) fit into multiple categories, and others (such as digoxin, adenosine, and magnesium) fit into none.



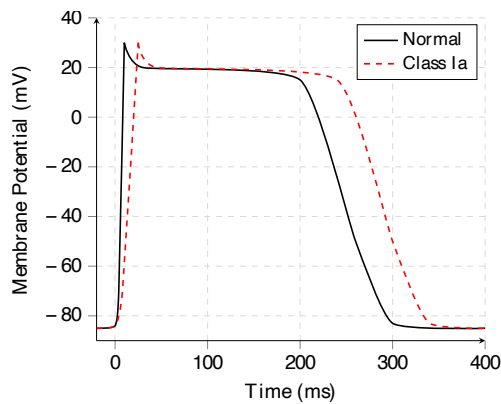
Class I

- Na^+ - channel blockade inhibits action potential prolongation by blocking **active and refractory** sodium channels in a **use-dependent fashion**
- This inhibits tachyarrhythmias whilst allowing normal conduction
- Extent of block depends on the heart rate, membrane potential, and the subclass of drug
- Sodium channel blockade **increases pacing threshold** and **defibrillation energy requirement**

Class Ia

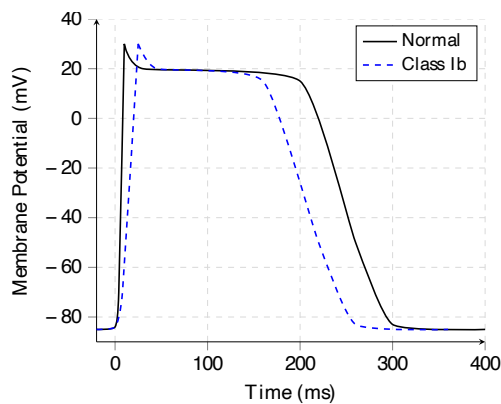
- Class Ia drugs have mixed properties of Ib and Ic, and also have Class III effects
- As they **prolong the AV conduction** and **prolong the action potential** they **increase both QRS duration and the QT interval**
- Examples include **procainamide**

Pro-arrhythmic effects may result because AV nodal conduction may be increased, so despite decreased atrial activity increased ventricular conductance results in a potentially fatal shortening of diastolic time



Class Ib

- Class Ib drugs bind to open sodium channel, and will **associate and dissociate** from a sodium channel **in the course of a normal beat**
- Tachyarrhythmias are prevented because dissociation occurs too slowly for a further action potential to be generated
- Class Ib drugs will bind selectively to refractory channels, such as occurs in ischaemia
- As they have **little effect on normal cardiac tissue** they have **little effect on the ECG**
- Examples of class Ib agents include include **phenytoin** and **lignocaine**



Class Ic

- Class Ic drugs **associate and dissociate slowly creating a steady-state level of block**
- This causes indiscriminate blockade and general reduction in excitability
- Class Ic agents are used to suppress unidirectional or intermittent conduction pathways
- As they markedly slow conduction velocity they **increase QRS duration**
- Examples of Class Ic agents include **flecainide**

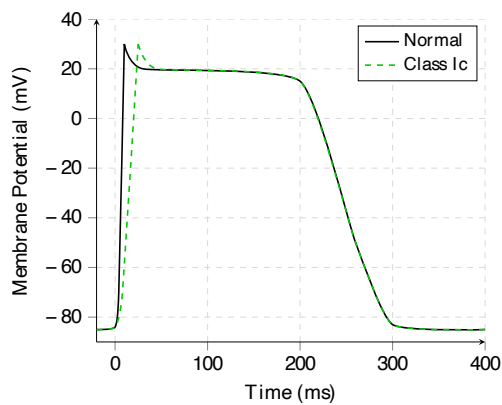


Table 1: Class I Antiarrhythmics

	Procainamide	Lignocaine	Phenytoin	Flecainide	Mexiletine
Aliases		Lidocaine Xylocard	Dilantin DBL Phenytoin Injection	Tambocor	Mexitil
Class	Class Ia antiar-rhythmic	Class Ib an-tiarrhythmic; aminoamide lo-cal anaesthetic	Anticonvulsant; class IB antiar-rhythmic	Class Ic antiar-rhythmic	Class Ib antiar-rhythmic
Chemistry	Aminobenza-mide	Aminoamide	Hydantoin	Monocarboxylic acid amide (benzamide); racemate	Lignocaine ana-logue
Dose					
Load (IV)	Up to 12mg/kg at ≤ 0.3 mg/kg/min; max 1g				
Infusion (IV)	50mcg/kg/min	2-4mg/min			
VT (IV)		1mg/kg over 1-2 minutes; max 100mg			
Status (IV)			10-15mg/kg; rate ≤ 50 mg/min		
Dysrhythmia (IV)			3-5mg/kg; re-peat if required		
SVT/VT (IV)				2mg/kg (up to 150mg) over 10-30 minutes; infusion 1.5mg/kg/hr	

	Procainamide	Lignocaine	Phenytoin	Flecainide	Mexiletine
SVT (PO)				50mg BD; max 300mg/day	
VT (PO)				100mg BD; max 400mg/day	
Ventricular arrhythmia (PO)					200-300mg Q8H; max 1.2g/day
Loading (PO)					400mg, then 200mg after 8 hours
Pharmaceutics					
Presentation	Clear 100mg/mL injection	Clear 20mg/mL injection; 100mg/mL infusion concentrate	50mg/mL clear colourless solution with propylene glycol and ethanol	50mg or 100mg tablet; clear 150mg/15mL injection	150mg, 200mg and 250mg capsules
Compatibility	5% glucose	5.5% glucose; 0.9% NaCl; Ringer's; sodium bicarbonate; dextran	Do not dilute; give into a large vein, flush with 0.9% NaCl; incompatible with glucose	5% glucose only; chloride or alkaline solutions may precipitate	
Solubility	Highly water-soluble	Water-soluble	pH 12; pKa 8.0-9.2; poor water solubility	Water-soluble	Freely water- and alcohol-soluble; pKa 9.2
Pharmacokinetics					
Absorption	75-95% oral bioavailability; Tmax 1.5-2 hours	Oral bioavailability 35%; IV only for arrhythmia	IV bioavailability 100%	90-95% oral bioavailability; Tmax ~3 hours; no first-pass effect	~90% oral bioavailability; peak 2-3 hours

	Procainamide	Lignocaine	Phenytoin	Flecainide	Mexiletine
Distribution	pKa: 9.23	pKa: 7.9	pKa: 8.0-9.2	pKa: 9.3	pKa: 9.2
	Vd: 1.5 - 2.5 L/kg	Vd: 0.9 L/kg	Vd: 1.6 - 2.5L	Vd: 8.7 L/kg	Vd: 5 - 7L/kg
	Protein binding: 15-25%	Protein binding: 70%	Protein binding: 90%	Protein binding: 40%	Protein binding: 50-60%
Metabolism	Half-life: 3-4 hours (parent drug)	Half-life: 10-20 minutes (IV bolus); 45-90 minutes (subcutaneous); 90-120 minutes (terminal)	Half-life: ~22 hours; range 7-42 hours	Half-life: 20 hours	Half-life: 10-12 hours
	Duration: 4-8 hours (including active metabolite)		Duration: IV effect up to 24 hours	Pathway: Hepatic (70%); active metabolites	Pathway: Hepatic CYP2D6 and CYP1A2
	Pathway: Hepatic acetylation (16-33%)	Pathway: Hepatic CYP1A2/3A4 (90-95%); active metabolites	Pathway: Hepatic CYP2C9 and CYP2C19 hydroxylation; saturable	Metabolites: Active metabolites	Metabolites: Inactive hydroxylated and conjugated metabolites
	Metabolites: NAPA (N-acetylprocainamide; active)	Metabolites: Monoethylglycinexylidide (active), Glycinexylidide (active)	Metabolites: HPPH (inactive)		

	Procainamide	Lignocaine	Phenytoin	Flecainide	Mexiletine
Elimination	Renal: 30-60% urine unchanged; NAPA renally eliminated	Renal: <10% excreted unchanged	Renal: Biliary metabolites are reabsorbed then excreted in urine	Renal: 30% excreted unchanged	Renal: ~10% unchanged; urinary acidification ↑ excretion
Mechanism					
Target	Nav1.5 sub-unit of voltage-gated sodium channel Potassium channel	Nav1.5 sub-unit of voltage-gated sodium channel	Voltage-gated Na ⁺ channels	Nav1.5 sub-unit of voltage-gated sodium channel	Activated and inactivated cardiac Na ⁺ channels
Action	Open Na ⁺ channel block → ↓ Phase 0/conduction + ↑ QRS; K ⁺ channel block → ↑ APD/refractory period/QT	Open/inactivated Na ⁺ channel block with rapid dissociation → ↓ Phase 0, automaticity and APD in depolarised/ischaemic ventricular tissue; minimal effect on normal tissue or QRS; ↓ QT	Stabilises the inactivated state → ↓ generation and propagation of action potentials	Slow Na ⁺ channel dissociation → use-dependent block → ↓ Phase 0 + conduction; minimal change in APD	Rapid use-dependent Na ⁺ channel blockade → ↓ phase 0 upstroke and action-potential duration in ventricular tissue

	Procainamide	Lignocaine	Phenytoin	Flecainide	Mexiletine
Contraindications	Second/complete AV block or hemiblock without pacemaker Systemic lupus erythematosus TdP	Severe SA, AV or intraventricular block Stokes-Adams syndrome Supraventricular arrhythmias	Sinus bradycardia SA block Second- or third-degree AV block Stokes-Adams syndrome	Second/third-degree AV block without pacemaker Bifascicular block without pacemaker Cardiogenic shock Post-MI asymptomatic PVCs or non-sustained VT Severe renal/hepatic impairment without plasma monitoring	Cardiogenic shock Second- or third-degree AV block without a pacemaker
Pregnancy Code Justification	B2	A	D Fetal hydantoin syndrome, neurodevelopmental disorder and fetal or neonatal bleeding	B3	B1
Pharmacodynamics Respiratory			Rapid IV administration → respiratory depression		

	Procainamide	Lignocaine	Phenytoin	Flecainide	Mexiletine
CVS	↓ SVR/BP/CO ↑ PR/QRS/QT; AV block TdP; proar- rhythmia AF/flutter → ↑ ventricular rate	AV block ↓ Inotropy/HR/ MAP; cardio- vascular col- lapse	↓ MAP Bradycardia, conduction block and car- diac arrest ↓ Ventricular automaticity and QT	↑ PR/QRS; min- imal ↑ QT; heart block ↓ Inotropy; may precipitate CCF Proarrhythmia ↑ Pacing threshold ↑ Defibrillation energy require- ment	↓ Ventricular automaticity Minimal effect on QRS and QT Proarrhythmia
CNS	Dizziness	Circumoral tin- gling; paraes- thesia Dizziness; visual disturbance; confusion Seizures; coma	Anticonvulsant Ataxia, nys- tagmus and dysarthria	Dizziness Visual distur- bance Paraesthesia Headache	Tremor Dizziness Ataxia Visual distur- bance
Metabolic			Chronic use → ↓ vitamin D		
Gastrointestinal	Nausea; diar- rhoea		Hepatotoxicity Gingival hyper- plasia	Nausea	N/V Heartburn
Haematological	Lupoid syn- drome (20- 30%; long- term) Agranulocytosis/ bone marrow suppression (~0.5%)		Folate de- ficiency and blood dyscrasias		

	Procainamide	Lignocaine	Phenytoin	Flecainide	Mexiletine
Considerations	Slow acetylators → earlier lupoid syndrome Renal impairment → procainamide/NAPA accumulation ↑ Neuromuscular blockade Reduces antimicrobial effect of sulfonamides	Shock, CCF, hepatic disease or prolonged infusion → ↓ clearance Propranolol, metoprolol or cimetidine → ↓ clearance	Therapeutic total concentration 10-20mcg/mL; free 0.8-2mcg/mL Hypoalbuminaemia or uraemia → measure or interpret free concentration Continuous ECG and BP monitoring during IV loading High pH and propylene glycol → purple-glove syndrome and tissue necrosis Potent CYP inducer → many interactions SJS, TEN and DRESS	↑ Mortality in heart failure patients Amiodarone → ↑ flecainide; halve flecainide dose Renal impairment or alkaline urine → ↓ elimination	Give with food or antacid Therapeutic plasma concentration 0.5-2µg/mL Severe hepatic impairment or right heart failure → ↑ half-life Initiate in hospital for life-threatening ventricular arrhythmia

Class II

Normal β -adrenergic stimulation has a number of pro-arrhythmic effects:

- Increased pacemaker potential current
- Increased slow-inward Ca^{2+} current
- Increased repolarising K^+ and Cl^- currents
- Increased Ca^{2+} stored in the sarcoplasmic reticulum, which may be spontaneously released causing a delayed-after-depolarisations
- Reduced serum $[\text{K}^+]$

β -blockers have an antiarrhythmic effect by antagonising these mechanisms. They are useful for treatment of arrhythmias occurring with sympathetic over-activation, such as post MI.

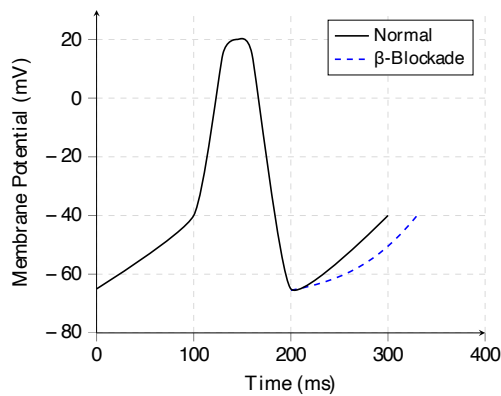


Table 2: Class II Antiarrhythmics

	Esmolol	Metoprolol	Atenolol	Propranolol
Aliases	Esmolol Noridem	Metoprolol IV Viatris	Tenormin; Noten	Inderal; Cardinol
Class	Ultra-short β_1 -selective blocker	β_1 -selective blocker	β_1 -selective blocker	Non-selective β blocker
Chemistry	Aryloxy-propanolamine	Aryloxy-propanolamine	Aryloxy-propanolamine	Aryloxy-propanolamine; racemate
Dose				
SVT (IV)	500 μ g/kg over 1 minute $\rightarrow$ 50-200 μ g/kg/min			
Arrhythmia (IV)		1-5mg at 1-2mg/min; repeat q5min; usual total 10-15mg	2.5mg boluses; max 10mg	1mg boluses at <1mg/min; usual total 1-3mg
Arrhythmia/HTN (PO, IR)		12.5-100mg BD	50-100mg OD	
HTN/angina (PO)				40mg BD; usual 120-320mg/day
HTN (PO)				10-40mg TDS-QID
Arrhythmia/thyrotoxicosis (PO)				
Paediatric arrhythmia (PO)				0.25-0.5mg/kg TDS-QID
Pharmaceutics				
Presentation	Clear 100mg/10mL injection (10mg/mL)	50mg or 100mg IR tablet; 23.75mg, 47.5mg, 95mg or 190mg CR tablet; clear 5mg/5mL injection	25mg, 50mg or 100mg tablet; syrup; clear injection	10mg or 40mg tablet; 160mg LA capsule; clear 1mg/mL injection

	Esmolol	Metoprolol	Atenolol	Propranolol
Compatibility	0.45% or 0.9% NaCl; 5% glucose; Hartmann's	0.9% NaCl; 5% or 10% glucose; glucose/saline		
Solubility	Minimal lipid solubility	Poor lipid solubility	Minimal lipid solubility	Highly lipid-soluble
Pharmacokinetics				
Absorption	IV only	50% oral bioavailability	45-55% oral bioavailability	30% oral bioavailability
Distribution	pKa: 9.5	pKa: 9.7	pKa: 9.6	pKa: 9.45
	Vd: 3.4 L/kg	Vd: 2.8-4.8 L/kg	Vd: 0.3-0.6 L/kg	Vd: 4 L/kg
	Protein binding: 60%	Protein binding: 12%	Protein binding: 5%	Protein binding: 95%
Metabolism	Half-life: 9 minutes	Half-life: 3-4 hours	Half-life: 6-9 hours	Half-life: 3-6 hours
	Pathway: Red-cell esterase hydrolysis → inactive acid metabolite + methanol	Pathway: Hepatic CYP2D6 metabolism	Pathway: Minimal hepatic metabolism	Pathway: Extensive first-pass hepatic metabolism
	Metabolites: Inactive acid metabolite, Methyl alcohol	Metabolites: 2-hydroxymetoprolol (active; minor contribution)		Metabolites: 4-hydroxypropranolol (active; oral only), Inactive metabolites
Elimination	Renal: <2% unchanged; 73-88% as acid metabolite in urine	Renal: ~3% unchanged; remainder as metabolites in urine	Renal: Excreted unchanged in urine	Renal: 95-100% as metabolites/conjugates in urine

	Esmolol	Metoprolol	Atenolol	Propranolol
Mechanism				
Target	$\beta 1$ adrenoreceptor	$\beta 1$ adrenoreceptor	$\beta 1$ adrenoreceptor	$\beta 1$ adrenoreceptor $\beta 2$ adrenoreceptor
Action	Competitive $\beta 1$ antagonism $\rightarrow$ $\downarrow$ Gs-mediated cAMP	Competitive $\beta 1$ antagonism $\rightarrow$ $\downarrow$ Gs-mediated cAMP	Competitive $\beta 1$ antagonism $\rightarrow$ $\downarrow$ Gs-mediated cAMP	Competitive $\beta 1/\beta 2$ antagonism $\rightarrow$ $\downarrow$ Gs-mediated cAMP; membrane-stabilising activity
Contraindications	Severe bradycardia or sick sinus Second/third-degree AV block Cardiogenic shock or overt CCF Severe hypotension Acute asthma Untreated phaeochromocytoma Inotrope/vasopressor dependence	Bronchospasm Sinus bradycardia or sick sinus Second/third-degree AV block Shock or uncontrolled CCF Hypotension Severe PVD Untreated phaeochromocytoma	Bronchospasm Sinus bradycardia or sick sinus Second/third-degree AV block Shock or uncontrolled CCF Severe PVD Metabolic acidosis	Asthma, bronchospasm or airway obstruction Sinus bradycardia or sick sinus Second/third-degree AV block Shock or CCF Hypotension Severe PVD Prolonged fasting/hypoglycaemia Untreated phaeochromocytoma
Pregnancy				
Code	C	C	C	C
Justification	Fetal/neonatal bradycardia	Fetal/neonatal bradycardia	Fetal/neonatal bradycardia	Fetal/neonatal bradycardia
Pharmacodynamics				
Respiratory				Bronchospasm

	Esmolol	Metoprolol	Atenolol	Propranolol
CVS	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP ↑ SVR (β ₂ effect) Bradycardia/AV block
CNS				Tiredness Nightmares Sleep disturbance
Endocrine				↓ T ₄ → T ₃ conversion
Metabolic	↓ Insulin release (β ₂ effect if selectivity lost) Masked hypoglycaemia	↓ Insulin release (β ₂ effect if selectivity lost) Masked hypoglycaemia	↓ Insulin release (β ₂ effect if selectivity lost) Masked hypoglycaemia	↓ Insulin release Blunted hypoglycaemic response
Renal	↓ Renin Venous irritant Effects resolve within ~30 minutes of cessation IV verapamil → cardiac arrest Incompatible with NaHCO ₃ , furosemide, diazepam or thiopental	↓ Renin CYP2D6 polymorphism → variable half-life β ₁ selectivity ↓ at higher doses Verapamil/diltiazem → severe bradycardia/AV block/↓ inotropy Abrupt withdrawal → angina, arrhythmia or MI	↓ Renin Renal impairment → dose reduction β ₁ selectivity ↓ at higher doses Verapamil/diltiazem → severe bradycardia/AV block/↓ inotropy Abrupt withdrawal → angina, arrhythmia or MI	Verapamil/diltiazem → severe bradycardia/AV block/↓ inotropy Abrupt withdrawal → angina, arrhythmia or MI
Considerations				

Class III

Blocking of outward K^+ channels slows cardiac repolarisation, which increases the cardiac refractory period. This has a number of beneficial effects:

- Decreased automaticity
- Decreased ectopy
- Reduced defibrillation energy requirement
- Increased inotropy

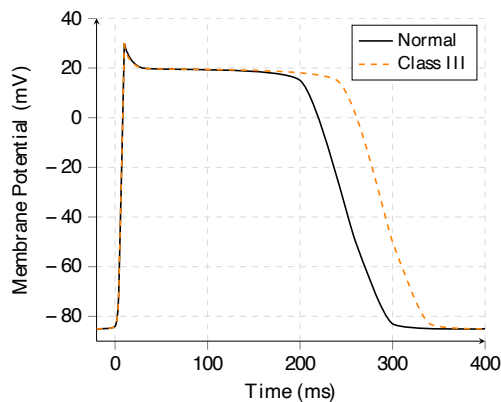


Table 3: Class III Antiarrhythmics

	Amiodarone	Sotalol
Aliases	Cordarone	Sotacor
Class	Class III antiar-rhythmic	Class II/III antiar-rhythmic
Chemistry	Iodinated benzofu-ran	Ethanolamine derivative
Dose		
Cardiac ar-rest (IV)	5mg/kg up to 300mg IV	
Arrhythmia (IV)	5mg/kg over 20 minutes - 2 hours; then 15mg/kg over 24 hours	0.5 - 1.5mg/kg over 10 minutes; max 320mg/day

	Amiodarone	Sotalol
Maintenance (PO)	200mg TDS for 1 week; then 200mg BD for 1 week; then 200mg OD	
Arrhythmia (PO)		40-160mg BD
Pharmaceutics		
Presentation	100mg or 200mg tablet; clear pale-yellow 150mg/3mL injection	80mg or 160mg tablet; clear 10mg/mL injection
Compatibility	5% glucose only; do not mix with other drugs	0.9% NaCl; 5% glucose
Solubility	Highly lipid-soluble; poorly water-soluble	Highly water-soluble
Pharmacokinetics		
Absorption	20-80% oral bioavailability; slow and erratic; Tmax ~4.5 hours	~100% oral bioavailability; Tmax 2-3 hours; ↓ with food or milk
Distribution	pKa: 6.56	pKa: 8.3
	Vd: 66 L/kg (adipose tissue and muscle accumulation)	Vd: 1.6-2.4 L/kg
	Protein binding: 96%	Protein binding: None

	Amiodarone	Sotalol
Metabolism	Half-life: 29 days (mean) Duration: Delayed by redistribution; maximum Class I/IV effects may take weeks Pathway: Hepatic CYP3A4 → desethylamiodarone Metabolites: Desethylamiodarone (active)	Half-life: 12.7 hours Pathway: Not metabolised
Elimination	Renal: <5% renal; predominantly biliary, skin and lacrimal; not dialysable	Renal: 75% urine unchanged within 72 hours; <10% faeces
Mechanism		
Target	IKr potassium channel Inactivated Na⁺ channel L-type Ca²⁺ channel β and α adrenoreceptors	β adrenoreceptor (non-selective; no intrinsic sympathomimetic activity) Potassium channel (repolarising K⁺ current, Phase 3)

	Amiodarone	Sotalol
Action	IKr block → ↑ APD/QT; Na ⁺ block → ↓ Phase 0; non-competitive β block + L-type Ca ²⁺ block → ↓ SA/AV conduction; α-blocker-like activity → ↓ SVR	D/L isomers → K ⁺ block → ↑ APD/QT; L-isomer → non-selective β block → ↓ AV conduction; no membrane-stabilising activity
Contraindications	Porphyria Sinus bradycardia or SA block Sick sinus or advanced AV block without pacemaker Thyroid dysfunction TdP-provoking drugs Severe ↓ BP or circulatory collapse (IV) Neonates (benzyl alcohol)	Bronchospasm, asthma or COPD Sinus bradycardia or sick sinus Second/third-degree AV block without pacemaker Shock or uncontrolled CCF CrCl <10mL/min Long QT, TdP or hypomagnesaemia
Pregnancy		
Code	C	C
Justification	Long t _{1/2} ; fetal thyroid dysfunction/bradycardia; check neonatal TFTs after exposure	Fetal/neonatal bradycardia
Pharmacodynamics		
Respiratory	Pneumonitis (10% 3-year risk) Pulmonary fibrosis Pleuritis	Bronchospasm

	Amiodarone	Sotalol
CVS	↓ HR/BP/SVR ↑ QT; TdP uncommon Peripheral venous irritation Rapid IV → ↓ BP (polysorbate 80)	↓ HR/BP/inotropy; AV block ↑ QT TdP; ↑ with high dose, long QT, electrolyte disturbance or renal impairment
CNS	Corneal microdeposits → blurred vision Sleep disturbance; vivid dreams Peripheral neuropathy	Masking of hypoglycaemia symptoms
Endocrine	Hyperthyroidism (1%) Hypothyroidism (6%)	
Musculoskeletal	Photosensitivity Grey skin discolouration	
Genitourinary		Sexual dysfunction
Gastrointestinal	Nausea; vomiting Hepatitis; cirrhosis; jaundice	
Considerations	CYP3A4/2C9/2D6 + P-gp inhibition → ↑ digoxin, statin, warfarin, phenytoin and antiarrhythmic exposure Also exhibits Class I, II, and IV activity	Renal impairment → accumulation + ↑ TdP risk Avoid Class Ia/III antiarrhythmics Abrupt withdrawal → angina, arrhythmia or MI

Due to the prolonged repolarisation, they will also cause a **long QT** (though in the case of amiodarone this is not

associated with an increased risk of TPD).

Class IV

Class IV drugs **inhibit L-type Ca^{2+} channels**, inhibiting the slow inward calcium current, which:

- Slows SA and AV nodal conduction
AV blockade slows transmission of supra-ventricular arrhythmias.
- Reduces inotropy
- Prevents after-depolarisations
This suppresses ectopy by reducing calcium leak from sarcoplasmic reticulum.

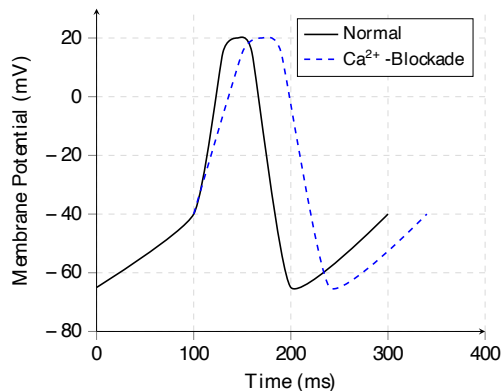


Table 4: Class IV Antiarrhythmics

	Verapamil	Diltiazem
Aliases	Isoptin	Cardizem; Dilzem
Class	Non-dihydropyridine calcium channel blocker	Non-dihydropyridine calcium channel blocker
Chemistry	Phenylalkylamine; racemate (D-isomer has local anaesthetic activity)	Benzothiazepine
Dose		

	Verapamil	Diltiazem
SVT (IV)	75-150mcg/kg (usually 5mg) over ≥2 minutes; repeat after 5-10 minutes	
Paediatric HTN crisis (IV)	0.05-0.1mg/kg/hr; usual max 1.5mg/ kg/day	
HTN/angina (PO)	40-120mg TDS	30-120mg TDS; max 360mg/day
Rate control (SR)	120-240mg BD	120-360mg OD
Rate control (IV)		0.25mg/kg over 2 minutes; repeat 0.35mg/kg after 15 minutes; infusion 5-15mg/hr
Pharmaceutics		
Presentation	5mg/2mL injection; tablet; oral solution	5mg/mL injec- tion; 30mg, 60mg, 120mg or 180mg tablet; SR formula- tions
Solubility	Excellent lipid solu- bility	Excellent lipid solu- bility
Pharmacokinetics		
Absorption	24% oral bioavail- ability	39% oral bioavail- ability
Distribution	pKa: 8.73	pKa: 7.5
	Vd: 3.8L/kg	Vd: 5.3L/kg
	Protein binding: 84-91%	Protein binding: 80-86%

	Verapamil	Diltiazem
Metabolism	Half-life: 4.5 - 12 hours Duration: Exceeds plasma half-life Pathway: Hepatic; CYP3A4 substrate and inhibitor Metabolites: Norverapamil (active; ~20% potency)	Half-life: 2 - 5 hours Duration: Exceeds plasma half-life Pathway: Hepatic; CYP3A4 substrate and inhibitor Metabolites: De-sacetyldiltiazem (active; 25 - 50% potency)
Elimination	Renal: Metabolites	Renal: Metabolites
Mechanism		
Target	Cardiac and vascular L-type Ca^{2+} channel	Cardiac and vascular L-type Ca^{2+} channel
Action	State-dependent block $\rightarrow$ $\downarrow$ AV conduction, contractility and arterial/coronary tone	State-dependent block $\rightarrow$ $\downarrow$ AV conduction and arterial/coronary vasodilation

	Verapamil	Diltiazem
Contraindications	Cardiogenic shock or decompensated heart failure Sick sinus syndrome; 2nd or 3rd degree AV block without pacemaker Severe ↓ MAP AF/flutter with an accessory pathway Ventricular tachycardia Concurrent IV beta blocker outside ICU	Sick sinus syndrome; 2nd or 3rd degree AV block without pacemaker Severe bradycardia Severe ↓ MAP or cardiogenic shock Severe heart failure Acute MI with pulmonary congestion AF/flutter with an accessory pathway Ventricular tachycardia IV beta blocker; IV dantrolene; ivabradine
Pregnancy		
Code	C	C
Justification	Maternal ↓ MAP → fetal hypoxia	Maternal ↓ MAP → fetal hypoxia
Pharmacodynamics		
CVS	↓ SVR ↓ MAP ↓ Afterload ↓ HR ↓ AV conduction ↓ Inotropy Coronary vasodilation	↓ SVR ↓ MAP ↓ Afterload ↓ HR ↓ AV conduction Mild ↓ Inotropy Coronary vasodilation
CNS	Headache	Headache
Endocrine	Gynaecomastia	
Musculoskeletal	Peripheral oedema Flushing	Peripheral oedema Flushing
Gastrointestinal	Constipation	Constipation ↓ Lower oesophageal sphincter tone

	Verapamil	Diltiazem
Considerations	Beta blockers → bradycardia/AV block ↑ Digoxin concentration Prolonged neuromuscular blockade	Beta blockers or digoxin → bradycardia/AV block Do not crush SR formulations

Alternatives to Vaughan Williams

As the Vaughan Williams classification system does not neatly divide agents, and some agents do not fit into any category, they may also be classified by their uses:

Table 5: Alternative Antiarrhythmic Selection by Indication

Indication	Examples
SVT	Digoxin, adenosine, verapamil, β -blockers
VT	Lignocaine, mexiletine
SVT/VT	Amiodarone, flecainide
Digoxin toxicity	procainamide, sotalol
	Phenytoin

Table 6: Non-Vaughan Williams Antiarrhythmics

	Digoxin	Adenosine	Magnesium sulfate
Aliases	Lanoxin	Adenocor	
Class	Cardiac glycoside	Antiarrhythmic	Electrolyte replacement
Chemistry	Cardiac glycoside	Endogenous purine nucleoside	Divalent cation salt
Dose			
Maintenance (PO)	62.5-250mcg OD		
Load (PO)	10-20mcg/kg		
Load (IV)	0.5-1mg total; 0.25-0.5mg Q4-6H, each over ≥ 5 minutes		
SVT (IV)		3mg, then 6mg, then 12mg; rapid bolus	
Hypomagnesaemia (IV)			10-20mmol over 10-20 minutes
Pharmaceutics			
Presentation	62.5mcg or 250mcg tablet; 50mcg/mL elixir; clear 250mcg/mL injection	Clear colourless 3mg/mL solution	Clear 2mmol/mL solution; dilute 10mmol in 100mL for peripheral IV
Compatibility	5% glucose; 0.9% NaCl; 4% glucose/0.18% NaCl		0.9% NaCl; Hartmann's; 5% glucose; glucose/NaCl
Solubility	Essentially insoluble in water	Slightly water-soluble	Water-soluble
Pharmacokinetics			
Absorption	70-80% oral bioavailability; P-glycoprotein efflux; onset 2-3 hours		30% oral bioavailability; $\uparrow$ with magnesium depletion

	Digoxin	Adenosine	Magnesium sulfate
Distribution	pKa: 7.15 Vd: 5-11 L/kg (dependent on lean body mass) Protein binding: 25%		pKa: -3 Protein binding: 30%
Metabolism	Half-life: 35-44 hours Pathway: Minimal hepatic (~16% of clearance) Metabolites: Dihydrodigoxin, Digoxigenin	Half-life: <10 seconds Pathway: Cellular uptake → inosine and AMP Metabolites: Inosine, AMP	Half-life: Approximately 4 hours Pathway: Not metabolised; cofactor in numerous metabolic processes
Elimination	Renal: 50-70% of IV dose excreted unchanged; half-life prolonged in renal failure		Renal: Renal; reabsorption mainly in TAL/DCT; ↑ plasma Mg ²⁺ → ↓ reabsorption
Mechanism			
Target	Na ⁺ /K ⁺ ATPase	A1 adenosine receptor	NMDA receptor L-type calcium channel
Action	Na ⁺ /K ⁺ ATPase inhibition → ↑ intracellular Na ⁺ → ↓ Na ⁺ /Ca ²⁺ exchange → ↑ Ca ²⁺ and inotropy; ↑ vagal tone → ↓ AV conduction; ↑ Phase 4 slope and automaticity	A1 activation → ↑ K ⁺ efflux + ↓ Ca ²⁺ current → SA/AV hyperpolarisation and transient AV block	↑ Sarcoplasmic Ca ²⁺ uptake → smooth muscle relaxation; ↓ acetylcholine release at NMJ; NMDA + L-type Ca ²⁺ channel inhibition → CNS depression and ↓ AV conduction; Na ⁺ /K ⁺ ATPase cofactor

	Digoxin	Adenosine	Magnesium sulfate
Contraindications	2nd/3rd-degree AV block VT/VF Digoxin-induced arrhythmia AF with accessory pathway Hypertrophic obstructive cardiomyopathy	Sick sinus syndrome 2nd/3rd-degree AV block without pacemaker Asthma or COPD Long QT syndrome Severe ↓ MAP or decompensated CCF	Heart block Severe renal failure
Pregnancy Code	A	B2	
Pharmacodynamics Respiratory		Bronchospasm Dyspnoea; ↑ RR/VT	Bronchodilation Respiratory depression
CVS	↓ HR; ↑ inotropy/CO/ coronary flow ↑ PR; ↓ QT Bigeminy; PVCs; AV block; SVT; VT Arterial vasoconstriction	Transient AV block/asystole AF/flutter; ventricular ectopy ↓ MAP	↓ SVR ↓ MAP ↓ HR
CNS	Red-green dyschromatopsia; visual disturbance Headache; ↓ LOC	Headache; dizziness	CNS depression Anticonvulsant
Musculoskeletal			Muscle weakness Areflexia
Metabolic Genitourinary	Gynaecomastia		↓ Uterine tone and contractility
Gastrointestinal	Nausea Anorexia		
Haematological	Eosinophilia Rash		

	Digoxin	Adenosine	Magnesium sulfate
Considerations	Narrow therapeutic index Renal impairment → ↓ elimination Hypokalaemia/ hypomagnesaemia/ hypercalcaemia → ↑ toxicity Toxicity + DC cardioversion → ↑ VF risk ↑ Levels: amiodarone, cap- topiril, erythromycin, vera- pamil ↓ Levels: antacids, cholestyramine, phenytoin, metoclopramide	Dipyridamole → ↑ effect Methylxanthines → ↓ effect Carbamazepine → ↑ AV block	Potentiates neuromuscular blockers and CNS depres- sants Nifedipine → exaggerated ↓ MAP

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Adrenergic Vasoactives

Adrenergic drugs:

- Act via:
 - Dopamine receptors (D)
 - Adrenoreceptors (α and β)
- Can be:
 - **Direct-acting**
Stimulate the receptor.
 - **Indirect-acting**
Stimulate the release of noradrenaline to cause effects.
- Classified as either:
 - **Naturally-occurring catecholamines**
 - **Synthetic catecholamines**
 - **Synthetic sympathomimetics**
Drugs which act on adrenoreceptors but are not classified as catecholamines due to their chemical structure.

This covers the pharmacology of specific catecholamines and sympathomimetics. The synthesis of endogenous catecholamines is covered under Chapter , whilst specifics of catecholamine receptor function is covered under Chapter .

Table 1: Naturally Occurring Adrenergic Vasoactives

	Adrenaline	Noradrenaline	Dopamine
Aliases	Epinephrine	Norepinephrine	
Class	Inotrope	Adrenergic vasopressor	Inotrope
Chemistry	Endogenous catecholamine	Endogenous catecholamine	Endogenous catecholamine
Dose			
Arrest	1mg IV Q3-5 min		
Shock	0.05 - 2µg/kg/min IV	0.05 - 1.5µg/kg/min IV	2 - 50µg/kg/min IV
Anaphylaxis	No IV access: 500µg IM Q5 min Grade II: 20µg IV Grade III: 100-200µg IV		
Airway oedema	5mg neb		
Push dose	5-10µg		
Pharmaceutics			
Presentation	Clear colourless 1mg/mL or 0.1mg/mL solution; light-sensitive; sodium metabisulfite	Clear colourless/yellow 1mg/mL concentrate; light-sensitive	Clear colourless/pale-yellow 40mg/mL concentrate; light-sensitive; potassium metabisulfite
Solubility	Minimal water solubility	Water-soluble	Freely water-soluble
Compatibility		Dilute in a glucose-containing solution; not saline alone	
Pharmacokinetics			
Absorption	No PO bioavailability due to destruction by COMT and MAO in the gut brush border Variable ETT and SC absorption	No PO bioavailability due to destruction by COMT and MAO in the gut brush border	No PO bioavailability

	Adrenaline	Noradrenaline	Dopamine
Distribution	pKa: 9.69 Vd: 0.1 - 0.2L/kg Protein binding: 12%	pKa: 8.85 Vd: 0.12L/kg Protein binding: 25%	Protein binding: Unknown
Metabolism	Half-life: 2 minutes Duration: Very short; rapid onset Pathway: Rapid COMT and MAO metabolism in liver, tissues and sympathetic nerves Metabolites: VMA, Meta-drenaline	Half-life: 2 minutes Duration: Very short; rapid onset Pathway: Rapid COMT and MAO metabolism in liver, tissues and sympathetic nerves Metabolites: VMA, Normetadrenaline	Half-life: 2 minutes Duration: Onset 5 minutes; duration <10 minutes Pathway: 25% of dose converted to noradrenaline, Remainder metabolised by MAO and COMT
Elimination	Renal: Metabolites	Renal: Metabolites Other: Pulmonary uptake of up to 25%	Renal: 97% as metabolites
Mechanism Target	α 1 adrenoreceptor α 2 adrenoreceptor β 1 adrenoreceptor β 2 adrenoreceptor	α 1 adrenoreceptor α 2 adrenoreceptor β 1 adrenoreceptor	D1 receptor β 1 adrenoreceptor α 1 adrenoreceptor

	Adrenaline	Noradrenaline	Dopamine
Action	$\beta > \alpha$ at low doses; $\alpha 1$ dominates at high doses	$\alpha 1/\alpha 2 \gg \beta 1$; negligible $\beta 2$ activity	D1 at 0.5 - 2 μ g/kg/min; $\beta 1$ at 2 - 10 μ g/kg/min; α at >10 μ g/kg/min
Pregnancy Code	A		B3
Pharmacodynamics			
Respiratory	↑ MV - Bronchodilation		
CVS	↑ Inotropy ↑ HR ↑ SVR and PVR ↑ BP ↑ CO ↑ Myocardial O₂ consumption - Coronary vasodilation - Arrhythmogenic	↑ SVR ↑ Preload ↑ MAP ↑ Myocardial O₂ consumption ↑ Coronary flow - Reflex ↓ HR - CO ↔/↓	↑ Inotropy ↑ HR ↑ CO - Coronary vasodilation ↑ SVR and PVR ↑ VR - Tachyarrhythmias
CNS	↑ Pain threshold ↑ MAC		- Inhibits prolactin secretion - Nausea
Musculoskeletal	Necrosis with extravasation	Necrosis with extravasation	Necrosis with extravasation
Metabolic	↑ BMR ↑ Lipolysis ↑ Gluconeogenesis and BSL ↑ Lactate - Initially ↑ insulin secretion (β), then ↓ (α)	Acidosis → ↓ effect	
Renal	↓ RBF	↓ RBF	↑ RBF and UO at low doses, not renoprotective
Genitourinary	↑ Sphincter tone	↓ Uterine blood flow - Foetal bradycardia	
Gastrointestinal		↓ Splanchnic blood flow	Mesenteric vasodilation

	Adrenaline	Noradrenaline	Dopamine
Considerations	Correct hypovolaemia, hypoxia, hypercapnia and acidosis TCA/MAOI → ↑ effect May worsen dynamic LVOT obstruction	MAOI/TCA → severe prolonged hypertension Extravasation → phentolamine infiltration Avoid leg veins in elderly/PVD	Correct hypovolaemia, hypoxia, hypercapnia and acidosis MAOI → start ≤1/10 usual dose Extravasation → phentolamine infiltration IV phenytoin → ↓ MAP/bradycardia
Contraindications		Uncorrected hypovolaemia Mesenteric or peripheral vascular thrombosis	Phaeochromocytoma Atrial/ventricular tachyarrhythmia or VF Dynamic LVOT obstruction Hyperthyroidism Cyclopropane or halogenated anaesthesia Ergotamine

Table 2: Synthetic Adrenergic Vasoactives

	Dobutamine	Isoprenaline	Ephedrine	Metaraminol	Phenylephrine
Aliases		Isoproterenol Isoprel		Aramine	Neo-Synephrine
Class	Inodilator	Non-selective β agonist	Direct and indirect sympathomimetic	Adrenergic vasopressor	Direct α_1 agonist
Chemistry	Synthetic catecholamine	Synthetic catecholamine	Non-catechol phenylisopropylamine	Phenylethylamine; structural analogue of noradrenaline	Non-catecholamine sympathomimetic
Dose					
Shock	2.5 - 20 μ g/kg/min IV				
Bradycardia (IV)		0.5 - 20 μ g/min			
Hypotension (IV)			3 - 6mg Q3-4min up to 30mg total		
Bolus				0.5 - 2mg IV	
Infusion				0.5 - 5mg/hr IV	
Bolus (IV)					50 - 100 μ g
Infusion (IV)					40 - 60 μ g/min maintenance
Pharmaceutics					
Presentation	Clear 12.5mg/mL concentrate; sodium metabisulfite	Clear colourless/slightly yellow 0.2mg/mL solution	Clear colourless 3mg/mL solution	Clear colourless/slightly yellow 10mg/mL solution; sodium metabisulfite	Clear colourless 10mg/mL solution; sodium metabisulfite

	Dobutamine	Isoprenaline	Ephedrine	Metaraminol	Phenylephrine
Compatibility	D5W or 0.9% NaCl; avoid alkaline solutions	NS, D5W, Hartmann's	0.9% NaCl, Hartmann's	0.9% NaCl, D5W, Hartmann's	NS or D5W
Solubility	Sparingly water-soluble			Freely water-soluble	
Pharmacokinetics					
Absorption	No PO bioavailability due to gut COMT and MAO	Rapid parenteral absorption	Complete IV bioavailability; onset 3-5 minutes	Oral absorption feasible but not used clinically. Good IM and SC absorption.	
Distribution	pKa: 10.14 Vd: 0.2L/kg Protein binding: Unknown; presumed minimal	pKa: 8.6	pKa: 9.6	pKa: 8.79 Vd: 4L/kg Protein binding: 45%	pKa: 9.8

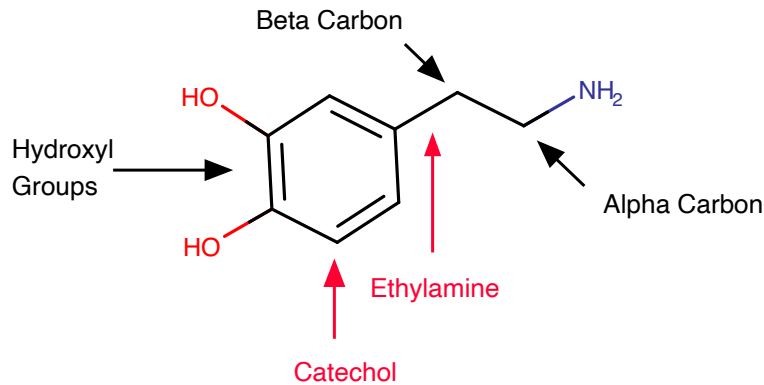
	Dobutamine	Isoprenaline	Ephedrine	Metaraminol	Phenylephrine
Metabolism	Half-life: ~2 minutes Duration: Very short; rapid onset Pathway: Metabolised by COMT and MAO Metabolites: 3-O-methyl-dobutamine (inactive)	Half-life: Few minutes IV; up to 2 hours SC Duration: Brief Pathway: Hepatic COMT; poor MAO substrate Metabolites: 3-O-methylisoprenaline (weak β antagonist)	Half-life: 3-7 hours; urine pH-dependent Duration: 10-15 minutes IV Pathway: Not metabolised by MAO/COMT; partial N-demethylation Metabolites: Norephedrine (active)	Half-life: Distribution half-life minutes; elimination over hours to days Duration: 20-60 minutes per dose Pathway: Not metabolised by MAO or COMT; persists in catecholamine storage vesicles	Duration: 20 minutes IV; up to 50 minutes SC Pathway: Hepatic MAO; not COMT
Elimination	Renal: Metabolites	Renal: 50% unchanged + metabolites	Renal: ~90% in 24h; mainly unchanged; $\uparrow$ with acidic urine	Renal: Slowly excreted unchanged over days	
Mechanism					

	Dobutamine	Isoprenaline	Ephedrine	Metaraminol	Phenylephrine
Target	$\beta 1$ adrenoreceptor (full agonist) $\beta 2$ adrenoreceptor (weak agonist) $\alpha 1$ adrenoreceptor (net partial agonist)	$\beta 1$ adrenoreceptor $\beta 2$ adrenoreceptor	α adrenoreceptors β adrenoreceptors	$\alpha 1$ adrenoreceptor $\beta 1$ adrenoreceptor	$\alpha 1$ adrenoreceptor
Action	$\beta 1 \rightarrow \uparrow \text{cAMP} \rightarrow \uparrow \text{intracellular } \text{Ca}^{2+} \rightarrow \uparrow \text{inotropy}$	$\beta 1 = \beta 2 \gg \alpha$; $\text{Gs} \rightarrow \uparrow \text{cAMP}$	Direct α/β agonism + $\uparrow$ noradrenaline release; repeated dosing $\rightarrow$ noradrenaline depletion/tachyphylaxis	Direct $\alpha 1/\beta 1$ agonist False transmitter; displaces stored noradrenaline $\rightarrow \uparrow$ release	Direct $\alpha 1$ agonism $\rightarrow \text{Gq/IP3/Ca}^{2+} \rightarrow$ vasoconstriction
Contraindications	Dynamic LVOT obstruction	Tachyarrhythmia Digitalis-induced tachycardia/heart block Ventricular dysrhythmia requiring inotropy Recent MI or angina	Closed-angle glaucoma Phaeochromocytoma HOCM MAOI within 14 days or linezolid Tachyarrhythmia, coronary thrombosis or VF Halothane/cyclopropane anaesthesia	Cyclopropane or halothane anaesthesia	Severe hypertension Ventricular tachycardia

	Dobutamine	Isoprenaline	Ephedrine	Metaraminol	Phenylephrine
Pregnancy					
Code	B2	A	A	C	B2
Justification				Uterine vaso- constriction → ↓ placental perfusion and fetal hypoxia	
Pharmacodynamics					
Respiratory		Bronchodilation	Bronchodilation		
CVS	↑ Inotropy ↑ Lusitropy ↑ HR ↑ CO ↓ SVR (net; β_2 $> \alpha_1$) ↓ PVR ↑ Myocardial O ₂ consumption Arrhythmogenic	↑ HR ↑ Inotropy ↑ CO ↓ SVR ↓↓ MAP Dysrhythmias ↑ Myocardial O ₂ consumption	↑ HR ↑ CO ↑ MAP Dysrhythmias	↑ SVR ↑ PVR ↑ BP Reflex ↓ HR ↓ CO ↑ Coronary flow	↑ SVR and PVR ↑ MAP Reflex ↓ HR ↓ CO ↑ Coronary flow Dysrhythmias rare
CNS		Headache/ dizziness	CNS stimulation ↑ MAC Mydriasis		Headache
Musculoskeletal		Tremor	Extravasation → necrosis ↓ K ⁺ ↓ RBF	Necrosis with extravasation	
Metabolic					
Renal	↑ RBF (sec- ondary to ↑ CO) ↑ UO; no renal benefit			↓ Renal blood flow	↓ RBF and UO

	Dobutamine	Isoprenaline	Ephedrine	Metaraminol	Phenylephrine
Genitourinary			Urinary retention ↓ Uterine activity		↓ Uterine blood flow
Gastrointestinal		↓ Smooth-muscle tone	↓ Tone and motility	↓ Splanchnic blood flow	↓ Splanchnic blood flow
Considerations	Correct hypovolaemia AF → ↑ AV conduction ↑ HR/MAP and ventricular ectopy ↑ Myocardial O ₂ demand β-blockade → unopposed α effects	May precipitate hypotension HR >130 → ventricular dysrhythmias Myocardial ischaemia/necrosis Adrenaline/digitalis/halothane → dysrhythmias MAOI → ↑ effect	Correct hypovolaemia, hypoxia, hypercapnia and acidosis TCA/clonidine → ↑ pressor effect Urinary alkalisation → ↑ t _{1/2} β-blockers → ↓ cardiac/bronchodilator effect	Acidosis → ↓ effect MAOI/TCA → ↑ effect Digitalis → arrhythmias Repeated dosing → cumulative effect	Not a catecholamine Low CO state → may further ↓ CO MAOI/TCA/oxytocics → ↑ pressor effect Sodium metabisulfite → bronchospasm in susceptible patients Repeated doses retain effect

Structure-Activity Relationships of Sympathomimetics

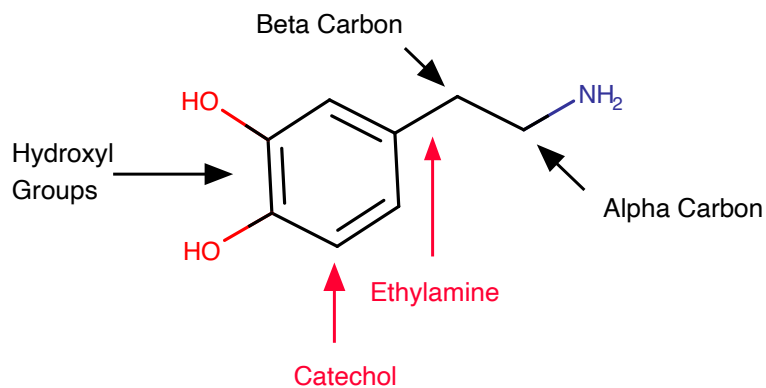


Catecholamines consist of:

- A catechol ring
A benzene ring with two hydroxyl groups in the 3 and 4 position.
 - Losing one hydroxyl group
 - Increases lipid solubility and decreases the potency 10-fold
 - Prevents metabolism by COMT, prolonging duration of action
 - Losing both hydroxyl groups decreases the potency 100-fold
 - Changing the hydroxyl groups to the 3 and 5 position increases beta-2 selectivity when there is also a large substitution present on the amine group
- An ethylamine tail
Consists of:
 - Beta carbon
The first carbon.
 - Adding a hydroxyl group decreases lipid solubility and CNS penetration
 - Adding any group increases alpha and beta selectivity

- Alpha carbon
The second carbon.
 - Adding a group prevents metabolism by MAO, prolonging duration of action
 - Methylation increases indirect activity
- Amine group
The terminal nitrogen.
 - Addition of a methyl group generally increases beta selectivityAs the chain length increases, so does the beta selectivity.

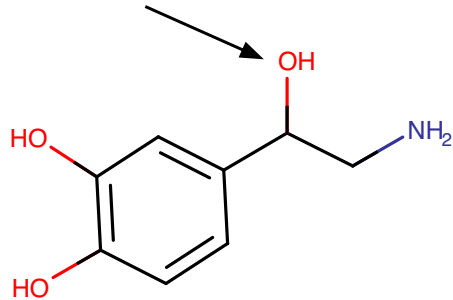
Dopamine



- Dopamine is the prototypical catecholamine, to which others are compared

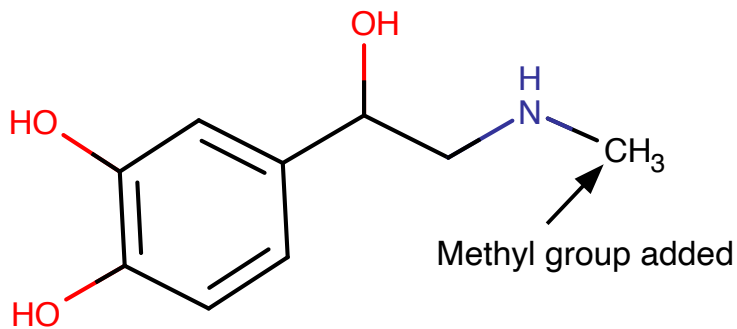
Noradrenaline

Hydroxyl Group added



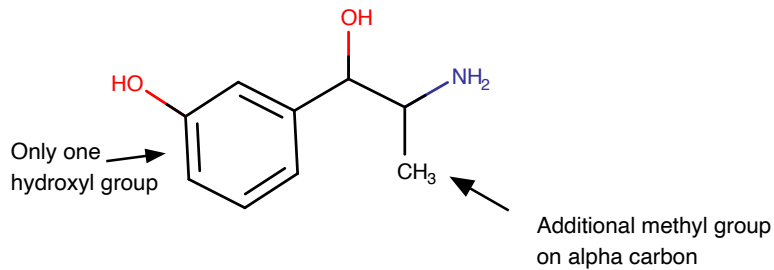
- Noradrenaline has a hydroxyl group added to the beta carbon, increasing its alpha selectivity

Adrenaline



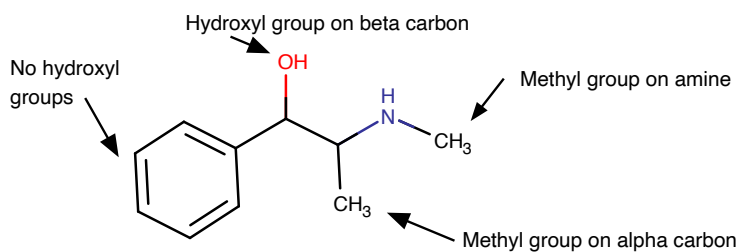
- Adrenaline is similar to noradrenaline with an additional hydroxyl group on the beta carbon
- Adrenaline also has a methyl group added to the terminal amine, increasing beta selectivity

Metaraminol



- Metaraminol has an additional hydroxyl group on the beta carbon
- Metaraminol has only one hydroxyl group on the phenol ring, so:
 - It is no longer classified as a catecholamine
 - It is not metabolised by COMT, prolonging its duration of action
 - It has reduced potency, requiring administration in higher doses
- Metaraminol has an additional methyl group on the alpha carbon, preventing metabolism by MAO and further prolonging its duration of action

Ephedrine



- Like metaraminol, ephedrine has a hydroxyl group on the beta carbon and a methyl group on the alpha carbon

- Ephedrine has no hydroxyl groups on the phenol ring, further reducing its potency and increasing its elimination half-life
 - Ephedrine has a methyl group on the amine, increasing its beta selectivity
-

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Non-Adrenergic Vasoactives

Key non-adrenergic cardiovascular drugs include **vaso-pressin** (and its analogues, **terlipressin** and **ornipressin**), **phosphodiesterase III inhibitors** such as milrinone, and **calcium sensitisers** such as levosimendan.

Table 1: Non-Adrenergic Vasoactives

	Vasopressin	Milrinone	Levosimendan
Aliases	ADH Argipressin Arginine vasopressin	Primacor	Simdax
Class	Non-adrenergic vasopressor	Inodilator	Inodilator
Chemistry	Cyclic nonapeptide	Bipyridine derivative	Pyridazinone-dinitrile derivative
Dose			
Shock	0.01-0.08 units/min IV		
Heart failure (load)		50µg/kg IV over 10min	
Heart failure (infusion)		0.1-0.75µg/kg/min	
Heart failure			0.05-0.2µg/kg/min IV for 24h
Pharmaceutics			
Presentation	20 units/mL solution	1mg/mL concentrate	2.5mg/mL concentrate
Compatibility	NS or D5W	NS, 0.45% NaCl or D5W; incompatible with bicarbonate/loop diuretics	D5W; compatible with furosemide, digoxin and GTN
Solubility	Water-soluble	Soluble at physiological pH	Minimally water-soluble
Pharmacokinetics			
Absorption	No PO bioavailability; destroyed by intestinal peptidases	92% PO bioavailability; IV-only clinical formulation	85% PO bioavailability; IV-only clinical formulation
Distribution	pKa: 10.26	pKa: 4.6 and 8.5	pKa: 6.3
	Vd: 0.14-0.2L/kg	Vd: 0.38L/kg	Vd: 0.3L/kg
	Protein binding: 30%	Protein binding: 70%	Protein binding: 99%

	Vasopressin	Milrinone	Levosimendan
Metabolism	Half-life: 17-35 minutes Duration: Peak pressor effect ~15 minutes; offset ~20 minutes Pathway: 35% metabolised by endothelial peptidases in the liver Metabolites: Inactive peptide fragments	Half-life: 2.3 hours; prolonged in renal impairment Duration: Onset 5-15 minutes Pathway: Predominantly renal; minor hepatic glucuronidation Metabolites: O-glucuronide (inactive)	Half-life: ~1 hour; OR-1896 >80 hours Duration: Develops over ~3 days; persists 2-3 weeks Pathway: 95% → inactive metabolites; 5% → OR-1855/OR-1896 Metabolites: OR-1855 (active), OR-1896 (active)
Elimination	Renal: 65% excreted unchanged	Renal: 83% unchanged; 12% O-glucuronide	Renal: 54% urine; 44% faeces; <0.05% unchanged
Mechanism			
Target	V1 receptor (Gq-coupled) V2 receptor (Gs-coupled)	Phosphodiesterase 3 (PDE3)	Troponin C Vascular ATP-sensitive K ⁺ channels
Action	V1 → ↑ IP3/Ca ²⁺ → vasoconstriction. V2 → ↑ cAMP → aquaporin-2 insertion → ↑ water reabsorption. Vasopressor affinity persists at low pH.	PDE3 inhibition → ↑ cAMP → ↑ myocardial inotropy/lusitropy and vascular smooth-muscle relaxation; cardiac/vascular > airway smooth muscle	Stabilises Ca ²⁺ -bound troponin C → ↑ contractility without ↑ intracellular Ca ²⁺ . Opens vascular KATP channels → vasodilation. PDE3 inhibition at high concentrations.

	Vasopressin	Milrinone	Levosimendan
Contraindications	Chronic nephritis with nitro-gen retention	Severe aortic stenosis Severe pulmonary stenosis Hypertrophic obstructive cardiomyopathy	Severe hypotension/ tachycardia Ventricular filling/outflow obstruction CrCl <30mL/min Severe hepatic impairment Previous TdP
Pregnancy			
Code	B2	B3	
Pharmacodynamics			
CVS	↑ SVR and PVR ↑ MAP Reflex ↓ HR ↓ CO Coronary vasoconstriction at high doses	↑ Inotropy and lusitropy ↑ CO ↓ SVR and PVR ↓ MAP Dysrhythmias	↑ Inotropy and lusitropy ↑ CO ↑ HR ↓ SVR and PVR ↓ MAP Dysrhythmias/QT prolongation
CNS		Headache	
Endocrine	↑ ACTH (V3)		
Metabolic		↓ K ⁺	↓ K ⁺
Renal	↑ Water reabsorption (V2)	↑ RBF secondary to ↑ CO	
Genitourinary	↑ Uterine contractility (V1)		
Gastrointestinal	↓ Mesenteric blood flow ↓ Portal pressure		
Haematological	↑ Platelet aggregation (V1) ↑ Factor VIII and vWF (V2)	Thrombocytopaenia	

	Vasopressin	Milrinone	Levosimendan
Considerations	Noradrenaline - sparing Coronary, mesenteric, skin and digital ischaemia Hyponatraemia/water intoxication DI after cessation	Loading dose often omitted if hypotensive ↓ Infusion in renal impairment Oral chronic use → ↑ mortality	Active metabolites peak ~48h after infusion Prolonged effect in renal/ hepatic impairment Not dialysable

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Centrally Acting Anti-Hypertensives

Table 1: Centrally Acting Antihypertensives

	Clonidine	Methyldopa
Aliases	Catapres; Dixarit	α -methyldopa Aldomet
Class	Central α_2 /I1 agonist	Central α_2 agonist
Chemistry	Imidazoline derivative	Phenylalanine derivative
Dose		
HTN (PO)	75 μ g BD-TDS; maintenance 150-300 μ g TDS; max 900 μ g/day	
Crisis (IV/IM)	150-300 μ g; IV over 5 minutes; repeat q3-6h	
Sedation/analgesia (IV)	10-200 μ g	
Neuraxial adjunct	1-2 μ g/kg	
Paediatric (PO/IV)	1-5 μ g/kg q8h	
Hypertension (PO)		250mg BD/TDS; max 3g/day
Pharmaceutics		
Presentation	150 μ g tablet; clear 150 μ g/mL injection	Yellow 250mg film-coated tablet

	Clonidine	Methyldopa
Compatibility	0.9% NaCl	
Solubility	Amphoteric; water- and lipid-soluble	Slightly water-soluble; pKa 9.85
Pharmacokinetics		
Absorption	70-100% oral bioavailability; minimal first-pass metabolism; rapid absorption	25% oral bioavailability (8-62%); peak 2-3 hours
Distribution	pKa: 8	pKa: 9.85
	Vd: 2.1 L/kg	Vd: 0.6L/kg
	Protein binding: 20-40%	Protein binding: <15%
Metabolism	Half-life: 5-7 hours	Half-life: 1-2 hours
	Duration: ~6 hours	Duration: ~10 hours
	Pathway: 30-50% hepatic to inactive metabolites; remainder excreted unchanged	Pathway: Intestinal, hepatic and neuronal metabolism
	Metabolites: Inactive metabolites	Metabolites: α -methyldopamine, α -methylnoradrenaline (active)
Elimination	Renal: 50% as unchanged drug	Renal: Up to 50% unchanged
Mechanism		
Target	α_2 adrenoreceptor Imidazoline receptor (I1)	Central presynaptic α_2 adrenoreceptors

	Clonidine	Methyldopa
Action	Central α_2/I_1 agonism $\rightarrow$ $\downarrow$ SNS outflow/NA release; peripheral α_2 agonism $\rightarrow$ transient $\uparrow$ MAP after IV bolus	Active metabolites replace noradrenaline and stimulate central α_2 receptors $\rightarrow$ $\downarrow$ sympathetic outflow
Contraindications	Severe bradyarrhythmia, sick sinus or second/third-degree AV block	Active hepatic disease Previous methyldopa-associated hepatic disorder MAOI therapy Catecholamine-secreting tumour Porphyria
Pregnancy		
Code	B3	A
Pharmacodynamics		
CVS	Transient $\uparrow$ MAP with IV bolus Prolonged $\downarrow$ MAP $\downarrow$ HR $\downarrow$ AV conduction $\uparrow$ Baroreceptor sensitivity	$\downarrow$ SVR and MAP HR and CO usually maintained Bradycardia Postural $\downarrow$ MAP
CNS	Sedation Analgesia/ $\downarrow$ opioid requirement Anxiolysis	Sedation Depression Nightmares $\downarrow$ MAC $\uparrow$ Prolactin
Endocrine		
Metabolic	$\downarrow$ Stress response	
Renal	$\downarrow$ ADH $\rightarrow$ diuresis	Renal blood flow maintained
Genitourinary		Impotence $\downarrow$ Libido
Gastrointestinal		Dry mouth N/V

	Clonidine	Methyldopa
Haematological		Positive Coombs test Haemolytic anaemia Drug-induced lupus
Considerations	Abrupt withdrawal → severe rebound HTN; taper over >7 days Concurrent β blocker → cease β blocker first TCAs → ↓ effect; alcohol/sedatives → ↑ CNS depression Opioid withdrawal → ↓ autonomic symptoms Renal/hepatic impairment → consider dose reduction	Preferred centrally acting antihypertensive in pregnancy Delayed peak effect despite short plasma half-life Renal impairment → prolonged effect and lower dose Hepatotoxicity

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Calcium Channel Blockers

Ca²⁺ - channel blockers:

- Have affinity for L -type calcium channels
L -type channels exist in myocardium, nodal, and vascular smooth muscle.
 - Variable affinity for each causes a preference for either nodal and inotropic, or vascular effects
- Prevent Ca²⁺ entry into cells in a use-dependent fashion

Table 1: Classification of Calcium Channel Blockers

Class	Chemical Structure	Drugs
Class I	Phenylalkylamines	Verapamil
Class II	Dihydropyridines	Nifedipine, amlodipine, nimodipine
Class III	Benzothiazepines	Diltiazem

Table 2: Comparison of Calcium-Channel Blockers

	Verapamil	Amlodipine	Diltiazem
Aliases	Isoptin	Norvasc	Cardizem; Dilzem
Class	Non-dihydropyridine calcium channel blocker	Dihydropyridine calcium channel blocker	Non-dihydropyridine calcium channel blocker
Chemistry	Phenylalkylamine; racemate (D-isomer has local anaesthetic activity)	1,4-dihydropyridine	Benzothiazepine
Dose			
SVT (IV)	75 - 150mcg/kg (usually 5mg) over ≥ 2 minutes; repeat after 5 - 10 minutes		
Paediatric HTN crisis (IV)	0.05 - 0.1mg/kg/hr; usual max 1.5mg/kg/day		
HTN/angina (PO)	40 - 120mg TDS	2.5 - 10mg OD	30 - 120mg TDS; max 360mg/day
Rate control (SR)	120 - 240mg BD		120 - 360mg OD
Rate control (IV)			0.25mg/kg over 2 minutes; repeat 0.35mg/kg after 15 minutes; infusion 5 - 15mg/hr
Pharmaceutics			
Presentation	5mg/2mL injection; tablet; oral solution	5mg or 10mg tablet	5mg/mL injection; 30mg, 60mg, 120mg or 180mg tablet; SR formulations
Solubility	Excellent lipid solubility	Excellent lipid solubility	Excellent lipid solubility
Pharmacokinetics			
Absorption	24% oral bioavailability	64% oral bioavailability; Tmax 6 - 12 hours	39% oral bioavailability

	Verapamil	Amlodipine	Diltiazem
Distribution	pKa: 8.73	pKa: 8.6	pKa: 7.5
	Vd: 3.8L/kg	Vd: 21.4L/kg	Vd: 5.3L/kg
	Protein binding: 84-91%	Protein binding: 95.5%	Protein binding: 80-86%
Metabolism	Half-life: 4.5-12 hours	Half-life: 30-50 hours	Half-life: 2-5 hours
	Duration: Exceeds plasma half-life	Duration: ≥ 24 hours	Duration: Exceeds plasma half-life
	Pathway: Hepatic; CYP3A4 substrate and inhibitor	Pathway: Hepatic; CYP3A4	Pathway: Hepatic; CYP3A4 substrate and inhibitor
	Metabolites: Norverapamil (active; ~20% potency)	Metabolites: Inactive metabolites	Metabolites: Desacetyldiltiazem (active; 25-50% potency)
Elimination	Renal: Metabolites	Renal: 10% unchanged; 60% as metabolites	Renal: Metabolites
Mechanism			
Target	Cardiac and vascular L-type Ca^{2+} channel	Vascular L-type Ca^{2+} channel	Cardiac and vascular L-type Ca^{2+} channel
Action	State-dependent block $\rightarrow$ $\downarrow$ AV conduction, contractility and arterial/coronary tone	$\downarrow$ Ca^{2+} influx $\rightarrow$ arterial and coronary vasodilation	State-dependent block $\rightarrow$ $\downarrow$ AV conduction and arterial/coronary vasodilation

	Verapamil	Amlodipine	Diltiazem
Contraindications	Cardiogenic shock or de-compensated heart failure Sick sinus syndrome; 2nd or 3rd degree AV block without pacemaker Severe ↓ MAP AF/flutter with an accessory pathway Ventricular tachycardia Concurrent IV beta blocker outside ICU		Sick sinus syndrome; 2nd or 3rd degree AV block without pacemaker Severe bradycardia Severe ↓ MAP or cardiogenic shock Severe heart failure Acute MI with pulmonary congestion AF/flutter with an accessory pathway Ventricular tachycardia IV beta blocker; IV dantrolene; ivabradine
Pregnancy			
Code	C	C	C
Justification	Maternal ↓ MAP → fetal hypoxia	Maternal ↓ MAP → fetal hypoxia	Maternal ↓ MAP → fetal hypoxia
Pharmacodynamics			
CVS	↓ SVR ↓ MAP ↓ Afterload ↓ HR ↓ AV conduction ↓ Inotropy Coronary vasodilation	↓ SVR ↓ MAP ↓ Afterload Minimal ↑ HR Coronary vasodilation	↓ SVR ↓ MAP ↓ Afterload ↓ HR ↓ AV conduction Mild ↓ Inotropy Coronary vasodilation
CNS	Headache	Headache	Headache
Endocrine	Gynaecomastia		
Musculoskeletal	Peripheral oedema Flushing	Peripheral oedema Flushing	Peripheral oedema Flushing

	Verapamil	Amlodipine	Diltiazem
Gastrointestinal	Constipation	Constipation	Constipation ↓ Lower oesophageal sphincter tone
Considerations	Beta blockers → bradycardia/AV block ↑ Digoxin concentration Prolonged neuromuscular blockade	Severe aortic stenosis (afterload reduction may be detrimental) Angina may worsen after initiation or dose increase	Beta blockers or digoxin → bradycardia/AV block Do not crush SR formulations

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Direct Vasodilators

Direct vasodilators include:

- Ca^{2+} channel blockers (see Calcium Channel Blockers)
- Nitrates
 - Increase production of NO:
 - NO activates guanylate cyclase, increasing cGMP
 - cGMP inhibits Ca^{2+} uptake into smooth muscle and enhances its sequestration into smooth endoplasmic reticulum
 - The decrease in cytoplasmic $[\text{Ca}^{2+}]$ causes smooth muscle relaxation and vasodilation
- Hydralazine
 - Multimodal mechanism of action, including:
 - Opens K^{+} channels, hyperpolarising vascular smooth muscle
 - Decreases intracellular Ca^{2+} in vascular smooth muscle
 - Activation of guanylate cyclase

Table 1: Direct Vasodilators

	Sodium nitroprusside	Glyceryl trinitrate	Hydralazine
Aliases	SNP	GTN Nitroglycerin	Apresoline
Class	Nitrate vasodilator	Nitrate vasodilator	Potassium channel activator
Chemistry	Inorganic ferrocyanide complex	Organic nitrate	Phthalazine derivative
Dose			
Infusion (IV)	0.25 - 10 µg/kg/min	5 - 200 µg/min	200 - 300 µg/min → 50 - 150 µg/min
Angina (SL)		300 - 600 µg; repeat q5min ×3	
Crisis (IV)			5 - 20mg
Pharmaceutics			
Presentation	50mg/2mL concentrate	5mg/mL IV concentrate 400 µg/dose SL spray Transdermal patch	20mg powder for injection
Compatibility	5% glucose only; protect bag from light	5% glucose or 0.9% NaCl; non-PVC tubing	Reconstitute with 1mL water; dilute in 0.9% NaCl; no glucose
Solubility	Highly water-soluble; minimally fat-soluble	Poorly water-soluble; highly fat-soluble	Moderate water solubility
Pharmacokinetics			
Absorption	0% oral bioavailability	Sublingual bioavailability ~40%, onset 1-3 minutes Oral bioavailability ~1% (extensive first-pass hepatic metabolism) Transdermal absorption variable	22-66% oral bioavailability (extensive first-pass hepatic metabolism); well absorbed from GIT

	Sodium nitroprusside	Glyceryl trinitrate	Hydralazine
Distribution	pKa: -3.3	pKa: -5.6	pKa: 7.3
	Vd: 0.25 L/kg	Vd: 3.3 L/kg	Vd: 1.34 L/kg
	Protein binding: None	Protein binding: 60%	Protein binding: 87%
Metabolism	Half-life: ~2 minutes	Half-life: ~30 minutes	Half-life: ~1 hour
	Duration: Onset within seconds	Duration: Sublingual ~35-45 minutes, IV offset within minutes	Duration: 2-4 hours
	Pathway: Spontaneous decomposition → ferrous nitrosyl (FeNO) + 5 cyanide ions (CN ⁻), Reaction with haemoglobin → cyanmethaemoglobin + 4 CN ⁻ , CN ⁻ converted to thiocyanate by hepatic rhodanese	Pathway: Hepatic reductions, vascular walls, erythrocytes	Pathway: Hepatic; polymorphic N-acetylation
	Metabolites: Cyanide (CN ⁻ ; toxic), Thiocyanate (less toxic; renally excreted)	Metabolites: Glyceryl dinitrates (active; less potent), Glyceryl mononitrates (inactive)	Metabolites: Inactive metabolites
Elimination	Renal: Thiocyanate excreted renally; accumulates in renal impairment	Renal: Metabolites	Renal: Inactive metabolites
Mechanism			
Target	Soluble guanylate cyclase	Soluble guanylate cyclase	KATP channel (proposed)

	Sodium nitroprusside	Glyceryl trinitrate	Hydralazine
Action	Spontaneous NO donor → activates sGC → ↑ cGMP → ↑ K ⁺ conductance → membrane hyperpolarisation and ↓ intracellular Ca ²⁺ → vascular smooth muscle relaxation. Balanced arterial and venous dilation (unlike GTN which preferentially dilates veins). Risk of coronary steal in coronary artery disease.	NO donor → activates sGC → ↑ cGMP → ↑ K ⁺ conductance → membrane hyperpolarisation and ↓ intracellular Ca ²⁺ → vascular smooth muscle relaxation.	K ⁺ channel opening/↓ intracellular Ca ²⁺ → direct arteriolar relaxation; mechanism uncertain
Contraindications	Compensatory HTN or high-output HF Severe renal disease Vitamin B12 deficiency Leber optic atrophy or tobacco amblyopia	PDE5 inhibitor or sGC stimulator ↑ ICP Severe ↓ MAP or uncorrected hypovolaemia Pericardial tamponade or constrictive pericarditis	SLE Severe tachycardia or high-output HF Mechanical cardiac obstruction Aortic dissection Porphyria
Pregnancy			
Code	C	B2	C
Justification	Short hypertensive-crisis use may be safe with maternal pH/cyanide monitoring		IV use in 3rd trimester → fetal distress/arrhythmia
Pharmacodynamics			
Respiratory	↑ Shunt (↓ HPV)		

	Sodium nitroprusside	Glyceryl trinitrate	Hydralazine
CVS	↓ Preload (venodilatation) ↓ Afterload (arterial dilation) Reflex ↑ HR ↓ MAP Coronary steal risk in CAD	↓ Preload (venodilatation) ↓ Afterload Coronary arterial vasodilation Reflex ↑ HR ↓ MAP	↓ SVR ↓ Afterload ↓ MAP Reflex ↑ HR ↑ CO ↑ Renin release (RAAS activation) Coronary vasodilation
CNS	↑ ICP (caution in neurological injury) Headache	↑ ICP (caution in neurological injury) Headache (common)	↑ CBF Headache Dizziness
Metabolic	Cyanide toxicity at doses >2µg/kg/min (lactic acidosis, elevated mixed venous O ₂)		
Renal	Thiocyanate accumulation in renal impairment → toxicity		
Gastrointestinal			Nausea Anorexia
Haematological	Methaemoglobinaemia More potent than GTN Infusion >2µg/kg/min → ↑ CN accumulation SCN half-life ~3 days; prolonged in renal failure Haemodialysis removes SCN, not CN Sustained infusion → tachyphylaxis Aortic dissection → β block first	Methaemoglobinaemia (rare) Continuous use → tachyphylaxis; use nitrate-free interval Venodilation > arteriolar dilation at low doses Remove patch before DC cardioversion	IV onset 5-30 minutes Slow acetylators → ↑ exposure/SLE risk Arteriolar only; no venodilation CAD → reflex tachycardia/myocardial ischaemia Chronic use → drug-induced SLE
Considerations			

Nitrate Toxicity

Nitrate toxicity can be related to:

- Cyanide
- Thiocyanate
- Methaemoglobinaemia

Cyanide Toxicity

Cyanide toxicity occurs only with SNP, as CN^- is produced as a byproduct of metabolism or the effect of light.

- **Kinetics**

- Rapid cellular uptake
- Small V_D
- Hepatically metabolised to **thiocyanate**, using thiosulfate as a substrate

- **Mechanism**

CN^- binds to cytochrome oxidase, preventing oxidative phosphorylation. This causes histotoxic hypoxia, and is characterised by:

- Rapid loss of consciousness and seizures
- Metabolic acidosis
- Lactataemia
- Arrhythmia
- Increased MVO_2
- Hypertension

Due to tachyphylaxis to SNP.

- Risk of cyanide toxicity from SNP is related to:

- Infusion rate
- Infusion duration

- **Management**

- Supportive care, including inotropes
- Cyanide chelators

Bind CN^- , removing it from the circulation. Include:

- Dicobalt edetate
- Hydroxycobalamin (Vitamin B_{12})

- Sulfur donors
Provide additional sulfhydryl groups, allowing further hepatic metabolism of CN^- to SCN^- .
Include:
 - **Thiosulfate**
- Nitrites
Converts Oxy-Hb to Met-Hb, which has a higher affinity for CN^- than cytochrome oxidase. Include:
 - Sodium nitrite
 - Amyl nitrite

Thiocyanate Toxicity

Thiocyanate is produced with hepatic metabolism of CN^- . Toxicity occurs when thiocyanate accumulates, which occurs in:

- Long duration SNP infusions
7-14 days.
- Patients with renal failure
Reduced clearance, may occur in 3-6 days.
- Patients given thiosulfate for management of CN^- toxicity.
- **Effects**
Multisystemic, including:
 - Rash
 - Abdominal pain
 - Weakness
 - CNS disturbance
- **Treatment**
 - Dialysis

Methaemoglobinaemia

Methaemoglobinaemia occurs when the Fe^{2+} (ferrous) ion in haemoglobin is oxidised to the Fe^{3+} (ferric) form, which

is **unable** to **bind oxygen**.

- Due to the high concentration of oxygen in erythrocytes, methaemoglobin is continually being formed
- Several endogenous reduction systems exist to keep MetHb levels stable at ~1%
 - Predominantly cytochrome-*b*₅ reductase
 - NADPH-MHb reductase

This reduces methaemoglobinaemia in the presence of a reducing agent, classically **methylene blue**.
 - Reduced glutathione

More important in preventing oxidative stress in other cells than the RBC.
- Disease occurs due to the loss in oxygen-carrying capacity from the loss of effective haemoglobin
 - e.g. a 20% MetHb level gives a *theoretical* oxygen carrying capacity of 80% of the actual haemoglobin

There is in fact a slight **left shift** of the **oxy-haemoglobin dissociation curve**, as oxygen binds more tightly to the partially-oxidised haemoglobin.

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Angiotensin Converting Enzyme Inhibitors

ACE inhibitors prevent the conversion of angiotensin I to angiotensin II by angiotensin converting enzyme (ACE) in the lungs, in turn reducing effects of angiotensin II. These effects include:

- Vasoconstriction
- Noradrenaline reuptake inhibition
- Thirst
- ADH release
- ACTH release
- Aldosterone release
- Reduces K_f, reducing GFR

Indications

- **Hypertension**
 - Particularly in insulin dependent diabetes with diabetic nephropathy
 - Less effective for this indication in the black population
 - Contribute to post-operative hypertension and may be withheld perioperatively
- **Cardiac failure**
All grades.
- **MI with LV dysfunction**
Improved prognosis.

Classification

Can be divided into three groups based on pharmacokinetics:

- Active drug with active metabolites
Captopril.
- Prodrug
Ramipril.
- Not metabolised and excreted unchanged in urine
Lisinopril.

Table 1: Common Effects of ACE Inhibitors

Property	Drug
Resp	Bradykinin cough
CVS	↓ SVR and BP. Unaffected HR and baroreceptor response.
Endocrine	Hypoglycaemia in diabetics
Renal	With a normal renal perfusion pressure, natriuresis results. However, a fall in renal perfusion pressure may cause pre-renal failure (e.g. renal artery stenosis).
Haeme	Agranulocytosis, thrombocytopenia
Immune	Angioedema
Metabolic	↑ Renin release.
Interactions	↓ Aldosterone release, which ↑ the efficacy of spironolactone and may precipitate hyperkalaemia. Pharmacodynamic interaction with NSAIDs to drop renal perfusion pressure.

Table 2: Individual ACE Inhibitors

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Aliases	Zedace	Acetec	Monopril	Zinopril	Coversyl	Accupril	Ramipril Sandoz
Class	ACE in- hibitor	ACE in- hibitor	ACE in- hibitor	ACE in- hibitor	ACE in- hibitor	ACE in- hibitor	ACE in- hibitor
Chemistry	L-proline derivative; sulfhydryl (-SH)	L-proline deriva- tive; dicar- boxylate (-COOH)	L-proline derivative; phosphoryl (-PO ₂)	L-proline deriva- tive; dicar- boxylate (-COOH)	L-proline deriva- tive; dicar- boxylate (-COOH)	L-proline deriva- tive; dicar- boxylate (-COOH)	L-proline deriva- tive; dicar- boxylate (-COOH)
Dose							
HTN (PO)	12.5mg initially; then 25- 50mg BD; max 75mg BD	5mg OD; usual 10- 40mg/day OD-BD	10mg OD; usual 20-40mg OD; up to 80mg/day	5-10mg OD; usual 10-20mg OD; max 40mg/day	5mg OD; max 10mg/ day	10-20mg OD; usual 20-80mg/ day OD- BD	2.5mg OD; max 10mg/ day
HF (PO)	6.25mg TDS; usual 25-75mg BD; max 150mg/ day	2.5mg OD; usual 10- 20mg/day OD-BD	10mg OD; usual 20- 40mg OD; max 40mg/ day	2.5mg OD; usual 5- 20mg OD	2.5mg OD → 5mg OD	5mg BD → 10-20mg BD	
Post-MI (PO)	6.25mg; then 25mg TDS → 50mg TDS						
Diabetic nephropathy (PO)	25mg TDS or 50mg BD						

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
LV dysfunction (PO)		2.5mg BD → 10mg BD					
Acute MI (PO)				5mg; 5mg after 24 hours; 10mg after 48 hours; then 10mg OD for 6 weeks			
Stable CAD (PO)					5mg OD for 2 weeks → 10mg OD		
Post-MI HF (PO)							2.5mg BD; max 5mg BD
Proteinuria (PO)							1.25mg OD; max 5mg/day
CV risk (PO)							2.5mg OD → 5mg after 1 week → 10mg after 3 weeks
Pharmaceutics							

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Presentation	25mg or 50mg tablet	2.5mg, 5mg, 10mg or 20mg tablet	10mg, 20mg or 40mg tablet	5mg, 10mg or 20mg tablet	2.5mg, 5mg or 10mg tablet	5mg, 10mg, 20mg or 40mg tablet	1.25mg, 2.5mg, 5mg or 10mg tablet
Solubility			Water-soluble			Freely water-soluble	
Pharmacokinetics							
Absorption	75% oral bioavailability; effect maximal ~15 minutes; food ↓ absorption	60% oral bioavailability; effect maximal ~1 hour; unaffected by food	32% oral bioavailability; effect maximal ~1 hour; food slows absorption	25% oral bioavailability; effect maximal ~1 hour; Tmax ~7 hours; unaffected by food	66% oral bioavailability; effect maximal ~1 hour; food ↓ conversion to perindoprilat	37% oral bioavailability; effect maximal ~1 hour; food ↓ absorption	28% oral bioavailability; effect maximal ~1-2 hours; food slows absorption
Distribution	pKa: 9.8	pKa: 3	pKa: 4.3	pKa: 2.5	pKa: 3.8	pKa: 3.7	pKa: 3.7
	Vd: 2L/kg Protein binding: 30%	Protein binding: 50%	Vd: 0.8-1.1L/kg Protein binding: Fosinoprilat 99.4%	Vd: 124L (~2L/kg) Protein binding: Minimal	Vd: 0.2L/kg Protein binding: Perindoprilat 20%	Vd: 3.6-7.8L/kg Protein binding: 97%	Vd: 0.1L/kg Protein binding: Ramipril 73%; ramiprilat 56%

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Metabolism	Half-life: 1-2 hours Duration: ~6 hours Pathway: 60% hep- atic; ac- tive drug with active metabo- lites Metabo- lites: Active metabo- lites	Half-life: Enalaprilat 11 hours Duration: 24 hours Pathway: 60% hep- atic; pro- drug → enalaprilat Metabo- lites: Enalaprilat (active)	Half-life: Fosino- prilat 11.5 hours Pathway: 50% hep- atic; pro- drug → fosinoprilat Metabo- lites: Fos- inopri- lat (ac- tive), Glu- curonide conjugate (inactive)	Half-life: 12.6 hours Path- way: Not metabolised; active drug	Half-life: Perindo- pril 1 hour; free perindo- prilat 17 hours Duration: 24 hours Pathway: 88% hep- atic; pro- drug → perindo- prilat Metabo- lites: Perindo- prilat (ac- tive), Five inactive metabo- lites	Half-life: Quinaprilat 2 hours; terminal 25 hours Pathway: ~99.9% hepatic; prodrug → quinaprilat Metabo- lites: Quinapri- lat (ac- tive), Minor inactive metabo- lites	Half-life: Ramiprilat 13-17 hours Duration: 24 hours Pathway: Mainly hepatic; prodrug → ramiprilat Metabo- lites: Ramipri- lat (ac- tive), Glu- curonides and dike- topiper- azine metabo- lites (inac- tive)

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Elimination	Renal: 40% ex- creted un- changed	Renal: 40% ex- creted un- changed; enalaprilat primarily renal	Renal: 50% excreted un- changed; balanced renal/ hepatobiliary clearance	Renal: 100% excreted unchanged	Renal: 12% ex- creted un- changed; free perindo- prilat renal	Renal: 0.1% ex- creted un- changed; quinapri- lat pre- dominantly renal	Renal: ~2% excreted un- changed; 60% parent/ metabolites in urine
Mechanism							
Target	An- giotensin- converting enzyme (ACE)	An- giotensin- converting enzyme (ACE)	An- giotensin- converting enzyme (ACE)	An- giotensin- converting enzyme (ACE)	An- giotensin- converting enzyme (ACE)	An- giotensin- converting enzyme (ACE)	An- giotensin- converting enzyme (ACE)
Action	ACE zinc- site inhi- bition → ↓ Ang II and aldos- terone; ↑ bradykinin	ACE zinc- site inhi- bition → ↓ Ang II and aldos- terone; ↑ bradykinin	ACE zinc- site inhi- bition → ↓ Ang II and aldos- terone; ↑ bradykinin	ACE zinc- site inhi- bition → ↓ Ang II and aldos- terone; ↑ bradykinin	ACE zinc- site inhi- bition → ↓ Ang II and aldos- terone; ↑ bradykinin	ACE zinc- site inhi- bition → ↓ Ang II and aldos- terone; ↑ bradykinin	ACE zinc- site inhi- bition → ↓ Ang II and aldos- terone; ↑ bradykinin

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Contraindications	Pregnancy Previous ACE in- hibitor-as- sociated, hereditary or idio- pathic an- gioedema Sacubitril/ valsartan within 36 hours	Pregnancy Previous ACE in- hibitor-as- sociated, hereditary or idio- pathic an- gioedema Sacubitril/ valsartan within 36 hours Aliskiren in diabetes	Pregnancy Previous ACE in- hibitor- associ- ated an- gioedema Aliskiren in diabetes	Pregnancy Previous ACE in- hibitor-as- sociated, hereditary or idio- pathic an- gioedema Aliskiren in diabetes or GFR <60mL/ min/ 1.73m ²	Pregnancy Previous ACE in- hibitor-as- sociated, hereditary or idio- pathic an- gioedema Renal artery stenosis Sacubitril/ valsartan within 36 hours Aliskiren in diabetes or GFR <60mL/ min/ 1.73m ²	Pregnancy Previous ACE in- hibitor- associ- ated an- gioedema Sacubitril/ valsartan within 36 hours Aliskiren in diabetes	Pregnancy Previous ACE in- hibitor-as- sociated, hereditary or idio- pathic an- gioedema Haemody- namically signifi- cant re- nal artery stenosis Sacubitril/ valsartan Aliskiren in diabetes or CrCl <60mL/ min
Pregnancy Code	D	D	D	D	D	D	D

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Justification	2nd/3rd trimester → fetal renal dys-function, oligohy-dramnios/death; 1st-trimester safety un-certain	2nd/3rd trimester → fetal renal dys-function, oligohy-dramnios/death; 1st-trimester safety un-certain	2nd/3rd trimester → fetal renal dys-function, oligohy-dramnios/death; 1st-trimester safety un-certain	2nd/3rd trimester → fetal renal dys-function, oligohy-dramnios/death; 1st-trimester safety un-certain	2nd/3rd trimester → fetal renal dys-function, oligohy-dramnios/death; 1st-trimester safety un-certain	2nd/3rd trimester → fetal renal dys-function, oligohy-dramnios/death; 1st-trimester safety un-certain	2nd/3rd trimester → fetal renal dys-function, oligohy-dramnios/death; 1st-trimester safety un-certain
Pharmacodynamics							
Respiratory	Dry cough	Dry cough	Dry cough	Dry cough	Dry cough	Dry cough	Dry cough
CVS	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and my-ocardial remod-elling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and my-ocardial remod-elling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and my-ocardial remod-elling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and my-ocardial remod-elling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and my-ocardial remod-elling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and my-ocardial remod-elling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and my-ocardial remod-elling
Endocrine	Hypogly-caemia (diabetes)	Hypogly-caemia (diabetes)	Hypogly-caemia (diabetes)	Hypogly-caemia (diabetes)	Hypogly-caemia (diabetes)	Hypogly-caemia (diabetes)	Hypogly-caemia (diabetes)
Metabolic	↑ Renin ↑ K+	↑ Renin ↑ K+	↑ Renin ↑ K+	↑ Renin ↑ K+	↑ Renin ↑ K+	↑ Renin ↑ K+	↑ Renin ↑ K+

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Renal	↑ Na - triuresis if renal perfusion maintained ↓ GFR/AKI if Ang II - dependent (e.g. re - nal artery stenosis)	↑ Na - triuresis if renal perfusion maintained ↓ GFR/AKI if Ang II - dependent (e.g. re - nal artery stenosis)	↑ Na - triuresis if renal perfusion maintained ↓ GFR/AKI if Ang II - dependent (e.g. re - nal artery stenosis)	↑ Na - triuresis if renal perfusion maintained ↓ GFR/AKI if Ang II - dependent (e.g. re - nal artery stenosis)	↑ Na - triuresis if renal perfusion maintained ↓ GFR/AKI if Ang II - dependent (e.g. re - nal artery stenosis)	↑ Na - triuresis if renal perfusion maintained ↓ GFR/AKI if Ang II - dependent (e.g. re - nal artery stenosis)	↑ Na - triuresis if renal perfusion maintained ↓ GFR/AKI if Ang II - dependent (e.g. re - nal artery stenosis)
Haematological	Agranulo - cytosis Thrombo - cytopenia	Agranulo - cytosis Thrombo - cytopenia	Agranulo - cytosis Thrombo - cytopenia	Agranulo - cytosis Thrombo - cytopenia	Agranulo - cytosis Thrombo - cytopenia	Agranulo - cytosis Thrombo - cytopenia	Agranulo - cytosis Thrombo - cytopenia

	Captopril	Enalapril	Fosinopril	Lisinopril	Perindopril	Quinapril	Ramipril
Considerations	An-gioedema (bradykinin-mediated) K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI ↓ Antihypertensive effect in Black patients	An-gioedema (bradykinin-mediated) K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI ↓ Antihypertensive effect in Black patients	An-gioedema (bradykinin-mediated) K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI ↓ Antihypertensive effect in Black patients Clearance partly preserved in renal failure	An-gioedema (bradykinin-mediated) K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI ↓ Antihypertensive effect in Black patients	An-gioedema (bradykinin-mediated) K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI ↓ Antihypertensive effect in Black patients	An-gioedema (bradykinin-mediated) K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI ↓ Antihypertensive effect in Black patients	An-gioedema (bradykinin-mediated) K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI ↓ Antihypertensive effect in Black patients

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Angiotensin Receptor Blockers

Angiotensin receptor antagonists are very similar to ACE inhibitors, except:

- Bradykinin does not accumulate as it is still broken down by ACE
Therefore there is no cough and patient compliance is improved.
- The AT₁ receptor in cardiac tissue is more comprehensively blocked which *may* improve cardiac outcomes
- The AT₂ receptor is not blocked, which *may* also improve cardiac outcomes

Table 1: Individual Angiotensin Receptor Blockers

	Candesartan	Irbesartan	Losartan	Olmesartan	Telmisartan	Valsartan
Aliases	Atacand	Avapro	Cozaar	Olmetec	Micardis	Diovan
Class	Angiotensin receptor blocker	Angiotensin receptor blocker	Angiotensin receptor blocker	Angiotensin receptor blocker	Angiotensin receptor blocker	Angiotensin receptor blocker
Chemistry	Imidazole analogue; candesartan cilexetil ester prodrug	Imidazole analogue	Imidazole analogue	Imidazole analogue; olmesartan medoxomil ester pro-drug	Imidazole analogue	Imidazole analogue
Dose						
HTN (PO)	8 - 16mg OD; max 32mg/day	150mg OD; max 300mg/day		20mg OD; max 40mg/day	40mg OD; max 80mg/day	80mg OD; max 320mg/day
HF (PO)	4mg OD; double at ≥ 2 -week intervals $\rightarrow$ 32mg OD					40mg BD $\rightarrow$ 160mg BD
Diabetic nephropathy (PO)		300mg OD				
HTN/diabetic nephropathy (PO)			50mg OD; 25 - 100mg/day OD-BD			
CV prevention (PO)					80mg OD	
Post-MI (PO)						20mg BD $\rightarrow$ 160mg BD
Pharmaceutics						

	Candesartan	Irbesartan	Losartan	Olmesartan	Telmisartan	Valsartan
Presentation	4mg, 8mg, 16mg or 32mg tablet	150mg or 300mg tablet	50mg tablet	10mg, 20mg or 40mg tablet	40mg or 80mg tablet	40mg, 80mg, 160mg or 320mg tablet
Compatibility	Tablet may be crushed for NG		Tablet may be crushed for NG			
Pharmacokinetics						
Absorption	14% oral tablet bioavailability; Tmax 3-4 hours; food has no clinically significant effect	60-80% oral bioavailability; Tmax 1.5-2 hours; unaffected by food	33% oral bioavailability; Tmax 1 hour (active metabolite 3-4 hours); food has no clinically significant effect	25.6% oral bioavailability; Tmax 2 hours; food has minimal effect	40% oral bioavailability at 40mg; Tmax 0.5-2 hours; food has no clinically significant effect	23% oral bioavailability; Tmax 2-4 hours; food ↓ AUC 48%
Distribution	pKa: 6	pKa: 4.1	pKa: 5.5	Vd: 16-29L	pKa: 4.45	pKa: 3.6
	Vd: 0.1L/kg	Vd: 53-93L	Vd: 34L	Protein binding: 99.7%	Vd: 6.6L/kg	Vd: 17L
	Protein binding: Over 99%	Protein binding: 90%	Protein binding: Over 99%		Protein binding: Over 99.5%	Protein binding: 94-97%

	Candesartan	Irbesartan	Losartan	Olmesartan	Telmisartan	Valsartan
Metabolism	Half-life: 9 hours Pathway: Prodrug hydrolysed during absorption; minor CYP2C9 metabolism Metabolites: Candesartan (active)	Half-life: 11-15 hours Pathway: Hepatic glucuronidation and CYP2C9 oxidation Metabolites: Irbesartan glucuronide	Half-life: 2 hours (active metabolite 6-9 hours) Pathway: Substantial first-pass CYP2C9/CYP3A4 metabolism; 14% → active metabolite Metabolites: E-3174 carboxylic acid (active)	Half-life: 10-15 hours Pathway: Prodrug hydrolysed by gut/portal esterases; virtually no further metabolism Metabolites: Olmesartan (active)	Half-life: 18.3-23 hours Pathway: Hepatic acylglucuronidation; not CYP-mediated Metabolites: Acylglucuronide (inactive)	Half-life: 9.5 hours Pathway: ~25% of absorbed dose metabolised Metabolites: Valeryl 4-hydroxy valsartan (inactive)
Elimination	Renal: 30% in urine; 70% in faeces; mainly unchanged via urine/bile	Renal: Under 2% unchanged; 20% of total dose in urine, remainder in faeces	Renal: 4% unchanged; 35% of total dose in urine and 58% in faeces	Renal: 30-50% of absorbed drug in urine; remainder via bile/faeces	Renal: Under 1% in urine; almost all in faeces, mainly unchanged	Renal: 83% in faeces; 13% in urine; mainly unchanged
Mechanism Target	AT1 receptor	AT1 receptor	AT1 receptor	AT1 receptor	AT1 receptor	AT1 receptor

	Candesartan	Irbesartan	Losartan	Olmesartan	Telmisartan	Valsartan
Action	Competitive AT1 antagonism → blocks Ang II-mediated RAAS effects	Competitive AT1 antagonism → blocks Ang II-mediated RAAS effects	Competitive AT1 antagonism → blocks Ang II-mediated RAAS effects	Competitive AT1 antagonism → blocks Ang II-mediated RAAS effects	Competitive AT1 antagonism → blocks Ang II-mediated RAAS effects	Competitive AT1 antagonism → blocks Ang II-mediated RAAS effects
Contraindications	Pregnancy Lactation Severe hepatic impairment or cholestasis Aliskiren in diabetes or GFR <60mL/min/1.73m ²	Pregnancy Aliskiren in diabetes or moderate-severe renal impairment ACE inhibitor in diabetic nephropathy	Pregnancy Aliskiren in diabetes	Pregnancy Severe renal impairment (CrCl <30mL/min) Severe hepatic impairment or biliary obstruction Aliskiren in diabetes	Pregnancy Lactation Biliary obstruction or severe hepatic impairment Aliskiren in diabetes or GFR <60mL/min/1.73m ²	Pregnancy Severe hepatic impairment, biliary cirrhosis or cholestasis Aliskiren in type 2 diabetes
Pregnancy						
Code	D	D	D	D	D	D
Justification	2nd/3rd-trimester RAAS blockade → fetal injury/death; 1st-trimester safety uncertain	2nd/3rd-trimester RAAS blockade → fetal injury/death; 1st-trimester safety uncertain	2nd/3rd-trimester RAAS blockade → fetal injury/death; 1st-trimester safety uncertain	2nd/3rd-trimester RAAS blockade → fetal injury/death; 1st-trimester safety uncertain	2nd/3rd-trimester RAAS blockade → fetal injury/death; 1st-trimester safety uncertain	2nd/3rd-trimester RAAS blockade → fetal injury/death; 1st-trimester safety uncertain
Pharmacodynamics						

	Candesartan	Irbesartan	Losartan	Olmesartan	Telmisartan	Valsartan
CVS	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and myocardial remodelling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and myocardial remodelling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and myocardial remodelling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and myocardial remodelling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and myocardial remodelling	↓ SVR ↓ MAP No reflex ↑ HR ↓ Vascular and myocardial remodelling
Metabolic	↑ Renin ↑ K ⁺	↑ Renin ↑ K ⁺	↑ Renin ↑ K ⁺	↑ Renin ↑ K ⁺	↑ Renin ↑ K ⁺	↑ Renin ↑ K ⁺
Renal	↑ Natriuresis ↓ GFR/AKI if Ang II-dependent (e.g. renal artery stenosis)	↑ Natriuresis ↓ GFR/AKI if Ang II-dependent (e.g. renal artery stenosis)	↑ Natriuresis ↓ GFR/AKI if Ang II-dependent (e.g. renal artery stenosis)	↑ Natriuresis ↓ GFR/AKI if Ang II-dependent (e.g. renal artery stenosis)	↑ Natriuresis ↓ GFR/AKI if Ang II-dependent (e.g. renal artery stenosis)	↑ Natriuresis ↓ GFR/AKI if Ang II-dependent (e.g. renal artery stenosis)
Gastrointestinal				Sprue-like enteropathy → chronic diarrhoea/ weight loss		
Haematological			↓ Platelet aggregation (TXA2 antagonism)			

	Candesartan	Irbesartan	Losartan	Olmesartan	Telmisartan	Valsartan
Considerations	No bradykinin accumulation → ↓ cough/angioedema vs ACE inhibitors More complete AT1 blockade vs ACE inhibitors Unopposed AT2 activation K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI	No bradykinin accumulation → ↓ cough/angioedema vs ACE inhibitors More complete AT1 blockade vs ACE inhibitors Unopposed AT2 activation K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI	No bradykinin accumulation → ↓ cough/angioedema vs ACE inhibitors More complete AT1 blockade vs ACE inhibitors Unopposed AT2 activation K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI	No bradykinin accumulation → ↓ cough/angioedema vs ACE inhibitors More complete AT1 blockade vs ACE inhibitors Unopposed AT2 activation K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI	Mild-moderate hepatic impairment: max 40mg/day No bradykinin accumulation → ↓ cough/angioedema vs ACE inhibitors More complete AT1 blockade vs ACE inhibitors Unopposed AT2 activation K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI Not dialysable	CrCl <30mL/min or mild-moderate hepatic impairment: max 80mg/day No bradykinin accumulation → ↓ cough/angioedema vs ACE inhibitors More complete AT1 blockade vs ACE inhibitors Unopposed AT2 activation K ⁺ -sparing diuretics/K ⁺ → hyperkalaemia NSAIDs → ↓ effect and AKI

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Angiotensin Receptor Neprilysin Inhibitors

Inhibit the neutral endopeptidase neprilysin, which catalyses the degradation of:

- Natriuretic peptides
 - ANP
 - BNP
 - Bradykinin
 - Angiotensin II
- Isolated neprilysin inhibition may not have a major effect on BP due to ↑ angiotensin II.

This inhibition:

- Leads to increased:
 - Natriuretic peptides
 - Bradykinin
 - Substance P
- Causes:
 - Natriuresis
 - Vasodilation
 - Ventricular unloading
 - ↓ Ventricular remodelling
 - ↓ Sympathetic tone

Indications

Used for:

- Hypertension
- Heart failure with reduced ejection fraction

Table 1: Common Effects of Neprilysin Inhibitors

Property	Drug
Resp	Bradykinin cough
CVS	↓ BP - synergistic effect when combined RAAS agents, ↓ CVS mortality
Renal	May be nephro-protective, ↑ K
Immune	Angioedema due to bradykinin accumulation, so usually formulated with an ACE-I or ARB

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Potassium Channel Activators

Potassium channel activators stimulate ATP sensitive K⁺ channels, causing an increase in intracellular cGMP and subsequent relaxation of smooth muscle in the:

- Heart
- Venous capacitance vessels
- Arterioles

Table 1: Nicorandil

Nicorandil	
Aliases	Ikotab
Class	Potassium channel activator
Chemistry	Pyrimidinecarboxamide; amphoteric
Dose	
Angina	10 - 20mg PO BD; start 5mg BD if headache-prone
Pharmaceutics	
Presentation	10mg, 20mg tablets
Solubility	Sparingly water-soluble
Pharmacokinetics	
Absorption	75% PO bioavailability; peak 30-60 minutes
Distribution	pKa: 3.18
	Vd: 1.0-1.4L/kg
	Protein binding: ~24%

	Nicorandil
Metabolism	Half-life: ~2 hours Duration: ~12 hours Pathway: Hepatic denitration → nicotinamide pathway Metabolites: Inactive metabolites
Elimination	Renal: >60% urine in 24h; ~1% unchanged
Mechanism	
Target	ATP-sensitive K ⁺ channels
Action	Opens vascular KATP channels → hyperpolarisation → arterial vasodilation; nitrate moiety → ↑ cGMP → venodilation
Contraindications	Cardiogenic shock Hypotension or hypovolaemia Acute MI with LV failure and low filling pressures PDE5 inhibitors sGC stimulators
Pregnancy	
Code	B3
Pharmacodynamics	
CVS	↓ SVR ↓ Afterload ↓ MAP ↓ Preload ↑ Coronary flow Reflex ↑ HR
CNS	Headache Dizziness
Metabolic	↑ K ⁺
Gastrointestinal	Ulceration → haemorrhage/perforation/fistula
Haematological	Inhibition of platelet aggregation Thrombocytopaenia

	Nicorandil
Considerations	Persistent ulceration → cease Headache often improves with continued use

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Sodium Channel Blockers

Sodium channel blockers include:

- Class Ia:
 - **Procainamide**
 - Quinidine
 - Disopyramide
- Class Ib:
 - **Lignocaine**
 - Mexiletine (lignocaine analogue)
- Class Ic:
 - **Flecainide**

In general:

- IV preparations are given for VT
- Good PO bioavailability and low protein binding
- Metabolites are renally cleared

Table 1: Sodium -Channel Blockers

	Procainamide	Lignocaine	Flecainide
Aliases		Lidocaine Xylocard	Tambocor
Class	Class Ia antiarrhythmic	Class Ib antiarrhythmic; aminoamide local anaes- thetic	Class Ic antiarrhythmic
Chemistry	Aminobenzamide	Aminoamide	Monocarboxylic acid amide (benzamide); racemate
Dose			
Load (IV)	Up to 12mg/kg at ≤ 0.3 mg/ kg/min; max 1g		
Infusion (IV)	50mcg/kg/min	2 - 4mg/min	
VT (IV)		1mg/kg over 1 - 2 minutes; max 100mg	
SVT/VT (IV)			2mg/kg (up to 150mg) over 10 - 30 minutes; infusion 1.5mg/kg/hr
SVT (PO)			50mg BD; max 300mg/day
VT (PO)			100mg BD; max 400mg/day
Pharmaceutics			
Presentation	Clear 100mg/mL injection	Clear 20mg/mL injection; 100mg/mL infusion con- centrate	50mg or 100mg tablet; clear 150mg/15mL injection
Compatibility	5% glucose	5.5% glucose; 0.9% NaCl; Ringer's; sodium bicarbon- ate; dextran	5% glucose only; chloride or alkaline solutions may precipitate
Solubility	Highly water - soluble	Water - soluble	Water - soluble
Pharmacokinetics			

	Procainamide	Lignocaine	Flecainide
Absorption	75-95% oral bioavailability; Tmax 1.5-2 hours	Oral bioavailability 35%; IV only for arrhythmia	90-95% oral bioavailability; Tmax ~3 hours; no first- pass effect
Distribution	pKa: 9.23 Vd: 1.5-2.5 L/kg Protein binding: 15-25%	pKa: 7.9 Vd: 0.9 L/kg Protein binding: 70%	pKa: 9.3 Vd: 8.7 L/kg Protein binding: 40%
Metabolism	Half-life: 3-4 hours (parent drug) Duration: 4-8 hours (in- cluding active metabolite) Pathway: Hepatic acetyla- tion (16-33%) Metabolites: NAPA (N- acetylprocainamide; active)	Half-life: 10-20 minutes (IV bolus); 45-90 minutes (subcutaneous); 90-120 minutes (terminal) Pathway: Hepatic CYP1A2/ 3A4 (90-95%); active metabolites Metabolites: Monoethyl- glycinexylidide (active), Glycinexylidide (active)	Half-life: 20 hours Pathway: Hepatic (70%); active metabolites Metabolites: Active metabo- lites
Elimination	Renal: 30-60% urine un- changed; NAPA renally elim- inated	Renal: <10% excreted un- changed	Renal: 30% excreted un- changed
Mechanism Target	Nav1.5 subunit of voltage- gated sodium channel Potassium channel	Nav1.5 subunit of voltage- gated sodium channel	Nav1.5 subunit of voltage- gated sodium channel

	Procainamide	Lignocaine	Flecainide
Action	Open Na ⁺ channel block → ↓ Phase 0/conduction + ↑ QRS; K ⁺ channel block → ↑ APD/refractory period/QT	Open/inactivated Na ⁺ chan- nel block with rapid disso- ciation → ↓ Phase 0, auto- maticity and APD in depo- larised/ischaemic ventricular tissue; minimal effect on normal tissue or QRS; ↓ QT	Slow Na ⁺ channel dissocia- tion → use-dependent block → ↓ Phase 0 + conduction; minimal change in APD
Contraindications	Second/complete AV block or hemiblock without pace- maker Systemic lupus erythemato- sus TdP	Severe SA, AV or intraven- tricular block Stokes-Adams syndrome Supraventricular arrhythmias	Second/third-degree AV block without pacemaker Bifascicular block without pacemaker Cardiogenic shock Post-MI asymptomatic PVCs or non-sustained VT Severe renal/hepatic im- pairment without plasma monitoring
Pregnancy Code	B2	A	B3
Pharmacodynamics CVS	↓ SVR/BP/CO ↑ PR/QRS/QT; AV block TdP; proarrhythmia AF/flutter → ↑ ventricular rate	AV block ↓ Inotropy/HR/MAP; cardio- vascular collapse	↑ PR/QRS; minimal ↑ QT; heart block ↓ Inotropy; may precipitate CCF Proarrhythmia ↑ Pacing threshold ↑ Defibrillation energy re- quirement

	Procainamide	Lignocaine	Flecainide
CNS	Dizziness	Circumoral tingling; paraesthesia Dizziness; visual disturbance; confusion Seizures; coma	Dizziness Visual disturbance Paraesthesia Headache
Gastrointestinal	Nausea; diarrhoea		Nausea
Haematological	Lupoid syndrome (20-30%; long-term) Agranulocytosis/bone marrow suppression (~0.5%)		
Considerations	Slow acetylators → earlier lupoid syndrome Renal impairment → procainamide/NAPA accumulation ↑ Neuromuscular blockade Reduces antimicrobial effect of sulfonamides	Shock, CCF, hepatic disease or prolonged infusion → ↓ clearance Propranolol, metoprolol or cimetidine → ↓ clearance	↑ Mortality in heart failure patients Amiodarone → ↑ flecainide; halve flecainide dose Renal impairment or alkaline urine → ↓ elimination

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Beta-Blockers

β -blockers are competitive (often highly selective) antagonists of β -adrenoreceptors. They are sub-classified into selective and non-selective agents:

- Selective (β_1 antagonism) (BEAM)
 - Bisoprolol
 - Esmolol
 - Atenolol
 - Metoprolol
- Non-selective (β_1 and β_2 antagonism)
 - Propranolol
 - Sotalol
 - Timolol
- Non-selective (β & α antagonism)
 - Carvedilol
 - Labetalol

Indications

- Cardio
 - Angina
 - Arrhythmia
 - Rate-control in AF
 - Paroxysmal SVT
 - Sinus tachycardia from $\uparrow$ catecholamines
 - Cardiac Failure
 - Secondary prevention for MI
- Vascular
 - Hypertension (2nd line)
Also useful for aggressive control of BP.
 - Hypotensive anaesthesia

- Attenuate hypertensive response to laryngoscopy
- Non-CVS
 - Thyrotoxicosis
 - Glaucoma (topically)
 - Anxiety
 - Migraine prophylaxis

Table 1: Common Effects of Beta Blockers

Property	Action
Kinetics	Variability primarily due to lipid solubility. Poor lipid solubility confers poor gut absorption and minimises need for hepatic metabolism. Lipid soluble agents will have CNS effects and be excreted in breast milk.
Respiratory	Bronchospasm.
CVS	↓ Inotropy, ↓ HR, ↓ MVO ₂ , ↓ BP, ↑ SVR (β ₂ effect), worsen arrhythmia.
CNS	Tiredness, nightmares, and sleep disturbance with lipid soluble agents. ↓ IOP.
Metabolic	↓ Insulin release and blunted hypoglycaemic response (β ₂ effect).
Interactions	Contraindicated with cardioselective Ca ²⁺ channel blockers. due to extreme ↓ HR & ↓ inotropy.

Table 2: Comparison of Beta-Blockers

	Esmolol	Metoprolol	Atenolol	Propranolol	Labetalol
Aliases	Esmolol Nori-dem	Metoprolol IV Viatris	Tenormin; Noten	Inderal; Cardinol	Labetalol SXP
Class	Ultra - short β 1 - selective blocker	β 1 - selective blocker	β 1 - selective blocker	Non - selective β blocker	Non - selective β blocker; α 1 blocker
Chemistry	Aryloxy - propanolamine	Aryloxy - propanolamine	Aryloxy - propanolamine	Aryloxy - propanolamine; racemate	Aryloxy - propanolamine; 4 stereoisomers (S,R α - blocker; R,R β - blocker; 2 relatively inactive)
Dose					
SVT (IV)	500 μ g/kg over 1 minute $\rightarrow$ 50-200 μ g/kg/min				
Arrhythmia (IV)		1 - 5mg at 1 - 2mg/min; repeat q5min; usual total 10 - 15mg	2.5mg boluses; max 10mg	1mg boluses at <1mg/min; usual total 1 - 3mg	
Arrhythmia/HTN (PO, IR)		12.5 - 100mg BD			
HTN/angina (PO)			50 - 100mg OD		
HTN (PO)				40mg BD; usual 120 - 320mg/day	100 - 800mg BD
Arrhythmia/thyrotoxicosis (PO)				10 - 40mg TDS - QID	

	Esmolol	Metoprolol	Atenolol	Propranolol	Labetalol
Paediatric arrhythmia (PO)				0.25 - 0.5mg/kg TDS - QID	
Severe HTN (IV)					20mg over ~2min → 40mg → 80mg q10- 20min; max 300mg/day
Severe HTN (IV infusion)					2mg/min; usual total 50- 200mg; max 300mg/day
Pregnancy HTN (IV infusion)					20mg/hour; ↑ by 20mg/hour q20min; max 160mg/hour and 300mg/day
Pharmaceutics					
Presentation	Clear 100mg/ 10mL injection (10mg/mL)	50mg or 100mg IR tablet; 23.75mg, 47.5mg, 95mg or 190mg CR tablet; clear 5mg/5mL in- jection	25mg, 50mg or 100mg tablet; syrup; clear in- jection	10mg or 40mg tablet; 160mg LA capsule; clear 1mg/mL injection	50mg, 100mg, 200mg or 400mg tablet; clear 100mg/ 20mL injection
Compatibility	0.45% or 0.9% NaCl; 5% glu- cose; Hart- mann's	0.9% NaCl; 5% or 10% glucose; glucose/saline			0.9% NaCl; 5% glucose; Hartmann's; in- compatible with 4.2% NaHCO ₃

	Esmolol	Metoprolol	Atenolol	Propranolol	Labetalol
Solubility	Minimal lipid solubility	Poor lipid solubility	Minimal lipid solubility	Highly lipid-soluble	Fair lipid solubility
Pharmacokinetics					
Absorption	IV only	50% oral bioavailability	45-55% oral bioavailability	30% oral bioavailability	10-80% oral bioavailability (highly variable due to first pass)
Distribution	pKa: 9.5	pKa: 9.7	pKa: 9.6	pKa: 9.45	pKa: 9.3
	Vd: 3.4 L/kg	Vd: 2.8-4.8 L/kg	Vd: 0.3-0.6 L/kg	Vd: 4 L/kg	Vd: 3-16 L/kg
	Protein binding: 60%	Protein binding: 12%	Protein binding: 5%	Protein binding: 95%	Protein binding: 50%
Metabolism	Half-life: 9 minutes	Half-life: 3-4 hours	Half-life: 6-9 hours	Half-life: 3-6 hours	Half-life: 3-8 hours
	Pathway: Red-cell esterase hydrolysis → inactive acid metabolite + methanol	Pathway: Hepatic CYP2D6 metabolism	Pathway: Minimal hepatic metabolism	Pathway: Extensive first-pass hepatic metabolism	Pathway: Hepatic glucuronidation; extensive first pass
	Metabolites: Inactive acid metabolite, Methyl alcohol	Metabolites: 2-hydroxymetoprolol (active; minor contribution)		Metabolites: 4-hydroxypropranolol (active; oral only), Inactive metabolites	Metabolites: Inactive glucuronides

	Esmolol	Metoprolol	Atenolol	Propranolol	Labetalol
Elimination	Renal: <2% unchanged; 73-88% as acid metabolite in urine	Renal: ~3% unchanged; remainder as metabolites in urine	Renal: Excreted unchanged in urine	Renal: 95-100% as metabolites/conjugates in urine	Renal: Glucuronides excreted in urine and bile/faeces; minimal unchanged drug
Mechanism					
Target	β_1 adrenoreceptor	β_1 adrenoreceptor	β_1 adrenoreceptor	β_1 adrenoreceptor β_2 adrenoreceptor	β_1 adrenoreceptor β_2 adrenoreceptor α_1 adrenoreceptor
Action	Competitive β_1 antagonism → ↓ Gs-mediated cAMP	Competitive β_1 antagonism → ↓ Gs-mediated cAMP	Competitive β_1 antagonism → ↓ Gs-mediated cAMP	Competitive β_1/β_2 antagonism → ↓ Gs-mediated cAMP; membrane-stabilising activity	Competitive β/α_1 antagonism → ↓ cAMP + vasodilation; $\beta:\alpha$ 3:1 PO/~7:1 IV; membrane-stabilising activity

	Esmolol	Metoprolol	Atenolol	Propranolol	Labetalol
Contraindications	Severe brady- cardia or sick sinus Second/third- degree AV block Cardiogenic shock or overt CCF Severe hy- potension Acute asthma Untreated phaeochro- mocytoma Inotrope/ vasopressor dependence	Bronchospasm Sinus brady- cardia or sick sinus Second/third- degree AV block Shock or un- controlled CCF Hypotension Severe PVD Untreated phaeochro- mocytoma	Bronchospasm Sinus brady- cardia or sick sinus Second/third- degree AV block Shock or un- controlled CCF Severe PVD Metabolic aci- dosis	Asthma, bron- chospasm or airway obstruc- tion Sinus brady- cardia or sick sinus Second/third- degree AV block Shock or CCF Hypotension Severe PVD Prolonged fasting/ hypoglycaemia Untreated phaeochro- mocytoma	Asthma or COPD Sinus brady- cardia or sick sinus Second/third- degree AV block Cardiogenic shock or symp- tomatic CCF Hypotension Severe PVD Metabolic aci- dosis Untreated phaeochro- mocytoma
Pregnancy					
Code	C	C	C	C	C
Justification	Fetal/neonatal bradycardia	Fetal/neonatal bradycardia	Fetal/neonatal bradycardia	Fetal/neonatal bradycardia	Fetal/neonatal bradycardia
Pharmacodynamics					
Respiratory				Bronchospasm	Bronchospasm

	Esmolol	Metoprolol	Atenolol	Propranolol	Labetalol
CVS	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP	↓ HR ↓ Inotropy ↓ MVO ₂ ↓ MAP ↑ SVR (β ₂ effect) Bradycardia/AV block	↓ SVR ↓ MAP HR/CO largely maintained acutely
CNS				Tiredness Nightmares Sleep disturbance	Orthostatic dizziness
Endocrine				↓ T ₄ → T ₃ conversion	
Metabolic	↓ Insulin release (β ₂ effect if selectivity lost) Masked hypoglycaemia	↓ Insulin release (β ₂ effect if selectivity lost) Masked hypoglycaemia	↓ Insulin release (β ₂ effect if selectivity lost) Masked hypoglycaemia	↓ Insulin release Blunted hypoglycaemic response	↓ Insulin release Masked hypoglycaemia
Renal	↓ Renin	↓ Renin	↓ Renin		↓ Renin

	Esmolol	Metoprolol	Atenolol	Propranolol	Labetalol
Considerations	Venous irritant Effects resolve within ~30 minutes of cessation IV verapamil → cardiac arrest Incompatible with NaHCO ₃ , furosemide, diazepam or thiopental β ₁ selectivity ↓ at higher doses	CYP2D6 polymorphism → variable half-life β ₁ selectivity ↓ at higher doses Verapamil/diltiazem → severe bradycardia/AV block/↓ inotropy Abrupt withdrawal → angina, arrhythmia or MI	Renal impairment → dose reduction β ₁ selectivity ↓ at higher doses Verapamil/diltiazem → severe bradycardia/AV block/↓ inotropy Abrupt withdrawal → angina, arrhythmia or MI	Verapamil/diltiazem → severe bradycardia/AV block/↓ inotropy Abrupt withdrawal → angina, arrhythmia or MI	Keep supine during IV administration Rare severe hepatocellular injury; do not re-start after injury Verapamil/diltiazem → severe bradycardia/AV block/↓ inotropy Abrupt withdrawal → angina, arrhythmia or MI

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Amiodarone

Amiodarone is an antiarrhythmic agent with a complex mechanism of action and many effects.

- K^+ channel blockade in cardiac myocytes, inhibiting the slow outward current and slowing repolarisation (Class III)
- β -blocker-like activity on SA and AV nodes, decreasing automaticity and slowing nodal conduction (Class II)
- Ca^{2+} channel blocker-like activity on L-type Ca^{2+} channels, decreasing the slow inward Ca^{2+} current, increasing depolarisation time and decreasing nodal conduction (Class IV)
- α -blocker-like activity, decreasing SVR

Table 1: Amiodarone

	Amiodarone
Aliases	Cordarone
Class	Class III antiarrhythmic
Chemistry	Iodinated benzofuran
Dose	
Cardiac arrest (IV)	5mg/kg up to 300mg IV
Arrhythmia (IV)	5mg/kg over 20 minutes - 2 hours; then 15mg/kg over 24 hours
Maintenance (PO)	200mg TDS for 1 week; then 200mg BD for 1 week; then 200mg OD
Pharmaceutics	
Presentation	100mg or 200mg tablet; clear pale-yellow 150mg/3mL injection
Compatibility	5% glucose only; do not mix with other drugs

Amiodarone	
Solubility	Highly lipid-soluble; poorly water-soluble
Pharmacokinetics	
Absorption	20-80% oral bioavailability; slow and erratic; T _{max} ~4.5 hours
Distribution	pK _a : 6.56 V _d : 66 L/kg (adipose tissue and muscle accumulation) Protein binding: 96%
Metabolism	Half-life: 29 days (mean) Duration: Delayed by redistribution; maximum Class I/IV effects may take weeks Pathway: Hepatic CYP3A4 → desethylamiodarone Metabolites: Desethylamiodarone (active)
Elimination	Renal: <5% renal; predominantly biliary, skin and lacrimal; not dialysable
Mechanism	
Target	IKr potassium channel Inactivated Na ⁺ channel L-type Ca ²⁺ channel β and α adrenoreceptors
Action	IKr block → ↑ APD/QT; Na ⁺ block → ↓ Phase 0; non-competitive β block + L-type Ca ²⁺ block → ↓ SA/AV conduction; α-blocker-like activity → ↓ SVR

	Amiodarone
Contraindications	Porphyria Sinus bradycardia or SA block Sick sinus or advanced AV block without pacemaker Thyroid dysfunction TdP-provoking drugs Severe ↓ BP or circulatory collapse (IV) Neonates (benzyl alcohol)
Pregnancy	
Code	C
Justification	Long t _{1/2} ; fetal thyroid dysfunction/bradycardia; check neonatal TFTs after exposure
Pharmacodynamics	
Respiratory	Pneumonitis (10% 3-year risk) Pulmonary fibrosis Pleuritis
CVS	↓ HR/BP/SVR ↑ QT; TdP uncommon Peripheral venous irritation Rapid IV → ↓ BP (polysorbate 80)
CNS	Corneal microdeposits → blurred vision Sleep disturbance; vivid dreams Peripheral neuropathy
Endocrine	Hyperthyroidism (1%) Hypothyroidism (6%)
Musculoskeletal	Photosensitivity Grey skin discolouration
Gastrointestinal	Nausea; vomiting Hepatitis; cirrhosis; jaundice
Considerations	CYP3A4/2C9/2D6 + P-gp inhibition → ↑ digoxin, statin, warfarin, phenytoin and antiarrhythmic exposure Also exhibits Class I, II, and IV activity

A **mnemonic** for some of the rarer effects is **BITCH**:

- **B**lue skin
- **I**nterstitial lung disease

- Thyroid
 - Corneal
 - Hepatic
-

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Sotalol

The D-isomer of sotalol is a **class III antiarrhythmic**, whilst the L-isomer also has **class II activity**.

Table 1: Sotalol

	Sotalol
Aliases	Sotacor
Class	Class II/III antiarrhythmic
Chemistry	Ethanolamine derivative
Dose	
Arrhythmia (PO)	40-160mg BD
Arrhythmia (IV)	0.5-1.5mg/kg over 10 minutes; max 320mg/day
Pharmaceutics	
Presentation	80mg or 160mg tablet; clear 10mg/mL injection
Compatibility	0.9% NaCl; 5% glucose
Solubility	Highly water-soluble
Pharmacokinetics	
Absorption	~100% oral bioavailability; Tmax 2-3 hours; ↓ with food or milk
Distribution	pKa: 8.3
	Vd: 1.6-2.4 L/kg
	Protein binding: None
Metabolism	Half-life: 12.7 hours
	Pathway: Not metabolised

	Sotalol
Elimination	Renal: 75% urine unchanged within 72 hours; <10% faeces
Mechanism	
Target	β adrenoreceptor (non-selective; no intrinsic sympathomimetic activity) Potassium channel (repolarising K^+ current, Phase 3)
Action	D/L isomers $\rightarrow K^+$ block $\rightarrow \uparrow$ APD/QT; L-isomer $\rightarrow$ non-selective β block $\rightarrow \downarrow$ AV conduction; no membrane-stabilising activity
Contraindications	Bronchospasm, asthma or COPD Sinus bradycardia or sick sinus Second/third-degree AV block without pacemaker Shock or uncontrolled CCF CrCl <10mL/min Long QT, TdP or hypomagnesaemia
Pregnancy	
Code	C
Justification	Fetal/neonatal bradycardia
Pharmacodynamics	
Respiratory	Bronchospasm
CVS	$\downarrow$ HR/BP/inotropy; AV block $\uparrow$ QT TdP; $\uparrow$ with high dose, long QT, electrolyte disturbance or renal impairment
CNS	Masking of hypoglycaemia symptoms
Genitourinary	Sexual dysfunction
Considerations	Renal impairment $\rightarrow$ accumulation + $\uparrow$ TdP risk Avoid Class Ia/III antiarrhythmics Abrupt withdrawal $\rightarrow$ angina, arrhythmia or MI

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Digoxin

Digoxin is a cardiac glycoside used in the treatment of atrial arrhythmias and in cardiac failure as a positive inotrope.

Digoxin has both a direct and indirect **mechanism of action**:

- Direct
Inhibits cardiac Na^+/K^+ ATPase, causing:
 - Increasing intracellular $[\text{Na}^+]$, decreasing activity of the $\text{Na}^+/\text{Ca}^{2+}$ pump
 - Increased intracellular Ca^{2+} increases inotropy
 - Decreased K^+ results prolongs refractory period of the AV node and bundle of His
- Indirect
Parasympathomimetic effects by increasing ACh release at cardiac muscarinic receptors. This is due to sensitisation of baroreceptors, and increased vagal output from the nucleus of tractus solitarius
 - Slows AV nodal conduction and ventricular response
This improves coronary blood flow, increasing time for ventricular filling, and improving cardiac output.

Table 1: Digoxin

	Digoxin
Aliases	Lanoxin
Class	Cardiac glycoside
Chemistry	Cardiac glycoside
Dose	
Maintenance (PO)	62.5 - 250mcg OD

	Digoxin
Load (PO)	10-20mcg/kg
Load (IV)	0.5-1mg total; 0.25-0.5mg Q4-6H, each over ≥ 5 minutes
Pharmaceutics	
Presentation	62.5mcg or 250mcg tablet; 50mcg/mL elixir; clear 250mcg/mL injection
Compatibility	5% glucose; 0.9% NaCl; 4% glucose/0.18% NaCl
Solubility	Essentially insoluble in water
Pharmacokinetics	
Absorption	70-80% oral bioavailability; P-glycoprotein efflux; onset 2-3 hours
Distribution	pKa: 7.15
	Vd: 5-11 L/kg (dependent on lean body mass)
	Protein binding: 25%
Metabolism	Half-life: 35-44 hours
	Pathway: Minimal hepatic (~16% of clearance)
	Metabolites: Dihydrodigoxin, Digoxigenin
Elimination	Renal: 50-70% of IV dose excreted unchanged; half-life prolonged in renal failure
Mechanism	
Target	Na ⁺ /K ⁺ ATPase
Action	Na ⁺ /K ⁺ ATPase inhibition $\rightarrow$ $\uparrow$ intracellular Na ⁺ $\rightarrow$ $\downarrow$ Na ⁺ /Ca ²⁺ exchange $\rightarrow$ $\uparrow$ Ca ²⁺ and inotropy; $\uparrow$ vagal tone $\rightarrow$ $\downarrow$ AV conduction; $\uparrow$ Phase 4 slope and automaticity

	Digoxin
Contraindications	2nd/3rd-degree AV block VT/VF Digoxin-induced arrhythmia AF with accessory pathway Hypertrophic obstructive cardiomyopathy
Pregnancy	
Code	A
Pharmacodynamics	
CVS	↓ HR; ↑ inotropy/CO/coronary flow ↑ PR; ↓ QT Bigeminy; PVCs; AV block; SVT; VT Arterial vasoconstriction
CNS	Red-green dyschromatopsia; visual disturbance Headache; ↓ LOC
Metabolic	Gynaecomastia
Gastrointestinal	Nausea Anorexia
Haematological	Eosinophilia Rash
Considerations	Narrow therapeutic index Renal impairment → ↓ elimination Hypokalaemia/hypomagnesaemia/hypercalcaemia → ↑ toxicity Toxicity + DC cardioversion → ↑ VF risk ↑ Levels: amiodarone, captopril, erythromycin, verapamil ↓ Levels: antacids, cholestyramine, phenytoin, metoclopramide

Table 2: Digoxin Drug Interactions

Interaction	Drug
Increased level	Amiodarone, captopril, erythromycin, verapamil
Decreased level	Antacids, cholestyramine, phenytoin, metoclopramide

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Adenosine

Adenosine acts via A₁ adenosine receptors in the SA and AV node, which when stimulated open K⁺ channels causing hyperpolarisation and a reduction in Ca²⁺ current, with subsequent blockade of AV nodal conduction.

Table 1: Adenosine

	Adenosine
Aliases	Adenocor
Class	Antiarrhythmic
Chemistry	Endogenous purine nucleoside
Dose	
SVT (IV)	3mg, then 6mg, then 12mg; rapid bolus
Pharmaceutics	
Presentation	Clear colourless 3mg/mL solution
Solubility	Slightly water-soluble
Pharmacokinetics	
Metabolism	Half-life: <10 seconds
	Pathway: Cellular uptake → inosine and AMP
	Metabolites: Inosine, AMP
Mechanism	
Target	A ₁ adenosine receptor
Action	A ₁ activation → ↑ K ⁺ efflux + ↓ Ca ²⁺ current → SA/AV hyperpolarisation and transient AV block

Adenosine	
Contraindications	Sick sinus syndrome 2nd/3rd-degree AV block without pacemaker Asthma or COPD Long QT syndrome Severe ↓ MAP or decompensated CCF
Pregnancy	
Code	B2
Pharmacodynamics	
Respiratory	Bronchospasm Dyspnoea; ↑ RR/VT
CVS	Transient AV block/asystole AF/flutter; ventricular ectopy ↓ MAP
CNS	Headache; dizziness
Considerations	Dipyridamole → ↑ effect Methylxanthines → ↓ effect Carbamazepine → ↑ AV block

Table 2: Adenosine Drug Interactions

Interaction	Drug
Increased effect	Dipyridamole
Decreased effect	Methylxanthines, such as aminophylline and caffeine

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Magnesium

Mg²⁺ is a cation that is important for neurotransmission and neuromuscular excitability. Magnesium:

- Inhibits ACh release at the NMJ
- Acts a cofactor in multiple enzyme systems
- Is important in the production of:
 - ATP
 - DNA
 - RNA

Table 1: Magnesium Sulfate

	Magnesium sulfate
Class	Electrolyte replacement
Chemistry	Divalent cation salt
Dose	
Hypomagne- saemia (IV)	10-20mmol over 10-20 minutes
Pharmaceutics	
Presentation	Clear 2mmol/mL solution; dilute 10mmol in 100mL for peripheral IV
Compatibility	0.9% NaCl; Hartmann's; 5% glucose; glucose/NaCl
Solubility	Water-soluble
Pharmacokinetics	
Absorption	30% oral bioavailability; ↑ with magne- sium depletion
Distribution	pKa: -3
	Protein binding: 30%

	Magnesium sulfate
Metabolism	Half-life: Approximately 4 hours
	Pathway: Not metabolised; cofactor in numerous metabolic processes
Elimination	Renal: Renal; reabsorption mainly in TAL/DCT; $\uparrow$ plasma Mg^{2+} $\rightarrow$ $\downarrow$ reabsorption
Mechanism	
Target	NMDA receptor L-type calcium channel
Action	$\uparrow$ Sarcoplasmic Ca^{2+} uptake $\rightarrow$ smooth muscle relaxation; $\downarrow$ acetylcholine release at NMJ; NMDA + L-type Ca^{2+} channel inhibition $\rightarrow$ CNS depression and $\downarrow$ AV conduction; Na^{+}/K^{+} ATPase cofactor
Contraindications	Heart block Severe renal failure
Pharmacodynamics	
Respiratory	Bronchodilation Respiratory depression
CVS	$\downarrow$ SVR $\downarrow$ MAP $\downarrow$ HR
CNS	CNS depression Anticonvulsant
Musculoskeletal	Muscle weakness Areflexia
Genitourinary	$\downarrow$ Uterine tone and contractility
Considerations	Potentiates neuromuscular blockers and CNS depressants Nifedipine $\rightarrow$ exaggerated $\downarrow$ MAP

Table 2: Serum Magnesium Concentration and Clinical Effects

[Plasma]	Effect
< 0.7 mmol.L ⁻¹	Arrhythmia
4-6 mmol.L ⁻¹	Nausea, hyporeflexia , speech impairment
6-10 mmol.L ⁻¹	Weakness, respiratory depression, bradycardia
> 10 mmol.L ⁻¹	Cardiac arrest

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Atropine

Naturally occurring tertiary amine which competitively antagonises ACh at the muscarinic receptor, causing parasympatholytic effects.

Table 1: Atropine

Atropine	
Class	Muscarinic antagonist
Chemistry	Racemic tertiary amine tropane alkaloid; L-isomer active
Dose	
Bradycardia	0.5 - 1mg IV Q5min; max 2mg
Premedication	300 - 600µg IM/SC 1h before anaesthesia, or IV before induction
Reversal	600µg - 1.2mg slow IV with anti- cholinesterase
Organophosphates	1 - 2mg IV/IM (up to 6mg if severe) Q5min; then 0.02 - 0.08mg/kg/h IV
Pharmaceutics	
Presentation	600µg/mL clear colourless solution
Solubility	Atropine sulfate is highly water-soluble
Pharmacokinetics	
Absorption	90% PO bioavailability; IV cardiac onset 2 - 4min
Distribution	pKa: 15.5 Vd: 1 - 6L/kg Protein binding: 50%

Atropine	
Metabolism	Half-life: 2 - 5h Duration: Cardiac 2 - 3h Pathway: ~50% hepatic hydrolysis Metabolites: Tropane, tropic acid, noratropine
Elimination	Renal: ~50% unchanged; remainder as metabolites
Mechanism	
Target	Muscarinic receptors (M1 - M5)
Action	Reversible, competitive, non-selective muscarinic antagonism → parasympatholysis
Pregnancy	
Code	A
Pharmacodynamics	
Respiratory	Bronchodilation ↓ Secretions
CVS	↑ HR ↑ AV nodal conduction Arrhythmias
CNS	Central anticholinergic syndrome; confusion/delirium/hallucinations Mydriasis; ↑ IOP ↑ CSF secretion in choroid Cerebral vasoconstriction
Musculoskeletal	↓ Sweating → hyperthermia
Genitourinary	Urinary retention
Gastrointestinal	↓ LoS tone ↓ Gastric acid secretion ↓ Intestinal motility Constipation Antiemetic effect

Atropine	
Considerations	Relative cautions: pyloric stenosis, glaucoma, prostatic hypertrophy Crosses BBB and placenta Doses <0.5mg may cause paradoxical bradycardia in adults ↓ Airway secretions → mucus plugging

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Diuretics

An understanding of the pharmacology of diuretics.

Diuretics are drugs that act on the kidney to increase urine production. They can be classified by their mechanism of action into:

- Thiazides
- Loop diuretics
- Potassium sparing
- Aldosterone antagonists
- Osmotic
- Carbonic Anhydrase inhibitors

Table 1: Common Pharmacological Features of Diuretics

Property	Diuretics
Absorption	Typically poor bioavailability (exception: acetazolamide)
Distribution	Variable protein binding
Metabolism	Generally not metabolised. Key exceptions: Spironolactone is extensively metabolised with active metabolites, and a small amount of furosemide is metabolised to glucuronide.
Elimination	Renal elimination of unchanged drug

Property	Diuretics
CVS	Reduced intra and extravascular volume
Renal	Any diuretic which inhibits sodium reabsorption can precipitate hypokalaemia (as a greater intra-luminal concentration of sodium results in exchange of sodium for potassium ions), hyponatraemia (as there is still a net loss of sodium), and alkalosis (from loss of hydrogen ions exchanged for sodium, or the overall raised strong ion difference).

Table 2: Comparison of Diuretics

	Hydrochlorothiazide	Furosemide	Amiloride	Spironolactone	Mannitol	Acetazolamide
Aliases	Dithiazide	Furosemide Lasix		Aldactone	Osmitrol	Diamox
Class	Thiazide diuretic	Loop diuretic	ENaC blocker; potassium-sparing diuretic	Mineralocorticoid receptor antagonist; potassium-sparing diuretic	Osmotic diuretic; osmotherapy agent	Carbonic anhydrase inhibitor; diuretic
Chemistry	Benzothiadiazine derivative	Anthranilic acid derivative	Pyrazinoylguanidine derivative	Steroid lactone	Six-carbon sugar alcohol	Sulfonamide derivative
Dose						
Hypertension	12.5-50mg PO daily; max 100mg/day			50-100mg PO daily		
Oedema	25-100mg PO once or twice daily; max 200mg/day	20-80mg PO; repeat or increase by 20-40mg after ≥6h; max 400mg/day (except advanced renal failure)				5mg/kg PO daily for 1-2 days; alternate with a rest day

	Hy- drochloroth- iazide	Frusemide	Amiloride	Spiro-nolac- tone	Mannitol	Acetazo- lamide
Oedema IV		20-40mg slow IV; re- peat or in- crease by 20mg after ≥2h				
Pulmonary oedema		40mg slow IV; may in- crease to 80mg after 1h				
Oedema or hypertension			5mg PO daily with a kali- uretic di- uretic; may ↑ to 10mg; 15mg then 20mg/day only if hy- pokalaemia persists at 10mg, with electrolyte monitoring			
CCF oedema				100mg PO daily; 25- 200mg/day maintenance		

	Hy- drochloroth- iazide	Frusemide	Amiloride	Spironolac- tone	Mannitol	Acetazo- lamide
Raised ICP					0.25 - 1g/kg IV	
Renal test					0.2g/kg IV over 3 - 5min	
Pharmaceutics						
Presentation	25mg tablets	10mg/mL oral solution; 10mg/mL solution for injection	5mg tablets	25mg and 100mg tablets	10% and 20% solu- tions for IV infusion	250mg tablets
Compatibility		Compatible with 0.9% NaCl and Hartmann's; alkaline so- lution			Do not co- infuse with blood; KCl/ NaCl may precipitate; some antibi- otics incom- patible	
Solubility	Practically insoluble in water	Practically insoluble in water; freely soluble in dilute alkali	Sparingly soluble in water	Practically insoluble in water	Water sol- uble; may crystallise at low temper- ature	Slightly solu- ble in water
Pharmacokinetics						
Absorption	65 - 75% PO bioavailability	10 - 100% PO bioavail- ability; mainly gastric	50% PO bioavailabil- ity; reduced by food	95% PO bioavailability with food	IV only	60 - 100% PO bioavail- ability

	Hy- drochloroth- iazide	Frusemide	Amiloride	Spironolac- tone	Mannitol	Acetazo- lamide
Distribution	pKa: 9.09	pKa: 3.6	pKa: 8.67	pKa: 18.01	pKa: 12.59	pKa: 7.2
	Vd: 1.5 - 4.2L/kg	Vd: 0.1 - 0.2L/kg	Vd: 5L/kg	Vd: ~2L/kg	Vd: 0.2L/kg	Vd: 0.3L/kg
	Protein bind- ing: 40-68%	Protein bind- ing: 95%	Protein bind- ing: 23%	Protein bind- ing: 90%		Protein bind- ing: 90%

	Hydrochlorothiazide	Furosemide	Amiloride	Spironolactone	Mannitol	Acetazolamide
Metabolism	Half-life: 6-9h Duration: 6-12h Pathway: None	Half-life: 30-120min Duration: ~6h PO; ~2h IV Pathway: ~35% glucuronidation, predominantly renal Metabolites: Active glucuronide; ~25% parent activity	Half-life: 6-9h Duration: ~24h Pathway: None	Half-life: 2-3h; active metabolites 10-12h Duration: Onset 3-4h; maximum effect ~2 days Pathway: Complete hepatic metabolism Metabolites: Canrenone, Sulfur-containing active metabolites	Pathway: Small amount metabolised	Half-life: 6-10h Duration: 8-12h Pathway: None

	Hydrochlorothiazide	Furosemide	Amiloride	Spiro-nolactone	Mannitol	Acetazolamide
Elimination	Renal: Virtually all unchanged	Renal: ~65% unchanged by filtration and proximal tubular secretion	Renal: ~50% of a 20mg dose unchanged in urine and ~40% in faeces within 72h	Renal: Metabolites (including inactive); some biliary excretion	Renal: Predominantly unchanged by glomerular filtration; <10% tubular reabsorption	Renal: All unchanged
Mechanism						
Target	NCC	NKCC2 chloride-binding site	ENaC	Mineralocorticoid receptor Androgen receptor		Carbonic anhydrase
Action	NCC inhibition in the distal convoluted tubule → ↑ Na ⁺ and Cl ⁻ excretion and ↑ Ca ²⁺ reabsorption	NKCC2 inhibition in the thick ascending limb → ↓ medullary gradient and ↑ Na ⁺ , K ⁺ , Cl ⁻ and water excretion	ENaC inhibition in the distal nephron → ↓ Na ⁺ reabsorption and ↓ K ⁺ and H ⁺ secretion	Competitive aldosterone antagonism in the distal nephron → ↓ ENaC expression, ↑ Na ⁺ excretion and ↓ K ⁺ secretion	↑ Extracellular and tubular osmolality → intracellular water shift, medullary washout and osmotic diuresis	Non-competitive inhibition in the proximal tubule → ↓ HCO ₃ ⁻ reabsorption and bicarbonate diuresis

	Hydrochlorothiazide	Furosemide	Amiloride	Spiro-nolactone	Mannitol	Acetazolamide
Contraindications	Anuria	Anuria or complete renal shutdown Severe hypokalaemia or hyponatraemia Hypovolaemia, dehydration or hypotension Hepatic coma or pre-coma Pregnancy or breastfeeding	Hyperkalaemia Anuria, renal insufficiency or diabetic nephropathy Concurrent potassium-sparing diuretic	Acute renal insufficiency, significant renal impairment or anuria Hyperkalaemia or Addison's disease Pregnancy Concurrent eplerenone	Pre-existing plasma hyperosmolality Severe heart failure or pulmonary oedema Anuria due to severe renal disease or no response to test dose Active intracranial bleeding except during craniotomy Severe dehydration Disrupted blood-brain barrier	Hyperchlo-raemic acidosis Hypokalaemia or hyponatraemia Adrenal insufficiency GFR <10mL/min Marked hepatic impairment or cirrhosis Severe glaucoma due to peripheral anterior synechiae or haemorrhagic glaucoma Long-term use in chronic non-congestive angle-closure glaucoma

	Hydrochlorothiazide	Furosemide	Amiloride	Spironolactone	Mannitol	Acetazolamide
Pregnancy						
Code	C	C	C	B3	B2	B3
Justification	May cause fetal electrolyte disturbance and neonatal thrombocytopenia	May cause fetal electrolyte disturbance and neonatal thrombocytopenia	May cause fetal electrolyte disturbance	Potential feminisation of the male fetus		
Pharmacodynamics						
Respiratory	Acute pulmonary oedema (uncommon); ARDS (very rare)	↓ Pulmonary oedema			Initial volume expansion may cause pulmonary oedema	↑ RR secondary to metabolic acidosis
CVS	↓ MAP, plasma volume and SVR	↓ Preload, circulating volume and MAP	↓ MAP and circulating volume	↓ MAP and circulating volume	Initial ↑ circulating volume; later hypovolaemia from diuresis	↓ MAP from diuresis
CNS	Dizziness and headache	Tinnitus and deafness, especially with rapid or high-dose IV use	Headache, dizziness and paraesthesia	Drowsiness, dizziness, confusion and ataxia	↓ ICP and brain volume Headache, confusion, seizures and rebound ↑ ICP	↓ IOP and ICP Paraesthesia, drowsiness and ataxia

	Hydrochlorothiazide	Furosemide	Amiloride	Spironolactone	Mannitol	Acetazolamide
Endocrine				Gynaecomastia, menstrual irregularity and impotence		
Musculoskeletal	Cramps and weakness with electrolyte depletion	Cramps and weakness with electrolyte depletion	Weakness and cramps with electrolyte disturbance		Cramps	
Metabolic	Hypokalaemic hypochloreaemic alkalosis and hyponatraemia Hyperuricaemia, hyperglycaemia and hypercalcaemia	Hypokalaemic hypochloreaemic alkalosis, hyponatraemia and hypomagnesaemia Hyperuricaemia and hyperglycaemia	Hyperkalaemia, hyponatraemia and metabolic acidosis	Hyperkalaemia, hyponatraemia and type 4 RTA	Hypo- or hypernatraemia, hypo- or hyperkalaemia and acid-base disturbance ↑ Serum osmolality	Hyperchloraemic metabolic acidosis, hypokalaemia and hyponatraemia

	Hydrochlorothiazide	Furosemide	Amiloride	Spironolactone	Mannitol	Acetazolamide
Renal	RBF usually unchanged; GFR may ↓ without significant volume depletion; moderate diuresis	May ↑ RBF if volume maintained; GFR usually unchanged; potent diuresis	Weak natriuresis with K ⁺ retention	Mild diuresis with K ⁺ retention	↑ RBF and osmotic diuresis Osmotic nephrosis and AKI	Urinary alkalisation and diuresis Crystalluria and nephrolithiasis
Genitourinary				Antiandrogenic effects		
Gastrointestinal	Nausea, vomiting and pancreatitis	Nausea, vomiting and pancreatitis	Nausea, diarrhoea and abdominal pain	Nausea, vomiting, diarrhoea and gastritis	Nausea, vomiting and thirst	Nausea, vomiting and diarrhoea
Haematological	Thrombocytopenia and other blood dyscrasias	Thrombocytopenia and other blood dyscrasias				Blood dyscrasias

	Hydrochlorothiazide	Furosemide	Amiloride	Spironolactone	Mannitol	Acetazolamide
Considerations	NSAIDs ↓ diuresis; thiazides ↑ lithium toxicity May cause acute angle-closure glaucoma; cumulative-dose association with non-melanoma skin cancer Avoid after hydrochlorothiazide-associated ARDS	IV ≤4mg/min; ≤2.5mg/min if creatinine >5mg/dL; rapid injection ↑ ototoxicity Aminoglycosides ↑ ototoxicity; NSAIDs ↓ natriuresis; diuretics ↑ lithium toxicity Hypoalbuminaemia ↑ Vd and ↓ tubular delivery	Monitor K ⁺ closely, especially with renal impairment, diabetes, ACE inhibitors or ARBs Avoid K ⁺ supplements and potassium-containing salt substitutes	Monitor K ⁺ and renal function; avoid K ⁺ supplements and other potassium-sparing drugs ACE inhibitors, ARBs and NSAIDs ↑ hyperkalaemia; ↑ lithium and digoxin exposure	Monitor osmolality, electrolytes, renal function and urine output Give renal test dose in marked oliguria; stop if urine output remains inadequate Inspect 20% solution for crystals; warm to redissolve, cool and reinspect; use a sterile filter-type set Selected PI does not recommend use in acute TBI or acute stroke	Reduce dose by half or extend interval if GFR >10mL/min with renal impairment Salicylates may cause severe toxicity; acetazolamide alters weak-acid/base clearance

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Intravenous Fluids

Intravenous fluids can be classified into:

- **Crystalloids**

Can pass freely through a semipermeable membrane.

Can be further classified into:

- ECF replacement solutions
Have a $[Na^+]$ similar to ECF, such that they are confined mostly to the ECF.
- Maintenance solutions
Designed to distribute throughout TBW.
- Special solutions
These solutions don't fit into the above two categories, and include:
 - Hypertonic saline
 - Mannitol
 - 8.4% Sodium Bicarbonate

- **Colloids**

Substance evenly dispersed throughout another solution in which it is insoluble. Can be classified into:

- Naturally occurring
 - Albumin
Heat-treated human albumin.
 - Produced at low pH but not technically sterile
Use within 3 hours of opening.
 - Contributes to plasma oncotic pressure
 - Contributes to drug and endogenous substance binding
- Synthetic
 - Dextrans
High molecular weight sugars synthesised from sucrose by bacteria.

- Interfere with haemostasis due to vWF inhibition
- Interfere with blood crossmatch
- Risk of anaphylaxis
- Gelatins
 - High molecular weight proteins produced by collagen hydrolysis.
 - Greatest anaphylaxis risk
 - Do not interfere with clotting
- Hydroxyl-ethyl starches
 - Risk of anaphylaxis
 - Risk of renal impairment
 - Accumulate in the reticuloendothelial system

Table 1: Composition of Common Crystalloid Solutions

Contents (mmol.L ⁻¹)	0.9% NaCl	Hartmann's	Plasmalyte
Na ⁺	154	130	140
Cl ⁻	154	109	98
K ⁺		4	5
Ca ²⁺		3	
Mg ²⁺			1.5
Lactate		28	
Acetate			27
Gluconate			23
pH	5.0	6.5	5.5

References

http://www.anaesthesiamcq.com/FluidBook/fl7_2.php

Neuropharmacology

Neurotransmitters

Describe the major neurotransmitters and their physiological role, with particular reference to GABA, excitatory and inhibitory amino acids, acetylcholine, noradrenaline, dopamine and serotonin and NMDA receptor

GABA

Gamma aminobutyric acid is the major inhibitory CNS neurotransmitter. GABA receptors have three subtypes:

- GABA_A
 - Inotropic receptor important for the action of many drugs.
 - Pentameric structure
 - 2 α
Bind GABA.
 - 2 β
 - 1 γ
 - Affected by many different drugs:
 - Benzodiazepines
Positive allosteric modulation at the α/γ interface.
 - General anaesthetic agents
Including propofol, barbiturates, halogenated volatiles, and etomidate.
 - Act at the β subunit
Cause a conformational change which

increases Cl^- opening time, hyperpolarising the cell.

- GABA_B
Metabotropic receptor.
- GABA_C
Inotropic receptor located only in the retina.

NMDA

N-methyl D-aspartate receptor is an inotropic receptor that is:

- Agonised by glutamate
 - Glycine is co-agonist
 - Voltage dependent
 - Central pore usually blocked by an Mg^{2+} ion
 - Becomes unblocked when partially depolarised
 - Important in the action of drugs which do not act at the GABA_A receptor
- Antagonised by:
- Ketamine
 - Xenon
 - N_2O
-

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Anticonvulsants

An understanding of the pharmacology of anti-depressant, anti-psychotic, anti-convulsant, and anti-Parkinsonian medication

Anticonvulsants work via a number of different mechanisms:

Sodium Channel Blockers

Sodium channel blockers:

- Stabilise the inactive state of the channel, preventing return to the active state and prevent generation of further action potentials
This halts post-tetanic potentiation and limits the development of seizure activity.
- May also have Class I antiarrhythmic properties
Due to Na^+ blocking effects.
- Include:
 - Phenytoin
 - Carbamazepine
 - Lamotrigine

GABA Mediators

GABA is the key inhibitory neurotransmitter in the CNS.
GABA mediators:

- Enhance the effect of GABA
Multiple potential mechanisms:
 - Direct GABA -receptor agonists
e.g. Benzodiazepines and phenobarbital.
 - Positive allosteric modulation
e.g. Propofol and thiopentone.
 - GABA reuptake inhibition
e.g. Tiagabine.
 - GABA transaminase inhibition
e.g. Vigabatrin.
 - Increase GABA synthesis
e.g. Sodium Valproate.

Glutamate Blockers

Glutamate is an important CNS excitatory neurotransmitter.
Glutamate antagonists:

- Are generally avoided due to their side effect profile, which includes psychosis and hallucinations
- Include topiramate

Other Agents

Gabapentin and pregabalin:

- Do not appear to mediate GABA
 - Inhibit of excitatory $\alpha_2\delta$ voltage-gated calcium channels in the CNS
This gives them anticonvulsant properties.
-

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Local Anaesthetics

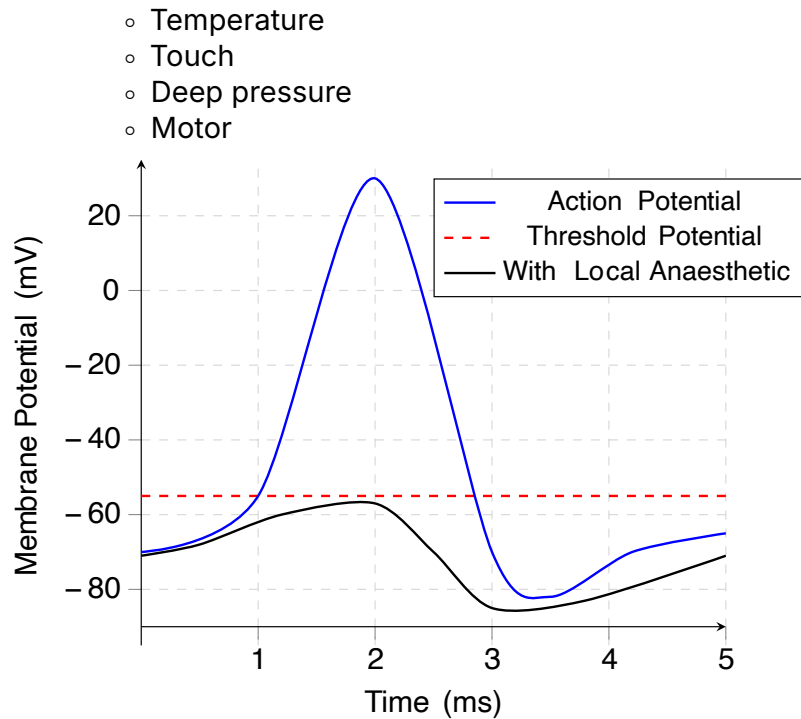
Understanding of the pharmacology of local anaesthetic drugs, including their toxicity

Local anaesthetic drugs create a use-dependent temporary blockade of neuronal transmission by blocking the voltage-gated sodium channel in the cell membrane, preventing depolarisation.

Mechanism of Action

Action is dependent on blockade of the sodium channel. Two theories exist:

- **Unionised** drug **passes through the cell membrane**, and then becomes **ionised intracellularly**
- The ionised drug is then able to bind to the **open** sodium channel, and prevent conduction of sodium and therefore generation of an action potential
 - Local anaesthetics also display reduced affinity for K^+ and L-type Ca^{2+} channels
 - This theory explains use-dependent blockade, as sodium channels can only be blocked in their open state
- An alternative suggested mechanism of action is the drug enters the cell membrane and mechanically distorts the channel, rendering it ineffective
- Onset is inversely proportional to the size of the fibre
 - From fastest to slowest:
 - Pain



Chemical Structure of Local Anaesthetics

All local anaesthetics are weak bases consisting of:

- A hydrophilic component
- A lipophilic aromatic ring
- An amide or ester link connecting the two

Chemical structure influences pharmacological behaviour:

- Hydrophilic portion
 - Typically the tertiary amine.
 - Determines ionisation
 - 3 bonds: Lipid soluble
 - 4 bonds: Water soluble
- Lipophilic portion
 - Typically aromatic ring.

- Determines lipid solubility, and therefore potency, toxicity, and duration of action
- Ester vs. amide
 - Amides
 - Hepatically metabolised (hydroxylation and N-de-alkylation)
This is slower, therefore there is a **greater risk** of systemic toxicity.
 - Stable in solution
 - Esters
 - Heat-sensitive
Cannot be autoclaved.
 - Rapidly hydrolysed in plasma
Organ independent elimination.
 - Have a greater incidence of allergy
Due to the inactive metabolite PABA.
- Amine group length
 - Potency and toxicity increase as carbon-chain increases
 - Toxicity (but not potency) continues to increase beyond 10 carbons
- Isomerism
 - Alters behaviour:
 - Levobupivacaine is less toxic
 - R-ropivacaine is less potent and more toxic

Key Characteristics of Local Anaesthetics

Characteristics are related to chemical structure. These include:

- **Potency**
 - Potency is expressed with the minimum effective concentration of local anaesthetic (C_m)
This is the concentration of LA that results in complete block of a nerve fibre in 50% of subjects in standard conditions. More potent agents have a lower C_m .

- Potency is a function of:
 - Lipid solubility
Potency (and also toxicity) increases with greater lipid solubility.
 - Vasodilator properties
In general, local anaesthetics cause vasodilation in low concentrations, and vasoconstriction at high concentrations (except cocaine, which causes vasoconstriction at all concentrations).

- **Duration of action**

Duration of action is a function of:

- Drug factors
 - Vasodilator properties
Vasoconstriction increases the duration of block.
 - Use of additives
Addition of adrenaline to lignocaine increases duration of block.
 - Lipid solubility
Increased lipid solubility increases duration of action, as agent remains in the nerve for longer.
 - Potency therefore has a positive correlation with duration of action
 - Duration of action is increased when pH increases, as the ionised portion falls
- Protein binding
Highly protein bound agents have an increased duration of action due to increased tissue binding.
 - Protein binding decreases with decreasing pH, increasing the fraction of unbound drug
This is why agents such as bupivacaine are more cardiotoxic in acidotic patients.
 - Local anaesthetics are predominantly bound to α -1-acid glycoprotein (AAG)
AAG is reduced in pregnancy, increasing the free drug fraction and therefore

reducing the toxic dose of LA in pregnant patients.

- Patient factors

- Tissue pH

- Decreased duration of block when tissue pH is low.

- Metabolic impairment

- Hepatic failure increases duration of action of aminosteroids
 - Butylcholinesterase deficiency increases duration of ester local anaesthetics

- Site of administration Well vascularised tissue (e.g. intercostal area) will have greater systemic uptake of drug than vessel poor tissue.

- **Onset**

Speed of onset is related to:

- Drug factors

- Dose

- Increasing the dose increases the speed of onset, as per Fick's Law.

- Increased concentration will increase speed of onset and block density
 - Increased volume (without increasing dose, resulting in decreased concentration) will decrease speed of onset

- Lipid solubility

- An ↑ lipid solubility:

- ↑ The speed at which the local anaesthetic enters the nerve
 - Correlates with ↑ potency

- Ionised portion

- Only unionised drug can cross cell membranes. Ionisation is a function of:

- pKa
 - Tissue pH
 - This is also why anaesthetics are ineffective in anaesthetising infected tissue, as the low pH makes the majority of the LA ionised and unable to cross

the cell membrane.

- Patient factors
 - Nerve activity
Local anaesthetics produce a **frequency dependent blockade**, meaning nerves firing frequently will be blocked more rapidly than quiescent nerves
 - Nerve fibre size
Larger nerves require an increased concentration of local anaesthetic to achieve blockade than smaller nerves.
 - Nerve type
Different nerve fibres are affected at different speeds, which is mostly (though not entirely) a function of critical length.
 - A γ (proprioceptive) are affected first
 - Small myelinated A δ (sharp pain, cold) fibres are affected second
 - Large myelinated nerves are affected third These include A α (motor) and A β (touch) fibres.
 - Unmyelinated nerves are affected last These include C (dull pain, heat) fibres.
 - Hyperkalaemia
Reduces onset of action.

Toxicity

Local anaesthetics are:

- Toxic to both the **CNS** and **CVS**
- Toxicity occurs when there is an excess plasma concentration
This occurs when the rate of drug entering the systemic circulation is greater than the drug leaving the systemic circulation due to redistribution and metabolism.

Toxicity is related to the:

- **Drug factors**

- **Drug used**

- Agents are compared using the **CC/CNS ratio**, which is the ratio of the dose of drug required to cause cardiovascular collapse (CC) compared to the dose required to cause seizure. It is a crude alternative to the therapeutic index.

- **Dose used**

- Continuous infusions are more likely to cause a delayed onset of local anaesthetic toxicity.

- **Block factors**

- **Site** of administration

- This affects the rate of uptake into the systemic circulation, and the likelihood of inadvertent intravascular injection. Ranked (from highest to lowest):

- Intravascular (obviously)

- This is the most common cause of LA toxicity.

- Site is also relevant here: an injection into the carotid artery will cause toxicity at a lower dose than if injected into a peripheral vein.

- Intercostal

- Caudal

- Epidural

- Brachial plexus

- Subcutaneous

- Use of **adjuncts**

- Adrenaline

- Vasoconstrictor properties reduce systemic absorption of LA.

- **Technique**

- Frequent aspiration

- Test dose

- Use of ultrasound

- **Patient factors**

- Anything that increases peak [plasma] can lead to an increased risk of LA toxicity.

- Blood flow to affected area

- α 1-acid glycoprotein

Low levels of this protein increase free drug fraction.

- Neonates and infants have half the level of AAG than adults.
- Hepatic disease
Reduces clearance of amides, which may cause toxicity with repeated doses or use of infusions.
- Age
Organ blood flow (and therefore clearance), as well as pharmacokinetic interactions may affect clearance of LA. Both **children and the elderly** have **reduced clearance** of LA.
- Acidosis
Increases unionised portion.
- Hypercarbia
Increases cerebral blood flow.

Cardiac Toxicity

Cardiac toxicity occurs due to:

- Blocking of the cardiac Na^+ channel (K^+ and Ca^{2+} channels may also be involved)
Severity of toxicity will vary depending on how long the agent binds to the channel, with **less toxicity** caused by agents spending **less time bound**:
 - Lignocaine
Spends the shortest time bound to the channel, so causes the least amount of toxicity. This is also why lignocaine can be used as an antiarrhythmic, but other agents can not.
 - Bupivacaine
Takes **10x as long** to dissociate as lignocaine. This can lead to re-entrant arrhythmias, and then VF. The **risk** of this is **increased in tachycardia** due to use-dependent blockade.
 - Ropivacaine
Dissociates more rapidly from cardiac channels than bupivacaine.

- Direct myocardial depressant effects
Reduces cAMP levels by disrupting metabotropic receptors.

Cardiac toxicity is **triphasic**:

- Initial phase
 - Hypertension
 - Tachycardia
- Intermediate phase:
 - Hypotension
 - Myocardial depression
- Terminal phase:
 - Severe hypotension
 - Vasodilation
 - Various arrhythmias
 - Sinus bradycardia
 - Variable degree heart block
 - VT
 - VF
 - Asystole

CNS Toxicity

Local anaesthetics in their unionised state can cross the BBB and interfere with CNS conduction. **CNS toxicity is biphasic**:

- Initially, inhibitory interneurons are blocked
This causes excitatory effects:
 - Perioral tingling
 - Slurred speech
 - Visual disturbances
 - Tremulousness
 - Dizziness
 - Confusion
 - ConvulsionsTypically signifies the end of the excitatory phase.

- Secondly, there is a general depression of all CNS neurons

This causes inhibitory effects:

- Coma
- Apnoea

Treatment

Toxicity is managed with an **ABC approach**, though definitive management uses **Intralipid emulsion**:

- Intralipid is an emulsion of soya oil, glycerol, and egg phospholipids.
Mechanism of action is uncertain, but theories include:
 - Lipid sink
ILE binds unionised LA, causing it to distribute off receptor sites.
 - Fatty acid metabolism
Cardiac fatty acid metabolism is interrupted by LA. ILE provides a source of fatty acids to allow metabolism to continue.
 - Competitive antagonism
ILE may directly inhibit LA binding.
 - Dosing of Intralipid 20%:
 - Bolus of 1.5ml.kg^{-1} over 1 minute
 - Infusion at $15\text{ml.kg}^{-1}.\text{hr}^{-1}$
 - Complications include **pancreatitis**
Note that ILE interferes with amylase and lipase assays, and so these will be unreliable.
 - Note that whilst propofol can be used to treat seizures, the amount of lipid contained in propofol is inadequate to bind LA
-

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Neuraxial Blockade

Describe the physiological consequences of a central neuraxial block

Central neuraxial blockade refers to blockade of fibres in the spinal cord by administration of intrathecal or epidural local anaesthetic.

Respiratory Responses

An increasing level of block will lead to greater effects:

- Thoracic
 - Impediment to active expiration and expectoration due to blockade of intercostals and abdominal wall musculature
 - Loss of vital capacity
 - Loss of some accessory muscle use
- Cervical
 - Impediment due to diaphragmatic blockade.

Cardiovascular Responses

Occur due to blockade of sympathetic chain fibres in the thoracolumbar region.

An increasing level of block will lead to greater effects:

- Sacral
Parasympathetic blockade only. Minimal CVS effects.
- Lower thoracic/lumbar
Arteriolar and venous vasodilation in lower abdomen and lower limbs, causing a fall in SVR, BP, and GFR.
- Upper thoracic
Loss of cardioaccelerator fibres above T5, causing a reduction in heart rate and contractility, compounding hypotension due to fall in SVR.
- Cranial Nerves
Vagal blockade will reduce PNS tone and attenuate some of the loss of SNS tone.
- Brainstem
Inhibition of vasomotor centre with profound fall in CVS parameters.

CNS Responses

An increasing level of block will lead to greater effects:

- Cervical
Horner's syndrome (miosis, anhydrosis, ptosis) due to loss of sympathetic trunks.
 - Cranial nerve Pupillary dilation due to CN III blockade.
 - Brainstem and Cerebral Cortex Anaesthesia due to blockade of the reticular activating system and thalamus.
-

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Opioids

Key definitions:

- **Opiates** are all naturally -occurring substances with morphine-like properties
- **Opioids** is a general term for substances with an affinity for opioid receptors
- **Opium** is a mixture of alkaloids from the poppy plant

Classification of Opioids

- **Naturally occurring**
 - Endogenous opioids
 - Endorphins
 - Enkephalins
 - Dynorphins
 - Opium derivatives
 - Phenanthrenes
 - Morphine
 - Codeine
- **Semisynthetic**
Simple modifications to morphine.
 - Diacetylmorphine
 - Buprenorphine
 - Oxycodone
- **Synthetic**
 - Phenylpiperidines
 - Fentanyl
 - Alfentanil
 - Remifentanil
 - Pethidine

- Diphenylpropylamines
 - Methadone

Opioid Receptor Classification

All opioid receptors are Gi receptors. Activation:

- Inhibits adenylyl cyclase, reducing cAMP
 - Pre-synaptically inhibits voltage-gated Ca^{2+} channels
 - Decreases Ca^{2+} influx
 - Reduces neurotransmitter release
 - Post-synaptically stimulates activates K^{+} channels
 - Causes K^{+} efflux
 - Leads to membrane hyperpolarisation

Table 1: Opioid Receptor Classification and Effects

Receptor	Actions	Notable Properties
MOP	Analgesia (spinal and brain), euphoria, meiosis (via stimulation of the Edinger-Westphal nucleus), nausea and vomiting (via CTZ), sedation, bradycardia, inhibition of gut motility, urinary retention, physical dependence	Only opioid receptor to cause nausea/vomiting

Receptor	Actions	Notable Properties
KOP	Analgesia (predominantly spinal), sedation, meiosis, dysphoria	Less respiratory depression
DOP	Analgesia, respiratory depression, urinary retention, physical dependence	Minimal constipation
NOP	Anxiety, depression, change in appetite	Hyperalgesia at low doses, analgesic at high doses

Mechanism of effects:

- **Respiratory depression**

Decreases central chemoreceptor sensitivity to CO₂.

- **Constipation**

Stimulation of opioid receptors in the gut.

- Normally activated by local endogenous opioids (used as neurotransmitters)
- Agonism of these receptors (μ , k , and to a smaller extent, δ) reduces GIT secretions and peristalsis

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Inhalational Anaesthetics

Structure-activity relationships of inhalational agents

Describe the uptake, distribution and elimination of inhalational anaesthetic agents and the factors which influence induction and recovery from inhalational anaesthesia including the:

- Concepts of partition coefficients, concentration effect and second gas effect
- Relationships between inhaled and alveolar concentration
- Significance of the distribution of cardiac output and tissue partition coefficients on uptake and distribution of volatile agents

Describe the concept and clinical application of MAC in relation to inhaled anaesthetic agents

Describe how the pharmacokinetics of drugs commonly used in anaesthesia in neonates and children differ from adults and the implications for anaesthesia

Properties of an ideal inhalational anaesthetic agent

Inhaled anaesthetics are chemicals with general anaesthetic properties that can be delivered by inhalation. They can be divided into:

- **Volatile** anaesthetic agents
Volatility refers to the tendency of a liquid to vapourise. Volatile agents include:
 - Sevoflurane
 - Isoflurane
 - Desflurane
 - Methoxyflurane
 - Enflurane
 - Halothane
 - Ether
- Anaesthetic gases
 - Nitrous oxide
 - Xenon

Key Principles of Inhalational Agents

Key principles:

- The **clinical effect** of an inhalational agent **is dependent on its partial pressure** within the CNS
- **At equilibrium**, the partial pressure in the CNS (P_B) **equals** the partial pressure in blood (P_a), and in the alveoli (P_A)
Reaching equilibrium is rarely achieved in practice as it takes many hours.
- Rate of onset and offset of an inhalational agent are dependent on both physiological and pharmacological factors affecting the transfer of agent:
 - Into the alveoli
 - From the alveoli into blood
 - From blood into the CNS

Minimum Alveolar Concentration (MAC)

MAC is **defined** as the minimum alveolar concentration at **steady state** which prevents a movement response to a standard surgical stimulus (1cm forearm incision) in 50% of a population.

Note that this definition:

- Does not reflect lack of awareness
- Does reflect the action of an agent on spinal cord reflexes
This is why 1 MAC is adequate for the majority of patients, as awareness and recall are suppressed at lower MAC values than are required for immobility.
- Consciousness is better estimated by MAC-awake
End-tidal concentration of agent that prevents appropriate responses to a verbal command in 50% of a population.
 - Note that this technically measures *awareness* rather than memory.
 - **MAC-awake** is typically **one-third** of **MAC** for commonly-used agents
- Is only valid at **sea-level**
The **clinical effect** of an agent is **dependent** on its **partial pressure not concentration**.
 - At 1atm, these are almost the same
1atm = 100kPa; therefore 2% sevoflurane is 2kPa
 - As altitude increases, the actual partial pressure will fall for any given concentration i.e. 2% sevoflurane at 0.5atm is 1kPa of sevoflurane.

MAC is:

- A measure of potency (i.e. the EC_{50} of the agent, where the outcome is movement)
The MAC of an agent is inversely proportional to potency; i.e. more potent agents require smaller alveolar concentrations to produce anaesthesia.

- This gives rise to the **Meyer-Overton hypothesis**, which suggests that anaesthesia requires a sufficient number of molecules to dissolve into the neuronal cell membrane.
 - If this was true, the product of the oil:gas partition coefficient and MAC would be constant, which is not the case.
- Additive

The MACs of different agents used simultaneously are additive.
- Normally-distributed

Not all patients will be unresponsive at 1 MAC.

 - The standard deviation is **0.1**, so 95% of patients will not move in response to a stimulus at 1.2 MAC
- Estimated clinically using **end-tidal** gas measurement

MAC is not based on arterial partial pressure (F_a) of agent.

 - This is an important difference, because **even at steady-state, $F_a \neq F_A$**
 - This occurs due to:
 - V/Q mismatch

Shunted alveoli will not absorb anaesthetic agent, and unperfused alveoli will contain agent that is not being absorbed.

 - This is worsened by the effects of anaesthesia
 - Volatile agents are heavy and have finite diffusibility
 - However, the difference between F_a and F_A for any agent is the same at steady state (and in absence of nitrous oxide)

This means that, **at steady-state**, MAC will be proportional to, and an accurate measure of, P_a .
- One of several related terms:
 - MAC awake

Concentration required to prevent response to a verbal stimuli in **absence of noxious stimuli**.

 - Typically $\sim 1/3^{\text{rd}}$ of MAC for most agents

- (sevoflurane, isoflurane, desflurane)
- Notably higher for nitrous oxide (MAC-awake ~2/3^{rds} of MAC)
- MAC-awake is typically less than MAC-asleep as:
 - Hysteresis between alveolar and effect site concentrations
During induction, alveolar concentration is higher than effect site concentration, and so overestimates effect. During wash out, alveolar concentration is less than effect site concentration, and the reverse effect occurs.
 - "Neural inertia"
Intrinsic resistance of nerve cells to a change in their state.
- MAC-BAR
Minimum alveolar concentration required to **block adrenergic response**, i.e. to prevent a rise in HR or BP following skin incision.
- MAC₉₅
The MAC required to prevent a movement response to a standard surgical stimulus in 95% of the population.
- MAC.hr⁻¹
The amount of time a patient is exposed to 1 MAC of an agent. Used to compare different agents.

Factors Affecting MAC

Table 1: Factors That Decrease and Increase MAC

Decreases MAC	Increases MAC
Age (~6%/10 years ↑) and neonates	Youth
Hypothermia	Hyperthermia
Hypocapnea	Hypercapnea
Hyponatraemia	Hypernatraemia

Decreases MAC	Increases MAC
Acute alcohol and other CNS depressant intoxication	Chronic ETOH and CNS depressant abuse
Chronic amphetamine intake	Acute amphetamine/stimulant intake
Hypovolaemia/Hypotension	
Hypoosmolality	
↓ pH	
Lithium	
Hypoxia	
Anaemia	
Pregnancy	
	SNS activation and anxiety
	Increased P_{atm}

Note that addition of other agents (e.g. opioids) will affect different MAC subtypes (e.g. MAC_{50} vs MAC_{BAR}) differently.

Partition Coefficients

A **partition coefficient** describes the **relative affinity** of an agent for two phases, and is **defined** as the **ratio of the concentration** of agent in each phase, when both **phases are of equal volume** and the partial pressures are in **equilibrium at STP**.

- The **blood:gas partition coefficient** describes the solubility of the agent in blood relative to air, when the two phases are of equal volume and in equilibrium at STP
A **low** blood:gas partition coefficient indicates a **rapid** onset and offset. This is because:
 - Poorly soluble agents generate a high P_a , which creates a steep gradient between P_a and P_B , giving a rapid onset of action

- Conversely, soluble agents dissolve easily into pulmonary blood without substantially increasing P_a
This causes leads to a slow onset due to:
 - A large fall in P_A as the agent leaves the alveolus, decreasing the gradient for further diffusion
 - A small gradient between P_a and P_B
- The **oil:gas partition coefficient** describes the solubility of the agent in fat relative to air, when both phases are of equal volume and in equilibrium at STP
A **high** oil:gas partition coefficient indicates a greater potency, and therefore a low MAC.

Pharmacokinetics of Inhalational Agents

Achieving the required P_B requires maintaining P_A at a high enough level. By increasing P_A , the pressure gradient for diffusion into blood, and therefore CNS, is increased.

As discussed above, rate of onset of an inhalational agent is dependent on rate of uptake:

- Into the alveoli
- From the alveoli into blood
- From blood into the CNS

Factors affecting **alveolar concentration of agent**:

- **Inspired concentration**

A high inspired concentration (F_i) will increase the rate of increase of alveolar concentration (F_A). Inspired concentration is dependent on:

- Delivered concentration in fresh gas
- Fresh gas flow

Increasing FGF (and the concentration of agent in the added gas) increases F_i .

- Volume of the breathing system

A **lower circuit volume** will increase the rate at which the patient reaches equilibrium with the circuit, and therefore **increase F_i** .

- Circuit absorption

Absorption of agent by the circuit will **decrease** F_i .

- V_A

Increased alveolar ventilation increases F_i , as it replenishes agent that has been taken up into the vasculature.

- Similarly, increased **dead space** will prolong induction, as anaesthetic gas will be delivered to non-perfused alveoli

- **FRC**

A **large FRC** will dilute the amount of agent inspired with each breath, and so **reduce** F_i .

- This is measured with the V_A/FRC ratio
Increased ratio increases speed of onset.
 - Normal in adults: 1.5:1
 - Normal in neonates: 5:1

- **Second gas effect**

Use of N_2O with another agent will increase the P_A of that agent. This is because:

- N_2O is **20x** as soluble in blood as either blood or nitrogen, and is administered in high concentrations, so it is rapidly absorbed from alveoli
- If nitrous oxide is delivered at high concentrations, it's rapid absorption means that alveoli will **shrink**, causing:

- An increase in the fractional concentration of all other gases
This is known as the **concentration effect**, and increases the pressure gradient driving diffusion into blood, increasing speed of onset.

- The concentration effect is the cause of the second gas effect
- The concentration effect is **more pronounced as $F_i\text{N}_2\text{O}$ increases**
- The concentration effect is more profound in lung units with moderately **low V/Q ratios**, causing in a large increase in F_a

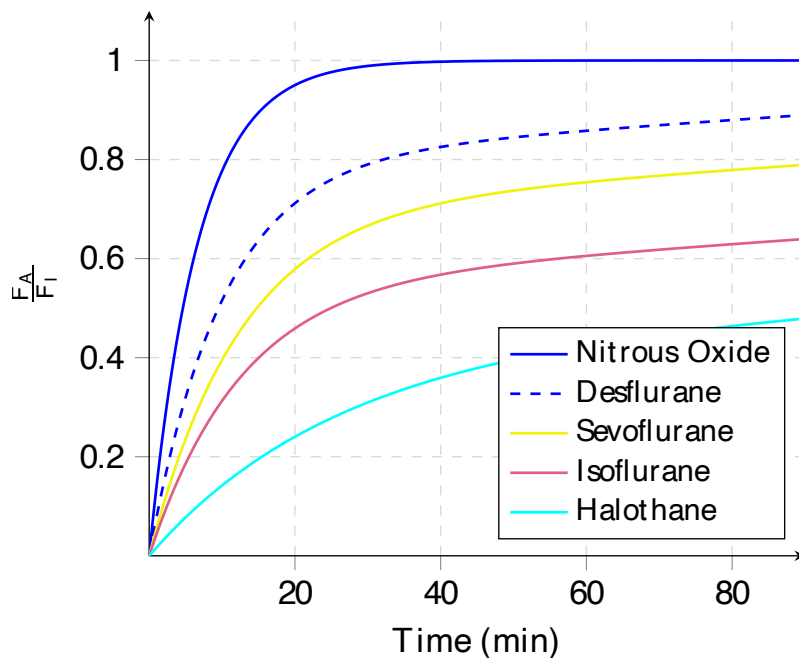
This results in a larger value of F_a for any

given F_A , even at steady state.

- Augmented ventilation as more inhalational agent is drawn in the alveoli from dead space gas
- The second gas effect also causes **diffusion hypoxia**

When inspired N_2O is reduced, N_2O will leave blood and enter the alveolus, displacing other gases in the alveolus.

- This can cause a reduction in PAO_2 , and therefore hypoxaemia
- Diffusion hypoxia is avoided by delivering 100% oxygen, which maintains an adequate PAO_2 as N_2O is removed



- Note that N_2O reaches a higher ratio faster than desflurane, despite its lower blood:gas partition coefficient, due to the concentration effect

Factors affecting **drug uptake from the lungs**:

- **Blood:gas partition coefficient**
Agents with a low blood:gas partition coefficient reach

$\frac{F_A}{F_I}$ equilibrium more rapidly. The blood:gas coefficient is affected by:

- Temperature
Blood:gas partition coefficients decrease as temperature increases.
- Haematocrit
Variable effect, which depends on the particular agents affinity for red cells or plasma (and serum constituents, e.g. albumin).
 - An agent that is less soluble in red cells (e.g. isoflurane) will have a decreased blood-gas partition coefficient in anaemia.
- Fat
Blood:gas partition coefficient increases following fat ingestion.

- **Alveolar blood flow**

Increased alveolar blood flow increases uptake and delivery to tissues, including the CNS.

- However, the increased uptake causes a reduction in P_A
Therefore, **rate of onset is reduced** when **alveolar blood flow is high**.
 - This effect is more pronounced with agents with a high blood:gas partition coefficient
 - Alveolar blood flow is a function of:
 - Cardiac output
 - Shunt

- **Alveolar - Venous partial pressure gradient**

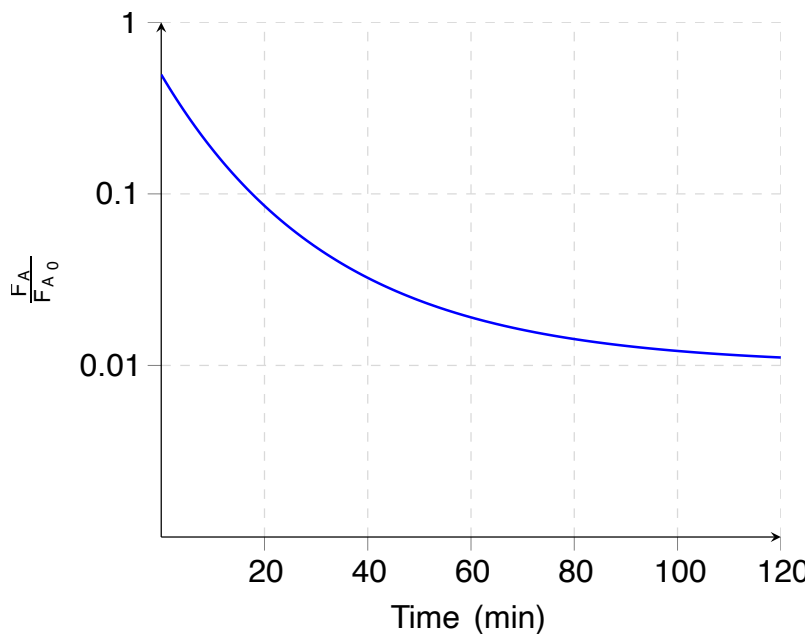
The difference in partial pressure of agent in the alveolus and venous blood is due to the uptake of drug in tissues. Tissue uptake is dependent on:

- Tissue blood flow
As the CNS has a high blood flow, it will equilibrate more quickly.
- Blood:tissue solubility coefficients
 - Muscle has similar affinity to blood, but equilibrates more slowly than the CNS due to lower blood flow
 - Fat has a much higher affinity for anaesthetic than muscle, but equilibrates very slowly

due to the very low blood flow
This is of greater importance in the obese,
especially during prolonged anaesthesia,
as they have a longer equilibration time and
therefore prolonged emergence.

Wash-out of Inhalational Agents

Recovery is dependent on how quickly an inhalational agent can be eliminated from the effect site, and can be graphed by the F_A/F_{A0} ratio over time:



Washout can be divided into:

- Rapid washout
Of agent in circuit and FRC.
 - The time constant for removal of agent from the circuit is a function of circuit volume and fresh gas flow, i.e.

$$\tau = \frac{CV}{FGF}$$
- Slow washout
Of agent in patient.

- The time constant for removal of agent from the patient is a function of FRC and minute ventilation, i.e.

$$\tau = \frac{FRC}{MV}$$

Factors affecting volatile washout:

- Brain-Blood and Tissue-Blood
 - Tissue:Blood coefficient of agent
 - Duration and depth of anaesthesia
Important for highly soluble agents used in long cases.
- Blood-Alveolus
 - Blood:gas coefficient of agent
Highly soluble agents will have an increased amount of drug dissolved in tissue, so a large reservoir of drug exists that will have to be removed.
 - Alveolar Cardiac output
Decreased cardiac output **increases** elimination.
 - Shunt
Decreases elimination.
- Alveolus-Air
 - MV_A/FRC
Increased alveolar ventilation increases elimination.
- Other factors
 - Metabolism of agent
Agents undergoing metabolism are eliminated more rapidly.
 - Absorption of agent into circuit
 - Percutaneous loss
Loss of agent by diffusion from tissues into external environment.

Alteration to Pharmacokinetics

Increased rate of induction in **children** due to:

- Increased V_A/FRC ratio
Increases P_A .
- Lower albumin and cholesterol
Reduced blood-gas solubility coefficients for some agents. Increased rate of induction in **elderly** due to:
- Lower MAC requirement
- Lower albumin
Reduces blood-gas solubility coefficients for some agents.
- Lower cardiac output
 P_a and therefore P_B is established more rapidly.

Altered rate of induction in **pregnancy** due to:

- Increased V_A/FRC ratio
 - Increased minute ventilation
This is of greater importance in spontaneous ventilation, as this is controlled by the anaesthetist during controlled ventilation.
 - Decreased FRC
Increases P_A , increasing P_B and speed of onset.
- Lower albumin
Reduces blood-gas solubility coefficients for some agents.
- Increased CO
Reduces rate of rise of P_A , reducing P_B and therefore speed of onset.
- Reduced MAC requirement
Progesterone has some sedative properties.

Alteration to Pharmacokinetics with Special Methods of Administration

In **target-controlled anaesthesia**, FGF and agent F_I are controlled by the machine to reach the target F_A rapidly at low concentrations. This causes:

- An initial over-pressure of F_I , in order to fill the FRC and reach the desired F_A

- A more rapid induction, as the target F_a is reached more rapidly

In **liquid injection**, anaesthetic agent is injected into the breathing system. This causes:

- A very large degree of overpressure
In this circumstance, the rate of rise of end-expired agent concentration is identical for different agents.
 - i.e. Onset is independent of the blood:gas coefficient

Mechanism of Action of Inhaled Anaesthetic Agents

Mechanisms of action can be divided into:

- **Macroscopic**
At the level of the brain and spinal cord.
 - In the spine by:
 - Decreasing transmission of noxious afferent signals at the thalamus
 - Inhibition of spinal efferents, decreasing motor responses
 - In the brain by:
 - Global depression of CBF and glucose metabolism
- **Microscopic**
Synapses and axons by:
 - Inhibiting pre-synaptic excitatory activity:
 - ACh
 - 5-HT
 - Glutamine
 - Augmenting post-synaptic inhibitory activity:
 - GABA_A
- **Molecular**
Anaesthetic agents may alter the function of molecules within the CNS. These include:

- Alteration of α -subunits of the GABA_A receptor
This prolongs the time it spends open once activated, prolonging the inhibitory Cl⁻ current and increasing the degree of hyperpolarisation.
- Enhance the activity of two-pore K⁺ channels
Increases the resting membrane potential of both pre-synaptic and post-synaptic CNS neurons.

Incomplete Theories of the Mechanism of Action of General Anaesthetic Agents

Meyer - Overton Hypothesis:

- Potency of anaesthetics relates to their lipid solubility
- Anaesthetic molecules dissolve into CNS membranes, disrupting their effect
- Flaws:
 - Not all lipid soluble drugs have general anaesthetic effects
 - Other factors disrupt cell membranes without causing anaesthesia

Volume Expansion, Pressure Reversal (Mullin's Critical Volume Hypothesis):

- CNS cell membranes expand with general anaesthetic agents
This distorts channels responsible for maintaining membrane potential and generating action potentials.
- Increased ambient pressure reverses general anaesthesia
- Flaws:
 - Does not account for stereo-selectivity of drug-receptor interactions
I.e. receptors select for one stereoisomer over others.

Structure-Activity Relationships of Inhaled Anaesthetics

- Chemical structures of different volatile anaesthetics are covered in the pharmacopeia.

Different chemical and physical properties alter the effect of inhalational agents:

- Physical
 - Molecular weight
A decrease in molecular weight decreases boiling point and therefore increases SVP.
- Chemical
 - H⁺ content
Greater hydrogen content:
 - Increases flammability
 - Increases potency
 - F⁻ content
Greater fluoride content:
 - Decreases flammability
 - Decreases oxidative metabolism
This decreases toxicity.
 - Decreases potency
 - Cl⁻ content
Increased chloride increases potency.
 - -CHF₂ (Di-fluor-methyl group)
 - Produces CO in the presence of dry soda lime

The Ideal Inhaled Anaesthetic Agent

From the properties discussed above, we can construct the following ideal agent:

- Physicochemical
 - Liquid at room temperature
 - High SVP

- Low specific heat capacity
 - Long shelf-life
 - Light stable
 - Heat stable
 - Does not react with the components in the breathing circuit
 - Rubber
 - Metal
 - Plastic
 - Soda lime
 - Not flammable/explosive
 - Smells nice
 - Preservative free
 - Environmentally friendly
 - Cheap
 - Pharmacokinetic
 - High oil:gas partition coefficient
Low MAC.
 - Low blood:gas partition coefficient
Rapid onset and offset.
 - Not metabolised
 - Non-toxic
 - Pharmacodynamic
 - Does not cause laryngospasm or airway hyper-reactivity
 - No effect on HDx parameters
 - Analgesic
 - Hypnotic
 - Amnestic
 - Anti-epileptic
 - No increase in ICP
 - Skeletal muscle relaxation
 - Anti-emetic
 - No tocolytic effects
 - Not teratogenic or otherwise toxic
 - No drug interactions
-

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Propofol

Propofol (2-6 di-isopropylphenol) is a phenolic derivative with effects on many receptors including:

- GABA_A
Potentiates the effect of GABA, prolonging Cl⁻ channel opening and hyperpolarising the cell.
- Glycine
- Nicotinic ACh
- D₂ receptors

Table 1: Propofol

	Propofol
Aliases	2,6-diisopropylphenol
Class	IV anaesthetic
Chemistry	Alkylphenol; phenolic derivative
Dose	
Induction (IV)	1.5 - 2.5mg/kg
Maintenance (IV)	4 - 12mg/kg/hr
ICU sedation (IV)	0.3 - 4mg/kg/hr
Procedural sedation (IV)	0.5 - 1mg/kg over 1 - 5 minutes then 1.5 - 4.5mg/kg/hr
Pharmaceutics	
Presentation	White oil-in-water emulsion at pH 7 - 8.5 and 10 or 20mg/mL; contains soya oil, MCT and egg lecithin; 1.02kcal/mL; no antimicrobial preservative
Solubility	pKa 11; almost entirely unionised at physiological pH; minimally water soluble

Propofol	
Pharmacokinetics	
Absorption	Onset ~30 seconds (IV)
Distribution	pKa: 11
	Vd: 4 L/kg
	Protein binding: 98%
Metabolism	Half-life: $t_{1/2\alpha}$ 2-4 minutes; rapid elimination $t_{1/2}$ 30-60 minutes; terminal $t_{1/2}$ 4-23 hours
	Duration: 5-10 minutes after induction dose
	Pathway: Hepatic and extra-hepatic glucuronidation; clearance 21-29mL/kg/min
	Metabolites: Inactive glucuronide and quinol conjugates
Elimination	Renal: Renal elimination of inactive glucuronide and sulfate metabolites; green urine
Mechanism	
Target	GABAA receptor Glycine receptor Nicotinic ACh receptor D2 receptor
Action	Binds to β -subunit of postsynaptic GABAA receptor $\rightarrow$ $\uparrow$ duration of Cl ⁻ channel opening $\rightarrow$ $\uparrow$ Cl ⁻ conductance $\rightarrow$ hyperpolarisation $\rightarrow$ $\downarrow$ neuronal depolarisation; also potentiates glycine, inhibits nicotinic ACh and D2 receptors
Pregnancy	
Code	C

	Propofol
Justification	Neonatal CNS/respiratory depression; greater concern if fetus compromised
Pharmacodynamics	
Respiratory	Respiratory depression Apnoea Strong suppression of laryngeal reflexes ↓ Response to hypoxia and hypercapnia Bronchodilation
CVS	↓ SVR (arterial and venous vasodilation via ↑ NO release) ↓ VR ↓ BP ↓ Inotropy (via ↓ SNS tone) ↓ MVO ₂
CNS	Depresses baroreceptor reflex Hypnosis; rapid loss of consciousness within one arm-brain circulation time ↓ CMRO ₂ ↓ CBF ↓ ICP Anticonvulsant ↓ IOP Paradoxical excitatory effects in approx- imately 10% (dystonic movements of subcortical origin)
Muscu- loskeletal	Pain on injection into small veins
Metabolic	Fat overload syndrome and lipaemia fol- lowing prolonged infusion Inhibits mitochondrial function
Renal	Green urine
Gastrointesti- nal	Antiemetic ↓ Hepatic blood flow

	Propofol
Considerations	Propofol infusion syndrome (typically >48 hours and >5mg/kg/hr): lactic acidosis, arrhythmia, cardiac failure, rhabdomyolysis, hyperkalaemia, lipaemia, hepatomegaly and renal failure No antimicrobial preservative — use immediately; replace infusion system ≤12 hours

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Barbiturates

Thiopentone is a positive allosteric modulator at GABA_A receptors (at a separate site to benzodiazepines) in the CNS. Barbiturates cause:

- Decreased rate of dissociation of GABA
Increases the duration of channel opening, causing effective hyperpolarisation due to increased Cl⁻ conductance.
 - Clinical effects differ from benzodiazepines as benzodiazepines increase *frequency* of opening, whilst barbiturates increase *duration*
- Direct activation of the channel at higher doses

Table 1: Thiopentone

	Thiopentone
Aliases	Thiopental Pentothal
Class	Ultrashort-acting IV barbiturate anaesthetic
Chemistry	Thiobarbiturate; sulfur analogue of pentobarbitone
Dose	
Induction (IV)	3-4mg/kg
Pharmaceutics	
Presentation	470mg yellow hygroscopic powder with 30mg sodium carbonate; equivalent to 500mg mixture
Compatibility	Reconstitute to 25mg/mL; do not mix with acidic drugs or administer precipitated solution
Solubility	pKa 7.4; reconstituted pH 10-11

Thiopentone	
Pharmacokinetics	
Absorption	IV
Distribution	pKa: 7.4
	Vd: 2.5L/kg
	Protein binding: 80%
Metabolism	Half-life: 3-8 hours
	Duration: 5-10 minutes after induction
	Pathway: Hepatic oxidation and desulfuration
	Metabolites: Predominantly inactive metabolites, Pentobarbitone (minor active metabolite)
Elimination	Renal: Metabolites; minimal unchanged
Mechanism	
Target	GABA-A receptor
Action	↑ Duration of GABA-mediated Cl ⁻ channel opening; high concentrations directly activate GABA-A and ↓ glutamate excitation
Contraindications	
	Acute or variegate porphyria
	Severe respiratory compromise or inability to maintain the airway
	Constrictive pericarditis
Pregnancy	
Code	A
Justification	General anaesthetics may depress neonatal CNS and respiration, particularly in a compromised fetus
Pharmacodynamics	

	Thiopentone
Respiratory	Apnoea Respiratory depression Bronchospasm and laryngospasm
CVS	Venodilation and vasodilation → ↓ MAP ↓ Contractility ↑ HR
CNS	Hypnosis; no analgesia Anticonvulsant ↓ CMRO ₂ , CBF and ICP Dose-dependent EEG suppression
Metabolic	Acute porphyric crisis
Renal	↓ RBF and UO
Gastrointestinal	↓ Lower oesophageal sphincter tone
Considerations	Onset 30-40 seconds; offset by redistribution Repeat doses or infusion → accumulation and prolonged recovery High pH → extravasation necrosis Intra-arterial injection → arteriospasm, ischaemia and gangrene ↑ Free drug in acidosis or hypoalbuminaemia Haemodynamic instability → reduce dose and inject slowly CNS depressants → synergistic depression

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Ketamine

Ketamine is a phencyclidine derivative used for induction, sedation, analgesia, and as a bronchodilator in severe asthma.

Ketamine acts via:

- **Non-competitive antagonist** of **NMDA** and **glutamate** receptors in the CNS
- Reduces presynaptic glutamate release
- Sodium channel inhibition
Local anaesthetic-like effect.
- Potential monoaminergic, muscarinic, and nicotinic antagonism

Table 1: Ketamine

	Ketamine
Aliases	Ketalar
Class	IV anaesthetic
Chemistry	Arylcyclohexylamine; phencyclidine derivative
Dose	
Induction (IV)	1 - 4.5mg/kg (usual 1 - 2mg/kg) over 60 seconds
Induction (IM)	6.5 - 13mg/kg (usual 5 - 10mg/kg)
Sedation (IV)	0.2 - 0.5mg/kg
Pharmaceutics	
Presentation	Clear colourless solution 100mg/mL; pH 3.5 - 5.5
Compatibility	Dilute 100mg/mL 1:1 with water for injection, 0.9% sodium chloride or 5% glucose before IV use; incompatible with barbiturates and diazepam

Ketamine	
Solubility	pKa 7.5; freely soluble in water
Pharmacokinetics	
Absorption	Up to 93% IM bioavailability
Distribution	pKa: 7.5
	Vd: 1-3 L/kg
	Protein binding: 20-50%
Metabolism	Half-life: $t_{1/2\alpha}$ 10-15 minutes; $t_{1/2\beta}$ 2.5 hours
	Duration: 5-10 minutes IV; 12-25 minutes IM
	Pathway: Hepatic N-dealkylation → norketamine (1/6 potency) → inactive metabolites
	Metabolites: Norketamine (active)
Elimination	Renal: 90% urinary recovery, mostly metabolites (2-4% unchanged); 5% faecal
Mechanism	
Target	NMDA receptor Voltage-gated Na ⁺ channel
Action	Non-competitive NMDA antagonism at phencyclidine site → ↓ excitatory transmission; ↓ presynaptic glutamate release; Na ⁺ channel blockade
Pregnancy	
Code	B3
Justification	Neonatal CNS/respiratory depression; greater concern if fetus compromised
Pharmacodynamics	

	Ketamine
Respiratory	Bronchodilation Tachypnoea Relative preservation of laryngeal reflexes Apnoea with rapid injection Preserved central response to CO ₂
CVS	↑ HR ↑ BP ↑ SVR ↑ MVO ₂ Direct myocardial depressant effect (unmasked in catecholamine-depleted states) Depresses baroreceptor reflex
CNS	Dissociative anaesthesia within 90 seconds Analgesia ↑ IOP Emergence phenomena (hallucinations, delirium) — reduced by concurrent benzodiazepine; risk ↑ with higher doses and rapid administration Purposeful movements unrelated to stimulus may occur during surgical anaesthesia
Musculoskeletal	↑ Skeletal muscle tone
Renal	Cystitis with long-term high-dose use
Gastrointestinal	Nausea Vomiting
Considerations	Racemic mixture or single S(+) enantiomer (2-3x more potent than R(-), but less bronchodilatory)

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Dexmedetomidine

Dexmedetomidine is a central α_2 -agonist ($\alpha_2:\alpha_1$ activity 1600:1) used for its sedation and analgesic properties.

Table 1: Dexmedetomidine

	Dexmedetomidine
Aliases	Precedex
Class	Alpha-2 adrenoreceptor agonist
Chemistry	Imidazole derivative
Dose	
ICU sedation loading (IV)	1µg/kg over 10-20 minutes if required
ICU sedation maintenance (IV)	0.2-1µg/kg/hr
Procedural sedation loading (IV)	1µg/kg over 10-20 minutes
Procedural sedation maintenance (IV)	0.2-1µg/kg/hr, usually initiated at 0.6µg/kg/hr
Pharmaceutics	
Presentation	Clear, colourless, preservative-free solution 100µg/mL
Compatibility	Dilute with 0.9% sodium chloride to 4µg/mL before administration; incompatible with amphotericin B and diazepam
Pharmacokinetics	
Absorption	Onset 10-15 minutes (IV)

	Dexmedetomidine
Distribution	Vd: 118 L Protein binding: 94%
Metabolism	Half-life: $t_{1/2\alpha}$ 6 minutes; $t_{1/2\beta}$ 2 hours Pathway: Hepatic N-glucuronidation, CYP2A6 hydroxylation and N-methylation Metabolites: Inactive
Elimination	Renal: 95% renal (metabolites); 4% faecal
Mechanism	
Target	α_2 adrenoreceptor
Action	Postsynaptic α_2 activation in locus coeruleus $\rightarrow$ Gi activation $\rightarrow$ $\uparrow$ K ⁺ conductance $\rightarrow$ hyperpolarisation $\rightarrow$ sedation; spinal α_2 activation $\rightarrow$ analgesia
Pregnancy	
Code	B1
Pharmacodynamics	
Respiratory	Minimal respiratory depression Apnoea Airway obstruction
CVS	Initial transient $\uparrow$ SVR and $\uparrow$ BP (peripheral vasoconstriction) Followed by $\downarrow$ MAP and $\downarrow$ HR Rebound $\uparrow$ BP on abrupt cessation $\downarrow$ CO (sympatholytic effect) Sinus arrest (rapid bolus/high vagal tone)

	Dexmedetomidine
CNS	Sedation resembling natural sleep Anxiolysis at low dose (anxiogenic at high dose) No reliable amnesia ↓ MAC ↓ SNS outflow Minor analgesic effect (likely related to opioid potentiation and anxiolysis)
Considerations	Rousable sedation Continuous ECG, BP and SpO ₂ monitoring ↓ Dose in hepatic impairment and age >65

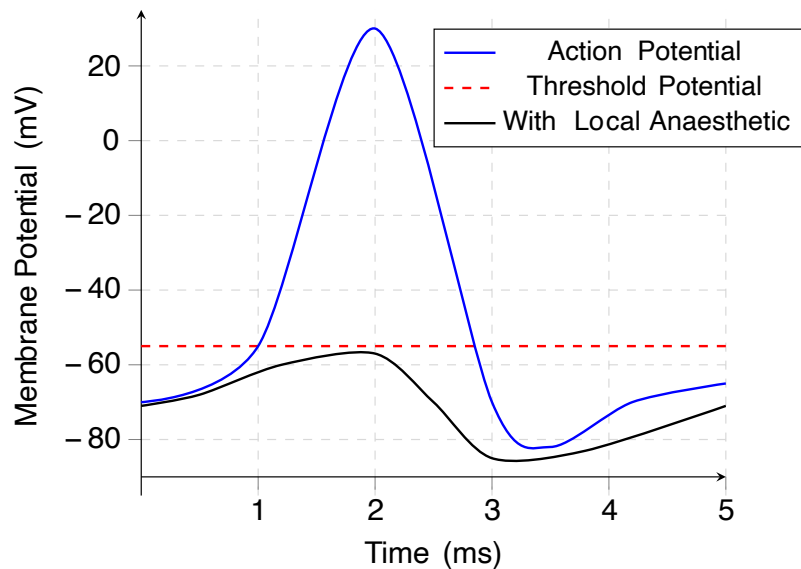
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Local Anaesthetic Agents

- Local anaesthetic drugs deliver a use-dependent, temporary blockade of neuronal transmission
- **Unionised** drug **passes through the cell membrane**, and then becomes **ionised intracellularly**
- The ionised drug is then able to bind to the ion channel, and prevent conduction of sodium and therefore generation of an action potential



- All local anaesthetics consist of:
 - A hydrophilic component
 - A lipophilic aromatic ring
 - An amide or ester link connecting the two

Table 1: Common Features of Local Anaesthetics

Property	Action
Class	Amide (-NHCO-) or Ester (-COOH-)
Pharmaceutics	Amides are stable in solution, esters are unstable in solution. All are formulated as a hydrochloride salt to ensure water solubility.
pKa	All are weak bases, and have a pKa > 7.4
Onset	Onset is related to dose (Fick's Law) and pKa, with a low pKa giving a faster onset as there is more unionised drug present and therefore more drug able to cross the cell membrane. This is why local anaesthetics are poor at anaesthetising infected tissues, as the tissue pH is low resulting in a greater proportion of ionised drug, and less drug reaching the effect site.
Duration of Action	Duration of action is related to protein binding, with greater protein binding giving a longer duration of action
Potency	Potency is related to lipid solubility (higher lipid solubility increases potency) and vasodilator properties (weaker vasodilators having greater potency)

Property	Action
Absorption	Systemic absorption varies with site of entry (from highest absorption to lowest: IV, intercostal, caudal epidural, lumbar epidural, brachial plexus, subcutaneous), dose , and presence of vasoconstrictors
Distribution	Amides are extensively protein bound, esters are minimally bound
Metabolism	Amides are hepatically metabolised, esters are hydrolysed by plasma cholinesterases (giving a much shorter $t_{1/2}$)
CVS	Vasodilatation at low concentrations, vasoconstriction at high concentrations. Inhibition of cardiac Na^+ channels, inhibiting maximum rate of rise of phase 0 of the cardiac action potential. Negative inotropy proportional to potency.
CNS	Dose-dependent CNS effects: circumoral tingling, visual disturbances, tinnitus, tremors, dizziness, slurred speech, convulsions, coma, apnoea. Potentiated by other CNS depressants and hypercarbia (due to $\uparrow$ CBF and $\downarrow$ seizure threshold).

Property	Action
Toxic Effects	Esters have a higher incidence of allergy due to their metabolite para-amino benzoic acid (PABA). Local anaesthetic toxicity is predominantly CNS and CVS.

Table 2: Comparison of Local Anaesthetics

	Lignocaine	Bupivacaine	Ropivacaine	Cocaine
Aliases	Lidocaine Xylocard	Marcaïn	Naropin	Numbrino
Class	Class Ib antiarrhythmic; aminoamide local anaesthetic	Amide local anaesthetic	Amide local anaesthetic	Ester local anaesthetic; indirect sympathomimetic
Chemistry	Aminoamide	Racemic aminoamide	Pure S(-)-enantiomer aminoamide	Tropane ester
Dose				
VT (IV)	1mg/kg over 1-2 minutes; max 100mg			
Infusion (IV)	2-4mg/min			
Block		Up to 2mg/kg	Up to 3mg/kg	
Lumbar epidural analgesia		15-60mg bolus; 12.5-18.75mg/hour infusion	20-40mg bolus; 12-28mg/hour infusion	
Nasal topical				40-160mg via pledgets for up to 20 minutes; max 3mg/kg
Pharmaceutics				
Presentation	Clear 20mg/mL injection; 100mg/mL infusion concentrate	0.25% and 0.5% injection ($\pm$ adrenaline); 0.125% epidural infusion; 0.375% registered but not marketed in selected PI	0.2%, 0.75% and 1% injection; 0.5% registered but not marketed in selected PI	4% nasal solution

	Lignocaine	Bupivacaine	Ropivacaine	Cocaine
Compatibility	5.5% glucose; 0.9% NaCl; Ringer's; sodium bicarbonate; dextran			
Solubility	Water - soluble	Lipid - soluble	Hydrochloride water solubility ~50mg/mL	Hydrochloride water - soluble
Pharmacokinetics				
Absorption	Oral bioavailability 35%; IV only for arrhythmia	Site - dependent; intercostal greatest and subcutaneous least; adrenaline slows absorption	Complete and biphasic from epidural space; absorption half - lives 14 minutes and 4 hours	~23% systemic absorption after 160mg via nasal pledgets; peak ~30 minutes from application
Distribution	pKa: 7.9	pKa: 8.1	pKa: 8.1	pKa: 8.6
	Vd: 0.9 L/kg	Vd: 73L	Vd: 47L	Vd: ~2L/kg
	Protein binding: 70%	Protein binding: 96%	Protein binding: 94%	Protein binding: 84 - 92%

	Lignocaine	Bupivacaine	Ropivacaine	Cocaine
Metabolism	Half-life: 10-20 minutes (IV bolus); 45-90 minutes (subcutaneous); 90-120 minutes (terminal) Pathway: Hepatic CYP1A2/3A4 (90-95%); active metabolites Metabolites: Monoethylglycinexylidide (active), Glycinexylidide (active)	Half-life: 2.7 hours (IV; longer after epidural administration) Duration: 0.5%: 2-5 hours epidural; up to 12 hours peripheral nerve block Pathway: Hepatic Metabolites: Pipecolylxylidine and 4-hydroxybupivacaine (inactive)	Half-life: 1.8 hours (IV; longer after epidural administration) Duration: Long-acting Pathway: Hepatic CYP1A2 aromatic hydroxylation; minor CYP3A4 N-dealkylation Metabolites: 3-hydroxyropivacaine (weak intrinsic activity; rapidly conjugated), 2,6-pipecoloxylidide (PPX; ~one-twelfth of ropivacaine activity)	Half-life: 1.54 hours (4% nasal pledgets) Duration: Topical anaesthesia at least 30 minutes Pathway: Plasma butyrylcholinesterase and hepatic carboxylesterases; minor CYP3A4 N-demethylation Metabolites: Benzoylecgonine and ecgonine methyl ester (inactive), Norcocaine (active; <1% of parent exposure)
Elimination	Renal: <10% excreted unchanged	Renal: Mainly metabolites; ~6% unchanged	Renal: 86% urine (~1% unchanged); 9% faeces after IV administration	Renal: Almost entirely metabolites; less than 5% unchanged
Mechanism				

	Lignocaine	Bupivacaine	Ropivacaine	Cocaine
Target	Nav1.5 subunit of voltage-gated sodium channel	Open and inactivated voltage-gated Na ⁺ channels	Open and inactivated voltage-gated Na ⁺ channels	Voltage-gated Na ⁺ channels Noradrenaline, dopamine and serotonin transporters
Action	Open/inactivated Na ⁺ channel block with rapid dissociation → ↓ Phase 0, automaticity and APD in depolarised/ischaemic ventricular tissue; minimal effect on normal tissue or QRS; ↓ QT	Use-dependent Na ⁺ channel blockade → reversible interruption of impulse conduction	Use-dependent Na ⁺ channel blockade → reversible interruption of impulse conduction	Reversible Na ⁺ channel blockade plus inhibition of monoamine reuptake
Contraindications	Severe SA, AV or intraventricular block Stokes-Adams syndrome Supraventricular arrhythmias	IV administration or Bier block Obstetric paracervical block Uncorrected hypotension for neuraxial anaesthesia Infection at injection site or septicaemia	IV administration or Bier block Uncorrected hypotension for neuraxial anaesthesia Infection at injection site or septicaemia	
Pregnancy Code	A	A	B1	
Pharmacodynamics				
Respiratory		Toxicity causes respiratory depression and apnoea	Toxicity causes respiratory depression and apnoea	Toxicity causes respiratory depression and apnoea

	Lignocaine	Bupivacaine	Ropivacaine	Cocaine
CVS	AV block ↓ Inotropy/HR/MAP; cardiovascular col- lapse	Neuraxial sympa- thetic block → va- sodilation and ↓ MAP Toxicity causes my- ocardial depression, conduction block and ventricular dys- rhythmias	Neuraxial sympa- thetic block → va- sodilation and ↓ MAP Toxicity causes my- ocardial depression, conduction block and ventricular dys- rhythmias	Vasoconstriction, ↑ MAP and HR Coronary vasocon- striction and QT pro- longation Toxicity causes my- ocardial depression and ventricular dys- rhythmias
CNS	Circumoral tingling; paraesthesia Dizziness; visual disturbance; confu- sion Seizures; coma	Local and regional anaesthesia Toxicity may progress from cir- cumoral paraesthe- sia and tinnitus to seizures, coma and apnoea	Local and regional anaesthesia Toxicity may progress from cir- cumoral paraesthe- sia and tinnitus to seizures, coma and apnoea	Local anaesthesia Abuse → euphoria and agitation Toxicity → hyper- thermia Toxicity causes seizures, coma and apnoea
Musculoskeletal		Concentration-de- pendent motor block	Less motor block than bupivacaine	
Gastrointestinal		Drug-induced liver injury		

	Lignocaine	Bupivacaine	Ropivacaine	Cocaine
Considerations	Shock, CCF, hepatic disease or prolonged infusion → ↓ clearance Propranolol, metoprolol or cimetidine → ↓ clearance	Most cardiotoxic commonly used local anaesthetic; slow dissociation from myocardial Na ⁺ channels Acidosis, hyperkalaemia, hypocalcaemia and hypoxia worsen toxicity Fractional injection with repeated aspiration reduces intravascular injection risk Continuous intra-articular infusion may cause chondrolysis Lipid emulsion for severe local-anaesthetic systemic toxicity	Less cardiotoxic than bupivacaine; faster dissociation from myocardial Na ⁺ channels Fractional injection with repeated aspiration reduces intravascular injection risk Continuous intra-articular infusion may cause chondrolysis CYP1A2 inhibitors may reduce clearance Lipid emulsion for severe local-anaesthetic systemic toxicity	High abuse and dependence potential Avoid damaged nasal mucosa Sympathomimetics may cause severe hypertension and dysrhythmias Disulfiram increases cocaine exposure Plasma cholinesterase deficiency may increase toxicity

Table 3: Clinical Phases of Lignocaine Toxicity by Serum Concentration

Serum concentration ($\mu\text{g.ml}^{-1}$)	Phase	Effect
2	Safe	Antiarrhythmic. May begin to have lightheadedness, circumoral tingling, numbness
5	Excitatory	Dysarthria
8	Excitatory	Visual changes
10	Excitatory	Seizures
12	Depressive	Loss of consciousness
20	Depressive	Respiratory depression
25	Depressive	CVS depression

Pharmaceutics of Topical Local Anaesthetics

Effect of topical local anaesthetics is governed by **Fick's Law**.

Table 4: Pharmaceutics of Topical Local Anaesthetics

Characteristic	Effect
Pharmaceutic Factors	
Presentation	Aerosol improves speed of onset by moisturising skin
Concentration of active component	Increase speed of onset

Characteristic	Effect
pH	↑ pH ensures more local anaesthetic is in the unionised form, ↑ absorption.
Additives	Affect pH and vasoconstrictor activity
Drug Factors	
Molecular weight	Small molecules will diffuse more easily
pKa	Affects ionisation and therefore lipid solubility
Lipid solubility	↑ lipid solubility improves speed of onset.
Potency	Determines amount of drug needed to produce an effect
Vasoconstrictor activity	Will affect both speed of onset and degree of systemic absorption
Patient Factors	
Site	Degree of vascularity of site
Skin	Skin thickness and area will affect onset

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Benzodiazepines

Benzodiazepines are double-ringed positive allosteric modulators of the GABA receptors in the CNS. They:

- Bind to the α/γ interface of the receptor, increasing affinity of the receptor for GABA
- This leads to **hyperpolarisation** of the **cell membranes** and decreased neuronal transmission

The mechanism varies between receptors:

- GABA_A is a **ligand gated** post-synaptic Cl⁻ ion channel
Activation **increases Cl⁻ conductance** via **increasing frequency** of channel opening.
- GABA_B is a pre- and post-synaptic **G-protein coupled** receptor
Activation **increases K⁺ conductance**.

Table 1: Common Features of Benzodiazepines

Property	Action
Uses	Sedation, anxiolysis, hypnotic, anticonvulsants, amnestic, muscle relaxation
Absorption	
Distribution	Highly lipid soluble and protein bound, very low V _D
Metabolism	Generally active metabolites.
Elimination	Renal elimination of active and inactive metabolites.
Resp	↓ V _T , ↑RR, apnoea.
CVS	↓ SVR, ↓SBP, ↓ DBP, ↑HR. Typically stable CO.

Property	Action
CNS	Hypnosis, sedation, anterograde amnesia, anticonvulsant, ↓ CBF. ↓ MAC.
MSK	Skeletal muscle relaxation.
Metabolic	↓ Adrenergic stress response.

Table 2: Comparison of Benzodiazepines

	Midazolam	Diazepam	Clonazepam
Aliases	Hypnovel	Valium; Propam; Stesolid	Rivotril Paxam
Class	Benzodiazepine	Benzodiazepine	Benzodiazepine anticonvulsant
Chemistry	Benzodiazepine	Benzodiazepine	7-nitro-1,4-benzodiazepine
Dose			
Sedation (IV)	0.01-0.05mg/kg		
Induction (IV)	0.15-0.2mg/kg		
ICU sedation (IV)	0.03-0.2mg/kg/hr		
Paediatric premedication (PO)	0.5mg/kg (max 20mg)		
Usual (IV/IM)		2-10mg every 3-4 hours; max 30mg/8h	
Status epilepticus (IV)		5-10mg; repeat every 10-15 minutes (max 30mg)	
Status (IV)			1mg slowly; repeat to control, max 10mg
Epilepsy (PO)			1mg/day initially; maintenance 4-8mg/day in 3-4 doses, max 20mg/day
Pharmaceutics			
Presentation	Clear solution 1 or 5mg/mL; pH 3.3	Clear colourless-pale yellow solution 5mg/mL; 53% propylene glycol; 31% ethanol	0.5mg tablet; 2.5mg/mL oral liquid; 1mg/mL injection with supplied diluent

	Midazolam	Diazepam	Clonazepam
Compatibility		Avoid dilution — precipitates and adsorbs to plastic	Mix injection 1:1 with diluent to 0.5mg/mL; for infusion add 1mg to at least 85mL compatible fluid
Solubility	pKa 6.5; water soluble at pH <4 as hydrochloride salt; becomes lipid soluble at physiological pH	pKa 3.4; practically insoluble in water	
Pharmacokinetics			
Absorption	50% oral bioavailability; >90% IM bioavailability	94% oral bioavailability; IM absorption slow and erratic	PO bioavailability 90%; peak 2-3 hours
Distribution	pKa: 6.5	pKa: 3.4	Vd: 3L/kg
	Vd: 0.6-1.9 L/kg	Vd: 0.8-1.0 L/kg	Protein binding: 82-86%
	Protein binding: 97%	Protein binding: 98%	
Metabolism	Half-life: $t_{1/2\beta}$ 1.4-2.4 hours; prolonged in elderly, critical illness, cirrhosis, CHF and obesity	Half-life: $t_{1/2\beta}$ 20-48 hours; nordiazepam ~96 hours	Half-life: 39 ± 8 hours
	Pathway: Hepatic CYP3A4/3A5 → α -hydroxymidazolam (active) → inactive glucuronide	Pathway: Hepatic CYP3A4/2C19 → nordiazepam, temazepam and oxazepam (active)	Pathway: Hepatic reduction (CYP3A4), hydroxylation and acetylation
	Metabolites: α -hydroxymidazolam (active); inactive glucuronide	Metabolites: Nordiazepam, temazepam and oxazepam (active)	Metabolites: 7-amino-clonazepam (inactive), 7-acetamido-clonazepam (inactive)

	Midazolam	Diazepam	Clonazepam
Elimination	Renal: Renal elimination of active and inactive metabolites; α -hydroxymidazolam accumulates in renal failure	Renal: 70% renal (metabolites)	Renal: 50-70% metabolites; <2% unchanged
Mechanism			
Target	GABAA receptor	GABAA receptor	GABA-A receptor benzodiazepine site
Action	Positive allosteric modulator at the α/γ interface of the GABAA receptor $\rightarrow$ $\uparrow$ affinity of GABAA for GABA $\rightarrow$ $\uparrow$ frequency of Cl ⁻ channel opening $\rightarrow$ $\uparrow$ Cl ⁻ conductance $\rightarrow$ hyperpolarisation and $\downarrow$ neuronal transmission	Positive allosteric modulator at the α/γ interface of the GABAA receptor $\rightarrow$ $\uparrow$ affinity of GABAA for GABA $\rightarrow$ $\uparrow$ frequency of Cl ⁻ channel opening $\rightarrow$ $\uparrow$ Cl ⁻ conductance $\rightarrow$ hyperpolarisation and $\downarrow$ neuronal transmission	Positive allosteric modulation $\rightarrow$ $\uparrow$ frequency of GABA-mediated Cl ⁻ channel opening
Contraindications			Chronic obstructive airways disease with incipient respiratory failure Severe hepatic impairment Dependence on alcohol, drugs of abuse or CNS depressants
Pregnancy			
Code	C	C	B3
Justification	High-dose labour use $\rightarrow$ neonatal hypotonia, respiratory depression and hypothermia; prolonged use $\rightarrow$ withdrawal	High-dose labour use $\rightarrow$ neonatal hypotonia, respiratory depression and hypothermia; prolonged use $\rightarrow$ withdrawal	High-dose labour exposure $\rightarrow$ neonatal hypotonia, respiratory depression and hypothermia; prolonged use $\rightarrow$ neonatal withdrawal
Pharmacodynamics			

	Midazolam	Diazepam	Clonazepam
Respiratory	↓ VT ↑ RR Apnoea Respiratory response to raised CO ₂ flattened	↓ VT ↑ RR Apnoea Respiratory response to raised CO ₂ flattened	↓ VT Apnoea
CVS	↓ SVR ↓ SBP ↓ DBP ↑ HR Typically stable CO	↓ SVR ↓ SBP ↓ DBP ↑ HR Typically stable CO	↓ MAP
CNS	Hypnosis Sedation Anterograde amnesia Anticonvulsant ↓ CBF ↓ MAC	Hypnosis Sedation Anterograde amnesia Anticonvulsant ↓ CBF ↓ MAC	Sedation and hypnosis Anticonvulsant Anterograde amnesia Ataxia
Endocrine	↓ Adrenergic stress response	↓ Adrenergic stress response	
Musculoskeletal	Skeletal muscle relaxation	Skeletal muscle relaxation	Skeletal muscle relaxation
Considerations	α-hydroxymidazolam accumulates in renal failure	IV rate ≤5mg/min Long-acting active metabolites — accumulation risk	May ↑ salivation Reversal with flumazenil

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Antidepressants

Symptoms and management of TCA overdose is covered under Tricyclic Antidepressant Overdose.

Antidepressant drugs include:

- Tricyclic Antidepressants (TCAs)
 - Mechanism of action by multiple effects, including:
 - Competitively inhibit reuptake of NA and 5-HT
 - Muscarinic antagonism
 - Leads to anticholinergic side effects (dry mouth, blurred vision, constipation, urinary retention).
 - H₁ and H₂ antagonism
 - α_1 antagonism
 - NMDA antagonism
- Selective Serotonin Reuptake Inhibitors (SSRIs)
 - Inhibit neural reuptake of 5-HT
 - Preferred over TCAs as:
 - Similar effectiveness
 - Better side effect profile
- Monoamine Oxidase Inhibitors (MAO-Is)
 - Inhibit monoamine oxidase on external mitochondrial membrane, increasing the level of amine neurotransmitters in the CNS and PNS
 - Two enzymes exist:
 - MAO-A
 - Dominant enzyme in CNS
 - Acts on serotonin, noradrenaline, adrenaline
 - MAO-B
 - Dominant in GIT and platelets
 - Responsible for 75% of MAO activity
 - Preferential metabolism of non-polar amines

- MAO-Is classified by their mechanism and selectivity
 - Non-selective, irreversible
 - Bind covalently to the enzyme, permanently inactivating it.
 - May lead to hypertensive crisis when catecholamine levels increased
 - Tyramine in food
 - Metabolised by MAO-B.
 - Indirectly acting **sympathomimetics**
 - Absolutely contraindicated.
 - Risk of serotonin syndrome with serotonin reuptake inhibitors
 - Include:
 - Phenelzine
 - Isocarboxazid
 - Tranylcypromine
 - Enzyme levels will take 2-3 weeks to recover following cessation
 - MAO-A selective, reversible
 - Hypertensive crisis is less common
 - MAO-B unaffected - tyramine is metabolised
 - Short acting
 - Enzyme levels normalise after 24 hours of cessation.
 - Include:
 - Moclobemide
 - MAO-B selective
 - Much lower risk of hypertensive crisis
 - Include:
 - Selegiline
- Discontinuation syndrome may occur if abruptly ceased

Table 1: Comparison of Antidepressant Classes

Property	Tricyclic Antidepressants	Selective Serotonin Reuptake Inhibitors	Monoamine Oxidase Inhibitors
Example Uses	Amitriptyline Depression, treatment of chronic pain and trigeminal neuralgia	Fluoxetine Depression, anxiety	Treatment resistant depression. Now largely superseded due to side-effect profile
Absorption Distribution	High PO bioavailability Highly lipid soluble with High V_D . Very highly protein bound - leads to interactions with warfarin, digoxin, and aspirin	High PO bioavailability Highly protein bound, high V_D	
Metabolism	Hepatic with active metabolites. Large interpatient variability	Hepatic with non-linear kinetics Venlafaxine does not affect CYP450 enzymes.	
Elimination		Unaffected by renal impairment	
Resp CVS	Dry mouth Postural hypotension, $\uparrow$ HR. QT prolongation and widening QRS in overdose, with arrhythmia more likely when QRS exceeds 0.16s.	Less cardiotoxic than TCAs, may precipitate serotonin syndrome	
CNS	Sedation, blurred vision, lowered seizure threshold. Excitation, followed by seizures and depression in overdose.	Identical antidepressant effect to TCAs. Less sedation	
Renal	Urinary retention		

Property	Tricyclic Antidepressants	Selective Serotonin Reuptake Inhibitors	Monoamine Oxidase Inhibitors
GU	Sexual dysfunction	Greater incidence of sexual dysfunction compared with TCAs	
GIT	Constipation	Greater incidence of N/V compared with TCAs	
Other	Multiple complex drug interactions, including arrhythmias and variable BP with sympathomimetics, central anticholinergic syndrome, serotonin syndrome, and seizures. ↑ Sensitivity to catecholamines - suggest avoiding: • Indirectly acting sympathomimetics • Ketamine • Surgical stress	Continue during perioperative period to avoid risk of discontinuation syndrome.	

Individual Antidepressants

Table 2: Tricyclic Antidepressants

Amitriptyline	
Aliases	Endep
Class	Tricyclic antidepressant
Chemistry	Tricyclic compound
Dose	
Neuropathic pain (ICU, PO/NG)	10mg nocte; increase to 25 - 50mg nocte
Depression (PO, outpatient)	75mg/day in divided doses; max 150mg/day
Depression (PO, inpatient)	100mg/day initially; usual max 200mg/day; exceptionally 300mg/day
Pharmaceutics	
Presentation	10mg, 25mg and 50mg tablets
Solubility	Water-insoluble and lipid-soluble
Pharmacokinetics	
Absorption	Rapid and complete; bioavailability 33 - 62% due to first-pass metabolism
Distribution	pKa: 9.76
	Vd: 19L/kg
	Protein binding: 84%
Metabolism	Half-life: ~20 hours
	Pathway: Hepatic N-demethylation and hydroxylation, mainly CYP2D6
	Metabolites: Nortriptyline (active), Hydroxy and N-oxide metabolites (some antidepressant activity)

Amitriptyline	
Elimination	Renal: Mainly metabolites and conjugates; very little unchanged
Mechanism	
Target	Noradrenaline and serotonin transporters Muscarinic, H1 and alpha1 receptors Cardiac Na ⁺ channels in overdose
Action	Inhibits noradrenaline and serotonin re-uptake; antimuscarinic, H1 and alpha1 antagonism
Contraindications	MAOI use within 14 days
	Cisapride
	Acute recovery after myocardial infarction
Pregnancy	
Code	C
Justification	Prolonged maternal use may cause neonatal withdrawal
Pharmacodynamics	
CVS	↓ MAP and ↑ HR Overdose causes QRS/QT prolongation and dysrhythmias
CNS	Antidepressant and neuropathic analgesia Sedation and blurred vision Decreased seizure threshold
Renal	Urinary retention
Gastrointestinal	Xerostomia and constipation
Considerations	Sympathomimetics and ketamine may cause dysrhythmias or variable BP
	Serotonergic medicines may cause serotonin syndrome
	Overdose causes cardiac Na ⁺ channel blockade Abrupt cessation may cause withdrawal

Table 3: Serotonin-Reuptake Inhibitors

	Fluoxetine	Venlafaxine
Aliases	Prozac Zactin	Efexor-XR
Class	Selective serotonin reuptake inhibitor	Serotonin-norepinephrine reuptake inhibitor
Chemistry	Phenoxypiphenyl-propylamine	Phenethylamine derivative; racemate
Dose		
Depression (PO)	20mg daily; doses >20mg/day in 2 divided doses; max 80mg/day	
OCD (PO)	20mg daily; usual 20-60mg/day; max 80mg/day	
Depression/anxiety (PO, MR)		75mg daily; max 225mg daily
Panic disorder (PO, MR)		37.5mg daily for 4-7 days, then 75mg daily; max 225mg daily
Pharmaceutics		
Presentation	20mg capsules	37.5mg, 75mg and 150mg modified-release capsules
Compatibility		Swallow whole; do not divide, crush, chew or dissolve
Solubility	Poorly water-soluble and lipid-soluble	Freely water-soluble
Pharmacokinetics		

	Fluoxetine	Venlafaxine
Absorption	80-95%; peak 6-8 hours; food may slightly delay but does not reduce absorption	At least 92%; bioavailability ~45% due to first-pass metabolism; MR peaks ~6 hours (parent) and ~9 hours (O-desmethylvenlafaxine)
Distribution	pKa: 9.8 Vd: 14-100L/kg Protein binding: 94%	pKa: ~10 Vd: 6-7L/kg Protein binding: 27%
Metabolism	Half-life: 1-3 days after acute dosing; 4-6 days with chronic dosing; norfluoxetine 4-16 days Pathway: Extensive hepatic metabolism by CYP2D6 and other CYP pathways Metabolites: Norfluoxetine (active), Unidentified inactive metabolites	Half-life: Disposition: 5 hours; O-desmethylvenlafaxine 11 hours; MR apparent: 15 hours (absorption-limited) Pathway: Hepatic CYP2D6 to O-desmethylvenlafaxine; CYP3A4 N-demethylation Metabolites: O-desmethylvenlafaxine (active), N-desmethylvenlafaxine and N,O-didesmethylvenlafaxine

	Fluoxetine	Venlafaxine
Elimination	Renal: Metabolites	Renal: 87% of dose in urine within 48 hours; 5% of dose unchanged, remainder metabolites
Mechanism		
Target	Serotonin transporter	Serotonin and noradrenaline transporters
Action	Selectively inhibits neuronal serotonin reuptake	Inhibits serotonin and noradrenaline reuptake; weak dopamine reuptake inhibition
Contraindications	MAOI use within 14 days; wait at least 5 weeks after fluoxetine before an MAOI Pimozide Metoprolol for cardiac failure	MAOI use within 14 days; wait at least 7 days after venlafaxine before an MAOI
Pregnancy		
Code	C	B2
Justification	Third-trimester exposure may cause neonatal withdrawal	
Pharmacodynamics		
CVS		↑ MAP and HR
CNS	Antidepressant Agitation, insomnia, anxiety and tremor	Antidepressant and anxiolytic Dizziness, somnolence or insomnia
Endocrine	SIADH and hyponatraemia	SIADH and hyponatraemia
Metabolic	Weight loss or gain	

	Fluoxetine	Venlafaxine
Genitouri- nary	Sexual dysfunction	Sexual dysfunction
Gastroin- testinal	Nausea, diarrhoea and anorexia	Nausea, xerostomia and constipation
Haematolog- ical	Increased bleeding	Increased bleeding
Considera- tions	Serotonergic medicines may cause serotonin syndrome Potent CYP2D6 inhibitor with prolonged interaction risk Long half-life requires at least a 5-week MAOI washout Discontinuation symptoms are rare with the long half-life Hepatic impairment requires a lower or less frequent dose	Serotonergic medicines may cause serotonin syndrome Abrupt cessation commonly causes discontinuation symptoms GFR 10-70mL/min: reduce dose by 25-50%; haemodialysis: reduce by 50% Mild-moderate hepatic impairment: reduce dose by 50%; consider further reduction if severe Overdose may cause seizures, QRS/QT prolongation and dysrhythmias

Table 4: Monoamine Oxidase Inhibitors

	Phenelzine	Isocarboxazid	Tranylcypromine	Moclobemide	Selegiline
Aliases	Nardil	Marplan	Parnate	Aurorix	Eldepryl L-deprenyl
Class	Irreversible non-selective monoamine oxi- dase inhibitor	Irreversible non-selective monoamine oxi- dase inhibitor	Irreversible non-selective monoamine oxi- dase inhibitor	Reversible selective monoamine oxi- dase inhibitor	Irreversible selective monoamine oxi- dase B inhibitor
Chemistry	Hydrazine derivative	Hydrazine derivative	Substituted am- phetamine	Benzamide derivative	Substituted am- phetamine
Dose					
Depression (PO)	15mg TDS; increase to at least 60mg/day; max 90mg/day	10mg BD; in- crease by 10mg every 2-4 days to 40mg/day; max 60mg/day	10mg morn- ing and after- noon; increase to 10mg TDS after 2 weeks; max 30mg/day	Initially 300 or 450mg/day in 2 divided doses after meals; usual 300- 600mg/day; no increase in the first week	
Parkinson disease (PO)					5mg with breakfast and 5mg with lunch; max 10mg/day
Pharmaceutics					
Presentation	15mg tablets	10mg tablets	10mg tablets	150mg and 300mg tablets	5mg tablets
Solubility			Reasonably wa- ter-soluble	Poorly water- soluble	Lipid-soluble; hydrochloride freely water- soluble
Pharmacokinetics					

	Phenelzine	Isocarboxazid	Tranylcypromine	Moclobemide	Selegiline
Absorption	Rapid		Rapid and complete; bioavailability ~50%	Complete; bioavailability ~50% due to first-pass metabolism and increases with repeated dosing	Rapid; bioavailability ~10% due to first-pass metabolism and increased up to 3-fold by high-fat food
Distribution			pKa: 9.6 Vd: 1.75L/kg Protein binding: Probably high	pKa: ~6.2 Vd: ~1.2L/kg (Vss) Protein binding: 50%	pKa: 8.6 Vd: ~25L/kg Protein binding: 96%

	Phenelzine	Isocarboxazid	Tranylcypromine	Moclobemide	Selegiline
Metabolism	Half-life: 1.2h	Duration: MAO inhibition persists after drug elimination	Half-life: ~2 hours	Half-life: ~2 hours	Half-life: ~1.5 hours
	Duration: MAO inhibition up to 2 weeks		Duration: MAO activity recovers 3-5 days after cessation	Duration: MAO inhibition ~24 hours	Duration: Prolonged by irreversible MAO-B inhibition
	Pathway: Acetylation		Pathway: N-acetylation and ring hydroxylation; probably hepatic	Pathway: Hepatic morpholine oxidation, hydroxylation and deamination; partly CYP2C19/2D6	Pathway: Hepatic CYP2B6; possible CYP3A4/2A6 contribution (in vitro)
	Metabolites: Phenylacetic acid, Parahydroxyphenylacetic acid		Metabolites: N-acetylated and ring-hydroxylated metabolites (weakly active)	Metabolites: N-oxide metabolite (slight activity; very low circulating concentrations), Other metabolites (some activity in vitro/animals; very low circulating concentrations)	Metabolites: N-desmethylelegiline (active), L-methamphetamine (active), L-amphetamine (active)

	Phenelzine	Isocarboxazid	Tranylcypromine	Moclobemide	Selegiline
Elimination	Renal: Up to 79% of dose as metabolites within 96h		Renal: Minimal; 4% unchanged	Renal: Metabolites; less than 1% unchanged	Renal: 86% of active metabolites recovered in urine
Mechanism					
Target	Monoamine oxidase A and B	Monoamine oxidase A and B	Monoamine oxidase A and B	Monoamine oxidase A	Monoamine oxidase B at doses 10mg/day or less
Action	Irreversibly inactivates MAO → reduced monoamine metabolism	Irreversibly inhibits MAO → increased central biogenic amines	Irreversibly inhibits MAO → increased central monoamines	Reversible preferential MAO-A inhibition reduces noradrenaline, serotonin and dopamine metabolism	Irreversible MAO-B inhibition → reduced dopamine metabolism

	Phenelzine	Isocarboxazid	Tranylcypromine	Moclobemide	Selegiline
Contraindications	Phaeochromocytoma Cerebrovascular disease or congestive heart failure Current/past hepatic disease or abnormal LFTs Guanethidine, sympathomimetics, other antidepressants, pethidine or dextromethorphan Elective GA; cease at least 10 days before surgery	Other MAOIs, dibenzazepines, sympathomimetics, pethidine, dextromethorphan or serotonergic antidepressants Phaeochromocytoma Cardiovascular or cerebrovascular disease Hypertension Hepatic disease or abnormal LFTs	Other MAOIs, TCAs, SSRIs, SNRIs, sympathomimetics, pethidine or dextromethorphan Phaeochromocytoma Cardiovascular or cerebrovascular disease Hypertension Hepatic disease	Acute confusional states Serotonin syndrome Clomipramine, selegiline, bupropion, tritans, pethidine, tramadol, dextromethorphan or linezolid	Pethidine SSRIs, SNRIs, TCAs, sympathomimetics or other MAOIs Active peptic ulcer
Pregnancy Code	B3		B2	B3	B2
Pharmacodynamics					
CVS	Orthostatic ↓ MAP Tyramine or sympathomimetics may cause hypertensive crisis	Orthostatic hypotension Tyramine or sympathomimetics may cause hypertensive crisis	Orthostatic hypotension Tyramine or sympathomimetics may cause hypertensive crisis	Hypertension	Orthostatic hypotension Hypertension and dysrhythmias

	Phenelzine	Isocarboxazid	Tranylcypromine	Moclobemide	Selegiline
CNS	Antidepressant Sedation, dizziness, insomnia and tremor Seizures in toxicity	Antidepressant Dizziness, insomnia, somnolence and headache Seizures in toxicity SIADH	Antidepressant Insomnia, agitation and dizziness Seizures and mania	Antidepressant Restlessness, agitation, insomnia and dizziness	Antiparkinsonian Insomnia, dizziness, confusion and hallucinations Dyskinesia with levodopa
Endocrine Metabolic Genitourinary	Weight gain Sexual dysfunction	Weight gain Sexual dysfunction	Hypoglycaemia Sexual dysfunction		
Gastrointestinal	Xerostomia, nausea and constipation Hepatotoxicity	Xerostomia, nausea and constipation	Hepatotoxicity	Nausea	Nausea, xerostomia and constipation

	Phenelzine	Isocarboxazid	Tranylcypromine	Moclobemide	Selegiline
Considerations	Tyramine-restricted diet during treatment and for 2 weeks after cessation Serotonergic medicines may cause serotonin syndrome Sympathomimetics may cause hypertensive crisis Cocaine and local anaesthetics containing sympathomimetic vasoconstrictors are contraindicated Abrupt cessation may cause withdrawal	Second-line antidepressant Tyramine-restricted diet during treatment and for 2 weeks after cessation Serotonergic medicines may cause serotonin syndrome Sympathomimetics may cause hypertensive crisis Abrupt cessation may cause withdrawal	Tyramine-restricted diet during treatment and for at least 1 week after cessation Serotonergic medicines may cause serotonin syndrome Sympathomimetics may cause hypertensive crisis Abrupt cessation may cause withdrawal	Serotonergic medicines may cause serotonin syndrome Sympathomimetics and large tyramine loads may cause hypertension MAO activity returns within ~24 hours of cessation Severe hepatic impairment requires half to one-third dose	MAO-B selectivity diminishes above 20mg/day → possible tyramine sensitivity Pethidine may cause fatal hyperpyrexia and rigidity Serotonergic medicines may cause serotonin syndrome Levodopa adverse effects may require a 10-30% levodopa reduction May cause impulse-control disorders

Serotonin Syndrome

Serotonin syndrome is excessive serotonin in the CNS, typically as a consequence of drug interactions. The syndrome may be mild, moderate, or severe, and presents with some or all of:

- Altered mental state
 - Confusion
- Motor changes
 - Myoclonus
 - Hyperreflexia
 - Tremor
- Autonomic instability
 - Diaphoresis
 - Shivering
 - Fever

Serotonin syndrome is typically self-limiting and resolves with cessation of the drug.

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Antipsychotics

Antipsychotics are drugs used for the management of psychoses and thought disorders. They have a complicated mechanism of action with effects on multiple receptors:

- Central dopamine (typically D_2 , but varies with agent) antagonism
Responsible for the antipsychotic properties
- 5-HT₂ antagonism
- Other receptors which are quantitatively less important:
 - H₁ antagonism
 - α_1 antagonism
 - Muscarinic ACh antagonism

Based on their affinity to various receptors, they are (loosely) classified as either:

- **Typical** or 1st generation antipsychotics
Higher affinity for D_2 receptors (subsequently less blockade of 5-HT₂), causing a greater effect on 'positive' symptoms' and a greater incidence of extrapyramidal side effects
- **Atypical** or 2nd generation, which typically have fewer motor effects
Have greater effect on negative symptoms.

Table 1: Common Features of Antipsychotics

Property	Drug
Uses	Behavioural emergencies, schizophrenia/psychosis
CVS	QT prolongation

Property	Drug
CNS	Apathy, ↓ initiative, ↓ response to external stimuli, ↓ aggression. No loss of intellectual function.
Endocrine	↑ Prolactin (typical antipsychotics)
Haeme	Leukopenia and agranulocytosis (predominantly clozapine, but can be all)
Metabolic	Weight gain, diabetes, hypercholesterolaemia (all atypical > typical)
Other Toxicities	Neuroleptic malignant syndrome, EPSE

Neuroleptic Malignant Syndrome

Antipsychotic Malignant Syndrome is rare and presents similarly to MH, with a rapid rise in body temperature and confusion. It has a high mortality (up to 20%).

Extra-Pyramidal Side Effects

Motor disturbances from antipsychotic use are termed EPSEs, and are divided into two main types:

- **Acute Dystonic Reactions** are involuntary movements and parkinsonian symptoms. They are:
 - More common with typical agents
 - Decline with ongoing use
 - Reversible with cessation of the agent
- **Tardive dyskinesia** is similar to ADR, except:
 - Involuntary movements are more pronounced and disabling
 - It occurs with long term use (10-20 years)

- They are **irreversible**, and worsen when therapy is stopped

Table 2: Comparison of Antipsychotics

	Haloperidol	Olanzapine	Clozapine
Aliases	Haldol Serenace	Zyprexa	Clozaril Clopine
Class	Typical antipsychotic	Atypical antipsychotic	Atypical antipsychotic
Chemistry	Butyrophenone	Thienobenzodiazepine	Tricyclic dibenzodiazepine
Dose			
Agitation (IM/IV)	0.5 - 10mg; repeat after ≥ 1 hour IM or ≥ 30 minutes IV; max 20mg/day		
Agitation (IM)		5 - 10mg; repeat ≤ 10 mg after ≥ 2 hours, then ≥ 4 hours later; max 3 doses/30mg in 24 hours	
Psychosis (PO/NG) Schizophrenia (PO)		5 - 20mg daily	12.5mg daily - BD initially; usual 200 - 450mg/day divided; max 600mg/day (exceptionally 900mg/day)
Pharmaceutics			
Presentation	5mg/mL injection; tablets and 2mg/mL oral liquid	10mg powder for IM injection; tablets and orally disintegrating wafers	25mg and 100mg yellow tablets; 50mg/mL oral suspension
Compatibility	IV undiluted or dilute in 5 - 10mL 0.9% NaCl; protect from light	Reconstitute IM vial with 2.1mL water for injection; use within 1 hour	
Solubility	Minimally water - soluble	Practically insoluble in water	
Pharmacokinetics			

	Haloperidol	Olanzapine	Clozapine
Absorption	Rapid and complete PO absorption; bioavailability ~60%; IM peak ~20 minutes	PO bioavailability ~60%, peak 5-8 hours; IM peak 15-45 minutes	PO absorption 90-95%; bioavailability 50-60%; steady-state peak 2.1 hours; food has no effect
Distribution	pKa: 8.66 Vd: 18L/kg Protein binding: 92%	Vd: 14L/kg Protein binding: 93%	Vd: 5.4±3.5L/kg Protein binding: 95%
Metabolism	Half-life: 14-36 hours Pathway: Hepatic CYP3A4 and CYP2D6; N-dealkylation and carbonyl reduction Metabolites: Piperidine metabolites (inactive), Hydroxyhaloperidol (weakly active)	Half-life: 33 hours (5th-95th percentile 21-54 hours) Pathway: Hepatic glucuronidation and CYP1A2/2D6 oxidation Metabolites: 10-N-glucuronide (does not cross BBB), N-desmethyl and 2-hydroxymethyl metabolites (weak activity in animal studies)	Half-life: 12 hours (range 4-66 hours) Pathway: Hepatic CYP1A2 and CYP3A4; minor CYP2C19 and CYP2D6 Metabolites: N-desmethyl-clozapine (active)
Elimination	Renal: Mainly metabolites; ~1% unchanged	Renal: 57% urine and 30% faeces; ~7% unchanged in urine	Renal: 50% urine and 30% faeces; trace unchanged
Mechanism			

	Haloperidol	Olanzapine	Clozapine
Target	Dopamine D2 receptors α -adrenoceptors	5-HT _{2A} and 5-HT _{2C} receptors Dopamine D1-D5 receptors Muscarinic receptors Histamine H1 receptors α ₁ -adrenoceptors	Dopamine D1 and D2 receptors 5-HT receptors Muscarinic receptors Histamine H1 receptors α -adrenoceptors
Action	Competitive postsynaptic D2 antagonism, predominantly in mesolimbic pathways	Monoaminergic antagonism with greater affinity for 5-HT ₂ than D2 receptors	Weak D1/D2 and potent serotonergic, muscarinic, histaminergic and α -adren- ergic antagonism
Contraindications	Parkinson disease or basal-ganglia lesions Coma or CNS depression Clinically significant cardiac disease, QTc prolongation or ventricular arrhythmia Clinically significant bradycardia, 2nd/3rd-degree heart block or uncorrected hypokalaemia Concomitant QT-prolonging drugs Prolactin-dependent tumour		Drug-induced granulocytopenia/agranulocytosis or bone-marrow disorder Unable to undergo regular blood tests Uncontrolled epilepsy Severe renal, cardiac or hepatic disease Circulatory collapse, CNS depression or coma Paralytic ileus
Pregnancy			
Code	C	C	C
Justification	High doses in late pregnancy → prolonged neonatal neurological disturbance	Third-trimester exposure → neonatal extrapyramidal or withdrawal symptoms	Adult pharmacological and toxicological effects may occur in the fetus
Pharmacodynamics			
Respiratory	Laryngospasm and bronchospasm	Respiratory depression (overdose)	

	Haloperidol	Olanzapine	Clozapine
CVS	QT prolongation and torsades de pointes Tachycardia and ↓ MAP	↓ MAP, bradycardia and tachycardia	Tachycardia and ↓ MAP Myocarditis and cardiomyopathy
CNS	Antipsychotic Extrapyramidal effects and tardive dyskinesia ↓ Seizure threshold	Antipsychotic Somnolence, dizziness and tremor ↓ Seizure threshold	Antipsychotic Sedation ↓ Seizure threshold
Endocrine Metabolic	↑ Prolactin	↑ Prolactin Weight gain, dyslipidaemia and hyperglycaemia	Weight gain, dyslipidaemia and hyperglycaemia
Genitourinary Gastrointestinal	Urinary retention Antiemetic Constipation and xerostomia	Xerostomia and constipation	Gastrointestinal hypomotility Hypersalivation Neutropenia and agranulocytosis
Haematological			WBC/ANC weekly for 18 weeks, monthly thereafter and for 1 month after cessation Interruption >2 days → restart 12.5mg daily - BD Myocarditis risk highest in first month Smoking cessation → ↑ clozapine concentration Concomitant myelosuppressants → avoid Neuroleptic malignant syndrome
Considerations	IM recommended; IV → continuous ECG monitoring Neuroleptic malignant syndrome Lithium → possible severe neurotoxicity CYP3A4/2D6 inhibitors or inducers alter exposure	IM + parenteral benzodiazepines → excessive sedation and cardiorespiratory depression Smoking and carbamazepine → ↑ CYP1A2 clearance Neuroleptic malignant syndrome Tardive dyskinesia Elderly dementia-related psychosis → ↑ mortality and cerebrovascular events	

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Anticonvulsants

In **general**, anticonvulsants are:

- Well absorbed orally
- Highly protein bound
- Hepatically metabolised by CYP450 enzymes, and induce their own metabolism (as well as that of other drugs)
- Renally eliminated
- Interact with each other

Table 1: Anticonvulsants

	Phenytoin	Sodium valproate	Carbamazepine	Levetiracetam
Aliases	Dilantin DBL Phenytoin In- jection	Valproate Valproic acid Epilim	Tegretol	Keppra
Class	Anticonvulsant; class IB antiarrhythmic	Broad-spectrum an- ticonvulsant; mood stabiliser	Anticonvulsant; class I antiarrhythmic	Anticonvulsant
Chemistry	Hydantoin	Branched-chain fatty acid	Tricyclic diben- zoazepine	S-enantiomer pyrrolidone; racetam
Dose				
Status (IV)	10-15mg/kg; rate ≤50mg/min			
Dysrhythmia (IV)	3-5mg/kg; repeat if required			
Load (IV)		Up to 10mg/kg over 3-5 minutes		
Infusion (IV)		1-2mg/kg/hour; max 2500mg/day		
Epilepsy (PO)			100-200mg 1-2 times/day initially; usually 400mg 2-3 times/day	
Monotherapy (PO/IV)				250mg BD for 2 weeks, then 500mg BD; max 1500mg BD
Adjunct (PO/IV)				500mg BD; max 1500mg BD
Pharmaceutics				

	Phenytoin	Sodium valproate	Carbamazepine	Levetiracetam
Presentation	50mg/mL clear colourless solution with propylene glycol and ethanol	400mg powder with 4mL water for injection; reconstitutes to 100mg/mL	100mg/5mL liquid; 200mg tablet; 200mg and 400mg CR tablets	250mg, 500mg and 1g tablets; 100mg/mL oral solution; IV solution
Compatibility	Do not dilute; give into a large vein, flush with 0.9% NaCl; incompatible with glucose	Infuse at 4mg/mL, max 8mg/mL, in 0.9% NaCl, glucose 5% or glucose-saline; separate line	Dilute liquid 1:1 with water for NG administration	IV dose may be diluted in 100mL 0.9% NaCl and infused over 15 minutes
Solubility	pH 12; pKa 8.0-9.2; poor water solubility	pKa 4.8; sodium salt is water-soluble	pKa 7-13; minimally water-soluble and highly lipid-soluble	pKa -2.0; very water-soluble
Pharmacokinetics				
Absorption	IV bioavailability 100%	IV 100%; PO nearly 100%	PO bioavailability ~85%; slow, peak 4-24 hours	Rapid and complete PO absorption; bioavailability ~100%; peak 1.3 hours
Distribution	pKa: 8.0-9.2	pKa: 4.8	pKa: 7-13	pKa: -2
	Vd: 1.6-2.5L	Vd: 0.2L/kg	Vd: 0.7-1.4L/kg	Vd: 0.5-0.7L/kg
	Protein binding: 90%	Protein binding: 90%; concentration-dependent	Protein binding: 75-80%	Protein binding: <10%

	Phenytoin	Sodium valproate	Carbamazepine	Levetiracetam
Metabolism	Half-life: ~22 hours; range 7-42 hours Duration: IV effect up to 24 hours Pathway: Hepatic CYP2C9 and CYP2C19 hydroxylation; saturable Metabolites: HPPH (inactive)	Half-life: 9-18 hours Pathway: Hepatic glucuronidation 40-60%; beta-oxidation 30-40%; CYP oxidation remainder Metabolites: Active and inactive metabolites	Half-life: 36 hours single dose; 16-24 hours after autoinduction Pathway: Hepatic CYP3A4 epoxidation; induces its own metabolism Metabolites: Carbamazepine-10,11-epoxide (active)	Half-life: 7 ± 1 hours Pathway: Non-CYP acetamide hydrolysis Metabolites: ucb L057 (inactive)
Elimination	Renal: Biliary metabolites are reabsorbed then excreted in urine	Renal: 1-3% unchanged; remainder metabolites	Renal: 72% mainly metabolites; 28% faeces	Renal: 95%; 66% unchanged and 24% ucb L057 within 48 hours
Mechanism				
Target	Voltage-gated Na ⁺ channels	Voltage-gated Na ⁺ channels GABA degradative enzymes	Voltage-gated Na ⁺ channels	Synaptic vesicle protein 2A N-type Ca ²⁺ channels
Action	Stabilises the inactivated state → ↓ generation and propagation of action potentials	Stabilises inactivated Na ⁺ channels and inhibits GABA catabolism → ↑ brain GABA	Stabilises the inactivated state → ↓ sustained repetitive neuronal firing	Mechanism incompletely defined; SV2A binding and partial ↓ N-type Ca ²⁺ currents contribute

	Phenytoin	Sodium valproate	Carbamazepine	Levetiracetam
Contraindications	Sinus bradycardia SA block Second- or third-degree AV block Stokes-Adams syndrome	Pregnancy unless no suitable alternative for epilepsy Acute or chronic hepatic dysfunction Urea-cycle disorder Uncorrected systemic carnitine deficiency Hepatic porphyria POLG-related mitochondrial disorder	AV block Hepatic porphyria Bone-marrow depression Significant hepatic dysfunction MAOI use within 2 weeks	
Pregnancy				
Code	D	D	D	B3
Justification	Fetal hydantoin syndrome, neurodevelopmental disorder and fetal or neonatal bleeding	Contraindicated; dose-dependent malformations, 1-2% neural-tube defects and developmental disorders	~1% spina bifida; craniofacial, nail and developmental abnormalities; fetal or neonatal bleeding	Risk relative to conventional anticonvulsants is unknown
Pharmacodynamics				
Respiratory	Rapid IV administration → respiratory depression			
CVS	↓ MAP Bradycardia, conduction block and cardiac arrest ↓ Ventricular automaticity and QT		↓ Ventricular automaticity Toxicity → QRS prolongation and dysrhythmias	

	Phenytoin	Sodium valproate	Carbamazepine	Levetiracetam
CNS	Anticonvulsant Ataxia, nystagmus and dysarthria	Anticonvulsant Mood stabilisation Hyperammonaemic encephalopathy	Anticonvulsant Ataxia, nystagmus and diplopia Anticholinergic toxicity	Anticonvulsant Somnolence, asthenia and dizziness Behavioural and psychiatric disturbance
Endocrine			ADH-like effect → fluid retention and hyponatraemia	
Metabolic	Chronic use → ↓ vitamin D	Hyperammonaemia Hypocarnitinaemia Fanconi syndrome		
Renal				
Gastrointestinal	Hepatotoxicity Gingival hyperplasia	Hepatotoxicity Pancreatitis	Hepatotoxicity	
Haematological	Folate deficiency and blood dyscrasias	Thrombocytopenia and platelet dysfunction Coagulopathy	Thrombocytopenia and leukopenia Aplastic anaemia and agranulocytosis	

	Phenytoin	Sodium valproate	Carbamazepine	Levetiracetam
Considerations	Therapeutic total concentration 10-20microg/mL; free 0.8-2microg/mL Hypoalbuminaemia or uraemia → measure or interpret free concentration Continuous ECG and BP monitoring during IV loading High pH and propylene glycol → purple-glove syndrome and tissue necrosis Potent CYP inducer → many interactions SJS, TEN and DRESS	Protein binding saturates → disproportionate ↑ free drug Monitor LFT, FBC and coagulation Altered consciousness, vomiting or ataxia → check ammonia Carbapenems markedly ↓ valproate concentration Inhibits CYP, glucuronidation and epoxide hydrolase; displaces protein-bound drugs Valproate ↑ lamotrigine half-life and rash risk Consider carnitine for symptomatic hypocarnitinaemia or hyperammonaemia	Autoinduction over 2-4 weeks → falling concentration and shorter half-life Potent CYP3A4 inducer → many interactions Therapeutic concentration 4-12microg/mL HLA-B1502 and HLA-A3101 → ↑ severe cutaneous reactions SJS, TEN and DRESS Antagonises non-depolarising neuromuscular blockers Monitor FBC, LFT and Na⁺	Renal impairment → dose reduction PO↔IV conversion uses the same total daily dose and frequency Minimal CYP-mediated interactions Aggression, depression, psychosis and suicidality

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GABA Analogues

Gabapentin and pregabalin:

- Are both **structural analogues** of GABA
- Have no direct action on the GABA_A receptor
- **Act on the $\alpha_2\delta$ subunit of voltage gated Ca^{2+} channels** in the CNS, inhibiting neurotransmitter release
- May have some NMDA receptor activity

Table 1: Comparison of GABA Analogues

	Gabapentin	Pregabalin
Aliases	Neurontin Nupentin	Lyrica
Class	Anticonvulsant; neuropathic anal- gesic	Anticonvulsant; neuropathic anal- gesic
Chemistry	GABA analogue; amino acid	S-enantiomer GABA analogue; amino acid
Dose		
Epilepsy (PO)	300mg TDS; usual 900-1800mg/day, max 2400mg/day	75mg BD; ↑ to 150mg BD after 1 week; max 300mg BD
Neuropathic pain (PO/NG)	300mg TDS; max 3600mg/day	
Neuropathic pain (PO)		75mg BD; ↑ to 150mg BD after 3- 7 days; max 300mg BD
Pharmaceutics		

	Gabapentin	Pregabalin
Presentation	100mg, 300mg and 400mg capsules; 600mg and 800mg tablets	25mg, 75mg, 150mg and 300mg capsules
Compatibility	Capsules may be opened and mixed with 10mL water for NG administration	
Solubility	Freely water-soluble	Freely water-soluble
Pharmacokinetics		
Absorption	Saturable PO absorption; bioavailability ~60% at 300-600mg TDS and ↓ with dose; food has no effect	Rapid; bioavailability ≥90%; peak 1 hour; food ↓ rate but not extent
Distribution	Vd: 58 ± 6L	Vd: 0.56L/kg
	Protein binding: <3%	Protein binding: Nil
Metabolism	Half-life: 5-7 hours	Half-life: 6.3 hours
	Pathway: Not appreciably metabolised	Pathway: Negligible metabolism
		Metabolites: N-methylpregabalin (0.9%)
Elimination	Renal: Unchanged	Renal: 98% unchanged
Mechanism		
Target	α2δ subunit of presynaptic voltage-gated Ca ²⁺ channels	α2δ subunit of presynaptic voltage-gated Ca ²⁺ channels

	Gabapentin	Pregabalin
Action	Precise mechanism uncertain; ↓ Ca ²⁺ entry and excitatory neurotransmitter release	↓ Release of glutamate, noradrenaline and substance P
Pregnancy		
Code	D	D
Justification	Risk relative to conventional anticonvulsants is unknown	↑ Major congenital malformations (5.9% vs 4.1% unexposed)
Pharmacodynamics		
Respiratory	Respiratory depression	Respiratory depression
CVS	Peripheral oedema	Peripheral oedema
CNS	Anticonvulsant Neuropathic analgesia Somnolence, dizziness, ataxia and confusion	Anticonvulsant Neuropathic analgesia Dizziness, somnolence and confusion
Metabolic	Weight gain	Weight gain
Gastrointestinal	Nausea, diarrhoea and constipation	Nausea, constipation and diarrhoea

	Gabapentin	Pregabalin
Considerations	Renal impairment → dose reduction; haemodialysable Opioids/CNS depressants → ↑ sedation, respiratory depression, coma and death Misuse, dependence and withdrawal Abrupt withdrawal → seizures or status epilepticus DRESS, SJS and TEN	Renal impairment → dose reduction; haemodialysis removes ~50% in 4 hours Opioids/CNS depressants → ↑ sedation, respiratory depression, coma and death Misuse, dependence and withdrawal Gradual withdrawal over ≥1 week

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Inhalational Anaesthetic Agents

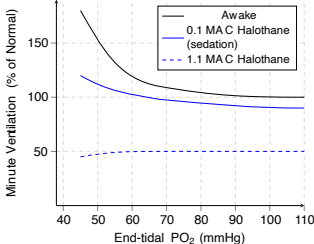
Describe the effects of inhalational agents on the cardiovascular, respiratory and central nervous systems

Describe the toxicity of inhalational agents

Describe the comparative pharmacology of nitrous oxide, halothane, enflurane, isoflurane, desflurane, sevoflurane, xenon and ether

This section covers features and structures of inhalational anaesthetics. Structure-activity relationships are covered under inhalational anaesthetics.

Table 1: Common Features of Inhalational Anaesthetic Agents

Property	Action
Metabolism	Hepatic CYP450 (CYP2E1) metabolises C-halogen bonds to release halogen ions (F^- , Cl^- , Br^-), which can be nephrotoxic and hepatotoxic. The C-F bond is minimally metabolised compared to the C-Cl, C-Br, and C-I bonds. All agents undergo hepatic oxidation, except for halothane which is reduced.
Resp	All halogenated agents $\downarrow V_T$ and $\uparrow RR$, with an overall $\downarrow$ in MV and therefore cause $PaCO_2$ to $\uparrow$; and $\downarrow$ sensitivity of central respiratory centres to CO_2. Impairment of HPV may worsen V/Q matching and $\uparrow$ shunt. 
CVS	$\downarrow$ MAP (predominantly by $\downarrow$ in SVR due to NO release and Ca^{2+} channel blockade), $\downarrow$ inotropy due to Ca^{2+} channel blockade.

Property	Action
CNS	Hypnosis. $\downarrow$ CMRO₂. Above 1 MAC there is uncoupling of the CBF - CMRO₂ relationship, and CBF $\uparrow$ despite $\downarrow$ CMRO₂ due to cerebral vasodilation. ICP may mirror CBF changes. All except halothane have some analgesic effect. $\downarrow$ EEG frequency such that θ- and δ-wave dominate the EEG as depth $\uparrow$. May cause burst suppression.
MSK	Muscle relaxation via blockade of Ca²⁺ channels. Additional augmentation of the effects of NMBD due to skeletal muscle vasodilation. May precipitate MH.
Renal	Dose dependent $\downarrow$ in RBF, GFR, and UO secondary to $\downarrow$ in MAP and CO. Fluorinated ethers produce F⁻ ions when hepatically metabolised, which may produce high-output renal failure at serum concentrations >50μmol/L. This is probably only a concern with methoxyflurane (as it has significant (>70%) hepatic metabolism) when used at anaesthetic doses.
GIT	$\downarrow$ Hepatic blood flow.
GU	Tocolysis.
Toxic Effects	Decreased fertility and increased risk of spontaneous abortion in operating theatre personnel.

Comparison of Common Inhalational Agents

Table 2: Comparison of Common Inhalational Anaesthetics

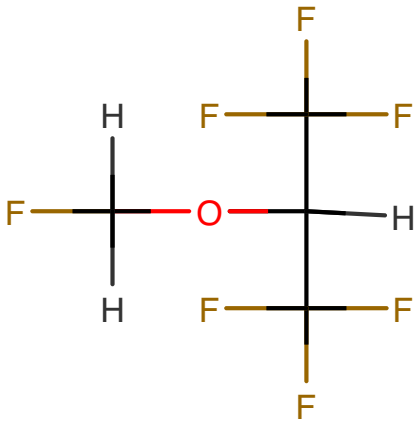
	Sevoflurane	Isoflurane	Desflurane
Aliases	Sevorane	Aerrane Forane	Suprane
Class	Volatile general anaesthetic	Volatile general anaesthetic	Volatile general anaesthetic
Chemistry	Fluorinated methyl isopropyl ether	Halogenated methyl ethyl ether; structural isomer of enflurane	Fluorinated methyl ethyl ether
Dose			
Induction	Inspired concentration up to 5% in adults or 7% in children	0.5%, increasing to 1.3-3% inspired	
Maintenance	0.5-3%; MAC ~2% (middle-aged, O ₂)	1-2.5% with N ₂ O/O ₂ or 1.5-3.5% in O ₂ ; MAC ~1.15% (middle-aged, O ₂)	2.5-8.5% end-tidal with O ₂ or O ₂ -enriched air; MAC ~6.6% (age-dependent)
Pharmaceutics			
Presentation	100% volatile liquid with ≥300ppm water	100% volatile liquid	100% volatile liquid
Compatibility	Requires an agent-specific vaporiser	Requires an agent-specific vaporiser	Requires a heated agent-specific vaporiser
Solubility	Blood:gas 0.63-0.69; oil:gas 47.2-53.9 (37°C)	Blood:gas 1.4; oil:gas 98	Blood:gas 0.42; oil:gas 29
Pharmacokinetics			
Absorption	Pulmonary; low blood solubility → rapid wash-in and wash-out	Pulmonary; moderate blood solubility → intermediate wash-in and wash-out	Pulmonary; very low blood solubility → rapid wash-in and wash-out

	Sevoflurane	Isoflurane	Desflurane
Metabolism	Duration: Rapid offset Pathway: ~5% hepatic CYP2E1 metabolism Metabolites: Hexafluoroisopropanol, inorganic fluoride and CO ₂	Pathway: ~0.2% hepatic metabolism Metabolites: Trifluoroacetic acid and inorganic fluoride	Duration: Rapid offset Pathway: Minimal hepatic metabolism (~0.02%)
Elimination	Renal: Predominantly exhaled unchanged; HFIP rapidly glucuronidated and eliminated	Renal: ~95% recovered in expired air; metabolites renal	Renal: Almost entirely exhaled unchanged
Mechanism			
Target	GABAA and glycine receptors	GABAA and glycine receptors	GABAA and glycine receptors
Action	NMDA (putative) and nicotinic acetylcholine receptors Enhances inhibitory and suppresses excitatory ion-channel transmission	NMDA (putative) and nicotinic acetylcholine receptors Enhances inhibitory and suppresses excitatory ion-channel transmission	NMDA (putative) and nicotinic acetylcholine receptors Enhances inhibitory and suppresses excitatory ion-channel transmission
Contraindications	Baralyme-containing re-breathing circuits Known or suspected malignant-hyperthermia susceptibility Previous hepatitis or unexplained hepatic dysfunction after a halogenated anaesthetic	Known or suspected malignant-hyperthermia susceptibility Previous hepatitis or unexplained hepatic dysfunction after a halogenated anaesthetic Non-selective MAOI use Obstetric surgery	Paediatric inhalational induction Known or suspected malignant-hyperthermia susceptibility Previous hepatitis or unexplained hepatic dysfunction after a halogenated anaesthetic

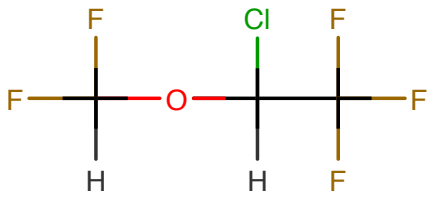
	Sevoflurane	Isoflurane	Desflurane
Pregnancy			
Code	B2	B3	B3
Justification	May cause neonatal CNS and respiratory depression	May cause neonatal CNS and respiratory depression	May cause neonatal CNS and respiratory depression
Pharmacodynamics			
Respiratory	Non-irritant bronchodilator ↓ VT and MV with ↑ RR and PaCO ₂ Impaired HPV	Airway irritation, coughing and laryngospasm ↓ VT and MV with ↑ RR and PaCO ₂ Bronchodilation and impaired HPV	Airway irritation, coughing, breath-holding and laryngospasm ↓ VT and ↑ RR; >1 MAC → ↓ MV and ↑ PaCO ₂ Bronchodilation and impaired HPV
CVS	↓ SVR and MAP without reflex tachycardia Minimal negative inotropy and QT prolongation	↓ SVR and MAP with reflex ↑ HR Mild negative inotropy	↓ SVR and MAP with minimal negative inotropy ↑ HR; rapid concentration increases cause marked sympathetic activation
CNS	Hypnosis; limited analgesic effect ↓ CMRO ₂ ; smallest ↑ CBF above 1.1 MAC Postoperative agitation in children	Hypnosis; limited analgesic effect ↓ CMRO ₂ ; ↑ CBF above 1 MAC	Hypnosis and amnesia ↓ CMRO ₂ ; ↑ CBF above 1 MAC
Musculoskeletal	Muscle relaxation and potentiation of non-depolarising NMBs	Muscle relaxation and potentiation of non-depolarising NMBs	Muscle relaxation and potentiation of non-depolarising NMBs
Metabolic	May trigger malignant hyperthermia	May trigger malignant hyperthermia	May trigger malignant hyperthermia

	Sevoflurane	Isoflurane	Desflurane
Renal	↓ RBF, GFR and urine output secondary to haemodynamic effects	↓ RBF, GFR and urine output secondary to haemodynamic effects	Renal blood flow generally preserved; haemodynamic depression may ↓ GFR and urine output
Genitourinary	Uterine relaxation	Uterine relaxation	Uterine relaxation
Gastrointestinal	Rare hepatitis	Rare hepatitis	Splanchnic blood flow generally preserved
Considerations	Soda lime degradation produces Compound A; nephrotoxic in rats but not established in humans Avoid desiccated CO ₂ absorbents; fresh gas flow <2L/min in circle absorber systems not recommended CYP2E1 induction by isoniazid or alcohol increases fluoride production	Coronary steal in selected animal models; human clinical significance uncertain Desiccated CO ₂ absorbent may produce carbon monoxide Repeat exposure within a short interval may increase hepatic risk	Unsuitable for inhalational induction because of airway irritation Rapid increases above 1 MAC → tachycardia and hypertension Desiccated CO ₂ absorbent may produce carbon monoxide Greater global-warming effect than isoflurane or sevoflurane

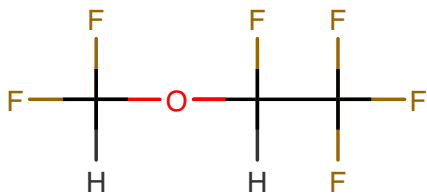
Sevoflurane:

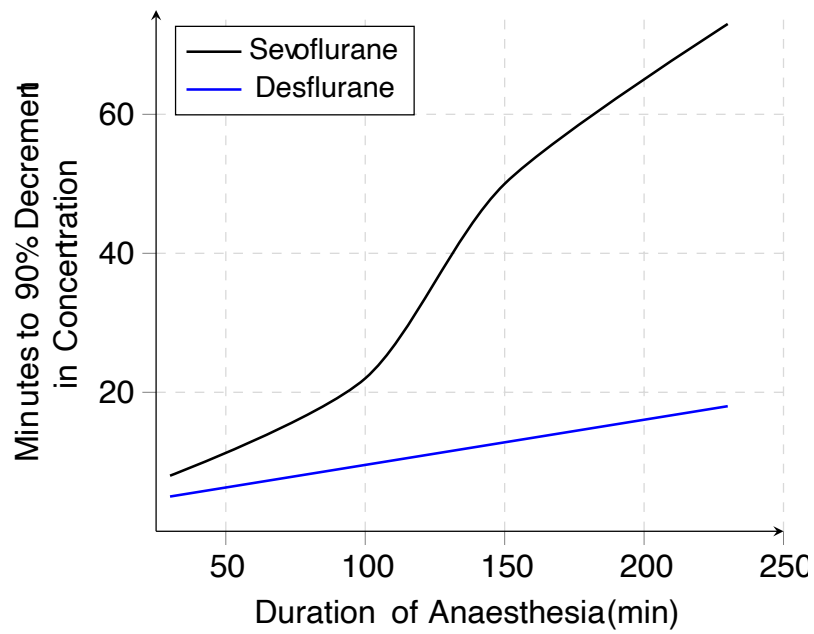


Isoflurane:



Desflurane:





Comparison of Uncommon Inhalational Agents

Table 3: Comparison of Uncommon Inhalational Anaesthetics

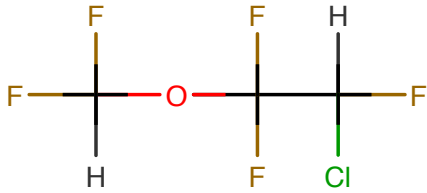
	Enflurane	Halothane	Xenon
Aliases	Ethrane	Fluothane	LENOXe
Class	Volatile general anaesthetic	Volatile general anaesthetic	Inhaled general anaesthetic gas
Chemistry	Halogenated methyl ethyl ether; structural isomer of isoflurane	Halogenated alkane	Inert gas
Dose			
Induction	2 - 4.5% inspired	0.5 - 1.5%; MAC ~0.75%	
Maintenance	0.5 - 3%; MAC ~1.7% in O ₂		
Maintenance (adults, ASA I-II)			51 - 69% with ≥30% O ₂ and opioids; MAC ~71%
Pharmaceutics			
Presentation	100% volatile liquid	100% volatile liquid stabilised with 0.01% thymol	100% liquefied medicinal gas
Compatibility	Requires an agent-specific vaporiser	Requires an agent-specific vaporiser; protect from light	Requires a xenon-calibrated anaesthetic system
Solubility	Blood:gas 1.8; oil:gas 98	Blood:gas 2.4 - 2.5; oil:gas 224	Blood:gas 0.14; oil:gas 1.9
Pharmacokinetics			
Absorption	Pulmonary; moderately blood-soluble → slower wash-in and wash-out	Pulmonary; high blood solubility → slow wash-in and wash-out	Pulmonary; very low blood solubility → rapid wash-in and wash-out

	Enflurane	Halothane	Xenon
Metabolism	Pathway: Hepatic CYP2E1 Metabolites: Inorganic fluoride	Duration: Slow offset Pathway: ~20-25% hepatic oxidative and reductive metabolism Metabolites: Trifluoroacetic acid and trifluoroacetylated protein adducts, Bromide and chloride ions	Duration: Rapid offset Pathway: Not metabolised
Elimination	Renal: Predominantly exhaled unchanged; metabolites renal	Renal: Predominantly exhaled unchanged; metabolites renal	Renal: Exhaled unchanged
Mechanism			
Target	GABAA and glycine receptors NMDA (putative) and nicotinic acetylcholine receptors	GABAA and glycine receptors NMDA and nicotinic acetylcholine receptors	NMDA receptors TREK two-pore K ⁺ channels
Action	Enhances inhibitory and suppresses excitatory ion-channel transmission	Enhances inhibitory and suppresses excitatory ion-channel transmission	Non-competitive NMDA antagonism and TREK activation → analgesia and anaesthesia
Contraindications	Seizure disorders Known or suspected malignant-hyperthermia susceptibility	Known or suspected malignant-hyperthermia susceptibility Previous unexplained jaundice or fever after halothane Porphyria Raised ICP	

	Enflurane	Halothane	Xenon
Pregnancy			
Code	A	A	
Justification	May cause neonatal CNS and respiratory depression	May cause neonatal CNS and respiratory depression	
Pharmacodynamics			
Respiratory	Marked ↓ VT and ↑ PaCO ₂ Bronchodilation and impaired HPV	Bronchodilation ↓ VT and MV; ↑ RR	↓ RR and ↑ VT; MV preserved Dense, viscous gas → ↑ airway resistance Less diffusion hypoxia than N ₂ O
CVS	↓ MAP from myocardial depression and some vasodilation	Greatest ↓ inotropy, HR and MAP among volatile agents Sensitises myocardium to catecholamines → ventricular dysrhythmias	MAP and contractility preserved; may ↓ HR
CNS	Hypnosis and analgesia High concentrations and hypocapnia cause 3-Hz spike-wave activity and seizures ↓ CMRO ₂ ; ↑ CBF above 1 MAC	Hypnosis without analgesia Greatest ↑ CBF above 1.1 MAC; may ↑ ICP	Analgesia and anaesthesia ↓ cerebral metabolism and CBF ↑ PONV
Musculoskeletal	Muscle relaxation and potentiation of non-depolarising NMBs	Muscle relaxation and potentiation of non-depolarising NMBs	Muscle relaxation at >60% Does not trigger malignant hyperthermia
Metabolic	May trigger malignant hyperthermia	May trigger malignant hyperthermia	
Renal	Inorganic fluoride may cause nephrotoxicity	↓ RBF, GFR and urine output secondary to haemodynamic effects	No reported nephrotoxicity

	Enflurane	Halothane	Xenon
Genitourinary	Uterine relaxation	Potent uterine relaxation	
Gastrointestinal	Rare hepatitis	Halothane hepatitis and fulminant hepatic necrosis	No reported hepatotoxicity
Considerations	Avoid hyperventilation because hypocapnia increases epileptiform activity Chronic isoniazid or other hydrazine-containing compounds → fluoride may exceed 50micromol/L, especially beyond 2 MAC-hours Desiccated CO2 absorbent may produce carbon monoxide	Previous exposure increases immune-mediated hepatitis risk Adrenaline and noradrenaline may cause ventricular tachycardia or fibrillation Thymol accumulates in vaporisers → drain and clean regularly	Historical LENOXe regimen (HAS 2008); maintenance only, not induction Cannot provide 1 MAC when >30-35% O2 is required Rare and expensive; low-flow circuits permit recycling

Enflurane:



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Nitrous Oxide

Describe the pharmacology of nitrous oxide

Describe the comparative pharmacology of nitrous oxide, ~~halothane, enflurane, isoflurane, desflurane, sevoflurane, xenon and ether~~

Table 1: Nitrous Oxide

	Nitrous oxide
Aliases	Laughing gas
Class	Inhaled analgesic and weak general anaesthetic
Chemistry	Inorganic oxide gas
Dose	
Analgesia	25-50% with O ₂
Anaesthesia (adjunct)	Up to 79% with ≥21% O ₂
Pharmaceutics	
Presentation	100% compressed medical gas stored as a liquefied gas
Compatibility	Supports combustion; use gas-specific regulators and scavenging
Solubility	Blood:gas 0.46-0.47; oil:gas 1.4
Pharmacokinetics	
Absorption	Rapid pulmonary uptake
Distribution	V _d : Rapidly enters and expands gas-filled spaces

	Nitrous oxide
Metabolism	Duration: Rapid onset and offset Pathway: Negligible metabolism
Elimination	Renal: Exhaled unchanged
Mechanism	
Target	NMDA receptors Dynorphin and κ -opioid signalling
Action	NMDA antagonism with proposed opioid signalling → analgesia and anaesthesia
Contraindications	Trapped gas where expansion is hazardous Vitamin B12/folate deficiency or genetic disorders of this pathway Severe cardiac dysfunction or pronounced cardiac failure Signs suggesting raised ICP Reduced consciousness or cooperation when used for analgesia
Pregnancy	
Code	A
Justification	May cause neonatal CNS and respiratory depression
Pharmacodynamics	
Respiratory	Small ↓ VT and ↑ RR; MV unchanged Diffusion hypoxia
CVS	↑ SNS tone offsets mild myocardial depression ↑ PVR
CNS	Analgesia at >20%; weak anaesthesia with MAC ~105% ↑ CBF and ICP ↑ PONV
Musculoskeletal	Does not trigger malignant hyperthermia
Metabolic	Inactivates vitamin B12 → ↑ homocysteine and impaired DNA synthesis
Genitourinary	Minimal uterine relaxation

	Nitrous oxide
Gastrointestinal	Expands bowel gas
Haematological	Prolonged or repeated exposure → megaloblastic change and leucopenia
Considerations	Avoid methotrexate; additive folate toxicity Prolonged or repeated exposure may cause myeloneuropathy Give supplemental O ₂ after anaesthesia Entonox is 50% N ₂ O/50% O ₂

Entonox

Entonox is a 50/50 mixture of nitrous oxide and oxygen, used as analgesia in labor and minor procedures.

Table 2: Properties of Entonox

Property	Entonox (50% O ₂ , 50% N ₂ O)
Pharmaceutics	The gases dissolve each other and behave differently than would be expected from their individual properties. This is the Poynting effect .
Critical Temperature/Pressure	<i>Pseudocritical</i> temperature of -6°C , below which it will separate into liquid 50% N ₂ O (with some dissolved O ₂), and gaseous O ₂ . This is most likely to occur at 117 bar, and can lead to delivery of a hypoxic mixture. Delivery of a hypoxic mix is prevented by: • Storing cylinders horizontally (↑ area for diffusion) • Storing cylinders at temperatures > 5°C • Using a dip tube so that liquid 50% N₂O is used before the gaseous mixture

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Opioids

Table 1: Common Pharmacological Features of Opioids

Property	Effect
Uses	Analgesia, sedation, elimination of sympathetic response to laryngoscopy/surgical stress response
Resp	↓ CNS sensitivity to CO ₂ causing respiratory depression (↓ RR > ↓ V _T) - ↑ reliance on hypoxic drive (therefore respiratory depression may be potentiated by high FiO ₂)
CVS	↓ HR. May ↓ BP due to histamine release (less with synthetic agents). ↑ PVR
CNS	Sedation, euphoria. Nausea and vomiting due to CTZ stimulation. Meiosis due to stimulation of the Edinger-Westphal nucleus.
Renal	↓ MAC up to 90% ↑ ADH, ↑ ureteric and sphincter tone
MSK	Muscle rigidity, pruritus (especially with intrathecal administration)

Property	Effect
Metabolic	↓ ACTH, prolactin, gonadotrophic hormone secretion. ↑ ADH secretion
GIT	↓ Peristalsis and GIT secretions with subsequent constipation
Immunological	Impaired: chemotaxis, lymphocyte proliferation, and antibody production

Table 2: Morphine

	Morphine
Class	Opioid
Chemistry	Natural phenanthrene opioid
Dose	
Analgesia (IV)	0.1 - 0.2mg/kg every 4 hours PRN; administer slowly
Analgesia (IV Infusion)	0.02 - 0.04mg/kg/hr
Pharmaceutics	
Presentation	Clear, slightly yellow to yellow, preservative-free solution; 5, 10, 15 or 30mg/mL; pH 3.2 - 4.0
Compatibility	Compatible with 0.9% NaCl, 5% glucose and Hartmann's; incompatible with alkaline solutions, aminophylline, barbiturates and phenytoin
Solubility	pKa 8.0; octanol-water partition coefficient ~1.42; 23% unionised at pH 7.4
Pharmacokinetics	
Absorption	PO bioavailability 30%; low lipid solubility → slower onset and SC absorption

Morphine	
Distribution	pKa: 8 Vd: 3.5 L/kg Protein binding: 35%
Metabolism	Half-life: 1.5 - 2 hours (healthy subjects) Duration: 3 - 4 hours Pathway: Hepatic glucuronidation to morphine-3-glucuronide (~70%; neuroexcitatory) and morphine-6-glucuronide (~10%; active) Metabolites: Morphine-3-glucuronide (neuroexcitatory); morphine-6-glucuronide (active)
Elimination	Renal: Renal elimination of glucuronides; accumulation in renal impairment
Mechanism	
Target	μ-opioid receptor (MOP) κ-opioid receptor (KOP)
Action	G_i/o activation $\rightarrow$ $\downarrow$ adenylyl cyclase and cAMP; $\uparrow$ postsynaptic K^+ conductance and $\downarrow$ presynaptic Ca^{2+} influx $\rightarrow$ hyperpolarisation and $\downarrow$ neurotransmitter release
Pregnancy	
Code	C
Justification	Neonatal respiratory depression; withdrawal with prolonged use
Pharmacodynamics	

	Morphine
Respiratory	↓ CNS sensitivity to CO₂ → ↓ RR and ↓ VT (↓ RR > ↓ VT) ↑ Reliance on hypoxic drive Respiratory depression potentiated by high FiO₂
CVS	↓ HR ↓ BP due to non-immune histamine release → vasodilation ↑ PVR
CNS	Sedation Euphoria Nausea and vomiting due to CTZ stimulation Miosis (stimulation of Edinger-Westphal nucleus) ↓ MAC up to 90%
Musculoskeletal	Muscle rigidity Pruritus (especially with intrathecal administration)
Metabolic	↓ ACTH, gonadotrophins, cortisol and testosterone; ↑ prolactin
Renal	↑ ADH ↑ Ureteric and bladder sphincter tone → urinary retention
Gastrointestinal	↓ Peristalsis and GIT secretions → constipation
Considerations	↑ Free drug in renal failure as uraemia displaces morphine from binding sites Morphine-3-glucuronide and morphine-6-glucuronide accumulate in renal impairment → neuroexcitation and prolonged opioid effect Equianalgesic dose (IV reference): 10mg

Table 3: Comparison of Semisynthetic Opioids

	Oxycodone	Buprenorphine
Aliases	OxyNorm	Temgesic
Class	Opioid	Partial opioid agonist
Chemistry	Semisynthetic phenanthrene opioid	Semisynthetic phenanthrene opioid
Dose		
Analgesia (PO IR)	5-10mg every 4-6 hours	
Analgesia (IV)	Initially 1-5mg over 1-2min; increments every 5-10min to establish analgesia; maintenance ≥ 4 h apart	
PCA (IV)	≤ 0.03 mg/kg; ≥ 5 min lockout	
Analgesia (IV/IM)		300-600 μ g every 6-8 hours; slow IV
Analgesia (SL)		200-400 μ g every 6-8 hours
Pharmaceutics		
Presentation	10mg/mL injection/infusion; 50mg/mL infusion only (opioid-tolerant palliative care)	Colourless 300 μ g/mL injection; 200 μ g sublingual tablet
Compatibility	0.9% NaCl, 5% glucose and water for injections; likely incompatible with alkaline formulations (precipitation); incompatible with prochlorperazine	

	Oxycodone	Buprenorphine
Solubility	pKa 8.5; quite hydrophilic; less than 10% unionised at physiological pH	Highly lipophilic; limited water solubility
Pharmacokinetics		
Absorption	PO bioavailability 60-80%; SC Tmax 15-30 minutes; equivalent IV and SC availability; morphine-like lipid solubility but faster onset, possibly due to active CNS uptake	PO bioavailability 15% due to first-pass metabolism; SL bioavailability 35%, peak 2-4 hours; parenteral bioavailability ~100%
Distribution	pKa: 8.5	pKa: 8.5
	Vd: 3 L/kg	Vd: 87-197 L/kg
	Protein binding: 45%	Protein binding: 96%

	Oxycodone	Buprenorphine
Metabolism	Half-life: 3-5 hours Duration: ~4h (IV) Pathway: Hepatic CYP3A4 → noroxycodone (~80%); CYP2D6 → oxymorphone (~20%); metabolites make little contribution to effect Metabolites: Noroxycodone; oxymorphone (minor contribution to effect)	Half-life: 5 hours (analgesia); 34.6 hours (high-dose dependence treatment) Duration: 4-8 hours Pathway: Hepatic CYP3A4 N-dealkylation and glucuronidation → norbuprenorphine and glucuronides Metabolites: Norbuprenorphine (active); glucuronides
Elimination	Renal: Renal elimination of parent drug and metabolites; accumulation in renal impairment	Renal: 70% biliary/faecal; 30% renal; enterohepatic cycling
Mechanism		
Target	μ-opioid receptor (MOP) κ-opioid receptor (KOP) δ-opioid receptor (DOP)	μ-opioid receptor (MOP) — partial agonist κ-opioid receptor (KOP) — antagonist

	Oxycodone	Buprenorphine
Action	Gi/o activation → ↓ adenylyl cyclase and cAMP; ↑ post- synaptic K ⁺ con- ductance and ↓ presynaptic Ca ²⁺ influx → hyper- polarisation and ↓ neurotransmitter release	High-affinity partial MOP agonism with slow dissociation; KOP antagonism; Gi/o → ↓ cAMP, ↑ K ⁺ conductance and ↓ Ca ²⁺ influx
Pregnancy		
Code	C	C
Justification	Neonatal respira- tory depression; withdrawal with prolonged use	Neonatal respira- tory depression; withdrawal with prolonged use
Pharmacodynamics		
Respiratory	↓ CNS sensitivity to CO ₂ → ↓ RR and ↓ VT (↓ RR > ↓ VT) ↑ Reliance on hy- poxic drive Respiratory de- pression potenti- ated by high FiO ₂	↓ CNS sensitivity to CO ₂ → ↓ RR and ↓ VT ↑ Reliance on hy- poxic drive Peak respiratory depression 3-5h in CO ₂ -response studies
CVS	↓ HR ↓ BP due to his- tamine release/ peripheral vasodi- lation ↑ PVR	Mild ↓ HR/BP; maximum observed 10-15% at thera- peutic doses

	Oxycodone	Buprenorphine
CNS	Sedation Euphoria Nausea and vom- iting due to CTZ stimulation Miosis (stimulation of Edinger-West- phal nucleus) ↓ MAC up to 90%	Sedation Nausea and vom- iting due to CTZ stimulation Miosis (stimulation of Edinger-West- phal nucleus)
Muscu- loskeletal	Muscle rigidity Pruritus (especially with intrathecal ad- ministration)	Muscle rigidity Pruritus (especially with intrathecal ad- ministration)
Metabolic	↓ ACTH, go- nadotrophins, cor- tisol and testos- terone; ↑ prolactin	↓ ACTH, go- nadotrophins, cor- tisol and testos- terone; ↑ prolactin
Renal	↑ ADH ↑ Ureteric and bladder sphinc- ter tone → urinary retention	↑ ADH ↑ Ureteric and bladder sphinc- ter tone → urinary retention
Gastroin- testinal	↓ Peristalsis and GIT secretions → constipation	↓ Peristalsis and GIT secretions → constipation

	Oxycodone	Buprenorphine
Considerations	↑ Oxycodone and noroxycodone in renal/hepatic impairment Equianalgesic dose (PO): 10mg = 15mg PO morphine Equianalgesic dose (IV): 10mg = 10mg IV morphine	Slow MOP dissociation → prolonged effect and poor naloxone reversal May antagonise full MOP agonists and precipitate withdrawal Plasma concentration poorly correlates with clinical effect Equianalgesic dose (IM): 0.4mg = 10mg morphine

Table 4: Comparison of Synthetic Opioids

	Fentanyl	Alfentanil	Remifentanyl
Aliases	Sublimaze	Rapifen	Ultiva
Class	Opioid	Opioid	Opioid
Chemistry	Synthetic phenylpiperidine opioid	Synthetic phenylpiperidine opioid	Synthetic phenylpiperidine opioid
Dose			
Laryngoscopy (IV)	1-3µg/kg over 3-5 minutes	20-50µg/kg	
Paediatric (IV; 2-12 years)	2-3µg/kg		
Analgesia (IV)		Up to 7µg/kg; increments 2-3µg/kg every 10-15 minutes	
Infusion (IV)		0.5-1µg/kg/min	
Induction (IV)			0.5-1µg/kg/min with or without 1µg/kg over 60 seconds
Analgesia (IV Infusion)			0.05-2µg/kg/min
ICU Sedation (IV)			0.1-0.15µg/kg/min; increase by 0.025µg/kg/min; no bolus
TCI (Minto)			3-8ng/mL; up to 15ng/mL for highly stimulating surgery
Pharmaceutics			
Presentation	Clear, colourless, preservative-free solution; 50µg/mL; pH 4.0-7.5	Clear, colourless solution; 0.5mg/mL in 2 or 10mL ampoules	White lyophilised powder containing glycine; 1, 2 or 5mg; clear and colourless after reconstitution

	Fentanyl	Alfentanil	Remifentanyl
Compatibility	Compatible with 0.9% NaCl and 5% glucose; incompatible with thiopental and methohexitone	Compatible with 0.9% NaCl, 5% glucose and compound sodium lactate	
Solubility	pKa 8.4; octanol-water partition coefficient 717; 600x more lipid soluble than morphine; 9% unionised at pH 7.4	pKa 6.5; octanol-water partition coefficient 128; 90x more lipid soluble than morphine; 90% unionised at pH 7.4	pKa 7.3; octanol-water partition coefficient 17.9; 20x more lipid soluble than morphine; 58% unionised at pH 7.4
Pharmacokinetics			
Absorption	Rapid IV onset (peak ~5 minutes); slow transdermal absorption; PO bioavailability 33%; significant first-pass pulmonary uptake	IV; maximal analgesia ~1 minute	IV; peak effect 1-3 minutes
Distribution	pKa: 8.4	pKa: 6.5	pKa: 7.3
	Vd: 4 L/kg	Vd: 0.4-1 L/kg	Vd: Central 0.1 L/kg; steady-state 0.35 L/kg
	Protein binding: 85%	Protein binding: 92%	Protein binding: 70%

	Fentanyl	Alfentanil	Remifentanyl
Metabolism	Half-life: ~3.5 hours Duration: Variable; with doses greater than 3 µg/kg tissues become saturated and duration is significantly prolonged Pathway: Hepatic CYP3A4 N-dealkylation → norfentanyl (inactive); also intestinal CYP3A4 metabolism Metabolites: Norfentanyl (inactive)	Half-life: 90-111 minutes Duration: 5-10 minutes Pathway: Hepatic N- and O-dealkylation to inactive metabolites Metabolites: Inactive	Half-life: Effective 3-10 minutes; terminal 10-20 minutes Duration: Offset 5-10 minutes from ceasing infusion; independent of infusion duration or dose Pathway: Rapid hydrolysis by non-specific blood and tissue esterases; independent of renal or hepatic function Metabolites: GR90291 (4,600x less potent; renally cleared)
Elimination	Renal: Renal elimination of inactive metabolites; ~10% of dose excreted unchanged in urine	Renal: Renal elimination of metabolites	Renal: 95% urine as GR90291; parent clearance independent of renal function
Mechanism			
Target	µ-opioid receptor (MOP)	µ-opioid receptor (MOP)	µ-opioid receptor (MOP)

	Fentanyl	Alfentanil	Remifentanyl
Action	Gi/o activation → ↓ adenylyl cyclase and cAMP; ↑ post-synaptic K ⁺ conductance and ↓ presynaptic Ca ²⁺ influx → hyperpolarisation and ↓ neurotransmitter release	Gi/o activation → ↓ adenylyl cyclase and cAMP; ↑ post-synaptic K ⁺ conductance and ↓ presynaptic Ca ²⁺ influx → hyperpolarisation and ↓ neurotransmitter release	Gi/o activation → ↓ adenylyl cyclase and cAMP; ↑ post-synaptic K ⁺ conductance and ↓ presynaptic Ca ²⁺ influx → hyperpolarisation and ↓ neurotransmitter release
Pregnancy			
Code	C	C	C
Justification	Neonatal respiratory depression; withdrawal with prolonged use	Neonatal respiratory depression; withdrawal with prolonged use	Neonatal respiratory depression; withdrawal with prolonged use
Pharmacodynamics			
Respiratory	↓ CNS sensitivity to CO ₂ → ↓ RR and ↓ VT (↓ RR > ↓ VT) ↑ Reliance on hypoxic drive Respiratory depression potentiated by high FiO ₂	↓ CNS sensitivity to CO ₂ → ↓ RR and ↓ VT (↓ RR > ↓ VT) ↑ Reliance on hypoxic drive Respiratory depression potentiated by high FiO ₂	↓ CNS sensitivity to CO ₂ → ↓ RR and ↓ VT (↓ RR > ↓ VT) ↑ Reliance on hypoxic drive Respiratory depression potentiated by high FiO ₂
CVS	Vagal bradycardia Blunted cardiovascular reflexes and ↓ sympathetic response to intubation ↑ PVR	↓ HR ↑ PVR	Vagal bradycardia ↓ MAP (~20%) ↑ PVR
CNS	Sedation Euphoria Nausea and vomiting due to CTZ stimulation Miosis (stimulation of Edinger-Westphal nucleus) ↓ MAC up to 90%	Sedation Euphoria Nausea and vomiting due to CTZ stimulation Miosis (stimulation of Edinger-Westphal nucleus) ↓ MAC up to 90%	Sedation Euphoria Nausea and vomiting due to CTZ stimulation Miosis (stimulation of Edinger-Westphal nucleus) ↓ MAC up to 90%

	Fentanyl	Alfentanil	Remifentanil
Musculoskeletal	Chest wall rigidity Pruritus (especially with intrathecal administration)	Muscle rigidity Pruritus	Chest wall rigidity Pruritus
Metabolic	↓ ACTH, gonadotrophins, cortisol and testosterone; ↑ prolactin	↓ ACTH, gonadotrophins, cortisol and testosterone; ↑ prolactin	↓ ACTH, gonadotrophins, cortisol and testosterone; ↑ prolactin
Renal	↑ ADH ↑ Ureteric and bladder sphincter tone → urinary retention	↑ ADH ↑ Ureteric and bladder sphincter tone → urinary retention	↑ ADH ↑ Ureteric and bladder sphincter tone → urinary retention
Gastrointestinal	↓ Peristalsis and GIT secretions → constipation	↓ Peristalsis and GIT secretions → constipation	↓ Gastrointestinal motility

	Fentanyl	Alfentanil	Remifentanyl
Considerations	Minimal histamine release Tissue saturation $>3\mu\text{g/kg}$ $\rightarrow$ prolonged duration; low-dose offset reflects redistribution Equianalgesic dose (IV): $100\mu\text{g} = 10\text{mg IV morphine}$	Shorter $t_{1/2}$ than fentanyl despite lower clearance due to smaller V_d Midazolam prolongs effect via CYP3A3/4 competition Low pK_a and larger dose $\rightarrow$ rapid onset despite lower lipid solubility than fentanyl (Bowman's Principle) Equianalgesic dose (IV): $1\text{mg} = 10\text{mg IV morphine}$	Contains glycine — must not be administered intrathecally or epidurally Constant context-sensitive half-time $\rightarrow$ predictable 5-10 minute offset independent of infusion duration Establish longer-acting analgesia before cessation GR90291 accumulates in renal impairment without clinically significant μ-opioid effect Minto TCI model uses age and lean body mass; may underestimate lean body mass if BMI $>35\text{kg/m}^2$ (female) or $>40\text{kg/m}^2$ (male) Equianalgesic dose (IV): $50\mu\text{g} = 10\text{mg IV morphine}$

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COX Inhibitors

Cyclo-oxygenase inhibitors are typically **used** to treat **mild to moderate pain**. Oral COX inhibitors typically have:

- **Rapid absorption**
- **High protein binding**
- **Low V_D**

Mechanism of Action

There are two(ish) **isoenzymes** of COX:

- **COX-1**
Important for homeostatic function.
- **COX-2**
Induced with tissue damage and contributes to inflammation. COX-2:
 - Exists in the vascular endothelium where it synthesises prostacyclin (which opposes the action of thromboxanes)
 - Inhibition may result in a relative abundance of thromboxane, causing platelet aggregation and vasoconstriction
- **COX-3**
Variant of COX-1 which exists centrally and mediates the analgesic and antipyretic effects of paracetamol.

Effects occur due to:

- Decrease in endoperoxidases
Inhibited by COX.

- Increase in other arachidonic-acid derived factors
Due to the diversion of arachidonic acid down other pathways.

COX inhibition has different effects in different tissues:

- Prevents subsequent conversion of prostaglandins to thromboxane A_2 and PGI_2
- Peripherally, inhibition of prostaglandin synthesis is anti-inflammatory
- Centrally, it is anti-pyretic
- In the stomach, it decreases mucous production and leads to mucosal ulceration
- Aspirin (a non-specific COX inhibitor), prevents production of both thromboxane A_2 and PGI_2
 - As platelets **have no nucleus**, the COX inhibition remains for the entirety of the platelet lifespan
 - Endothelial cells will produce new COX within hours, and so its anti-inflammatory effects are temporary

Adverse Effects

- **Asthma/Bronchospasm**
Secondary to increased leukotriene synthesis due to increased arachidonic acid levels. Occurs in 20% of asthmatics with NSAID use.
- **Platelet dysfunction**
A consequence of COX-1 inhibition only, and may result in increased perioperative bleeding risk (though decreased AMI and CVA risk).
- **Thrombotic events**, including MI and CVA
Risk is greater with **COX-2** inhibitors, due to **selective inhibition of prostacyclin**. with NNH for non-fatal MI being 500 patient-years, and NNH for fatal MI being 1000 patient-years.
- **Impaired GFR**
Occurs as a consequence of uninhibited afferent arteriolar constriction. Worse with concurrent hypo-

volaemia, renal artery stenosis, or concurrent ACE-I use.

- **Gastric erosion**

A consequence of impaired mucosal secretion through COX-1 inhibition. This can result in pain, anaemia, or fatal bleed. In general, risk of gastric erosion is (from highest to lowest risk):

- Ketorolac
- Diclofenac/naproxen
- Ibuprofen (<1.2g/day)
- COX-2 Inhibitors

- **Transaminitis** may occur following NSAID use

Table 1: Comparison of COX Inhibitors

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Aliases		Voltaren	Toradol	Brufen	Celebrex	Dynastat
Class	Non - se - lective COX inhibitor	Non - se - lective COX inhibitor	Non - se - lective COX inhibitor	Non - se - lective COX inhibitor	Selective COX - 2 in - hibitor	Selective COX - 2 in - hibitor
Chemistry	Aromatic acetate	Heteroaryl acetic acid	Heteroaryl acetic acid	Aryl - propi - onic acid	Diarylhetere - rocyclic sul - fonamide	Water - sol - uble sul - fonamide prodrug of valdecoxib
Dose						
Antiplatelet	100mg PO daily					
Analgesia/migraine	300 - 900mg PO Q4 - 6H; max 3.6g/ day					
Analgesia (PO/PR)		50mg BD/ TDS; max 150mg/day		400 - 800mg TDS		
Analgesia (IM)			10 - 30mg 4 - 6 - hourly; max 90mg/ day; ≤5 days			
Analgesia (IV, off - label)			15 - 30mg 6 - hourly; ≤5 days			
Paediatric (PO/PR)				10mg/kg		

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Analgesia (PO)					100 - 200mg PO BD ≤5 days	
Analgesia (IV)						1mg/kg up to 40mg once
Pharmaceutics						
Presentation	100mg, 300mg and 500mg tablets	25mg or 50mg EC tablet; 12.5 - 100mg sup- pository	30mg/mL injection	400mg tablet	100mg or 200mg cap- sule	40mg pow- der; recon- stitute with 2mL
Compatibility						Reconsti- tute: 0.9% NaCl, 5% glucose, or 0.45% NaCl/ 5% glucose; Hartmann's line-com- patible
Solubility	Slightly wa- ter-soluble	pKa 4.2; quite lipid soluble and minimally water soluble	pKa 3.54; water-solu- ble	pKa 4.9; quite lipid soluble and poorly water soluble	pKa 11.1; practically insoluble in water	pKa 6.7; highly wa- ter-soluble
Pharmacokinetics						

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Absorption	50% PO bioavailability; rapid absorption, mainly from upper small intestine; Tmax ~1h	Rapid; 50-65% oral bioavailability	100% IM bioavailability; Tmax ~50 minutes	~100% oral bioavailability; Tmax 45 minutes fasting	60-80% oral bioavailability; Tmax 2-3 hours	Rapid, near-complete conversion to valdecoxib; onset 7-14 minutes
Distribution	pKa: 2.97	pKa: 4.2	pKa: 3.54	pKa: 4.9	pKa: 11.1	pKa: 6.7
	Vd: 0.1-0.2L/kg	Vd: 1.4 L/kg	Vd: 13 L	Vd: 0.1-0.14 L/kg	Vd: 5-6 L/kg	Vd: 0.8 L/kg (parecoxib); 55 L (valdecoxib)
	Protein binding: 58% (aspirin); salicylate 80-90%	Protein binding: 99%	Protein binding: >99%	Protein binding: 99%	Protein binding: 97%	Protein binding: 98%

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Metabolism	Half-life: Aspirin ~20min; salicylate 2-12h (dose-dependent) Pathway: ~80% hepatic → salicyluric acid and glucuronides; saturable in overdose Metabolites: Salicylic acid (active); salicyluric acid; glucuronides	Half-life: 1-2 hours Pathway: CYP2C9 → 4-hydroxydiclofenac; glucuronidation and sulfation Metabolites: 4-hydroxydiclofenac	Half-life: 5-6 hours Pathway: Hepatic glucuronidation; minor p-hydroxylation Metabolites: Ketorolac glucuronide; p-hydroxyketorolac (inactive)	Half-life: 1.9-2.2 hours Pathway: Hepatic oxidation → hydroxyibuprofen and carboxyibuprofen; inactive Metabolites: Hydroxyibuprofen; carboxyibuprofen (inactive)	Half-life: 8-12 hours Pathway: CYP2C9 → hydroxycyclooxygenase → carboxycyclooxygenase and glucuronide; inactive Metabolites: Hydroxycyclooxygenase; carboxycyclooxygenase; glucuronide (inactive)	Half-life: Parecoxib <1 hour; valdecoxib ~8 hours Duration: 6 to >24 hours Pathway: Hydrolysis → valdecoxib; CYP3A4/CYP2C9 and glucuronidation Metabolites: Valdecoxib (active); hydroxylated valdecoxib (minor active)

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Elimination	Renal: Urine; alkalinisation ↑ unchanged salicylate clearance	Renal: 60-70% urine; 30-35% biliary/faecal; enterohepatic recirculation	Renal: 91% urine; 6% faeces	Renal: 95% urine; ~1% unchanged	Renal: 27% urine; 57% faeces; <1% unchanged	Renal: 70% urine as inactive metabolites; <5% unchanged valdecoxib
Mechanism						
Target	COX-1 COX-2	COX-1 COX-2	COX-1 COX-2	COX-1 COX-2	COX-2	COX-2
Action	Irreversible COX acetylation → ↓ TxA ₂ /PGI ₂ ; platelet COX-1 inhibition → ↓ aggregation for platelet lifespan; endothelium regenerates COX within hours	Reversible COX-1/2 inhibition → ↓ prostaglandin synthesis → ↓ inflammation and nociceptor sensitisation	Inhibition of COX-1 and COX-2 → ↓ prostaglandin synthesis → ↓ sensitivity of peripheral nociceptors	Reversible COX-1/2 inhibition → ↓ prostaglandin synthesis → ↓ inflammation and nociceptor sensitisation	Selective COX-2 inhibition (30:1) → ↓ prostaglandin synthesis at sites of inflammation	Conversion to valdecoxib → selective COX-2 inhibition (61:1) → ↓ prostaglandin synthesis

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Contraindications	Active GI bleeding Viral fever in children	NSAID-induced asthma, angioedema or urticaria Active GI ulcer, bleeding or perforation Severe cardiac, renal or hepatic failure CABG perioperative pain Third-trimester pregnancy Proctitis (PR)	Moderate/severe renal impairment; dehydration or hypovolaemia Active/recurrent GI ulcer or bleeding Haemorrhagic diathesis; anticoagulation; high bleeding-risk surgery Prophylactic pre-operative or intra-operative use Neuraxial administration Concurrent aspirin, NSAIDs, oxpentifylline, probenecid or lithium Pregnancy, labour, delivery or lactation	NSAID-induced asthma, rhinitis or urticaria Active or recurrent GI ulcer/bleeding; inflammatory bowel disease Severe cardiac, renal or hepatic failure Active bleeding CABG perioperative pain Third-trimester pregnancy	NSAID-induced asthma or urticaria Sulfonamide reaction Active peptic ulcer or GI bleeding Unstable IHD; MI or CVA within 3 months NYHA II-IV heart failure CABG perioperative pain Severe renal or hepatic impairment	CABG or major vascular surgery Significant IHD, PAD or cerebrovascular disease NSAID-induced asthma or urticaria Sulfonamide reaction Severe hepatic impairment Third-trimester pregnancy or breast-feeding

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Pregnancy						
Code	C	C	C	C	B3	C
Justification	Late use → duct clo- sure, de- layed labour and mater- nal/neonatal bleeding; 100mg/day does not ↑ bleeding time	Possible ↑ miscarriage; late use → duct clo- sure, oligo- hydramnios, renal impair- ment, platelet inhibition and delayed labour	Possible ↑ miscarriage; late use → duct clo- sure, oligo- hydramnios, renal impair- ment, platelet inhibition and delayed labour	Possible ↑ miscarriage; late use → duct clo- sure, oligo- hydramnios, renal impair- ment, platelet inhibition and delayed labour	Possible ↑ miscarriage; late use → duct clo- sure, oligo- hydramnios, renal impair- ment, platelet inhibition and delayed labour	Possible ↑ miscarriage; late use → duct clo- sure, oligo- hydramnios, renal impair- ment, platelet inhibition and delayed labour
Pharmacodynamics						

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Respiratory	Uncouples oxidative phosphorylation → ↑ O ₂ consumption and CO ₂ production Stimulates, then depresses, respiratory centre in toxicity Bronchospasm (~20% of asthmatics with NSAID use)	Bronchospasm (~20% of asthmatics)	Bronchospasm in approximately 20% of asthmatics	Bronchospasm (~20% of asthmatics; ↑ leukotrienes)	Bronchospasm in NSAID-sensitive asthma	Bronchospasm in NSAID-sensitive asthma
CVS	↓ MI/CVA risk ↑ MAP; CCF exacerbation	↑ MI/CVA risk; similar to COX-2 inhibitors IV → local thrombosis ↑ BP; oedema; CCF exacerbation	↑ BP; oedema; CCF exacerbation ↑ MI/CVA risk	Low dose → minimal pro-thrombotic effect; 2.4g/day or prolonged use → ↑ arterial thrombosis ↑ BP; oedema; CCF exacerbation	↑ MI/CVA risk ↑ BP; oedema; CCF exacerbation	↑ MI/CVA risk ↑ BP; oedema; CCF exacerbation

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Metabolic	Toxicity: respiratory alkalosis/ metabolic acidosis; respiratory depression → respiratory acidosis Reye syn- drome (<12 years): mi- tochondrial injury → hepatic failure/ cerebral oedema/ encephalopathy; mortality ~40%	Transaminitis				
Renal	AKI	↓ GFR; AKI	↓ GFR; AKI	↓ GFR; AKI	↓ GFR; AKI	↓ GFR; AKI
Gastrointestinal	GI ulceration	GI ulceration/ bleeding	Highest GI ulcer/ bleeding risk among com- monly used NSAIDs	GI ulcera- tion; lowest risk among non-selec- tive NSAIDs at <1.2g/day	↓ GI ulcera- tion vs non- selective NSAIDs	↓ GI ulcera- tion vs non- selective NSAIDs

	Aspirin	Diclofenac	Ketorolac	Ibuprofen	Celecoxib	Parecoxib
Haematological	Bleeding	Platelet dysfunction ↑ Perioperative bleeding risk	Platelet dysfunction	Platelet dysfunction (COX-1 inhibition) ↑ Perioperative bleeding risk	No platelet inhibition at therapeutic concentrations	No platelet inhibition at therapeutic concentrations
Considerations	Zero-order elimination in overdose Urinary alkalisation ↑ elimination	Synovial persistence → effect outlasts plasma $t_{1/2}$	IV route: off-label in Australia Potent analgesic; minimal anti-inflammatory effect No opioid receptor activity, sedation or respiratory depression	May impair aspirin antiplatelet effect	CYP2C9 poor metabolisers → ↑ exposure	Current Australian PI permits a single perioperative dose

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Tramadol

Tramadol is an analgesic agent with a complicated mechanism of action:

- Action at all opioid receptors, but particularly MOP, causing analgesia as well as nausea and vomiting
- Inhibits 5-HT reuptake which provides descending inhibitory analgesia
- Inhibits NA reuptake descending inhibitory analgesia
- NMDA receptor antagonist

Table 1: Tramadol

	Tramadol
Aliases	Tramal
Class	Opioid
Chemistry	Aminocyclohexanol; racemate
Dose	
Analgesia (PO/IV/IM)	50-100mg Q4-6H (PO immediate-release); IV over 2-3min; max 400mg/day
Pharmaceutics	
Presentation	50mg capsule; 50-200mg SR tablet; 100mg/mL oral drops; clear 50mg/mL injection
Compatibility	0.9% NaCl; 5% glucose; 4.2% or 8.4% sodium bicarbonate; Hartmann's
Solubility	pKa 9.41; highly lipophilic
Pharmacokinetics	
Absorption	68-72% PO bioavailability (IR capsules); >90% with repeated Q6H capsules due to first-pass saturation; IM ~100%

	Tramadol
Distribution	pKa: 9.41 Vd: 2-3 L/kg (young adults) Protein binding: 20%
Metabolism	Half-life: 5-7h (young adults) Pathway: Hepatic CYP2D6 O-demethylation and CYP3A4 N-demethylation; glucuronidation and sulfation Metabolites: O-desmethyltramadol (active)
Elimination	Renal: ~95% urine as parent/metabolites; 15-19% unchanged (young adults)
Mechanism	
Target	μ-opioid receptor (MOP) Serotonin reuptake transporter Noradrenaline reuptake transporter NMDA receptor
Action	(+)-tramadol → weak MOP agonism + ↓ 5-HT reuptake; (-)-tramadol → ↓ NA reuptake; O-desmethyltramadol → stronger MOP agonism; NMDA antagonism
Contraindications	MAO-I use within 14 days
Pregnancy	Uncontrolled epilepsy
Code	C
Justification	Neonatal respiratory depression; withdrawal with prolonged use
Pharmacodynamics	
Respiratory	↓ RR (less than morphine)
CVS	Postural ↓ BP; tachycardia
CNS	↑ Seizure risk (epilepsy; SSRI/SNRI/TCA) Dizziness; sedation Dependence; withdrawal

	Tramadol
Endocrine	↑ ADH → hyponatraemia
Genitourinary	Urinary retention
Gastrointestinal	Nausea; vomiting; constipation; dry mouth
Considerations	Serotonergic drugs → serotonin syndrome CYP2D6 poor metaboliser → ↓ O-desmethyltramadol + ↓ analgesia ~1/5 morphine potency

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Paracetamol

Paracetamol an analgesic and antipyretic which is typically classed as an NSAID, though it is unique and important enough to get its own page. It has a number of mechanisms of action:

- **Non-selective COX inhibition**, including COX-3
This confers some of the analgesic properties
- **Inhibition of central prostaglandin synthesis**
This confers the antipyretic effect by inhibiting prostaglandin E synthesis in the anterior hypothalamus in response to pyrogens
- **Serotonergic inhibition**
Provides some additional analgesic action
- **Cannabinoid inhibition**
Provides some additional analgesic action via endocannabinoid reuptake inhibition.

Table 1: Paracetamol

	Paracetamol
Aliases	Acetaminophen
Class	Analgesic and antipyretic
Chemistry	Acetanilide derivative
Dose	
Adult analgesia (PO/PR/IV)	1g Q4-6H; max 4g/day
Adult antipyresis (PO/PR/IV)	1g Q4-6H; max 4g/day
Paediatric analgesia (PO/IV)	20mg/kg stat, then 15mg/kg Q4H; max 90mg/kg/day for 48h, then 60mg/kg/day

	Paracetamol
Paediatric antipyresis (PO/IV)	20mg/kg stat, then 15mg/kg Q4H; max 90mg/kg/day for 48h, then 60mg/kg/day
Pharmaceutics	
Presentation	Tablet; capsule; syrup; clear 10mg/mL IV solution
Compatibility	0.9% NaCl; 5% glucose
Solubility	pKa 9.5; moderately soluble in both water and lipid
Pharmacokinetics	
Absorption	~80-90% oral bioavailability; PO/PR on-set ~40 minutes; IV onset 5-10 minutes; peak ~1 hour
Distribution	pKa: 9.5 Vd: 0.5-1 L/kg Protein binding: 10-20%
Metabolism	Half-life: 2-3 hours Pathway: Hepatic glucuronidation and sulfation; minor CYP oxidation → NAPQI Metabolites: NAPQI (toxic); glucuronide and sulfate conjugates (inactive)
Elimination	Renal: 90% urine, mainly conjugates; <5% unchanged; active tubular secretion
Mechanism	
Target	Prostaglandin pathways Serotonergic pathways Endocannabinoid pathways
Action	Uncertain; central and peripheral actions → analgesia and antipyresis
Contraindications	Severe hepatocellular insufficiency or hepatic failure
Pregnancy	

Paracetamol	
Code	A
Pharmacodynamics	
Respiratory	May exacerbate analgesic asthma due to glutathione depletion
CVS	Vasodilation (IV)
CNS	Analgesia Opioid-sparing
Metabolic	Antipyretic
Haematological	Cytopenias (rare)
Considerations	NAPQI + glutathione depletion → centrilobular hepatic necrosis Toxic doses — single ingestion over 200mg/kg; repeated ingestion over 150mg/kg/day for two days; over 100mg/kg/day for three days Risk factors for toxicity — glutathione deficiency (extremes of age, malnutrition, hepatic dysfunction); enzyme inducers (carbamazepine, phenytoin, phenobarbital, rifampicin, alcohol, OCP) PO overdose — activated charcoal ≤1 hour; IV N-acetylcysteine guided by paracetamol nomogram Flucloxacillin coadministration → ↑ high anion gap metabolic acidosis risk

Toxicity

- Paracetamol is partially metabolised to the toxic N-acetyl-p-amino-benzoquinone imine (**NAPQI**)
 - In normal circumstances this rapidly conjugated with glutathione
 - In toxicity, glutathione is exhausted
 - NAPQI then covalently binds to critical proteins in hepatocytes, causing centrilobular hepatic

- necrosis and cell death
- Toxic doses:
 - **>200mg.kg⁻¹** in a single ingestion
 - Repeated ingestion of **>150mg.kg⁻¹.day⁻¹** for two days
 - **>100mg.kg⁻¹.day⁻¹** for three days
- Risk factors for toxicity:
 - Glutathione deficiency
 - Extremes of age
 - Malnutrition
 - Hepatic dysfunction
 - Enzyme inducers:
 - Anti-epileptics
 - Carbamazepine
 - Phenytoin
 - Phenobarbitone
 - Rifampicin
 - ETOH
 - OCP

Features of Overdose

- Conscious
- Nausea, vomiting, and epigastric pain
- Haemolytic anaemia
- Distributive shock
- Hyperglycaemia
- Late (>48 hours) hepatic failure
- Later (3-5 days) coagulopathy
- Fulminant hepatic failure (3-7 days)

Treatment of Overdose

- Activated charcoal with tablet ingestion if seen within 1 hour of ingestion.
- Serum paracetamol level to determine requirement for NAC (N-acetylcysteine) based on the nomogram

- IV NAC is used as it is a glutathione precursor, replenishing depleted glutathione and facilitating further conjugation of NAPQI
-

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Anticholinesterases

Anticholinesterases antagonise AChE, decreasing the breakdown of ACh and therefore increasing its availability at the:

- Nicotinic receptor
Increases muscle strength.
 - Reversal of non-depolarising neuromuscular blockade
- Muscarinic receptor
Increases parasympathetic tone.

Anticholinesterases can be:

- Reversible
- Form a carbamylated enzyme complex
- Irreversible

Table 1: Neostigmine

	Neostigmine
Aliases	Prostigmin
Class	Acetylcholinesterase inhibitor
Chemistry	Quaternary ammonium compound
Dose	
Reversal (IV)	0.05mg/kg (max 5mg)
Ileus (IV)	2.5mg over ≥5 hours
Myasthenia gravis (PO)	15-30mg Q2-4H
Pharmaceutics	
Presentation	Clear colourless 2.5mg/mL solution
Compatibility	NS, D5W; may be mixed with glycopyrrolate
Solubility	Water-soluble; minimally lipid-soluble
Pharmacokinetics	

	Neostigmine
Absorption	<5% oral bioavailability
Distribution	pKa: 12
	Vd: 0.12L/kg
	Protein binding: 15 - 25%
Metabolism	Half-life: 70 minutes
	Duration: 20 - 30 minutes
	Pathway: Acetylcholinesterase and plasma esterase hydrolysis; hepatic mitochondrial metabolism
	Metabolites: Quaternary alcohol
Elimination	Renal: 70% unchanged
Mechanism	
Target	Acetylcholinesterase
Action	Competing substrate → carbamylated enzyme → ↓ acetylcholine hydrolysis
Contraindications	Mechanical GI or urinary obstruction
	Doubtful bowel viability
	Peritonitis (for peristalsis)
	Bronchial asthma
	Depolarising neuromuscular blockade
Pregnancy	
Code	B2
Pharmacodynamics	
Respiratory	Bronchospasm ↑ Secretions
CVS	↓ HR ↓ CO ↓ MAP
CNS	Intrathecal analgesia and N/V Cerebral vasoconstriction

	Neostigmine
Muscu- skeletal	Reverses non-depolarising neuromuscu- lar blockade Fasciculations High doses → depolarising block and paralysis ↑ Sweating
Genitourinary	↑ Urination
Gastrointesti- nal	↑ Peristalsis ↑ LoS tone N/V Diarrhoea
Considera- tions	Does not cross the blood-brain barrier at usual systemic doses Reversal: give with atropine or glycopy- rrolate Muscarinic effects at lower doses; nico- tinic effects at higher doses Renal impairment → prolonged effect

Table 2: Effects of Organophosphate Anticholinesterases

Property	Organophosphates
Class	Irreversible anticholinesterase
Uses	Insecticides, pesticides, chemical weapons
Presentation	
Route of Administration	Topical
Dosing	
Absorption	Rapid topical absorption due to high lipid solubility
Distribution	Crosses BBB
Metabolism	Not metabolised
Elimination	$t_{1/2}$ α of weeks
Duration	Until new AChE is synthesised
Resp	Bronchospasm, ↑ secretion
CVS	↓ HR, ↓ CO
CNS	Central cholinergic syndrome

Property	Organophosphates
MSK	Paralysis
GIT	↑ Peristalsis, ↑ LoS tone, N/V
Other	May be reversed in initial stages (before organophosphate - AChE complex has 'aged') with pralidoxime

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Antimuscarinics (Cardiac)

Antimuscarinics used for bronchodilation are covered under Antimuscarinics (Respiratory), whilst atropine is covered separately.

Antimuscarinics are as competitive, reversible antagonists of ACh at the muscarinic receptor. They are divided into:

- Naturally occurring **tertiary amines**
These can cross the blood-brain barrier, and have central effects.
 - Atropine
 - Hyoscine
- Synthetic **quaternary amines**
Do not cross the blood-brain barrier.
 - Glycopyrrolate

Table 1: Glycopyrrolate and Hyoscine

	Glycopyrrolate	Hyoscine
Aliases	Glycopyrronium Robinul	Scopolamine
Class	Muscarinic antago- nist	Anticholinergic
Chemistry	Quaternary ammo- nium compound	Tropane alkaloid
Dose		
Premedica- tion (IV/IM)	4 - 5µg/kg (max 400µg)	
Bradycardia (IV)	4 - 5µg/kg (max 400µg)	
Reversal (IV)	200µg per 1mg neostigmine	

	Glycopyrrolate	Hyoscine
Motion sickness (IM/SC)		200µg
Premedication (IM/SC)		200-600µg
Secretions (IV/IM/SC)		300-600µg Q6-8H
Pharmaceutics		
Presentation	Clear colourless 200µg/mL solution	Clear colourless 600µg/mL solution
Compatibility	NS, D5W; incompatible with diazepam and thiopentone	
Solubility	Water- and alcohol-soluble	Highly water-soluble
Pharmacokinetics		
Absorption	Onset within 1 minute IV; peak effect 30-45 minutes IM	10-50% oral bioavailability
Distribution	Vd: 0.16L/kg	pKa: 7.55 Vd: 2L/kg Protein binding: 30%
Metabolism	Half-life: 1.7 hours Duration: Vagal blockade 2-3 hours; antisialagogue effect up to 7 hours Pathway: Minimal hepatic hydrolysis	Half-life: 9 hours Duration: Vestibular effects may persist for days Pathway: Extensive hepatic metabolism to inactive metabolites

	Glycopyrrolate	Hyoscine
Elimination	Renal: 85% urinary recovery within 48 hours; >80% of urinary recovery unchanged	Renal: 2% unchanged; remainder as metabolites
Mechanism		
Target	Muscarinic receptors	Muscarinic receptors (M1-M5)
Action	Competitive antagonism of acetylcholine at peripheral muscarinic receptors	Competitive antagonism of acetylcholine at muscarinic receptors
Contraindications		
		Myasthenia gravis Megacolon Angle-closure glaucoma Tachycardia Prostatic enlargement with urinary retention GI obstruction, stenosis or paralytic ileus Anticoagulation (IM route)
Pregnancy		
Code	B2	B2
Pharmacodynamics		
Respiratory	Bronchodilation ↓ Secretions	Bronchodilation ↓ Secretions
CVS	Reverses vagal ↓ HR ↑ HR at doses >200µg	↑ HR

	Glycopyrrolate	Hyoscine
CNS		Central anticholin- ergic syndrome Delirium Hallucinations Mydriasis Amnesia
Muscu- loskeletal	↓ Sweating	↓ Sweating
Genitouri- nary	Urinary retention	Urinary retention
Gastroin- testinal	↓ Oral and gastric secretions ↓ Motility Constipation	↓ Gastric acid se- cretion ↓ Intestinal motility Constipation Antiemetic effect
Considera- tions	Does not cross the blood-brain barrier Renal impairment → prolonged effect Large doses may block the neuro- muscular junction	Crosses the blood- brain barrier and placenta Avoid in elderly pa- tients

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Neuromuscular Blockers

Understanding of the pharmacology of neuro-muscular blocking drugs

The neuromuscular junction is a **chemical communication** between the motor neuron and the muscle cell, and relies upon:

- Release of ACh-containing vesicles from the motor nerve in response to influx of Ca^{2+}
- Binding of ACh to nicotinic ACh receptors on the muscle cell

Neuromuscular blockers generally act here, by antagonising the action of ACh.

Nicotinic ACh receptors sit on the shoulders of junctional folds of muscle cells, whilst acetylcholinesterase is buried in the clefts.

Factors Affecting Neuromuscular Blockade

Table 1: Patient Factors Affecting Neuromuscular Blockade

Factor	Effect	Mechanism
Hepatic Disease	↑ Duration of aminos-teroids and suxamethonium	↓ Metabolism ↓ Production of pseudocholinesterase in severe disease

Factor	Effect	Mechanism
Pseudo-cholinesterase deficiency	↑ Duration of suxamethonium	↓ Metabolism
Age	↑ Sensitivity in neonates Particularly premature infants.	Incomplete maturation of NMJ
Hypokalaemia	Potentiates non-depolarising blockade Inhibits depolarising blockade	↑ Depolarisation threshold
Hyperkalaemia	Potentiates depolarising blockade Inhibits non-depolarising blockade	↓ Depolarisation threshold
Hypermagnesaemia	Potentiates blockade	↓ ACh release ↓ Sensitivity of post-synaptic membrane
Hypocalcaemia	Potentiates blockade	↓ Presynaptic ACH release ↓ Sensitivity of post-synaptic membrane
Respiratory acidosis	Potentiates blockade	Enhances effect of NMB agents
Hypothermia	Potentiates blockade	↓ Hepatic metabolism ↓ Renal elimination ↓ Hoffman degradation
Hypovolaemia	↓ Rate of block onset ↑ Duration of block	Prolonged circulation time

Factor	Effect	Mechanism
Myasthenia Gravis	↑ Sensitivity to non-depolarising agents	Autoimmune blockade of receptors gives pre-existing level of block
Eaton-Lambert Syndrome	↑ Sensitivity to all NMBs	Autoimmune destruction of voltage-gated Ca^{2+} channels prevent ACh vesicle exocytosis

Table 2: Drug Factors Affecting Neuromuscular Blockade

Drugs	Effect	Mechanism
Frusemide	Potentiates blockade at low dose Inhibits blockade at high dose	Inhibits protein kinases (reducing AMP/ATP synthesis) at low dose Inhibits PDE at high doses, which ↑ ACh release
Inhalational anaesthetics	Potentiates blockade	Stabilise post-junctional membrane Presynaptic ACh receptor blockade
Antibiotics	Potentiates blockade	Aminoglycosides and tetracyclines prolong blockade
Local anaesthetics	Potentiates blockade	↓ ACh release Stabilises post-junctional membrane
Anti-cholinesterases OCP	Inhibits blockade Potentiates depolarising blockade	↑ ACh levels at the NMJ by ↓ ACh breakdown Competes for binding sites on plasma cholinesterases
Ca^{2+} -channel blockers	Potentiates blockade	Inhibit Ca^{2+} dependent ACh release
Lithium	Potentiates blockade	Augments action of NMBs

Depolarisation threshold refers to the magnitude of stimulus required to depolarise cell.

Table 3: Additional Factors Affecting the Onset of Neuro-muscular Blockade

Factor	Effect	Mechanism
Potency	Low potency decreases time to onset	Bowman's principle
Dose	Increased dose decreases time to onset	Greater concentration gradient
Cardiac Output	High output decreases time to onset	Increased drug delivery
Muscle group blood flow	High muscular flow decreases time to onset	Increased drug delivery
Priming Principle	(May) decrease time to onset	Pre-induction blockade of a portion of ACh receptors using a sub-paralytic dose

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Most additional factors affecting the onset of blockade can be directly attributed to **Fick's Law**:

Bowman's principle describes the relationship between dose and potency. In brief: Less potent drugs must be administered in higher doses to have their clinical effect, and so have a greater concentration gradient driving diffusion to the effect site.

A 'priming' dose of non-depolarising blocker should occupy <70% of ACh receptors, and so should not cause significant neuromuscular blockade in the awake patient.

After induction, a second dose is given to occupy the remaining receptors and complete blockade.

Depolarising NMBs

Succinylcholine binds to the nicotinic ACh receptor causing depolarisation. It cannot be hydrolysed by acetylcholinesterase in the NMJ, and so remains bound to the receptor. This:

- Produces a sustained depolarisation which keeps voltage-gated sodium channels in their inactive state
- Prevents the post-junctional membrane from responding to further ACh release

Table 1: Suxamethonium

	Suxamethonium
Aliases	Succinylcholine Succinyl choline Suxamethonium Medsurge
Class	Depolarising neuromuscular blocker
Chemistry	Bis-choline ester; two ACh molecules linked at their acetyl groups
Dose	
Intubation (IV)	0.6mg/kg; range 0.3 - 1.1mg/kg
Paediatric (IV)	1 - 2mg/kg
IM	Up to 2.5mg/kg; max 150mg
Pharmaceutics	
Presentation	50mg/mL clear colourless solution; 100mg/2mL
Compatibility	Do not mix with other drugs; incompatible with thiopental
Solubility	pH 3 - 5
Pharmacokinetics	

	Suxamethonium
Absorption	IV or IM
Distribution	pKa: ≥ 13
	Vd: 0.14L/kg
	Protein binding: 20%
Metabolism	Half-life: 1-2 minutes
	Duration: IV 2-3 minutes; offset within 10 minutes
	Pathway: Rapid hydrolysis by plasma butyrylcholinesterase
	Metabolites: Succinylmonocholine, Choline
Elimination	Renal: ~10% unchanged
Mechanism	
Target	Muscle-type nicotinic acetylcholine receptor
Action	ACh agonism $\rightarrow$ persistent end-plate depolarisation and voltage-gated Na ⁺ channel inactivation
Contraindications	MH susceptibility Risk of $\uparrow$ K ⁺ if severe hyperkalaemia, burns (post-acute phase), or upper/lower motor-neurone lesion with major muscle wasting Myopathy or congenital myotonia Atypical or low plasma butyrylcholinesterase Acute narrow-angle glaucoma or penetrating eye injury
Pregnancy	
Code	A
Pharmacodynamics	

	Suxamethonium
Respiratory	Apnoea Prolonged respiratory failure
CVS	Bradycardia Dysrhythmias and cardiac arrest
CNS	↑ ICP
Musculoskeletal	Fasciculation → myalgia Masseter spasm Flaccid paralysis
Metabolic	↑ K ⁺ Malignant hyperthermia
Renal	Rhabdomyolysis and myoglobinuria → AKI
Gastrointestinal	↑ Intra gastric pressure
Considerations	Rapid onset; ultrashort duration ~20% reaches the NMJ because of rapid plasma hydrolysis Butyrylcholinesterase deficiency → prolonged apnoea ↑ IOP; avoid penetrating eye injury High anaphylaxis risk Refrigerate at 2-8°C; do not freeze

Adverse Effects

The adverse effects of suxamethonium can be remembered as **three major, three minor, and three pressures**:

- **Major**
 - Anaphylaxis
 - Suxamethonium Apnoea
 - Malignant hyperthermia
- **Minor**
 - Hyperkalaemia
 - Myalgias
 - Bradycardia
- **Pressure**
 - IOP

- ICP
- Intra gastric pressure

Phase I and Phase II Blockade

Initial blockade is termed **Phase I**, which is a partial depolarising block. Sustained use of suxamethonium may cause a Phase II block which:

- Appears similar to a non-depolarising block
- May be due to:
 - Presynaptic inhibition of ACh synthesis and release
 - Desensitisation of the post-junctional receptor

Key differences include:

Table 2: Comparison of Phase I and Phase II Neuromuscular Block

Property	Phase I Block	Phase II Block
Block Amplitude	Reduced	Reduced
Train-of-four ratio	>0.7	< 0.7
Post-tetanic potentiation	No	Yes
Effect of anti-cholinesterases	Block augmented	Block inhibited

Malignant Hyperthermia

- Rare autosomal dominant genetic condition
- Triggered by suxamethonium and volatile anaesthetic agents
- Mutation of the ryanodine receptor causes excessive amounts of calcium to leave the sarcoplasmic reticulum, causing continual muscle contraction

- Results in greatly increased carbon dioxide, lactate, and heat production
- Cell lysis with myoglobulinaemia and hyperkalaemia results

Suxamethonium Apnoea

- A deficiency of butylcholinesterase causes suxamethonium to not be metabolised
 - May be congenital (genetic) or acquired (hepatic failure)
 - Can be treated with fresh frozen plasma
-

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Non-Depolarising Neuromuscular Blockers

Non-depolarising NMBs are muscle relaxants used to:

- Facilitate laryngoscopy and tracheal intubation
- Control ICP
- Improve respiratory system compliance
- Improve patient safety on transportation

Mechanism of action is by competitive antagonism of ACh at the NMJ, preventing generation of end-plate potentials. Effective pharmacodynamic response requires >70% receptor occupation.

Table 1: Common Features of Non-Depolarising Neuromuscular Blockers

Property	Action
Route of Administration	IV/IM
Distribution	Small V_D as they are polar and unable to cross lipid membranes
Elimination	Reduced urinary clearance which prolongs the mechanism of action of aminosteroids in renal failure
Resp	Apnoea
MSK	↑ Duration in hypothermia

Property	Action
Renal	↑ Duration in acidosis, ↑ duration in hypokalaemia, ↓ duration in hyperkalaemia, ↑ duration in hypermagnesaemia
Metabolic	Critical Illness Myopathy in patients with long-term relaxant use

The ED₉₅ is:

- The dose of a neuromuscular blocking drug required to produce a 95% reduction in twitch height in 50% of the population
- A commonly-used therapeutic end-point for neuro-muscular blocking drugs
Typically, induction doses used are 2-5x the ED₉₅.

Table 2: Comparison of Non-Depolarising Neuromuscular Blockers

	Rocuronium	Vecuronium	Pancuronium	Atracurium	Cisatracurium
Aliases	Rocuronium Baxter Esmeron	Vecuronium SUN Norcuron	Pancuronium Bromide Juno Pavulon	Atracurium Be- silate Medsurge Tracrium	Nimbex
Class	Non-depolaris- ing neuromus- cular blocker	Non-depolaris- ing neuromus- cular blocker	Non-depolaris- ing neuromus- cular blocker	Non-depolaris- ing neuromus- cular blocker	Non-depolaris- ing neuromus- cular blocker
Chemistry	Quaternary aminosteroid; vecuronium analogue	Monoquaternary aminosteroid	Bisquaternary aminosteroid	Benzylisoquino- linium diester; mixture of 10 stereoisomers	Benzylisoquino- linium diester; R-cis,R'-cis atracurium stereoisomer
Dose					
Intubation (IV)	0.6mg/kg	0.1mg/kg	0.05-0.1mg/kg	0.5mg/kg	0.15mg/kg
RSI (IV)	1mg/kg				
Maintenance (IV)	0.15mg/kg	0.02-0.04mg/ kg	0.01-0.02mg/ kg	0.08-0.1mg/kg	0.03mg/kg
Infusion (IV)	0.3-0.6mg/kg/ hour	0.8-1.4microg/ kg/min		5-10microg/kg/ min	1-3microg/kg/ min
ICU (IV)			0.06mg/kg Q1- 1.5H		
Pharmaceutics					
Presentation	10mg/mL solu- tion for IV injec- tion or infusion	10mg powder for IV injection or infusion	2mg/mL solu- tion for IV injec- tion	10mg/mL solu- tion for IV injec- tion or infusion	2mg/mL or 5mg/mL solu- tion for IV injec- tion or infusion

	Rocuronium	Vecuronium	Pancuronium	Atracurium	Cisatracurium
Compatibility	Compatible with 0.9% NaCl, glucose 5% and Hartmann's; flush shared lines between drugs	Reconstitute with water for injection; incompatible with thiopentone	Do not mix with other drugs or infusion solutions; altered pH may precipitate	Compatible with 0.9% NaCl, glucose 5% and Hartmann's (4h at $\geq 0.5\text{mg/mL}$, $\leq 25^\circ\text{C}$); do not mix with alkaline solutions	Do not mix with alkaline solutions, thiopentone, ketorolac or propofol emulsion
Solubility	Freely water-soluble; $>200\text{mg/mL}$	16mg/mL in water; hydrolyses in aqueous solution	Water-soluble bromide salt	Water-soluble	Water-soluble
Pharmacokinetics					
Absorption	Poor PO; IV only	Poor PO; IV only	Poor PO; IV only	Poor PO; IV only	Poor PO; IV only
Distribution	pKa: 7.96	pKa: 8.97	pKa: 7.4	pKa: 19.02	pKa: 19.02
	Vd: 0.27L/kg	Vd: 0.2L/kg	Vd: 0.2L/kg	Vd: 0.12L/kg	Vd: 0.2L/kg
	Protein binding: 46%	Protein binding: 77%	Protein binding: 82%	Protein binding: 38%	Protein binding: 38%

	Rocuronium	Vecuronium	Pancuronium	Atracurium	Cisatracurium
Metabolism	Half-life: 60-100 minutes	Half-life: 110 minutes	Half-life: 110 minutes	Half-life: 21 minutes	Half-life: 26 minutes
	Duration: 30-40 minutes	Duration: 20-40 minutes	Duration: 45-60 minutes	Duration: 15-35 minutes	Duration: 25-30 minutes
	Pathway: <1% hepatic metabolism	Pathway: ~5% hepatic metabolism Metabolites: 3-hydroxyvecuronium; ~80% potency	Pathway: <1% hepatic metabolism	Pathway: Plasma ester hydrolysis and Hofmann elimination Metabolites: Laudanosine, Quaternary acrylate	Pathway: Hofmann elimination; metabolite ester hydrolysis Metabolites: Laudanosine, Monoquaternary acrylate
Elimination	Renal: ~10%; ~90% biliary	Renal: ~30%; 30-40% biliary	Renal: 80-90% unchanged; 10-20% hepatic or biliary	Renal: Minimal unchanged drug; metabolites in urine and bile	Renal: Minimal unchanged drug; renal and hepatic metabolite clearance
Mechanism Target	Muscle-type nicotinic acetylcholine receptor	Muscle-type nicotinic acetylcholine receptor	Muscle-type nicotinic acetylcholine receptor	Muscle-type nicotinic acetylcholine receptor	Muscle-type nicotinic acetylcholine receptor

	Rocuronium	Vecuronium	Pancuronium	Atracurium	Cisatracurium
Action	Competitive ACh antagonism at the motor end-plate prevents channel opening and depolarisation	Competitive ACh antagonism at the motor end-plate prevents channel opening and depolarisation	Competitive ACh antagonism at the motor end-plate prevents channel opening and depolarisation Concurrent depolarising neuromuscular blocker	Competitive ACh antagonism at the motor end-plate prevents channel opening and depolarisation	Competitive ACh antagonism at the motor end-plate prevents channel opening and depolarisation
Contraindications					
Pregnancy					
Code	B2	C	B2	C	C
Justification		No demonstrated fetal or neonatal adverse effects		No demonstrated fetal or neonatal adverse effects	No demonstrated fetal or neonatal adverse effects
Pharmacodynamics					
Respiratory	Apnoea	Apnoea	Apnoea	Apnoea Histamine release → bronchospasm	Apnoea
CVS	High dose → ↑ HR	No vagolysis → bradycardia may persist	Vagal muscarinic block → ↑ HR, MAP and CO Arrhythmias	Histamine release → flushing; ↓ MAP	Bradycardia ↓ MAP; flushing
CNS				Laudanosine → ↓ seizure threshold	Laudanosine → ↓ seizure threshold

	Rocuronium	Vecuronium	Pancuronium	Atracurium	Cisatracurium
Musculoskeletal	Flaccid paralysis	Flaccid paralysis	Flaccid paralysis	Flaccid paralysis	Flaccid paralysis
Considerations	Rapid onset; intermediate duration Higher anaphylaxis risk than other neuromuscular blockers Hepatic, biliary or renal impairment → prolonged block Reverse with sugammadex	Minimal histamine release Reverse with sugammadex	Long duration; cumulative with repeat dosing Minimal histamine release Renal or hepatobiliary impairment → prolonged block	Organ-independent inactivation → suitable in renal or hepatic failure Hypothermia → ↓ infusion requirement Laudanosine accumulation with prolonged ICU infusion	Organ-independent inactivation → suitable in renal or hepatic failure Less histamine release and laudanosine than atracurium Hypothermia → ↓ infusion requirement

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Dantrolene

Dantrolene is a ryanodine (RYR1) receptor antagonist, which prevents release of Ca^{2+} from the sarcoplasmic reticulum, uncoupling the process of excitation-contraction.

Table 1: Dantrolene

	Dantrolene
Aliases	Dantrium
Class	Direct-acting skeletal muscle relaxant
Chemistry	Hydantoin derivative; sodium hemihydrate
Dose	
MH (IV)	2.5mg/kg Q10-15 minutes up to 10mg/kg
Pharmaceutics	
Presentation	20mg light-yellow powder with 3g mannitol; reconstitute with 60mL water for injection to 0.333mg/mL
Compatibility	Compatible only with water for injection; protect from light and use within 6 hours
Solubility	pH ~9.5 after reconstitution
Pharmacokinetics	
Absorption	IV
Distribution	Protein binding: 85%
Metabolism	Half-life: 5 hours
	Pathway: Hepatic
	Metabolites: 5-hydroxydantrolene (active), Acetamido analogue

	Dantrolene
Elimination	Renal: Dantrolene and metabolites
Mechanism	
Target	Ryanodine receptor 1
Action	Inhibits Ca ²⁺ release from the sarcoplasmic reticulum → uncouples excitation-contraction
Pregnancy	
Code	B2
Pharmacodynamics	
Respiratory	Skeletal muscle weakness → respiratory failure
CVS	Fluid load → pulmonary oedema
CNS	Drowsiness Dizziness
Musculoskeletal	Skeletal muscle relaxation
Renal	Mannitol → diuresis
Gastrointestinal	Hepatotoxicity
Considerations	High pH → extravasation necrosis Each 20mg vial requires 60mL diluent → substantial fluid load

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Sugammadex

Table 1: Sugammadex

	Sugammadex
Aliases	Sugammadex Juno Bridion
Class	Selective relaxant binding agent
Chemistry	Modified gamma-cyclodextrin
Dose	
Moderate block (IV)	2mg/kg at reappearance of T2
Deep block (IV)	4mg/kg at 1 - 2 post-tetanic counts
Immediate (IV)	16mg/kg 3 minutes after rocuronium 1.2mg/kg
Pharmaceutics	
Presentation	100mg/mL clear colourless to slightly yellow-brown solution; 200mg/2mL or 500mg/5mL
Solubility	pH 7 - 8; osmolality 300 - 500mOsm/kg
Pharmacokinetics	
Absorption	IV
Distribution	pKa: 2.82
	Vd: 0.16-0.20L/kg
	Protein binding: Minimal

	Sugammadex
Metabolism	Half-life: ~2 hours
	Duration: 1.5 - 3 minutes to TOF ratio 0.9
	Pathway: Not metabolised
Elimination	Renal: 96% of dose in urine; $\geq 95\%$ of urinary recovery unchanged
Mechanism	
Target	Rocuronium Vecuronium
Action	Encapsulates free aminosteroid blocker → plasma - NMJ concentration gradient and reversal
Pregnancy	
Code	B2
Pharmacodynamics	
Respiratory	Residual or recurrent neuromuscular block
CVS	Bradycardia → cardiac arrest
Musculoskeletal	Reverses rocuronium and vecuronium block
Haematological	Transient $\uparrow$ aPTT and PT(INR)
Considerations	Dose to quantitative depth of block

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Haematological Pharmacology

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Anticoagulants

Table 1: Anticoagulants

	Warfarin	Heparin	Enoxaparin
Class	Vitamin K antagonist	Parenteral anticoagulant	Low molecular weight heparin
Chemistry	Coumarin derivative	Glycosaminoglycan; molecular weight 5,000-25,000 Da	Glycosaminoglycan; low molecular weight heparin fragments; mean molecular weight 5,000 Da
Pharmaceutics			
Solubility	Practically insoluble in water	Excellent water solubility	Excellent water solubility
Pharmacokinetics			
Absorption	~100% oral bioavailability	Minimal oral bioavailability (~1%); IV onset immediate	Minimal oral bioavailability (~1%)

	Warfarin	Heparin	Enoxaparin
Distribution	pKa: 5.87 Vd: 0.08 - 0.12 L/kg Protein binding: 99%	pKa: - 2.0 to - 4.0 Protein binding: Highly protein-bound, mainly to lipoproteins (LDL)	pKa: - 2.8 Vd: 0.05 L/kg Protein binding: Less than UFH; only chains under 6,000 Da are protein-bound; does not bind to heparin-binding proteins
Metabolism	Half-life: 36 - 42 hours Pathway: Complete hepatic metabolism; S-isomer metabolised by CYP2C9 to 7-hydroxywarfarin (inactive); also via reductases to warfarin alcohols (minimal anticoagulant activity) Metabolites: 7-hydroxywarfarin; warfarin alcohols	Half-life: Dose-dependent — approximately 30 min at 25 U/kg, 60 min at 100 U/kg, 150 min at 400 U/kg Pathway: Sequestered into reticuloendothelial cells and degraded gradually into inactive metabolites; biphasic saturable clearance — rapid at low doses, slower when reticuloendothelial system is saturated at high doses	Half-life: 4 - 7 hours Pathway: Mainly hepatic desulfation and depolymerization to lower molecular weight fragments (less potent or inactive)
Elimination	Renal: Renal and faecal elimination of metabolites; 92% of dose recovered in urine within 1 week; minimal free warfarin renally excreted	Renal: Renal elimination of inactive metabolites	Renal: Approximately 40% of active and inactive fragments combined excreted renally; monitoring by anti-Xa activity

	Warfarin	Heparin	Enoxaparin
Mechanism			
Target	Vitamin K epoxide reductase complex 1 (VKORC1)	Antithrombin III (ATIII)	Antithrombin III (ATIII)
Action	Interferes with the cyclic interconversion of vitamin K and its 2,3-epoxide, preventing γ -carboxylation of Gla residues on the N-terminal regions of vitamin K-dependent coagulation factors II, VII, IX, and X, as well as anticoagulant proteins C and S, resulting in hepatic synthesis of functionally inactive factors	Binds antithrombin III causing a conformational change of the active site, increasing its affinity for factor Xa and thrombin (1:1 anti-IIa and anti-Xa effect); at higher concentrations also inhibits IXa, XIa, XIIa and platelet aggregation	Binds antithrombin III causing a conformational change $\rightarrow$ $\uparrow$ affinity for factor Xa but not thrombin (4:1 anti-Xa to anti-IIa ratio); more predictable effect on Xa than UFH
Pregnancy			
Code	D	C	C
Justification	6-9 weeks $\rightarrow$ embryopathy; later exposure $\rightarrow$ fetal bleeding/CNS damage; $\uparrow$ miscarriage/perinatal bleeding	Placental haemorrhage $\rightarrow$ prematurity/fetal loss	Placental haemorrhage $\rightarrow$ prematurity/fetal loss
Pharmacodynamics			
CVS	Microthrombi	Hypotension with rapid IV administration	
Metabolic	Less osteoporosis than heparin due to reduced protein binding	Osteoporosis	
Renal		Inhibits aldosterone secretion	

	Warfarin	Heparin	Enoxaparin
Gastrointestinal	Nausea Vomiting	↑ LFTs	
Haematological	Haemorrhage Skin necrosis Systemic microemboli Calciphylaxis	Haemorrhage HITTs Alopecia	Haemorrhage Lower risk of HITTs than UFH; less thrombocytopenia
Considerations	Hypersensitivity reactions Onset of anticoagulant effect at 8-12 hours Peak effect at 72 hours due to half-life of existing clotting factors Requires INR monitoring; complicated pharmacokinetics Significant interactions with CYP enzyme inducers and inhibitors Reversal agents — vitamin K, FFP, Prothrombinex Teratogenic	1 unit is the amount required to prevent 1mL of cats blood clotting for 24 hours at 0°C Risk of HITTs Monitoring by APTT or ACT Large interpatient variability due to variable levels of ATIII and heparin binding proteins Reversal — protamine 1mg per 100 units of heparin	Monitored with anti Xa Incomplete reversal with protamine — only the anti-IIa effect is inhibited Caution in renal failure — approximately 40% renally excreted

HITTs

Heparin-Induced Thrombotic Thrombocytopenia comes in two flavours:

- **Type I:**
 - Is non-immune mediated
 - Occurs within 4 days of anticoagulant doses
 - Is an isolated thrombocytopenia without clinical significance
- **Type II:**
 - Is immune mediated
 - Occurs within 4 - 14 days
 - Is associated with serious thrombosis and high mortality (typically from PE) and morbidity (from CVA and limb ischaemia)

Protamine

Protamine is:

- A basic cationic protein derived from salmon sperm which combines with the acidic anionic heparin to form a stable, inactive salt in solution
- Cleared more rapidly than heparin
Rebound anticoagulation may occur.

Adverse effects from protamine include:

- Anaphylaxis
- Direct (non-anaphylaxis) histamine release
 - Bronchospasm
 - Hypotension
- Pulmonary hypertension
This can be profound and result in a dramatic increase in RV afterload and EDV, with a corresponding fall in LV preload (interventricular interdependence), leading to dramatic hypotension and arrest.
 - Mediated by thromboxanes

- Due to **protamine-heparin complexes**, rather than protamine alone
Administration of protamine in absence of heparin does not lead to pulmonary hypertension.
 - Anticoagulation
When given in excess.
-

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Direct Thrombin Inhibitors

Direct thrombin inhibitors prevent cleavage of fibrinogen to fibrin, and are therefore very effective anticoagulants.

Table 1: Dabigatran

	Dabigatran
Class	Direct thrombin inhibitor
Chemistry	Small molecule polypeptide
Dose	
VTE prophylaxis	220mg PO daily
Atrial fibrillation	150mg PO BD
Pharmaceutics	
Presentation	75mg capsules 110mg capsules
Solubility	Poor water solubility
Pharmacokinetics	
Absorption	6.5% oral bioavailability; etexilate rapidly/ completely converted to dabigatran; Tmax 90-180 minutes
Distribution	pKa: 4 Vd: 1.0 L/kg Protein binding: 35%

	Dabigatran
Metabolism	Half-life: 12 - 14 hours
	Pathway: Prodrug activated by plasma and hepatic esterases; approximately 20% conjugated with glucuronic acid to form acylglucuronides (pharmacologically active, similar properties to unconjugated dabigatran)
	Metabolites: Acylglucuronides (active)
Elimination	Renal: 80% renal excretion of unchanged drug; 20% biliary excretion of acylglucuronides
Mechanism	
Target	Thrombin
Action	Reversible competitive inhibitor of thrombin (including fibrin-bound thrombin); prevents cleavage of fibrinogen to fibrin and inhibits other thrombin-driven haemostasis
Pregnancy	
Code	C
Justification	Placental haemorrhage → prematurity/fetal loss
Pharmacodynamics	
CVS	Peripheral oedema
CNS	Insomnia
Metabolic	Fever
Haematological	Haemorrhage
Considerations	Dialysable Reversible with idarucizumab Significant interactions with amiodarone, quinidine, St John's Wort, and other anti-coagulant and antiplatelet agents Allergy reported

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Antifibrinolytics

Antifibrinolytics include aprotinin, aminocaproic acid, and **tranexamic acid**. All prevent the breakdown of fibrin (!) by various mechanisms. TXA competitively inhibits plasminogen activator, reducing rate of fibrinolysis.

Table 1: Tranexamic Acid

	Tranexamic acid
Aliases	TXA
Class	Serine protease inhibitor
Chemistry	Monocarboxylic acid; synthetic derivative of lysine
Dose	
IV	1g slow IV; may be followed by infusion of 1g over 8 hours
Pharmaceutics	
Presentation	Tablets Syrup Clear colourless solution for injection
Solubility	Highly water-soluble; minimally fat-soluble
Pharmacokinetics	
Absorption	50% oral bioavailability
Distribution	pKa: 10.22 Vd: 0.18 L/kg Protein binding: 3% (all to plasminogen)

	Tranexamic acid
Metabolism	Half-life: 2 hours
	Pathway: Minimally metabolised
Elimination	Renal: 95% excreted unchanged in urine; dose reduction required in renal impairment
Mechanism	
Target	Plasminogen
Action	Competitive inhibitor of plasminogen activation by binding to the 5-lysine site on plasminogen, inhibiting formation of plasmin and displacing plasminogen from the surface of fibrin
Pregnancy	
Code	B1
Pharmacodynamics	
Gastrointestinal	Nausea Vomiting
Haematological	Reduces fibrinolysis Possible ↑ DVT/PE risk
Considerations	Dose reduction required in renal impairment Allergic dermatitis reported

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Antiplatelets

Note aspirin is included under COX inhibitors.

Classification of Antiplatelet Agents

Antiplatelet agents can be classified by which stage of platelet function they affect:

- Adhesion
 - vWF inhibitors
e.g. Dextran 70.
- Activation
 - Prostacyclins
e.g. Epoprostenol.
 - Phosphodiesterase inhibition
e.g. Dipyridamole.
 - COX inhibitors
Prevent thromboxane A2 production, e.g. aspirin.
- Aggregation
 - ADP receptor antagonists
Prevent activation of GP IIb/IIIa receptors,
e.g. clopidogrel.
 - GP IIb/IIIa receptor antagonists
Prevent platelet aggregation via fibrin linkages
between GP IIb/IIIa receptors, e.g. tirofiban.

Table 1: Comparison of Common Antiplatelets

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
Aliases		Persantin Pytazen			Brilinta
Class	Antiplatelet agent; ADP receptor antagonist	Phosphodiesterase inhibitor; adenosine uptake inhibitor	Antiplatelet agent; GP IIb/IIIa antagonist	P2Y12 inhibitor	P2Y12 inhibitor
Chemistry	Thienopyridine	Pyrimido-pyrimidine	Tyrosine derivative	Thienopyridine	Nucleoside (adenosine) analogue
Dose					
Load	300mg PO			60mg PO	180mg PO
Maintenance	75mg PO daily			10mg PO daily; 5mg if <60kg or consider if ≥75 years	90mg PO BD
CVA (PO)		200mg BD			
Antithrombotic adjunct (PO)		150mg SR BD			
Perfusion imaging (IV)		0.14mg/kg/min for 4 minutes; max 60mg			
Stress echo (IV)		0.56mg/kg over 4 minutes, then 0.28mg/kg over 2 minutes after 4 minutes; max 0.84mg/kg			

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
UA/NSTEMI (IV)			0.4microg/kg/ min for 30min, then 0.1microg/ kg/min for 48 - 108h		
CrCl <30mL/min			Halve loading and mainte- nance infusions		
Pharmaceutics					
Presentation		25mg tablet; 150mg SR tablet/capsule; clear yellow 5mg/mL so- lution		5mg and 10mg tablets	90mg film- coated or orodispersible tablets
Compatibility		Dilute IV $\geq 1:2$ with 0.45% or 0.9% NaCl or 5% glucose; do not mix with other drugs			May crush/ disperse for PO/ NG administra- tion
Solubility	Basically insol- uble in water	Lipophilic; log P 3.92	Slightly soluble in water	Practically in- soluble in water	Practically in- soluble in water
Pharmacokinetics					

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
Absorption	50% absorbed; only 2% of oral dose converted to active metabolite; rapid absorption with onset within 2 hours	Variable PO absorption	Minimal oral bioavailability	Rapid, complete absorption; 80% PO bioavailability	25-65% PO bioavailability; absorption within 2-4h
Distribution	pKa: 3.5	Vd: 100-140L at steady state	pKa: 3.7	pKa: 5.1	pKa: 12.9
	Vd: 550 L/kg		Vd: 0.4-1.0 L/kg	Vd: 1L/kg	Vd: 1.2L/kg
	Protein binding: 98%	Protein binding: 97-99%	Protein binding: 64%	Protein binding: 98% (mainly albumin)	Protein binding: 99.8%

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
Metabolism	Half-life: Approximately 8 hours	Half-life: IV: distribution 3 minutes; elimination 40 minutes; terminal 15 hours	Half-life: Approximately 2 hours	Half-life: Active metabolite 5-7h	Half-life: ~8h
	Duration: 7-10 days (irreversible platelet binding)	Pathway: Hepatic glucuronidation; possible enterohepatic recirculation	Duration: Platelet aggregation returns to baseline 4-8 hours after cessation	Duration: 7-10 days	Duration: 48-72h; ~60% platelet inhibition at 24h and ~10% at 5 days
	Pathway: Pro-drug; major activity hydrolysed by carboxylesterase 1 to inactive carboxylic acid metabolite; only 2% converted by CYP450 to clop-AM (active form)	Metabolites: Dipyridamole monoglucuronide	Pathway: Minimal metabolism	Pathway: Pro-drug; intestinal hydrolysis → thiolactone → active metabolite via CYP3A4/2B6	Pathway: Active parent; CYP3A → active AR-C124910XX
	Metabolites: Clop-AM (active); inactive carboxylic acid metabolite			Metabolites: Active thiol; inactive metabolites	Metabolites: AR-C124910XX (active)

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
Elimination	Renal: 50% urinary, 50% faecal elimination of metabolites	Renal: ~5% after IV; faecal elimination also occurs	Renal: Cleared renally as unchanged drug	Renal: 68% of metabolites in urine; remainder in faeces	Renal: Inactive metabolites renal; active metabolite biliary; majority faecal
Mechanism					
Target	P2Y ₁₂ ADP receptor	Cyclic-nucleotide phosphodiesterases Adenosine uptake	GP IIb/IIIa receptor	P2Y ₁₂ receptor	P2Y ₁₂ receptor
Action	Irreversibly inhibits ADP binding to P2Y ₁₂ receptor on platelets, preventing platelet activation and the subsequent ADP-mediated activation of the GP IIb/IIIa complex	PDE inhibition and ↓ adenosine uptake → ↑ platelet cAMP/cGMP → ↓ intracellular Ca ²⁺ and platelet activation; coronary vasodilation	Reversible antagonism of the GP IIb/IIIa receptor, preventing binding of fibrinogen, fibrin, and von Willebrand factor, thus preventing platelet crosslinking and aggregation	Irreversible P2Y ₁₂ inhibition → prevents ADP-mediated platelet activation and GP IIb/IIIa activation; ↑ potency vs clopidogrel	Reversible non-competitive P2Y ₁₂ inhibition → prevents ADP-mediated platelet activation and GP IIb/IIIa activation

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
Contraindications		Acute MI or unstable angina (IV stress) Symptomatic severe AS (IV stress) Haemodynamically significant arrhythmia or uncontrolled symptomatic CCF (IV stress) Acute PE or pulmonary infarction, myocarditis, pericarditis, active endocarditis or aortic dissection (IV stress) Regular oral dipyridamole (additional IV contraindicated)		Active bleeding Previous TIA/CVA Severe hepatic impairment	Active bleeding Previous ICH Moderate-severe hepatic impairment Strong CYP3A4 inhibitors
Pregnancy					
Code	B1	B1	B1	B1	B1
Justification	Placental haemorrhage → prematurity/fetal loss			Placental haemorrhage → prematurity/fetal loss	
Pharmacodynamics					

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
Respiratory		Bronchospasm			Dyspnoea
CVS		Vasodilatation → ↓ MAP and re- flex tachycardia Coronary steal → angina or MI	Coronary artery dissection	↓ MI/stent thrombosis/ revascularisation vs clopidogrel	Bradyarrhythmia; AV block
CNS		Headache Dizziness			
Metabolic					↑ Urate; gout
Renal					↑ Creatinine
Gastrointestinal	Mucosal irritation	Nausea Diarrhoea			
Haematological	Haemorrhage Aplastic anaemia Thrombocytopenia Neutropenia	↓ Platelet adhesion and aggregation Thrombocytopenia Haemorrhage	Haemorrhage	Bleeding; ↑ vs clopidogrel Aplastic anaemia; thrombocytopenia; neutropenia TTP	Bleeding Aplastic anaemia; thrombocytopenia; neutropenia TTP

	Clopidogrel	Dipyridamole	Tirofiban	Prasugrel	Ticagrelor
Considerations	Many pharmacokinetic interactions including genetic variability in CYP450 metabolism	↓ Cholinesterase-inhibitor effect → worsening myasthenia gravis Aminophylline reverses most IV effects; variant angina → nitrates as aminophylline may worsen	Onset within 10 minutes of IV administration	≥75 years → generally avoid Stop ≥7 days before elective surgery Greater benefit in diabetes vs clopidogrel/ticagrelor	Stop ≥5 days before elective surgery Platelet transfusion unlikely to reverse effect Not dialysable

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Antiseptics

Outline the pharmacology of antiseptics and disinfectants

Key Definitions

Relevant definitions for antiseptics include:

- **Cleaning**
Physical removal of foreign material.
 - Used for **non-critical items**, which come into contact with healthy skin but not mucous membranes (e.g. blood pressure cuff)
- **Decontamination**
Destruction of contaminants such that they cannot reach a susceptible site in sufficient number to cause harm.
- **Disinfection**
Elimination of all pathological organisms, excluding spores.
 - Used for **semi-critical items**, which are those that contact mucous membranes but do not break the blood barrier (e.g. endoscopes, laryngoscopes)
- **Sterilisation**
Elimination of all forms of microbial life, including spores.
 - Used for **critical items**, which are those that enter sterile or vascular tissue and pose a high risk of

infection (e.g. surgical instruments, vascular and urinary catheters)

Table 1: Properties of Common Antiseptic Agents

Drug	Isopropyl Alcohol	Chlorhexidine	Povidone iodine
Pharmaceuticals	Typically 60-90% - requires some water to denature protein. Flammable.	May be aqueous or combined with isopropyl alcohol.	Iodine combined with a polymer (povidone) to enhance water solubility
Antiviral Properties	Poor antiviral	Poor antiviral	Good antiviral
Antibacterial Properties	Broad spectrum antibacterial	Broad spectrum antibacterial and antifungal	Broad spectrum including fungi, spores (unlike iodine), and tuberculosis
Toxic	Irritant on mucous membranes and open wounds	Hypersensitivity	Hypersensitivity
Other		Persistent antiseptic effect	Requires continual release of iodine to achieve effect. Inactivated by organic substances. Stains.

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Penicillins

- Penicillins are **bactericidal** antibiotics that prevent cell-wall synthesis by preventing cross-linking of peptidoglycans by replacing the natural substrate with their β -lactam ring
- Penicillins bind to penicillin binding proteins (**PBPs**) in the bacterial wall
- Penicillins only rarely achieve complete eradication of sensitive organisms without addition of a **synergistic antibiotic** (such as gentamicin)

Table 1: Common Pharmacological Features of Penicillins

Property	Effect
Absorption	Typically well absorbed orally. IM dosing tend to cause localised pain and irritation.
Distribution	Typically have good tissue penetration. Only cross the blood-brain barrier and enter bone if it is inflamed. Typically low protein binding (exception is flucloxacillin, which is 95% protein bound).
Metabolism	Typically small proportion is hepatically metabolised.

Property	Effect
Elimination	Majority (60-90%) is eliminated unchanged in urine predominantly by active tubular secretion, with renal clearance proportional to total renal plasma flow. A small quantity is secreted in bile.

Mechanisms of Resistance

- Alteration or protection of PBPs
 - Gram negative bacteria may have altered permeability of porins in their outer membrane, which protects the PBP
- Hydrolysis by **β -lactamase**-producing bacteria
 - **Clavulanic acid** and **tazobactam inhibit β -lactamase**, which can render otherwise resistant bacteria sensitive
 - Notably, flucloxacillin has a modified beta-lactam ring that is not sensitive to β -lactamases

Table 2: Comparison of Penicillin Classes

	Narrow spec-trum, naturally occurring	Narrow spec-trum, synthetic	Extended-spectrum	An-tipseu-domonal
Exam-ples	Ben-zylni-cillin , phe-noxymethylpeni-cillin	Flu-cloxacillin	Ampi-cillin , amoxacillin	Piperacillin , ticarcillin

	Narrow spec- trum, naturally occurring	Narrow spec- trum, synthetic	Extended- spectrum	An- tipseu- domonal
Indica- tions	Gram positives and anaer- obes, particu- larly <i>strepto- cocci</i> and meningo- cocci. Also <i>listeria</i> , <i>Clostridia</i> , and <i>Tre- ponema</i> .	Gram positive cocci, particu- larly <i>staphy- lococci</i> but also strepto- cocci.	Gram positive, particu- larly <i>entero- cocci</i> . Some gram negative.	Gram positive, gram negative including pseu- domonas.
Other bits	Highly bacteri- cidal	Less active than ben- zylpeni- cillin on organ- isms sensitive to both.	Can penetrate some gram- negatives.	Gram negative cover.

Individual Penicillins

Table 3: Naturally Occurring Penicillins

	Benzylpenicillin	Phe- noxymethylpeni- cillin
Aliases	BenPen Penicillin G	LPV Penicillin V
Class	Narrow - spectrum penicillin	Narrow - spectrum penicillin
Chemistry	Natural penicillin; β -lactam sodium salt	Natural penicillin; β -lactam potassium salt
Dose		
Infection (IV/ IM)	1.2 - 2.4g Q4 - 6H	
Meningitis (IV)	2.4g Q4H	
Infection (PO)		250 - 500mg Q6H
Rheumatic fever pro- phylaxis (PO)		125 - 250mg BD
Pharmaceutics		
Presentation	600mg, 1.2g and 3g powder for in- jection	250mg and 500mg capsules
Compatibility	Reconstitute with water for injection; dilute IV in water or 0.9% NaCl	
Solubility	Water - soluble	Freely water - sol- uble; practically ethanol - insoluble
Pharmacokinetics		
Absorption		~25% absorbed; slightly $\uparrow$ fasting
Distribution	Vd: ~0.35L/kg Protein binding: ~60%	Protein binding: 80%

	Benzylpenicillin	Phe- noxymethylpeni- cillin
Metabolism	Half-life: 30 minutes Duration: 4-6 hours after 600mg IM Pathway: <30% hepatic inactivation to penicilloic acid Metabolites: Penicilloic acid	
Elimination	Renal: >70% within 6 hours; mainly tubular secretion	Renal: Rapid; delayed in renal impairment
Mechanism		
Target	Penicillin-binding proteins	Penicillin-binding proteins
Action	Inhibits peptidoglycan cross-linking → defective bacterial cell wall and lysis	Inhibits peptidoglycan cross-linking → defective bacterial cell wall and lysis
Pregnancy		
Code	A	A
Pharmacodynamics		
CNS	Convulsions and coma with massive doses, especially renal failure	
Metabolic	Hypernatraemia	
Renal	Interstitial nephritis	Interstitial nephritis
Gastrointestinal	Diarrhoea; C. difficile colitis	Nausea Diarrhoea; C. difficile colitis

	Benzylpenicillin	Phe- noxmethylpeni- cillin
Haematolog- ical	Cytopenias	Cytopenias
Considera- tions	Bactericidal; time- dependent killing Narrow Gram-pos- itive and anaerobic activity; inactivated by penicillinases Poor CSF pen- etration without meningeal inflam- mation	Bactericidal; time- dependent killing Oral treatment of mild-moderate in- fections; unsuitable for acute severe infection Narrow Gram-pos- itive and anaerobic activity; inactivated by penicillinases

Table 4: Anti-Staphylococcal Penicillins

	Flucloxacillin
Aliases	Flopen Staphlex
Class	Antistaphylococcal penicillin
Chemistry	Isoxazolyl penicillin; β -lactam
Dose	
Infection (PO)	500mg Q6H
Infection (IV)	1-2g Q6H
Pharmaceutics	
Presentation	250mg and 500mg capsules; 250mg, 500mg and 1g powder for injection
Compatibility	IV compatible with water, 0.9% NaCl, Hartmann's and 5% glucose; do not mix with aminoglycosides
Solubility	Water-soluble
Pharmacokinetics	
Absorption	Food delays and reduces oral absorption; give 30-60 minutes before meals
Distribution	Protein binding: 95%

	Flucloxacillin
Metabolism	Pathway: Minimal hepatic metabolism
Elimination	Renal: Major route; $\geq 10\%$ active metabolite, up to 50% in renal failure
Mechanism	
Target	Penicillin-binding proteins
Action	Inhibits peptidoglycan cross-linking $\rightarrow$ defective bacterial cell wall and lysis
Contraindications	Previous flucloxacillin-associated jaundice/hepatic dysfunction
Pregnancy	
Code	B1
Pharmacodynamics	
CNS	Convulsions at high concentrations
Metabolic	Paracetamol $\rightarrow$ 5-oxoprolineaemia/HAGMA; $\uparrow$ risk with sepsis, malnutrition or renal impairment High doses $\rightarrow$ $\downarrow$ K ⁺ , $\uparrow$ Na ⁺
Renal	Interstitial nephritis
Gastrointestinal	Diarrhoea; C. difficile colitis
Haematological	Cholestatic hepatitis Cytopenias
Considerations	Bactericidal; time-dependent killing Penicillinase-stable; active against susceptible staphylococci, not MRSA Probenecid $\downarrow$ tubular secretion; flucloxacillin $\downarrow$ tacrolimus concentration Age >55 or >14 days $\rightarrow$ $\uparrow$ cholestatic hepatitis

Table 5: Aminopenicillins

	Ampicillin	Amoxicillin
Aliases	Austrapen	Amoxil Amoxycillin

	Ampicillin	Amoxicillin
Class	Extended-spectrum penicillin	Extended-spectrum penicillin
Chemistry	Aminopenicillin; β -lactam	Aminopenicillin; β -lactam
Dose		
Infection (IV/IM)	250-500mg Q6H	
Meningitis/septicaemia (IV)	200mg/kg/day divided Q4-6H; max 12g/day	
Infection (PO)		250-500mg Q8H
Infection (IV)		1-2g Q6H
Pharmaceutics		
Presentation	1g powder for injection	250mg and 500mg capsules; 125mg/5mL, 250mg/5mL and 100mg/mL oral liquids; 1g powder for injection
Compatibility	Reconstitute with water for injection; stable in 0.9% NaCl or Ringer's for 24 hours	IV stable 6 hours in 0.9% NaCl, 3 hours in Hartmann's and 1 hour in glucose solutions
Solubility	Freely water-soluble	Slightly water-soluble
Pharmacokinetics		
Absorption		Rapid and well absorbed PO; acid-stable; food has little effect
Distribution	pKa: 2.5 Protein binding: 29 $\pm$ 12%	Protein binding: 17%

	Ampicillin	Amoxicillin
Metabolism	Half-life: 1 hour; up to 20 hours in anuria	Half-life: 61 minutes; 16-20 hours in anuria
	Pathway: Limited hydrolysis to penicilloic acid	Pathway: Limited hydrolysis to penicilloic acid
	Metabolites: Penicilloic acid	Metabolites: Penicilloic acid
Elimination	Renal: 66-73% within 6 hours; unchanged and penicilloic acid	Renal: 75% within 6 hours; 60% unchanged and 15% penicilloic acid
Mechanism		
Target	Penicillin-binding proteins	Penicillin-binding proteins
Action	Inhibits peptidoglycan cross-linking → defective bacterial cell wall and lysis	Inhibits peptidoglycan cross-linking → defective bacterial cell wall and lysis
Pregnancy		
Code	A	A
Pharmacodynamics		
CNS	Convulsions at high concentrations	Convulsions at high concentrations
Renal	Interstitial nephritis	Interstitial nephritis
Gastrointestinal	Nausea Diarrhoea; C. difficile colitis	Nausea Diarrhoea; C. difficile colitis
Haematological	Cytopenias	Cytopenias

	Ampicillin	Amoxicillin
Considerations	Bactericidal; time-dependent killing Broad Gram-positive and limited Gram-negative activity; susceptible to β -lactamases CSF penetration improves with meningeal inflammation Probenecid $\downarrow$ tubular secretion; allopurinol or infectious mononucleosis $\rightarrow$ $\uparrow$ rash	Bactericidal; time-dependent killing Broad Gram-positive and limited Gram-negative activity; susceptible to β -lactamases CSF penetration improves with meningeal inflammation Probenecid $\downarrow$ tubular secretion; allopurinol or infectious mononucleosis $\rightarrow$ $\uparrow$ rash Renal impairment $\rightarrow$ dose reduction

Table 6: Antipseudomonal Penicillins

	Piperacillin	Ticarcillin
Aliases	Tazocin Piperacillin/tazobactam	Timentin Ticarcillin/clavulanate
Class	Antipseudomonal penicillin	Antipseudomonal penicillin
Chemistry	Ureidopenicillin; β -lactam	Carboxypenicillin; β -lactam disodium salt
Dose		
Infection (IV)	4g/0.5g piperacillin/tazobactam Q8H	
Infection ≥ 60 kg (IV)		3g/0.1g ticarcillin/clavulanate Q4-6H
Infection < 60 kg (IV)		200-300mg/kg/day ticarcillin divided Q4-6H

	Piperacillin	Ticarcillin
Pharmaceutics		
Presentation	2g/0.25g and 4g/0.5g piperacillin/tazobactam powder for infusion	3g/0.1g ticarcillin/clavulanate powder for infusion
Compatibility	Reconstitute with water, 0.9% NaCl or 5% glucose; in-compatible with Hartmann's and sodium bicarbonate	Reconstitute with water; dilute in 0.9% NaCl, 5% glucose or Hartmann's
Solubility		Very water-soluble
Pharmacokinetics		
Distribution	Protein binding: 21%	Vd: 9L Protein binding: 45%
Metabolism	Half-life: 0.7 - 1.2 hours Pathway: No bio-transformation	Half-life: 1.1 hours Pathway: Minimal
Elimination	Renal: 69% unchanged; filtration and tubular secretion; biliary secretion	Renal: ~70% unchanged within 12 hours; filtration and tubular secretion
Mechanism		
Target	Penicillin-binding proteins β -lactamases (tazobactam)	Penicillin-binding proteins β -lactamases (clavulanate)

	Piperacillin	Ticarcillin
Action	Inhibits peptidoglycan cross-linking → defective bacterial cell wall and lysis; tazobactam inhibits susceptible β -lactamases	Inhibits peptidoglycan cross-linking → defective bacterial cell wall and lysis; clavulanate inhibits susceptible β -lactamases
Pregnancy		
Code	B1	B2
Pharmacodynamics		
CNS	Encephalopathy and convulsions, especially renal failure	Myoclonus, depressed consciousness and convulsions, especially renal failure
Metabolic	Hypokalaemia	Hypokalaemia Hypernatraemia
Renal	Interstitial nephritis AKI	Haemorrhagic cystitis
Gastrointestinal	Diarrhoea; C. difficile colitis ↑ Transaminases	Nausea Diarrhoea; C. difficile colitis ↑ Transaminases
Haematological	Impaired platelet aggregation → bleeding Cytopenias	Impaired platelet aggregation → bleeding Cytopenias

	Piperacillin	Ticarcillin
Considerations	Bactericidal; time-dependent killing Broad Gram-positive, Gram-negative, Pseudomonas and anaerobic activity; tazobactam restores activity against susceptible β -lactamases 224mg sodium/4g piperacillin vial Renal impairment → dose reduction Do not mix with aminoglycosides; administer separately Probenecid ↓ tubular secretion; piperacillin ↓ methotrexate excretion	Bactericidal; time-dependent killing Gram-negative and Pseudomonas activity; clavulanate restores activity against susceptible β -lactamases Less active than piperacillin against Pseudomonas 359mg sodium/3.1g vial Renal impairment → dose reduction Probenecid ↓ ticarcillin tubular secretion

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Glycopeptides

Non- β -lactam agents that inhibit cell wall synthesis. They are:

- Active against gram-positive anaerobes and aerobes
- Bacteriostatic against enterococci and streptococci
- Bacteriocidal against staphylococci

Table 1: Vancomycin

Vancomycin	
Aliases	Vancomycin Baxter
Class	Glycopeptide antibiotic
Chemistry	Glycosylated tricyclic heptapeptide
Dose	
Load (IV)	15-20mg/kg
C. difficile (PO)	500mg-2g/day divided TDS-QID
Pharmaceutics	
Presentation	500mg and 1g powder for IV infusion or oral solution
Compatibility	β -lactams precipitate; flush line between drugs; dilute to $\leq 5\text{mg/mL}$
Solubility	Water-soluble hydrochloride salt
Pharmacokinetics	
Absorption	Poor PO; systemic absorption may occur with colitis
Distribution	Vd: 0.3-0.69L/kg
	Protein binding: 55%

	Vancomycin
Metabolism	Half-life: 4-6 hours; 7.5 days in anuria
	Pathway: None
Elimination	Renal: 75% unchanged within 24 hours by glomerular filtration
Mechanism	
Target	D-Ala-D-Ala terminus of peptidoglycan precursors
Action	Inhibits peptidoglycan polymerisation and cross-linking → defective bacterial cell wall
Pregnancy	
Code	B2
Pharmacodynamics	
CVS	Rapid infusion → histamine release, flushing, ↓ MAP, ↑ HR
CNS	Ototoxicity
Renal	Nephrotoxicity
Gastrointestinal	Oral solution is extremely unpalatable
Haematological	Neutropenia Thrombocytopenia
Considerations	Bactericidal; time-dependent killing Gram-positive activity only; synergistic with aminoglycosides Poor CSF penetration; improves with meningeal inflammation Renal impairment → dose adjustment and therapeutic drug monitoring IV phlebitis

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Aminoglycosides

Bactericidal antimicrobials that prevent protein synthesis by irreversible binding to the 30S ribosomal subunit, preventing mRNA transcription.

- As they are large, polar molecules, they must be actively transported into the cell
 - This occurs with an oxygen dependent transporter
 Therefore they are **not effective against anaerobes**.
- Transport is inhibited by increased Ca^{2+} , Mg^{2+} , low pH, and low O_2
- Aminoglycoside killing is dependent on the peak concentration over MIC

Typically peak concentration must be 8 - 10x MIC.

- Exposure to aminoglycosides causes bacteria to down-regulate aminoglycoside uptake, and therefore increases MIC

This effect disappears after ~24 hours, and is one justification for daily dosing of aminoglycosides.

Additional justifications include:

- Allows larger single doses to be used, increasing bactericidal effect
- Aminoglycosides exhibit a post-antibiotic effect

Ongoing bactericidal activity even after concentration falls.

Table 1: Gentamicin

	Gentamicin
Aliases	Gentamicin Noridem
Class	Aminoglycoside antibiotic

	Gentamicin
Chemistry	Gentamicin sulfate; mixture of related aminoglycosides
Dose	
Infection (IV)	7mg/kg daily
Pharmaceutics	
Presentation	80mg/2mL solution for IV or IM injection
Compatibility	Dilute in 100-200mL 0.9% NaCl or glucose 5%; do not mix with other drugs
Solubility	Water-soluble
Pharmacokinetics	
Absorption	Poor PO; rapid and complete IM
Distribution	Vd: 0.2L/kg
	Protein binding: 0-3%
Metabolism	Half-life: 2-3 hours; up to 70 hours in renal failure
	Pathway: None
Elimination	Renal: Almost entirely unchanged by glomerular filtration
Mechanism	
Target	30S ribosomal subunit
Action	Irreversible binding disrupts initiation and translocation and causes mRNA misreading → defective proteins
Contraindications	Previous aminoglycoside ototoxicity or nephrotoxicity
Pregnancy	
Code	D
Justification	Fetal nephrotoxicity and eighth-nerve toxicity
Pharmacodynamics	
Respiratory	Neuromuscular blockade → apnoea
CNS	Irreversible vestibular or auditory toxicity
Musculoskeletal	Weakness

	Gentamicin
Renal Considerations	Usually reversible nephrotoxicity Bactericidal; concentration-dependent killing with post-antibiotic effect Oxygen-dependent uptake → inactive against anaerobes Peak 8–10× MIC Narrow therapeutic index → therapeutic drug monitoring Renal impairment → extend dose interval Furosemide and other ototoxic or nephrotoxic drugs ↑ toxicity Potentiates neuromuscular blockers

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Lincosamides

Inhibit protein synthesis by disrupting the 50S ribosomal subunit. May be bacteriostatic or bacteriocidal, depending on the concentration and the particular organism.

Table 1: Clindamycin

Clindamycin	
Aliases	Dalacin C
Class	Lincosamide antibiotic
Chemistry	Semisynthetic chlorinated lincomycin derivative
Dose	
Infection (IV)	600mg Q6H
Pharmaceutics	
Presentation	150mg/mL solution for IV or IM injection; 2mL and 4mL ampoules
Compatibility	Compatible with 0.9% NaCl and glucose solutions; incompatible with ampicillin, phenytoin, barbiturates, aminophylline, calcium gluconate, magnesium sulfate, ceftriaxone and ciprofloxacin
Solubility	Water-soluble phosphate ester
Pharmacokinetics	
Metabolism	Half-life: 3 hours
	Pathway: CYP3A4 and CYP3A5 oxidation
	Metabolites: Clindamycin sulfoxide, N-desmethylclindamycin
Elimination	Renal: Drug and metabolites

Clindamycin	
Mechanism	
Target	50S ribosomal subunit
Action	Inhibits bacterial protein synthesis and toxin production
Pregnancy	
Code	A
Pharmacodynamics	
Respiratory	Neuromuscular blockade
CVS	Rapid IV injection → ↓ MAP, cardiopulmonary arrest
Renal	AKI
Gastrointestinal	Nausea Diarrhoea; <i>C. difficile</i> colitis ↑ LFTs; jaundice
Haematological	Cytopenias
Considerations	Bacteriostatic or bactericidal depending on concentration and organism Gram-positive cocci and anaerobes; no Gram-negative aerobic activity Excellent bone penetration; negligible CSF penetration despite meningeal inflammation Infuse ≤30mg/min at ≤12mg/mL Potentiates nondepolarising neuromuscular blockade Erythromycin antagonism

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Metronidazole

Metronidazole interrupts cellular metabolism by **preferential reduction**, capturing electrons that would be usually transferred to other molecules. This leads to a build up of cytotoxic intermediate metabolic compounds and free radicals, that result in DNA breakage and subsequent cell death.

Table 1: Metronidazole

	Metronidazole
Aliases	Flagyl
Class	Nitroimidazole antibiotic
Chemistry	5 - nitroimidazole
Dose	
Infection (IV)	500mg Q8H
Infection (PO)	400mg Q8H
Pharmaceutics	
Presentation	500mg/100mL ready-to-use IV solution; contains 310mg sodium
Compatibility	Compatible with 0.9% NaCl and glucose solutions; incompatible with Hartmann's and Ringer's solutions
Pharmacokinetics	
Absorption	~100% bioavailability PO; oral and IV exposure equivalent
Distribution	Protein binding: Insignificant

	Metronidazole
Metabolism	Half-life: 7.3 hours Pathway: Hepatic hydroxylation, side-chain oxidation and glucuronidation Metabolites: Active hydroxymethyl metabolite
Elimination	Renal: ~75% as unchanged drug and metabolites within 5 days Faecal: Most of remainder
Mechanism	
Target	DNA
Action	Anaerobic reduction forms cytotoxic intermediates and free radicals → DNA strand breaks and cell death
Contraindications	Blood dyscrasia Active organic CNS disease Disulfiram within 2 weeks Alcohol or propylene glycol during and for 3 days after treatment Cockayne syndrome (IV)
Pregnancy	
Code	B2
Pharmacodynamics	
CNS	Peripheral neuropathy Seizures; encephalopathy
Renal	Dark urine
Gastrointestinal	Nausea; vomiting Metallic taste
Haematological	Cytopenias

	Metronidazole
Considerations	Bactericidal against anaerobes; antiprotozoal Crosses BBB and placenta Severe hepatic impairment → dose reduction Haemodialysis removes drug and active metabolite → supplementary dose Alcohol → disulfiram-like reaction ↑ warfarin effect; cimetidine ↑ and enzyme inducers ↓ metronidazole exposure

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Antifungals

Antimicrobial agents targeting eukaryotic and heterotrophic microbes. Can be divided by class into:

- Azoles
Inhibit ergosterol synthesis. Subdivided into:
 - Triazoles
 - Fluconazole
 - Itraconazole
 - Voriconazole
 - Posaconazole
 - Imidazoles
 - Ketoconazole
- Echinocandins
Inhibit glucan synthesis.
 - Caspofungin
 - Micafungin
 - Anidulafungin
- Polyenes
Disrupt cell membrane.
 - Amphotericin B
 - Nystatin

Mechanisms of Antifungal Resistance

Three broad mechanisms:

- Increased efflux
Increased expression of transport proteins removing drug from cell.

- Alteration of target enzyme
Changes to protein target prevent drug binding or inactivation.
 - Typically only requires changes in a few amino acids
- Alteration of drug metabolism
Reduced enzyme activity prevents accumulation of toxic product.

Amphotericin resistance is rare *in vivo*, and is typically via different mechanisms:

- Decreased ergosterol content
- Altered sterol:phospholipid ratio

Table 1: Comparison of Antifungals

	Fluconazole	Voriconazole	Caspofungin	Amphotericin B
Aliases	Diflucan	Vfend	Cancidas Caspofungin AN	AmBisome
Class	Triazole antifungal	Triazole antifungal	Echinocandin anti-fungal	Polyene antifungal
Chemistry	Fluorinated bis-triazole	Fluoropyrimidine triazole	Semisynthetic lipopeptide	Amphoteric hep-taene macrolide; liposomal formulation
Dose				
Severe infection (IV/PO)	400mg load, then 200-400mg daily			
Load (IV)		6mg/kg Q12H for 2 doses	70mg	
Mould infection (IV)		4mg/kg Q12H		
Candida infection (IV)		3mg/kg Q12H		
Infection (PO, ≥40kg)		200mg BD		
Infection (IV)			50mg daily	
Systemic mycosis (IV)				3-5mg/kg daily
Mucormycosis (IV)				5-10mg/kg daily
Pharmaceutics				
Presentation	100mg/50mL, 200mg/100mL and 400mg/200mL IV solutions; oral tablets, capsules and suspension	50mg and 200mg tablets; 40mg/mL oral suspension; 200mg powder for IV infusion	50mg and 70mg powder for IV infusion	50mg liposomal powder for IV infusion

	Fluconazole	Voriconazole	Caspofungin	Amphotericin B
Compatibility	IV solution compatible with 0.9% NaCl and Hartmann's; do not mix with other drugs	Do not mix with other drugs, blood products, concentrated electrolytes or sodium bicarbonate; separate line for other infusions	0.9%, 0.45% or 0.225% NaCl or Hartmann's; glucose solutions → instability; do not co-infuse with other drugs	Reconstitute with water, then dilute only in glucose 5%; saline and other drugs → precipitation
Solubility	Sparingly water-soluble	0.7mg/mL in water; IV formulation contains sulfobutyl betadex	Freely water-soluble	Poorly water-soluble; incorporated into liposomes
Pharmacokinetics				
Absorption	90% bioavailability PO; unaffected by gastric pH	96% bioavailability PO; high-fat meals ↓ absorption	IV only	Negligible PO
Distribution	Vd: Approximates total body water Protein binding: 11-12%	Vd: 4.6L/kg Protein binding: 58%	Protein binding: 97%	Vd: 18.9-49.1L

	Fluconazole	Voriconazole	Caspofungin	Amphotericin B
Metabolism	Half-life: 30 hours Pathway: Limited	Half-life: ~6 hours; dose-dependent Pathway: Hepatic CYP2C19, CYP2C9 and CYP3A4 Metabolites: Minimally active voriconazole N-oxide	Half-life: 9-11 hours; terminal phase 40-50 hours Pathway: Slow hydrolysis, N-acetylation and spontaneous degradation Metabolites: L-747969, Amino acid degradates	Half-life: 26-38.2 hours Pathway: Not established
Elimination	Renal: 80% unchanged; 11% as metabolites	Renal: <2% unchanged; ~80% as metabolites	Renal: 41% of radioactivity; 1.4% unchanged Faecal: 34% of radioactivity	Renal: Negligible urinary excretion
Mechanism				
Target	Fungal CYP51 14 α -sterol demethylase	Fungal CYP51 14 α -sterol demethylase	β -(1,3)-D-glucan synthase	Ergosterol in fungal cell membranes
Action	Inhibits ergosterol synthesis → abnormal fungal cell membrane	Inhibits ergosterol synthesis → abnormal fungal cell membrane	Inhibits β -glucan synthesis → defective fungal cell wall and osmotic instability	Forms membrane pores → leakage of intracellular contents and cell death

	Fluconazole	Voriconazole	Caspofungin	Amphotericin B
Contraindications	Cisapride, astemizole, pimozide, quinidine or erythromycin Terfenadine with $\geq 400\text{mg/day}$	Rifampicin, carbamazepine, phenobarbitone or St John's wort Sirolimus or ergot alkaloids QT-prolonging CYP3A4 substrates		
Pregnancy				
Code	D	B3	B3	B3
Justification	Spontaneous abortion and congenital anomalies; repeated $400-800\text{mg/day} \rightarrow$ characteristic birth-defect pattern			
Pharmacodynamics				
Respiratory			Histamine release $\rightarrow$ bronchospasm Histamine release $\rightarrow$ flushing, $\uparrow$ HR	Infusion reaction $\rightarrow$ dyspnoea, hypoxia Infusion reaction $\rightarrow$ $\downarrow$ or $\uparrow$ MAP, $\uparrow$ HR
CVS	QT prolongation; torsades de pointes	QT prolongation; torsades de pointes $\downarrow$ MAP; $\uparrow$ HR		
CNS	Headache Seizures	Visual disturbance Hallucinations; confusion Headache	Headache	Headache
Musculoskeletal		Long-term use $\rightarrow$ periostitis		Infusion reaction $\rightarrow$ back, chest or abdominal pain; rigors

	Fluconazole	Voriconazole	Caspofungin	Amphotericin B
Metabolic		↓ K ⁺ , Mg ²⁺ and Ca ²⁺	↓ K ⁺	↓ K ⁺ ↓ Mg ²⁺ Renal tubular acidosis
Renal		AKI		AKI
Gastrointestinal	Nausea; diarrhoea Hepatitis; hepatic failure	Nausea; diarrhoea Hepatitis; hepatic failure	Nausea; diarrhoea ↑ LFTs; hepatitis	Nausea; vomiting ↑ LFTs; cholestasis
Haematological	Leukopaenia Thrombocytopaenia	Cytopenias	Anaemia	Normocytic anaemia

	Fluconazole	Voriconazole	Caspofungin	Amphotericin B
Considerations	Fungistatic Candida, Crypto- coccus and endemic mycoses; no useful Aspergillus activity Good CSF penetra- tion independent of meningeal inflam- mation Potent CYP2C9/ 2C19 and moder- ate CYP3A4 inhibitor → numerous inter- actions Renal impairment → dose reduction; haemodialysis re- moves ~50% Monitor LFTs; severe cutaneous reactions	Broad-spectrum; Aspergillus, Candida including C. krusei, Scedosporium and Fusarium Non-linear phar- macokinetics; CYP2C19 genotype → marked exposure variability Inhibits CYP2C19, CYP2C9 and CYP3A4 → numer- ous interactions Therapeutic drug monitoring Take PO ≥1 hour before or after food CrCl <50mL/min → prefer PO because IV vehicle accumu- lates Child-Pugh A-B → halve maintenance dose Monitor LFTs, elec- trolytes and ECG; phototoxicity with long-term use	Fungicidal against Candida; fungistatic against Aspergillus Candida includ- ing azole-resistant strains and biofilms; Aspergillus No activity against Cryptococcus, Fusarium or Tri- chosporon Infuse over ~1 hour; rapid infusion ↑ his- tamine release Moderate hepatic impairment → 35mg daily after 70mg load Renal impairment and dialysis → no dose adjustment Rifampicin and other clearance inducers → consider 70mg daily ↓ tacrolimus ex- posure → monitor concentration IV phlebitis	Broad-spectrum and fungicidal; resis- tance is uncommon Liposomal formula- tion ↓ nephrotoxicity versus conventional amphotericin B Poor CSF penetra- tion Infuse over 30- 60 minutes; doses >5mg/kg over 2 hours Monitor renal func- tion, K⁺, Mg²⁺, LFTs and FBC Other nephrotoxins ↑ nephrotoxicity Corticosteroids ↑ hypokalaemia; hypokalaemia ↑ digoxin toxicity

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Insulins

Insulins are synthetic polypeptide hormones. They:

- Have a similar mechanism of action and pharmacodynamics of endogenous insulin
- One unit of insulin is defined as the amount required to make a previously healthy 2kg rabbit hypoglycaemic

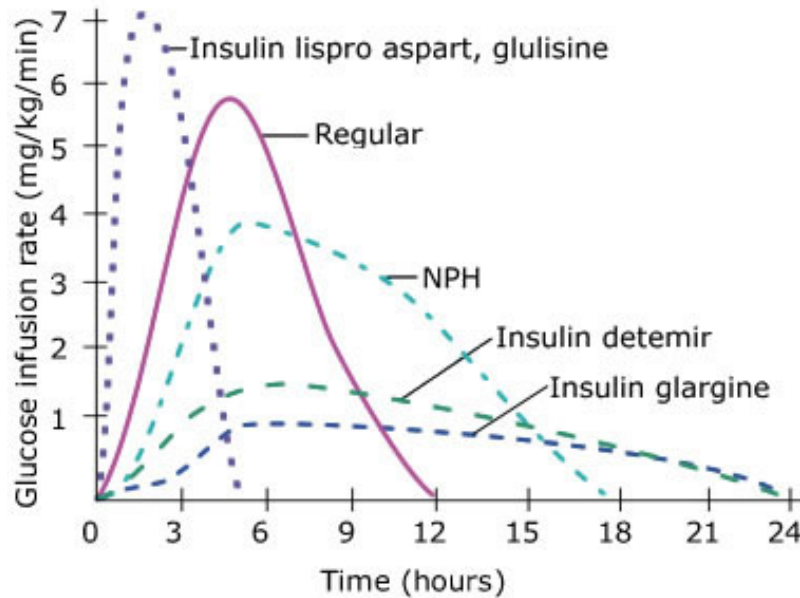
Types of Insulin

Different insulins are categorised by their time of **onset**, **peak**, and **duration**, and are classified as either:

- Fast acting
- Intermediate acting
- Long acting

Activity Profiles of Different Types of Insulin

Figure 1: Activity profiles of different types of insulin. Source: UCSF Diabetes Teaching Center.



Fast Acting

Fast acting insulins are used for controlling BSL spikes post meals, and for control of hyperglycaemia. Administered subcutaneously they have an:

- Onset of 5-15 minutes
- Peak at 1-2 hours
- Last 4-6 hours.

Fast acting insulins include:

- Insulin Aspart (Novorapid)
- Insulin Lispro (Humalog)

Intermediate Acting

Intermediate acting insulins are used for control of BSL between meals as a pseudo-basal bolus. Administered subcutaneously they have an:

- Onset of 1 - 2 hours
- Peak at 4 - 6 hours
- Last >12 hours

Intermediate acting insulins include:

- NPH
- Protophane

Long Acting

Long acting insulins are used for creating a baseline insulin level. Administered subcutaneously they have an:

- Onset of 1 - 1.5 hours
- Peak at 5 hours
- Last 24 hours

Long-acting insulins include:

- Insulin glargine (Lantus)
- Insulin detemir (Levemir)

Table 1: Exogenous Insulin Preparations

	Insulin aspart	Insulin lispro	Insulin isophane	Insulin glargine	Insulin detemir
Aliases	NovoRapid	Humalog	NPH insulin Protaphane	Optisulin Lantus	Levemir
Class	Rapid-acting insulin analogue	Rapid-acting insulin analogue	Intermediate- acting human insulin	Long-acting insulin analogue	Long-acting insulin analogue
Chemistry	Human in- sulin analogue; ProB28 → Asp	Human in- sulin analogue; LysB28 and ProB29 re- versed	Recombinant human insulin complexed with protamine and zinc	Human in- sulin analogue; AsnA21 → Gly with ArgB31 and ArgB32 added	Human in- sulin analogue; ThrB30 deleted and LysB29 myristoylated
Dose					
T1/T2DM (SC)	0.5 - 1units/kg/ day total; 50 - 70% as prandial insulin				
T1DM (SC)		~0.55units/kg/ day total; ~50% as prandial in- sulin	0.5 - 1units/kg/ day total	0.2 - 1units/kg/ day total; 40 - 50% as basal insulin	0.2 - 1units/kg/ day total; 40 - 50% as basal insulin
T2DM (SC)			0.3 - 0.6units/kg/ day initially		0.1 - 0.2units/kg daily initially
Pharmaceutics					
Presentation	Clear colour- less 100units/ mL solution in 10mL vial, 3mL cartridge or pen	Clear colour- less 100units/ mL solution in 10mL vial, 3mL cartridge or pen	Cloudy white 100units/mL suspension in 10mL vial, 3mL cartridge or pen	Clear colourless 100units/mL solution in 3mL cartridge or pen	Clear colourless 100units/mL solution in 3mL cartridge or pen

	Insulin aspart	Insulin lispro	Insulin isophane	Insulin glargine	Insulin detemir
Compatibility	NPH insulin immediately before SC injection	Humulin NPH immediately before SC injection	May mix with soluble insulin immediately before injection	Do not mix or dilute	
Solubility	Highly soluble; reduced self-association	Highly soluble; reduced self-association	Protamine-zinc suspension	Soluble at pH 4; precipitates at physiological pH	Soluble; self-associates and binds albumin
Pharmacokinetics					
Absorption	SC onset 10-20 minutes; peak 1-3 hours	SC onset 15-30 minutes; peak 0.5-2.5 hours	SC onset 1.5-4 hours; peak 6-14 hours	Slow release from SC microprecipitates; onset 1-3 hours; no peak	SC onset 50-120 minutes; no peak effect; Cmax 6-8 hours
Distribution	Vd: 0.44L/kg	Vd: 0.44L/kg	Vd: 0.44L/kg	Vd: 0.44L/kg	Vd: 0.1L/kg
	Protein binding: 0-9%	Protein binding: Minimal	Protein binding: Minimal	Protein binding: Minimal	Protein binding: >97%

	Insulin aspart	Insulin lispro	Insulin isophane	Insulin glargine	Insulin detemir
Metabolism	Half-life: 81 minutes after SC injection	Duration: 3-6.5 hours	Duration: 16-24 hours	Duration: ~24 hours	Half-life: 5-7 hours
	Duration: 3-5 hours	Pathway: Hepatic, renal and peripheral proteolysis; receptor-mediated endocytosis	Pathway: Protamine degradation; hepatic, renal and peripheral proteolysis; receptor-mediated endocytosis	Pathway: Rapid β -chain carboxyl-terminal proteolysis	Duration: Up to 24 hours; dose-dependent
	Pathway: Hepatic, renal and peripheral proteolysis; receptor-mediated endocytosis	Metabolites: Inactive peptides	Metabolites: Inactive peptides	Metabolites: M1 (active), M2 (active)	Pathway: Proteolysis similar to human insulin
	Metabolites: Inactive peptides				Metabolites: Inactive peptides
Elimination	Renal: ~1% unchanged	Renal: ~1% unchanged	Renal: ~1% unchanged; inactive metabolites	Renal: ~1% unchanged	Renal: ~1% unchanged
Mechanism					
Target	Insulin receptor tyrosine kinase	Insulin receptor tyrosine kinase	Insulin receptor tyrosine kinase	Insulin receptor tyrosine kinase	Insulin receptor tyrosine kinase

	Insulin aspart	Insulin lispro	Insulin isophane	Insulin glargine	Insulin detemir
Action	Receptor autophosphorylation → IRS/PI3K/Akt signalling → GLUT4 translocation and altered enzyme activity	Reduced self-association accelerates SC absorption; receptor activation → IRS/PI3K/Akt signalling → GLUT4 translocation and altered enzyme activity	Tissue proteases degrade protamine → delayed insulin absorption; receptor activation → IRS/PI3K/Akt signalling → GLUT4 translocation and altered enzyme activity	Neutralisation after SC injection → micro-precipitates → sustained release; receptor activation → IRS/PI3K/Akt signalling → GLUT4 translocation and altered enzyme activity	SC self-association and reversible albumin binding prolong delivery; receptor activation → IRS/PI3K/Akt signalling → GLUT4 translocation and altered enzyme activity
Contraindications	Hypoglycaemia	Hypoglycaemia	Hypoglycaemia		
Pregnancy					
Code	A	A		B3	A
Pharmacodynamics					
CVS	Positive inotropy	Positive inotropy	Positive inotropy	Positive inotropy	Positive inotropy
CNS	Severe hypoglycaemia → seizures or coma	Severe hypoglycaemia → seizures or coma	Severe hypoglycaemia → seizures or coma	Severe hypoglycaemia → seizures or coma	Severe hypoglycaemia → seizures or coma
Endocrine	↓ BGL ↓ Glucagon release Hypoglycaemia	↓ BGL ↓ Glucagon release Hypoglycaemia	↓ BGL ↓ Glucagon release Hypoglycaemia	↓ BGL ↓ Glucagon release Hypoglycaemia	↓ BGL ↓ Glucagon release Hypoglycaemia

	Insulin aspart	Insulin lispro	Insulin isophane	Insulin glargine	Insulin detemir
Musculoskeletal	↑ Glucose and amino-acid uptake ↑ Glycogen and protein synthesis	↑ Glucose and amino-acid uptake ↑ Glycogen and protein synthesis	↑ Glucose and amino-acid uptake ↑ Glycogen and protein synthesis	↑ Glucose and amino-acid uptake ↑ Glycogen and protein synthesis	↑ Glucose and amino-acid uptake ↑ Glycogen and protein synthesis
Metabolic	↓ K ⁺ ↓ Lipolysis and proteolysis ↑ Lipogenesis	↓ K ⁺ ↓ Lipolysis and proteolysis ↑ Lipogenesis	↓ K ⁺ ↓ Lipolysis and proteolysis ↑ Lipogenesis	↓ K ⁺ ↓ Lipolysis and proteolysis ↑ Lipogenesis	↓ K ⁺ ↓ Lipolysis and proteolysis ↑ Lipogenesis
Considerations	Give immediately before or soon after starting a meal IV use requires medical supervision; no advantage over soluble human insulin IV Rotate injection sites Renal or hepatic impairment → adjust dose	Give within 15 minutes before or 20 minutes after starting a meal IV use requires medical supervision U100 solution may be used in compatible SC pumps Rotate injection sites Renal or hepatic impairment → ↓ insulin requirement	Resuspend before use Once or twice daily Protamine is fish-derived Variable absorption and pronounced peak Rotate injection sites Renal or hepatic impairment → ↓ insulin requirement	SC only; IV administration may cause severe hypoglycaemia Once daily at the same time Rotate injection sites Renal or hepatic impairment → ↓ insulin requirement	SC only; do not add to infusion fluids Lower doses may require twice-daily dosing Rotate injection sites Severe hypoalbuminaemia → limited data Renal or hepatic impairment → adjust dose

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Oral Hypoglycaemics

Table 1: Oral Hypoglycaemics

	Metformin	Gliclazide	Pioglitazone	Dapagliflozin
Aliases	Diabex	Diamicron	Actos	Forxiga
Class	Biguanide	Sulfonylurea	Thiazolidinedione	SGLT2 inhibitor
Chemistry	Biguanide	Second-generation sulfonylurea	Thiazolidinedione	C-aryl glucoside
Dose				
T2DM (PO)	500mg daily or BD; titrate to max 1g TDS	30mg daily; titrate by 30mg; max 120mg/day	15-30mg daily; increase after 4 weeks; max 45mg/day	
T2DM/HF/CKD (PO)				5-10mg daily
Pharmaceutics				
Presentation	White 500mg, 850mg and 1g film-coated tablets	White scored 60mg modified-release tablet	White to off-white 15mg, 30mg and 45mg tablets	Yellow 10mg film-coated tablet
Solubility	Water-soluble; pKa 12.4	Practically water-insoluble; pKa 5.8	Practically water-insoluble	
Pharmacokinetics				

	Metformin	Gliclazide	Pioglitazone	Dapagliflozin
Absorption	Saturable and incomplete; 40-60% oral bioavailability	Completely absorbed; 97% oral bioavailability; MR plateau 6-12 hours	83% oral bioavailability; peak ~2 hours	78% oral bioavailability; peak ~2 hours
Distribution	pKa: 12.4	pKa: 5.8	Vd: 0.25L/kg	Protein binding: 91%
	Vd: 1-4L/kg	Vd: 0.2-0.4L/kg	Protein binding: >99%	
	Protein binding: Minimal	Protein binding: 85-97%		
Metabolism	Half-life: 6 hours	Half-life: 11 hours	Half-life: 5-6 hours; active metabolites	Half-life: 12.9 hours
	Pathway: Minimal metabolism	Duration: 24 hours	16-23 hours	Duration: 24 hours
		Pathway: Extensive hepatic metabolism	Duration: Slow onset; maximal effect 1-3 months	Pathway: UGT1A9 glucuronidation; minor CYP metabolism
		Metabolites: Inactive metabolites	Pathway: Hepatic CYP2C8 and CYP3A4 Metabolites: M-IV (active; major), Other active and inactive metabolites	Metabolites: Dapagliflozin 3-O-glucuronide (inactive)

	Metformin	Gliclazide	Pioglitazone	Dapagliflozin
Elimination	Renal: 90% cleared renally; unchanged drug	Renal: 4% renally eliminated; metabolites extensively excreted in urine	Renal: 45% in urine; 55% in faeces	Renal: 75% in urine, <2% unchanged; 21% in faeces, ~15% unchanged
Mechanism				
Target	Mitochondrial complex I AMPK	SUR1/SUR2 subunits of β -cell KATP channels	PPAR γ nuclear receptor	Sodium-glucose cotransporter 2
Action	$\downarrow$ Hepatocyte ATP $\rightarrow$ AMPK activation $\rightarrow$ $\downarrow$ gluconeogenesis and glycogenolysis; $\uparrow$ peripheral insulin sensitivity and $\downarrow$ intestinal glucose absorption	KATP channel closure $\rightarrow$ $\downarrow$ K $^{+}$ efflux $\rightarrow$ β -cell depolarisation $\rightarrow$ Ca $^{2+}$ influx $\rightarrow$ insulin release	PPAR γ agonism $\rightarrow$ altered transcription of insulin-responsive genes $\rightarrow$ $\uparrow$ insulin-dependent glucose disposal and $\downarrow$ hepatic glucose output	Reversible competitive SGLT2 inhibition in the proximal tubule $\rightarrow$ $\downarrow$ glucose and Na $^{+}$ reabsorption $\rightarrow$ glucosuria, osmotic diuresis and $\uparrow$ tubuloglomerular feedback

	Metformin	Gliclazide	Pioglitazone	Dapagliflozin
Contraindications	Metabolic acidosis, DKA or diabetic precoma CrCl <60mL/min Acute renal impairment, dehydration, sepsis or shock Tissue hypoxia Severe hepatic impairment Alcohol intoxication or alcoholism Elective major surgery Lactation	Type 1 DM, DKA, diabetic precoma or coma Severe renal or hepatic impairment Miconazole Pregnancy or lactation	NYHA class II-IV HF	
Pregnancy				
Code	C	C	B3	D
Justification	Strict normoglycaemia is important during pregnancy.	May enter fetal circulation and cause neonatal hypoglycaemia.		Animal studies show developing-kidney toxicity corresponding to the second and third trimesters.
Pharmacodynamics				
CVS			Fluid retention/ oedema HF exacerbation	↓ MAP, preload and afterload ↓ Volume overload
Endocrine	↓ Hepatic glucose output ↑ Insulin sensitivity No hypoglycaemia alone	↑ Insulin secretion ↑ Insulin sensitivity Hypoglycaemia Secondary treatment failure	↑ Insulin sensitivity ↓ BGL and HbA1c May restore ovulation	↓ BGL and HbA1c ↓ Insulin requirements Euglycaemic DKA

	Metformin	Gliclazide	Pioglitazone	Dapagliflozin
Musculoskeletal		Photosensitivity Erythema multiforme or exfoliative der- matitis	↑ Fractures in women	
Metabolic	Lactic acidosis	Hypokalaemia ↑ Appetite and weight	Weight gain	Weight loss
Renal				Glucosuria Osmotic diuresis Transient natriuresis ↓ Intraglomerular pressure
Genitourinary			Possible ↑ bladder cancer risk	Genital candidiasis UTI
Gastrointestinal	N/V Diarrhoea Abdominal discom- fort Anorexia Taste disturbance	N/V Abdominal pain Diarrhoea or consti- pation Cholestasis		
Haematological	↓ Vitamin B12	Rare cytopenias	↓ Hb and Hct from haemodilution	↑ Haematocrit

	Metformin	Gliclazide	Pioglitazone	Dapagliflozin
Considerations	Take in divided doses with meals Renal impairment → accumulation and lactic acidosis Iodinated contrast → cease 48h before if renal impairment, otherwise at procedure; restart after 48h if renal function normal Major surgery → cease 48h before; restart after 48h if renal function normal Illness with ↓ renal function → temporarily cease	Take MR dose with breakfast; do not crush or chew Fasting or missed meals → ↑ hypoglycaemia G6PD deficiency → haemolytic anaemia Alcohol → hypoglycaemia and disulfiram-like reaction Non-ionic albumin binding → little drug displacement	Depends on endogenous insulin; not for type 1 DM or DKA Women, HF or insulin use → start 15mg Do not use with current or previous bladder cancer Monitor fluid retention and HF after initiation or dose increase ALT >2.5× ULN → do not initiate Visual change → assess for macular oedema	Risk of euglycaemic DKA. Cease before major surgery or prolonged fasting; restart when stable and eating eGFR <25mL/min/1.73m² → do not initiate eGFR <45mL/min/1.73m² → ↓ glucose-lowering effect Volume depletion → ↓ MAP Type 1 DM → avoid (↑ DKA) Severe hepatic impairment → avoid

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Oxytocics

Oxytocics are agents which increase the force of uterine contraction.

Table 1: Oxytocics

	Oxytocin	Ergometrine	Dinoprost	Carboprost
Aliases	Syntocinon	Ergonovine	Prostin F2 α Prostaglandin F2 α	Hemabate
Class	Posterior pituitary hormone; oxytocic	Ergot alkaloid; oxy-tocic	Endogenous prostaglandin; oxy-tocic	Prostaglandin F2 α analogue; oxytocic
Chemistry	Synthetic cyclic nonapeptide	Amide ergot alkaloid; maleate salt	Prostaglandin F2 α tromethamine salt	15-methyl prostaglandin F2 α trometamol salt
Dose				
Labour	1-4mU/min IV; $\uparrow$ by 1-2mU/min at ≥ 20 min intervals; max 20mU/min			
PPH	5-10IU IM or 5IU slow IV	200microg IM; 200microg slow IV in emergency		
Caesarean	5IU slow IV after fetal delivery			
PPH (intramyometrial)			0.5-1mg; repeat if required, max 3mg total	
Refractory atonic PPH				250microg deep IM; repeat $\sim$ q90min (≥ 15 min in severe cases); max 2mg
Pharmaceutics				
Presentation	5IU/mL and 10IU/mL solutions for injection	500microg/mL solution for IM or IV injection	5mg/mL injection	250microg/mL solution for IM injection

	Oxytocin	Ergometrine	Dinoprost	Carboprost
Compatibility	Physically compatible with 0.9% NaCl/5% glucose; use electrolyte-containing diluent; incompatible with bisulphites/metabisulphites	Protect from light; refrigerate at 2-8°C	Store at 15-30°C; dilute to 0.5mg/mL with 0.9% NaCl	Refrigerate at 2-8°C; do not freeze
Solubility		Water-soluble (BP); sparingly soluble (USP)		
Pharmacokinetics				
Absorption	IV injection <1min; IM injection 2-4min	Rapid and complete after IM injection		Peak plasma concentration 15-60min after IM injection
Distribution	Vd: 0.17L/kg (healthy men, IV) Protein binding: Very low			
Metabolism	Half-life: 3-20min Duration: 30-60min IM; possibly shorter IV Pathway: Oxytocinases in plasma, liver and kidney	Duration: 45min IV; ≥3h IM Pathway: Probably predominantly hepatic	Pathway: Rapid systemic metabolism	Duration: Longer acting than dinoprost

	Oxytocin	Ergometrine	Dinoprost	Carboprost
Elimination	Renal: <1% unchanged; hepatic and renal clearance	Renal: Probably predominantly non-renal; faecal excretion	Renal: Metabolites predominantly renal	
Mechanism				
Target	Oxytocin receptors Vasopressin V2 receptors at high doses	α -adrenergic receptors 5-HT ₂ receptors	Prostaglandin FP receptors	Prostaglandin FP receptors
Action	Oxytocin-receptor Gq activation → ↑ intracellular Ca ²⁺ and myometrial contraction	Partial agonism → uterine contraction and vasoconstriction; sustained uterine contraction at high doses	Stimulates myometrium → uterine contraction	Stimulates myometrial smooth muscle → uterine contraction and placental-site haemostasis
Contraindications	Foetal distress Obstructed labour or contraindication to vaginal delivery Placenta praevia, vasa praevia or placental abruption Hypertonic uterine contractions Severe toxemia Within 6h of vaginal prostaglandins	Pre-eclampsia, eclampsia or hypertension (including history) Peripheral vascular or heart disease Hepatic or renal impairment Before placental delivery or with suspected retained placenta Severe or persistent sepsis	Active cardiac, pulmonary, renal or hepatic disease Acute pelvic inflammatory disease Severe asthma	Active cardiac, pulmonary, renal or hepatic disease Acute pelvic inflammatory disease Pregnancy
Pregnancy Code	A	C	C	D

	Oxytocin	Ergometrine	Dinoprost	Carboprost
Justification	Idiosyncratic uterine sensitivity may cause fetal anoxia	May cause premature or hypertonic labour		Uterine stimulation may cause fetal loss, irreversible injury or death
Pharmacodynamics				
Respiratory	Fluid overload may cause pulmonary oedema	Dyspnoea and pulmonary oedema	Bronchoconstriction	Bronchoconstriction and hypoxia Rare pulmonary oedema
CVS	Rapid IV bolus → ↓ MAP, flushing and reflex ↑ HR QTc prolongation and myocardial ischaemia	↑ SVR, CVP and MAP; ↓ HR Coronary vasospasm, myocardial ischaemia and dysrhythmias	Large doses → ↑ SVR and MAP	↑ SVR, MAP and HR
CNS	Headache; water intoxication may cause seizures or coma	Headache, dizziness and seizures with severe hypertension	Headache	Headache and dizziness
Endocrine	↑ Milk ejection	Repeated doses → ↓ prolactin and lactation		
Metabolic	ADH-like effect → water retention and hyponatraemia		Fever and chills	Transient fever and flushing
Renal	↓ Urine output with prolonged high-dose infusion			
Genitourinary	↑ Uterine contraction frequency and tone	↑ Uterine contraction frequency and tone; fetal distress	↑ Uterine contraction frequency and tone	↑ Uterine tone and contraction

	Oxytocin	Ergometrine	Dinoprost	Carboprost
Gastrointestinal	Nausea and vomiting	Nausea, vomiting and diarrhoea	Nausea, vomiting and diarrhoea	Diarrhoea, nausea, vomiting and abdominal cramping
Considerations	Avoid rapid IV bolus Electrolyte-free fluid ↑ water-intoxication risk Blood products may contain oxytocinase; avoid co-administration through the same line	IV only in emergency; give over ≥1min Sympathomimetics and β-blockers may ↑ vasoconstriction Prolonged use may cause ergotism and gangrene	Second-line severe atonic PPH; off-label (Queensland 2013) Aspirate before intramyometrial injection; avoid systemic injection	Do not give IV Use cautiously in asthma, glaucoma, hypertension, epilepsy or scarred uterus Potentiates other oxytocics; concomitant use not recommended

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Tocolytics

Tocolytics are agents which decrease uterine tone. Tocolytics include:

- β_2 - agonists
- Ca^{2+} - channel antagonists
- COX Inhibitors
- MgSO_4
- Nitrates
- Volatile anaesthetic agents

All tocolytics are discussed in more detail elsewhere - this covers just the mechanism of action of their uterine effects.

Table 1: Comparison of Tocolytic Drug Classes

Drug	β_2 - agonists	Ca^{2+} - channel antagonists	COX Inhibitors
Example	Salbutamol, Terbutaline	Nifedipine	Indomethacin
Mechanism of Action	Activate GPCR, $\uparrow$ cAMP, which activates protein kinase A and leads to inhibition of myosin light chain kinase and relaxation	Block L - type Ca^{2+} channels, causing relaxation	Inhibit prostaglandin synthesis, which are vital for uterine contraction

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Gastrointestinal Pharmacology

Acid Suppression

Table 1: Acid-Suppressions

	Sodium citrate	Aluminium hydroxide	Calcium carbonate	Omeprazole	Ranitidine
Aliases		Gastrogel	Caltrate Tums	Losec	Zantac
Class	Non-particulate antacid	Antacid	Antacid and calcium salt	Proton-pump inhibitor	H ₂ -receptor antagonist
Chemistry	Trisodium citrate salt	Trivalent-cation alkali	Divalent-cation carbonate	Substituted benzimidazole prodrug	Guanidine derivative
Dose					
Aspiration (PO)	30mL of 0.3mol/L immediately before anaesthesia				
Dyspepsia (PO)		640mg 5-6 times daily after meals and nocte; max 3.84g/day	1-2g; repeat hourly PRN; max 7.5g/day		

	Sodium citrate	Aluminium hydroxide	Calcium carbonate	Omeprazole	Ranitidine
Calcium (PO)			500-1800mg elemental calcium/day		
GORD/PUD (PO)				20mg daily	150mg BD or 300mg nocte
Prophylaxis (PO/IV)				40mg daily	
UGIB (IV)				40mg Q6-12H	
Prophylaxis (IV)					50mg Q8H
Pharmaceutics					
Presentation	Clear colourless 0.3mol/L oral solution; 2.647g/30mL	320mg/5mL oral suspension	500mg chewable tablet containing 200mg elemental calcium	Light pink 10mg and pink 20mg enteric-coated tablets; 40mg powder for IV injection	White 150mg round and 300mg capsule-shaped film-coated tablets
Compatibility					IV formulation compatible with NS, D5W, D10W and Hartmann's
Solubility	Highly water-soluble	Essentially water-insoluble; pKa 15.7	Poorly water-soluble	Slightly water-soluble; acid-labile; pKa 3.97	Highly water-soluble; pKa 7.8
Pharmacokinetics					
Absorption	Rapid systemic absorption	Minimal; up to 0.2% with citrate	~15% of oral calcium absorbed	Small-intestinal; bioavailability 35% initially and 60% with repeat dosing	50% oral bioavailability; peak 1-3 hours

	Sodium citrate	Aluminium hydroxide	Calcium carbonate	Omeprazole	Ranitidine
Distribution		pKa: 15.7		pKa: 3.97	pKa: 7.8
				Vd: 0.3L/kg	Vd: 1.4L/kg
				Protein binding: 97%	Protein binding: 13-25%

	Sodium citrate	Aluminium hydroxide	Calcium carbonate	Omeprazole	Ranitidine
Metabolism	Duration: Short	Duration: 1-3 hours	Duration: 2-3 hours with food	Half-life: 0.5-1 hour	Half-life: 2 hours
	Metabolites: Bicarbonate	Pathway: None	Pathway: Dissociates to calcium and carbonate	Duration: Proton-pump inhibition up to 72 hours Pathway: Hepatic CYP2C19 and CYP3A4; acid activation in parietal-cell canaliculi Metabolites: Sulfenamide (active), Hydroxyomeprazole (inactive), Omeprazole sulphone (inactive)	Duration: 4-10 hours Pathway: ~50% hepatic N-oxidation, S-oxidation and demethylation Metabolites: Ranitidine N-oxide, Ranitidine S-oxide, Desmethylranitidine

	Sodium citrate	Aluminium hydroxide	Calcium carbonate	Omeprazole	Ranitidine
Elimination	Renal: Systemically absorbed citrate renally eliminated	Renal: Absorbed aluminium renally eliminated; unabsorbed salts in faeces	Renal: Absorbed calcium renally regulated; unabsorbed calcium in faeces	Renal: 80% as metabolites; remainder in faeces	Renal: ~50% parent and metabolites; tubular secretion
Mechanism					
Target	Gastric hydrochloric acid	Gastric hydrochloric acid	Gastric hydrochloric acid	Luminal gastric H ⁺ /K ⁺ -ATPase	Gastric parietal-cell H ₂ receptor
Action	Citrate alkalinity neutralises gastric acid → ↑ gastric pH	Al(OH) ₃ + 3HCl → AlCl ₃ + 3H ₂ O	CaCO ₃ + 2HCl → CaCl ₂ + H ₂ O + CO ₂	Acid-activated sulfenamide covalently binds active proton pumps → irreversible inhibition of gastric-acid secretion	Competitive H ₂ antagonism → ↓ adenylyl cyclase/cAMP → ↓ H ⁺ /K ⁺ -ATPase recruitment and gastric-acid secretion
Contraindications	Metabolic or respiratory alkalosis Hypocalcaemia Hypochlorhydria		Hypercalcaemia Digitalis toxicity	Cilostazol	Acute porphyria
Pregnancy					
Code				B3	B1
Pharmacodynamics					
Respiratory	↓ Aspiration-pneumonitis severity	Particulate aspiration → pneumonitis	Particulate aspiration → pneumonitis	↑ Pneumonia risk	↑ Pneumonia risk

	Sodium citrate	Aluminium hydroxide	Calcium carbonate	Omeprazole	Ranitidine
CVS			Hypercalcaemia → shortened QT and arrhythmia		Rapid IV administration → ↓ HR, ↓ MAP and arrhythmia
CNS		Accumulation → encephalopathy	Severe hypercalcaemia → confusion or coma	Headache	Confusion or hallucinations
Musculoskeletal		Accumulation → osteomalacia and proximal myopathy			
Metabolic	Metabolic alkalosis	Hypophosphataemia Systemic alkalosis	Hypercalcaemia Metabolic alkalosis	Hypomagnesaemia Vitamin B12 and iron deficiency	
Renal	Urinary alkalisation	Aluminium accumulation in renal impairment	Nephrolithiasis Renal dysfunction	Interstitial nephritis	

	Sodium citrate	Aluminium hydroxide	Calcium carbonate	Omeprazole	Ranitidine
Gastrointestinal	Rapid ↑ gastric pH Unpleasant taste	↑ Gastric pH Constipation	↑ Gastric pH Belching and flatulence Constipation Rebound acid secretion	↑ Gastric pH and ↓ secretion volume N/V, abdominal pain and diarrhoea ↑ C. difficile infection Rebound acid hypersecretion after withdrawal	↓ Gastric acid and pepsin secretion ↓ Gastric alcohol dehydrogenase N/V, abdominal pain and altered bowel habit
Haematological Considerations	30mL contains 27mmol Na ⁺ HF, oedema, renal impairment or hypertension → sodium load Citrate ↑ intestinal aluminium absorption Alters absorption and renal clearance of pH-dependent drugs	Separate interacting oral drugs by ≥2 hours Renal impairment → aluminium toxicity Citrate ↑ aluminium absorption Shake suspension	Separate tetracyclines, quinolones and levothyroxine Excess calcium plus alkali → milk-alkali syndrome Thiazides → ↑ hypercalcaemia risk Chew antacid tablets	Swallow or disperse enteric-coated pellets; do not crush or chew CYP2C19 variability and interactions ↓ Clopidogrel active-metabolite exposure Consider withholding during high-dose methotrexate	Rare cytopenias Renal impairment → reduce dose Exclude gastric malignancy before treating ulcer Raised gastric pH → altered drug absorption Sucralfate ≥2g ↓ ranitidine absorption; separate by 2 hours

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Antiemetics

Antiemetic drugs can be classified by their mechanism of action:

- Serotonin antagonists
 - Ondansetron
- Corticosteroids
 - Dexamethasone
 - Has additional effects on postsurgical pain and fatigue.
- Dopamine antagonists
 - Phenothiazines
 - Chlorpromazine
 - Prochlorperazine
 - Butyrophenones
 - Droperidol
 - Benzamides
 - Metoclopramide
- Anticholinergics
 - Hyoscine
 - Atropine
- Antihistamines
 - Cyclizine
- NK₁ antagonists
 - Aprepitant
- Others
 - Benzodiazepines
 - Cannabinoids
 - Propofol

Table 1: Comparison of Antiemetics

	Ondansetron	Droperidol	Metoclopramide	Cyclizine
Aliases	Zofran	Droleptan	Maxolon Metamide	Valoid Nausicalm
Class	5-HT ₃ antagonist	Butyrophenone dopamine antag- onist	Benzamide dopamine antag- onist	First-generation H ₁ antagonist
Chemistry	Carbazole; pKa 7.4	Butyrophenone; pKa 7.46	Substituted benza- mide; pKa 9.27	Piperazine deriva- tive; pKa 8.5
Dose				
PONV (IV/IM)	4mg at induction; established PONV 4mg, may ↑ to 8mg 4-8mg Q6H			
Nausea (PO/IV)				
Induction (IV)		0.25mg/kg		
Maintenance (IV)		1.25-2.5mg		
Regional (IV/IM)		2.5-5mg		
Nausea (PO)			10mg Q6-8H	25-50mg Q8H
Nausea (IV/IM)			10-20mg TDS; max 0.5mg/kg/day or 30mg/day for 5 days	
Nausea (IV)				25-50mg up to TDS
Nausea (IM)				50mg up to TDS
Paediatric (PO/IV)				1mg/kg Q8H
PONV (IV)				50mg 20 min- utes before end of surgery
Pharmaceutics				

	Ondansetron	Droperidol	Metoclopramide	Cyclizine
Presentation	4mg and 8mg tablets or wafers; 0.8mg/mL oral liquid; clear colourless 2mg/mL solution	Clear 2.5mg/mL solution in brown ampoules	10mg tablet; clear colourless 5mg/mL solution	50mg tablet; clear colourless light-sensitive 50mg/mL solution
Compatibility	Compatible with 0.9% NaCl, 5% glucose, Hartmann's and 10% mannitol	Compatible with 0.9% NaCl, 5% glucose and lactated Ringer's; incompatible with thiopentone and methohexital	Compatible with 0.9% NaCl, 5% glucose and Hartmann's	Do not mix with other drugs
Solubility	Sparingly water-soluble	Practically water-insoluble; light-sensitive	Highly water-soluble	Water-soluble
Pharmacokinetics				
Absorption	Oral bioavailability ~60%; T _{max} ~1.5 hours	Oral bioavailability ~75%; rapid after IM injection; onset 3-10 minutes	Oral bioavailability ~80%; IV onset 1-3 minutes; IM onset 10-15 minutes	Oral bioavailability 50-80%; T _{max} ~2 hours
Distribution	pKa: 7.4	pKa: 7.46	pKa: 9.27	pKa: 8.5
	Vd: 2.5L/kg	Vd: 1.5L/kg	Vd: 3.5L/kg	Vd: 16-20L/kg
	Protein binding: 70-76%	Protein binding: 85-90%	Protein binding: 13-22%	Protein binding: 59-76%

	Ondansetron	Droperidol	Metoclopramide	Cyclizine
Metabolism	Half-life: 3.8 hours Duration: 4-8 hours Pathway: Hepatic CYP1A2, CYP2D6 and CYP3A4; hydroxylation then glucuronide/sulphate conjugation Metabolites: Inactive conjugates	Half-life: 2 hours Duration: Sedation 2-4 hours; altered consciousness up to 12 hours Pathway: Extensive hepatic CYP3A4 metabolism Metabolites: Inactive metabolites	Half-life: 4-6 hours Duration: 1-2 hours Pathway: Hepatic CYP2D6 and glucuronide/sulphate conjugation Metabolites: Glucuronide and sulphate conjugates	Half-life: 20 hours Duration: 4-6 hours Pathway: Extensive hepatic CYP2D6 metabolism Metabolites: Norcyclizine (weak H1 activity)
Elimination	Renal: ~5% unchanged; predominantly metabolites	Renal: ~75% of radiolabel recovered in urine; <1% unchanged drug	Renal: 20-50% unchanged; conjugates in urine	Renal: <1% within 24 hours
Mechanism				
Target	Central and peripheral 5-HT ₃ ligand-gated cation channels	Central D ₂ receptors α ₁ receptors Muscarinic receptors	Central and peripheral D ₂ receptors Peripheral muscarinic receptors	Central H ₁ receptors Muscarinic receptors
Action	Competitive antagonism on vagal afferents and in the area postrema → ↓ emetic signalling	D ₂ antagonism → antiemesis (CTZ) and sedation; α ₁ and muscarinic antagonism → autonomic effects	D ₂ antagonism → antiemesis; muscarinic agonism → ↑ upper-GI motility	H ₁ and muscarinic antagonism in vestibular and brainstem emetic pathways

	Ondansetron	Droperidol	Metoclopramide	Cyclizine
Contraindications	Apomorphine	QT prolongation or congenital long-QT syndrome Hypokalaemia or hypomagnesaemia Clinically significant bradycardia Coma Parkinson's disease Pheochromocytoma	GI haemorrhage, obstruction or perforation Pheochromocytoma Porphyria Epilepsy Drugs causing extrapyramidal reactions	Acute alcohol intoxication
Pregnancy Code Justification	B1	C High doses in late pregnancy → prolonged neonatal neurological disturbance	A	B3
Pharmacodynamics Respiratory			Acute dystonia → upper-airway obstruction	↓ Airway secretions
CVS	QT prolongation Rapid IV administration → bradycardia Arrhythmias	QT prolongation → torsades de pointes α1 antagonism → ↓ MAP	↑/↓ HR ↑/↓ BP	↑ HR Severe heart failure → ↑ MAP and ↓ CO
CNS	Headache Rapid IV administration → dizziness or transient visual disturbance	Sedation Extrapyramidal effects Neuroleptic malignant syndrome	Acute dystonia Akathisia Tardive dyskinesia Neuroleptic malignant syndrome Sedation	Sedation Delirium Hallucinations

	Ondansetron	Droperidol	Metoclopramide	Cyclizine
Endocrine		↑ Prolactin	↑ Prolactin → galactorrhoea	
Musculoskeletal				IV use → paralysis reported
Genitourinary				Urinary retention
Gastrointestinal	Antiemetic Constipation	Antiemetic Constipation	Antiemetic ↑ Gastric emptying ↑ Lower oesophageal sphincter tone	Antiemetic ↑ Lower oesophageal sphincter tone Constipation
Haematological			Methaemoglobinemia	
Considerations	Avoid in 1st trimester Avoid in congenital long-QT syndrome Correct hypokalaemia and hypomagnesaemia Moderate/severe hepatic impairment → max 8mg/day Serotonergic drugs → serotonin syndrome Does not treat motion sickness or dopamine-agonist emesis	Adult anaesthetic adjunct doses; give regional IV doses slowly ECG before use; monitor for ≥7 hours after the procedure Correct electrolyte abnormalities before use Potentiates opioids and other CNS depressants CYP1A2/3A4 inhibitors → prolonged effect	Young patients and higher doses → ↑ extrapyramidal side effects Renal impairment → ↓ clearance and t _{1/2} up to 15 hours Antimuscarinics and opioids antagonise prokinetic effect	Pregnancy → use not advised Glaucoma, urinary retention, GI obstruction or prostatic hypertrophy Severe heart failure or acute MI Porphyria Additive CNS depression and antimuscarinic effects Abuse for euphoric/hallucinatory effects

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Intravenous Contrast

Intravenous contrast may be divided into:

- **X-ray Contrast**

These agents are all based on a tri-iodinated benzene ring, which absorbs x-ray radiation. Alterations to this ring alter toxicity, lipophilicity, and elimination.

- Agents are classified by these structural differences into:

- **Ionic**

Ionic substances are strong acids and are water soluble due to ionisation. They are further divided into:

- **Monomers**

Typically high molecular weight.

- **Dimers**

- **Non-Ionic**

Water soluble due to hydrophilic side chains. Lower molecular weight than ionic contrast agents.

- **Monomer**

Agent of choice for angiography.

- Easy to inject

- Water soluble at physiologic pH

- **Dimer**

Harder to inject than monomers due to higher viscosity. Typically used for urography.

- All are renally eliminated, and may be retained in renal dysfunction

- **Gadolinium Contrast**

Gd^{3+} , due to its seven unpaired electrons, is paramagnetic and will alter the magnetic field of an MRI machine.

- Free gadolinium is nephrotoxic and must be chelated
- This increases its solubility and allows it to be renally eliminated
- Gadolinium also attenuates x-rays, but is not used as x-ray contrast as doses required would be toxic

Adverse Reactions

Adverse reactions to low-osmolarity agents are uncommon (3%), with severe reactions being very rare (0.04%) and fatal reactions being extremely rare (1:170,000).

General Adverse Reactions

Adverse reactions include:

- Chemotoxicity
 - Platelet inhibition
 - Increased vagal tone
 - Negative inotropy
 - Negative chronotropy
- Ionic toxicity
 - Cellular membrane dysfunction
 - May worsen myasthenia gravis.
- Osmotoxicity
 - Pain
 - Emesis
 - Increased PAP
 - Decreased PVR
- Hypersensitivity reaction
 - Typically occur within 20 minutes of injection.

Risk factors include:

- Asthma or atopy
- Critically ill
- Cardiac disease

- Renal disease

Contrast Nephropathy

Defined as an increase in creatinine by 25% above baseline within three days of IV contrast administration.

- It is theorised that osmotic stress and direct tubular toxic effects lead to renal tubular injury, and may cause acute tubular necrosis
 - Typically is benign, with creatinine returning to baseline within 10-14 days
 - Significant uncertainty remains about whether contrast media cause acute kidney injury. If this risk is present, it is probably only relevant in patients who have:
 - Impaired renal function
 - Arterial contrast
 - Rehydration and volume correction are effective in preventing a rise in creatinine
-

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Definitions

This appendix is a list of key definitions that are common to many topics.

A

- **Absolute Humidity**

Mass of water vapour in a given volume of air. Measured in mg.L^{-1} .

- **Absorption**

The rate at which a drug leaves its site of administration and the extent to which this occurs.

- **Accuracy**

The ability of a measuring device to match the actual value of the quantity being measured.

- **Acid**

A proton donor.

- **Acidaemia**

Arterial blood $\text{pH} < 7.35$.

- **Acidosis**

A process which leads to an excess of hydrogen ions, and *may* lead to acidaemia if there is inadequate compensation. Can be subdivided into:

- **Respiratory acidosis:** $\text{PaCO}_2 > 45$
- **Metabolic acidosis:** $\text{HCO}_3^- < 22$

- **Activity**

The *effective* concentration of a substance in a reacting system.

- **Acute Pain**

Defined as pain of:

- Recent onset

- Limited probable duration
- Identifiable causal and temporal relationship to injury or disease
- **Adiabatic**
A process that occurs without transfer of heat or matter. For example, gases heat up when compressed (greater than the energy used to compress them), and cool when allowed to expand (adiabatic cooling).
- **Affinity**
Ability of a drug to bind to a receptor.
- **Afterload**
Ventricular wall stress at the onset of systole.
- **Aging**
Naturally occurring, physiological decline in the structure and functional reserve of all organ systems.
- **Agonist**
Drug which produces a maximal response at receptor site.
- **Alkalaemia**
Arterial blood pH > 7.45.
- **Alkalosis**
A process which leads to a deficit of hydrogen ions, and may lead to alkalaemia if there is inadequate compensation. Can be subdivided into:
 - **Respiratory alkalosis:** $\text{PaCO}_2 < 35$
 - **Metabolic alkalosis:** $\text{HCO}_3^- > 26$
- **Allodynia**
Pain caused by a previously non-painful stimulus.
- **Allosteric Modulator**
Substance which binds a receptor distant to the ligand-binding site, and modifies (positively or negatively) the effect of the ligand. Has no activity in absence of a ligand.
- **Anaesthesia**
Without sensation.
- **Analogue Signal**
Where the output of the transducer varies with the input signal.
- **Anion**
Negatively charged ion.
- **Anode**

The electrode which conventional current flows into.

- **Anrep effect**

Method of myocardial autoregulation in which an increase in afterload causes an increase in contractility.

- **Antagonist**

Drug which produces no response at the receptor, but prevents other ligands binding.

- **Autoregulation**

Ability of an organ to maintain homeostasis in the presence of dynamic physiological conditions.

- **Azeotrope**

A mixture of two substances that cannot be separated by fractional distillation, as each component shares same boiling point. This is typically temperature dependent.

B

- **Base**

Proton acceptor.

- **Base Excess**

Amount of acid that must be added to a solution to lower its pH to 7.4, at 37°C and with a PaCO₂ of 40mmHg.

- **Bathmotropy**

Degree of myocardial excitability. Used with either positive or negative bathmotropy.

- **Bell - Magendie Law**

The principle that in the spinal cord the dorsal roots are sensory and the ventral roots are motor.

- **Bias**

The systematic distortion of the estimated intervention effect away from the "truth", caused by inadequacies in the design, conduct, or analysis of a trial.

- **Black-body radiation**

Electromagnetic radiation given off by all bodies at greater than 0°K. Wavelength of radiation emitted depends on the temperature of the body.

- **Bohr Effect**

An increase in $[H^+]$ or $PaCO_2$ decreases Hb affinity for O_2 .

- **Boiling Point**

The temperature at which the vapour pressure of a liquid equals the environmental pressure surrounding the liquid.

- Therefore boiling point decreases as environmental pressure falls, as there is less external pressure keeping molecules in their liquid state
- Boiling differs from evaporation as molecules anywhere in the liquid may enter the gaseous phase, whilst evaporation occurs only at the surface

- **Bowditch Effect**

Increase in contractility seen with an increase in HR. Also known as the Treppe effect.

- **Boyle's Law**

Pressure of a gas is inversely proportional to volume.

- **Buffer**

Solution containing a weak acid and its conjugate base and will resist a change in pH when a stronger acid or base is added.

C

- **Calibration**

A process of checking a monitoring device for linearity of correlation between actual and measured values over a given measurement range.

- **Capacitance**

Ability of a system to store electrical charge. Measured in Farads.

- **Central Blood Volume**

Volume of blood in heart and lungs.

- **Central Sensitisation**

Increased responsiveness of nociceptive neurons in the central nervous system (i.e., post-synaptic) to their normal or subthreshold afferent input.

- **Chemotaxis**

Movement of cells along a gradient of increasing concentration of an attracting molecule.

- **Chronic Pain**

Pain that:

- Persists beyond the time of tissue healing
- Frequently has no clearly identifiable cause

- **Clearance**

Volume of plasma completely cleared of a substance per unit time.

- **Coronary Blood Flow**

At rest is ~5% of CO, or 225 ml.min⁻¹, and may increase 3-4x during exercise.

- **Colloid**

Substance evenly dispersed throughout another solution in which it is insoluble.

- **Colligative Properties**

The properties of a solution that depend on the ratio of solute to solvent, and not on the type of molecules present. These include:

- Vapour pressure
- Boiling point
- Freezing point
- Osmotic pressure

- **Compliance**

Distensibility of a system. Expressed as the change in volume for a given change in pressure.

- **Concentration Effect**

Describes the disproportionately rapid rise in Fi/FA ratio of nitrous oxide, as its rapid diffusion across the alveolar membrane increases the concentration of alveolar gas, and also augments respiration by drawing in dead space gas.

- **Context - Sensitive Half - Time**

Time taken for plasma drug concentration to fall to 50% of its starting value after cessation of a drug infusion aimed to maintain a constant plasma concentration. Varies with the context, or duration, of drug infusion.

- **Contractility**

Factors affecting myocardial performance, indepen-

dent of preload and afterload.

- **Critical Length**

The length of axon which must be blocked in order to prevent action potential transmission. It is dependent on myelination and fibre diameter.

- **Critical Point**

The point on a phase diagram where the liquid and gas phases of a substance have the same density, and are therefore indistinguishable.

- This point is where a substance is at both its critical temperature and critical pressure

- **Critical Pressure**

Pressure required to liquefy a vapour at its critical temperature.

- **Critical Temperature**

Temperature above which a substance cannot be liquified, irrespective of how much pressure is applied.

- **Critical Volume**

The volume occupied by a given amount of substance at its critical point.

D

- **Dalton**

Unit of mass equal to $1/12^{\text{th}}$ of the mass of Carbon-12.

- **Dalton's Law**

The partial pressure of a gas in a mixture is equal to the pressure that gas would exert if it occupied the volume alone.

- **Dead Space**

Inspired gas not participating in gas exchange. Includes:

- Apparatus dead space
Gas in the ventilator or breathing circuit.

- Anatomical dead space
Gas in the conducting zone of the lung.
- Alveolar dead space
Alveolar gas not participating in gas exchange.
Also known as West Zone 1.
- Physiological dead space
Sum of alveolar and anatomical dead space.
- **Density**
Mass per unit of volume.
- **Dependence**
When a characteristic withdrawal syndrome occurs when a drug is withdrawn, or an antagonist administered.
- **Diffusion**
Passive movement of a substance down an activity gradient by Brownian motion.
- **Diffusion Hypoxia**
Fall in alveolar PAO_2 due to dilution of alveolar gas by N_2O diffusing from blood to alveoli.
- **Digestion**
Process of breaking down macromolecules into readily absorbed compounds.
- **Doppler Effect**
Alteration in frequency of a signal due to a relative difference in velocity between the emitter and observer. Detected frequencies will be:
 - Higher if the emitter is moving toward the observer
 - Lower if the emitter is moving away from the observer
- **Down regulation**
Decrease in receptor number due to chronic agonist exposure.
- **Drift**
A fixed deviation from the true value at all points in the measured range.
- **Drug**
Substance administered to cause a change in a physiological system.
- **Duplicate Publication**
Where the same set of results are published in multiple

journals. Academically unethical, and will cause a systematic bias in a meta-analyses as the same set of patients are included twice.

- **Dyne**
Force required to accelerate 1g by $1\text{cm}\cdot\text{sec}^{-2}$.

E

- **Efficacy**
Maximal effect produced by a drug. Analogous to intrinsic activity.
- **Electrocardiogram**
Graphical recording of the vector sum of cardiac electrical activity, as measured by electrodes on the skin.
- **Emulsion**
A fine dispersion of minute droplets of one liquid in another in which it is not soluble or miscible.
- **Enzyme**
Biological catalyst.
- **Eutectic**
A mixture of substances with the lowest possible melting point than any other mixture of the same substances (and lower than that of either substance).
- **Excitability**
How rapidly an excitable cell depolarises. Given by the gradient of phase 0 of the action potential, and is dependent on the function of voltage-gated sodium channels.
- **Exponential Function**
Mathematical function where the rate of change is proportional to the current value.
- **External Validity**
How well findings from one setting can be applied to another.

F

- **Fahraeus-Lindqvist effect**
Decrease in apparent viscosity that occurs when a suspension (e.g. blood) flows through a tube of smaller diameter.
- **Fasting**
Metabolic state achieved after complete digestion and absorption of a meal prior to the onset of starvation.
- **Fick Principle**
Blood flow to an organ equals the uptake of a tracer substance by that organ, divided by the arterio-venous concentration difference.
- **Flow**
Quantity of fluid passing a point per unit time.
- **Fourier Analysis**
Deconstruction of a complex waveform by separating it into its constituent sine waves. The slowest component is known as the **fundamental frequency**.
- **Free radical**
Extremely reactive molecular constituent carrying an unpaired electron.
- **Freezing point**
Temperature at which molecular movement begins.
- **Functional Residual Capacity**
Volume of gas in the lungs at the end of a normal tidal expiration, when the recoil pressure of the lungs equals the expansile pressure of the chest wall.

G

- **Galvanometer**
Device to measure electrical current, usually via deflection of a wire in a magnetic field.
- **Gas**
Substance above its critical temperature.

- **General anaesthesia**
Drug induced, controlled, and reversible production of unconsciousness.
- **Gibbs - Donnan Effect**
Describes the tendency of diffusible ions to distribute themselves such that the ratios of the concentrations are equal when they are in the presence of non-diffusible ions.
- **Grahams Law**
The speed of diffusion of a gas through a membrane is inversely proportional to the square root of the molecular weight.

H

- **Haldane effect**
Deoxygenated blood forms carbamino compounds and buffers H^+ better than oxygenated blood.
- **Half - Life**
Time taken for drug concentration (typically in plasma) to fall by half.
- **Heat**
Describes the transfer of thermal energy from one body to another body; measured in Joules.
- **Henry's Law**
Amount of gas dissolved in a substance is directly proportional to the partial pressure of gas at the gas-liquid interface.
- **Heterometric autoregulation**
Change in ventricular function based on myocardial fibre length. Also known as **Starling's Law**.
- **Homeometric autoregulation**
Mechanisms which alter myocardial performance independent of fibre length.
- **Hormone**
Chemical messenger secreted by a **ductless gland** and has action on a distant target cell.

- **Hyperalgesia**
Greater than normal amount of pain from a noxious stimulus. May be:
 - Primary
Occurring in the region of tissue damage, e.g. in an inflamed area around a wound.
 - Secondary
Extending beyond the region of tissue damage.
- **Hypoxaemia**
When PaO_2 is less than 60mmHg.
- **Hypoxia**
The point at which inadequate oxygenation of tissues results in anaerobic metabolism.
- **Hysteresis**
When the future state of a system depends not only on its current state, but on the states preceding it.

I

- **Ideal Gas**
A gas which will obey the ideal gas law. An ideal gas must have:
 - Negligible intermolecular attraction
 - A small molecular volume compared to the space between the molecules
- **Idiosyncrasy**
An effect of a drug affecting only a small number of patients, typically due to the action of a particular metabolite.
- **Inductance**
Property of a conductor by which a change in current induces an electromotive force in the conductor and any nearby conductors.
- **Inotrope**
Drug which alters myocardial contractility.
- **Intrinsic Activity**
Maximal effect produced by a drug. Analogous to efficacy.

- **Impedance**
Resistance to alternating current.
- **Internal Validity**
Where a causal relationship between variables has been properly demonstrated, i.e. a lack of bias.
- **Irritability**
How easily an excitable cell can be stimulated. Given by how close the resting membrane potential is to threshold potential.
- **Isomer**
Compound with the same chemical formula, but different chemical structure or arrangement of atoms.
- **Isotherm**
Line of constant temperature drawn on a pressure-volume graph for a gas, which describes the relationship between pressure, temperature, and volume for a particular gas.

J

- **Joule**
Energy transferred to an object when it is acted on by 1N for 1m.

L

- **Laminar Flow**
Flow occurring smoothly and without turbulence.
- **Local Anaesthetic**
Drug which reversibly prevents the conduction of the nerve impulse in the region to which it is applied, without affecting consciousness.

M

- **MAC**
The minimal alveolar concentration (measured in % of 1 atm) at steady state which prevents a movement response to a standard surgical stimulus (midline incision) in 50% of a population.
- **Manometer**
Device which measures gas pressure.
- **Mean Systemic Filling Pressure**
The pressure measured anywhere in the systemic circulation when all flow of blood is stopped.
- **Mixed Venous Blood**
Blood from the IVC, SVC and coronary sinus, which has been mixed by the pumping action of the RV and is typically sampled from the pulmonary artery.
- **Mole**
Amount of a substance which contains as many representative particles as there are atoms in 12g of carbon-12.
- **Molality**
Number of moles of solute per kg of solvent.
- **Molarity**
Number of moles of solute per L of solvent. Varies with:
 - Temperature
 - Solvent density
 - Solute volume

N

- **Natural Frequency**
Frequency at which a system will oscillate at if disturbed and left alone.
- **Nausea**
Unpleasant subjective sensation associated with urge to vomit.

- **Neuropathic Pain**
Pain caused by a lesion or disease of the somatosensory nervous system.
- **Nociception**
Neural process of encoding a noxious stimulus.

O

- **Odds Ratio**
Estimate of risk, where the OR is the ratio of odds of an outcome in those treated vs. those not treated. OR = 1 suggests no effect, ≤ 1 suggests reduced risk > 1 suggests increased risk.
- **Ohm**
Resistance which will allow one ampere of current to flow per volt of potential difference.
- **Opiate**
Naturally occurring substance with morphine-like properties.
- **Oncotic Pressure**
Proportion of osmotic pressure due to colloid.
- **Opioid**
Describes any substance with activity at opioid receptors, and which can be reversed by naloxone.
- **Osmosis**
Movement of a solvent across a semipermeable membrane to an area of greater solute concentration.
- **Osmotic Pressure**
Pressure that must be applied to a solution to prevent the movement of a solvent from entering a solution with higher osmolality.
- **Oxygen Flux**
Volume of oxygen delivered to the tissues per minute.

P

- **p50**
The partial pressure at which an oxygen-carrying protein is 50% saturated.
- **Pain**
Unpleasant sensory and emotional experience associated with actual or potential tissue damage, or expressed in terms of such damage.
- **Partition Coefficient**
Describe the relative affinity of an agent for two phases. It is defined as the ratio of the concentration of agent in each phase, when both phases are of equal volume and the partial pressures are in equilibrium at STP.
- **Pasteur Point**
 PO_2 at which oxidative phosphorylation ceases.
- **PEEP**
Supra-atmospheric airway pressure at the end of expiration.
- **pH**
The power of hydrogen. Describes the activity of hydrogen ions in a solution, and is expressed as $pH = -\log_{10}[H^+]$.
- **Preload**
Load imposed on a muscle before contraction, and measured as the average myocardial fibre length at the onset of systole. May be approximated clinically using EDV.
- **Precision**
The ability of a measurement device to provide reproducible results upon repeated measurement.
- **Pseudo-critical temperature**
Temperature at which a gas mixture will separate into its constituent components.

R

- **Radiation**
Transfer of energy via electromagnetic radiation.
- **Receptor**
Component of a cell which binds to a ligand and results in a change in function.
- **Reduction**
Reaction which results in a gain of an electron.
- **Reflex**
Unconscious, predictable response to a stimulus.
- **Regurgitation**
Passive passage of gastric contents into the mouth.
- **Relative Humidity**
Ratio of mass of water vapour in a given volume of air, to the mass required to saturate that volume at that temperature. Expressed as a percentage.
- **Respiratory Exchange Ratio**
Ratio of CO₂ produced to O₂ consumed at any given point.
- **Respiratory Quotient**
Ratio of CO₂ produced to O₂ consumed at steady-state.
- **Reynolds Number**
Dimensionless index which predicts the likelihood of turbulent flow.

S

- **Saturated Vapour**
Vapour which is in equilibrium with its own liquid state, i.e. there are as many molecules entering the vapour phase as there are those condensing into the liquid phase.
 - A saturated vapour contains the least amount of energy possible without condensing
- **Saturated Vapour Pressure**
Pressure exerted by a vapour which is in equilibrium

with its liquid state. Increases with temperature, since as the kinetic energy (heat) content of molecules increase, more of them enter the vapour phase.

- **Second Gas Effect**

Disproportionately rapid rise in FA/Fi ratio seen when an anaesthetic agent is co-administered with nitrous oxide.

- **Seebeck effect**

The generation of a potential difference at the junction of two dissimilar metals, with its value dependent on the temperature of the junction.

- **Shivering**

Involuntary, oscillatory, muscular activity that augments metabolic heat production.

- **Shunt**

Blood entering the left side of the circulation without being oxygenated via passage through the lungs.

- **Specific Gravity**

Density of a liquid, in mass per unit volume.

- **Specific Heat Capacity**

Amount of heat energy required to raise the temperature of 1kg of a substance by 1°K without a change in state.

- **Standard Base Excess**

The base excess calculated for an Hb of 5g.L⁻¹, and which gives a better representation of ECF pH.

- **Surface Tension**

Describes the tendency of a fluid to minimise its surface area.

- **Suspension**

Particles of any phase dispersed in a liquid.

- **Synergism**

When two drugs interact to produce a greater effect than would be expected.

T

- **Temperature**

Average kinetic energy of molecules within a material,

measured in degrees.

- **Thirst**
Conscious sensation of the physiological urge to drink.
- **Time constant**
Time it would take for an exponential function to complete if the initial rate of change continued. A process is:
 - 63% complete at 1T
 - 86.5% complete at 2T
 - 95% complete at 3T
- **Tonicity**
Effective osmolality of a solution. Given by the osmolality, minus the concentration of freely diffusable osmoles (in plasma, these are urea and glucose).
- **Tonometer**
Device which measures pressure of liquid.
- **Transducer**
Device which changes a signal from one energy form to another.
- **Treppe Effect**
Increase in contractility with an increase in HR. Also known as the Bowditch effect.
- **Turbulent Flow**
Irregular movement in radial, axial, and circumferential axes.

V

- **Valsalva Manoeuvre**
Forced expiration against a closed glottis.
- **Vapour**
Substance in a gaseous phase below its critical temperature.
- **Vapour pressure**
Pressure exerted by a vapour.
- **Venous admixture**
Amount of mixed venous blood that must be added to pulmonary end-capillary blood to give the observed arterial oxygen **content**.

- **Viscosity**
Describes the tendency of a fluid to resist flow.
 - **Volt**
Potential difference which dissipates 1W of energy per 1A of current.
 - **Volume of Distribution**
Apparent volume into which a drug is distributed to produce the identified plasma concentration.
-

Key Graphs

Graphs:

- Help you to convey knowledge and understanding efficiently in the written
- Are often a feature of the viva as they allow examiners to assess depth of understanding
 - You will be asked to demonstrate how they change under different physiological states

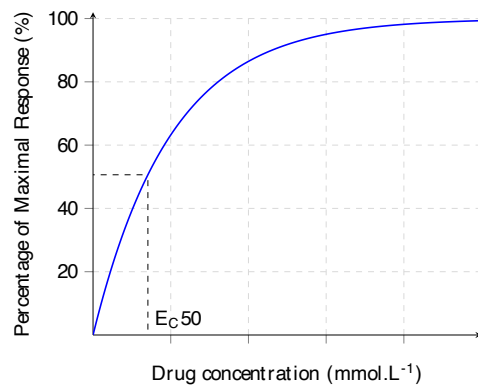
It is easy to get distracted by the curve, and forget the basics (especially in the written). To avoid this, approach them in the same way each time:

- **Axis**
First draw the axis.
 - If the axis is continuous (e.g. PaO_2), ensure you place an arrow at the far end
 - If the axis ends at a fixed point (e.g. SpO_2), ensure you place a bar at the end to signify it does not continue indefinitely
- **Labels**
Label each axis with what it is representing.
- **Units**
Give each label **appropriate units**.
 - In the viva, you can just say this out loud as you're drawing the axes
- **Curve**
Draw the curve.
- **Special Points**
Identify the key points of the curve and label these points. These include:
 - Intercepts
 - Inflection points

- Important values
e.g. The mixed venous point.

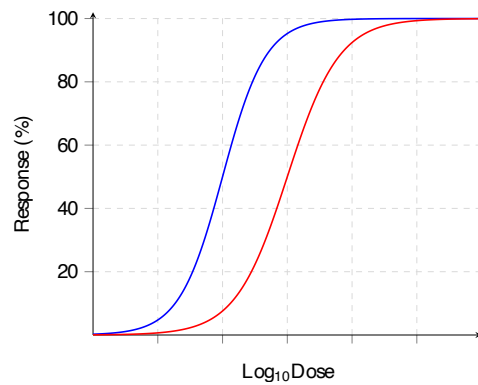
Pharmacology

Dose - Response:



- Dose response curve is a wash-in exponential
- Difficult to compare different drugs using this curve

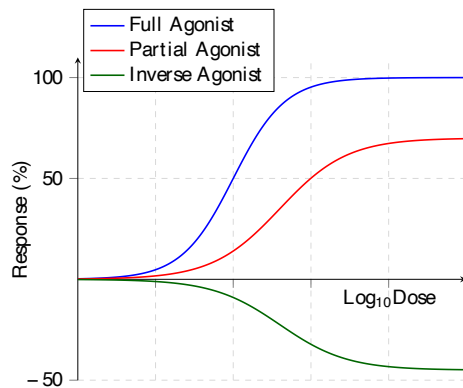
Log_{Dose} - Response:



- Log-transform of dose allows different drugs to be compared

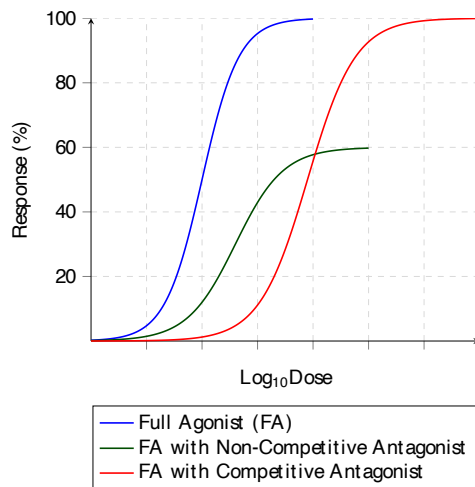
- Both red and blue drugs are full agonists (as they both reach 100% response), however the blue drug is more potent as it has a lower E_{D50}

Agonists:



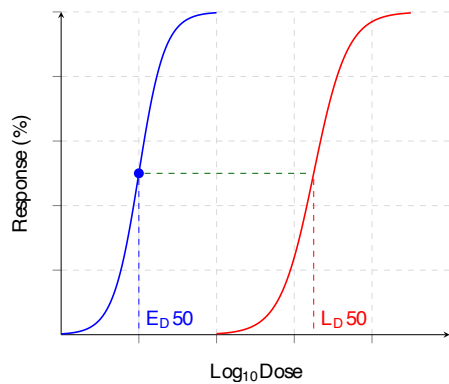
- Partial agonists do not reach 100% response
- Inverse agonists have a negative response

Antagonists:



- Non-competitive antagonists prevent maximal response being reached
- Competitive antagonists right shift the curve, as they can be overcome with increasing dose of agonist

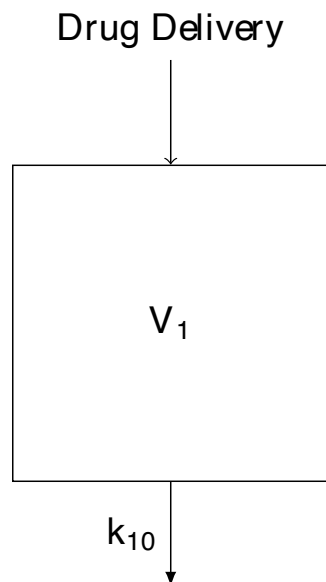
Therapeutic Index:



- Can be calculated from the ratio of the LD₅₀ and ED₅₀

Models

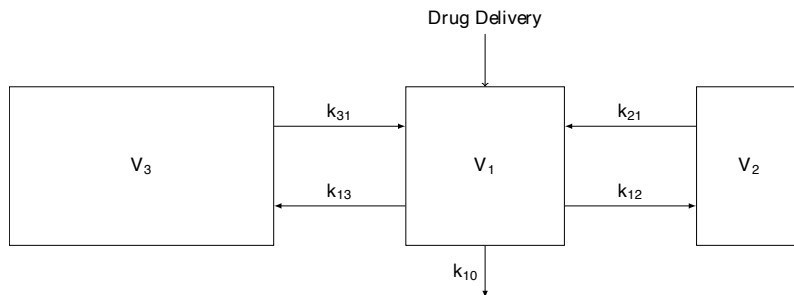
The One-Compartment Model:



- Drug is added to and removed from the single central compartment
There is no distribution possible.

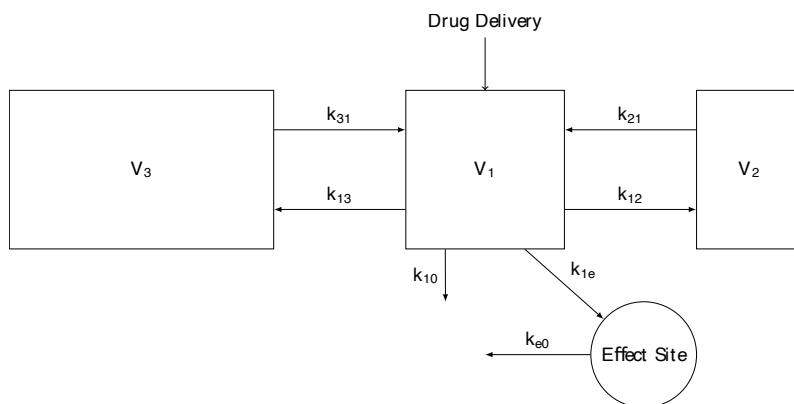
- V_1 is equal to the volume of distribution
- k_{10} is the rate constant for elimination

Three - Compartment Model:



- Drug is added to and removed from the central compartment
- Drug will also distribute to (and redistribute from) the peripheral compartments
- Plasma concentration will depend on:
 - Rate of drug delivery
 - Rate of drug distribution and redistribution
 - Rate of drug elimination

Effect - Site:

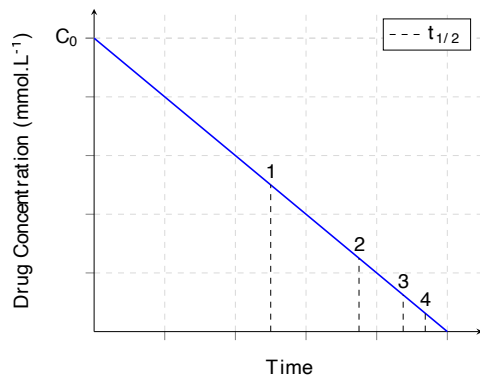


- Drug distributes to the effect site from the central compartment
- Effect site has no volume, but does have rate constants

- $t_{1/2}$ ke0 is generally drawn with drug being eliminated from the effect site, however in reality this does not occur as drug should only be eliminated from the central compartment

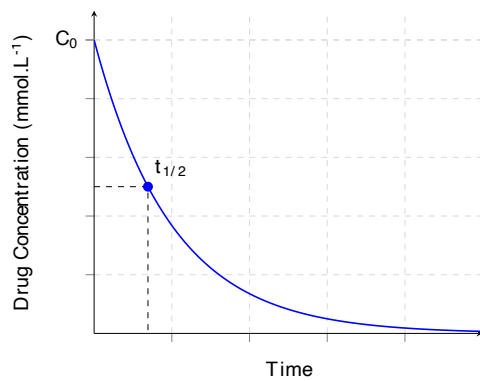
Pharmacokinetics

Zero-order kinetics:



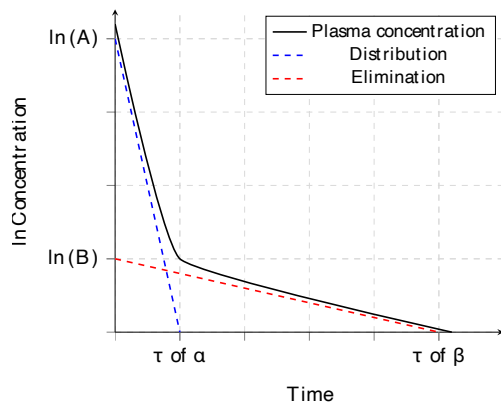
- A constant amount of drug is eliminated per unit time
- Half-life is not a constant value
Half-life progressively shortens, as the time taken to go from 50% to 25% is half the time it took to go from 100% to 50%.

First-Order Kinetics:



- A constant proportion of drug is eliminated per unit time
- Half-life is a constant value

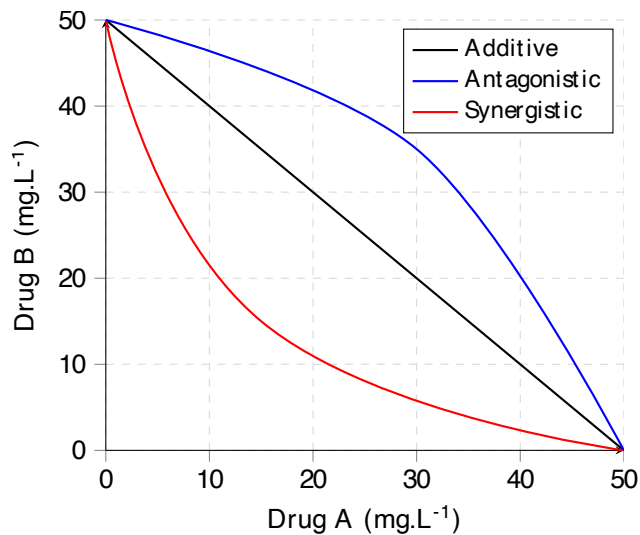
Biexponential elimination:



- Note that concentration has been log-transformed
- This describes the elimination of drug from a two compartment model

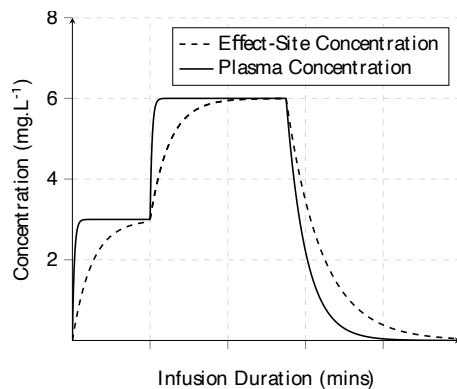
Pharmacodynamics

Isobologram:



- Plots lines of equal activity versus concentration of two drugs

Plasma - Site Targeting:



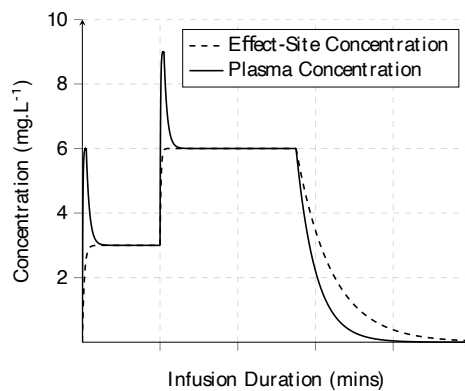
TCI graphs are easy to draw if you remember that:

- The pump aims to achieve the targeted concentration:
 - As rapidly as possible
 - Without overshoot
- Effect site concentrations fall slower than plasma site concentrations Drug can only redistribute back to plasma when effect site concentration is greater than plasma concentration.

Therefore in plasma site targeting:

- Plasma concentration rises rapidly with initial bolus dose
- Does not overshoot
- Effect site concentration rises more slowly
 - Exponential wash in curve as the concentration gradient between plasma and effect site falls over time

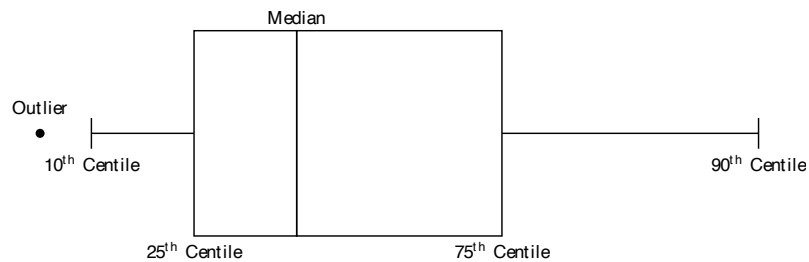
Effect-Site Targeting:



- Plasma concentration overshoots effect-site target and then declines rapidly
- Effect site concentration rises rapidly, and is achieved more quickly compared with plasma-site targeted model

Statistics

Boxplot:

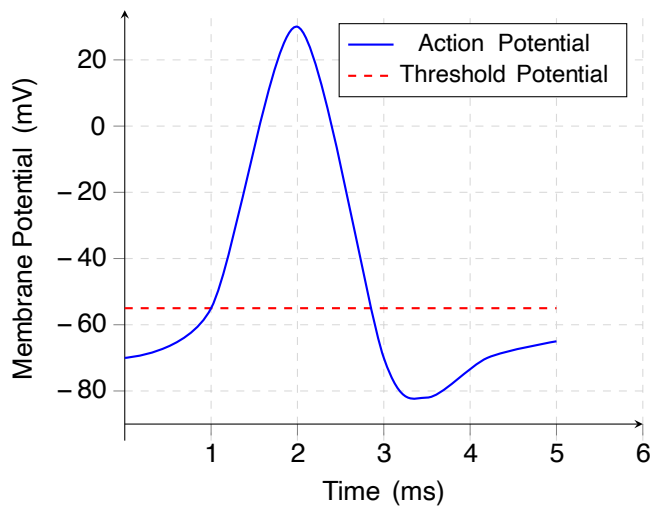


- Box is defined by the 25th and 75th centiles
- Line in the middle of the box is the median ("50th centile")
- "Whiskers" either side of the box define the 10th and 90th centiles
 - These may also refer to the 5th and 95th centiles
- Results outside of whiskers are defined as outliers, and are represented by single dots

Cellular Physiology

Action Potential

Action Potential:



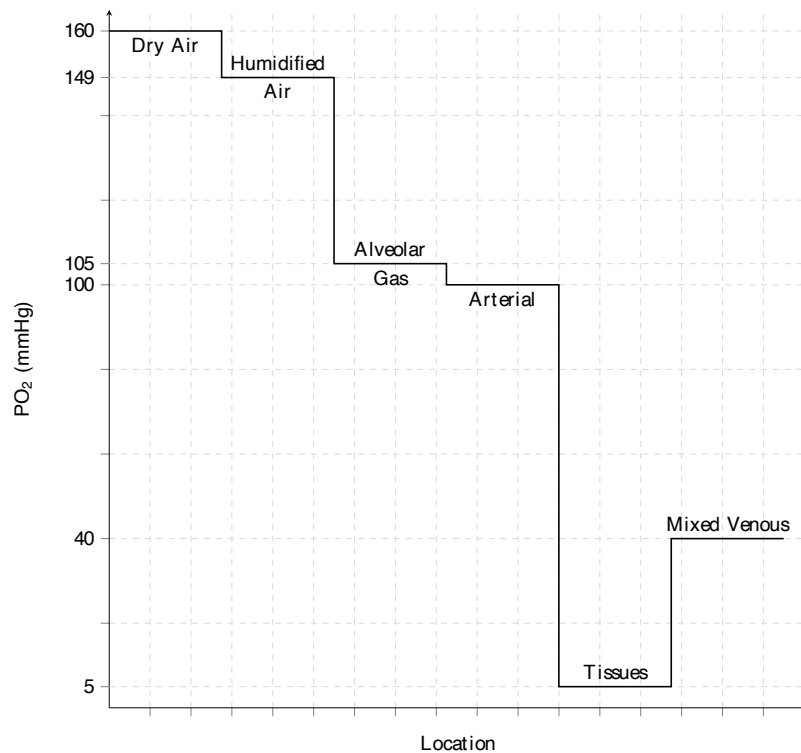
- Graph of membrane potential versus time
- Resting membrane potential is $\sim -70\text{mV}$

- Consists of **five phases**:
 - Rising Phase
A stimulus which rises above the threshold potential opens fast Na^+ channels, increasing Na^+ influx.
 - Peak Phase
Inactivation of fast-channels and delayed activation of K^+ channels slows depolarisation, which peaks at 30mV.
 - Falling Phase
Membrane potential falls with potassium efflux.
 - Hyperpolarisation
As potassium channels close slowly, the membrane potential slightly undershoots resting potential - this is the relative refractory period which lasts 10-15ms.
 - Resting
Cell is stable at resting membrane potential.

Respiratory

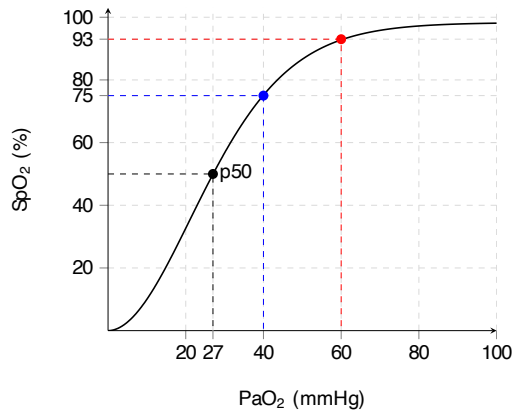
Oxygen

Oxygen Cascade:

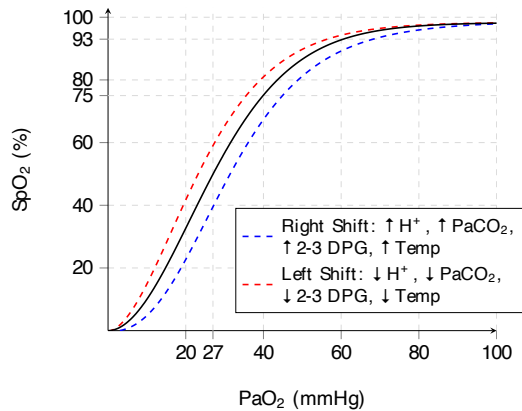


- Graph of location versus oxygen partial pressure
- Atmospheric (dry) air has a PO₂ of 160mmHg
- Tracheal (humidified) gas has a PO₂ of 149mmHg
Reduced due to saturated vapour pressure of water.
- Alveolar gas has a PO₂ of 105mmHg
Reduced to the presence of CO₂, as per the alveolar gas equation.
- Arterial blood has a PO₂ of ~100mmHg
Reduced to the Alveolar - arterial oxygen gradient.
- Tissues have a PO₂ of ~5mmHg
- Mixed venous blood has a PO₂ of ~40mmHg
Greater than tissue PO₂ as not all oxygen in blood diffuses into or is consumed by tissues.

Oxyhaemoglobin Dissociation Curve:

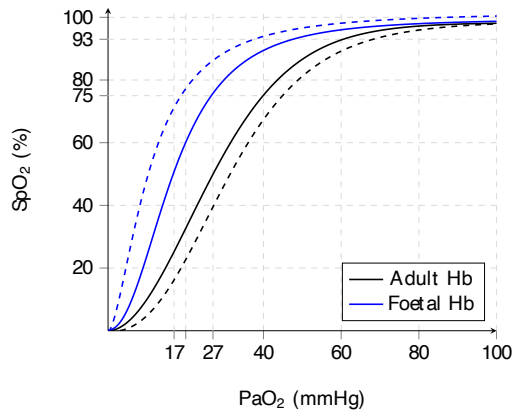


- Graph of PaO₂ versus oxygen saturation
- Note that PaO₂ is continuous, and so an arrow should be drawn at the tip of the x-axis, whilst saturation is finite and so the y-axis should be capped at 100%
- The curve is a sigmoid shape
- Key points:
 - At 10mmHg, saturation is 10%
 - The p50 is at 27mmHg
 - The mixed venous point is at 40mmHg, where haemoglobin is 75% saturated
Note that due to the Haldane effect, the mixed venous point does not technically exist on the arterial curve. This is a small point and is ignored in most graphs (including this one), but may be worth stating if you're feeling confident in the viva.
 - The "ICU point" (the upper inflection) is at 60mmHg where haemoglobin is 93% saturated
 - The arterial point is 97% saturated at 100mmHg



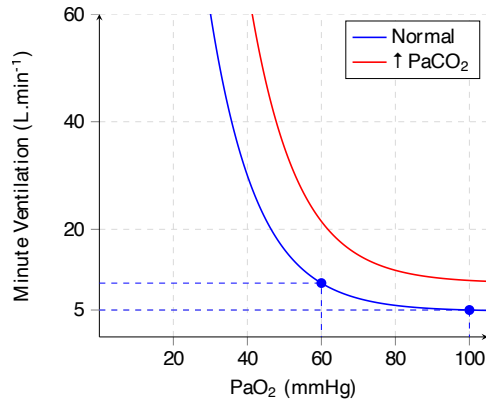
- The curve may be right-shifted by:
 - Increased H⁺
 - Increased PaCO₂
 - Increased temperature
 - Increased 2-3 DPG
- These shifts are defined by a movement of the p50

Double-Bohr Effect:

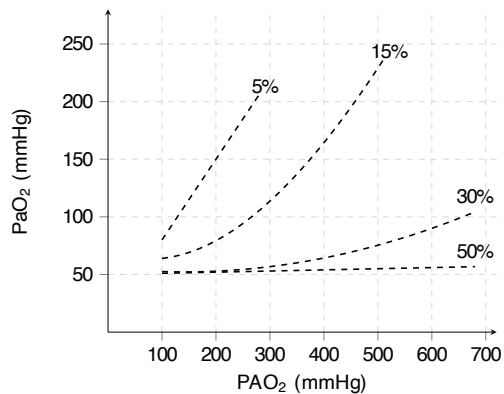


- The double Bohr effect can easily become confusing, especially when you are under pressure and only allowed one colour (as in the written exam)
Here is a straightforward method which minimises the confusion:
 1. Draw an adult curve with a p50 of 27mmHg
 2. Draw a foetal curve with a p50 of 17mmHg
 3. Draw a right-shifted adult curve

4. Draw a left-shifted foetal curve

PaO₂ and Minute Ventilation:

- Exponential curve
- Minute ventilation doubles as PaO₂ decreases from 100mmHg to 60mmHg
- Inflection point is ~50-60mmHg
Below this there is a large increase in ventilation.
- Hypercapnea leads to a greater minute ventilation for any given PaO₂

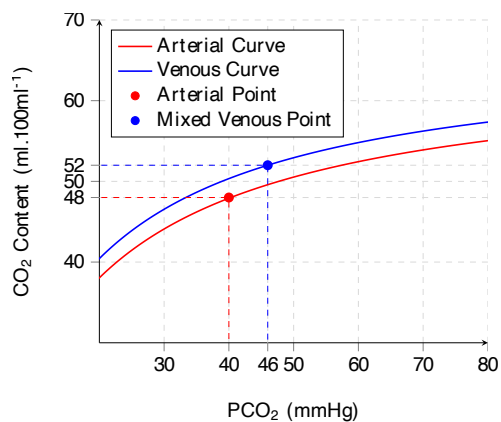
Isoshunt Diagram:

- Plots the relationship between PAO₂ versus PaO₂ for different (fixed) shunt fractions
- These are known as isoshunt lines
- Key isoshunt lines are:

- At 50% shunt, PaO₂ is essentially independent of PAO₂
- At 30% shunt, PaO₂ will not increase above 100mmHg on 100% oxygen at atmospheric pressure

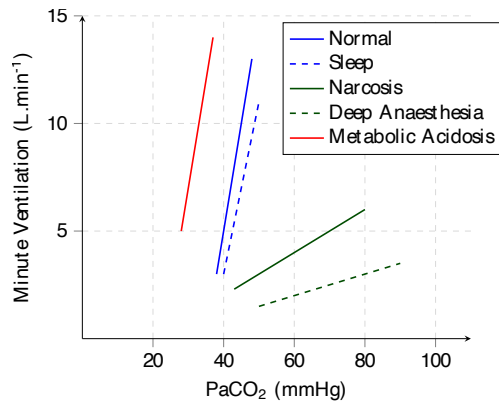
Carbon Dioxide

Carbon Dioxide Dissociation Curve:



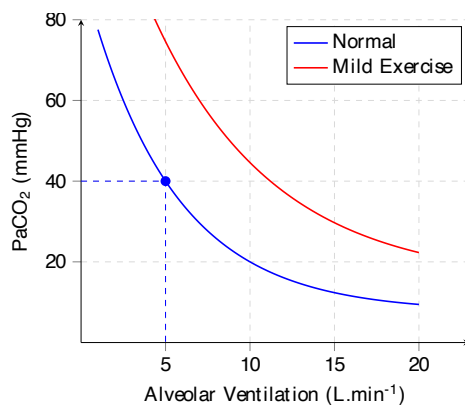
- Graph of carbon dioxide content versus partial pressure
- Key points on this curve:
 - Arterial CO₂ content is 48mls.100ml⁻¹ of blood at 40mmHg
 - Mixed venous CO₂ content is 52.mls.100ml⁻¹ of blood at 46mmHg
- Note that the mixed venous curve is up-shifted due to the Haldane effect
Remember that 50% of the difference in CO₂ content is due to the Haldane effect. Therefore:
 - The mixed venous curve should be drawn such that CO₂ content is 50mls.100ml⁻¹ at 40mmHg
 - The arterial curve should be drawn such that CO₂ content is 50mls.100ml⁻¹ at 46mmHg

PaCO₂ and Minute Ventilation:



- Graphs the change in minute ventilation for **primary change in PaCO₂**
- Remember that minute ventilation increases by ~3L.min⁻¹ for every 1mmHg increase in PaCO₂
From this, the relationship to other states can be derived:
 - Minute ventilation is reduced during sleep, but the central response to CO₂ is only minimally affected
 - The central response to CO₂ is heavily affected during anaesthesia
 - Minute ventilation is increased for any given PaCO₂ in the setting of acidosis

Alveolar Ventilation and PaCO₂:

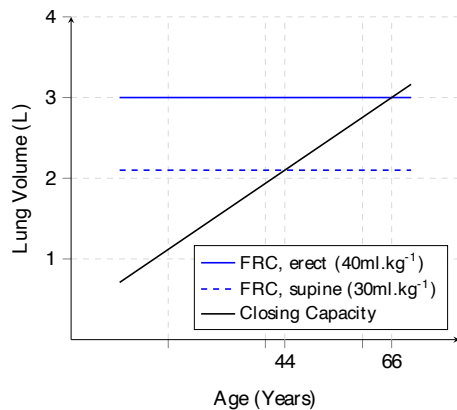


- Graphs the change in PaCO₂ for a **primary change in minute ventilation**

- Exponential curve as PaCO_2 is inversely proportional to minute ventilation
- Minute ventilation is increased for any given PaCO_2 during exercise

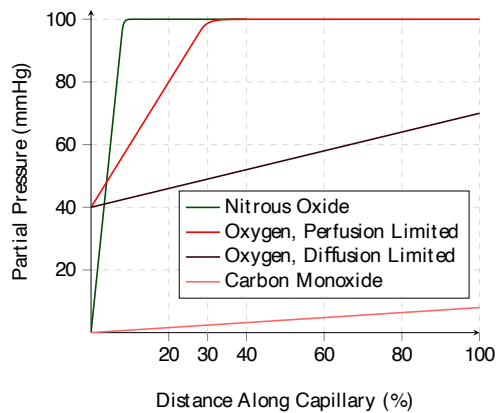
Anatomical and Physiological Interactions

Closing Capacity and Age:



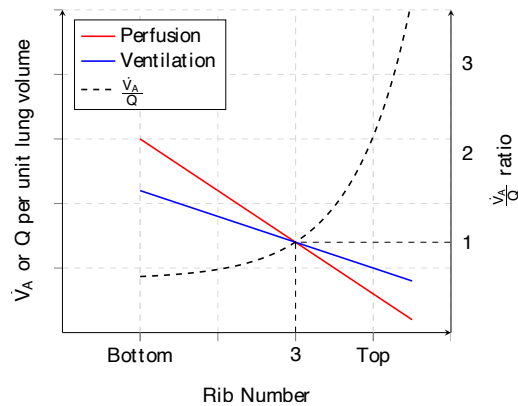
- Note that although FRC increases slightly with age, this is not generally shown on this graph
 - Closing capacity increases with increasing age
- Key intersections are:
- Greater than FRC when supine at 44 years of age
 - Greater than FRC when erect at 66 years of age

Diffusion and Perfusion Limitation:



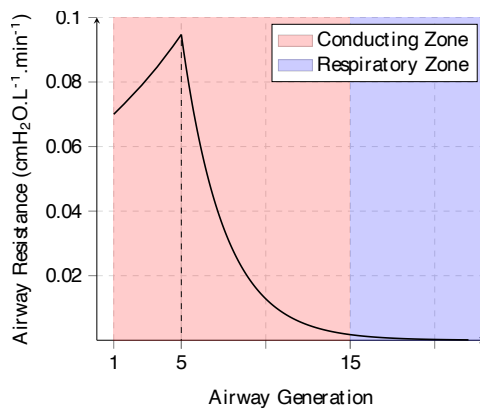
- Classically drawn as partial pressure versus distance along the capillary
Time along capillary may also be used, however note that total transit time will change with cardiac output.
- Note that at the beginning of the capillary, oxygen partial pressure will be equal to that of mixed venous blood
 - In perfusion limitation, PaO_2 will equal PAO_2 before the end of the capillary
 - In diffusion limitation, partial pressures will not be equal at the end of the capillary
 - In normal circumstances, PaO_2 equals PAO_2 at $\sim 1/3^2$ of the distance along the capillary
If time is being graphed on the x-axis, then this will occur at $\sim 0.25\text{s}$, as total capillary transit time is $\sim 0.75\text{s}$.
- Nitrous oxide rapidly diffuses into blood and is and not typically present in mixed venous blood, so this curve begins at the origin and PaN_2O will rapidly reach PAN_2O (in this instance 100mmHg)
- Carbon monoxide binds avidly to haemoglobin and so PaCO increases slowly, resulting in diffusion limitation

Regional Ventilation and Perfusion:



- Graph of alveolar ventilation and alveolar blood flow versus rib number in the **erect** person
- Basal alveoli have greater perfusion and ventilation than apical alveoli
- Note the perfusion gradient is steeper than the ventilation gradient
- Note that the V/Q ratio is:
 - ~ 1 at the 3rd rib
 - ~ 3.3 at the apex
 - ~ 0.63 at the base

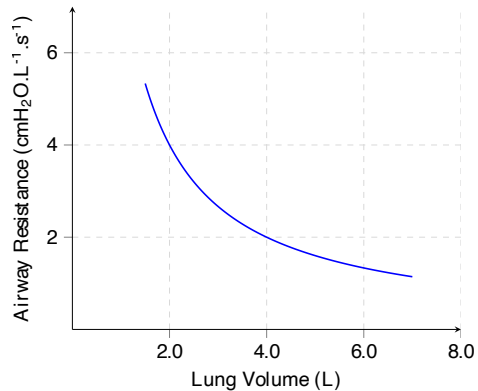
Airway Resistance and Airway Generation:



- Graph of airway resistance versus airway generation
- Airway generations are from 1 to 23, and so this graph should not extend outside these values

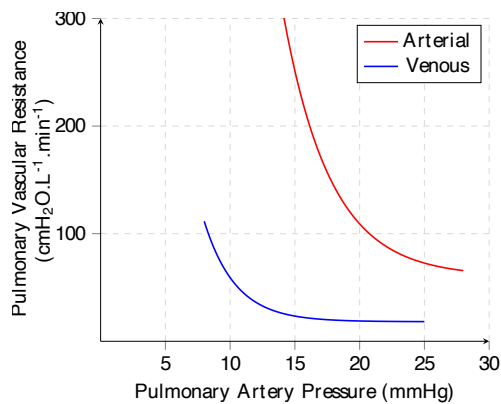
- Airway resistance is maximal at the 5th generation
This has the lowest total cross-sectional area.
- Airway resistance is negligible in the respiratory zone,
which exists after the 15th generation

Airway Resistance and Lung Volume:



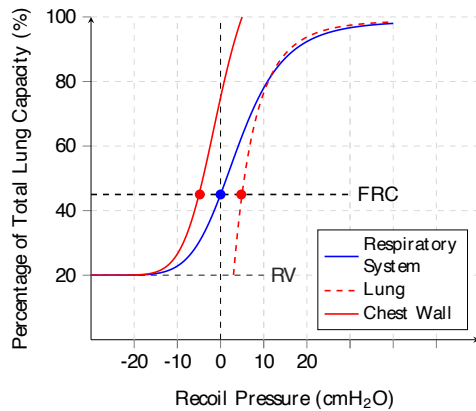
- Airway resistance decreases as lung volume increases
as radial tension distends airways, increasing their
cross-sectional area and lowering airway resistance

Pulmonary Vascular Resistance and Pulmonary Artery Pressure:



- Pulmonary vascular resistance decreases as pulmonary artery pressure increases
- Arterial pressure has a greater effect on PVR than venous pressure

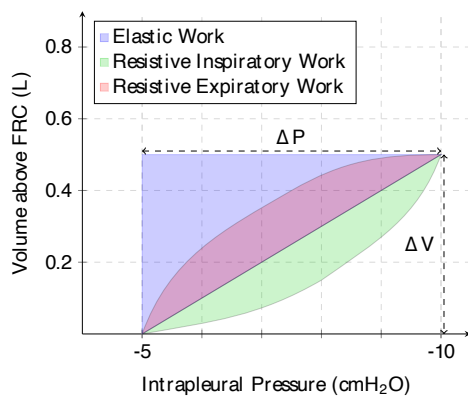
Lung and Chest Wall Volume and Pressure Relationships:



- Graph of lung volume versus **recoil pressure**
 - Expressing lung volume as a percentage of total lung capacity may make it easier to remember the key points on this graph
 - Note that recoil pressure is the pressure generated **between the lung and the chest wall** when they are distended, it is not intrapleural pressure
- This graph is complex and it is easy to draw incorrectly
This is an approach to make it as easy as possible:
 1. Draw a sigmoid graph for the pressure-volume relationship of the respiratory system as a whole
 - As recoil pressure is 0 at FRC this will be the y-intercept
 - The graph will asymptote at residual volume, as volume (by definition) cannot become lower than this volume
 2. Draw a steep run-away exponential for the pressure-volume relationship of the chest wall
 - Recoil pressure should be $-5\text{cmH}_2\text{O}$ at FRC
 - Recoil pressure should be $0\text{cmH}_2\text{O}$ at $\sim 75\%$ of TLC
 - Recoil pressure should not exceed $5\text{cmH}_2\text{O}$ at TLC
 3. Draw a steep wash-in exponential for the pressure-volume relationship of the lung

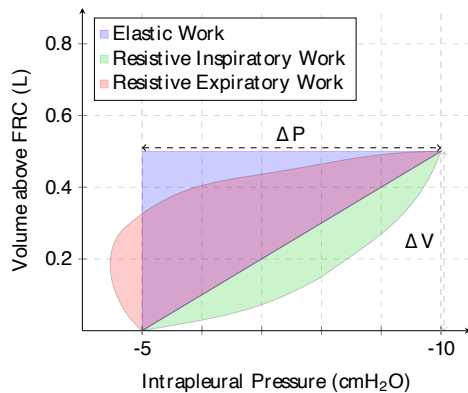
- Remember lung volume cannot fall below residual volume
- Recoil pressure should be $5\text{cmH}_2\text{O}$ at FRC
This should be equal and opposite to the recoil pressure for the chest wall, as the sum of these must be 0 at FRC.
- Note that this curve should slightly exceed the curve for the respiratory system as recoil pressure increases

Work of Breathing:



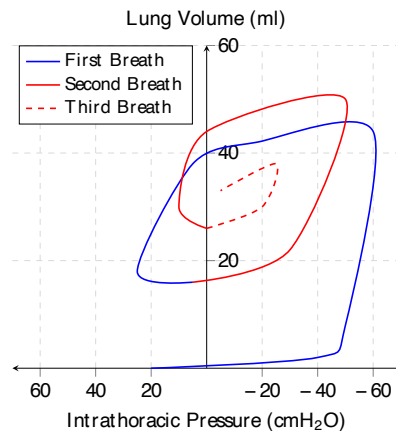
- Graph of lung volume (above FRC) versus intrapleural pressure
Note that intrapleural pressure becomes more negative along the x-axis.
- The area under different sections of this curve give the work of breathing
 - Elastic inspiratory work of breathing is given the blue triangle
 - Resistive work of expiration is given by the red area
Note that as this is entirely contained within the area of elastic inspiratory work, expiration is passive and does not require additional energy expenditure.
 - Resistive work of inspiration is given by the green area

Work of Breathing - Active Expiration:



- When resistive expiratory work exceeds elastic inspiratory work, active expiration must occur
- In this graph, active expiration is given by the red area not contained within the blue triangle

Neonatal First Breath:

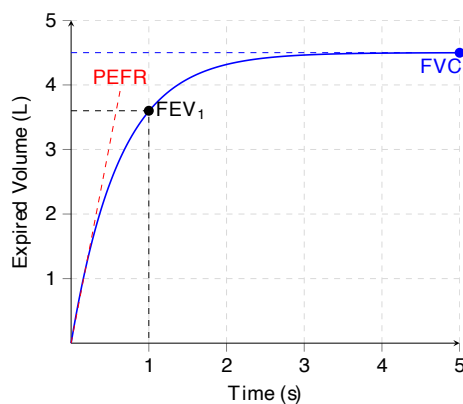


- This graph describes the pressure - volume changes of the neonate as it takes its first breaths and establishes FRC
- This graph is easy to draw provided you remember that:
 - Prior to the first breath, lung volume is 0
 - As the lung initially has very poor compliance, the intrapleural pressure must become very negative more lung volume increases substantially

- At the end of each breath, intrathoracic pressure is close to 0
- With each subsequent breath:
 - Lung compliance improves
Therefore the magnitude of pressure swings is reduced.
 - FRC increases
Lung volume at end-inspiration is increased.

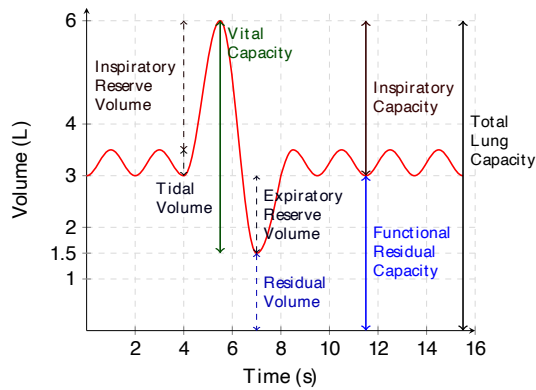
Spirometry

Forced Vital Capacity:



- Graph of expired volume (vital capacity) over time
- ~80% of total volume is expired within the first second (FEV₁)
- Total FVC is 4.5L in the 70kg Guyton Man
- The gradient of initial expiration is the peak expiratory flow rate

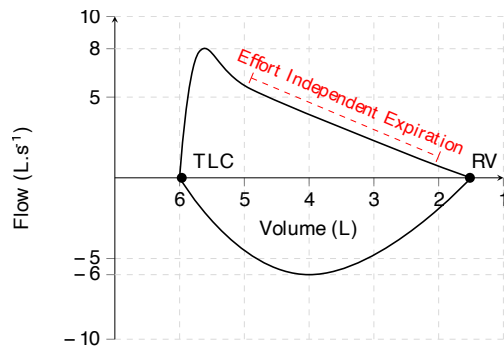
Spirometry:



- Graph of lung volume over time
- Includes a normal tidal breath and a vital capacity breath

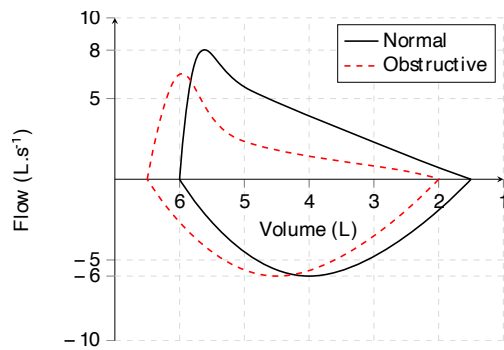
Flow-Volume Loops

Normal loop:



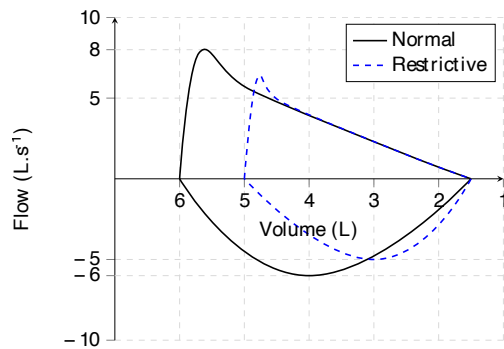
- Peak expiratory flow is $\sim 8 \text{ L.s}^{-1}$
- Peak inspiratory flow is $\sim 6 \text{ L.s}^{-1}$
- Effort independent expiration occurs during expiration

Obstructive Disease:



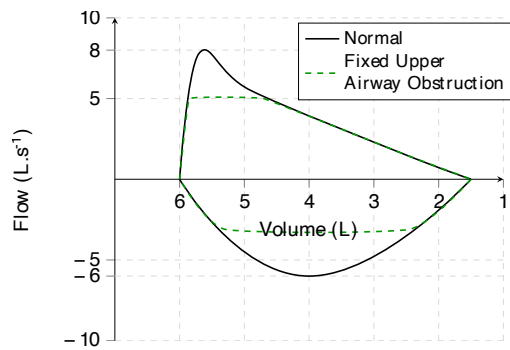
- Residual volume and total lung capacity are increased due to gas trapping
- Peak expiratory flow is reduced
- There is scalloping of the effort-independent portion of the curve
Also known as a concave curve.

Restrictive Disease:



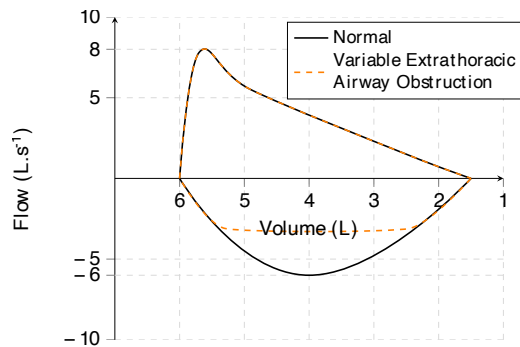
- Total lung capacity is reduced
- Residual volume is normal
- Peak expiratory flow *may* be reduced (as seen here)
However the FEV₁:FVC ratio will be normal in purely restrictive lung disease.
- Effort-independent expiration is linear and will join with the normal curve

Fixed Upper Airway Obstruction:



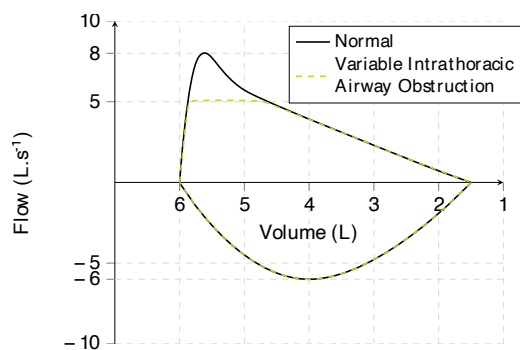
- Obstruction that does not change calibre throughout the respiratory cycle
- Peak expiratory and inspiratory flow rates are limited

Extrathoracic Obstruction



- Obstruction worsens during inspiration as it is 'pulled in' by negative intrathoracic pressure

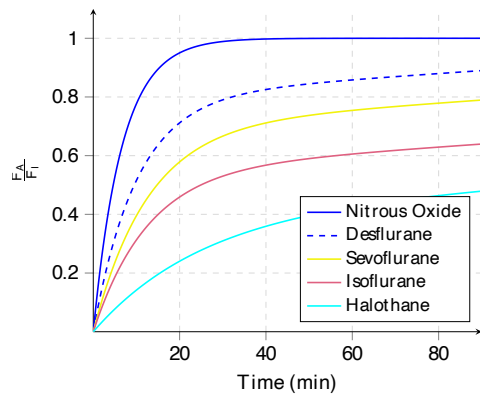
Intrathoracic Obstruction



- Obstruction worsens during expiration as it is compressed by dynamic airways compression

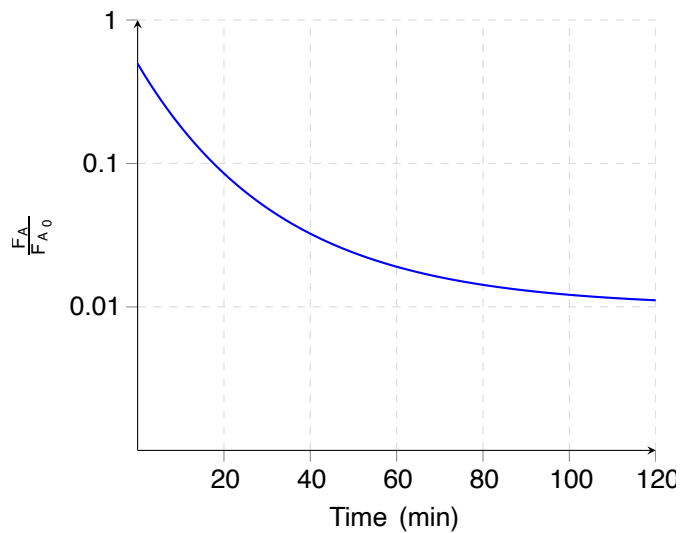
Anaesthetic Agents

F_A/F_I :



- Graph of the alveolar over inspired agent fraction versus time for various volatile agents
- Indicates the relative speed of onset of different agents
- Uptake of agent is proportional to solubility in blood, and therefore is in order of their blood:gas coefficients
 - The exception is nitrous oxide, which has a faster rate of rise than desflurane despite its greater blood:gas coefficient due to the concentration effect

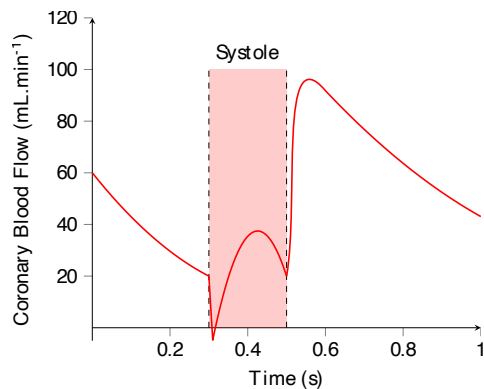
F_A/F_{A0} :



- Graph of alveolar agent fraction versus time for a volatile agent
- Note the logarithmic scale on the y-axis
- Exponential washout curve
- Function of two separate washout curves
 - Rapid washout with removal of agent from circuit and FRC
 - Slow washout due to diffusion of agent from tissues into blood, and then alveolus

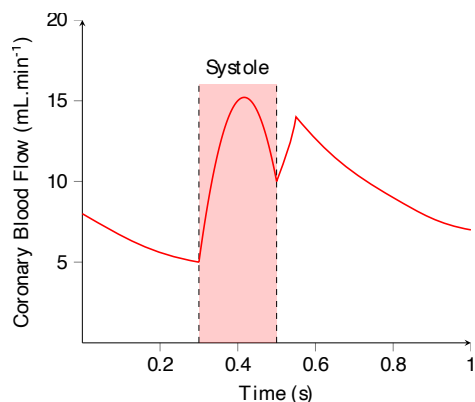
Cardiovascular

Left Ventricular Coronary Blood Flow:



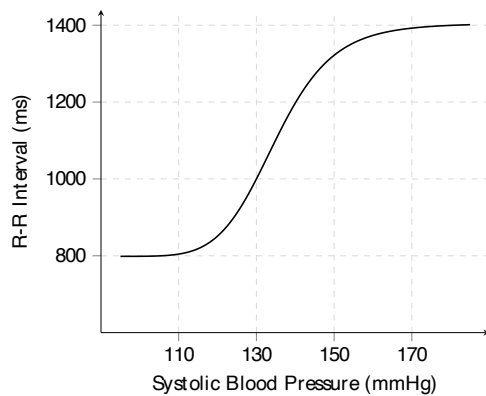
- Graph of blood flow to the left *ventricle* over time
Systole should be clearly identified.
- Left ventricular flow occurs predominantly in diastole
Peak flow is $\sim 115 \text{ mL} \cdot \text{min}^{-1}$.
- There is a brief period of flow reversal during isovolumetric contraction

Right Ventricular Coronary Blood Flow:



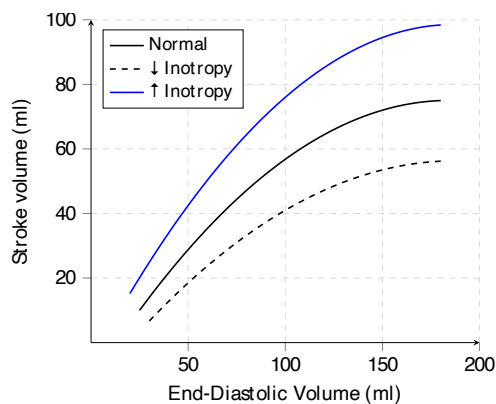
- Graph of blood flow to the right *ventricle* over time
- Right ventricular flow occurs throughout the cardiac cycle
This is because aortic root pressure exceeds cavity pressure throughout the cardiac cycle.
- Peak flow is $\sim 15 \text{ mL} \cdot \text{min}^{-1}$

Baroreceptor Response:



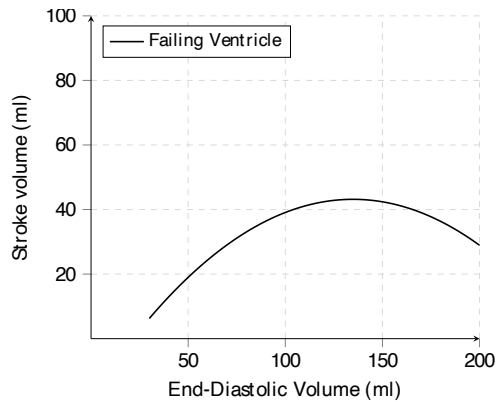
- Graph of heart rate versus systolic blood pressure
Note that the RR interval is inversely proportional to heart rate.
- Heart rate responses asymptote at extremes of blood pressure

Starling Curve:



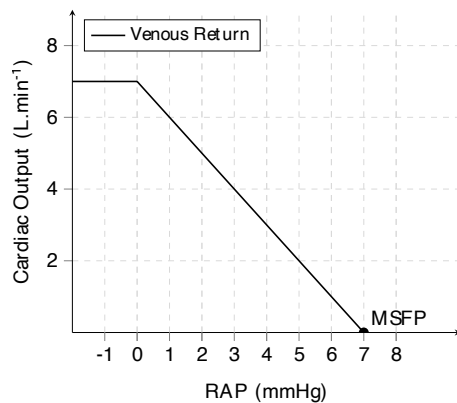
- Typically drawn as a graph of stroke volume (or cardiac output, assuming a constant heart rate) versus preload (typically estimated as end-diastolic volume, but may also be end-diastolic pressure)
- Graph does not cross the origin as EDV is never 0ml

Starling Curve - Failing:



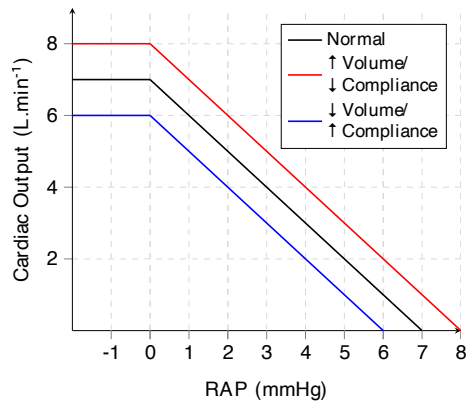
- Myocardium that has been overloaded by high end-diastolic volumes may lead to a decrease in tension generated by the myocardium

Venous Return:



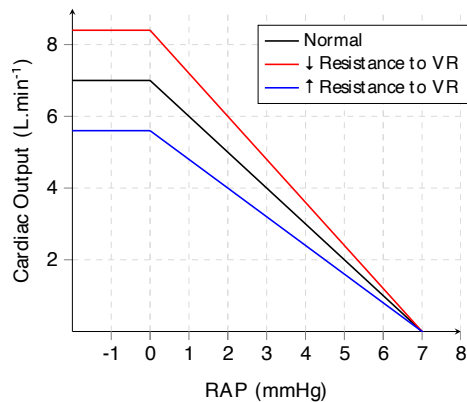
- Graph of venous return versus right atrial pressure
- The x-intercept is the point of no flow within the circulation (as VR = CO), and therefore is the mean systemic filling pressure
- The curve flattens when RAP becomes negative, as external tissues act as a Starling resistor and prevent further increases in flow

Venous Return - Compliance and Volume:



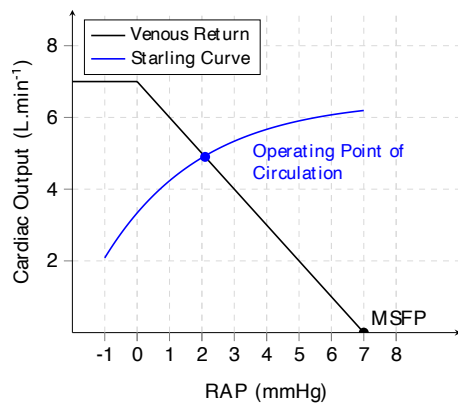
- Decreasing venous compliance or increasing circulating volume results in an increase in mean systemic filling pressure (as for any given compliance, pressure must increase if volume increases) and an increase in venous return for any given right atrial pressure
- The opposite occurs with a decrease in circulating volume or an increase in venous compliance

Venous Return - Resistance to Venous Return:



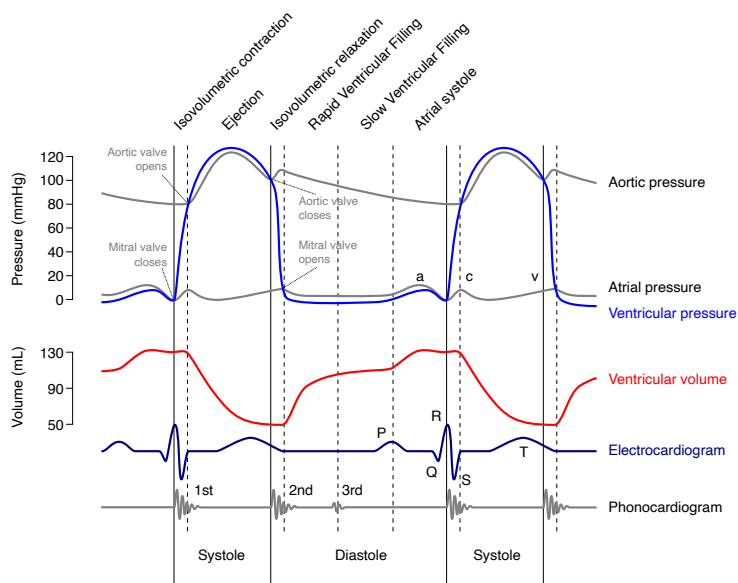
- Altering resistance to venous return (e.g. during pregnancy, or laparoscopic surgery) will alter venous return without changing mean systemic filling pressure

Circulatory Function Curve:



- Plotting the venous return curve and the Starling curve on the same axes generates this graph
 - This is only valid at steady state, i.e. when CO = VR
- Note that as steady-state exists when CO=VR, the intercept of these two curves is the operating point of the circulation

Wiggers Diagram:

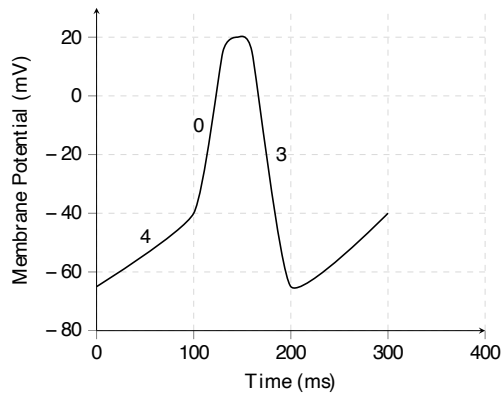


- Wiggers diagram is a graphical representation of the events during each phase of the cardiac cycle

- Key points to note:
 - Aortic diastolic pressure occurs just prior to aortic valve opening
A common mistake is to label diastolic pressure at the aortic notch.
 - Ventricular pressure exceeds aortic pressure during ejection
 - Aortic pressure will slightly exceed ventricular pressure during the last part of ejection
This is due to the inertia of ejected blood causing ongoing forward flow despite the pressure gradient.
 - The aortic notch occurs on the aortic pressure curve
A common mistake is to draw this on the ventricular curve.
 - CVP *slightly* exceeds ventricular pressure during ventricular filling
 - The C wave occurs during isovolumetric contraction
 - The V wave begins prior to the T wave, but peaks after the T wave has finished
 - Electrical events slightly precede ventricular mechanical events

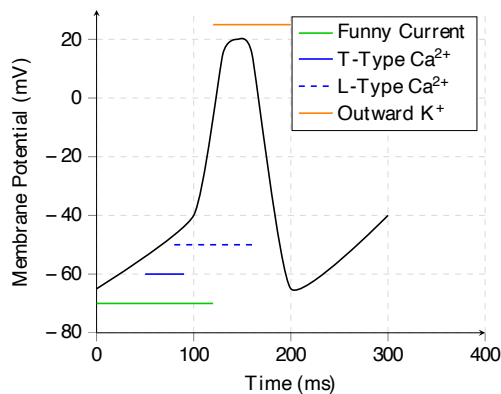
Action Potentials

Pacemaker Potential:



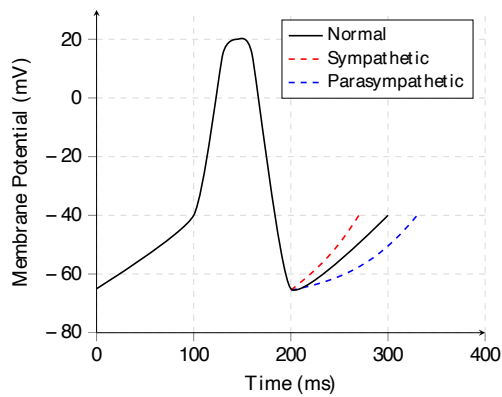
- The pacemaker potential has only three phases, and notably no 'resting phase'
This is due to the funny current.
- Maximal diastolic potential is -65mV
- Peak membrane potential is ~20mV

Pacemaker Potential - Ion Flux:



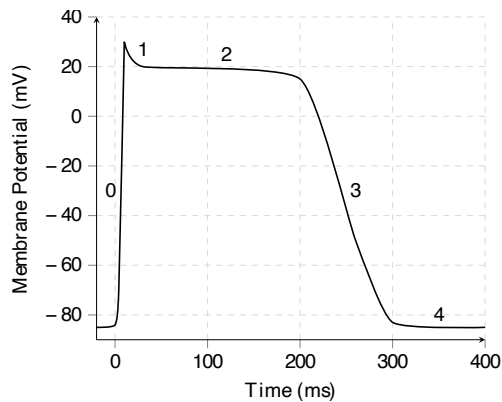
- Demonstrates the timing of electrolyte passage across the cell membrane
- Funny current occurs throughout phase 4 and the early part of phase 0
- T-type calcium current begins in late phase 4 and terminates prior to the onset of phase 0
- L-type calcium current overlaps with the T-type current and continues throughout phase 3
- Outward rectifying potassium current begins during phase 3 and continues during phase 4, restoring membrane potential

Pacemaker Potential - Autonomic Tone:



- Alteration to autonomic tone alters the slope of the funny-current
(Some sources also note a change to maximal diastolic potential, although this is not shown here).

Ventricular Action Potential:



- The ventricular action potential consists of 5 phases
 - 0: Rapid depolarisation
 - 1: Partial repolarisation
Due to initial efflux of potassium without proportional calcium influx.
 - 2: Plateau
Outward potassium current is matched by inward calcium current.

- 3: Repolarisation

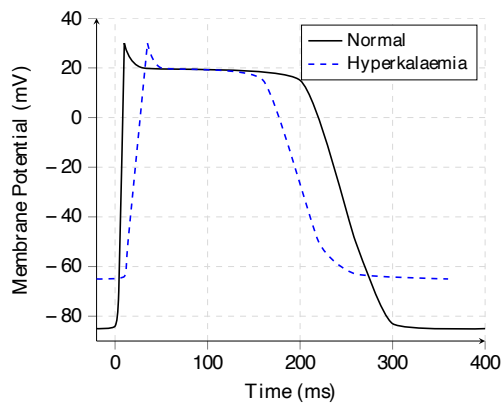
Note that the absolute refractory period ends when resting membrane potential falls below -50mV, which typically occurs at ~250ms.

- 4: Resting membrane potential

Note that:

- Resting Membrane Potential is typically ~ -85mV
- The relative refractory period ends when the membrane potential is at its resting state

Ventricular Action Potential - Hyperkalaemia:



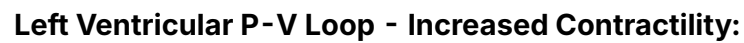
- In hyperkalaemia:

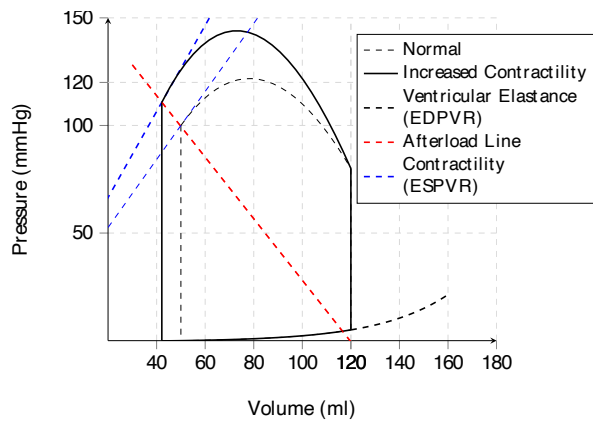
- The ventricle is more *irritable* as resting membrane potential is less negative, bringing it closer to threshold potential
- The duration of the action potential is shorter, increasing the chance for a re-entrant arrhythmia

Basic Pressure-Volume Loops

Pressure-volume loops are covered in detail under pressure-volume relationships.

Left Ventricular P-V Loop:





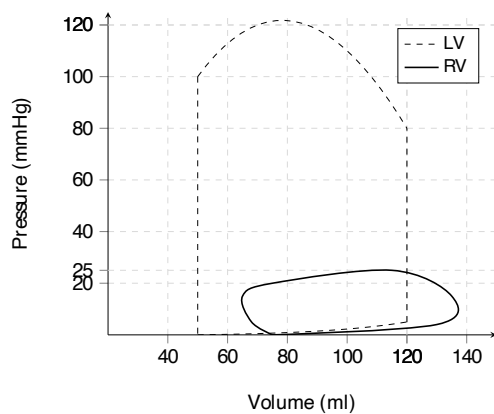
Advanced-Pressure Volume Loops

When drawing changes to more left-field pressure-volume loops which you may not have seen before approach them in the following way:

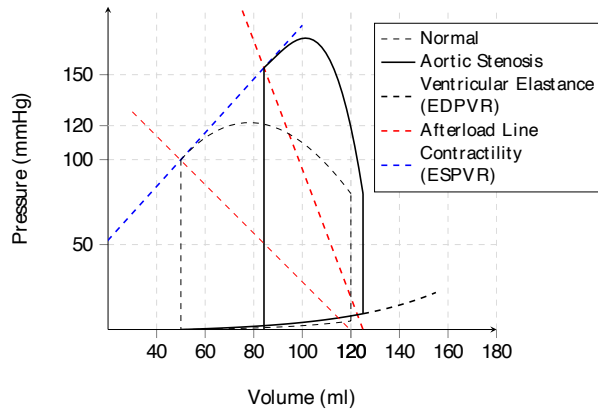
- How is preload changed?
- How is afterload changed?
- How is contractility changed?
- How are isovolumetric contraction and isovolumetric relaxation changed?

Advanced pressure-volume loops are covered in detail under pressure-volume relationships.

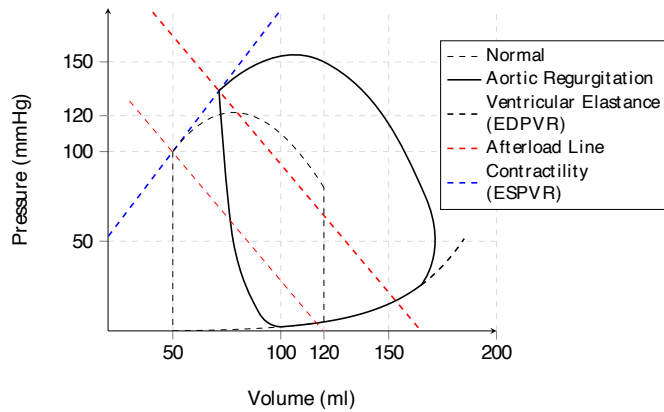
Right Ventricular P-V Loop:



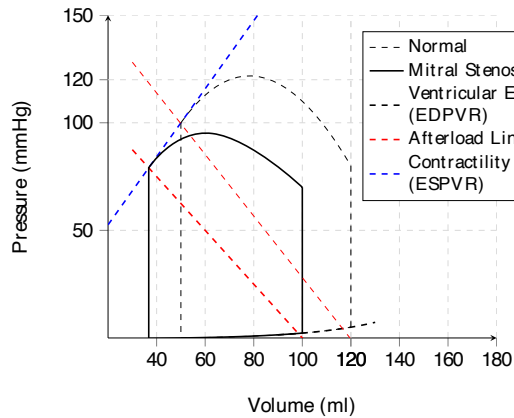
Left Ventricular P-V Loop - Aortic Stenosis:



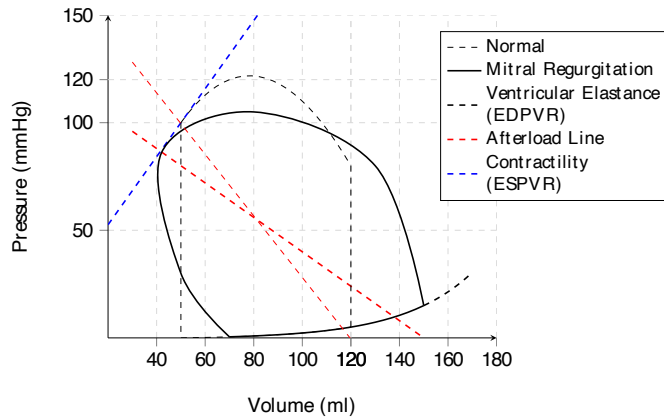
Left Ventricular P-V Loop - Aortic Regurgitation:



Left Ventricular P-V Loop - Mitral Stenosis:

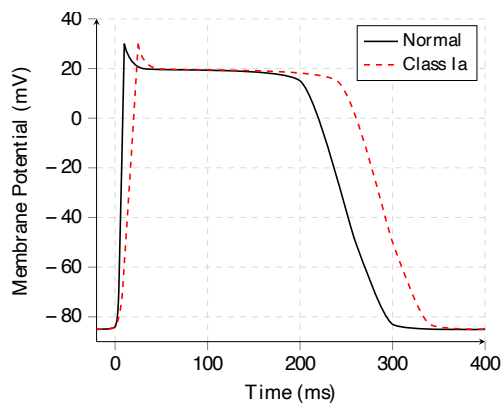


Left Ventricular P-V Loop - Mitral Regurgitation:



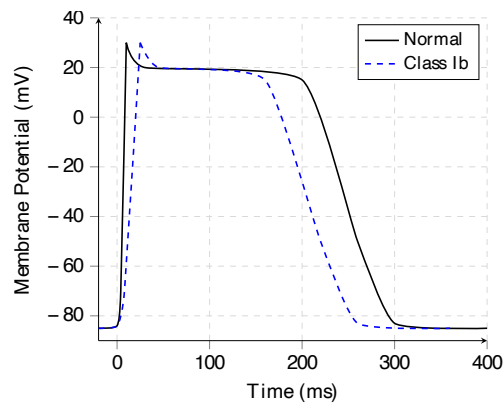
Antiarrhythmics

Ventricular Action Potential - Class Ia:



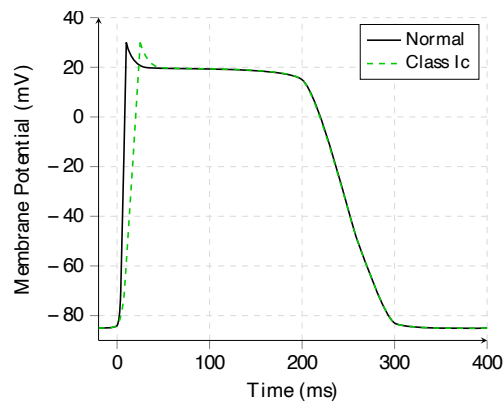
- Prolong the rate of rise of phase 0
- Lengthen the myocardial action potential

Ventricular Action Potential - Class Ib:



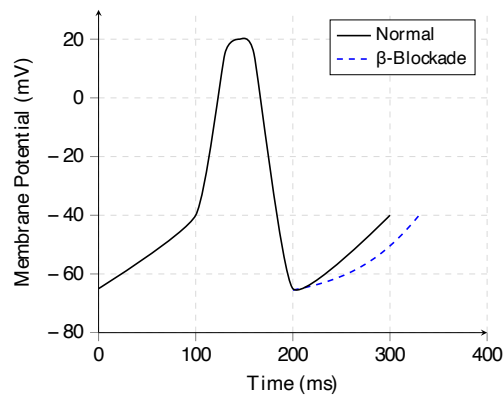
- Prolong the rate of rise of phase 0
- Shorten the myocardial action potential

Ventricular Action Potential - Class Ic:



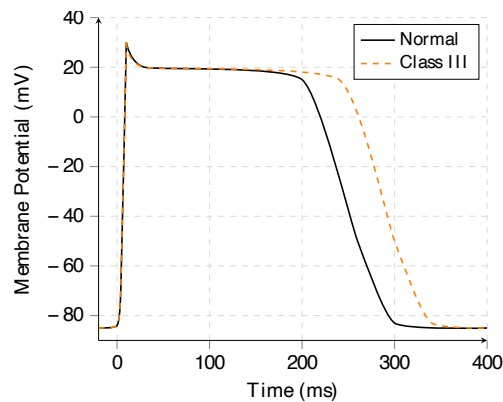
- Prolong the rate of rise of phase 0
- Do not alter the length of the myocardial action potential

Pacemaker Potential - Class II (Beta-Blockade):



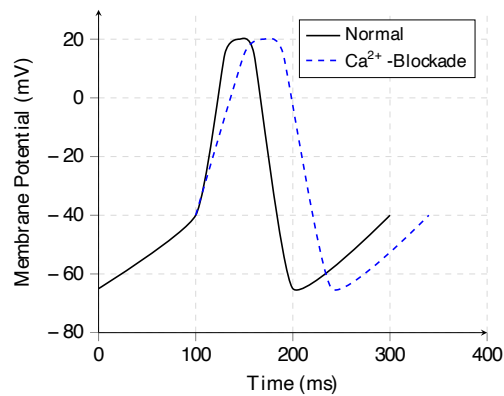
- Sympatholytic effect reduces the magnitude of the funny current

Ventricular Action Potential - Class III:



- Prolong duration of phase 3 of the myocardial action potential
This prolongs the refractory period and reduces the chance of a re-entry circuit occurring, and therefore reduces tachyarrhythmias but may increase the risk of *torsade de pointes* due to an increased risk of after depolarisations.

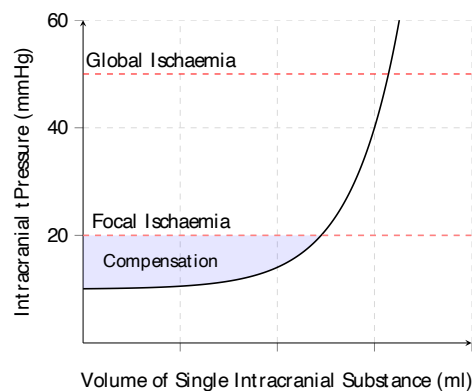
Pacemaker Potential - Class IV (Calcium Channel Blockade):



- In the pacemaker cell, reduce the magnitude of T-type and L-type calcium currents, reducing the rate of rise of phase 0 of the pacemaker action potential

CNS

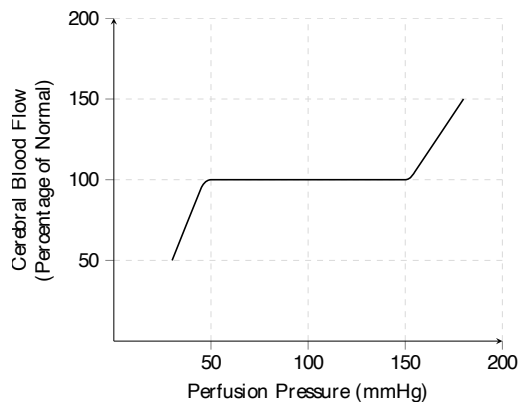
Monroe-Kellie Doctrine:



- Graphs the intracranial pressure versus the volume of a component (blood, brain, or CSF) in the cranial vault
Note that overall volume is *not* correct, as this is unchanged - if overall volume increased the pressure would reduce (e.g. such as a decompressive craniectomy).

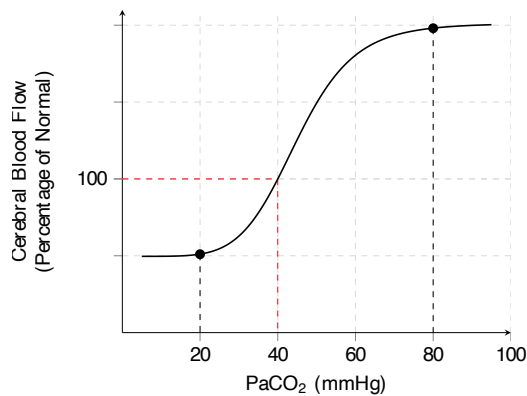
- Note the initial period of compensation, which occurs due to displacement of CSF to the spinal sub-arachnoid, decreased cerebral blood volume, and a decrease in CSF volume.
- Once compensatory responses are exhausted ICP will increase rapidly due to the poor elastance of the cranial vault
- Focal ischaemia occurs when ICP exceeds 20mmHg
- Global cerebral ischaemia occurs when ICP exceeds 50mmHg

Cerebral Blood Flow and Cerebral Perfusion Pressure:



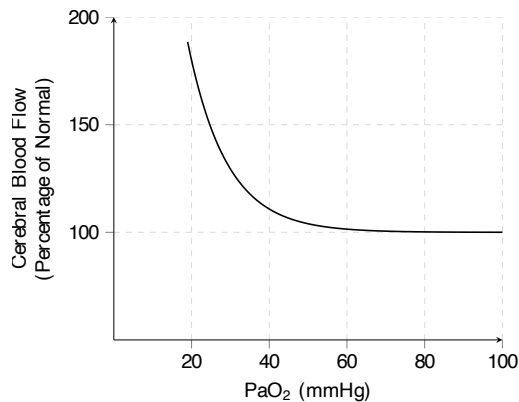
- Cerebral blood flow is autoregulated for a CPP of 50-150mmHg
(Note that this classic relationship is probably incorrect, and that CBF is probably only autoregulated across a narrow range of blood pressures).

Cerebral Blood Flow and PaCO₂:



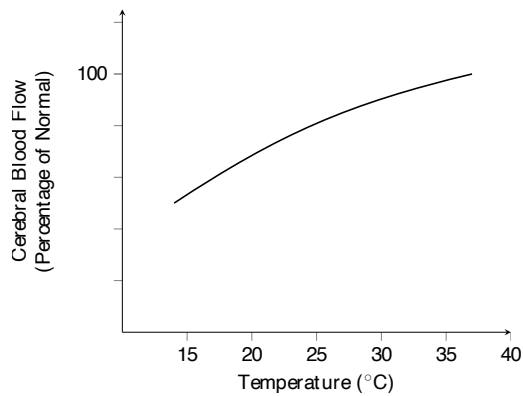
- CBF increases by ~3% for every 1mmHg increase in CO₂
- Below a PaCO₂ of 20mmHg, CBF cannot decrease further as the reduced flow results in tissue hypoxia, and metabolic autoregulatory responses
- Above a PaCO₂ of 80mmHg, CBF cannot increase further as vessels are maximally dilated

Cerebral Blood Flow and PaO₂:



- Above a PaO₂ of 60mmHg, CBF is essentially independent of PaO₂
- Below a PaO₂ of 60mmHg, CBF increases rapidly

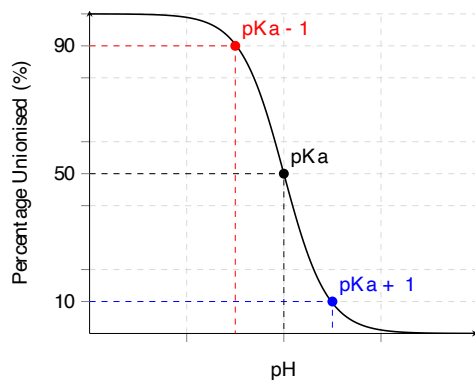
Cerebral Blood Flow and Temperature:



- Cerebral metabolic rate falls by ~6% per °C decrease in temperature
- This results in a concomitant reduction in CBF
This is an *almost* linear response.

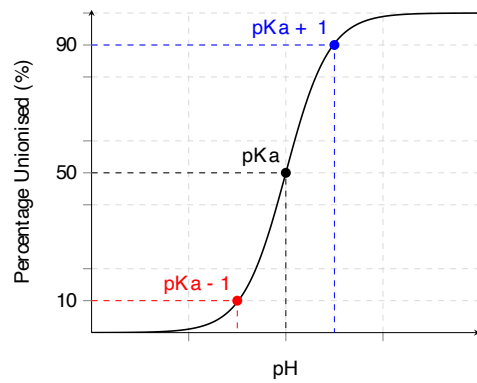
Renal & Acid-Base

Ionised potential vs pH - Acids:



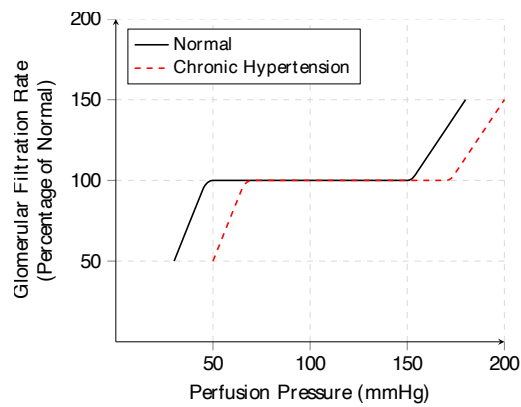
- **Acids** are ionised **above** their pKa

Ionised potential vs pH - Bases:



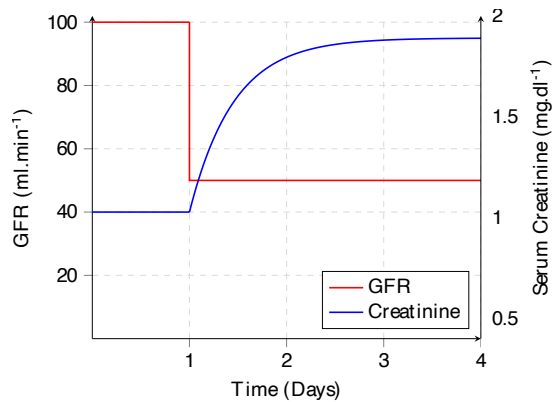
- Bases are ionised below their pK_a

Glomerular Filtration and Mean Arterial Pressure:



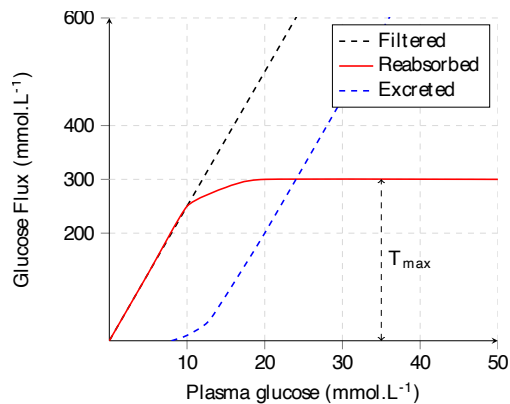
- GFR is autoregulated for a MAP between 60 and 160mmHg

Glomerular Filtration Rate and Serum Creatinine:



- At steady-state, GFR and serum creatinine are inversely proportional
- Following a step-change in GFR, it will take ~48 hours before steady-state is achieved again
During this period, estimates of GFR using serum creatinine will be less accurate.

Glucose Flux:



- As glucose is freely filtered at the glomerulus, filtered plasma glucose will be directly proportional to serum glucose
This relationship is given by the dotted black line.
- Under normal circumstances, all filtered glucose will be reabsorbed
This relationship is given by the overlap of the red and dotted black lines.

- When glucose filtration exceeds glucose reabsorption, glucose will begin to be excreted in urine.

This is given by the dotted blue line.

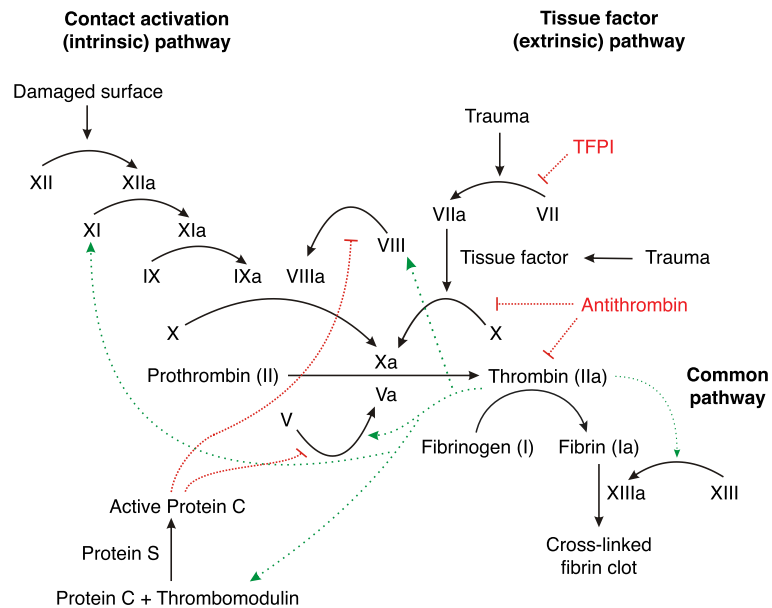
- The serum concentration of a substance at which this occurs is known as the transport maximum, or $T_{\max}$

In reality, some glucose will be filtered before $T_{\max}$ is reached. This is due to the different affinity of S-GLUT channels, and is the cause of the gentle curves seen on the plots of reabsorption and excretion.

- The serum concentration at which glucose starts to appear in urine is known as the threshold concentration
- The difference between threshold concentration and $T_{\max}$ is known as **splay**

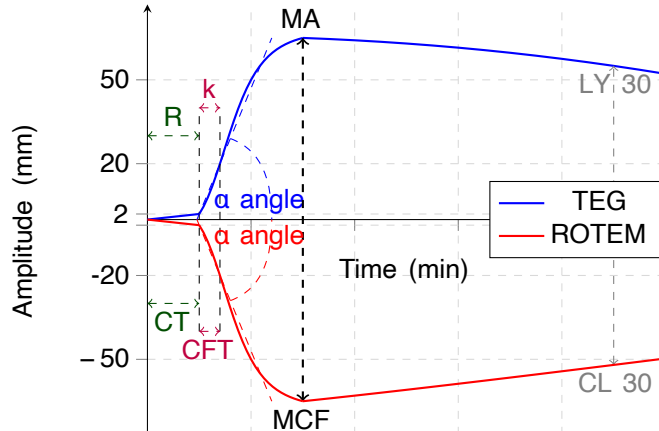
Haematology

Coagulation Cascade:



- The coagulation cascade is covered in detail under clotting

Thromboelastography:

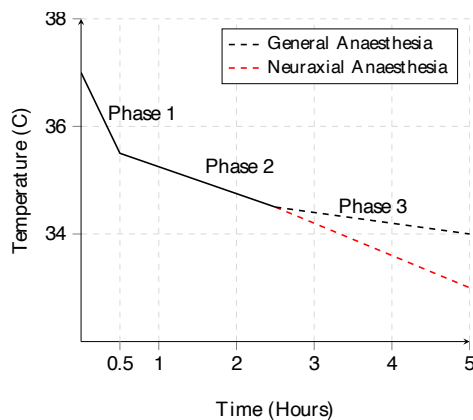


- TEG/ROTEM can be used to guide coagulopathy treatment as:
 - Prolonged R time
Indicates decreased clotting factor concentration; **give FFP**.

- Decreased α -angle/prolonged K time
Decreased rapidity of fibrinogen cross-linking;
give fibrinogen.
- Decreased MA (may be associated with prolonged K time)
Decreased maximal clot strength; **give platelets**
or **DDAVP.**
- Decreased CL 30/CL 60
Fibrinolysis; **give antifibrinolytic.**

Other

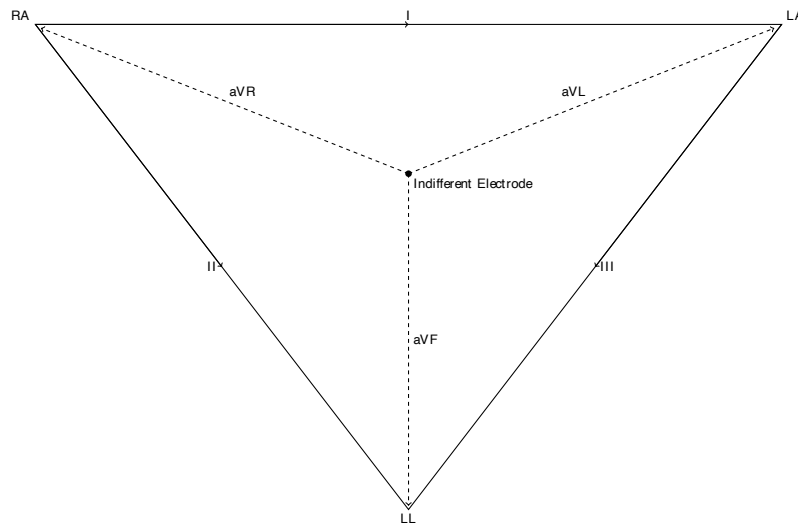
Heat Loss Under Anaesthesia:



- Heat loss under anaesthesia occurs in three phases:
 1. Rapid reduction: 1-1.5°C in 30 minutes
 2. Gradual reduction: 1°C over 2-3 hours
 3. Plateau: Further heat loss attenuated by metabolic heat reduction
 - Does not occur in neuraxial anaesthesia as vasoconstrictive responses are prevented by sympathectomy

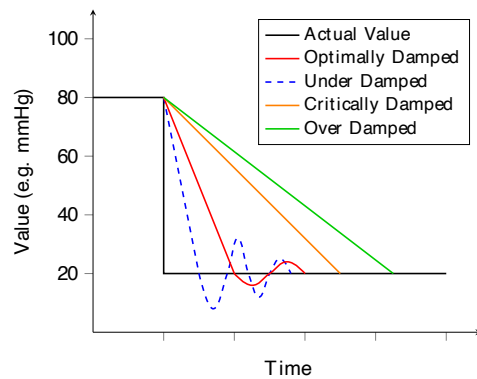
Equipment & Measurement

Einthoven's Triangle:



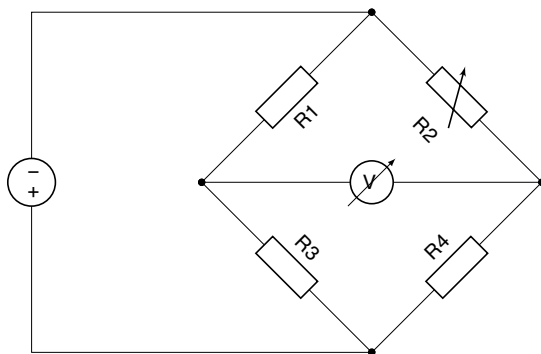
- Einthoven's triangle demonstrates the relationship between different limb leads and augmented leads on the ECG
- Understanding the triangle means one can identify misplaced ECG electrodes by the changes in ECG morphology
 - e.g. if the RA and LA electrodes are switched:
 - Lead I will invert its polarity
 - Lead II and III will be switched
 - Leads aVL and aVR will be switched
 - Lead aVF will be unchanged

Damping Coefficients:



- Following a step-change:
 - An optimally-damped waveform will return to baseline with one overshoot and one undershoot
 - An under-damped waveform returns to baseline rapidly but overshoots and undershoots several times
 - A critically damped waveform returns to baseline as fast as possible without overshooting
 - An over-damped waveform returns to baseline slower than a critically damped waveform, and does not overshoot

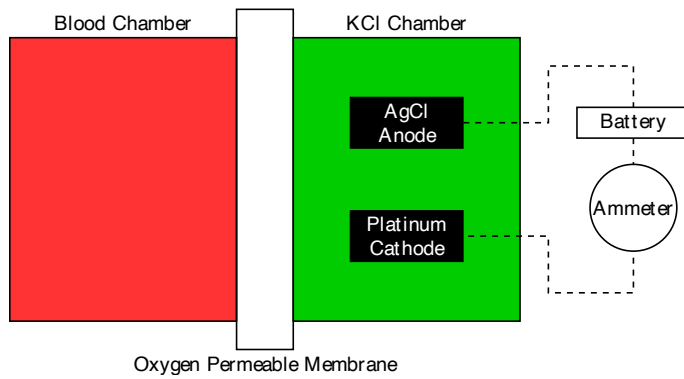
Wheatstone Bridge:



- Covered in detail under Wheatstone bridge

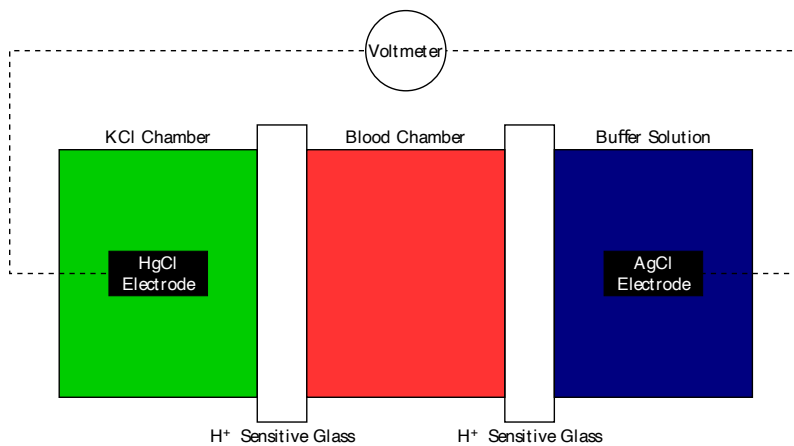
Gas Analysis

Clark Electrode:



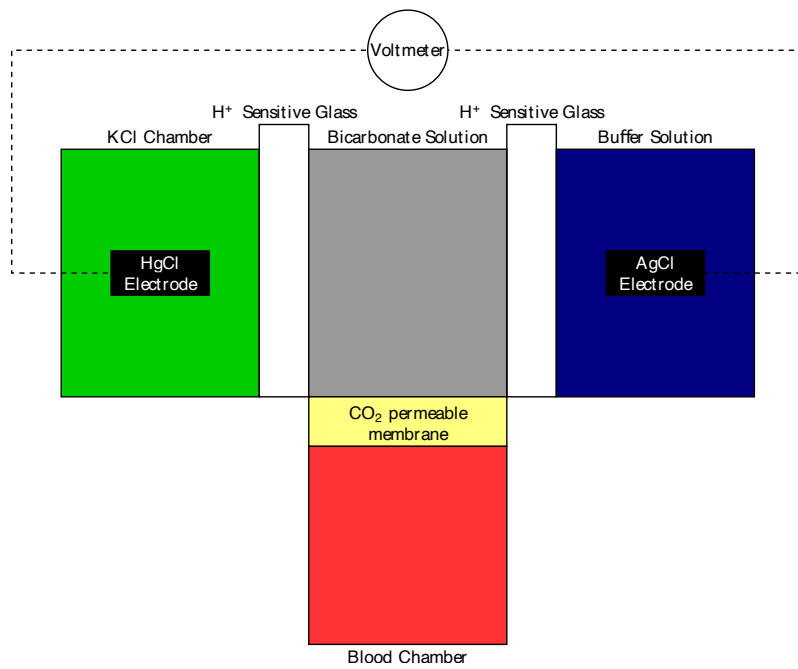
- Covered in detail under Oxygen Tension

pH Electrode:



- Covered in detail under pH Measurement

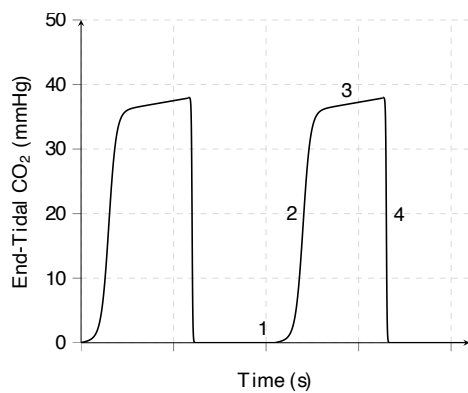
Severinghaus Electrode:



- Covered in detail under Carbon Dioxide Tension

Capnography

Capnograph:



- The capnograph waveform consists of four components:

1. Baseline
Inspiration and early dead-space expiration (containing no CO_2).
 2. Alveolar exhalation
 3. Alveolar plateau
Highest point is defined as $E_T\text{CO}_2$.
 4. Inspiration
- Variations on the waveform are covered under $E_T\text{CO}_2$
Waveform Variations
-

References

1. Wigger's Diagram (with some modifications) from Wigger's Diagram. 21/3/2012. (Image). By DanielChangMD (revised original work of DestinyQx); Redrawn as SVG by xavax. CC BY 3.0, via Wikimedia Commons.
2. Clotting Cascade 22/4/2007. (Image). By Joe D (Own work). CC BY 3.0, via Wikimedia Commons.

Laws and Equations

This appendix is a list of the key laws and equations common to many topics:

General Laws

- **Fick's Law of Diffusion**

Diffusion of a substance across a membrane is given by:

$$\dot{V} = \frac{A \times D \times \Delta P}{T}, \text{ where:}$$

- A = Area of the sheet
- D = Diffusion constant, which is proportional to the **solubility** of the gas and inversely proportional to the square root of the molecular weight, i.e. $D \propto \frac{Sol}{\sqrt{MW}}$
- ΔP = Pressure gradient
- T = Thickness of the sheet

- **Hagan-Poiseuille Equation**

Calculates the flow for a given pressure different of a particular fluid. May also be rearranged to calculate pressure or resistance.

- Given by the equation:

$$Q = \frac{\pi P r^4}{8 \eta l}, \text{ where:}$$

- Q is the flow
- P is the driving pressure
- η is the dynamic viscosity
- L is the length of tubing
- r is the radius
- Has several limitations:
 - Only models laminar flow

- Fluid must be incompressible
Not technically valid for air, but provides a good approximation when used clinically.
- Fluid must be Newtonian
- Fluid must be in a cylindrical pipe of uniform cross-section

- **Reynolds Number**

Reynolds Number is a dimensionless index used to predict the likelihood of turbulent flow. $R < 2000$ is likely to be laminar, $R > 2000$ is likely to be turbulent. Given by the equation:

- *Reynolds Number* = $\frac{2rvd}{n}$, where:
 - v is the linear velocity of fluid in m/s
 - d is the fluid density in PaS
 - r is the radius in m
 - n is the viscosity in m^2/s

Cell Physiology

- **Nernst Equation**

Calculates the electrochemical equilibrium for a given ion:

$$E (mV) = \frac{R.T}{z.F} \ln \frac{[ion]_{outside}}{[ion]_{inside}}, \text{ where:}$$

- E is the equilibrium potential for the ion
- R is the gas constant ($8.314 \text{ J.deg}^{-1}.\text{mol}^{-1}$)
- T is the temperature in Kelvin
- F is Faraday's Constant
- z is the ionic valency (e.g. +2 for Mg^{+2} , -1 for Cl^{-})

- **Goldman-Hodgkin-Katz Equation**

Calculates the membrane potential for given values of intracellular and extracellular ionic concentrations:

$$E (mV) = \frac{R.T}{F} \ln \frac{P_K[K^+]_o + P_{Na}[Na^+]_o + P_{Cl}[Cl^-]_i}{P_K[K^+]_i + P_{Na}[Na^+]_i + P_{Cl}[Cl^-]_o}, \text{ where:}$$

- P_x is the permeability constant for the ion, x
If the membrane is impermeable to x , then $P_x = 0$.

- **Henderson-Hasselbalch**

Calculates the pH of a buffer solution:

$pH = pK_a + \log \frac{[A^-]}{[HA]}$, where:

- pH is the pH of the solution
- pK_a is the pKa of the buffer
- A^- is the concentration of base
- HA is the concentration of acid

Respiratory Laws

- **Modified Bohr Equation**

The ratio of dead space to tidal volume ventilation equations the arterial - mixed-expired CO₂ difference, over the arterial CO₂. $\frac{V_D}{V_T} = \frac{PaCO_2 - PE'CO_2}{PaCO_2}$

- **Laplace's Law**

The larger the vessel radius, the larger the wall tension required to withstand a given internal fluid pressure. For a thin-walled sphere, Wall Tension (T) is half the product of pressure and radius, i.e. $T = \frac{P \cdot r}{2}$

- **Alveolar Gas Equation**

The alveolar PO₂ is equal to the PiO₂ minus the alveolar CO₂/the respiratory quotient, i.e.: $PA_{O_2} = Pi_{O_2} - \frac{PA_{CO_2}}{R}$

Gas Laws

- **Boyle's Law**

$PV = K$, i.e. pressure and volume are inversely related at constant temperature and pressure.

- Boyles Law can be used to work out how many litres of gas are remaining in gas cylinder, e.g.:
 - A standard C cylinder is 1.2L in size
 - Normal cylinder pressure is ~137bar, and atmospheric pressure is ~1bar
$$P_1 \cdot V_1 = P_2 \cdot V_2$$

$$137 \cdot 1.2 = 164$$
- Therefore, the cylinder contains ~164L of oxygen

- This can be used to calculate the volume of gas remaining in the cylinder during use, using the volume of the cylinder (fixed) and the current pressure as measured at the regulator
- **Charle's Law**
 $V = KT$, i.e. volume and temperature are linearly related when pressure is constant.
- **Gay-Lussac's Law/The Third Gas Law** $P = k.T$, i.e. pressure and temperature are linearly related when volume is constant.
- **The Universal Gas Equation**
 $PV = nRT$, i.e. combination of Boyle's, Charle's law combining each variable and the universal gas constant, R (8.13).
- **Henry's Law**
 The number of molecules of dissolved gas is proportional to the partial pressure of the gas at the surface of the liquid
- **Graham's Law of Diffusion**
 Diffusion rates through orifices are inversely proportional to the square root of the molecular weight
- **Dalton's Law of Partial Pressures**
 In a mixture of gases, each gas exerts the pressure that it would exert if it occupied the volume alone.

Cardiovascular Equations

- **Fick's Principle**
 Flow of blood through an organ equals the uptake of a tracer substance by the organ divided by the concentration difference of the substance across it, i.e.:

$$\dot{Q} = \frac{\dot{V}_{O_2}}{C_{aO_2} - C_{vO_2}}$$
- **Starling's Law of Fluid Exchange**
 Flow of fluid across the capillaries is proportional to the hydrostatic pressure difference and the oncotic pressure difference (times the reflection coefficient),

all times by the filtration coefficient, i.e.:

$$\text{Net flow out} = K[(P_c - P_i) - \sigma(\pi_c - \pi_i)]$$

- **Venous Admixture**

Calculates the shunt fraction by identifying how much mixed venous blood must be added to ideal pulmonary capillary blood to produce the identified arterial oxygen content.

$$\frac{\dot{Q}_S}{\dot{Q}_T} = \frac{C'_{O_2} - C_{a'O_2}}{C'_{O_2} - C_{\bar{v}'O_2}}$$

Equipment

- **Doppler equation**

Calculates the velocity of an object based on the change in observed frequency when a wave is reflected off (or emitted from) the object:

$$V = \frac{\Delta F s}{2F_0 \cos \theta} \text{ where:}$$

- V = Velocity of object
- F = Frequency shift
- s = Speed of sound (in blood)
- F_0 = Frequency of the emitted sound
- θ = Angle between the sound wave and the object

References

1. Davis & Kenny. Basic Physics and Measurement in Anaesthesia, 5th Edition.
2. Gorman. RAH Diving and Hyperbaric Medicine. Chapter 3: The physics of diving.

Structures for SAQs

Structured answers are vital:

- Structure aids both learning and recall of information
- A structured format is easy to follow less likely to irritate the drunk/tired/hungover mind of the examiner
- You may get marks for incomplete answers if your structure demonstrates you have an understanding of the topic, even if the details are not filled in

Some questions lend themselves more easily to a particular structure than others, but all questions can be made to fit even a basic structure.

Regulation of Physiological Responses

- Sensor
- Integration
- Effector

Change in Level of a Substance

- Intake
- Distribution
- Elimination

Compare and Contrast (Drugs)

- **Class**
- **Pharmaceutics**
 - Uses
 - Chemical
 - Presentation
 - Heat/light stability
 - Routes of administration
 - Doses
- **Pharmacokinetics**
 - **A**bsorption
 - **D**istribution
 - **M**etabolism
 - **E**limination
- **Pharmacodynamics**
 - Main Action
 - Mode of Action
 - Effects (See the physiological approach)
 - Side Effects
 - Toxic effects
 - Allergic
 - Dose-dependent
 - Idiopathic
 - Drug interactions

Describe Why Two Drugs are Used in Combination

- **Definition**
Brief, of each drug.
 - Consider the components of anaesthesia that each provides
- **Pharmacokinetics**
 - Interactions
 - Relative onset
 - Metabolism

- No effect/Induces/Inhibits
 - Elimination
- Pharmacodynamics
 - Isobologram
 - Synergistic/additive/antagonistic.
 - Then list the effects of each drug, and how they are modified by the other
 - e.g. if drugs are synergistic, then decreased doses will be required
 - Increases beneficial effects
 - Decreases adverse effects

Describe the Physiology of...

- Respiratory
 - Bronchodilation/constriction
 - Vasodilation/constriction
 - V_T
 - RR
 - Secretion
 - Laryngeal reflexes
- CVS
 - Preload
 - Contractility/Pump effects
 - Inotropy
 - Chronotropy/rhythm
 - Dromotropy
 - Lusitropy
 - Bathmotropy
 - Nodal effects
 - Coronary Blood Flow
 - Afterload/Pipe effects
 - SBP
 - DBP
 - MAP
 - SVR
 - PVR

- Intraarterial injection
- CNS
 - Sedation
 - Analgesia
 - Pro/anticonvulsant
 - Amnestic
 - Cerebral Metabolic Rate
 - Cerebral Blood Flow
 - ICP
 - IOP
- Musculocutaneous
 - Blood Flow
 - NMJ
- Endocrine
 - Gynaecomastia
 - Hair
 - Bone
- Renal and GU
 - Renal Blood Flow
 - Nephrotoxicity
 - Bladder tone
 - Uterine tone
- GIT and Hepatic
 - Hepatotoxicity/LFTs
 - Secretions
 - Gastric emptying
 - N/V/D/C
- Haematological
 - G6PD
 - Porphyrias
 - Bone marrow effects
- Immunological
 - Anaphylaxis
 - Histaminergic
 - Neutrophil function
- Metabolic

Anatomical Structure

- Anatomy of the structure
- Relationships
- Relevant surface anatomy
- Layers of dissection

Physics and Measurement

- Definition
 - Uses
 - Physical principles
 - Components
 - Calibration
 - Advantages/Disadvantages
-

References

- Dr. Podcast
- Wisdom from drunk, tired, and/or hungover examiners.